PART 7 — STRUCTURED ORAL EXAMINATION: 51 CLINICAL SCENARIOS

Read the three set-pieces first. They precede the cases and cover the technical ground that recurs in almost every scenario: placement timing and loading protocol; bone grafting materials with their advantages, disadvantages and complications; and incision and flap design. The 51 cases that follow include RCSEd's own published sample case (Appendix E), worked in full, and cover the breadth of the blueprint. Each is structured across the four marked domains with a full examiner question flow — enough material to sustain the 15 minutes.

Format reminder: 8 cases, each **10 minutes preparation + 15 minutes oral**, two examiners, **4 domains** × **5 marks each = 20 marks per examiner per case**, pass = **mean of 3** across all domains and cases.

How to use the 10 minutes of preparation

Write four headings on your paper and fill each with bullets. Do this for every case, every time, until it is automatic.

Heading	What to write
1. INTERPRET	Summary in one sentence: *"a* [*age*] [*sex*] *with* [*problem*] *wanting* [*expectation*]*."* Then: relevant medical/social/dental factors and **why each matters**; key clinical findings; key radiographic findings; **what I still don't know**; differential and working diagnosis.
2. PLAN & CONSENT	Prognosis of each relevant tooth; **all** options including no treatment; my recommendation and why; **SAC classification**; sequencing; what I'd consent for; whether I'd refer.
3. PROCEDURES	Surgical: timing, flap, site prep, augmentation, one/two stage, closure. Prosthetic: retention, abutment, occlusion, loading protocol, laboratory.
4. OUTCOMES	What could go wrong and how I'd manage it; what I'd measure; maintenance regime and interval; how I'd know it had succeeded; what I'd do if the patient were unhappy.

Opening line that scores. Never begin with a fact. Begin with a *framed summary*: *"This is a 45-year-old woman with suppuration around two linked mandibular implants placed three years ago, who has not seen a hygienist for two years and wants to know why her gum is sore and whether her implants are all right. My working diagnosis is peri-implantitis affecting both implants, and my priority is to establish the diagnosis, identify the risk factors and decide whether the implants are retainable."*

Nine phrases that raise your mark in any case: 1. "On the SAC classification I would grade this as..." 2. "The findings I would want, that I don't have, are..." 3. "The options, including no treatment, are..." 4. "This is what I would tell the patient in their own words..." 5. "The material risks for *this* patient are..., because of their occupation/expectations." 6. "The evidence for this is..., and I would grade it as..." 7. "This is beyond what I would do as a general dentist with an interest in implants, so I would..." 8. "I would record..." 9. "I would review at... and measure..."

Where the marks are in the SOE

The structured oral examination is not a viva about implant dentistry in general. It is a structured assessment of whether you reason safely about a specific patient, across four declared domains, at the level of a general dentist with an interest in implants. Seven observations about how cases at this level are built should shape your preparation.

1. Expect single-tooth and small-span problems, not full-arch reconstruction. Full-arch and complex reconstructive work is explicitly outside the scope of this Membership. The examinable material is the set of decisions a general dentist makes about **one tooth, one site, one existing implant**. Prepare depth on the single unit rather than breadth on the complex case.

2. Expect cases where something has already gone wrong. A large proportion of realistic scenarios present *after* treatment — a malpositioned implant placed by someone else, recession and metal show-through, a deeply placed fixture, a failing restoration, peri-implant disease. These test whether you can **diagnose a failure, identify its cause, decide whether the situation is retrievable, and have an honest conversation about it**. That is a different skill from planning a virgin case, and it is examined at least as often.

3. If an occupation is given, it is not decoration. A patient described as a lawyer, a teacher, a musician or someone working in the entertainment industry is there to test **Montgomery**: materiality of risk is judged from the position of *this* patient. Build the consent discussion around the occupation explicitly — what this particular person needs to know, and why a risk that is immaterial to one patient is material to another.

4. The medical history is either the pivot of the case or deliberately clear. When a complex history is given — antiresorptives, radiotherapy, poorly controlled diabetes, anticoagulation — that is the case. When you are told the medical history is clear, that escape route has been closed deliberately, and the examiners want your **local, occlusal and prosthetic** reasoning instead. Notice which kind of case you are in.

5. Occlusion is examined in its own right. Occlusion, interocclusal space and parafunction appear repeatedly across the blueprint and are routinely under-prepared. You must be able to describe an occlusal examination, name and justify an occlusal scheme, explain implant-protective occlusion and its biological basis, and manage a bruxist prosthodontically — including what you would say to him about prognosis.

6. Periodontal disease is examined in its own right. Be fluent in the **2017 staging and grading**, the **EFP stepwise approach** and its stability criteria, and the relationship between a history of periodontitis and peri-implantitis. A failing periodontal dentition is a legitimate implant case, and the examinable question is usually *whether, when and on what conditions* — not *which implant*.

7. Interim prostheses are part of the case data. If you are told what the patient is wearing while the site heals, you are expected to comment on it: what it is doing to the healing tissues, whether it is the right choice, and what you would change. The same applies to anything else stated in passing — a smile line, an attendance pattern, a stated expectation.

> KEY POINT
>
> **Preparation priority for the SOE, in order:** the single missing anterior tooth in a demanding patient → the failing or malpositioned existing implant → occlusion and parafunction → the periodontally compromised dentition → the edentulous and overdenture patient → the medically complex patient → everything else.

How the fifteen minutes is spent, and what the examiner is holding

The cases in this part are written to be **worked from**, so each carries more questions than any single oral will use. This short section is the other half of that: how the time is actually divided, what the examiner is looking at while you speak, and how to rehearse a case against the clock rather than reading it.

Every case in this part then carries its own run sheet, printed after the question flow: a time check that says plainly whether the printed material fits fifteen minutes; a **clock table** setting the four domains against the minutes they are actually given; **the spine** — the twelve answers to have ready if the case had to be run in half the time; and **where the marks are lost in that particular case**. Rehearse from those, not from the full flow.

The shape of the fifteen minutes

Clock	What happens	What it is testing
Before you enter — 10 minutes	Preparation with the case material: the scenario, the radiographs and any photographs or cone beam sections	Whether you arrive with a **position** rather than a summary. Use the last two minutes to decide your working diagnosis, your SAC grading and the three things you must establish.
0:00–0:20	Settling, confirming who you are and which case	Nothing. Do not start talking into it.
0:20–1:20	**Your opening presentation — about sixty seconds, uninterrupted**	Structure. Who the patient is, what they want, the key findings, the working diagnosis, and what you still need to establish. This is the only monologue you get; script it.
1:20–14:00	**Roughly 30 to 40 short questions**, drilled down in chains, moving as soon as a point is made	Everything else. Answers of **twenty to forty seconds**, in the four marked domains.
14:00–15:00	Closing questions, usually outcomes, records or candour	The endings candidates run out of time for. Keep something in reserve here.

The single most useful number in that table is **thirty to forty questions**. It tells you that the unit of currency is a **short answer**, not a discourse — an answer of ninety seconds costs you three questions you will never be asked, and each of those was a mark you might have had. If you have said your point, stop talking.

The four domains, and how a case moves through them

An examiner works the case across the four published domains, and each of the cases in this part is laid out in the same order so that you can rehearse the transitions:

Domain 1 — clinical interpretation and reasoning. Nearly always opens with **groundwork below the implant material**: surface anatomy, denture principles, occlusal schemes, periodontal staging, healing biology, reading the images. Candidates who have revised only implantology are caught here.

Domain 2 — clinical decision making and treatment planning. The options, the argument between them, the SAC classification, and the point at which you refer.

Domain 3 — decision making in relation to treatment procedures. How you would actually do it, in sequence, with the specific numbers and materials.

Domain 4 — interpretation and problem solving of clinical outcomes. What goes wrong, how often, how you would detect it, how you would prevent it, what you would record and what you would tell the patient.

Each domain carries **five marks**, marked independently; the descriptors and the difference between a 3 and a 5 are set out in §1.6. The practical consequence is that a case is not one performance but four, and **a weak Domain 4 cannot be rescued by a strong Domain 1**. If you notice that ten minutes have gone and nobody has asked you about complications, expect them now and have the prevalence figures ready.

How to rehearse a case properly

> KEY POINT
>
> **Work a case out loud, timed, with a colleague holding the page.** Give them these instructions: *let me open for sixty seconds without interrupting; then ask questions from the flow as fast as I answer them; move me on as soon as I have made the point; and mark down every time you had to prompt me.* Then compare the number of prompts against the descriptors — a prompted answer is a pass, an unprompted one is a good mark, and the count of prompts is the most honest feedback available to you before the day.

Three things to watch for in your own rehearsal, in order of how often they cost marks:

Not volunteering the risk. The competent-sounding answer that never mentions what could go wrong is a 3 at best.

Not stopping at the competence line. Every case in this part contains a point beyond which the right answer is *"that is beyond what I would do; I would refer, and this is who to."* Failing to see it is the commonest single reason a candidate does not pass.

Not closing. Almost every answer improves by ending with **what you would record and what you would tell the patient, in the patient's words**. Practise saying both in one breath, so that they survive the pressure of the clock.

THREE SET-PIECE ANSWERS TO PREPARE BEFORE ANY CASE

Whatever the presenting problem, an SOE case tends to travel through the same three technical territories. Prepare them once, properly, and you will have a fluent answer available in almost every case.

- *"When would you place the implant, and when would you load it?"*
- *"What would you graft with, and why that material?"*
- *"Describe the flap you would raise."*

These are **set pieces**. They are asked in nearly every case, they are worth marks in domain 3 every time, and they are the questions on which candidates most often produce a vague answer to a precise question. Each of the three below is written as a **spoken answer** you can deliver in 60 to 90 seconds, followed by the follow-up questions the examiners actually use and the traps in each.

Learn these three until you can deliver them cold. Then rehearse them **inside** the cases listed at the end of each section, because the marks are for applying them to the patient in front of you, not for reciting them.

Set-piece 1 — Placement timing and loading protocol

1.1 The single most important thing to say first

> **"Placement timing and loading protocol are two separate decisions. Type 1 to 4 describes *when I place the implant relative to the extraction*. Immediate, early and conventional loading describes *when I attach a restoration relative to the placement*. They are chosen independently, and any placement type can in principle be combined with any loading protocol — but the combinations are not equally well documented."**

Candidates routinely say "immediate implant" when they mean "immediate loading", or answer a question about loading by describing placement. Saying the sentence above at the start of the answer immediately separates you from that, and it frames everything that follows.

1.2 Placement timing — Types 1 to 4

Classification from the ITI/Hämmerle consensus (Hämmerle, Chen & Wilson 2004), which describes placement by the **state of healing of the extraction socket**, not simply by elapsed days.

Type	Name	Timing	State of the socket	Principal indication
Type 1	**Immediate**	Same day as extraction	No bone or soft tissue healing	Intact socket walls, facial wall ≥1 mm, no infection, thick phenotype, primary stability achievable apically/palatally, ideal 3-D position achievable
Type 2	**Early, with soft tissue healing**	**4–8 weeks**	Soft tissue healed and closed; no appreciable bone fill	The commonest compromise in the aesthetic zone: mucosa can be advanced for tension-free closure, infection has resolved, major resorption not yet complete
Type 3	**Early, with partial bone healing**	**12–16 weeks**	Substantial bone fill of the socket	Where more bone volume is wanted before placing; larger sockets; molars
Type 4	**Late**	**≥6 months**	Fully healed ridge	A healed site; after ridge preservation; after staged augmentation; where the patient could not proceed earlier

What to say about each, with the trade-off — this is what earns the mark:

Type 1 — immediate. *Advantages:* fewest surgical episodes; shortest overall treatment time; the socket guides the osteotomy; may preserve soft tissue architecture; high patient satisfaction. *Disadvantages:* **mid-facial recession is the principal risk**, greatest with a thin phenotype and a buccally positioned implant; the drill skids off the palatal wall so ideal position is harder to achieve; primary stability is less predictable; **it does not prevent resorption of the buccal bundle bone**, which is ligament-dependent and disappears with the tooth; the highest aesthetic complication rate; technique-sensitive. *Absolute pre-requisites:* no acute infection with suppuration, intact socket walls with a facial wall of at least 1 mm, sufficient apical and palatal bone for primary stability, the socket within the alveolar bone envelope, and a horizontal "jumping" gap of at least 2 mm at the implant shoulder which is grafted with a slowly resorbing xenograft.

Type 2 — early with soft tissue healing. *Advantages:* the soft tissue has healed and can be advanced, so **tension-free primary closure over a simultaneous graft is achievable**; any infection has resolved and can be assessed; a substantial proportion of the ridge is still present because most of the horizontal resorption has not yet completed. **This is the workhorse protocol in the aesthetic zone.** *Disadvantages:* two surgical episodes; the buccal plate may already have begun to resorb; a graft is often needed.

Type 3 — early with partial bone healing. *Advantages:* more bone in the socket, so better primary stability and less need for grafting than Type 2; useful in large or multi-rooted sockets. *Disadvantages:* more ridge resorption has occurred, particularly buccally; longer for the patient.

Type 4 — late. *Advantages:* a fully healed, predictable site; **the best-documented protocol**; no compromise from residual infection; simple surgery. *Disadvantages:* maximum ridge resorption — roughly 50% of the width is lost by 12 months, of which about two-thirds occurs in the first three months, so **augmentation is frequently required**; the longest treatment time; the patient wears an interim prosthesis throughout.

> **The examiner's favourite trap.** "Why not place immediately, it saves time?" The scoring answer names the pre-requisites and then says: *"Immediate placement does not preserve the buccal plate. The bundle bone is dependent on the periodontal ligament and resorbs whatever I do. What immediate placement saves is time and surgical episodes — it does not save bone, and it costs me control of the three-dimensional position, which in the aesthetic zone is the thing I am least willing to lose."*

1.3 Loading protocols

Definitions from the ITI consensus (Gallucci et al.). **Note the distinction between *restoration* and *loading*** — this is a favourite examiner question.

Protocol	Definition
Immediate restoration (Type A, non-occluding)	Restoration fitted within **1 week** of placement and deliberately kept **out of occlusal contact** in intercuspation and in all excursions
Immediate loading (Type A, occluding)	Restoration fitted **in occlusal contact** within **1 week** of placement
Early loading (Type B)	Restoration in occlusal contact between **1 week and 2 months**
Conventional loading (Type C)	Restoration in occlusal contact **later than 2 months**
(*Delayed loading*)	An older term for loading later than a conventional healing period; avoid it, and use the classification above

Conventional healing times to quote: classically **3 months in the mandible and 6 months in the maxilla**; modern moderately rough and hydrophilic surfaces have shortened this, and 6 to 8 weeks in favourable mandibular D2 bone is now well supported. In **grafted bone**, extend it — typically **4 to 6 months after simultaneous guided bone regeneration** and **6 to 9 months after staged augmentation**, because regenerated bone is less dense, less mature and has a lower initial bone-to-implant contact.

Criteria that must be met before immediate restoration or loading — all of them:

1. **Insertion torque above 35 Ncm, or ISQ above 70**
2. **No direct occlusal contacts** for an immediate restoration; controlled, light contacts only if loading
3. **No parafunction**
4. Adequate bone quality and quantity — **D4 bone is a contraindication**
5. No simultaneous major augmentation
6. Patient compliance with a soft diet and with not loading the provisional
7. For multiple implants, **cross-arch splinting** to distribute load

The biological basis, which you should be able to give in one sentence:

> *"The threshold is micromotion. Above roughly 50 to 150 micrometres at the interface, fibrous repair replaces bone formation, so every criterion for immediate loading — the torque, the ISQ, the absence of occlusal contact, the absence of parafunction — exists to keep micromotion below that threshold during the vulnerable period."*

And the vulnerable period itself:

> *"Total stability is primary mechanical stability plus secondary biological stability. Primary stability falls as the bone at the interface remodels, and secondary stability rises as new bone forms. The trough is at about **two to four weeks**, which is why an implant that felt solid at placement can be mobile at three weeks, and why early loading protocols must be selected rather than assumed."*

1.4 The combination grid — and its evidence grading

This is the grid from the ITI decision tree. Being able to reproduce it, and to grade your own plan against it, is one of the most efficient ways to demonstrate evidence-based practice in an SOE.

Placement	Immediate restoration/loading (A)	Early loading (B)	Conventional loading (C)
Immediate placement (Type 1)	Type 1A — **CD**	Type 1B — **CD**	Type 1C — **SCV**
Early placement (Type 2–3)	Type 2–3A — **CID**	Type 2–3B — **CID**	Type 2–3C — **SCV**
Late placement (Type 4)	Type 4A — **CD**	Type 4B — **SCV**	Type 4C — **SCV**

The grading scale, best to worst:

- **SCV** — scientifically and clinically validated
- **CWD** — clinically well documented
- **CD** — clinically documented
- **CID** — clinically insufficiently documented (*includes protocols that are simply not documented*)

What this grid tells you, and what to say:

> *"Conventional loading is scientifically and clinically validated whichever placement protocol I use. Immediate placement with immediate restoration is only clinically documented — it is a legitimate protocol in selected cases, but I would tell the patient it is less well validated than the alternative. And early placement combined with immediate or early loading is **clinically insufficiently documented**, so if a patient asked me for it I would explain that the evidence does not support it and I would not offer it."*

1.5 Delivering it in the SOE

When asked *"when would you place and load this implant?"*, answer in this order and you cannot go wrong:

1. **Separate the two decisions** in one sentence.
2. **State the placement type with its number and its timing**, and say what in *this* case determines it — socket integrity, buccal wall thickness, infection, phenotype, aesthetic risk.
3. **State the loading protocol**, and check the criteria out loud against this patient — torque, ISQ, occlusion, parafunction, bone quality.
4. **Grade it** against the ITI evidence table.
5. **State your fallback** — what you would do if the intra-operative findings differ from the plan.

> **Model sentence.** *"I would plan a **Type 2, early placement at six to eight weeks**, because the buccal wall is under 1 mm so immediate placement is excluded by the decision tree, and the soft tissue will have healed enough for tension-free closure over a simultaneous graft. I would use **conventional loading beyond two months**, which is Type 2–3C and scientifically and clinically validated — early or immediate loading here would be Type 2–3B or A, which are clinically insufficiently documented. My fallback is that if at surgery I cannot achieve the correct three-dimensional position with at least 2 mm of buccal bone, I stop, graft, and place at six to nine months."*

1.6 The follow-up questions examiners actually ask

"What is the difference between immediate restoration and immediate loading?" — Occlusal contact. Restoration is out of occlusion; loading is in occlusion. Both are within one week.

"You said 35 Ncm. What if you get 25?" — I would not restore immediately. I would place a healing abutment or submerge, and load conventionally, confirming stability with resonance frequency analysis before restoring, and looking for a rising rather than falling ISQ trend.

"Why can't you load an implant in D4 bone immediately?" — Because primary stability depends on mechanical interlock with the bone, and D4 has a thin cortex over sparse trabeculae, giving the lowest insertion torque and the least resistance to micromotion. It is the bone type with the highest failure rate, and immediate loading would exceed the micromotion threshold.

"What is progressive loading?" — A Misch concept: in soft bone, occlusal load is increased gradually, through a provisional in acrylic kept out of or in light occlusion and then in progressively firmer contact over weeks, to allow the bone to mature and remodel before full function. I would say I know the concept and that the evidence supporting it is modest.

"When would you use a two-stage rather than a one-stage protocol?" — Where a membrane or graft must be protected from exposure; where primary stability is compromised; where the soft tissue is thin or needs to be manipulated at second stage; and where the patient must wear a mucosa-borne denture over the site.

"How long after a sinus floor elevation would you load?" — Following the ITI decision tree: with simultaneous placement, prosthetic loading at 3 to 6 months. Where placement is staged, implant placement 4 to 8 months after grafting with particulate autogenous bone, or 9 to 12 months with bone substitute alone, and then loading 3 months after placement.

"The patient wants teeth on the day. What do you say?" — Explain the criteria, assess whether they are met, grade the protocol on the evidence, and offer a well-made interim prosthesis that does not load the site. Patient autonomy allows refusal of treatment; it does not compel me to provide a protocol that is not in their interest.

Rehearse this set-piece in: 2, 4, 9, 14, 21, 22, 24, 32, 41 and 51.

Set-piece 2 — Bone grafting: materials, indications, and complications

2.1 Start with the three mechanisms

> **"Every graft works by one or more of three mechanisms. Osteogenesis — the graft contains living cells that form bone, and only autogenous bone does this. Osteoinduction — the graft induces host mesenchymal cells to differentiate into osteoblasts, mediated by bone morphogenetic proteins; autogenous bone and, variably, demineralised allograft. Osteoconduction — the graft acts as a scaffold along which bone grows from the host; all graft materials do this. Some authors add osteopromotion, meaning the enhancement of regeneration without the material itself being osteoinductive — which is how a barrier membrane and, arguably, platelet concentrates act."**

Saying that first tells the examiner you understand *why* you are choosing a material, not just which one you happen to buy.

2.2 The four categories — pros, cons and consent

Autograft — the patient's own bone

Sources within a general dentist's scope: bone collected from the osteotomy in a **sterile bone trap**; **bone scraper** shavings from the external oblique ridge or the buccal cortex adjacent to the site; **rongeur** collection during alveoloplasty; and the **maxillary tuberosity**. **Sources outside it:** mandibular **ramus** and **symphysis block** grafts, and all extraoral sites (iliac crest, calvarium, tibia). *Recognise them, do not plan them.*

Advantages	Disadvantages
The only **osteogenic** material — contains viable osteoblasts and osteoprogenitor cells	**Donor site morbidity** — the single biggest drawback
Also **osteoinductive** and osteoconductive	**Limited volume**, particularly from intraoral particulate sources
Fastest incorporation and revascularisation	**Rapid and unpredictable resorption**, sometimes substantial
No disease transmission, no immunological risk	Extra surgical time; a second surgical site
No ethical, religious or cultural objection	Contamination with saliva and blood when collected in a trap
No cost	Technique-sensitive; particle size affects revascularisation

Allograft — human donor bone, processed

Forms: FDBA (freeze-dried bone allograft — mineralised, better space maintenance), **DFDBA** (demineralised — the demineralisation exposes bone morphogenetic proteins, so it is more osteoinductive but less space-maintaining), cortical or cancellous chips, and mineralised cortical/cancellous blends.

Advantages	Disadvantages
No donor site and no donor morbidity	**Human origin — must be disclosed at consent**; some patients decline
Unlimited supply, wide range of particle sizes	Theoretical disease transmission risk (effectively eliminated by processing; no reported cases from modern products)
Osteoconductive, and DFDBA variably osteoinductive	Osteoinductive potency **varies between donors and between batches**
Handles well; well documented in periodontal and socket applications	Regulated as human tissue — sourcing, traceability and record-keeping obligations
Resorbs and is replaced more predictably than xenograft	Slower and less complete incorporation than autogenous bone

Xenograft — another species

The standard material is deproteinised bovine bone mineral (DBBM); porcine and equine products also exist. The organic component is removed by heat and chemical treatment, leaving an **anorganic mineral scaffold** with a trabecular architecture very similar to human bone.

Advantages	Disadvantages
Outstanding volume and contour stability — resorption is very slow or negligible, which is why it is the material of choice for the jumping gap, ridge preservation and buccal contour augmentation in the aesthetic zone	**Bovine or porcine origin — must be disclosed**; religious, dietary, cultural and vegetarian objections are common and legitimate
Excellent osteoconduction, highly documented, the most-studied material in implant dentistry	**Osteoconductive only** — no osteogenic or osteoinductive capacity
Unlimited supply, no donor site	Residual particles persist for years and are visible histologically and radiographically, and reduce the proportion of vital bone
Predictable, easy handling	Theoretical prion transmission risk — regarded as eliminated by processing, but must be mentioned
Radiopaque, so the graft can be seen on follow-up radiographs	Slower and less complete turnover than allograft

Alloplast — synthetic

Examples: β-tricalcium phosphate (resorbable), **hydroxyapatite** (slowly or non-resorbable), **biphasic calcium phosphate** (a blend of the two, allowing the resorption rate to be tuned), **bioactive glass**, **calcium sulphate** (rapidly resorbing, useful as a binder or barrier), and polymers.

Advantages	Disadvantages
No donor site, no animal or human origin — **avoids all ethical, religious and cultural objection**	**Osteoconductive only**
Unlimited supply, low cost, long shelf life	Generally **less well documented in implant sites** than DBBM
Resorption rate can be selected by choosing the material	Resorption may be too fast (β-TCP) or too slow (dense HA) for the application
No disease transmission risk whatsoever	Particle migration and, occasionally, sequestration

Biologics and adjuncts

Autologous platelet concentrates (PRP, PRF) — the best-documented benefit is **soft tissue healing and reduced post-operative morbidity**; evidence for hard tissue gain is limited. **Enamel matrix derivative. rhBMP-2** — a potent osteoinductive growth factor used in selected specialist applications, associated with marked post-operative oedema. **rhPDGF.** Know what each is; be measured about the claims.

2.3 Composite grafting — the answer that usually wins

> *"In practice I use a **composite graft**. I place **autogenous chips collected from the osteotomy directly against the implant surface**, where their osteogenic cells are most valuable, and layer **deproteinised bovine bone mineral** over the outer surface, where its very slow resorption maintains the buccal contour for years. That combines the biology of autogenous bone with the volume stability of a xenograft, and it addresses the principal weakness of each used alone."*

A typical ratio is 1:1, and the construct is covered with a barrier membrane extending 2 to 3 mm onto sound bone.

2.4 Choosing the material — the reasoning to say out loud

Clinical situation	Material of choice	Why
Jumping gap at immediate placement	**DBBM alone**	Volume stability against buccal resorption; no space-maintenance problem because the socket walls contain it
Ridge preservation	**DBBM ± membrane or socket seal**	Slow resorption maintains dimension over the 4–6 month healing period
Buccal dehiscence/fenestration at placement	**Composite: autogenous chips + DBBM + collagen membrane**	Biology against the implant, contour stability outside
Staged horizontal augmentation	Composite ± tacked membrane; consider referral	Requires space maintenance and rigid stabilisation
Transcrestal sinus elevation	DBBM, or autogenous, or a composite	Contained space; the graft elevates the membrane
Patient declines animal-derived material	**Autogenous + alloplast + synthetic membrane**	Ethically acceptable; explain the difference in documentation

Clinical situation	Material of choice	Why
Vertical augmentation	—	**Refer.** Least predictable of all augmentation procedures

2.5 Complications of bone grafting — the part candidates omit

Examiners ask this precisely because most candidates can list materials and cannot list what goes wrong. Organise it as **donor site / recipient site / early / late / material-specific**, and it is impossible to dry up.

Donor site complications

Site	Complications
Osteotomy bone trap	Contamination with saliva and blood; very limited volume; no separate morbidity
Bone scraper — external oblique ridge / buccal cortex	Bleeding; perforation into the marrow space or the lingual plate; **injury to the mental nerve** if too anterior; heat generation if used dry; post-operative discomfort at the harvest site
Mandibular ramus (block — out of scope, know the risks)	**Inferior alveolar nerve injury**; lingual nerve injury; trismus; mandibular fracture; limited volume; difficult access
Mandibular symphysis (block — out of scope, know the risks)	**Altered sensation of the lower incisors and chin, which is common and may be permanent**; devitalisation of incisor teeth; **ptosis of the chin ("witch's chin") if the mentalis is not correctly repositioned**; haematoma; scarring; contour deformity
Maxillary tuberosity	Low-density bone of limited structural value; sinus perforation; bleeding
Palate (for soft tissue grafts)	**Haemorrhage from the greater palatine artery**; significant post-operative pain; delayed healing by secondary intention; necrosis if the flap is left too thin; sensory disturbance

Recipient site — intra-operative

Haemorrhage; perforation of the lingual or buccal plate; **injury to the mental or inferior alveolar nerve during a periosteal releasing incision**, which is a specific and under-recognised risk in the premolar region; **Schneiderian membrane perforation**; inability to achieve tension-free closure; flap perforation while releasing; inadequate space maintenance so the membrane collapses; failure to stabilise the graft.

Early post-operative

- **Wound dehiscence and membrane exposure — the commonest complication of all.** Causes: tension at closure, a suture line lying over the membrane, denture pressure, and smoking.
- **Infection** of the graft, which may require removal of the graft and membrane.
- Excessive swelling, haematoma and bruising, greater than for placement alone.
- Pain, particularly at a donor site.
- Suture failure; flap necrosis if the blood supply was compromised by flap design.
- **Loss of vestibular depth and of keratinised tissue** after extensive coronal advancement — a real and permanent cost of achieving tension-free closure.

Management of exposure, by membrane type: an exposed **collagen** membrane without infection is managed conservatively with chlorhexidine, no brushing over the site and close review, and often heals; an exposed **ePTFE or infected** membrane should be **removed early** to preserve whatever bone has formed. **Dense PTFE** may be left exposed intentionally over a socket and removed at 3 to 4 weeks.

Late

- **Graft resorption and volume loss**, with failure to achieve the planned dimension.
- **Insufficient bone for the planned implant**, requiring a narrower or shorter implant, a repeat graft, or abandonment.
- **Implant failure in grafted bone** — regenerated bone is less dense with a lower bone-to-implant contact, which is why healing periods are extended.
- **Sequestration** of graft particles through the mucosa months later.
- **Sinusitis, oro-antral communication, or graft displacement into the sinus** after elevation procedures.
- **MRONJ** in patients on antiresorptives.

- Scarring and mucogingival distortion; a visible or palpable ridge irregularity.

Material-specific

Slow-resorbing xenograft leaving a low proportion of vital bone; rapid-resorbing alloplast losing volume before bone forms; particle migration; and, with any allograft or xenograft, the **consequences of inadequate consent** if the patient later discovers the origin of the material.

2.6 Consent for grafting — the blueprint requires this specifically

Appendix A requires you to *"recognise and relate the importance of consent in the context of undertaking surgical procedures and the use of grafting materials and barriers."* Consent must include:

1. **The origin of the material** — autogenous, human donor, **bovine or porcine**, or synthetic — and the origin of the **membrane**, which is usually porcine collagen
2. The processing, and the **theoretical and extremely low** disease transmission risk
3. That **alternatives exist** if animal-derived or human-derived material is unacceptable, and what choosing them means for predictability
4. **Donor site morbidity** if autogenous bone is to be harvested
5. That **grafts can fail**, that exposure is the commonest complication, and what happens if the graft is lost
6. The **additional cost, time and morbidity**
7. That **smoking markedly reduces success**
8. That the material, brand and **lot number will be recorded** and given to the patient

2.7 Delivering it in the SOE

> **Model sentence.** *"I would use a **composite graft** — autogenous chips from the osteotomy against the implant surface for their osteogenic potential, with a **deproteinised bovine bone mineral** layered outside for volume stability, covered by a **native collagen membrane** extending 2 to 3 mm onto sound bone. I would decorticate the buccal plate first to open the marrow spaces, and achieve tension-free primary closure with a periosteal releasing incision. I would consent the patient specifically to the **bovine origin of the graft and the porcine origin of the membrane**, and offer an autogenous-plus-synthetic alternative if that were unacceptable. The complication I would warn about, and the one most likely to occur, is **membrane exposure** — which I would manage conservatively for a collagen membrane and by early removal for a PTFE one."*

2.8 The follow-up questions examiners actually ask

"Why not just use the patient's own bone?" — Volume, morbidity and resorption. Intraoral particulate sources give small volumes; block harvesting has real morbidity including sensory disturbance; and autogenous bone resorbs rapidly and unpredictably, so used alone it does not maintain contour.

"Which material has the best evidence?" — Deproteinised bovine bone mineral is the most extensively documented material in implant dentistry, particularly for ridge preservation, jumping gaps and contour augmentation.

"What are the PASS principles?" — **P**rimary closure, **A**ngiogenesis, **S**pace maintenance, **S**tability. I would say which of the four is most at risk in this particular case.

"Why decorticate?" — To open the marrow spaces so that blood, growth factors and osteoprogenitor cells can reach the graft. It is the "A" of PASS.

"How long before you place / load?" — Simultaneous GBR: 4 to 6 months before restoring. Staged augmentation: 6 to 9 months before placing.

"The patient is vegetarian and refuses bovine material. What now?" — Autogenous bone from the osteotomy and a scraper, with a synthetic alloplast for volume and a synthetic membrane; and I would explain honestly that this combination is less well documented in this application, and that the contour may be less stable.

"What if the membrane exposes at two weeks?" — Determine the membrane type and whether there is infection. Collagen, no infection: chlorhexidine, no brushing, close review, and it will often heal. PTFE or infected: remove early to salvage the bone already formed. Then identify the cause — tension, suture line position, denture pressure or smoking — and record it.

Rehearse this set-piece in: 2, 9, 14, 17, 21, 22, 32, 38, 49 and 51.

Set-piece 3 — Incision and flap design

3.1 Start with the principles

> **"I design every flap against six principles: adequate access for the whole procedure, including the graft and the membrane, not just the implant; a base wider than the apex so the blood supply is preserved; incisions over sound bone, not over the defect and not over the graft or membrane edge; preservation of the papillae where aesthetics matter; a design that permits tension-free closure; and atraumatic handling throughout, because the flap's blood supply and the periosteum determine how the graft beneath it heals."**

The blood supply point is the one to explain if asked. A mucoperiosteal flap is perfused from its base by supraperiosteal vessels running in an apico-coronal direction. A flap that is narrower at its base than at its apex, or that is criss-crossed by badly placed releasing incisions, has its supply interrupted — the consequences are marginal necrosis, dehiscence and, over a graft, loss of the graft.

3.2 The designs — what they are and when to use them

Design	Description	Use it when
Mid-crestal incision, no releasing incisions	A single incision along the crest, through keratinised tissue, with intrasulcular extension to adjacent teeth	Straightforward placement, adequate keratinised tissue, no or minimal grafting. Least morbidity
Palatally/lingually displaced crestal incision	The crestal incision is placed 1–2 mm palatal to the crest centre	You want to **move keratinised tissue buccally**, or you need buccal tissue for closure over a graft. The default in the aesthetic zone
Buccally displaced crestal incision	The reverse	Rarely — to move keratinised tissue palatally
Crestal + one or two vertical releasing incisions	Reliefs placed **at the line angles** of adjacent teeth, extending beyond the mucogingival junction, diverging apically	You need more access or coronal advancement. **Never through a papilla, never over the graft, never directly over a membrane edge**
Envelope (sulcular) flap	Intrasulcular incisions only, no vertical reliefs	Limited elevation; second-stage surgery; where you wish to avoid any scarring
Papilla-preserving / papilla-sparing (Palacci)	The incision is designed to leave the papilla intact on the adjacent tooth, or to raise it as a separate pedicle	**Aesthetic zone** — the papilla, once lost, is largely unrecoverable
Split-thickness (partial-thickness) flap	Sharp dissection leaving the periosteum on the bone	Repositioning soft tissue — apically positioned flaps, connective tissue graft recipient beds, free gingival graft beds where the periosteum is the vascular bed
Full-thickness (mucoperiosteal) flap	Blunt elevation of mucosa and periosteum together	Whenever bone must be seen or grafted
Flapless	Tissue punch or guided access with no elevation	Adequate ridge width and keratinised tissue, CBCT-planned, guided, no grafting anticipated. **Preserves periosteal blood supply, less morbidity — but you cannot see the bone**
Apically positioned flap	Split-thickness flap repositioned apically and sutured to periosteum	**Increasing the width of keratinised mucosa** at second stage; resective peri-implantitis surgery
Coronally advanced flap	Flap released and advanced coronally	Recession coverage, usually combined with a connective tissue graft
Roll flap / pouch	De-epithelialised palatal tissue rolled buccally into a pouch	Small buccal **contour** augmentation at second stage
Tunnel technique	Subperiosteal envelope created without vertical incisions	Connective tissue grafting with maximum preservation of blood supply
Periosteal releasing incision	A horizontal scoring of the inelastic periosteum at the base of the buccal flap	To achieve **tension-free primary closure** over a graft. Not a flap design in itself, but the manoeuvre that makes one work

3.3 Design by clinical situation — the answer to "what flap would you raise?"

Routine posterior placement, adequate keratinised tissue, no graft: Mid-crestal incision with short intrasulcular extensions; full thickness; usually **no releasing incision**. One-stage with a healing abutment.

Aesthetic zone, single implant with simultaneous contour GBR: Palatally displaced crestal incision, papilla-sparing intrasulcular extensions to the adjacent teeth, **one vertical relieving incision at the mesial line angle of the tooth one further along** — away from the graft. Full thickness, atraumatic. **Periosteal releasing incision** for tension-free closure. Two-stage submerged.

Second stage where keratinised tissue is abundant: Tissue punch or a small crestal incision.

Second stage where keratinised tissue is scarce: Palatally displaced crestal incision with an apically positioned split-thickness flap, to reposition the keratinised band buccally rather than excise it. Consider a simultaneous connective tissue graft if the phenotype is thin.

Ridge preservation after extraction: Flapless. Raising a flap to close a socket sacrifices vestibular depth and keratinised tissue and further compromises the buccal plate's blood supply. Use a socket seal or membrane and a figure-of-eight suture.

Peri-implantitis — resective, non-aesthetic site: Intrasulcular incisions with relieving incisions for access; full thickness; granulation tissue removed; **apically positioned flap** so the decontaminated and smoothed surface is accessible for cleaning.

Peri-implantitis — reconstructive, intrabony defect: Access flap preserving as much keratinised tissue as possible; **primary closure over the graft** is the goal, so a periosteal release may be needed.

Severely atrophic mandible / near the mental foramen: Crestal incision placed **slightly buccal or lingual to the crest** as the anatomy dictates, careful blunt identification and protection of the **mental neurovascular bundle**, and **no deep periosteal releasing incision in the premolar region**.

3.4 Atraumatic elevation — the exact words to use

> *"I incise **firmly to bone in a single stroke**, because repeated shallow passes shred the tissue. I elevate with a **sharp periosteal elevator working against bone, not against tissue**, starting at a corner or at the vertical incision so the periosteum lifts cleanly and completely. I **retract with a broad retractor resting on bone**, never crushing the flap margin, and I handle the flap only at its base with tissue forceps. I keep the flap **moist with saline-soaked gauze** throughout and I minimise the duration of retraction. When I close, I **approximate without strangulating** — tension causes ischaemia, necrosis and dehiscence, which is the commonest cause of membrane exposure."*

3.5 The periosteal releasing incision — describe it safely

> *"I hold the blade almost **parallel to the inner surface of the flap** and score the **periosteum horizontally at the base of the buccal flap**, through the periosteum only, then complete the release by **blunt dissection**. I test the release by lifting the flap passively with tissue forceps and confirming it reaches **3 to 4 mm beyond the closure line without tension**. The risks are **bleeding, perforation of the flap**, and — in the mandibular premolar region — **injury to the mental nerve**, so I identify the foramen first and keep the incision shallow there."*

3.6 Matching the closure to the flap

- **Over a graft or membrane: horizontal or vertical mattress sutures** to evert and carry the tension, with **interrupted sutures between them** to adapt the margins; **monofilament or ePTFE**, never braided silk, which wicks bacteria; **5/0** in the aesthetic zone, 4/0 elsewhere; removed at **10 to 14 days**.
- **Simple placement:** interrupted 4/0 monofilament, removed at **7 to 10 days**.
- **Apically positioned flap: periosteal sutures** to hold the flap at its new level, plus **sling sutures** around teeth or healing abutments.
- **Over an extraction socket with a graft: figure-of-eight or crossed mattress** suture.
- **Rule:** pass from **mobile to fixed** and from **thin to thick**; enter perpendicular about 3 mm from the margin; equal bites both sides; knot **off the incision line**.

3.7 Complications attributable to flap design

Wound dehiscence and **membrane exposure**; marginal flap necrosis from an inadequate base or a compromised blood supply; **papilla loss** from an incision through a papilla; **gingival recession** on adjacent teeth from sulcular incisions in thin phenotypes; **loss of vestibular depth and keratinised tissue** after coronal advancement; **visible scarring** from vertical incisions placed in the aesthetic zone; **nerve injury** from vertical or periosteal releasing incisions near the mental foramen and from lingual incisions in the molar region; **haemorrhage**; and delayed healing where the flap was crushed or allowed to dry.

3.8 Delivering it in the SOE

> **Model sentence.** *"I would use a **palatally displaced crestal incision** across the edentulous site with **papilla-sparing** intrasulcular extensions to the adjacent teeth, and a **single vertical relieving incision at the mesial line angle of the tooth one further along**, placed over sound bone and away from where the membrane will sit. Full thickness, raised atraumatically with a sharp elevator working against bone, retracted on bone and kept moist. Because I am grafting, I would add a **periosteal releasing incision** at the base of the buccal flap and test that the flap reaches passively beyond the closure line before I place the membrane. I would close with **horizontal mattress and interrupted 5/0 monofilament**, with the suture line lying over sound bone rather than over the membrane edge, and remove them at 10 to 14 days."*

3.9 The follow-up questions examiners actually ask

"Why palatally displaced?" — To carry the keratinised band buccally, where it is needed both for the aesthetic result and to give a robust, well-vascularised closure over the graft.

"Why not simply raise a big flap for better access?" — Because every millimetre of elevation costs blood supply, increases morbidity and swelling, and in the aesthetic zone risks papilla and marginal recession. I raise the smallest flap that gives adequate access for the whole procedure.

"Where exactly would you place the vertical incision, and why?" — At a **line angle**, never through a papilla and never at the mid-facial aspect of a tooth; extended beyond the mucogingival junction so the flap can be advanced; diverging slightly apically so the base is wider than the apex; and positioned so that the finished suture line lies over sound bone, away from the graft and membrane.

"When would you avoid a vertical releasing incision altogether?" — In the aesthetic zone where scarring is a concern and access can be obtained by an envelope design; in thin phenotypes; and where the incision would approach the mental foramen.

"What is the difference between a full- and a split-thickness flap, and when do you use each?" — Full thickness includes the periosteum and is used whenever bone must be exposed or grafted. Split thickness leaves the periosteum on the bone as a vascular bed, and is used when soft tissue is being repositioned or when a graft needs a periosteal bed — a free gingival graft, a connective tissue graft recipient site, or an apically positioned flap.

"You are grafting and cannot close without tension. What do you do?" — Perform a periosteal releasing incision and re-test passively. If it still will not close, reduce the graft volume, reconsider whether the augmentation is achievable, and be prepared to abandon and stage rather than close under tension — because closure under tension will lose the graft anyway.

"How does the flap design change if you are placing flapless?" — There is no flap, so I lose visualisation of the bone entirely. That means the anatomy must be certain from CBCT, the ridge width and keratinised tissue must be ample, no grafting must be anticipated, and I must be prepared to **raise a flap intra-operatively** if I find a dehiscence.

Rehearse this set-piece in: 2, 5, 9, 10, 14, 21, 22, 32, 37, 49 and 51.

How to drill the three set-pieces

Work one case at a time from Part 7C. In each case, force yourself through all three:

1. *"State the placement type and the loading protocol for this patient, with the criteria and the evidence grading."*
2. *"State the graft material, the reason for that material, the consent points and the most likely complication."*
3. *"Describe the incision, the flap, the release and the closure."*

If you can do that in **three minutes for any case in this notebook**, you have covered the majority of the domain 3 marks in the structured oral examination before the examiners have asked anything case-specific.

SOE Case 1 — The bruxist, the posterior maxilla and the occlusal scheme

THE CASE

59-year-old male. Clear medical history. Recent extraction of the **15 and 16** for cracked teeth. The **46 is over-erupted**. OPG shows generalised attritive wear and flattened condylar contours suggestive of **bruxism**. Cross-sectional CBCT slices are provided: at the **15**, ridge width is **8 mm coronally, 10 mm apically, with 20 mm of height**; at the **16**, there is a marked **buccal sinus drop with an oblique floor and a pendulum-shaped, "key-ring" cross-section — approximately 6 mm buccally and 12 mm palatally**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 59-year-old man, medically fit, who has recently lost the 15 and 16 to cracked teeth in a mouth showing generalised attritive wear and flattened condyles — so my working diagnosis is **bruxism**, and the cracks are a symptom of it rather than bad luck. The 46 has over-erupted into the 16 space. Radiographically these are two quite different sites: the 15 has generous bone in all three dimensions, while the 16 shows a **pendulum or key-ring cross-section** with an oblique, buccally dipping sinus floor giving about 6 mm buccally against 12 mm palatally. My working position is that the 15 is straightforward and the 16 is where the decision lies. What I need to establish is whether there is restorative space above the over-erupted 46, whether the 16 can be implanted without a lateral window, and how I protect whatever I place from his parafunction."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical groundwork

Q Describe the anatomy of the maxillary sinus.

"A pyramidal air-filled cavity in the body of the maxilla, lined by **respiratory epithelium**. Its base is the lateral nasal wall, its apex points into the zygomatic process, its roof is the orbital floor carrying the infraorbital nerve, its floor is the alveolar process — often lowest around the first molar — and its anterior wall carries the infraorbital foramen. It drains through the **ostium in the middle meatus**, high on the medial wall, which is why drainage is against gravity."

Q What is the Schneiderian membrane, and what thickness concerns you?

"The mucoperiosteal lining of the sinus — thin ciliated respiratory epithelium with periosteum, typically under **1 mm**. A very thin membrane tears easily; a membrane thickened beyond about **2 mm** suggests inflammation or an odontogenic source and warrants investigation first. Complete opacification, an air-fluid level, or anything obstructing the **osteomeatal complex** changes the plan, because an obstructed sinus becomes an infected one."

Q What is a septum, and why do you look for one?

"A **Underwood's septum** is a bony ridge projecting from the sinus floor, commonest between the second premolar and first molar. It matters because it compartmentalises the sinus, makes membrane elevation far more likely to tear, and complicates any window design. Finding one on the CBCT is often the point at which I decide the case is not mine."

Q What bone quality do you expect in the posterior maxilla, and what follows?

"Characteristically **D3 or D4** on Misch — thin cortex with fine trabecular bone. That means poor primary stability, so I under-prepare the osteotomy, stop a drill short, avoid tapping, and lean towards **conventional loading** rather than early or immediate. It is the opposite problem to the anterior mandible, where the risk is overheating dense bone."

Chain: reading this patient

Q Why did these teeth crack?

> "Heavily restored teeth carrying **parafunctional load**. That matters for what comes next, because an implant has **no periodontal ligament** — no proprioceptive protective reflex and only 3 to 5 micrometres of axial movement against 25 to 100 for a tooth. So parafunction is a **mechanical** risk factor, not a biological one: I expect screw loosening, chipping and component fracture rather than loss of integration."

Q What clinical signs would confirm the bruxism?

> "**Wear facets that match in excursion**, masseteric hypertrophy, linea alba, a scalloped tongue, cervical abfraction and repeatedly fractured restorations. Historically: morning jaw stiffness or headache, and a partner's report of night-time grinding, which is more reliable than the patient's own account."

Q Interpret the cross-section at the 16.

> "The **key-ring or pendulum shape** means the sinus floor is oblique and dips buccally, so usable height is asymmetric — around **6 mm buccally against 12 mm palatally**. An implant in the correct prosthetic position meets far less bone buccally than palatally. A transcrestal elevation is unreliable here, because the ITI decision tree assumes a **flat floor and more than 6 mm of residual height**; an oblique floor tips the osteotome and invites perforation."

Q And the 15?

> "Generous — **8 mm coronal width, 10 mm apical, 20 mm height**. That comfortably accommodates a standard-diameter implant with more than **1 mm of bone circumferentially**, and the height means the sinus is not a factor. Surgically this site is unremarkable; the challenge at the 15 is prosthetic, not surgical."

Q What further information do you want?

> "**Mounted study casts** to measure the restorative space against the over-erupted 46; a full occlusal analysis in centric relation and in excursion; sensibility and periodontal status of the 14, 17 and 47; confirmation of sinus health, membrane thickness and septa on the CBCT; and, most importantly, **what he actually wants** — function, appearance, or simply to avoid a denture."

Domain 2 — Clinical decision making and treatment planning

Chain: the occlusal groundwork

Q Name the recognised occlusal schemes.

> "**Mutually protected occlusion**, also called canine-guided; **group function**; and **bilateral balanced occlusion**. Each describes how load is distributed in intercuspation and, separately, in excursion — and the distinction between those two states is the thing to keep clear."

Q Describe mutually protected occlusion.

> "In intercuspation the **posterior teeth take the axial load** and protect the anterior teeth, which are held out of heavy contact. In excursion the **canine guidance discludes the posterior teeth**, protecting them from lateral load. It is my default wherever a sound natural canine exists, and it is the scheme I would provide here."

Q When would you choose group function instead?

> "Where the canine is periodontally compromised, worn flat, or is itself an implant. Then I would **distribute the excursive load** across several working-side posterior teeth rather than concentrate it on one unit — particularly on an implant, which cannot feel the overload and cannot move away from it."

Q Why is bilateral balanced occlusion wrong for a fixed implant restoration?

"Because it deliberately creates **non-working side contacts**, and those generate the largest bending moments of any contact in the mouth. It exists to stop a complete denture tipping — a free body needs contact on both sides. A fixed implant restoration is not a free body, so balance buys nothing and costs a great deal mechanically."

Chain: the options for these two sites

Q What are the options here?

"**No treatment, accepting a shortened arch** — legitimate in principle, but he has a two-unit gap and an over-erupting 46 that will worsen. **Removable partial denture** — cheap and non-surgical, poorly tolerated for two units in a bruxist. **Conventional bridge from 14 to 17** — a long span, heavy loading and a questionable abutment. **Implants** — straightforward at the 15; at the 16 the sub-options are a **short implant**, a transcrestal elevation, a lateral window, or restoring the 15 alone."

Q Which do you choose?

"A standard implant at the 15 and a **wide, short implant at the 16** using the buccal height available. Systematic reviews show short implants of 6 mm or less in the posterior jaws achieve survival **comparable to longer implants placed with augmentation** over three to five years, with **fewer biological complications and less morbidity** — and it keeps the case within my competence."

Q You have chosen a short implant — argue against yourself.

"The honest counter-argument is the **crown-to-implant ratio**: a short implant under a tall crown in a bruxist has an unfavourable lever arm, and the medium-term data on short implants specifically in parafunction is thin. Someone who augments would say I am accepting a mechanical compromise to avoid a procedure. I mitigate it with a wide diameter, a narrow occlusal table, no cantilever and a splint — but I accept the trade-off is real."

Q Would you splint the two implants?

"**Yes** — joined as two units, screw-retained so it can be retrieved. Splinting a short implant to a standard one shares load, which matters in a bruxist. The cost is cleansability, so I would design the connector and emergence profile to admit an **interdental brush** and would show him how to use it before the restoration is fitted."

Q You mentioned splitting — is a ridge split relevant here?

"No. Ridge splitting addresses **horizontal** deficiency; it needs about 3 mm of width with cancellous bone between two cortical plates, and is far more predictable in the maxilla than the mandible. Here the width is adequate at both sites and the deficiency at the 16 is **vertical**. It is the wrong operation for this problem."

Q SAC classification?

"Site 15: surgically **Straightforward**, prosthodontically **Straightforward to Advanced** because of the parafunction. Site 16: surgically **Advanced** because of the oblique sinus floor and the short-implant decision; prosthodontically **Advanced** for the crown-to-implant ratio, the parafunction and the opposing over-eruption."

Q What is beyond your competence here, and when would you refer?

"A **lateral window sinus elevation** is outside the scope of a general dentist with an interest in implants, and so is vertical block grafting. If the CBCT showed a septum, established sinus pathology, or if intra-operatively I met a frank perforation, I would stop and refer rather than improvise. I would also refer if he insisted on treating the 16 by augmentation."

Q What about the over-erupted 46?

"It is dealt with **before** the definitive restoration, never after. I measure the space on mounted casts, then in escalating order: **enameloplasty**; a new crown at reduced height, which may require elective endodontics; **orthodontic intrusion**; or accepting a compromised scheme. I would also consider modest crest reduction at the 16 if it gains space and a flatter platform — provided it does not cost me height I need."

Q What would you tell him, in your own words to him?

> " 'Your teeth cracked because you grind, and the grinding is not going away — so the new teeth are being built to survive it rather than to be perfect. The front site is straightforward. The back site has your sinus sitting low over it, so I plan to use a shorter, wider implant there; if when I open it the sinus is not where I expect, I will stop and just do the front one rather than take a risk. You will need a hard night guard for as long as you have these, and screws and porcelain will need attention over the years — that is maintenance, not failure.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: implant-protective occlusion

Q Define implant-protective occlusion.

> "The deliberate modification of the occlusal scheme to compensate for the **absence of a periodontal ligament** around an implant. It is not a separate scheme — it sits inside whichever scheme the natural dentition dictates, and modifies only how the implant restorations themselves are contacted."

Q What is its biological basis?

> "An implant moves about **3 to 5 micrometres axially** against 25 to 100 for a tooth, and 10 to 50 laterally against 56 to 108. A tooth moves immediately and in two phases; an implant moves gradually and linearly. And there are **no periodontal mechanoreceptors**, so the reflex that makes a patient unload a painful tooth is absent — the tactile threshold is far higher, and only osseoperception from periosteal, muscular and joint receptors remains."

Q What are the practical rules that follow?

> "**Light centric contacts** — the restoration holds shimstock only on heavy clenching, roughly 25 to 30 micrometres lighter than the adjacent teeth. **Contacts centralised over the long axis**, with a **narrowed occlusal table** and reduced cusp inclination, converting lateral force to axial. **No excursive contact on the implant** where a natural tooth can guide. **No non-working interferences**. **No cantilevers**. And never connect an implant rigidly to a natural tooth."

Q And what changes specifically because he is a bruxist?

> "**More or wider implants rather than fewer**; **monolithic rather than layered ceramic** to remove the chipping interface; splinting where it is cleansable; a **hard full-coverage splint**; and explicit consent to a higher rate of mechanical complications with a cost attached. I would also shorten the recall interval."

Chain: the procedure

Q Take me through the surgery at the 15.

> "Pre-operative chlorhexidine rinse, aseptic technique. Buccal and palatal infiltration. **Mid-crestal incision** from the 14 to the 17 with intrasulcular extension and a distal relieving incision if access demands it; full-thickness flap raised atraumatically. Graded drilling under copious sterile saline with intermittent pressure, direction indicator check, and — because this is **D3 bone** — I stop one drill short to under-prepare and raise primary stability."

Q And at the 16?

> "I plan the exact **buccal** height from the CBCT and treat that, not the palatal measurement, as my working depth. Drill short of the floor, verify with a depth gauge and a periapical, and place a **wide, short implant**. I would not attempt to gain height with an osteotome against an oblique floor — that is precisely the anatomy in which transcrestal elevation fails."

Q You breach the membrane. What do you do?

> "Stop drilling and assess. **Valsalva** to test the membrane: no air or bubbling and I may proceed, placing a small collagen membrane and graft over an intact elevation. If there is a **frank perforation**, I abandon that site, close primarily, place the patient on antibiotics and decongestant advice, and re-plan — either restoring the 15 alone or referring. Improvising a repair is beyond my scope."

Q What insertion torque do you want, and what if you achieve 15 Ncm at the 16?

> "**25 to 45 Ncm**. At 15 Ncm with a short implant in D4 bone I would not load early. I would **submerge it**, extend healing to four to six months, and re-assess with resonance frequency analysis before restoring — an ISQ above 70 for confidence, and a **falling ISQ** telling me it is failing."

Q One stage or two?

> "**One stage** with healing abutments where primary stability is good, because it avoids a second surgery and avoids repeatedly disrupting the mucosal seal. **Two stage** at the 16 if stability is marginal, if I have grafted, or if soft tissue closure is compromised. I am content to run a one-stage 15 alongside a submerged 16."

Q Describe the restorative phase.

> "**Conventional loading** — no immediate or early loading in a bruxist with soft posterior maxillary bone. Implant-level **open-tray impression with splinted copings** since the units are being joined, in polyether in a custom tray. **Screw-retained monolithic zirconia** on titanium bases. Occlusion set to implant-protective rules, then a **hard full-coverage acrylic night guard** at fit."

Q Why monolithic zirconia rather than layered porcelain?

> "Because the commonest technical complication in a parafunctional patient is **chipping of the veneering ceramic**, and monolithic material removes that interface entirely. The trade-off is aesthetics and wear of the opposing dentition, but in a posterior quadrant opposing a heavily worn dentition I accept that. I would polish rather than glaze any adjusted surface."

Q How would you manage the bruxism itself?

> "As a condition in its own right. **Characterise it** from the signs and history. **Identify contributors** — stress, caffeine, alcohol, smoking, recreational drugs, SSRIs. **Screen for obstructive sleep apnoea** — snoring, witnessed apnoeas, daytime somnolence — and refer to his GP if present, because sleep bruxism is associated with it and a mandibular advancement device must never precede that assessment."

Q Describe the splint you would provide.

> "A **hard, heat-cured or milled full-coverage acrylic splint**, usually maxillary, covering every tooth so none can over-erupt. Adjusted to **even bilateral contacts in intercuspation** with smooth canine guidance so the posterior teeth disclude in excursion. I review its **wear pattern at every recall** — it is the only objective measure I have of the force he generates, and it warns me when the restorations are at rising risk."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: what goes wrong and how you answer it

Q What will go wrong, and roughly when?

> "In order of likelihood: **screw loosening**; ceramic or acrylic chipping; wear of the night guard; and, if hygiene lapses, **peri-implant mucositis**. Less likely but more serious: failure of the short implant at the 16 during healing, and **sinusitis** if the membrane was compromised. Mechanical complications dominate in this patient — that is the shape of the risk."

Q He returns at 18 months with a loose restoration and no pain. What do you do?

> "I would not simply retighten. **Remove the restoration**, inspect the screw, the abutment and the implant connection for damage, and exclude **implant fracture** — pain-free looseness can be a fractured screw or a fractured fixture. Fit a **new screw**, find the cause — occlusion, an excursive interference or a non-passive fit — correct it, torque to the manufacturer's value and **re-torque at ten minutes**."

Q Your colleague would simply retighten it in five minutes. Why might they be right?

"Because a single early loosening after a torque that was never re-checked is often exactly what it appears to be, and dismantling a joined screw-retained restoration carries its own risk of damaging the connection. If it were the **first** episode, within months of fitting, with a passive fit I had verified myself and an occlusion I had checked, retightening with a new screw is defensible. It is repetition that makes it negligent."

Q At three weeks he has unilateral nasal discharge and cheek pain. Differential and action?

"**Acute sinusitis**, most likely iatrogenic from the 16, against a displaced graft particle, an oro-antral communication, or a coincidental viral upper respiratory infection. I would examine for a communication, test with Valsalva, image if indicated, prescribe according to local antimicrobial guidance with decongestant advice, and refer to **ENT** if it fails to settle or recurs. And I would record it as a significant event."

Q What would you record?

"The consent discussion including the named material risks and the fallback plan; the CBCT justification under **IR(ME)R** and the report; implant make, diameter, length, site and **lot number**; insertion torque and any ISQ; the surgical detail including flap design and any membrane contact; the occlusal scheme prescribed and the shimstock findings; the splint provision; baseline **periapical radiographs and probing depths** at fitting; and the recall interval agreed."

Q How would you know this treatment had succeeded?

"Survival of both implants; **marginal bone levels stable against the baseline** taken at fitting; probing depths and bleeding on probing acceptable with no suppuration; a stable, cleansable prosthesis; a splint that is being worn; and his own report of comfort and function measured against **what he told me at the start that he wanted**."

Q Maintenance?

"**Three to six monthly**, at the shorter end for parafunction and a joined restoration. At each visit: plaque and bleeding scores, six-point peri-implant probing with a plastic probe at gentle force, screw and prosthesis check, night-guard wear assessment, and professional biofilm removal with titanium or plastic instruments. Radiographs at one year, then only when clinically indicated."

WATCH OUT

Ask yourself whether the problem in front of you is **biological or mechanical** — in this patient it is nearly always mechanical, and saying so out loud buys you a sentence while you assemble the answer.

Running this case to time

Time check. The flow above holds 38 questions and about 1,905 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that is a little under **seventeen minutes**, so as written it **overruns by nearly two**. An examiner short of time drops the far end of Domain 3 — bruxism as a condition in its own right, and the splint description — to protect Domain 4, where the marks in this case sit. If your answers run past **thirty seconds** you will still be discussing occlusal schemes when the time ends.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the OPG and cross-sections in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Cracked 15 and 16 in a mouth with generalised attrition and flattened condyles, so **bruxism** is the diagnosis and the cracks its symptom; the over-erupted 46; a generous 15 against a **key-ring cross-section** at the 16, 6 mm buccally to 12 mm palatally; and the working position that the decision lies at the 16.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the anatomical groundwork, then reading this patient	Sinus anatomy with the **ostium in the middle meatus** and drainage against gravity; the **Schneiderian membrane** under 1 mm, and what thickening beyond 2 mm means; **Underwood's septum** and why finding one ends the case; **D3/D4** bone and what follows; then the oblique floor read against the ITI transcrestal criteria.
4:30–7:30	Domain 2: the occlusal groundwork, then the options for these two sites	**Mutually protected, group function, bilateral balanced** — distinguished, with why balance is wrong on a fixed restoration; the short implant with the survival evidence in one clause; the **argument against yourself** on crown-to-implant ratio; **SAC** site by site; and the **lateral window and block grafting named as the referral line**.
7:30–11:30	Domain 3: implant-protective occlusion, then the procedure	The **3–5 μm axial movement and absent mechanoreceptors** as the basis, then the rules — light centric contacts, narrowed table, no excursive contact, no cantilever; working to the **buccal** height at the 16; **25–45 Ncm**, and what you do at 15 Ncm; **conventional loading** and **monolithic zirconia**, argued from chipping.
11:30–14:30	Domain 4: what goes wrong, and the two scenarios	Mechanical complications dominating, **screw loosening first**; the 18-month loose restoration removed rather than retightened, with **screw and implant fracture excluded**; the three-week sinusitis differential and **ENT** referral; the record, with lot number and torque.
14:30–15:00	Closes on the patient	**In his words**: the teeth cracked because he grinds, the grinding is not going away, the back site may be abandoned intra-operatively if the sinus is not where you expect, and a hard night guard for as long as he has these.

The spine — the twelve answers to have ready if nothing else

1 Sinus anatomy and why drainage is against gravity · 2 Membrane thickness that concerns you, and septa · 3 Bone quality in the posterior maxilla, and what follows · 4 What the key-ring cross-section means for usable height · 5 Why these teeth cracked — PDL against no PDL · 6 The three occlusal schemes, named · 7 Why bilateral balance is wrong on a fixed restoration · 8 The short implant, and the argument against yourself · 9 SAC, and the lateral window as the referral line · 10 Implant-protective occlusion and its biological basis · 11 Torque wanted, and what you do at 15 Ncm · 12 The loose restoration at eighteen months.

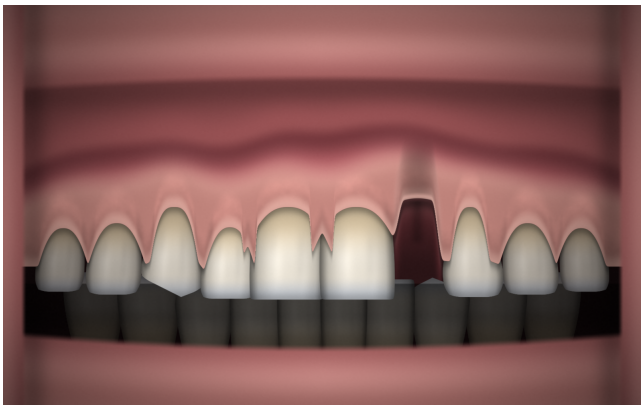
Where the marks are lost in this case

Treating the 16 as a sinus lift. The oblique, buccally dipping floor is why transcrestal elevation fails here and why a lateral window is outside your scope. Offering augmentation you could not deliver loses Domain 2; the wide short implant *plus* the referral threshold is the answer.

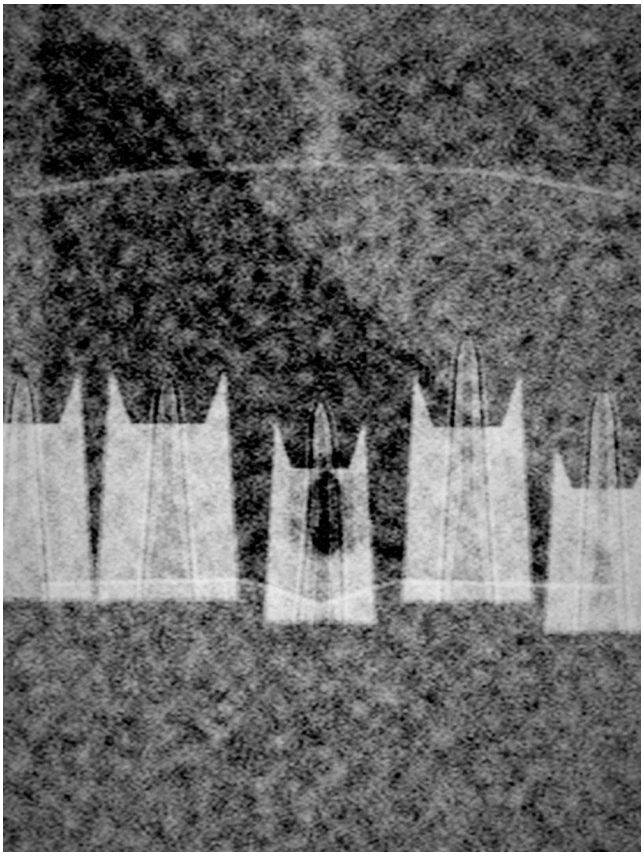
Skipping the occlusal groundwork. Domain 2 opens below the implant material. A candidate who cannot distinguish the three schemes, or who explains balance without saying it exists to stop a *denture* tipping, has lost marks before the plan is reached.

Forgetting the 46. The over-erupted 46 is dealt with before the definitive restoration, never after, and the escalating order — enameloplasty, crown at reduced height, orthodontic intrusion — must be volunteered. Unprompted, it turns a list of implants into a plan.

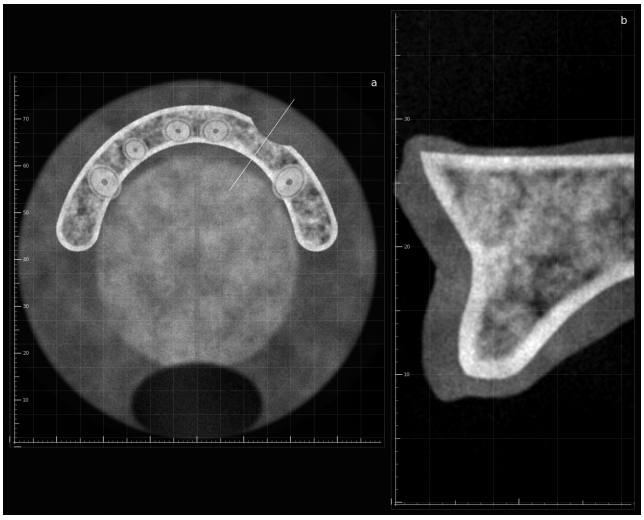
SOE Case 2 — Missing 22 in a patient who works in the entertainment industry



Anterior view, interim retainer removed — simulated illustration, not a clinical image



The pre-extraction periapical of the 22 — simulated illustration, not a clinical image



CBCT at the 22 site — simulated illustration, not a clinical image

THE CASE

26-year-old who works in the **entertainment industry** as a television presenter. The **22 was lost 6 weeks ago**. She is currently wearing an **Essix retainer with a pontic**. The pre-extraction periapical showed **external root resorption** in a virgin, unrestored tooth with an associated radiolucency. Clinically: a **narrow ridge** at 22; the **23 has a triangular crown form**, the **21 rectangular**; **normal (medium/thick) biotype with adequate keratinised gingiva**. CBCT shows a **buccal concavity** at the site.

Pace: one structured opening of about 60 seconds, then roughly 37 questions in the remaining 13 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 26-year-old television presenter who lost the 22 six weeks ago to **external root resorption** in an otherwise virgin tooth. Her occupation puts her face on camera in high definition, so the aesthetic outcome is unusually material to her, and that shapes both the consent conversation and my threshold for referral. The site has a **narrow ridge with a buccal concavity**, a favourable medium-to-thick biotype with adequate keratinised tissue, and a **crown form discrepancy** — a triangular 23 against a rectangular 21. My working position is that this is an aesthetically demanding single-unit site that will need augmentation, and the three things I must establish before committing are her **smile line**, the cause of the resorption and the prognosis of the neighbours, and what she actually expects."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical and biological groundwork

Q Describe the anatomy of the anterior maxilla that matters for implant placement.

"The **labial plate** over the root prominences with the canine eminence, a **labial concavity** above the apices deepening towards the lateral, the **nasopalatine canal** at the midline, the nasal floor above, and scalloped gingiva whose **papillae depend on the bony crest**."

Q How thick is the labial plate at a lateral incisor, and what is it made of?

"Under **1 mm** in most anterior sites, often under 0.5 at a lateral. It is largely **bundle bone** — ligament-dependent, with no cancellous core to revascularise from — so it resorbs when the tooth goes, and that loss shows as mid-facial recession."

Q Take me through socket healing and the ridge change that goes with it.

"**Clot**, granulation tissue within a week, soft tissue closure by four to eight weeks, **woven bone** from four to six weeks, lamellar by six months. Meanwhile the ridge loses **half its width in a year**, two-thirds by three months, mostly buccally."

Q How do you classify her periodontal phenotype, and why do you care?

"**Phenotype** — gingival thickness, keratinised width, bone morphotype — is the 2017 World Workshop term replacing biotype, judged by **probe transparency**. Thin tissue recedes and shows titanium; hers is **thick and flat**, which resists recession and masks the abutment."

Chain: reading this patient

Q Why would a virgin tooth develop external root resorption, and why does the cause matter?

"Usually **previous trauma**; also orthodontic resorption, adjacent pathology or invasive cervical resorption. It matters because trauma means the **neighbours may share the prognosis**, and I will not place beside a tooth I expect to lose — so I **sensibility test the 21 and 23**."

Q What else do you need to examine?

"**Smile line at full smile and in speech** — the biggest aesthetic risk factor. Then lip dynamics, midline, incisal plane, papilla heights and gingival margins from 21 to 23, probing depths and shade, with photographs and an **ITI Aesthetic Risk Assessment**."

Q What is the significance of the triangular 23 against the rectangular 21?

"Its contact point sits **more coronally**, so contact-to-crest distance is greater and the papilla has more to fill. **Tarnow**: almost always present at 5 mm, **56% at 6 mm, 27% at 7 mm**. And one crown must reconcile two tooth forms."

Q What does the buccal concavity on the CBCT tell you?

"That the bone lies **palatal to the ideal prosthetic axis**, so an implant following the ridge either perforates the plate or emerges too buccally. This site needs **augmentation — at placement or staged**."

Domain 2 — Clinical decision making and treatment planning

Chain: the options and the argument between them

Q What are her options?

"**No treatment**, which must be offered, with a warning about drift. A denture as an interim. A **resin-bonded cantilever from the 23**. A conventional bridge. An **implant crown with augmentation**. **Orthodontic space closure**. Each with its biological cost and maintenance."

Q Your colleague would place a resin-bonded cantilever and never touch this ridge. Why might they be right?

"Minimally invasive, excellent survival for **single anterior cantilevers**, delivered in weeks, reversible in a 26-year-old, and it avoids nine months of treatment in a woman whose living depends on how she looks meanwhile. Against it: **debonding**, and a pontic over a concavity. **It turns on her smile line, not my preference.**"

Q Why not a conventional bridge — and what about orthodontic space closure?

"Preparing two **unrestored** teeth in a 26-year-old is irreversible, risks pulpal death and buys nothing over a cantilever. Space closure deserves raising — the **23 substituted for the 22**, guidance moved to the premolar — but needs an **orthodontic opinion**."

Q Has growth ceased at 26?

"Almost certainly, but I would confirm — stable height, shoe size and dentition, and **serial cephalograms a year apart** in real doubt. Implants behave as **ankylosed teeth**, and infraposition in an on-camera professional would be a serious failure."

Q What is alveolar ridge preservation, and since the tooth is already out, what is your equivalent now?

"Grafting the socket at extraction with a **slowly resorbing xenograft**; it roughly halves horizontal loss but **cannot prevent bundle bone loss**, so it is damage limitation. That moment has gone: my equivalent is **guided bone regeneration**, without socket walls to contain the graft."

Q SAC classification, and what part of this is within your competence?

"Surgically **Advanced**, **Complex** if her smile line is high — aesthetic zone, ridge deficiency, concavity. Prosthodontically **Advanced to Complex**. Simultaneous contour grafting is within my competence; **a high smile line, staged horizontal GBR or soft tissue reconstruction I refer** — said at the outset."

Q What must she be told, specifically because of who she is?

"Materiality under **Montgomery** is judged from her position. 'The gum edge may shrink and show a grey line, a black triangle may appear beside the canine. Six to nine months, mostly in a temporary — I need your filming dates. Even done well it may not match exactly.' Then a **mock-up** before committing."

Domain 3 — Decision making in relation to treatment procedures

Chain: placement protocol and loading

Q When would you place, and when would you load?

"Those are **two separate decisions**. Types 1 to 4 describe when I place relative to the extraction; immediate, early and conventional describe when a restoration goes into occlusion relative to placement. They are chosen independently, but the combinations are not equally well documented."

Q Give me the four placement types, and say which this site is.

"**Type 1** immediate, same day. **Type 2** early with soft tissue healing, four to eight weeks. **Type 3** early with partial bone healing, twelve to sixteen weeks. **Type 4** late, six months or more. At six weeks with healed mucosa she is **Type 2**."

Q Why is Type 2 the workhorse of the aesthetic zone?

"The mucosa has healed and can be advanced, giving **tension-free primary closure over a simultaneous graft**; infection has resolved; and much of the ridge remains. The price is two surgical episodes and a graft I will almost certainly need."

Q Would immediate placement have preserved her buccal plate?

"No. The buccal wall is **bundle bone**, dependent on the periodontal ligament, and resorbs whatever I do. Immediate placement saves time and surgical episodes, not bone, and costs me control of the **three-dimensional position** — what I least want to lose here."

Q So what loading protocol, and what does she wear meanwhile?

"**Conventional loading**, four to six months after simultaneous grafting, submerged; early loading after early placement is **clinically insufficiently documented**. Meanwhile a bonded pontic rather than the Essix, so nothing presses on the graft. My provisional is **restoration, not loading** — free of contact in intercuspation and every excursion, checked with shim stock."

Chain: choosing an augmentation material

Q What are the four categories of graft material, and what can each actually do?

"**Autograft, allograft, xenograft, alloplast.** All four are **osteoconductive**. Only autogenous bone is **osteogenic**, because only it carries living cells, and it is osteoinductive too; demineralised allograft is variably **osteoinductive**; xenograft and alloplast are scaffolds only."

Q Their advantages and disadvantages?

"Autograft: best biology and fastest incorporation, no ethical objection, but limited volume, donor morbidity, unpredictable resorption. Allograft: no donor site, unlimited, but **human origin** and batch variability. Xenograft: **outstanding contour stability**, best documented, but conductive only. Alloplast: no origin objection, cheap, least documented here."

Q What would you use to correct this buccal concavity, and why?

"A **composite graft** — autogenous chips from the osteotomy against the implant surface for their osteogenic cells, roughly **50:50 with deproteinised bovine bone mineral** outside, because a slowly resorbing **xenograft** holds buccal contour for years. Covered by a **resorbable collagen membrane** onto decorticated bone."

Q Simultaneous or staged augmentation — and where is the limit of your competence?

"**Simultaneous** where I can place in the correct prosthetic position and the defect is a contained dehiscence — my plan here; **staged** where the ridge cannot house the implant. Placing where bone dictates rather than the prosthesis is the commonest unrecoverable failure. Staged horizontal GBR, blocks or vertical augmentation I **refer**."

Q What must she consent to that is specific to the graft materials?

"That the graft is **bovine**, the membrane **porcine collagen**, and allograft **human donor** tissue — raised at planning, never on the day, because **religious, cultural, dietary and vegetarian objections** are common and legitimate. Then the very low transmission risk, the **autogenous-plus-synthetic alternative** and its weaker documentation, and that brand and **lot number** are recorded."

Chain: timing, position and the graft

Q Describe your flap.

"A **palatally displaced crestal incision** with papilla-sparing intrasulcular extensions to the 21 and 23, and one vertical relief at the mesial line angle of the 24, away from the graft. Full thickness, atraumatic, carrying keratinised tissue buccally."

Q Where exactly does the implant go?

"Shoulder **3 mm apical to the planned mid-facial margin**, **1 to 2 mm palatal** to the neighbours' emergence line, **1.5 mm from each adjacent root**, angled through the **cingulum**. Given the concavity I angle more palatally and accept an angulated screw channel."

Q What diameter, and why?

"A **narrow-diameter implant** around 3.3 mm in **titanium-zirconium alloy**, which gives it strength narrow titanium would lack. A standard diameter would consume the buccal bone I am protecting and over-contour the emergence."

Q You expect a buccal dehiscence over the concavity. Take me through the closure.

"Graft deliberately over-contoured, membrane 2 to 3 mm beyond the defect onto sound bone, then a **periosteal releasing incision** and tension-free closure with horizontal mattress and interrupted 5/0 monofilament — tension causes exposure, and exposure loses the graft."

Q Describe the restorative phase, and the occlusion you want.

> "Second stage with a **palatally repositioned flap**, a **connective tissue graft** if the phenotype is thin, a provisional developing emergence over 8 to 12 weeks, then a **customised coping** copying it. Definitive: **screw-retained**, titanium base with milled ceramic, reconciling the two crown forms. **Light contact in intercuspation**, **none in excursion** — the canine guides."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: what goes wrong and what you do about it

Q What are the most likely complications?

> "Aesthetic first — **mid-facial recession**, a **black triangle**, shade mismatch, and change as she ages while the implant does not. Biological — **graft or membrane exposure**, and mucositis. Mechanical — screw loosening or chipping, least likely in a lateral."

Q At two years she has 1.5 mm of mid-facial recession with visible greyness. Diagnose it before you treat it.

> "Four questions. Is the implant **too buccal** — commonest and least recoverable? Is the **phenotype thin** there after all? Is **inflammation** driving it? Has the abutment been **repeatedly disconnected**? That decides whether this is treatable or fundamental."

Q Give me the options.

> "**Connective tissue graft** with a coronally advanced flap. A **zirconia or ceramic abutment** to remove the grey. Recontouring or pink ceramic. **Accept and monitor** photographically. Or **explantation and re-treatment** if the position is fundamentally wrong — discussed frankly."

Q You have offered her a connective tissue graft. Argue against it.

> "Around an **established** implant it is far less predictable than the same graft at placement — no ligament, no interproximal bone to advance against, so complete coverage is not realistic. If the implant is too buccal it treats appearance, not cause. **Better said than over-promised.**"

Q How would you monitor her?

> "**Standardised photographs at every review** — non-negotiable here. **Periapical at fitting and one year**. Plastic probing, bleeding, recession from a fixed reference, a papilla index, and her satisfaction asked directly. Review at two weeks, second stage, fitting, three and twelve months, then six-monthly."

Q What would you record?

> "The **aesthetic risk assessment** and smile line; every option including no treatment; the **risks discussed with her occupation in mind**, in her words as well as mine; the mock-up; the surgical record with **graft lot numbers**; and baseline photographs and radiographs."

Q How will you know this has succeeded?

> "By her measure — that she works on camera without thinking about her lateral incisor. Objectively: **stable bone levels against baseline**, no bleeding or suppuration, margin and papillae unchanged against the photographic record, and a crown a stranger would not identify."

WATCH OUT

Come back to the aesthetic risk assessment — smile line, phenotype, crown form, ridge, expectation — and say which of those this question belongs to. It buys you a sentence and it is nearly always the right frame in an aesthetic-zone case.

Running this case to time

Time check. The flow above holds 38 questions and about 1,693 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that comes to a little over **fifteen minutes**: it fits, but with nothing to spare. Domain 3 carries three chains, so it compresses first — an examiner takes the graft categories as one question rather than four. If your answers run past **thirty seconds**, you will never reach the aesthetic complications in Domain 4.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapical and the CBCT slice in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 26-year-old **television presenter**, 22 lost six weeks ago to **external root resorption** in a virgin tooth; the occupation stated as material to consent and to the referral threshold; a **narrow ridge with a buccal concavity**, a medium-to-thick phenotype, and a **triangular 23 against a rectangular 21**; the working position that this is an aesthetically demanding single unit needing augmentation, and the three things still to establish.
1:20–4:30	Domain 1: the anatomical and biological groundwork, then reading this patient	The **labial plate under 1 mm and largely bundle bone**; socket healing, **half the ridge width lost in a year**; **phenotype** as the 2017 term, judged by probe transparency; why a virgin tooth resorbs, and why that means **sensibility testing the 21 and 23**; **Tarnow** — 56% at 6 mm, 27% at 7 — against the triangular crown form; and the concavity read as bone palatal to the prosthetic axis.
4:30–7:30	Domain 2: the options and the argument between them	Every option offered including **no treatment**; the **resin-bonded cantilever argued at its strongest**, then settled on her smile line rather than your preference; growth confirmed at 26 because implants behave as **ankylosed teeth**; **SAC Advanced, Complex with a high smile line**; and the competence line — simultaneous contour grafting yours, **a high smile line, staged horizontal GBR or soft tissue reconstruction referred, said at the outset**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: placement and loading, the graft material, then doing it	**Type 2** justified as the workhorse; **conventional loading**, the provisional as restoration and not loading; the four graft categories, **only autograft osteogenic**; **50:50 autogenous chips with deproteinised bovine mineral** under collagen; then position — **3 mm apical, 1–2 mm palatal, 1.5 mm from each root, through the cingulum** — a **3.3 mm titanium-zirconium** implant, and periosteal release for tension-free closure.
11:30–14:30	Domain 4: what goes wrong, and the recession scenario	**Aesthetic complications first**; the four diagnostic questions before treating 1.5 mm of mid-facial recession — too buccal, thin phenotype, inflammation, repeated disconnection; the options including **explantation**; the connective tissue graft **argued against honestly**; standardised photographs as monitoring that is non-negotiable here.
14:30–15:00	Closes on the patient and the record	**Montgomery materiality judged from her position**: the grey line, the black triangle, six to nine months mostly in a temporary, her filming dates, and that even done well it may not match exactly.

The spine — the twelve answers to have ready if nothing else

1 Labial plate thickness, and what bundle bone is · 2 Socket healing and the dimensional change · 3 Phenotype, and how you assess it · 4 Why a virgin tooth resorbed, and what you test next door · 5 Tarnow, and the triangular 23 · 6 What the buccal concavity means for the axis · 7 The resin-bonded cantilever, argued at its strongest · 8 SAC, and the smile line as the referral line · 9 The four placement types, and why Type 2 here · 10 The four graft categories, and what each can do · 11 Implant position in three dimensions · 12 The four questions before treating recession.

Where the marks are lost in this case

Answering as though she were any 26-year-old. Her occupation is the case: it sets materiality under Montgomery, it sets the referral threshold, and it makes her filming schedule part of the plan. Volunteering that unprompted is the difference between a 3 and a 5 in Domain 2.

Claiming immediate placement would have saved the plate. It would not — the buccal wall is bundle bone and resorbs regardless. Say so, and say what immediate placement really costs: control of the three-dimensional position, the one thing you cannot lose at a lateral incisor.

Never naming the smile line as the referral line. Simultaneous contour grafting at a contained dehiscence is within competence; a high smile line, staged horizontal GBR or soft tissue reconstruction is not. Candidates plan the surgery beautifully and never say where they would stop.

SOE Case 3 — A perio-endo lesion in a medically complex patient

THE CASE

Middle-aged woman. Medical history: **breast cancer treated with radiotherapy**, **uncontrolled diabetes**, on **hormone replacement therapy**. Presenting with the **23**: **10 mm mid-buccal pocket, 8 mm mesiobuccal**; periapical shows a **radiolucency extending from the lateral aspect of the root to the apex**; a **sinus tract is suspected but hard to identify** on a blurred photograph; the tooth is **sensitive to cold**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A middle-aged woman with three medical issues that all bear on healing — **breast cancer treated with radiotherapy**, **uncontrolled diabetes** and **HRT** — presenting with a 23 that has a **10 mm mid-buccal and 8 mm mesiobuccal pocket**, a radiolucency running from the lateral aspect of the root to the apex, a **suspected sinus tract**, and, importantly, a tooth that still **responds to cold**. My working diagnosis is a lesion of **combined periodontal and endodontic origin**, but the cold response argues against a straightforwardly necrotic pulp, so I would not commit yet. The three things I must establish are the true medical picture — particularly whether she has had an **antiresorptive** — her current **HbA1c**, and where that sinus tract actually leads."

Domain 1 — Clinical interpretation and reasoning

Chain: the diagnostic groundwork

Q By what routes can pulp and periodontium communicate?

"Four. The **apical foramen**, the main pathway in both directions. **Lateral and accessory canals**, most numerous in the apical third and in furcations. **Dentinal tubules**, exposed where cementum is absent or lost, particularly at the cemento-enamel junction and in cervical areas. And **developmental anomalies** — palatogingival grooves, most classically on maxillary incisors, and enamel projections. Iatrogenic routes such as perforations and root fractures are the fifth."

Q Classify the pulpal and periapical diagnoses.

"Pulpally: **normal**, **reversible pulpitis**, **symptomatic** or **asymptomatic irreversible pulpitis**, **pulp necrosis**, and previously treated or previously initiated. Periapically: **normal**, **symptomatic** or **asymptomatic apical periodontitis**, **acute apical abscess**, **chronic apical abscess** — which is the one that presents with a **sinus tract** — and condensing osteitis. Naming both halves forces me to be specific rather than saying 'an infected tooth'."

Q What does a cold test actually test, and what are its limits?

"It tests **A-delta fibre response**, so it tells me there is vital, innervated tissue somewhere in the canal system — not that the whole pulp is healthy. **False negatives** occur with calcification, recent trauma, heavy restorations, immature apices and anxious patients. **False positives** occur with pulp stones, periodontal ligament stimulation and multi-rooted teeth where one canal remains vital. It must be interpreted against contralateral and adjacent controls, never in isolation."

Q How would you stage and grade her periodontal condition?

"Under the **2017 World Workshop**: **stage** by severity and complexity, from interdental attachment loss and bone loss — I would need full charting and radiographs, but a 10 mm pocket suggests at least **Stage III**. **Grade** by rate of progression, using bone loss as a percentage of age, with **smoking and diabetes as grade modifiers**. **Extent** as localised, generalised or molar-incisor pattern. And I would state whether it is currently **stable, in remission or unstable**."

Q What does her diabetes do to that grade?

> "It raises it. An **HbA1c of 7.0% or above in a diagnosed diabetic is a grade modifier that takes her to Grade C** under the Workshop criteria, meaning rapid progression and a worse response to treatment. The relationship is **bidirectional** — hyperglycaemia impairs neutrophil function, collagen turnover and healing, while periodontal inflammation worsens glycaemic control. That is the single most useful thing I can tell her."

Chain: reading this patient

Q What is your differential for the 23?

> "A **combined perio-endo lesion**; a **primary endodontic lesion with secondary periodontal involvement** draining through the sulcus; a **primary periodontal lesion** in a diabetic with generalised periodontitis; a **vertical root fracture**; or **invasive cervical resorption** or other local pathology. A palatogingival groove would also be on the list at a maxillary anterior tooth."

Q Which finding discriminates most, and how?

> "The **cold sensitivity**. A tooth that responds to cold has vital pulp tissue, which argues against a primary necrotic endodontic lesion — although a partially necrotic or multi-canalled tooth can still respond. A **single deep narrow isolated pocket** in an otherwise reasonable mouth would point me towards a **vertical root fracture**; a broad pocket with generalised bone loss elsewhere points to periodontal disease as the primary problem."

Q What further investigations?

> "**Sensibility testing** of the 23, 22 and 24 with cold and electric pulp testing, against contralateral controls. **Full six-point charting**, mobility, plaque and bleeding scores, and a BPE. **Trace the sinus tract with a gutta-percha point** and radiograph with it in place. Good **long-cone paralleling periapicals at two horizontal angles**. Transillumination and staining for a crack. And a current **HbA1c and full drug list** from her GP and oncology team."

Q Why is the gutta-percha point the most useful single test here?

> "Because it converts a suspicion into an anatomical answer. The point follows the tract to its origin, and the radiograph then tells me whether the source is the **apex** — endodontic — the **lateral aspect of the root** — a lateral canal or a fracture — or the **sulcus and crest** — periodontal. It costs nothing, takes two minutes, and changes the whole treatment plan."

Q The photograph of the tract is blurred. What do you do?

> "I say so. **An inadequate record is not a basis for a diagnosis**, in the clinic or in the examination. Clinically I would dry the mucosa, palpate along the sulcus and look for a bead of pus on gentle pressure, place a gutta-percha point, and repeat the photograph properly. I would not commit to a diagnosis on a poor image and I would record that I had not."

Domain 2 — Clinical decision making and treatment planning

Chain: taking the medical history apart

Q The breast cancer and radiotherapy — what exactly do you need to know?

> "The **field**. Radiotherapy to the breast and axilla does not irradiate the jaws and has essentially no bearing on oral surgery. **Head and neck radiotherapy is an entirely different matter**, with a risk of **osteoradionecrosis** rising above about **50 Gy**, in which case elective implants in irradiated bone would need specialist assessment. I also want the date, whether she had chemotherapy, her current disease status and who is following her up."

Q Why do you ask about bisphosphonates and denosumab, and does the route matter?

> "Because both are commonly given in breast cancer — **intravenously for bone metastases**, or orally or subcutaneously for treatment-induced bone loss — and the woman herself often does not call them cancer drugs. Route and indication matter enormously: an **intravenous antiresorptive given for malignancy** puts her in the

high-risk MRONJ group, and I would regard **elective implant placement as contraindicated**. A low-dose oral bisphosphonate for osteoporosis is a much lower risk."

Q How would you risk-assess and manage her for MRONJ?

"Following **SDCEP**: identify the drug, dose, route, indication and duration; place her in the **higher-risk group** if it is intravenous for malignancy, if she has had it more than five years, or if there is concurrent steroid or antiresorptive-plus-antiangiogenic therapy. Then: avoid extractions where a tooth can reasonably be saved, use **atraumatic technique**, give clear warning of the risk, and **review healing at eight weeks**, referring if it has not healed."

Q The uncontrolled diabetes — what do you need and what does it change?

"I need the **HbA1c**, the type, the treatment, who manages it and whether she has complications. Poor control impairs **neutrophil chemotaxis and phagocytosis**, collagen synthesis and microvascular perfusion, so it increases infection risk, slows healing and accelerates periodontal destruction. It partly explains her 10 mm pocket. It is a **relative contraindication to elective implant surgery until improved**, not an absolute one."

Q What figure would you want before elective implant surgery?

"I would want her **stable and under her diabetes team's control**, and in practical terms an **HbA1c below about 7%**, or at minimum a clear downward trend agreed with her physician. I would not fix on a single number as a barrier — the evidence is that **well-controlled diabetics have implant survival close to non-diabetics**, while poorly controlled patients heal slowly and have more early failures and more peri-implant disease."

Q And the HRT?

"Largely a marker of **menopausal status and bone metabolism** rather than a problem in itself. There is some association with **xerostomia** and with gingival changes. It is **not a contraindication** to any of this treatment. What I would check is that HRT is not being confused with, or given alongside, an antiresorptive."

Chain: the plan and the argument about it

Q Suppose the 23 is unrestorable. What is your plan?

"Phased. **Phase 1** — relieve pain, control infection, and get her **diabetes controlled** in liaison with her GP or diabetes team, while obtaining a full oncology and drug history and stabilising her periodontal condition with non-surgical therapy and oral hygiene instruction. **Phase 2** — atraumatic extraction with ridge preservation if an implant is contemplated, plus an interim prosthesis. **Phase 3** — re-evaluate. **Phase 4** — only then decide the definitive replacement."

Q Would you plan an implant at all, and on what conditions?

"Potentially, but **not now and not unconditionally**. My conditions would be: confirmed **no intravenous antiresorptive for malignancy**; **HbA1c controlled** and stable; **periodontal disease treated and stable** with good plaque control, since a history of periodontitis is a risk factor for peri-implantitis; no smoking; and a healed, adequate ridge. If she remains uncontrolled, the honest plan is a **resin-bonded bridge or a denture**, and I would say so."

Q A colleague says the evidence shows controlled diabetics do as well as anyone, so why are you delaying?

"They are right about **controlled** diabetics — that is exactly why I am delaying, because she is not one yet. The delay is not a punishment, it is a change of the variable that predicts failure. Where they might be right against me is on **timing of the extraction**: delaying the extraction of an infected tooth does not help her glycaemic control and may worsen it, so I would **treat the infection promptly and delay only the implant**."

Q SAC classification, and where is the boundary of your competence?

"Even a single anterior unit becomes **Advanced** here, because the complexity is medical rather than anatomical — an uncontrolled diabetic with a possible antiresorptive history and active periodontal disease. The **diagnosis, the medical liaison, the extraction and the interim prosthesis are all within my scope**. If she proves to be on

an intravenous antiresorptive, or if head and neck radiotherapy is confirmed, **implant treatment is not mine to provide** and I would refer for specialist assessment."

Q How would you explain the delay to her, in your own words to her?

" 'Your gum and bone heal much more slowly when your blood sugar is high, and an implant depends entirely on bone healing around it. If we put one in now it is much more likely to fail, and I don't think that would be fair to you. What I'd like to do is settle the infection, give you something to fill the gap, and work with your doctor on your blood sugar. Then we look at the permanent options from a much stronger position.' Then I would ask her to **tell me back** what we have agreed."

Domain 3 — Decision making in relation to treatment procedures

Chain: the acute phase and the pharmacology

Q Does she need emergency treatment today?

"If she has pain, a spreading infection, systemic upset, trismus, or swelling threatening the airway or the eye, then yes. The treatment for a **localised fluctuant swelling is drainage** — through the tooth if the source is endodontic, through the pocket or by incision if it is periodontal — with debridement. **Drainage plus removal of the source is the treatment; everything else is an adjunct.**"

Q Does she need antibiotics?

"Only if there are **systemic signs** — fever, malaise, lymphadenopathy — **spreading infection**, or significant immunocompromise. Her uncontrolled diabetes lowers my threshold, because hyperglycaemia impairs neutrophil function, but antibiotics are **never a substitute for drainage**. Prescribing them instead of treating the source is the commonest antimicrobial stewardship failure in dentistry."

Q If you did prescribe, what and for how long?

"**Amoxicillin**, or **metronidazole** where anaerobes predominate, for a **short course of about five days with a definite review** — not a repeat. If penicillin-allergic, **clarithromycin** or **clindamycin**, checking interactions against her oncology and diabetes medication. I would record the **indication** in the notes, not just the drug, and I would arrange to see her rather than leaving the course to run unsupervised."

Q What would you tell her about the infection and her diabetes?

"That the two feed each other: 'An infection like this pushes your blood sugar up, and high blood sugar makes it harder for you to fight the infection. So please check your levels more often than usual over the next few days, and let your diabetes team know that you've got a dental infection and what we've given you.' That is a safety instruction, not general advice, and I would record giving it."

Chain: the operative decisions

Q If the tooth were treatable, what determines whether you save it?

"**Prognosis, and specifically the periodontal component.** Treating the endodontic half of a true perio-endo lesion often resolves the apical portion, but the periodontal attachment loss determines the long-term outcome. A **10 mm mid-buccal pocket on a vital tooth in an uncontrolled diabetic is a poor prognosis**, and a vertical root fracture is hopeless. I would treat any endodontic component first, re-evaluate at three months, then decide."

Q Describe the extraction.

"Confirm the medical position first, **antiresorptives specifically**. Then **periotomes and luxators** to preserve the buccal plate, minimal elevation, forceps with controlled rotation given a conical canine root, thorough **curettage of granulation tissue**, inspection of all four socket walls, and irrigation. I would examine the extracted root under magnification for a **fracture line**, because that answers the diagnostic question retrospectively and informs what I do next door."

Q Would you preserve the ridge?

"If the buccal plate is intact and an implant is genuinely contemplated later, yes — a **slowly resorbing xenograft with a collagen membrane or socket seal** and a **figure-of-eight suture**. If the site is actively infected with a large lesion, or if implant treatment is unlikely because of her medical position, grafting adds cost and risk for no return, and I would simply debride and let it heal."

Q She turns out to have had intravenous zoledronate. How does the extraction change?

"Under **SDCEP** she is high risk. I would first re-examine whether the tooth can be **retained instead** — root treatment and decoronation rather than extraction is a legitimate alternative in this group. If extraction is unavoidable: liaise with her oncologist, minimal trauma, **round off sharp bony edges, primary closure where achievable**, explicit warning of MRONJ, and **review at eight weeks with referral if unhealed**. No implants, now or later."

Q Post-operative instructions and analgesia?

"Written as well as verbal instructions with an emergency contact. **Paracetamol first line**; I would check before recommending **NSAIDs** because of renal function in a long-standing diabetic and any interaction with her other medication. Standard advice on rinsing from the next day, and specific advice to **monitor blood glucose** and to eat as normally as she can rather than skipping meals while sore."

Q What interim prosthesis?

"An **acrylic removable partial denture** as the immediate option — cheap, quick, and easily relieved over a healing socket — or a **bonded pontic** on the adjacent teeth if the site allows, which puts no load on the healing site at all. I would relieve either generously over the graft or socket and review it, since a prosthesis pressing on a healing site in a diabetic is a genuine risk."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: judging the result and knowing when to stop

Q If you treat the 23 rather than extract it, how do you monitor?

"**Reduction in probing depth and bleeding**, resolution of the sinus tract, and **radiographic healing of the radiolucency at six and twelve months** against a proper baseline. Alongside that, her **HbA1c trend**, because if that does not move, nothing else will. Standardised radiographs, same technique and angulation, or the comparison is worthless."

Q You said you would set a threshold. What is it?

"That I decide **in advance**, and write it down: if at **three months** the pocket has not reduced and the sinus tract persists, the tooth is not being saved and I move to extraction. The failure mode in these cases is repeating the same treatment hoping for a different result while she loses bone that I will later want for an implant."

Q At three months the pocket is 8 mm and the tract is still there. What now?

"I follow my own threshold. I would re-examine for a **vertical root fracture** — raise a small flap if necessary and inspect the root surface directly, which is the definitive test — and re-check her HbA1c. Then extract, with the technique dictated by her antiresorptive status, and move to the interim prosthesis. I would explain that this was the agreed plan, not a failure of the last three months."

Q Her HbA1c comes back at 9.5% and she tells you she cannot change it. What do you offer?

"The same care with different expectations. I would **treat her periodontal disease and keep her stable**, because that will help her glycaemia as well as her mouth; extract what is hopeless; and provide a **resin-bonded bridge or a denture**, which do not depend on osseointegration. I would not simply withhold treatment. What I would not do is place an implant and pretend the risk is the same."

Q She asks whether she will ever be able to have an implant. What do you say?

> " 'Possibly, yes — but not while things are as they are. If your blood sugar comes down and stays down, and your gums are stable, then an implant becomes a reasonable option and I'd be happy to look at it again. If your cancer team has you on a bone-strengthening drug through a drip, then implants aren't safe for you and I'd want us to plan around that rather than risk it.' Honest, specific, and with the door left open."

Q What would you record?

> "The **medical liaison in full** — who I wrote to, what I asked, what I was told and when; the diagnosis and the evidence for it, including that I would not diagnose from an inadequate photograph; the options discussed including no treatment; the **reasons for deferring implant treatment**; any emergency treatment and any prescription with its indication; the threshold I set and when I would re-evaluate; and the review plan. If she later complains of delay, that record is my defence — and more importantly it is what makes her care coherent for whoever sees her next."

WATCH OUT

Come back to the sequence: control the infection, control the medical risk, control the periodontal disease, and only then replace the tooth. Saying which of those four the question belongs to buys you a sentence and is almost always the right frame in a medically complex case.

Running this case to time

Time check. The flow above holds 35 questions and about 1,838 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that is a little over **sixteen minutes**, so it **overruns by about one**. The first thing an examiner drops is the pharmacology sub-chain in Domain 3 — which antibiotic, for how long. If your answers run past **thirty seconds**, you will still be reciting the medical history when the three-month threshold question should be asked.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapical and the photograph in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Three medical issues that all bear on healing — **radiotherapy, uncontrolled diabetes, HRT**; the 23 with **10 mm mid-buccal and 8 mm mesiobuccal** pockets, a radiolucency from the lateral root surface to the apex, a suspected tract; and the pivot — **it still responds to cold**. Then the three things to establish: an **antiresorptive**, her **HbA1c**, where the tract leads.
1:20–4:30	Domain 1: the diagnostic groundwork, then reading this patient	The **four routes between pulp and periodontium**; both halves of the diagnosis, with **chronic apical abscess** as the one presenting with a sinus tract; what a cold test tests, and its false results; **2017 staging and grading**, with **HbA1c $\geq$ 7.0% as a Grade C modifier**, **bidirectional**; then the differential, with **vertical root fracture** kept live.
4:30–7:30	Domain 2: the medical history taken apart, then the plan	**The radiotherapy field** — breast and axilla is not head and neck, osteoradionecrosis above about **50 Gy**; **antiresorptives asked for by route and indication**; **SDCEP** risk grouping; diabetes as a **relative, not absolute** contraindication; the four-phase plan; and the competence line — **intravenous antiresorptive for malignancy, or head and neck radiotherapy, and implant treatment is not yours**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the acute phase, then the operative decisions	**Drainage plus removal of the source is the treatment; everything else is an adjunct**; antibiotics only for systemic signs; extraction with **periotomes and luxators**, the root inspected under magnification for a **fracture line**; ridge preservation only if an implant is genuinely contemplated; and how zoledronate changes it — **retention and decoronation instead**.
11:30–14:30	Domain 4: judging the result, and knowing when to stop	Monitoring against a **standardised baseline** at six and twelve months, alongside the **HbA1c trend**; the **three-month threshold set in advance and written down**; and what you offer at 9.5% — a **resin-bonded bridge or denture**, not withholding treatment.
14:30–15:00	Closes on the patient	**In her words**: gum and bone heal slowly when blood sugar is high and an implant depends entirely on that healing — settle the infection, fill the gap, work with her doctor. Then ask her to **tell it back to you**.

The spine — the twelve answers to have ready if nothing else

1 The four routes between pulp and periodontium · 2 The pulpal and periapical diagnoses, both halves · 3 What a cold test tests, and its limits · 4 Staging and grading, with diabetes as the modifier · 5 Why the gutta-percha point is the single most useful test · 6 The radiotherapy field, and why it matters · 7 Antiresorptives by route and indication · 8 The HbA1c you want before elective surgery · 9 SAC, and the point at which this is not yours · 10 Drainage before antibiotics · 11 How zoledronate changes the extraction · 12 The three-month threshold, set in advance.

Where the marks are lost in this case

Diagnosing from the blurred photograph. The tract cannot be identified on the image given, and saying so is a mark rather than an evasion. **An inadequate record is not a basis for a diagnosis** — trace it with gutta-percha, radiograph it in place, and record that you would not commit until you had.

Treating "radiotherapy" as one thing. Breast and axillary fields do not irradiate the jaws and change nothing surgically. Candidates who invoke osteoradionecrosis without asking the field look as though they have memorised a risk rather than understood it — and then miss the antiresorptive, which is the real hazard here.

Delaying the extraction along with the implant. Poor glycaemic control defers the *implant*; it does not defer treating an infected tooth, which will worsen her control if left. Volunteer that separation, and volunteer the fallback — a resin-bonded bridge or denture — if her HbA1c never moves.

SOE Case 4 — The very deeply placed 46 implant, with mesial drifting of the 47

THE CASE

Middle-aged man, clear medical history. Complains of **pus discharge** and **bleeding on brushing**. The **46 implant** was placed **12 months ago** and carries a **cemented crown**. Periapical shows **residual cement**, a **very deeply positioned** implant, an **abutment that is not fully seated**, and the **47 drifting mesially**. The implant is a tissue-level design.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A medically fit middle-aged man with **suppuration and bleeding** around a twelve-month-old cemented crown on a 46 implant. The radiograph shows **residual submucosal cement**, a **very deeply placed implant**, an **abutment that is not fully seated**, and the **47 drifting mesially**. My working diagnosis is **peri-implantitis**, with at least three identifiable contributing causes: the cement, the depth of placement — which made cement removal impossible and created a pocket neither he nor I can clean — and the microgap and plaque trap at the unseated abutment. The drifting 47 suggests an open contact or an occlusal change. The question I have to answer before anything else is whether it is **rational to keep this implant at all**."

Domain 1 — Clinical interpretation and reasoning

Chain: the peri-implant groundwork

Q Compare the soft tissue attachment around an implant with that around a tooth.

"Both have a **junctional or barrier epithelium** of similar length. The difference is below it. Around a tooth, **connective tissue fibres insert into cementum perpendicularly**; around an implant there is no cementum, so the fibres run **parallel or circularly** around the abutment and are not attached to it. There is **no periodontal ligament**, so no ligament blood supply and no proprioception — the vascular supply is from periosteum and bone alone."

Q What are the clinical consequences of those differences?

"The seal is **weaker and more easily penetrated**, so a probe passes closer to bone and probing depths read deeper for the same health. Inflammation **spreads more readily into bone** and the lesion is larger and less well encapsulated than in periodontitis, so **progression is faster and often non-linear**. There is **no proprioceptive feedback**, so occlusal overload is not felt. And healing after treatment is less predictable because there is no ligament to regenerate from."

Q Define peri-implant health, mucositis and peri-implantitis.

"Under the **2017 World Workshop**: **health** is no erythema, no bleeding on gentle probing, no suppuration and no bone loss beyond initial crestal remodelling. **Mucositis** is **bleeding on gentle probing** with those same stable bone levels — an inflammatory lesion confined to the mucosa, and reversible. **Peri-implantitis** is bleeding and/or **suppuration**, **increased probing depth** compared with previous examinations, and **bone loss beyond initial crestal remodelling**."

Q You have no baseline radiograph. How do you make the diagnosis?

"By the Workshop's alternative threshold: **bleeding or suppuration on probing, plus a probing depth of 6 mm or more, plus a bone level 3 mm or more apical to the most coronal intraosseous part of the implant**. And I would make a real effort to obtain the original records from whoever placed it, because a genuine baseline changes both the diagnosis and the medico-legal picture."

Q How do you probe an implant, and with what?

"**Six sites**, gently — around **0.2 to 0.25 newtons**, which is lighter than most people use — with a **plastic or blunt periodontal probe** to avoid scoring the titanium surface. I record depth, bleeding, suppuration, recession and the width of keratinised mucosa. Probing an implant is a comparison against previous readings more than an absolute number, which is why serial charting matters so much."

Q What crestal bone change is normal in the first year?

"Around **1 to 1.5 mm in the first year** after loading, and thereafter no more than about **0.2 mm per year** — the classical **Albrektsson** success criteria. That is the yardstick against which 'beyond initial remodelling' is judged. Modern platform-switched designs typically remodel less than that, so I would not use 1.5 mm as an allowance to excuse loss I have found."

Chain: reading this radiograph and this mouth

Q What can this periapical not tell you?

"It is a **two-dimensional image of a three-dimensional defect** — it shows me mesial and distal bone and tells me nothing about the **buccal and lingual walls**, which is where the defect morphology that governs treatment actually lies. It will not show **non-radiopaque cement** at all. And without a standardised, reproducible paralleling technique I cannot compare it reliably with any future film."

Q Why does residual cement cause this?

"Because set cement submucosally is a **rough, plaque-retentive foreign body** in intimate contact with the implant surface, sitting in a space that cannot be cleaned by the patient or the hygienist. It provokes both a **plaque-mediated** and a **foreign-body** inflammatory response. The association with peri-implantitis is strong, and the deeper the margin, the more cement is left and the less likely it is ever to be found."

Q Why is the 47 drifting, and what does that tell you?

"Most likely because the **proximal contact has opened** — loss of contact between implant restorations and adjacent natural teeth is a well-recognised and common phenomenon, because the tooth continues to erupt and migrate while the ankylosed implant does not. It tells me there is now a **food trap** feeding the problem, that the occlusion has changed, and that the 47 needs assessing in its own right, not just as a contact to be closed."

Q What else do you assess clinically?

"Whether the **implant itself or only the prosthesis is mobile** — a mobile implant has lost integration and the conversation becomes removal. The **occlusion**, including whether the drifted 47 has created a heavy or eccentric contact. The **contour and cleansability** of the crown and whether he can physically reach the area. His **plaque control, smoking, glycaemic status and periodontal status** elsewhere. And what he was told at the time and what maintenance he has actually had."

Q Why does his periodontal status elsewhere matter so much?

"Because a **history of periodontitis is one of the strongest risk factors for peri-implantitis**, along with **poor plaque control, smoking and lack of supportive maintenance**. If he has untreated periodontitis, treating this implant in isolation will fail — the same pathogens and the same host response are in the rest of his mouth. So a full periodontal assessment is part of treating the implant, not a separate exercise."

Domain 2 — Clinical decision making and treatment planning

Chain: keep it or lose it

Q What is the first decision, and what determines it?

"Whether it is **rational or irrational to keep this implant**, as the ITI decision tree frames it. That turns on three things: the **extent of bone loss** and whether the implant retains adequate support; whether the **position permits a restoration that can be cleaned**; and whether **decontamination of the exposed surface is actually feasible**. If the answer to any of those is no, treating it is postponing an inevitable removal at the patient's expense."

Q Assume it is rational to keep it. Outline your treatment.

"**Step one, cause-related therapy** — remove the crown, remove all cement, seat or replace the abutment, and modify or remake the prosthesis so it is cleansable. **Step two, non-surgical therapy** with debridement and oral hygiene instruction. **Step three, re-assess at about a month**. **Step four, surgery** if suppuration or bleeding persists with probing depths of 6 mm or more. Then lifelong supportive care."

Q Why is removing the crown not optional here?

"Because I cannot remove submucosal cement, assess whether the abutment is seated, or judge the cleansability of the emergence profile with the restoration in place. **Every one of the identified causes sits under that crown.** Non-surgical debridement around a crown that is itself the cause is treatment of the symptom, and it will fail."

Q How do you consent him for crown removal?

"Explicitly, because it is a **cemented** crown and removal may destroy it. 'To treat this properly I need to get underneath the crown, because I think there's cement left behind that's causing the infection. The crown is glued on, so there's a real chance it'll be damaged or need remaking, and I need you to agree to that before I start. If we don't do it, the infection carries on and you may lose the implant.' And I would record that in those terms."

Q What else must that consent conversation cover?

"That the underlying implant position may prove **unrestorable**, so we may be removing the crown to find out that the implant must come out. That treatment **may not arrest the bone loss**. That **soft tissue recession is a likely consequence of successful treatment**, which the ITI tree specifically requires me to discuss. The cost of a new restoration. And the alternative — **removal of the implant** — with what would replace it."

Q What does the non-surgical phase actually consist of?

"**Mechanical debridement** with titanium or plastic-tipped curettes and **glycine air-polishing** to reach the threads without scoring the surface. **Individualised oral hygiene instruction** with correctly sized interdental brushes and a demonstration in his mouth rather than a leaflet. **Risk factor modification** — smoking, glycaemic control, and treatment of periodontitis elsewhere. Local antimicrobials or antiseptics as adjuncts only. Then re-assess at about **one month**."

Q The implant is very deeply positioned. What does that mean for the outcome?

"It is the fundamental problem. A deep implant creates a **peri-implant sulcus that neither he nor I can clean**, and it is why the cement was never retrievable. Even with textbook treatment the site remains vulnerable, and I would tell him so honestly. It also argues strongly for converting to a **screw-retained restoration** to eliminate cement entirely and give me retrievability at every maintenance visit."

Q A colleague would simply remove the implant now. Why might they be right?

"Because they would be treating the cause rather than its consequences. If the position is so deep that no restoration will ever be cleansable, then every step I have described buys twelve or eighteen months at real cost, with recession and possibly implantoplasty along the way, before he ends up in the same place with **less bone than he has today**. **Early removal preserves bone for a properly placed replacement.** My counter is that I do not yet know the defect morphology, and that decision should follow crown removal, not precede it."

Q SAC classification, and what would you refer?

"Managing established peri-implantitis around a malpositioned implant with a cemented restoration is at least **Advanced**, and **surgical reconstructive management of a peri-implantitis defect is Complex**. I would undertake the **cause-related and non-surgical phase, the crown removal, the cement removal and conversion to a cleansable screw-retained restoration** myself. **Surgical management, implantoplasty or explantation I would refer** if beyond my experience — and in the posterior mandible, explantation near the inferior alveolar nerve is not a procedure to learn on."

Q What do you tell him about the likely outcome, in his words?

> " 'You've got an infection in the gum and bone around the implant. Some of it is caused by cement left behind when the crown was fitted, and some by how deep the implant sits, which I can't change. I think we can settle the infection and I'd like to try — but I have to be straight with you: the gum will shrink back and it may look longer, and there's a chance this comes back and the implant has to come out. If that happens, I'd rather we plan for it than be surprised by it.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: getting the crown off and the surface clean

Q How would you remove the cemented crown?

> "First the conservative attempts: a **crown-removal instrument** or an **ultrasonic tip applied to the margin** to fatigue the cement lute. If it is firmly cemented, I would **section it with a bur**, cutting from occlusal to buccal, protecting the implant shoulder and the soft tissue, and accepting that I am likely to destroy it. I would have consented for that, and I would have a plan for a provisional before I started."

Q And how do you remove the cement?

> "Irrigate, then work under **magnification and good illumination** with **titanium or plastic scalers** and **floss knotted and drawn circumferentially** around the abutment, followed by **glycine air-polishing** to reach the threads. Then a **periapical to confirm no radiopaque cement remains** — while remembering that a non-radiopaque cement will not appear at all, which is precisely why radiopaque cements should be chosen in the first place."

Q How would you have prevented this at the fitting appointment?

> "By keeping the **margin at or no more than 1 to 1.5 mm below the mucosa**, ideally supramucosal. By using **minimal cement** in a vented crown, seating it on an **abutment replica extraoradly** first to express the excess, and using a **radiopaque cement**. By taking a **post-cementation radiograph**. Or simplest of all, by making it **screw-retained**, which removes the cause completely."

Q If you proceed to surgery, describe it.

> "A **full-thickness mucoperiosteal flap** for access, extended enough to see the whole defect. Complete **removal of granulation tissue** with curettes so I can read the defect morphology. **Decontamination of the exposed implant surface**. Then the reconstructive or resective decision based on the defect. Careful flap adaptation and suturing, post-operative **chlorhexidine**, and systemic antimicrobials only where genuinely indicated."

Q How do you decontaminate the surface?

> "**Mechanically** with titanium brushes, titanium curettes or ultrasonic tips, and **air-polishing with glycine or erythritol powder**, which reaches thread depths that instruments cannot. **Chemically** with saline, chlorhexidine or hydrogen peroxide. **Er:YAG laser** and photodynamic therapy are alternatives. The honest position is that **no method reliably re-establishes a biologically clean surface**, and that uncertainty is part of the prognosis I give him."

Q Reconstructive or resective — how do you choose?

> "By **defect morphology**, which is why I remove granulation tissue before deciding. A **contained intrabony defect with several walls** is a reconstructive indication — a bone substitute with a barrier membrane, submerged where possible. A **supracrestal or horizontal defect with no containment** cannot be reconstructed, so it is **resective**: an apically positioned flap, osseous recontouring, and implantoplasty of the exposed threads. Combined defects get a combined approach."

Q You have said you would do implantoplasty here. Argue against it.

> "It is irreversible; it **weakens the implant**, which matters in a narrow implant or where bone loss is already extensive; it releases **titanium particles** into the tissues, whose long-term significance is not settled; and it guarantees a **long, exposed, visible implant surface** with recession. Against a posterior non-aesthetic site with

a supracrestal defect and good evidence of reduced re-infection, I still favour it — but if the defect turns out to be contained, reconstruction is the better answer, and I would not have committed beforehand."

Q Would you give systemic antibiotics?

"Not routinely. There is **no convincing evidence that systemic antibiotics improve outcomes in non-surgical peri-implantitis therapy**, and the stewardship argument against them is strong. I would consider a peri-operative course in surgical reconstructive treatment, or where there is spreading infection or systemic upset. The decisive interventions here are **mechanical debridement, decontamination and making the site cleansable**."

Q How would you restore it afterwards?

"**Screw-retained, without question** — no cement, retrievable, and removable by me at maintenance visits. A **custom abutment shaped for cleansability**, with generous embrasures for an interdental brush and no over-contour, and the margin brought as supramucosal as the depth allows. I would correct the occlusion, **close the open contact with the 47**, and treat the 47 on its own merits."

Q What if you have to remove the implant?

"**Reverse-torque** removal first — it is the least destructive method and works well in the first years or where bone loss has already occurred. Trephine only if that fails, accepting the bone it costs, and with the **inferior alveolar canal identified on a CBCT beforehand** in this region. Then either spontaneous healing or grafting, and a discussion about a replacement implant later, a bridge, or accepting a shortened dental arch."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcomes, maintenance and the difficult conversation

Q How will you judge success?

"**Absence of suppuration**, absence of bleeding on gentle probing, **probing depths that have not increased**, and **radiographic bone levels stable against a new baseline** taken after treatment — because the old level is gone and the honest measure is stability from here. I would re-assess at one month, three months, and then at each maintenance visit."

Q How successful is treatment of peri-implantitis, honestly?

"Less successful than treatment of periodontitis, and he should be told that. **Disease resolution is achieved in a proportion of cases but recurrence is common**, and the strongest predictors of a good result are **removal of the cause, a cleansable restoration and adherence to maintenance** — which is exactly why the crown must come off. I would present this as controlling a condition, not curing it."

Q What maintenance would you provide?

"**Every three to five months**, as the ITI tree specifies after treating peri-implant disease: professional biofilm removal, reinforcement and re-demonstration of hygiene at every visit, **peri-implant probing at six sites compared against the new baseline**, occlusal check, and radiographs when there is a clinical indication rather than routinely. Non-attendance at maintenance is itself a reason to reconsider whether the implant should stay."

Q At one year the suppuration is back. What now?

"I would treat that as the answer to the question I asked at the start. Re-examine, re-radiograph against the post-treatment baseline, and check whether the cause really was removed — is any cement left, is the restoration genuinely cleansable, is his plaque control adequate, has his periodontitis been treated? If the cause has been removed and it has still recurred, then it is **irrational to keep the implant** and I would recommend removal and discuss what replaces it."

Q He asks whether the previous dentist was negligent. What do you say?

"I would **not comment on another clinician's care**. I would explain what I have found and what I think is causing it, factually and without attributing blame. I would tell him he is entitled to raise concerns **with the practice that treated him, or with the GDC**, and give him what he needs to do that. I would record exactly what I found and what I said, and **notify my indemnity provider** if I were asked to comment formally. If he wanted a report, I would report only on my own findings."

Q What would you record?

"The presenting complaint and his account of the history; **six-point probing, bleeding, suppuration and mobility**; the radiograph and my interpretation of its limitations; the diagnosis and the criteria I used; the **causes identified**; the options including removal; the **consent discussion in the words used**, particularly the risk to the crown and the likelihood of recession; what I actually did; the **new baseline** radiograph and chart; the maintenance interval agreed; and, verbatim, what I said when he asked about the previous dentist."

WATCH OUT

Go back to the first question — is it rational or irrational to keep this implant — and then to the three-step sequence of removing the cause, treating the disease, and making the site maintainable. Saying which of those the question belongs to buys you a sentence and is nearly always the right frame in a peri-implantitis case.

Running this case to time

Time check. The flow above holds 33 questions and about 1,731 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that comes to just over **fifteen minutes**, so it fits the slot almost exactly as written. There is no slack: Domain 3 runs from crown removal to explantation without a pause, and an examiner behind time drops the prevention question rather than the negligence question. If your answers run past **thirty seconds** you will be arguing about implantoplasty when you are asked what you would say to the patient.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapical in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Suppuration and bleeding at a twelve-month-old cemented crown on the 46; **peri-implantitis**, with **three causes named** — residual submucosal cement, a very deeply placed implant that made removing it impossible, and an **unseated abutment** with its microgap; the **47 drifting** as an open contact and a food trap; and the question that governs everything — **is it rational to keep this implant at all?**
1:20–4:30	Domain 1: the peri-implant groundwork, then reading this radiograph	**Fibres parallel and circumferential, not inserting**, no ligament, no proprioception — hence a weaker seal and faster, less linear progression; the 2017 definitions of **health, mucositis and peri-implantitis**; the no-baseline threshold — **bleeding or suppuration, ≥ 6 mm, ≥ 3 mm bone loss**; **0.2 to 0.25 N** probing with a plastic probe; **Albrektsson**; then what a two-dimensional film cannot show — the buccal and lingual walls, and non-radiopaque cement.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: keep it or lose it	The **rational-or-irrational** decision as the ITI tree frames it — bone loss, cleansable position, feasible decontamination; the four-step plan with **cause-related therapy first**; why **crown removal is not optional**; consent naming damage to the crown, possible unrestorability and **recession as a consequence of success**; the colleague who would explant now **argued at their strongest**; and the referral line — **surgery, implantoplasty and explantation near the inferior alveolar canal**.
7:30–11:30	Domain 3: getting the crown off, and the surface clean	Conservative removal then sectioning, consented for; cement cleared under **magnification with titanium instruments, knotted floss and glycine air-polishing**; prevention named — **margin within 1 to 1.5 mm, radiopaque cement, or screw retention**; granulation tissue removed **before** choosing; the honest admission that **no method reliably re-establishes a clean surface**; contained defect reconstructive, supracrestal resective; **implantoplasty argued against yourself**; **reverse-torque** removal with the canal identified on CBCT.
11:30–14:30	Domain 4: outcomes, maintenance and the difficult conversation	Success judged against a **new baseline**, because the old level is gone; the honest statement that **recurrence is common** and this is control rather than cure; **three- to five-monthly** maintenance with six-site probing; the one-year recurrence answered by your own opening question; and the negligence question — **no comment on another clinician**, the practice and the **GDC** named, indemnifier notified.
14:30–15:00	Closes on the patient	**In his words**: some of this is cement left behind and some is how deep the implant sits, which cannot be changed; the gum will shrink and may look longer; it may come back and the implant may have to come out — and better planned for than a surprise.

The spine — the twelve answers to have ready if nothing else

1 Implant versus tooth soft tissue attachment · 2 Health, mucositis, peri-implantitis defined · 3 The diagnosis with no baseline radiograph · 4 How you probe an implant, and with what · 5 Normal first-year crestal remodelling · 6 Why residual cement causes this · 7 Why the 47 has drifted · 8 Rational or irrational to keep it · 9 Why the crown must come off · 10 Reconstructive against resective, by defect morphology · 11 Implantoplasty, argued against yourself · 12 What you say when he asks about the previous dentist.

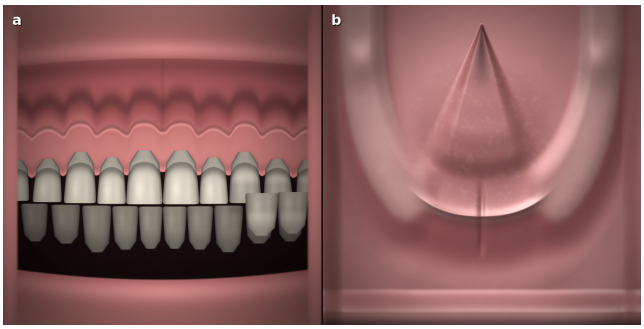
Where the marks are lost in this case

Debriding around the crown. Candidates describe glycine air-polishing and interdental brushes while leaving in place the restoration that is causing the disease. Every identified cause sits under that crown, so removing it — and consenting for its destruction beforehand — comes before any talk of surgery.

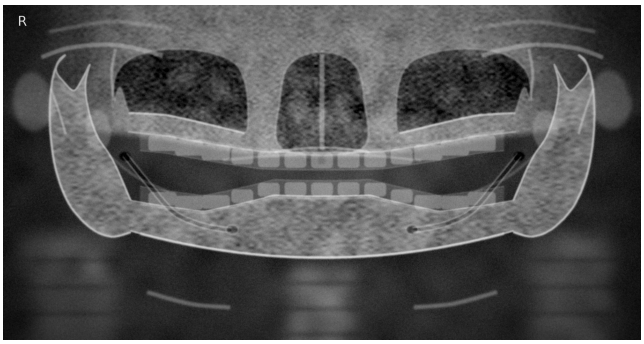
Treating the depth as a detail. The very deep placement is the reason the cement was unretrievable and the reason the sulcus can never be cleaned. Say plainly that even textbook treatment leaves this site vulnerable, and convert to **screw retention**; a candidate who promises resolution has misread the case.

Ducking the negligence question. It will be asked, and the answer is a competence line of its own: comment on your own findings only, tell him he may raise concerns with the practice or the **GDC**, notify your indemnifier, and record what you said verbatim.

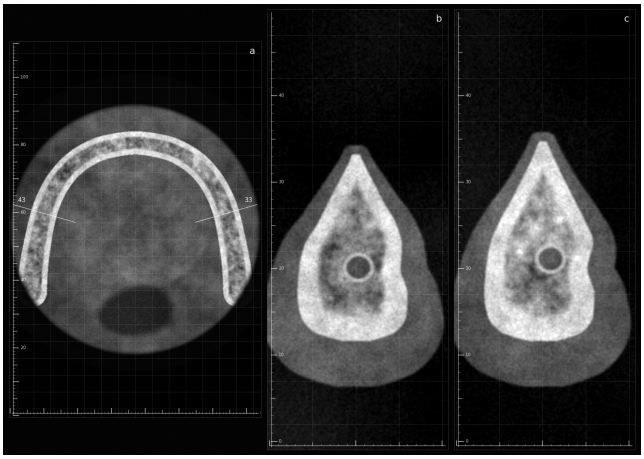
SOE Case 5 — The long-term denture wearer and the mandibular implant overdenture



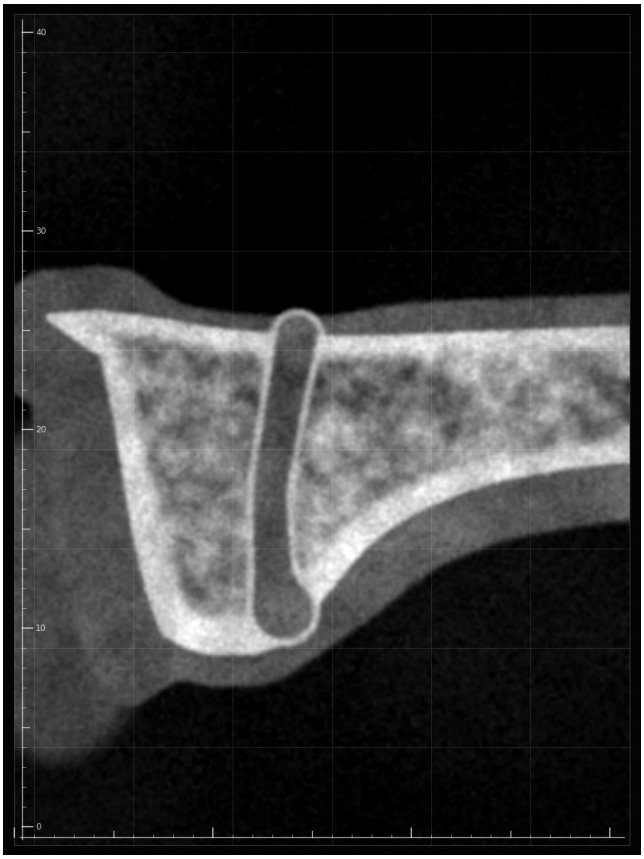
Intra-oral views, with the dentures in and the lower denture out — simulated illustration, not a clinical image



Panoramic view, dentures in situ — simulated illustration, not a clinical image



CBCT: axial locator and cross-sections at the 33 and 43 regions — simulated illustration, not a clinical image



CBCT: anterior maxilla at the midline — simulated illustration, not a clinical image

THE CASE

Middle-aged man reporting a **loose lower denture**, worn for **18 years**. The denture has a **well-extended** flange. Intra-oral photographs are provided with and without the denture. There is a **crossbite in the second and third quadrants**. CBCT shows the **incisive canal at the maxillary midline**, a **knife-edge mandibular ridge** with **posterior resorption**, imaged with the denture in place. Cross-sectional slices are provided at the **33 and 43** regions.

Pace: one structured opening of about 60 seconds, then 38 rapid questions in the remaining 13 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A middle-aged man with an 18-year-old lower complete denture, now loose. The flange is well extended, so this is not a border-moulding failure but **progressive residual ridge resorption**, which is faster in the mandible and continues lifelong. The CBCT confirms it — a knife-edge anterior ridge with posterior resorption. The posterior crossbite is acquired from the differing resorption patterns of the two jaws, not developmental. My working position is that a **two-implant mandibular overdenture** is the treatment most likely to solve his problem. What I need to establish is whether the anterior mandible will safely accept two implants, whether his existing prostheses are otherwise sound, what maintenance he can commit to, and what he actually wants."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical groundwork

Q Describe the surface anatomy of the edentulous maxilla.

"Front to back: **labial and buccal sulci** and frena; residual ridge and crest; **maxillary tuberosities**; **hamular notches**; incisive papilla; rugae; median palatine raphe; palatine fovea; and the **vibrating line**, the denture's posterior limit."

Q Which give support, and which give retention?

"**Support** — hard palate and firm ridge crest. **Relief** — incisive papilla, median raphe, flabby ridge. **Retention** — peripheral seal plus a **posterior palatal seal** at the vibrating line, holding as the soft palate moves."

Q Now the mandible, and its primary stress-bearing area.

"Labial, buccal and lingual sulci and frena; residual ridge; **buccal shelf**; external and internal oblique ridges; **mylohyoid ridge**; **retromolar pad**; genial tubercles; **mental foramina**. The buccal shelf bears load — dense cortical bone perpendicular to it. The resorbing crest is the worst place to load."

Q Why does the lower denture always give more trouble?

"A quarter of the bearing area, no palatal vault, a **mobile floor of mouth and tongue**, active muscle attachments, continuing resorption. Cawood and Howell: the mandible resorbs about **four times** as fast as the maxilla."

Q What is the significance of the incisive canal here?

"It carries the nasopalatine bundle: relieved in a denture, **avoided by any implant**, which will not integrate in it. It widens with resorption — a note for the future, as I am not planning maxillary implants."

Chain: reading this patient's radiographs and jaws

Q Why the posterior crossbite, and what does it mean for the new denture?

"**Acquired.** The maxilla resorbs upwards and inwards, the mandible downwards and outwards, so the lower ridge ends up anterior and lateral to the upper. So I set the posterior teeth **over the ridges, in crossbite**; teeth outside the crest lever the denture."

Q Describe the cross-sections at the 33 and 43, and classify the ridge.

"A **knife-edge crest**, narrow coronally and wider basally. I would record crest width, width 5 mm apical, height above the canal, and angulation, often lingual. **Cawood and Howell Class IV** — adequate height, inadequate width, so a horizontal deficiency."

Q What about bone density?

"**D1 or D2** on Misch. Excellent primary stability, hence the classic overdenture site, but poorly vascular — **overheating is the real risk**: sharp drills, slow speed, copious irrigation."

Q Limitations of a CBCT taken with the denture in place?

"Useful if the denture carries radiopaque markers, relating tooth position to bone. Limiting because the base creates artefact and **the mucosa is displaced**, misrepresenting tissue thickness. It must still be justified under IR(ME)R."

Q What else would you examine clinically?

"Lip support, smile line, **vertical dimension and freeway space**, TMJ; then ridge form, **keratinised width**, sulcus depth, floor-of-mouth height, tongue, saliva, stomatitis; and both dentures judged on retention, stability, support and occlusion."

Q How do you measure the freeway space?

"Marks on nose and chin, patient upright. Measure in occlusion; establish rest position by relaxation, swallowing and the closest speaking space; measure again. Mean of three. The difference is the freeway space — normally **2 to 4 mm**."

Domain 2 — Clinical decision making and treatment planning

Chain: complete denture fundamentals before implants

Q The three requirements of a complete denture?

> "**Support** — resistance to vertical seating force. **Retention** — resistance to dislodging force, from peripheral seal, adhesion, cohesion and muscular control. **Stability** — resistance to horizontal and rotational force, from ridge form, neutral-zone tooth position and balanced occlusion."

Q What occlusal scheme, and what are the alternatives?

> "**Balanced articulation** — bilateral simultaneous contact in centric and excursion — because a denture is a free body and any single contact tips it; delivered through Hanau's quint. Alternatives: anatomical 30-degree teeth, semi-anatomical, monoplane for an unpredictable jaw relationship, and **lingualised occlusion**, my choice on a resorbed ridge."

Q Does that change once the denture is implant-retained?

> "It matters more. **Lingualised occlusion with bilateral balance**, narrow occlusal table, anterior teeth free in centric, denture **tissue-supported posteriorly** — otherwise the load cantilevers onto two anterior implants."

Q What are the options for this man?

> "**Remake conventional dentures** with restored vertical dimension and a neutral-zone technique — must be offered, but the ridge is the limit. A **copy denture**. A **two-implant overdenture**, which **McGill and York** call the minimum standard for the edentulous mandible. Four implants and a bar. A fixed prosthesis. Or nothing."

Q Which do you recommend?

> "A copy-technique lower denture **retained by two implants at the 33 and 43 with stud attachments**, plus a new or relined upper. Strong long-term evidence, dense bone away from the mental foramina, modest cost and morbidity. **SAC: Straightforward to Advanced** — within my competence."

Q He has read about fixed teeth on implants. Argue against yourself.

> "**Lip support** — his flange is doing real work after 18 years of resorption and a fixed prosthesis cannot reproduce it; I would show him with a trial set-up. Then cleansability. And **full-arch fixed work is beyond a general dentist with an interest in implants** — I would refer."

Domain 3 — Decision making in relation to treatment procedures

Chain: overdenture design and its specific requirements

Q Take me through the attachment options and the space each needs.

> "**Studs** such as Locators — independent, cheap, cleanable, retention changed by swapping a nylon insert, tolerating about 20 degrees of divergence, needing **8 to 10 mm** of vertical space. **Balls** are bulkier. A **bar** splints and shares load and suits divergent implants, but needs 12 to 15 mm and is harder to clean."

Q How does the knife-edge ridge affect implant choice?

> "I need **1 mm of bone circumferentially**, so a 4 mm implant wants 6 mm of crest, which he lacks. Options: **crest reduction** to a wider basal level; a **narrow 3.3 mm implant**, acceptable here but with higher fracture risk under a stud, which transmits load unsplinted; ridge splitting; or GBR."

Q Where exactly do you place them?

> "**Canine regions, 33 and 43**, at least **5 mm anterior to the mental foramen or its anterior loop**, **20 to 22 mm apart**, as parallel as possible, following a ridge that is often lingually inclined, with room above for the housing plus **2 mm of acrylic**."

Q What goes wrong if they are too far forward, or too far back?

"Too far forward, the rotational axis becomes a point: the denture rocks, inserts wear, the base fractures where acrylic is thinnest. Too far back, the **mental nerve** is at risk, the posterior extension shortens and cantilevers the saddles, and a straight implant may perforate the submandibular fossa."

Q Take me through the surgery.

"Aseptic technique, **infiltration rather than a block** so he can report a sharp sensation. Mid-crestal incision canine to canine, full thickness, both **mental bundles identified**. Flatten the crest. Graded sharp drills, slow speed, copious irrigation. **25 to 45 Ncm**, one stage with healing abutments. Denture out 48 hours, then relieved and soft-relined."

Chain: placement protocol and loading

Q When would you place, and when would you load? Separate the two.

"**Two separate decisions.** Placement timing, Types 1 to 4, is when I place relative to the extraction. Loading — immediate, early, conventional — is when the restoration comes into occlusal contact. Any placement type can pair with any loading protocol, but the pairings are not equally documented."

Q Which placement type applies to this ridge, and why?

"**Type 4, late placement.** The classification describes socket healing, and after 18 years there is no socket — the ridge is fully healed. Best documented and most predictable, but maximum resorption has already happened: that is why the crest is knife-edge."

Q And the loading protocol?

"**Conventional loading** — pick-up beyond two months, about three months in the mandible, confirmed by clinical stability and ISQ where available. Type 4C is **scientifically and clinically validated**; no reason to take risk in a man who has coped for 18 years."

Q A colleague loads two mandibular implants on the day. What would have to be true?

"All of: **torque above 35 Ncm or ISQ above 70**, D1 or D2 bone, no parafunction, no simultaneous graft, a compliant patient on a soft diet. **Splinting the implants with a bar**, or placing four, strengthens the case; immediate loading of two independent studs is less well supported. Every criterion exists to control **micromotion**, above which fibrous repair replaces bone."

Q What would make you abandon early loading on the day?

"Torque under 35 Ncm, a low or falling ISQ, the drill dropping through soft bone, a dehiscence, or a crest needing reduction or grafting. Then healing abutments, a soft-relined denture and conventional loading — the stability trough at **two to four weeks** is unforgiving."

Chain: choosing an augmentation material

Q Name the four categories, and be honest about what each does.

"**Autograft** — osteogenic, osteoinductive and osteoconductive; the only one with living cells. **Allograft** — osteoconductive, demineralised freeze-dried bone variably osteoinductive. **Xenograft**, deproteinised bovine mineral — **osteoconductive only**. **Alloplast**, β-tricalcium phosphate or hydroxyapatite — **osteoconductive only**. A membrane is osteopromotive."

Q Advantages and drawbacks of each?

"**Autograft**: best biology, fastest incorporation; donor morbidity, small volume, unpredictable resorption. **Allograft**: no donor site; human origin, batch variability. **Xenograft**: outstanding volume stability, most documented; animal origin, particles persisting for years. **Alloplast**: cheap, no ethical objection; least documented, resorption too fast or too slow."

Q For this knife-edge ridge — graft it, or reduce it?

"**Reduce it.** Alveoloplasty to a wider basal level avoids grafting altogether, and he has height to spare provided I clear the canal and keep adequate implant length. I would graft only a residual dehiscence — **autogenous chips against the implant, bovine mineral outside**. Ridge splitting, blocks and vertical augmentation are **beyond me; I refer**."

Q Which barrier membrane, and why?

"**Resorbable native collagen** for a small contained dehiscence — no second surgery, and if it exposes without infection, chlorhexidine and no brushing usually sees it heal. **Non-resorbable PTFE** holds space better but must be removed, and exposure means removing it early — and a defect that size I would refer."

Q What must the consent cover for the graft material specifically?

"The **origin** — bovine graft, **porcine collagen** membrane, human if allograft — raised well before the day, because religious, cultural, dietary and vegetarian objections are legitimate. The theoretical, very low transmission risk. That autogenous-plus-synthetic alternatives exist and are less documented. And that **material, brand and lot number are recorded**."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: complications, their frequency and their prevention

Q Survival of implants used for a mandibular overdenture?

"**Around 95% or better at 10 years** in the anterior mandible, among the best-documented figures in implant dentistry. The consent point is that survival and satisfaction are excellent but **maintenance is frequent** — the failures are mechanical, not biological."

Q So the prosthetic complications, in order of frequency?

"**Worn nylon inserts** by a wide margin, typically annually. Then **denture base fracture** over the housings; abutment loosening; **relining** as the posterior ridge resorbs; tooth wear with loss of vertical dimension. Mucositis and peri-implantitis are less frequent but need the same monitoring."

Q How do you prevent each?

"Inserts — lowest retention that satisfies him, implants parallel, replaced by schedule. Base fracture — **2 mm of acrylic over the housing**, cobalt-chromium or mesh reinforcement, denture kept tissue-supported. Loosening — correct torque, rechecked. Relining — expected, so the fit surface is reviewed at every recall."

Q At two years it is loose and rocks. Differential?

"**Continued posterior ridge resorption**, so it is no longer tissue-supported — commonest, and the answer is a reline, not new inserts. Then worn inserts, a fractured base, a loose abutment, a lost implant, peri-implant disease, an occlusal error. I check each implant individually, the fit surface with pressure-indicating paste, then the occlusion."

Q What would you monitor, and what would you record?

"Six-monthly: **probing at six sites with a plastic probe**, bleeding, suppuration, plaque control, denture fit, extension, occlusion, attachment retention. Recorded: component lot numbers, insertion torque, the consent discussion, baseline radiographs at fitting, every insert change. Radiographs at one year, then as indicated."

Q Explain the maintenance to him, in his words.

" 'The two supports should last many years and the denture will feel secure. But it still rests on your gum, and the gum keeps shrinking slowly — so expect new clips about once a year and a reline every few years. If it clicks, rocks or feels loose, come in early.' "

WATCH OUT

Fall back to the three requirements — support, retention, stability — and say which one this problem belongs to. It buys you a sentence of thinking time and it is almost always the right frame for a denture question.

Running this case to time

Time check. The flow above holds 39 questions and about 1,760 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that is close to **fifteen minutes exactly**, which is the point: it is written to fill the slot at the speed the slot is actually run. If your answers are running longer than about **thirty seconds**, you will reach the end of Domain 2 and no further, and the marks in Domains 3 and 4 will be awarded on nothing.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the images where you can see them	Nothing. Do not start talking into this.
0:20–1:20	**Lets you present, uninterrupted**	The opening: an 18-year denture, well extended, therefore **progressive residual ridge resorption** rather than a border-moulding failure; the knife-edge anterior mandible and posterior resorption; an acquired crossbite; and a stated working position — a **two-implant mandibular overdenture** — with the three things you must still establish.
1:20–4:30	Domain 1: the anatomical groundwork, then the radiographs. Roughly ten questions, moving fast	Named landmarks of both jaws and **why each matters for denture support**; the incisive canal; reading the cross-sections at 33 and 43 — knife-edge form, cortical-dominant **D1/D2** density, height above the canal; and what the CBCT cannot tell you.
4:30–7:30	Domain 2: complete denture fundamentals before implants; then the options and the argument between them	Support, retention and stability defined separately; the **occlusal scheme** named and justified; **McGill and York** quoted in one clause; SAC classification of the case; and the honest alternative — a well-made new conventional denture first.
7:30–11:30	Domain 3: overdenture design, then placement protocol and loading, then the graft question	Locator versus bar with the specific requirements — **implant diameter, acrylic thickness over the attachment, fracture risk, cleansability**; **Type 4 placement** kept separate from **conventional loading**, with the criteria that would permit early loading; and alveoloplasty versus grafting, with the referral line named.
11:30–14:30	Domain 4: what goes wrong, how often, how you prevent it	Prosthetic complications by **type and frequency** — attachment wear and replacement, denture fracture over the attachment, relines, screw loosening — set against a high implant survival; then prevention, and **what you would record**.
14:30–15:00	Closes, usually on the patient	**What you would say to him, in his words**: what will improve, what will not, and what he must come back for. Candidates lose this mark by still being in Domain 3.

The spine — the twelve answers to have ready if nothing else

1 Why the denture is loose despite good extension · 2 The landmarks that give a lower denture its support · 3 What "knife-edge" means for implant placement · 4 Bone density here, and why it matters · 5 The mental foramen and where it puts your implants · 6 Support, retention, stability — defined · 7 The occlusal scheme, named and justified · 8 Two implants, and the McGill statement in one sentence · 9 Locator versus bar, with the specific requirements of each · 10 Placement type and loading protocol, kept apart · 11 The commonest prosthetic complications and roughly how often · 12 What you record and what you tell him.

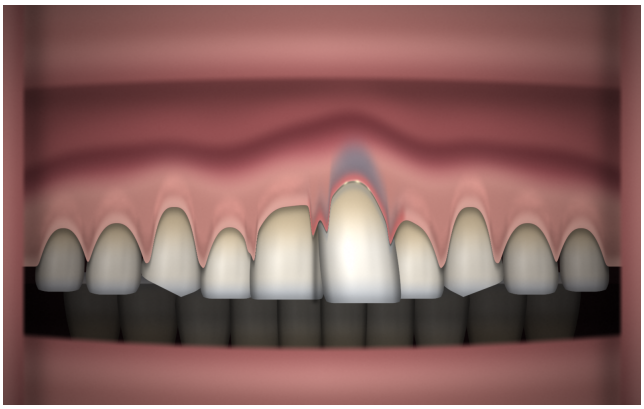
Where the marks are lost in this case

Going straight to implants. The first third of this case is undergraduate prosthodontics, and it is marked. A candidate who cannot define retention or name an occlusal scheme has failed Domain 1 before the CBCT appears.

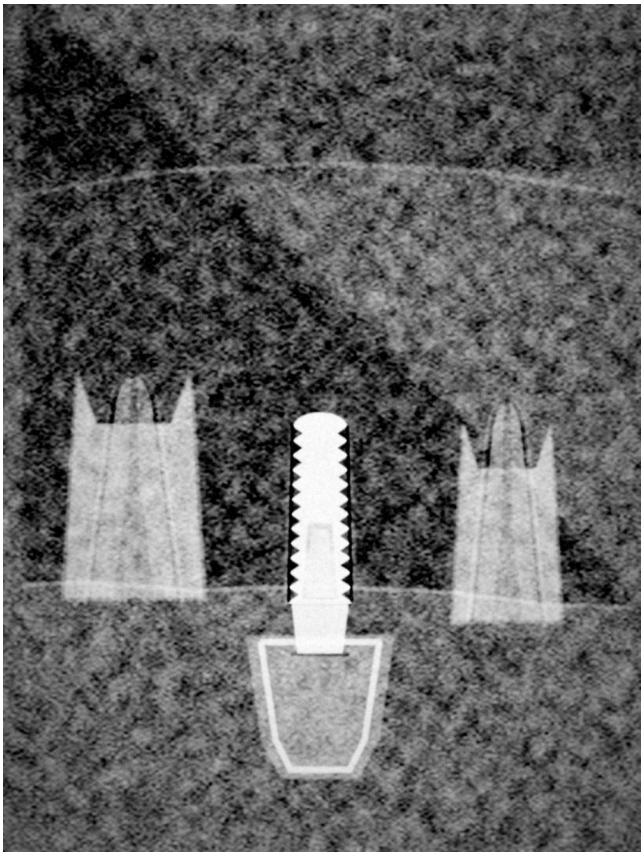
Confusing the two protocols. Placement timing and loading protocol are separate decisions. Saying "immediate" without saying *immediate what* reads as not knowing the difference.

Under-selling the maintenance. The overdenture is not finished at fit. Volunteer the attachment wear, the reline interval and the recall — unprompted, this is the difference between a 3 and a 5.

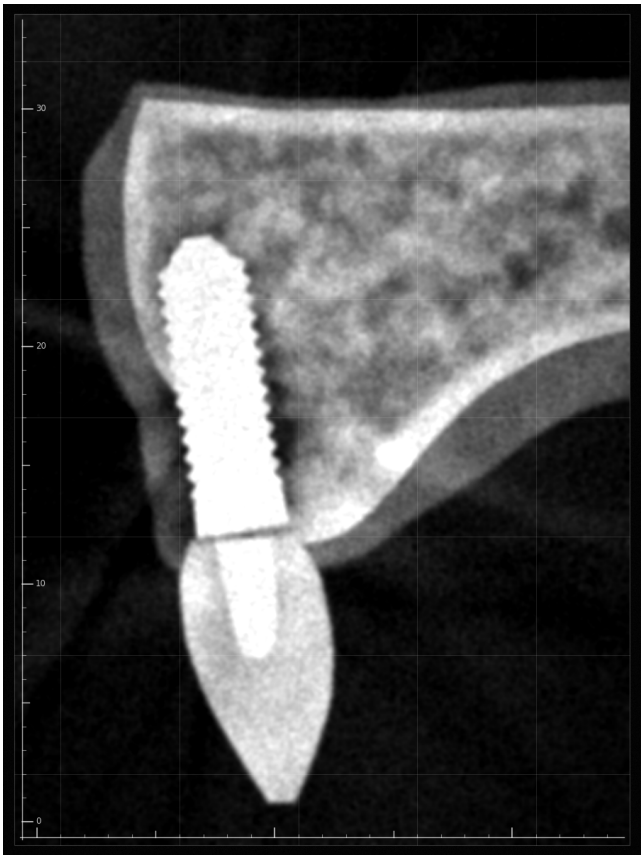
SOE Case 6 — Recession and grey show-through at a malpositioned anterior implant



Anterior view of the implant restoration at the 21 — simulated illustration, not a clinical image



Periapical of the 21 implant — simulated illustration, not a clinical image



CBCT cross-section at the 21 — simulated illustration, not a clinical image

THE CASE

Patient with an implant at the **21**, placed **immediately after extraction 10 years ago**. There is **recession with metal show-through**, and the implant is **facially malpositioned**. There is a history of **recurrent soft tissue infection** at the site.

Pace: one structured opening of about 60 seconds, then 36 questions in the remaining 13 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"An implant at the 21, placed immediately into the socket ten years ago, now facially malpositioned with mid-facial recession, visible metal and recurrent soft tissue infection. My reading is a chain that began at surgery: a fixture placed too far buccally where the labial plate is under 1 mm thick and is **bundle bone**, which resorbs once the ligament is lost. With no bone buccal to the implant, thin mucosa has receded over a decade and exposed the shoulder. The recurrent infection may be **peri-implantitis**, an uncleansable deep sulcus, or **residual cement**. What I need to establish is the three-dimensional position on a small-field CBCT, whether there is active disease, his **smile line**, and what actually troubles him — because a malpositioned implant cannot be restored into the right place."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical and biological groundwork

Q Describe the anatomy relevant to placing an implant at the 21.

"A **thin labial cortical plate**, often dehisced or fenestrated; the **incisive canal** palatally; the nasal floor above; and a **scalloped margin** whose papillae depend on the crestal bone of adjacent teeth. The alveolus exists only because the tooth did."

Q What is bundle bone, and how thick is that plate at the 21?

"The socket wall into which the periodontal ligament's **Sharpey's fibres** insert; being ligament-supplied, it resorbs when the tooth goes. Labially at the 21 that plate is usually **1 mm or less** and largely bundle bone — **Araújo and Lindhe's** explanation for buccal collapse."

Q What happens dimensionally to the socket in the first year?

"Roughly **50% of the ridge width**, two-thirds within three months, and greater buccally than palatally, so the crest migrates palatally."

Q What is gingival phenotype, and how do you assess it?

"Gingival thickness plus keratinised width plus bone morphotype; the **2017 World Workshop** replaced 'biotype' because tissue can be modified. I assess it by **probe transparency** — if the probe shows through the sulcus it is **thin**, under 1 mm. Thin phenotypes recede; thick ones pocket."

Q Define peri-implant health, mucositis and peri-implantitis.

"**Health** — no bleeding or suppuration and no bone loss beyond initial remodelling. **Mucositis** — bleeding on gentle probing without that loss, and reversible. **Peri-implantitis** — adds increased probing depth and **progressive bone loss**; without baselines, 6 mm depth with 3 mm bone loss."

Chain: reading this patient

Q So what has actually happened here?

"A buccally placed immediate implant, a thin or absent labial plate, a thin phenotype. The bone was never there to support the mucosa, so tissue crept apically until the **shoulder** was exposed. The infection is an uncleansable sulcus or **peri-implantitis**."

Q Why is the smile line pivotal, and what else must you establish?

"It decides whether this is functional with a cosmetic component or purely aesthetic — if his lip covers the margin, **maintenance is legitimate**. I also need six-point probing, retention type, keratinised width, papilla heights against the 11 and 22, and **what actually bothers him**."

Q Would you take a CBCT? Justify it.

"Yes, **small field of view**. Restoring versus explanting turns on buccal bone volume and angulation, which no two-dimensional film shows — a diagnostic yield that changes management, which is the **IR(ME)R** justification."

Domain 2 — Clinical decision making and treatment planning

Chain: what is realistically achievable

Q How would you record the recession, and do the classifications help you?

"Restoration margin to mucosal margin in millimetres, papilla heights, keratinised width and **standardised photographs**. **Cairo's** and **Miller's** classifications rest on interdental attachment, which around an implant is set by the adjacent teeth — they predict coverage at teeth, not implants."

Q Is coverage around an implant as predictable as around a tooth? Why not?

"No. A tooth has a **periodontal ligament** with inserting fibres to which a graft reattaches. An implant has **circumferential fibres**, no ligament and a poorer supply; the graft lives on periosteum alone, and outside the bony envelope there is nothing to advance over."

Q Start with the least interventive option.

> "**Treat the infection and maintain** — debridement with titanium or PEEK instruments, glycine air-polishing, single-tufted brush instruction, review. Legitimate where the smile line is low and recession stable — a real option, not a failure to treat."

Q Next — prosthetically, and then soft tissue?

> "A **ceramic or titanium-base ceramic abutment** to take the grey out from under thin mucosa, a recontoured crown and **pink ceramic** to mask the deficiency — cheap and reversible, but it does not move the implant. Then a **connective tissue graft**: complete coverage is unlikely, but thicker tissue stops further recession."

Q And the definitive option? Must the replacement be another implant?

> "**Explantation and reconstruction** — remove the fixture, let soft tissue heal, augment, then re-place correctly or restore without an implant. It need not be an implant: a **cantilever resin-bonded bridge from the 11** over a grafted ridge is often more reliable aesthetically."

Q Which would you recommend?

> "It depends on the CBCT and his smile line, and I would say so. Mildly buccal, stable, lip covering it — infection control, abutment change, thickening, maintenance. Grossly malpositioned, active disease or a high smile line — **explantation**."

Q A colleague would explant at the first visit. Why might they be right?

> "Because everything short of explantation spends his money and tissue on a fixture in the wrong place, and each intervention scars the site. If he ends up there anyway, the **least destructive time to remove it is now**."

Q SAC classification, and what would you refer?

> "**Complex**, surgically and prosthodontically. Within my competence: diagnosis, the CBCT request, infection control, abutment change and the discussion. **Beyond it**: explantation, staged augmentation and re-implantation in the aesthetic zone. I would do the diagnostic work and refer."

Q How do you have that conversation with him?

> " 'The implant sits too far forward in the bone, so the gum has always been thin and has shrunk back until the metal shows. I can change the part underneath and add gum tissue, but that won't move the implant. Fixing it properly means taking it out and rebuilding the area — most of a year, and neither is guaranteed.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: placement protocol and loading

Q This implant was placed immediately. What exactly does "immediate" describe?

> "**Placement timing and loading are two separate decisions.** Types 1 to 4 describe when the implant is placed relative to the extraction — this was **Type 1**, the same day. Immediate, early and conventional describe when a restoration is attached. They are chosen independently."

Q What are the pre-requisites for Type 1 placement, and were they met here?

> "No acute infection, intact socket walls with a **facial wall of at least 1 mm**, palatal bone for primary stability, the socket within the alveolar envelope, and a grafted **jumping gap of 2 mm or more**. Here the wall was thin and the fixture went buccally — several were unmet."

Q A colleague says immediate placement preserves the buccal plate. Argue against that.

"It does not. The buccal plate is largely **bundle bone**, dependent on the ligament, and resorbs whatever I put in the socket. Immediate placement saves time and surgical episodes — not bone — and costs control of the **three-dimensional position**."

Q If you explanted and re-treated, which placement type, and why not immediate?

"**Type 4, late placement** into a healed, staged-augmented ridge. Not immediate: an explantation defect has no intact socket walls, often residual infection and no bony envelope, so neither primary stability nor the correct position could be guaranteed."

Q And the loading protocol on the second attempt?

"**Conventional loading beyond two months** — Type 4C, scientifically and clinically validated — extended to six to nine months in staged grafted bone. I would stage it for torque under **35 Ncm** or ISQ under 70, D4 bone, parafunction, or simultaneous augmentation."

Chain: choosing an augmentation material

Q Name the four categories of graft material and the properties each has.

"**Autograft** — osteogenic, osteoinductive and osteoconductive, the only living material, but limited volume, donor morbidity, unpredictable resorption. **Allograft** — osteoconductive, variably osteoinductive, human origin. **Xenograft** and **alloplast** are osteoconductive only; xenograft gives outstanding volume stability, alloplast avoids all human and animal origin."

Q What would you graft an explantation defect with, and why that combination?

"Particulate **autograft or allograft** against the bone where biology matters, layered with a slowly resorbing **xenograft** outside for contour, because deproteinised bovine bone mineral barely resorbs and holds the buccal profile for years. It marries the biology of one to the stability of the other."

Q And the membrane?

"A **resorbable collagen membrane** onto sound bone for a contained defect — no second surgery, and if it exposes without infection, chlorhexidine and no brushing often see it heal. **Non-resorbable PTFE** maintains space better, but exposure means early removal."

Q Is that reconstruction within your competence?

"No. A large three-dimensional defect in the aesthetic zone, needing horizontal and often **vertical augmentation**, is beyond a general dentist with an interest in implants — vertical augmentation is the least predictable procedure of all. I would control the infection and **refer**."

Q What must you consent him to specifically about the materials?

"The **origin of every material** — bovine or porcine xenograft, human donor allograft, porcine collagen membrane — the theoretical transmission risk, that autogenous and synthetic alternatives exist, less well documented, and that lot numbers are recorded. **Religious, cultural and dietary objections must be raised well before the day of surgery.**"

Chain: doing it

Q Whichever route you take, what comes first?

"**Control the infection.** Retrieve the restoration if possible, debride with titanium or PEEK curettes and a plastic-sheathed ultrasonic tip, subgingival **glycine air-polishing**, re-establish a cleansable emergence. Reassess at four to six weeks — you do not graft into an inflamed field."

Q You find residual cement. What do you do?

"Remove all of it — scaler, air-polishing, flap access if I cannot see it. **Residual cement is a recognised cause of peri-implant disease**, and only radiopaque cements show on a film, so a clear radiograph proves nothing. Then convert to **screw retention**."

Q If you take the graft route, describe the harvest and the recipient site.

"A palatal **connective tissue graft**, single-incision, between distal of the canine and mesial of the first molar, **5 to 7 mm apical to the gingival margin** to spare the greater palatine bundle, 1 to 1.5 mm thick. Recipient: a split-thickness envelope advanced **without tension**, under a screw-retained provisional with a deliberately **concave subgingival contour**."

Q Describe explantation, and what if reverse-torque fails?

"**Reverse-torque** first, unscrewing the fixture out through the internal connection; it removes almost no bone. If it fails, **piezoelectric surgery** around the circumference, then re-attempt. A **trephine** only as a last resort — it sacrifices the ring of bone I need for reconstruction."

Q What do you do immediately after removal, and what interim prosthesis?

"Debride the granulation tissue and read the defect. If **contained**, graft immediately; if the buccal plate is absent, do not chase it — let soft tissue heal six to eight weeks, then stage it. Interim cover is a **bonded pontic**, never a denture flange."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcome, record and candour

Q What outcome would you promise him, and how would you judge it?

"None. Partial improvement is likely, **complete coverage unlikely**. I would judge it on **standardised photographs** at baseline and each review, millimetre measurements of mucosal margin, papilla height and keratinised width, the **Pink Aesthetic Score**, and his own rating in his words."

Q What would you record?

"His concern in his own words; the peri-implant chart with bleeding and suppuration; smile line, phenotype, keratinised width; retention type; the CBCT justification; photographs; **every option including no treatment**, with risks, costs and prognosis; and his decision."

Q He is dissatisfied, and then complains formally. How do you handle it?

"Listen, examine, review the records with him. If the outcome was consented, I say so and offer a second opinion; if not, **duty of candour** — tell him promptly, apologise, put right what I can. Then the practice complaints procedure, notify my indemnifier, record a **significant event**."

Q If he elects to do nothing, what would you expect in five years?

> "Slow further recession: the driver — a fixture outside the bony envelope under a thin phenotype — is unchanged. Recurrent mucositis and real risk of **peri-implantitis**, since the sulcus cannot be cleaned. The implant will probably stay integrated — **survival is not success**."

WATCH OUT

Ask yourself whether the problem is **bone, soft tissue, or implant position** — and say which. Position problems cannot be restored away, and naming that buys you a sentence and is almost always the right frame in the aesthetic zone.

Running this case to time

Time check. The flow above holds 37 questions and about 1,646 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that comes to just under **fifteen minutes**, so it fits, with perhaps ten seconds in hand. The answers are short because the question count is high: this is a case examined in rapid chains, not in paragraphs. If your answers run past **thirty seconds** you will still be on graft materials when the candour question arrives, and Domain 4 will be marked on nothing.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	An implant at the 21 placed immediately ten years ago, now facially malpositioned with recession, visible metal and recurrent infection — presented as **a chain that began at surgery**: a fixture too buccal where the plate is under 1 mm of **bundle bone**, no bone buccal to it, thin mucosa receding over a decade. Then the working position — **a malpositioned implant cannot be restored into the right place** — and what to establish: a small-field CBCT, active disease, his **smile line**, and what actually troubles him.
1:20–4:30	Domain 1: the anatomical and biological groundwork, then reading this patient	The thin labial plate, the incisive canal, papillae dependent on **adjacent** crestal bone; **bundle bone** and Sharpey's fibres, with **Araújo and Lindhe** named; **50% of ridge width in a year**; **phenotype** by probe transparency, thin recedes and thick pockets; the 2017 definitions with the no-baseline threshold; then the smile line as the pivot, and the **IR(ME)R** justification for the CBCT.
4:30–7:30	Domain 2: what is realistically achievable	Recession in millimetres against **standardised photographs**, with **Cairo and Miller** named *and* dismissed — they predict coverage at teeth, not implants; then the ladder — **infection control and maintenance as a real option**, ceramic abutment and pink ceramic, a graft to thicken rather than to cover, **explantation and reconstruction**, with the replacement **not necessarily an implant**; **SAC Complex**, and **explantation, staged augmentation and re-implantation in the aesthetic zone referred**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: placement and loading, the graft material, then doing it	**Placement timing and loading are separate decisions**; the **Type 1 pre-requisites listed and shown to have been unmet**; immediate placement refused as a means of preserving the plate; **Type 4** on any second attempt; the four graft categories, **autograft or allograft inside, slowly resorbing xenograft outside**; then the sequence that matters — **control the infection first, you do not graft into an inflamed field** — and **reverse-torque before piezo before trephine**.
11:30–14:30	Domain 4: outcome, record and candour	**Promise no outcome** — partial improvement likely, complete coverage unlikely; judged on standardised photographs, millimetre measurements, the **Pink Aesthetic Score** and his own rating; **duty of candour**, complaints procedure, indemnifier, significant event; and if he does nothing, slow further recession because the driver is unchanged — **survival is not success**.
14:30–15:00	Closes on the patient	**In his words**: the implant sits too far forward, so the gum has always been thin and has shrunk until the metal shows; the part underneath can be changed and tissue added, but **that will not move the implant**; fixing it properly means taking it out and rebuilding — most of a year, neither route guaranteed.

The spine — the twelve answers to have ready if nothing else

1 What bundle bone is, and the plate thickness at the 21 · 2 The dimensional change in the first year · 3 Phenotype, and how you assess it · 4 Health, mucositis, peri-implantitis defined · 5 Why the smile line is pivotal · 6 The CBCT justified under IR(ME)R · 7 Why Cairo and Miller do not help you here · 8 The least interventive option, offered seriously · 9 The Type 1 pre-requisites, and which were unmet · 10 Why immediate placement did not preserve the plate · 11 Control the infection before anything else · 12 What outcome you promise — and that survival is not success.

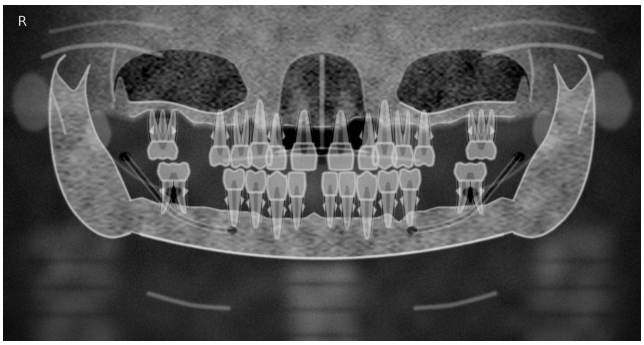
Where the marks are lost in this case

Restoring the position away. Candidates offer a ceramic abutment, pink ceramic and a connective tissue graft as though they solved the problem. They mask it. Say once, plainly, that the fixture sits outside the bony envelope and nothing prosthetic will move it — that sentence is the case.

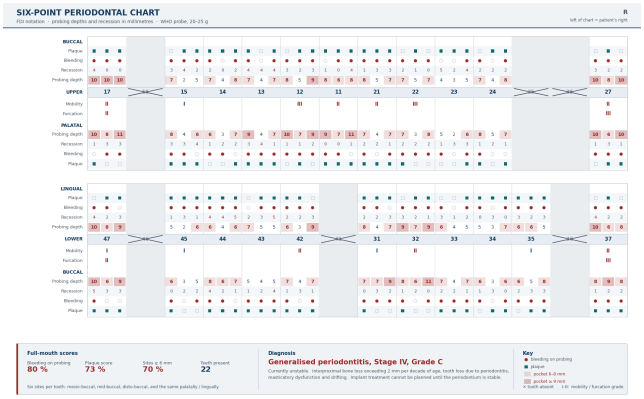
Dismissing maintenance as doing nothing. If his lip covers the margin and the recession is stable, infection control and review is a legitimate treatment, not a failure to treat. Offering it seriously, alongside explantation, is what shows judgement rather than enthusiasm.

Planning the reconstruction you would not do. Explantation, staged augmentation and re-implantation in the aesthetic zone — vertical augmentation above all — are beyond a general dentist with an interest in implants. Do the diagnostic work, control the infection, and **name the referral** before you are asked for it.

SOE Case 7 — Stage IV Grade C periodontitis and a failing dentition



Panoramic radiograph — simulated illustration, not a clinical image



The six-point periodontal chart — simulated illustration, not a clinical image

THE CASE

A patient with **generalised periodontitis, Stage IV, Grade C** — sometimes loosely called "grade 4C" — with **multiple failing teeth**. The question put to you is how to plan tooth replacement.

Pace: one structured opening of about 60 seconds, then roughly 39 questions in the remaining 13 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A patient with **generalised periodontitis, Stage IV, Grade C** and a collapsing dentition — advanced attachment loss, teeth already lost to the disease, and progression rapid for his age and for the plaque present. The question asked is tooth replacement, but **replacement is the last decision, not the first**. A history of periodontitis is the strongest patient-level risk factor for peri-implantitis, and Grade C means the destruction is active now. So my plan is periodontal first: diagnosis, risk factor control, extraction of hopeless teeth, non-surgical therapy, and re-evaluation against the **EFP stability criteria**. Only if he reaches and holds stability, with attendance demonstrated rather than promised, would I discuss implants — and even then a well-designed removable prosthesis or a shortened arch may serve him better."

Domain 1 — Clinical interpretation and reasoning

Chain: the periodontal groundwork

Q Name the components of the periodontium.

"**Gingiva**, **periodontal ligament**, **root cementum** and **alveolar bone**. The dentogingival junction is sulcular and **junctional epithelium**, attached by hemidesmosomes. The ligament fibres — crestal, horizontal, oblique, apical, interradicular — suspend the tooth and carry its proprioception."

Q What are the supracrestal attached tissues, and why do they matter?

"The 2017 term for **biologic width** — junctional epithelium plus connective tissue attachment, about **2 mm** above the crest. A margin within it causes inflammation and attachment loss. The same principle governs crown margin and implant platform depth."

Q How does the peri-implant attachment differ?

"**No ligament and no cementum**, so no inserting Sharpey's fibres. Fibres run **parallel and circumferentially**, the tissue collagen-rich and fibroblast-poor, the supply periosteal only. A **weaker seal** — peri-implantitis progresses faster and less predictably."

Q Take me through staging, and what makes him Stage IV?

"**Staging** is severity and complexity, from **interdental attachment loss at the worst site**: Stage I one to two millimetres, II three to four, III five or more with up to four teeth lost. **Stage IV** is that loss with **five or more lost**, plus bite collapse, drifting or degree 2 mobility."

Q And grading?

"**Grading** is rate of progression — **bone loss percentage over age** — A under 0.25, B 0.25 to 1.0, **C over 1.0** — or destruction exceeding the deposits. Modifiers: **smoking ten a day**, **HbA1c 7.0% or more**."

Q What completes the diagnosis?

"**Extent** — localised under 30% of teeth, **generalised** at 30% or more — and **status**: stable, in remission or unstable. So: 'generalised periodontitis, Stage IV, Grade C, currently unstable'. That last word stops the implant conversation."

Chain: what that means for this patient

Q Why is that diagnosis the single most important fact here?

"Because **previous periodontitis is the strongest patient-level risk factor for peri-implantitis**, and Grade C means active destruction now. Implants transplant the same dysbiotic biofilm onto a surface with a weaker seal and no ligament, and the disease recurs faster."

Q What must you know before planning?

"**Six-point charting** with recession, mobility, furcation and bleeding; plaque scores; full-mouth radiographs; **smoking in pack-years**; **HbA1c**; medical history; his priorities and finances; and what treatment he has had and how he responded."

Domain 2 — Clinical decision making and treatment planning

Chain: sequencing

Q Outline your plan in one sentence.

"**Periodontal first, prosthetic later** — cause-related therapy, risk factor control, extraction of hopeless teeth with an interim prosthesis, re-evaluation at three months, replacement only in a stable attending mouth, then **three-monthly supportive therapy indefinitely**."

Q What does Phase 1 consist of?

"Explain the diagnosis. **Smoking cessation** — very brief advice, then refer. Liaise with the GP on **glycaemic control**. Hygiene instruction with **plaque and bleeding scores every visit**, so behaviour change is measured. Extract hopeless teeth, then full-mouth **subgingival instrumentation** per **EFP S3**."

Q Which teeth are hopeless?

"**Grade 3 mobility**; bone loss beyond **75% of root length**; an unmaintainable **Grade III furcation**; unrestorable caries, vertical root fracture or persistent suppuration. Strategic value counts — a hopeless tooth under a denture saddle goes, a marginal overdenture abutment may stay."

Q How do you grade furcation and mobility?

"**Hamp** — Grade I horizontal loss under 3 mm, II over 3 mm, **III through-and-through**. **Miller** — Grade 1 up to 1 mm horizontally, 2 over 1 mm, **3 plus vertical depressibility**. Both recorded at charting; both drive extractions."

Q When do you re-evaluate, and against what endpoint?

"At **three months**, re-charting fully. The endpoint is the **EFP definition of stability**: **pockets 4 mm or less, no bleeding at any 4 mm site, full-mouth bleeding under 10%** — an objective number I can share with him."

Q He has not reached it. What then?

"**Step 3** — repeated instrumentation of residual pockets, surgery for deep sites, or referral. First re-check the obvious: cleaning, smoking, HbA1c, missed calculus or a furcation. **Non-response is usually compliance or diagnosis, not instrumentation.**"

Chain: whether implants at all

Q Would you ever place implants in this patient?

"Potentially, on four conditions: periodontitis **treated and objectively stable**; **smoking stopped** or substantially reduced; adequate **glycaemic control**; and three-monthly compliance **demonstrated over 6 to 12 months, not promised**. Even then I consent him to a peri-implantitis risk well above average."

Q What would you say to him in his own words?

" 'Your gum disease is why you are losing teeth, and it is moving fast. Implants don't decay, but they catch the same infection — and around an implant it does more damage, more quickly. So we settle the gums first, then talk about replacing teeth.' "

Q What are the alternatives?

"A **hygienic cobalt-chromium partial denture** — maximal gingival clearance, rests on sound teeth — **added to as more teeth are lost**. Complete dentures if the dentition is failing. The well-evidenced **shortened dental arch**. Or **two implants retaining a mandibular overdenture**."

Q A colleague says extract everything and place implants — the disease then has nothing to live in. Why might they be wrong?

"Because the pathogens are not confined to pockets — they persist on the **tongue, tonsils and mucosa** and recolonise peri-implant sulci within weeks. And the real drivers — **host response, smoking, glycaemic control, plaque control** — follow the patient, not the teeth."

Q SAC classification, and what is beyond you?

"At least **Advanced**, and with ridge defects, altered vertical dimension and multiple missing units, likely **Complex**. Within my scope: diagnosis, cause-related therapy, extractions, interim prostheses, maintenance. **Beyond it**: periodontal surgery and the definitive reconstruction — I refer or co-manage."

Domain 3 — Decision making in relation to treatment procedures

Chain: doing the work

Q What would you do at the extraction visits?

"Extract **atraumatically** — periotomes, luxators, and sectioning rather than expanding the socket — preserving every wall, and debride granulation tissue. Sequence so he is **never without anterior teeth or posterior support**. Alveoloplasty only where an undercut would compromise the denture."

Q Describe your subgingival instrumentation — and would you use adjunctive antimicrobials?

"Full-mouth, quadrant by quadrant under local anaesthesia, **hand instruments and ultrasonics together** — no meaningful difference between them, so access and thoroughness decide. Endpoint: a smooth root, no detectable deposit. **Systemic antimicrobials are not indicated routinely**; local agents suit isolated non-responding sites."

Q If you eventually restore implants, what would you design differently?

"**Screw-retained without exception** — retrievable, because I should assume I will remove it to treat peri-implant disease, and **no cement** left behind. Emergence profiles **cleansable with an interdental brush**, margins supragingival, and **three-monthly recall from fit**."

Q Does he need keratinised mucosa?

"In a plaque-susceptible patient, yes — at least **2 mm**, grafted if necessary. The evidence it is essential to survival is weak; the evidence it makes brushing comfortable and reduces plaque is reasonable. Comfort while cleaning determines his outcome."

Q How would you design his interim prosthesis?

"**Acrylic, hygienic, clear of the gingival margins**, easily added to and relined, resting on sound teeth. He will wear it a year or more, so it must not become the plaque trap that undoes the therapy."

Chain: placement protocol and loading

Q When would you place, and when would you load?

"Two separate decisions. **Types 1 to 4 describe when I place relative to the extraction**; **immediate, early and conventional describe when the restoration comes into occlusion**. They are chosen independently, and I grade the combination against the ITI evidence table."

Q Why not Type 1 — immediate placement into a socket lost to periodontitis?

"The pre-requisites are absent: **no acute infection, intact socket walls with a facial wall of at least 1 mm, apical and palatal bone for primary stability, and a jumping gap of 2 mm grafted with xenograft.** Periodontitis has destroyed the walls and infected the site. And immediate placement **does not preserve the buccal plate** — bundle bone is ligament-dependent."

Q Which type would you use, and how long would you wait?

"**Type 4 — late placement at six months or more** into a healed ridge; the best-documented protocol, and the honest choice where infection must resolve. At extraction I would do **ridge preservation** only where the site is a realistic future implant site."

Q And the loading protocol?

"**Conventional loading beyond two months** — Type 4C, **scientifically and clinically validated**; three months mandible, six maxilla, longer in grafted bone. I would not chase immediate loading: it needs 35 Ncm or ISQ over 70, and saved weeks do not justify the risk here."

Chain: choosing an augmentation material

Q What are the four categories of graft material, and what does each do?

> "**Autograft** — the only **osteogenic** material, also osteoinductive, but limited volume, donor morbidity, unpredictable resorption. **Allograft** — human donor, osteoconductive, variably osteoinductive, batch-dependent. **Xenograft** — deproteinised bovine mineral, **osteoconductive only**, but outstanding volume stability and best documented. **Alloplast** — synthetic, osteoconductive only, no ethical objection, less documented."

Q What would you use for socket preservation, and why?

> "A **slowly resorbing xenograft** — deproteinised bovine bone mineral — with a socket seal or membrane, placed flaplessly. Its very slow resorption holds the ridge dimension across four to six months of healing, which autogenous bone or a fast-resorbing alloplast would not."

Q The socket walls are already destroyed by the disease. What then?

> "Then space maintenance and graft stability are lost: this is staged augmentation, not socket preservation — and I would not graft an infected site. Debride, heal, reassess on CBCT. **Vertical and large horizontal augmentation in a Stage IV mouth are beyond a general dentist with an interest in implants** — I refer."

Q Would you use a barrier membrane, and which?

> "Yes — a **native porcine collagen membrane** for a socket or small dehiscence: resorbable, forgiving if it exposes, extended 2 to 3 mm onto sound bone. **Dense PTFE** may be left deliberately exposed and removed at three to four weeks. Titanium-reinforced membranes go with the referral."

Q What must you consent him to specifically about the graft?

> "The **origin — bovine graft, porcine membrane, human donor if allograft** — because **religious, cultural, dietary and vegetarian objections are legitimate**, raised well before the day of surgery, with an autogenous-plus-synthetic alternative offered. Also that **exposure is the commonest complication**, and that the **lot number is recorded**."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: monitoring and the long view

Q How would you monitor him?

> "**Plaque and bleeding scores every visit**, six-point charting at re-evaluation, radiographs against baseline, and once implants exist **six-site peri-implant probing** with bleeding and suppuration. I set the intervention threshold in advance and share it."

Q What would you record?

> "The full 2017 diagnosis with extent and status; charting and scores throughout; **smoking, HbA1c and the conversations about them**; every option including no treatment, with risks, costs and prognosis; that his **elevated peri-implantitis risk and lifelong maintenance** were explained; his decision."

Q What is your single greatest concern at five years?

> "**That he stops attending.** Every biological outcome rests on supportive care, and non-attendance is the strongest predictor of implant loss in treated periodontitis. I build it into the consent and the recall, and say plainly: 'The treatment is not the day I place the implant.' "

Q Two years on, one implant probes 7 mm with bleeding, suppuration and 4 mm of bone loss. Your assessment?

> "**Peri-implantitis**, here the disease recurring, not a local accident. Check compliance, smoking, HbA1c and prosthesis cleansability; remove the restoration; debride with titanium instruments and **glycine air-polishing**; reassess at one month. Persistent suppuration with 6 mm or more means surgery — I refer."

Q He tells you he has seen the hygienist twice in two years. What does that change?

> "Everything. In a Grade C patient that is the principal risk factor realising itself. I would explore **why** rather than lecture — cost, time, anxiety, lost sense of benefit — then consider whether **further implant treatment is contraindicated** and a removable design more honest."

Q How would you know the whole treatment had succeeded?

> "By his measure and mine. **His**: he eats what he wants, loses no more teeth, is not embarrassed. **Mine**: EFP stability held, no further teeth or implants lost, stable bone, and maintenance he attends. Prosthesis survival alone is not success."

WATCH OUT

Say where the patient is in the sequence — **diagnosis, cause-related therapy, re-evaluation, definitive treatment, or maintenance** — and answer from there. In a Stage IV Grade C case the answer is almost always that the periodontal phase is not finished yet.

Running this case to time

Time check. The flow above holds 40 questions and about 1,721 words of model answer — the highest question count in the notebook. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that is about **fifteen minutes and forty seconds**, so it **overruns by rather less than a minute**. An examiner behind time compresses the three chains of Domain 3 into two, usually by taking the graft categories as one question. If your answers run past **thirty seconds** you will still be staging and grading when the maintenance questions decide Domain 4.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the charting and radiographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	**Generalised periodontitis, Stage IV, Grade C** with a collapsing dentition; then the reframing that earns the mark — the question asked is tooth replacement, but **replacement is the last decision, not the first**; a history of periodontitis as the strongest patient-level risk factor for peri-implantitis, and Grade C meaning destruction is active now; so periodontal first, re-evaluated against the **EFP stability criteria**, with **attendance demonstrated rather than promised**.
1:20–4:30	Domain 1: the periodontal groundwork, then what it means for him	The four components; **supracrestal attached tissues** as the 2017 term for biologic width, about **2 mm**, governing crown margin and implant platform depth alike; the peri-implant attachment with **no ligament, no cementum, parallel fibres, a weaker seal**; **staging from interdental attachment loss at the worst site**, and what makes him **Stage IV**; **grading by bone loss over age, C above 1.0**; then extent and **status — "currently unstable"**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: sequencing, then whether implants at all	**Periodontal first, prosthetic later**, in one sentence; Phase 1 with **smoking cessation, GP liaison on glycaemic control, scores every visit**; which teeth are hopeless, with **Hamp and Miller** named; re-evaluation at **three months against pockets ≤ 4 mm, no bleeding at 4 mm sites, full-mouth bleeding under 10%**; the four conditions before any implant; the alternatives, including the **shortened dental arch**; **SAC Advanced to Complex**, with **periodontal surgery and the definitive reconstruction referred**.
7:30–11:30	Domain 3: doing the work, then placement and loading, then the graft	Extractions sequenced so he is **never without anterior teeth or posterior support**; **no routine systemic antimicrobials**; **screw-retained without exception**, because you should assume you will remove it; **2 mm of keratinised mucosa**; then **placement and loading as separate decisions**, **Type 1 refused** because periodontitis has destroyed the walls, **Type 4 at six months**, **conventional loading beyond two months**; and the limit — **you do not graft an infected site**, and **vertical or large horizontal augmentation here is referred**.
11:30–14:30	Domain 4: monitoring and the long view	Scores every visit, six-site peri-implant probing once implants exist, thresholds **set in advance and shared**; the greatest five-year concern named as **non-attendance**; the 7 mm implant with suppuration read as **the disease recurring, not a local accident**; and two hygienist visits in two years explored — **why, not a lecture** — with removable design offered as the more honest plan.
14:30–15:00	Closes on the patient and the record	**In his words**: the gum disease is why he is losing teeth and it is moving fast; implants do not decay but they catch the same infection, and around an implant it does more damage more quickly. Record the diagnosis with extent and status, and that **elevated peri-implantitis risk and lifelong maintenance were explained**.

The spine — the twelve answers to have ready if nothing else

1 The components of the periodontium · 2 Supracrestal attached tissues, and why they matter · 3 How the peri-implant attachment differs · 4 Staging, and what makes him Stage IV · 5 Grading, and the two modifiers · 6 Extent and status, and why "unstable" ends the conversation · 7 Which teeth are hopeless · 8 The EFP stability endpoint, in numbers · 9 The four conditions before any implant · 10 Why extracting everything does not remove the disease · 11 Why not Type 1 into a periodontitis socket · 12 The single greatest concern at five years.

Where the marks are lost in this case

Answering the question that was asked. He is presented as a tooth-replacement problem and candidates dutifully plan replacements. The mark is in refusing the framing: say in the opening minute that replacement is the last decision, and that **"currently unstable"** stops the implant conversation.

Quoting stability without the numbers. "Periodontally stable" is worth nothing; **pockets 4 mm or less, no bleeding at any 4 mm site, full-mouth bleeding under 10%** is an endpoint you can share with him and re-chart against. Volunteer the three-month re-evaluation, and what you do if he has not reached it.

Trusting a promise of attendance. Non-attendance is the principal risk factor realising itself, so compliance must be **demonstrated over six to twelve months before implants are considered** — and when he has seen the hygienist twice in two years, the honest answer is that further implant treatment may be contraindicated.

SOE Case 8 — Peri-implantitis at linked LR6 and LR7

THE CASE

(*The published RCSEd sample scenario, Appendix E.*) 45-year-old female, first attendance, routine check-up. **Occasional soreness of the gum near the implants**, sometimes aware of **discharge**, otherwise asymptomatic — **no pain, never any bleeding**. Brings radiographs from when the implants were restored. Implants placed **3 years ago** by a dentist who has retired; aware of soreness for **1 year, worsening**; lost the molars because of **failed crowns**. **No relevant medical history.** Lives alone; **never smoked**; unaware of grinding or clenching; moderate stress. Regular attender **until 2 years ago**; sees the hygienist periodically, **last visit over 2 years ago**. She wants to know **why she has the soreness and whether her implants are okay**. EO: NAD. Soft tissues NAD; oral hygiene **fair to good**; gingival tissues healthy **except at LR6/7 where the marginal tissues are inflamed with bleeding and pus discharge**. Probing depths generally up to 3 mm; **6 mm around the LR6/7 implants**. **BPE: upper 1/1/1; lower 4/1/1.** Intact, unrestored dentition. **Implants LR6 and LR7 linked by the restoration.** Class I occlusion. Post-surgical radiographs (patient's own) and radiographs taken at presentation are provided.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 45-year-old woman attending for the first time with a one-year history of worsening soreness and intermittent discharge around two linked implants placed three years ago. She has an otherwise intact, unrestored, periodontally healthy mouth with fair-to-good hygiene, has never smoked and has no relevant medical history — so the patient-level risk factors are largely absent. What stands out is that she has had **no supportive maintenance for over two years**, and that the two implants are **splinted by a single restoration** with 6 mm pockets, bleeding and suppuration confined to that sextant. My working diagnosis is **peri-implantitis at the LR6 and LR7**. My priorities are to confirm it against her own baseline radiographs, to establish why it has happened, and to decide whether the implants are retainable."

Domain 1 — Clinical interpretation and reasoning

Chain: the periodontal groundwork

Q Describe the attachment apparatus of a natural tooth.

"From the sulcus apically: **junctional epithelium**, attached to enamel by hemidesmosomes and a basal lamina, about 1 mm; then a **connective tissue attachment** of about 1 mm in which the supracrestal fibre groups — dentogingival, dentoperiosteal, circular and transseptal — insert perpendicularly into **cementum** as Sharpey's fibres. Below that, the **periodontal ligament** and alveolar bone. The epithelial plus connective tissue component is the supracrestal attachment, historically the biologic width, around 2 mm."

Q How does the peri-implant soft tissue differ?

"There is no cementum and **no periodontal ligament**, so nothing inserts into the implant. The junctional epithelium is longer, and the connective tissue fibres run **parallel or circumferential** to the implant surface rather than perpendicular. The tissue is more collagenous and less vascular, and the blood supply comes only from the periosteum and bone, not from a ligament. The result is a **weaker seal with poorer defence**."

Q What does that mean for this lesion?

"That peri-implantitis behaves worse than periodontitis. The lesion extends closer to the bone crest, is not encapsulated by a fibrous barrier, and **progresses faster and in a non-linear, accelerating pattern**. There is no periodontal ligament proprioception, so she has no pain and no awareness — which is exactly her history: soreness and discharge but never any bleeding she noticed, over a year."

Q How is the BPE recorded?

"Six sextants, using a **WHO probe with a 0.5 mm ball end** and black band from 3.5 to 5.5 mm, at a force of **20 to 25 grams**, walking around every tooth and recording the highest code per sextant. Code 0 healthy, 1 bleeding, 2 plaque retentive factor or calculus, 3 black band partly visible so 4 to 5 mm, 4 black band disappears so 6 mm or more, and * for furcation involvement or 7 mm attachment loss."

Q What does her code 4 tell you?

"That the lower right sextant has probing depths of 6 mm or more while every other sextant is a code 1. That distribution is the diagnosis in itself: a **localised, site-specific problem centred on the implants**, not generalised periodontitis. A code 4 obliges me to record a **full-mouth six-point pocket chart** with recession, plaque and bleeding scores, and radiographs where indicated — not just that sextant."

Q Define peri-implant health, mucositis and peri-implantitis.

"On the **2017 World Workshop** definitions: **health** — no erythema, no bleeding on probing, no suppuration, no increase in probing depth, no bone loss beyond initial crestal remodelling. **Mucositis** — bleeding on gentle probing with or without erythema, swelling or suppuration, and **no bone loss beyond crestal remodelling**; it is plaque-induced and reversible. **Peri-implantitis** — bleeding and/or suppuration, increased probing depth compared with previous examinations, and **bone loss beyond crestal remodelling**."

Q Which does she have, and on what evidence?

"**Peri-implantitis** at the LR6 and LR7 — suppuration, bleeding, 6 mm probing depths against 3 mm elsewhere, and I expect radiographic bone loss beyond remodelling when I compare with her baseline. The rest of the mouth is **peri-implant and periodontally healthy**. Formally I would say localised peri-implantitis at two implants in a patient with clinical gingival health on an intact periodontium."

Chain: reading her radiographs and her history

Q Why do her own post-restoration radiographs matter so much?

"Because the case definition is built on **change from a previous examination**. With her baseline I can apply it directly and I can estimate the **rate** of loss, which is what determines urgency and prognosis. Without a baseline I fall back on the surrogate thresholds: bleeding or suppuration, probing depth of **6 mm or more**, and bone level **3 mm or more apical to the most coronal intraosseous part of the implant**."

Q What are the risk factors in this history?

"The dominant one is **absence of supportive maintenance for over two years**. She has never smoked, has no medical history, denies parafunction and cleans reasonably — so the patient-level suspects are largely absent. That directs me to **prosthesis-related factors**: two implants **splinted by one restoration** so an interdental brush cannot pass, possible over-contour or ridge-lap, possible **residual cement** if it is cement-retained, implant depth, emergence profile, and the width of keratinised mucosa."

Q How do you probe an implant, and what are the pitfalls?

"**Six sites per implant** with a plastic, PEEK or titanium probe at a light force of about **0.2 to 0.25 newtons**, recording depth, bleeding and suppuration, plus recession and keratinised width. The pitfalls are that the restoration contour often prevents the probe reaching the true base, so depths are underestimated; and that absolute depth means little without a **baseline**, because a deeply placed implant reads deep when healthy."

Q What else would you examine and investigate?

"Mobility — testing the **implants themselves**, removing the restoration if necessary to distinguish a loose prosthesis from a mobile implant. A **standardised paralleling periapical** compared directly with her films. The prosthesis: retention type, fit, contour, whether a brush passes, residual cement, abutment seating. Occlusion, including excursive contacts on the implants. Full-mouth charting with plaque and bleeding scores. And with her consent I would try to **obtain the original records** from the retired dentist's practice."

Domain 2 — Clinical decision making and treatment planning

Chain: telling her

Q How do you explain the diagnosis to her, in her words?

> " 'Implants can get a gum infection just like teeth can, and that is what has happened here. There is an infection in the gum and bone around both of these implants — that is why it is sore and why you get the discharge. The rest of your mouth is healthy and you clean well. The two problems are that these implants are **joined together**, so you cannot clean between them however hard you try, and that nobody has been checking them for over two years. I need measurements and an X-ray to compare with yours before I tell you how much bone has been lost.' "

Q She asks whether it is her fault. What do you say?

> " 'No. You have been cleaning a restoration that is very difficult to clean, and you were not told this needed checking every few months. What matters now is what we change.' I would then be honest that going forward her cleaning and her attendance are the two things that decide the outcome — so it is not blame, but she does own part of the solution."

Chain: the plan

Q What is your treatment plan, in outline?

> "Following the **ITI structure**: cause-related therapy — explain the disease, discuss costs and adverse effects including recession, modify risk factors, and above all **make the prosthesis cleansable**. Then non-surgical therapy, then **re-assessment at about one month**. If it has resolved, straight to supportive care. If suppuration or bleeding persists with depths of 6 mm or more, surgical therapy. Then **supportive therapy every three to five months, indefinitely**."

Q Why is the linked restoration so central?

> "Because the disease is biofilm-driven, and no debridement I perform survives contact with a site she cannot clean at home. If an interdental brush cannot pass between the LR6 and LR7, the embrasure is a permanent reservoir. **Cleansability is the treatment**, not an adjunct to it."

Q Would you separate the two implants?

> "I would strongly consider it — two individual **screw-retained crowns** with open, cleansable embrasures. Splinting is sometimes justified mechanically, but here it is very likely causal. The counter-argument is the cost of a remake, which I would discuss openly. If that is not affordable now, **opening the embrasure on the existing restoration** is a legitimate interim measure."

Q Your colleague would leave the restoration alone and simply debride and review. Why might he be right?

> "Because removing a three-year-old restoration risks fracturing it, stripping a screw or damaging the abutment, and he may be left with a worse problem and an unrestored implant. He would also argue that debridement plus proper hygiene instruction resolves a proportion of cases without any prosthetic work, so the reversible step should be taken first. That is a fair position — I would test cleansability with a brush before condemning the restoration."

Q If it is cement-retained, what changes?

> "**Residual cement** moves to the top of my differential — it is a well-recognised cause of peri-implant disease, particularly with deep, subgingival margins. I would attempt retrieval, inspect for cement radiographically and directly, remove every trace, and if I re-restore I would convert to **screw retention** or use a cementable-screw-retained design so no cement sits subgingivally."

Q What is the SAC classification of this case?

> "The prosthetic re-treatment and the non-surgical management are **Advanced**. **Surgical management of peri-implantitis, and explantation, are Complex** — the outcome is unpredictable, the aesthetic and functional consequences are significant, and the evidence for reconstructive techniques is uncertain."

Q So what is within your competence, and when do you refer?

"Within: the diagnosis, the full charting and radiographic comparison, the prosthetic assessment, restoration removal, converting to a cleansable screw-retained design, non-surgical therapy, and lifelong supportive care. Beyond: **surgical peri-implantitis therapy, particularly reconstructive surgery, and explantation** — I would refer for those unless specifically trained. I would complete the diagnostic work first so the referral is useful rather than a hand-off."

Q What prognosis do you give her?

"Not one until I have the radiograph and the charting. Then I frame it by proportion of bone lost, whether the disease arrests at the one-month review, and her ability to clean and attend. I would be honest that the aim is to **arrest the disease, not regrow bone**, that **recession is a likely consequence of successful treatment**, and that with extensive bone loss the right answer may be removal."

Domain 3 — Decision making in relation to treatment procedures

Chain: the non-surgical appointment

Q Talk me through that appointment.

"Update the history and record plaque and bleeding scores. **Remove the restoration if it is retrievable** — that transforms access and lets me inspect for cement and for an unseated abutment. Debride with **titanium or PEEK curettes** and ultrasonics with a plastic-sheathed tip, then **glycine or erythritol air-polishing delivered subgingivally**. Irrigate. Modify contours where I can. Re-seat and re-torque. Then oral hygiene instruction, and book the one-month re-assessment."

Q Why air-polishing rather than curettes alone?

"Because the implant surface is **threaded and often roughened**, and a curette blade cannot reach into the threads or the microtopography. A low-abrasive **glycine or erythritol powder** disrupts biofilm in those recesses without gouging the titanium. The evidence shows it improves biofilm removal; it does not by itself change the need to make the site cleansable."

Q What must you not use on the implant surface, and why?

"**Stainless steel curettes and unsheathed metal ultrasonic tips** — they scratch titanium, leaving a rougher surface that accumulates more biofilm, and they shed metal particles. Also avoid **conventional abrasive prophylaxis pastes** and, on titanium, acidulated fluoride, which can corrode the surface."

Q Would you prescribe systemic antibiotics?

"Not routinely for non-surgical therapy — the evidence for benefit is weak and the antimicrobial stewardship argument is strong. I would consider them for **spreading infection, systemic upset, or as a peri-operative adjunct at reconstructive surgery**. Local delivery agents and photodynamic therapy are options, but I would present the evidence for them as modest."

Q What oral hygiene do you actually prescribe her?

"Size an **interdental brush** to each embrasure and have her demonstrate it in a mirror, not just watch me. Add a **single-tufted brush** angled at the implant necks, and superfloss or a floss threader beneath the linked unit while it remains linked. A powered toothbrush. I would give written instructions and re-check technique at every visit, because instruction given once is not instruction."

Q What would you record?

"The full six-point chart with bleeding and suppuration, plaque and bleeding scores, keratinised width, mobility of each implant tested separately, the radiographic comparison with her baseline and the bone loss measured in millimetres, the prosthesis assessment, the diagnosis in 2017 terms, the options discussed with **risks, costs and alternatives**, what she chose, the hygiene instruction given, and the review interval. Plus my attempt to obtain the original records."

Chain: if it goes to surgery

Q What would the surgical procedure involve?

> "A **full-thickness mucoperiosteal flap** for access, complete removal of granulation tissue, and **decontamination of the exposed implant surface** — mechanically with a titanium brush or ultrasonics, chemically with saline, chlorhexidine or hydrogen peroxide, or with an Er:YAG laser. Then treatment according to defect morphology, careful flap adaptation, antiseptic mouthrinse post-operatively, and systemic antimicrobials where indicated."

Q What decides reconstructive versus resective?

> "**Defect morphology.** A contained, circumferential intrabony defect with intact walls is the case for reconstruction with a bone substitute, with or without a membrane. A **supracrestal or dehiscence-type defect with lost walls** will not hold a graft, so the answer is resective — recontouring, implantoplasty and an apically positioned flap. Aesthetic risk is the other determinant."

Q Why is this particular site suited to a resective approach?

> "Because it is a **lower right molar region** — not aesthetically visible, with a thick tissue biotype and good access. The recession and exposed threads that resective surgery produces are far more acceptable here than they would be anteriorly, and the resulting profile is easier for her to clean, which is the whole point."

Q What is implantoplasty, and what are its drawbacks?

> "Mechanically **smoothing and polishing the exposed threaded surface** with burs so it no longer retains biofilm. Drawbacks: titanium particles shed into the tissues, overheating, weakening of the implant body — a real fracture concern in a narrow implant — and it is irreversible. It also demands meticulous soft tissue and suction management to limit particle spread."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: measuring the outcome and dealing with failure

Q What determines whether this treatment succeeds?

> "Three things: whether the **prosthesis can be made cleansable**, whether **she can and does clean it**, and whether she **attends three- to five-monthly maintenance**. The decontamination technique matters, but it fails without all three of those."

Q What outcomes will you measure, and when?

> "At one month: bleeding on probing, suppuration and probing depths. At three months and each maintenance visit: the same plus plaque scores. Radiographically: a **new baseline after treatment**, compared at 12 months. And her own symptoms — the soreness and discharge she presented with — because that is her measure, not mine."

Q Define a successful outcome in one sentence.

> "**Absence of suppuration, absence of bleeding on gentle probing, no further increase in probing depth and stable bone levels against the post-treatment baseline** — in a patient who is comfortable, can clean the restoration and attends her maintenance."

Q What happens at a maintenance visit?

> "Update the history, plaque and bleeding scores, **six-point peri-implant probing with bleeding and suppuration**, check restoration integrity, screw or cement margins and occlusion, professional biofilm removal with air-polishing, re-instruction, and radiographs only when clinically indicated against baseline. Then reset the interval by risk — three months if she is not stable, five if she is."

Q One implant continues to lose bone despite surgery. What now?

"Then it is **irrational to retain it** and I would recommend removal, by **reverse-torque** if possible to preserve bone. After healing, the options are to convert to a shorter span or a cantilever from the remaining implant if it is sound and the occlusion permits, to graft and replace, or to accept a shortened arch. I would explain the position honestly and document the discussion."

Q She asks whether the implants should ever have been placed at all.

"I would decline to speculate on another clinician's treatment planning, explain in general terms what I have found and why I think it has happened, and set out her options from here. If she wants to pursue it formally I would tell her how. I would record what I found and what I said, and I would **not put a critical opinion of a colleague into her notes as fact**. **Duty of candour applies to my own care**, not to a retrospective judgement of someone else's."

Q With hindsight, what would have prevented this?

"**Two separate screw-retained crowns** with cleansable embrasures rather than a linked cemented unit; adequate keratinised mucosa and an emergence profile that allows a brush; a documented **maintenance programme with the interval set at handover**; and a written recall the patient understands is not optional. Every one of those is a planning decision, not a surgical one — which is the lesson of this case."

WATCH OUT

Come back to the three drivers of peri-implant disease — **biofilm, access for cleaning, and maintenance** — and say which of the three this question belongs to. It buys a sentence of thinking time and it is almost always the right frame here.

Running this case to time

Time check. The flow above holds 39 questions and about 2,496 words of model answer. At examination pace — 140 words a minute, plus roughly five seconds for each question to be asked — that is close to **twenty-one minutes**, so it overruns by about **six**. What goes first is the chain **"if it goes to surgery"** — defect morphology, resective versus reconstructive and implantoplasty collapse into one question, because that surgery is contingent on a re-assessment you have not yet done. If your answers run past **thirty seconds**, you will never reach Domain 4 at all.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts her own films beside the new ones	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 45-year-old at **first attendance**, a year of worsening soreness and discharge at two implants placed three years ago; an intact, unrestored, periodontally healthy mouth, **never smoked, no medical history**, so the patient-level risks are absent. Then the two findings that are the case — **no maintenance for over two years**, and **two implants splinted by one restoration** — with pus and 6 mm pockets confined to that sextant. Working diagnosis **peri-implantitis at LR6 and LR7**.
1:20–4:30	Domain 1: periodontal groundwork, then her films and history	Undergraduate ground first: the **attachment apparatus**, Sharpey's fibres into cementum, the ~2 mm supracrestal attachment; then the peri-implant seal — **no cementum, no ligament, fibres parallel** — therefore **faster, non-linear progression with no proprioception**, which is her painless year. **BPE at 20–25 g**, her code 4 read as **localised and site-specific**. The **2017 World Workshop** triad. Why her own baseline gives the **rate**, and the surrogates without one — **6 mm, and 3 mm apical**. **Six sites, plastic probe, 0.2–0.25 N.**

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: telling her, then the plan	The diagnosis **in her words**, and the answer to "is it my fault" — **no, but her cleaning and attendance now decide the outcome**. The **ITI sequence**: cause-related therapy, non-surgical, **re-assessment at one month**, surgery only if suppuration or ≥6 mm persists, then **supportive care three- to five-monthly, indefinitely**. **Cleansability is the treatment, not an adjunct.** Separation into two screw-retained crowns, with the colleague's counter-argument taken seriously. Cement retention moves **residual cement** to the top of the differential. **SAC: re-treatment Advanced, surgery and explantation Complex.**
7:30–11:30	Domain 3: the non-surgical appointment, then surgery	The appointment in order — **remove the restoration if retrievable**, inspect for cement and an unseated abutment, **titanium or PEEK curettes**, sheathed ultrasonics, **subgingival glycine or erythritol air-polishing**, re-torque, review at one month. Why air-polishing: the surface is **threaded and roughened** and a blade cannot enter it. What must never touch titanium — **steel curettes, unsheathed tips**, abrasive paste, acidulated fluoride. Antibiotics **not routinely**. If surgery is reached, **defect morphology decides**, and this **lower right molar** suits resective.
11:30–14:30	Domain 4: measuring the outcome, and failure	The three determinants — **a cleansable prosthesis, her cleaning it, three- to five-monthly attendance** — and that no decontamination technique survives the loss of any one. Outcomes at one and three months, then twelve against a **new post-treatment baseline**. Continued loss after surgery: **removal by reverse-torque**, then shortened span, graft or shortened arch. And the colleague question — **decline to speculate, record what you said, never write a critical opinion as fact; candour applies to your own care**.
14:30–15:00	Closes on the patient and the record	What you say to her: the aim is to **arrest the disease, not regrow bone**, and **recession is a likely consequence of successful treatment**. Then what would have prevented it — two separate screw-retained crowns, a cleansable emergence, and a **maintenance interval set at handover**. Planning decisions, not surgical ones.

The spine — the twelve answers to have ready if nothing else

1 The attachment apparatus of a natural tooth · 2 How the peri-implant seal differs, and why the disease is worse · 3 BPE technique, and what her code 4 means · 4 The three 2017 definitions · 5 Why her own baseline decides the case, and the thresholds without one · 6 Probing an implant — sites, probe, force, pitfalls · 7 The diagnosis in her words, and "is it my fault" · 8 The ITI sequence and the one-month re-assessment · 9 Why the linked restoration is the disease · 10 The non-surgical appointment in order, and what must not touch titanium · 11 Success defined, and the maintenance visit · 12 When one implant keeps losing bone.

Where the marks are lost in this case

Treating this as an implant question. Domain 1 opens below the implant material and stays there — junctional epithelium, Sharpey's fibres, BPE technique and probing force. A candidate who reaches for the 2017 definitions before he can describe a natural attachment apparatus has thrown away free marks.

Debriding and reviewing without touching the prosthesis. The disease here is a splinted restoration a brush cannot pass. Volunteer that **cleansability is the treatment**, then volunteer the honest counter-argument about the risk of removing a three-year-old unit — that pairing, unprompted, is the difference between a 3 and a 5 in Domain 2.

Never naming the surgical line. Diagnosis, charting, prosthetic assessment, conversion to a cleansable screw-retained design, non-surgical therapy and lifelong maintenance are yours. **Surgical peri-implantitis therapy — reconstructive especially — and explantation are Complex and referred**, with the diagnostic work finished first so the referral is useful. Candidates describe implantoplasty in detail and never say they would not be doing it.

SOE Case 9 — The atrophic posterior maxilla

THE CASE

62-year-old woman, non-smoker, well-controlled hypertension. Missing **16 and 17** for eight years; wears no prosthesis. Wants a fixed replacement. CBCT: residual bone height beneath the sinus of **5 mm at the 16 and 4 mm at the 17**; ridge width 7 mm; **flat sinus floor, no septa, membrane 1 mm, patent ostium**. Interarch space is generous — the **46 and 47 have over-erupted by 2 mm**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A healthy 62-year-old non-smoker with well-controlled hypertension, missing the 16 and 17 for eight years and wearing nothing in the space, who now wants a fixed replacement. The CBCT shows an **atrophic posterior maxilla** — 5 mm of residual height at the 16 and 4 mm at the 17 — but with several favourable features: a 7 mm ridge width, a flat sinus floor with no septa, a healthy 1 mm membrane and a patent ostium. The complication is that the interarch space is generous only because the **46 and 47 have over-erupted by 2 mm** into the space. So there are two decisions: how to manage the vertical bone deficiency, and how to manage the over-erupted antagonists. And a prior question — whether she needs upper molars at all."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomy of the posterior maxilla

Q Describe the maxillary sinus.

"A pyramidal air-filled cavity in the maxillary body, roughly **15 millilitres** in the adult, with its base the lateral nasal wall and its apex towards the zygoma. The roof is the orbital floor carrying the infraorbital nerve; the floor is the alveolar process, often dipping between the molar roots; the anterior wall is the facial surface of the maxilla. It drains through the **ostium on the medial wall, high up, into the middle meatus** via the osteomeatal complex."

Q What lines it, and what is its nerve and blood supply?

"The **Schneiderian membrane** — pseudostratified ciliated columnar respiratory epithelium on a thin periosteum, normally **0.8 to 1.5 mm** thick. Sensory supply is the **posterior, middle and anterior superior alveolar nerves** and the infraorbital nerve, all branches of the maxillary division of the trigeminal. Blood supply is the **posterior superior alveolar and infraorbital arteries**, both from the maxillary artery."

Q Why does that blood supply matter surgically?

"Because the posterior superior alveolar and infraorbital arteries anastomose in the **lateral wall of the sinus** as the alveolar antral artery, with an intraosseous branch typically running around **15 to 20 mm above the crest**. A lateral window is cut straight across it, so it is the source of the troublesome bleed in that procedure. On a CBCT I can see it and plan the window around it — which is one of the arguments for CBCT before any lateral approach."

Q What does a 1 mm membrane and a patent ostium tell you?

"That the sinus is **healthy** — a normal-thickness membrane with no mucosal thickening, no polyp, no fluid level, and clearance working normally. It matters because the mucociliary escalator sweeps towards the ostium; if the ostium is blocked, anything I introduce into the sinus cannot clear and I create a **post-operative sinusitis**. A thickened membrane of more than 2 mm, or a blocked ostium, would send me to an ENT opinion before any elevation."

Q Would a thin membrane worry you at all?

"Yes — thin membranes **perforate more readily** during elevation. So 1 mm is favourable for sinus health but not a free pass surgically. A slightly thicker membrane is actually easier to lift intact. The other favourable features here are the **flat floor and the absence of septa**, since septa are the commonest anatomical cause of perforation."

Q Why is she down to 4 to 5 mm of bone?

"Two processes acting from opposite directions. From below, **pneumatisation of the sinus** after the molars were lost — the floor descends as the sinus expands into the space once occupied by the roots. From above, **crestal residual ridge resorption**, greatest in the first year and continuing. Eight years of both leaves the thin residual plate she has. The 17 is worse than the 16 because pneumatisation is typically greatest distally."

Q What bone quality do you expect, and why does it matter?

"**D4 on Misch, or Lekholm and Zarb type IV** — a thin, sometimes absent cortex over sparse, fine trabecular bone. It matters because it gives the **poorest primary stability, the longest healing time and the highest failure rate** of any site. Practically it means under-preparing the osteotomy, choosing a tapered implant with an aggressive thread, conventional loading, and no immediate loading."

Q What exactly do you measure on the CBCT?

"**Residual height beneath the sinus floor at each planned site**; **ridge width at the crest and 5 mm apical**; membrane thickness; presence and position of **septa**; the slope of the floor, since a sloping floor makes transcrestal elevation unpredictable; the **thickness of the lateral wall** and the course of the alveolar antral artery; patency of the ostium; and any sinus pathology. I would also check the exposure is justified under IR(ME)R and report the whole volume, not just my field of interest."

Q What would contraindicate any sinus procedure?

"**Acute sinusitis** — absolute until resolved. Then chronic rhinosinusitis, nasal polyposis, a **mucocele or retention cyst obstructing the ostium**, previous Caldwell-Luc surgery with a scarred sinus, and heavy smoking. Relatives include uncontrolled diabetes, bisphosphonate or antiresorptive therapy, and radiotherapy to the field. Here the sinus is clean, she does not smoke and her hypertension is controlled — so the sinus itself is not the problem."

Q Does her hypertension change anything?

"Not materially if it is genuinely well controlled. I would record a recent **blood pressure**, take a full drug history — a calcium channel blocker can cause gingival overgrowth, and some patients are on an antiplatelet — and use a local anaesthetic with adrenaline in normal dental doses, which is safe in controlled hypertension. She is **ASA II**. I would defer elective surgery if the pressure were uncontrolled."

Domain 2 — Clinical decision making and treatment planning

Chain: the occlusal groundwork

Q What is the shortened dental arch concept?

"**Käyser's** proposition that a dentition of the anterior teeth and premolars — around **20 teeth, or ten occluding pairs** — provides adequate function, comfort and aesthetics for most patients, without an increase in temporomandibular disorder, drifting or periodontal breakdown in a stable mouth. It is endorsed as a legitimate treatment goal rather than a failure of care, and it is the reason 'replace the molars' is not automatically the right answer."

Q Is a shortened arch acceptable in her?

"Potentially — she has occlusion to the second premolar and has managed eight years without any prosthesis, which is strong evidence she functions. The condition for a shortened arch is a **stable occlusion**, and hers is not entirely stable: the **46 and 47 have over-erupted 2 mm**. So the concept applies, but it needs the antagonists managing whichever route we take."

Q What are the over-erupted lower molars actually telling you?

"That the space has not been passively tolerated. Unopposed teeth **erupt with their alveolar bone**, so the occlusal plane is now disturbed, and the 'generous' interarch space is generous *because of* the over-eruption rather than despite it. If I restore the 16 and 17 at the current lower level I build a prosthesis into a deranged occlusal plane. They must be addressed before any definitive restoration."

Q How would you manage them?

"In escalating order: **enameloplasty** if 2 mm can be removed within enamel and the teeth are unrestored; **crowns at a reduced occlusal height**, possibly requiring elective endodontics if the reduction exposes pulp; or **orthodontic intrusion**, which is the most conservative of tooth tissue but the slowest and needs anchorage, often temporary anchorage devices. If she chose a shortened arch I would still manage them, to stop further drift and interference."

Chain: the options for the 16 and 17

Q Apply the ITI decision tree for the atrophic posterior maxilla.

"The first branch asks about the **interarch relationship and crest width**. Hers is favourable on both — generous space, 7 mm width — so the branch is **sinus floor elevation alone**, not a combined vertical augmentation. Then the sinus elevation tree keys on residual height: with **6 mm or less** the indicated approach is the **lateral window**; a transcrestal approach needs more than 6 mm with a horizontal floor. So the algorithm gives me a lateral window at both sites — which is **explicitly beyond the scope of this level of practice**, so I would refer for it."

Q What are your other options?

"**Short implants** — 6 mm fixtures at both sites with no sinus procedure. **A transcrestal elevation** of 2 to 3 mm with simultaneous placement, which might be feasible at the 16 with its 5 mm, though the tree favours a lateral window below 6 mm. **A cantilever from a single implant at the 15. A conventional or resin-bonded bridge. A removable partial denture.** Or **accept a shortened dental arch**."

Q Make the case for short implants.

"Systematic reviews report **survival comparable to longer implants placed with augmentation at three to five years**, with **fewer complications, less morbidity, lower cost and shorter treatment time**. A short implant is usually taken as 6 mm or less of intrabony length. With 4 to 5 mm of native bone I still need a small gain at both sites, so it is not augmentation-free here, but it converts a lateral window into a transcrestal procedure — a much smaller intervention."

Q You have argued for short implants. Now argue against them.

"The comparable-survival data are mostly **three to five years, not fifteen**, and this woman may need forty. A short implant has an **unfavourable crown-to-implant ratio**, less surface area in the poorest bone quality in the mouth, and less reserve — if it loses 2 mm of marginal bone it has lost a third of its length, where a 10 mm implant has lost a fifth. The technical complication rate is the real cost. My colleague who augments properly and places a longer implant is buying reserve."

Q Why not a cantilever from the 15?

"Because a **molar-sized cantilever from a single implant** in D4 bone applies a large bending moment to one fixture at the position of greatest occlusal load. That combination — poorest bone, highest force, no splinting — is the classic setting for screw loosening, screw fracture and marginal bone loss. I would not do it here."

Q Why not a bridge or a denture?

"A **conventional bridge** from the 15 to the 18 is too long a span and the 18 is very likely absent, so there is no distal abutment — it would be a two-unit cantilever from an unrestored premolar. A **resin-bonded bridge** cannot carry a molar span. A **removable partial denture** is a genuine option and must be offered, but she has worn nothing for eight years and specifically wants fixed, so I expect her to decline it."

Q So what do you recommend?

"I would put the **shortened arch honestly first** as the low-risk option, given eight years of comfortable function — with the lower molars adjusted. If she wants a fixed replacement, I would **refer for a lateral window elevation** with staged or simultaneous placement, having done the diagnostic and restorative planning myself. If the referral centre offered short implants instead, that is also reasonable, and I would be comfortable restoring either."

Q What is the SAC classification?

"The **surgical phase is Complex** — augmentation of an atrophic posterior maxilla, D4 bone, adjacent to the sinus. The **restorative phase is Advanced** — a two-unit posterior implant reconstruction in a modified occlusion. Naming it that way is how I justify referring the surgery while retaining the restoration."

Q What would you tell her, in her own words?

" 'You have lost quite a lot of the bone that used to hold those two teeth, and the sinus — the air space above them — has expanded down into it. There is about four to five millimetres of bone left where I would need ten or more. There are three honest choices: leave the gap, which you have managed well for eight years; have a specialist lift the sinus lining and add bone, then place implants — a bigger procedure with a recovery and a real chance of a blocked or infected sinus; or use shorter implants, which is a smaller operation but with less long-term evidence. Either way, your lower back teeth have grown up into the space and need reshaping first.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: sinus floor elevation

Q When is a transcrestal elevation indicated?

"Residual height **greater than 6 mm**, a **flat sinus floor**, a gain of only 3 to 4 mm required, adequate ridge width, a healthy sinus with a patent ostium, and no septa in the field. It is a blind procedure, so every one of those conditions is doing work — a sloping floor in particular makes the osteotome slide and tear the membrane."

Q Describe it.

"Flap, then prepare the osteotomy to **1 to 2 mm short of the sinus floor**, confirmed radiographically. Then **concave osteotomes of increasing diameter** with gentle malleting to green-stick fracture the floor, elevating the membrane with a plug of graft material pushed ahead of the osteotome as a hydraulic and mechanical cushion. Confirm membrane integrity, place the implant, close. Modern alternatives are **hydraulic elevation** or sequential drills with depth stops, both gentler than malleting."

Q How do you confirm the membrane is intact?

"The **Valsalva manoeuvre** — the patient pinches the nose and blows gently while I watch and listen at the osteotomy. **No air movement and no membrane movement** means intact. I would also look directly with good light and probe gently with a blunt instrument. If it is perforated, I would abandon elevation and either place without a gain or close and allow four to six months of healing before reattempting."

Q What are the complications of the transcrestal approach?

"**Membrane perforation** is the commonest. **Benign paroxysmal positional vertigo** from the malleting, caused by displacement of otoliths — an under-recognised and distressing complication. Then graft migration, **sinusitis**, oro-antral communication, loss of the implant into the sinus, and failure to achieve the intended gain."

Q What are your post-operative sinus precautions?

"**No nose-blowing**, sneeze with the mouth open, **no straws or smoking**, avoid air travel and diving, and a soft diet. Analgesia, a chlorhexidine rinse, and antibiotics with a **decongestant such as a topical or systemic nasal decongestant** where appropriate. I would warn her that a small amount of blood-stained nasal discharge on the first day is expected but that increasing pain, swelling or purulent discharge means she rings me."

Q In outline, what would the lateral window involve — so you can consent her for the referral?

"A buccal full-thickness flap, an **antrostomy cut in the lateral wall** with a round bur or piezosurgery, the window either hinged inwards or removed. The membrane is elevated from the floor and walls with curved elevators to create a contained compartment, which is filled with graft — usually a xenograft, alone or mixed with autogenous bone — a membrane is placed over the window, and it is closed tension-free. Implants are placed simultaneously if primary stability is achievable, otherwise staged at **six to nine months**."

Q What if the membrane perforates during that procedure?

"Small perforations are managed by **folding the membrane on itself or patching with a resorbable collagen membrane** and continuing. Large ones may mean abandoning the graft and closing, then reattempting after healing. The risk is that graft escapes into the sinus, obstructs the ostium and causes sinusitis or graft infection — which is why the ostium and membrane assessment beforehand matter so much."

Q If short implants were chosen instead, what changes surgically?

"**Under-preparation of the osteotomy** — stopping short of the final diameter, or using osteotomes rather than drills, to condense the sparse trabecular bone; a **tapered implant with an aggressive thread** and a moderately rough surface; low drill speed with irrigation; and accepting a lower insertion torque than I would in the mandible. **No immediate or early loading.** I would also consider **splinting the two crowns** to share load."

Q When would you load in this bone?

"**Conventional loading at around six months** in a grafted or D4 posterior maxilla, against three months in the mandible, confirmed clinically and with ISQ if available. Six to eight weeks would be early loading, which I would not choose here. If a sinus graft has been placed I would follow the surgeon's protocol, usually longer again."

Q What would you record?

"The medical history and blood pressure, the CBCT findings site by site with **measured residual heights, width, membrane thickness and ostium patency**, the justification for the exposure, the SAC assessment, the options discussed with **risks, benefits, costs and alternatives including no treatment and the shortened arch**, her decision, the referral and what it was for, the occlusal management plan for the 46 and 47, and the consent conversation about sinusitis and altered sensation."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: monitoring and dealing with what goes wrong

Q What would you monitor, and how often?

"Implant survival and **marginal bone levels against a baseline film taken at fitting**; peri-implant probing, bleeding and suppuration; **sinus symptoms**; the crown-to-implant ratio and any mechanical complications; and the occlusion, particularly where I have adjusted the opposing teeth. **Six-monthly initially**, then set by risk, with radiographs at one year and thereafter as clinically indicated."

Q How do you design the prosthesis to protect a short implant?

"Keep the **occlusal table narrow** so the load stays over the implant axis, **avoid excursive contacts** on the implant crowns entirely, **no cantilevers**, a flatter cusp angle, and **splint the two units** so they share load. I would also consent her specifically that a short implant has a less favourable lever arm, so a night guard is likely if any parafunction emerges."

Q At six months she reports a blocked nose and one-sided discharge. What now?

"That is a **unilateral sinusitis on the operated side until proved otherwise** — likely graft-related or from ostium obstruction. History, tenderness over the antrum, nasal endoscopy findings if available, and imaging. I would prescribe antibiotics and a decongestant, and **refer to ENT early** rather than treating it repeatedly myself. A displaced implant or migrated graft in the sinus needs removal."

Q The implant at the 17 fails to integrate. What do you do?

> "Remove it, curette the site, and be honest with her under **duty of candour**. Then decide whether to replace it after three to six months of healing — accepting that a second failure at the same site is more likely — or to **restore the 16 alone and accept a shortened arch to the first molar**, which in this patient is a perfectly good outcome. I would not cantilever a second molar from a single short implant."

Q She asks why implants fail more often up here than in her lower jaw.

> " 'The bone at the back of the upper jaw is the softest in the mouth — more like a sponge than a brick — and there is less of it because the sinus has expanded into it. The implant has less to grip while it is healing. That is why we take longer before we put the tooth on it, and why the success rate here is a few per cent lower than at the front of the lower jaw.' "

Q How would you know this treatment had succeeded?

> "By her measure first — that she chews comfortably on that side and does not think about it. Objectively: **implants stable with no bleeding or suppuration, marginal bone levels stable against baseline, no sinus symptoms, an occlusion that is even and free of excursive contacts on the implants**, and a maintenance pattern she attends. And if she chose the shortened arch, success is exactly the same measure with no surgery at all."

WATCH OUT

Come back to the two questions this case is really asking — **how much bone is there, and does she actually need molars?** Naming those buys a sentence of thinking time, and the shortened arch is a legitimate answer to almost anything here.

Running this case to time

Time check. The flow above holds 40 questions and about 2,864 words of model answer. At 140 words a minute plus roughly five seconds a question, that is nearly **twenty-four minutes** — an overrun of about **nine**, the largest in this section. The examiner drops from the middle of Domain 3: the **step-by-step transcrestal description and the lateral window mechanics** go first, compressed into one question about what you would consent her for, since you are referring that surgery anyway. Indications, Valsalva and the sinus precautions stay. If your answers run past **thirty seconds**, you will never reach Domain 4.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT slices up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A healthy 62-year-old non-smoker, 16 and 17 missing eight years with **nothing worn in the space**, now wanting fixed. An **atrophic posterior maxilla** — 5 mm at the 16, 4 mm at the 17 — with the favourable features named: **7 mm width, flat floor, no septa, 1 mm membrane, patent ostium**. Then the twist: the interarch space is generous **because the 46 and 47 have over-erupted 2 mm**. Two decisions, and one prior question: **does she need upper molars at all?**

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the anatomy of the posterior maxilla, then her CBCT	Undergraduate anatomy first: the **ostium high on the medial wall into the middle meatus**, the **Schneiderian membrane** at 0.8–1.5 mm, and the **arteries anastomosing in the lateral wall 15–20 mm above the crest** — the bleed in a lateral window. Then reading her: a patent ostium means **clearance works**, but a **thin membrane perforates more readily**, so 1 mm is not a free pass. Why she is at 4–5 mm — **pneumatisation from below, crestal resorption from above**, worse distally. **D4 / Lekholm and Zarb IV**: poorest stability, longest healing, highest failure. The absolute contraindication is **acute sinusitis**.
4:30–7:30	Domain 2: the occlusal groundwork, then the options	**Käyser's shortened dental arch** — 20 teeth, ten occluding pairs — named as a legitimate goal, not a failure of care, then applied: eight years of function, but the condition is a **stable occlusion** and hers is not. The over-erupted molars read correctly — **unopposed teeth erupt with their alveolar bone** — and managed by **enameloplasty, crowns at reduced height, or orthodontic intrusion**. The **ITI tree**: **6 mm or less indicates the lateral window**. Short implants argued for, then against — **short-term data, poor crown-to-implant ratio, 2 mm of loss is a third of the implant**.
7:30–11:30	Domain 3: sinus floor elevation, and what short implants change	Transcrestal indications, with every condition doing work — **>6 mm, flat floor, 3–4 mm gain, healthy sinus, no septa** — because it is blind. Then what you consent her for in the lateral window: antrostomy, a contained compartment, xenograft, staged at **six to nine months**. **Valsalva** to confirm integrity, and the complications including **benign paroxysmal positional vertigo from malleting**. For short implants: **under-preparation, tapered aggressive thread**, **no immediate or early loading**, splint, **conventional loading at around six months**.
11:30–14:30	Domain 4: monitoring, and what goes wrong	Monitoring against a **baseline film taken at fitting**, with **sinus symptoms** on the list. Protecting a short implant prosthetically — **narrow occlusal table, no excursive contacts, no cantilevers, splinted**. Then the scenario: a blocked nose and one-sided discharge at six months is **unilateral sinusitis on the operated side until proved otherwise**, and it goes to **ENT early**. Failure at the 17 — remove, curette, **duty of candour**, and restore the 16 alone rather than cantilever a second molar.
14:30–15:00	Closes on the patient and the record	Her three choices in her own words: leave the gap she has managed for eight years; a specialist lifts the lining and adds bone, with a real chance of a blocked or infected sinus; or shorter implants, a smaller operation with less evidence. **Either way the lower molars are reshaped first.** Success by her measure — she chews and does not think about it — is identical if she chooses the shortened arch.

The spine — the twelve answers to have ready if nothing else

1 The sinus, and where it drains · 2 The membrane, and the artery in the lateral wall · 3 What a patent ostium and a 1 mm membrane each tell you · 4 Why she is down to 4–5 mm · 5 D4 bone, and what it changes · 6 What contraindicates any sinus procedure · 7 The shortened dental arch, and whether it applies · 8 What the over-erupted 46 and 47 mean · 9 The ITI tree and the 6 mm threshold · 10 Short implants argued both ways · 11 Transcrestal indications, Valsalva, and the precautions · 12 One-sided discharge at six months.

Where the marks are lost in this case

Planning the surgery and ignoring the lower jaw. The 46 and 47 have over-erupted 2 mm and taken their alveolar bone with them. Restore the 16 and 17 to that level and you have built a prosthesis into a deranged occlusal plane. Volunteer the antagonists before the sinus, and volunteer that they need managing **even if she chooses the shortened arch**.

Never offering no molars at all. She has worn nothing for eight years and functions. Käyser's shortened arch is the low-risk option and should be put honestly first, not produced as an afterthought when the surgery is questioned. Candidates who only ever argue for treatment lose Domain 2.

Describing a lateral window as though they would do it. This is the competence line and it must be said out loud: **lateral window elevation in an atrophic posterior maxilla is Complex and beyond a general dentist with an interest in implants — it is referred.** What stays yours is the diagnostic work, the occlusal plan, the consent conversation and the **Advanced** restorative phase.

SOE Case 10 — Nerve proximity and the consent conversation

THE CASE

48-year-old male teacher, fit and well. Missing **36** for two years. OPT suggests **9 mm from crest to the superior border of the mandibular canal**. Ridge width appears adequate. He is a keen wind musician who plays clarinet in an orchestra.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A fit 48-year-old wanting a single lower first molar implant, with an apparent 9 mm of bone above the canal on a panoramic film and adequate ridge width. Two features dominate. First, **9 mm on an OPT is not 9 mm of usable bone** — the film magnifies, gives no buccolingual information and may not show the true superior border, so I cannot plan a length from it. Second, he is a **clarinettist and a teacher**, so altered lip and chin sensation would be far more than an inconvenience to him. Under **Montgomery** that makes nerve injury a highly material risk for this particular patient, and it will shape both the consent conversation and the plan. My tasks are to image him properly, establish what he wants, and decide whether the safety margin I can guarantee is one he is willing to accept."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical groundwork

Q Describe the course of the inferior alveolar nerve.

"It is the largest branch of the **posterior division of the mandibular nerve**, V3. It descends between the sphenomandibular ligament and the ramus, enters the **mandibular foramen** below the lingula, and runs forward and downward in the mandibular canal, usually lingual then more buccal as it comes forward. It divides at about the second premolar into the **mental nerve**, which exits the mental foramen, and the **incisive branch**, which continues in bone to the anterior teeth."

Q What does it supply?

"The mandibular molars and premolars and their periodontium via the dental branches. The **mental nerve** supplies the skin and mucosa of the lower lip, the labial gingivae from canine to canine, and the skin of the chin — all **unilaterally**. It is purely sensory at this point; the **mylohyoid nerve**, which branches off before the foramen, is motor to mylohyoid and anterior digastric and can also carry accessory sensation to the chin."

Q What is the anterior loop, and why does it matter here?

"The **anterior loop** is the segment where the incisive branch continues forward beyond the point at which the mental nerve turns back and up to exit the foramen — so the nerve extends anterior to the radiographic foramen. It is variable, commonly a couple of millimetres but sometimes considerably more. It matters because it is **not reliably visible on a panoramic film**, so a foramen-based safety margin taken from an OPT can be falsely reassuring."

Q What canal variations would you look for?

"A **bifid or trifid canal**, which is uncommon but well described and means a second bundle may sit outside my planned margin. **Retromolar and accessory foramina**. An **anterior looping** of the mental nerve. And a canal with an indistinct or absent cortication, which is a reason to be more, not less, conservative — I cannot measure to a border I cannot see."

Q Describe the cross-sectional shape of the posterior mandible in the 36 region.

"Typically a crest that is narrower than the base, with a **lingual concavity — the submandibular fossa** — below the mylohyoid ridge, and a buccal fullness towards the external oblique ridge. That concavity is the reason a drill following the apparent long axis of the ridge can **perforate the lingual plate** even when the crest looks wide, which risks the sublingual and submental vessels."

Chain: reading the imaging

Q Is the OPT measurement reliable?

"No, not on its own. A panoramic magnifies by roughly **20 to 30% and not uniformly**, it is a tomographic image with distortion outside the focal trough, it gives **no buccolingual information**, and the superior border of the canal is often indistinct. It is a good screening film for gross anatomy and pathology, but it is not a planning film when I am working close to the canal."

Q So justify the CBCT.

"Under **IR(ME)R** every exposure must be justified by a specific clinical question that a lower-dose alternative cannot answer. Mine is: what is the **true vertical height** above the canal at the 36, what is the **buccolingual position** of the canal, is there a **lingual concavity**, and is the canal bifid? None of those can be answered by a panoramic. I would use a **small field of view** at the smallest volume and resolution that answers the question."

Q What exactly would you measure on it?

"Crestal bone width at the crest and 5 mm apical; **vertical height from the crest to the superior cortex of the canal** at the planned site, not at the mean of the region; the buccolingual position of the canal relative to my planned axis; the depth of the submandibular fossa; and the bone density. I would record those numbers in the notes with the site they were taken at."

Q What else would you examine clinically before planning?

"Over-eruption of the **26** and tilting of the **35 and 37** after two years, which may have taken the restorative space. Inter-arch space and the occlusal scheme. Mouth opening and access to a lower second premolar–molar region. Periodontal status and plaque control. Smoking and any parafunction. And a **baseline sensory examination of both mental nerve distributions**, recorded before I touch him."

Q Why does that baseline matter so much?

"Because if he reports altered sensation afterwards I need to know whether it is new. A **pre-operative record of normal light touch, sharp-blunt discrimination and thermal sensation on both sides**, documented and ideally with the patient's own confirmation, is a few minutes' work that transforms my ability to manage a complaint honestly. In a patient for whom this risk is material it is not optional."

Domain 2 — Clinical decision making and treatment planning

Chain: the options, and the groundwork behind them

Q What are the options for a single missing 36?

"**No treatment or a shortened dental arch**; a **removable partial denture**; a **conventional fixed bridge** on the 35 and 37; a **cantilever or resin-bonded bridge**, which is poorly supported in a molar site; or a **single implant-retained crown**. Each has to be offered with its risks, benefits and costs, including doing nothing, before I can say his consent is valid."

Q Make the case for the shortened dental arch.

"The **shortened dental arch** concept holds that occlusal function, comfort and stability are generally maintained with around **20 teeth — the anteriors and premolars in occluding pairs**. Long-term studies show acceptable function and no major increase in TMD or periodontal breakdown. Here the 36 is a first molar, so he would still have a second molar contact; the argument is weaker than for a missing 37, but it is still a legitimate option to place before him."

Q You have offered an implant. Argue against it.

"That it introduces a **surgical risk to a nerve that currently works perfectly**, in a man whose livelihood and principal pleasure depend on lip sensation, to replace a single tooth he has managed without for two years. It commits him to lifelong maintenance and a restoration that will need components replaced. The **null option is genuinely reasonable here**, and a doctrine that every gap must be filled is exactly the reasoning that gets criticised in a complaint."

Q Your colleague would place a conventional bridge instead. Why might they be right?

"Because it avoids the nerve entirely. If the **35 and 37 are already heavily restored**, a bridge sacrifices little additional tooth tissue and can be delivered predictably in weeks, with a well-understood failure mode. The counter-argument is that the abutments here may be sound, and preparing sound teeth carries its own long-term biological cost — but if he places overwhelming value on avoiding nerve risk, a bridge is a defensible and possibly better answer."

Q What is the legal standard for consent?

"**Montgomery** — the test of materiality is no longer what a body of doctors would disclose but whether **a reasonable person in this patient's position, or this particular patient, would attach significance to the risk**. So I must know enough about him to know what matters to him. His clarinet playing converts a small risk of altered lip sensation from a nuisance into a career-relevant one, and I must therefore discuss it in proportion."

Q Tell me what you would say to him, in his words.

" 'There is a nerve running through your lower jaw that gives feeling to your lip and chin on that side. I'll take a 3D scan first so I know exactly where it is, and I plan to keep at least two millimetres away from it. Even so, there's a small chance of bruising or damaging it, which can leave numbness, tingling or an unpleasant altered feeling in your lip and chin. Most of these settle over weeks or months, but occasionally it's permanent. For most people that's an inconvenience — for you, with the clarinet, it could be a real problem, so I want you to weigh it carefully against simply leaving the gap.' "

Q What would you record?

"The options discussed including no treatment, the specific risks and the fact that I discussed **nerve injury in the context of his clarinet playing** — quoting the words, not just the heading. His questions and answers. The **baseline sensory examination**. The CBCT justification, the measurements and the planned clearance. The written information given and the cooling-off period. And his signature, on the understanding that consent is the conversation and not the form."

Q Would his hobby change your plan?

"Possibly. I would consider a **shorter, wider implant** to increase the safety margin, since contemporary evidence for short implants in the posterior mandible is good over the medium term and avoids grafting altogether. I would consider a **fully guided approach**. And I would explicitly re-offer the bridge and the shortened arch, because a plan chosen after hearing his priorities is a different plan from one chosen for him."

Q SAC classification?

"Normatively **Straightforward** — a single posterior implant in a healed site with adequate bone volume, non-aesthetic. But the **anatomical risk modifier** of nerve proximity and the **patient expectation modifier** created by his occupation and hobby raise it, and I would treat it as **Advanced**."

Q Is this within your competence, and when would you refer?

"A single posterior mandibular implant with a CBCT-verified clearance is within the scope of a general dentist with implant training, so yes — provided I have the CBCT and, ideally, a guide. I would refer if the clearance were **under about 2 mm at any point**, if the canal were bifid or uncorticated, if he wanted a guarantee I cannot give, or if he wished the case to be done somewhere that guided surgery and immediate specialist support are routine. That is a legitimate reason to refer even when the case itself is not complex."

Domain 3 — Decision making in relation to treatment procedures

Chain: designing the safety margin

Q How much clearance do you want from the canal, and why that figure?

"At least **2 mm from the superior cortex of the canal to the implant apex**. That covers the accuracy limits of the imaging, the guide or free-hand technique and the drill itself, and it allows for the fact that the neurovascular bundle can be compressed or heated without ever being touched. If my CBCT height is 9 mm true, I am planning an **implant of no more than about 7 mm** at that site."

Q Are you comfortable with a 7 mm implant in a molar site?

"Yes. **Short implants of 6 to 8 mm in the posterior mandible** perform comparably to longer implants with vertical augmentation over the medium term, with fewer complications, because they avoid a graft procedure entirely. The trade-off is that I must protect them: a **wider diameter**, a favourable crown-to-implant ratio, a narrow occlusal table and no cantilever."

Q Guided or free-hand?

"**Fully guided** is my preference here, because it carries the CBCT plan into the mouth and controls both depth and angulation — which is exactly what I am worried about. Free-hand with drill stops is acceptable in experienced hands, but if a small angular error could put me into the canal in a patient for whom that is career-altering, the guide is the proportionate answer."

Q What are the limitations of a guide?

"It is only as good as the **scan, the merge and the seating**. A poorly fitting guide, an unstable mucosa-borne guide, or a merge error between the CBCT and the intra-oral scan all introduce error, and **the guide gives false confidence** because it feels definitive. Reported deviations of a millimetre or more at the apex are not unusual, which is another reason the 2 mm margin exists rather than being spent."

Q Would you use an inferior alveolar block?

"I would prefer **infiltration with an articaine-based solution**, so that he retains enough lip sensation to report a sharp electric sensation if the drill approaches the bundle. Articaine infiltration in the posterior mandible is often sufficient for implant placement."

Q Your colleague blocks every posterior mandibular case. Why might they be right?

"Because **patient movement from inadequate anaesthesia is itself a hazard** when a drill is 2 mm from a nerve, and because the 'warning sign' argument is weak — by the time the patient reports a sensation, the injury may already have occurred, so it should never be relied on as a safety mechanism. The honest position is that a block is acceptable if the **planning is sound**; the infiltration is a supplement to good planning, not a substitute for it."

Q Take me through the drilling.

"Aseptic technique, chlorhexidine rinse. **Mid-crestal incision** with the releasing incision kept well away from the mental foramen, full-thickness flap, and I palpate and visualise the **lingual concavity**. Then graded drilling with **new sharp drills, slow speed, copious irrigation and intermittent pressure**, using drills with physical stops or a guided sleeve system, and a depth gauge check at each stage rather than only at the end."

Q How do you verify depth intra-operatively?

"A **direction indicator and a periapical after the pilot drill**, with a paralleling technique so the geometry is interpretable, comparing the indicator tip against the canal. Then depth gauges at each increment. If the periapical is ambiguous I stop and reassess rather than take one more drill — the point at which you are tempted to press on is exactly the point at which you should not."

Q How do you avoid perforating the lingual plate?

"Plan the axis on the **cross-sectional CBCT slice**, not on the crest form. Angle the implant slightly buccally where the fossa is deep, keep the flap raised so I can palpate the lingual plate directly with a finger, and be alert to the drill suddenly losing resistance. Perforation risks bleeding into the **sublingual and submandibular spaces**, which is a genuine airway emergency."

Q What would you do if you did get a floor-of-mouth bleed?

"Direct pressure, **bimanual compression** with a finger in the floor of mouth, and place the implant or a haemostatic agent to tamponade the site. Sit him up, keep him talking, and observe closely. If there is any **swelling of the floor of mouth, tongue elevation or difficulty swallowing**, that is an evolving airway obstruction and I call 999 and transfer to hospital — this can develop over hours, so I would not discharge him early."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: the nerve injury, if it happens

Q He telephones the next day reporting numbness. What do you do?

"**See him that day** — this is the one complication where delay changes the outcome. Take a history: is it anaesthesia, paraesthesia or **painful dysaesthesia**, when did it start, and is it changing. Examine and map it. Then image the site immediately to establish whether the implant is in or on the canal, because that determines whether I act surgically in the next 24 to 36 hours."

Q How do you map and record it?

"**Draw the affected area on a facial diagram with measurements**, and photograph it. Test **light touch with a cotton wisp, sharp-blunt discrimination, two-point discrimination and thermal sensation**, comparing with the contralateral side and with his recorded baseline. Ask him to score it, for example on a visual analogue scale, and note the functional effect — speech, drooling, and specifically his clarinet embouchure."

Q Classify nerve injuries for me.

"**Seddon** — **neurapraxia**, a conduction block with an intact axon and full recovery; **axonotmesis**, axonal damage with the endoneurium intact and recovery over weeks to months; **neurotmesis**, complete disruption with poor spontaneous recovery. **Sunderland** subdivides this into five degrees by which layer is disrupted, first degree being neurapraxia and fifth being complete transection. The clinical value is prognostic: the further down the scale, the less likely spontaneous recovery becomes."

Q The radiograph shows the implant apex in the canal. Now what?

"**Remove or back the implant out immediately** — ideally within 24 to 36 hours, and certainly within days. Compression and a foreign body in the canal produce ongoing injury, so the priority is decompression rather than preserving the fixture. I would explain this honestly, do it under local anaesthetic, and plan a shorter implant later once sensation has been reassessed."

Q And if the implant is clear of the canal?

"Then it is more likely a **drill, retraction or injection injury**, or oedema and haematoma compressing the bundle — which carries a better prognosis, and I would manage it conservatively with close review. A **short course of corticosteroid** and an NSAID in the first few days are commonly suggested to reduce perineural oedema and are reasonable if started early, though I would be honest that the evidence is limited. I would not simply reassure and discharge, because the review pattern is what lets me refer at the right moment."

Q When and where do you refer?

"**Urgently to oral and maxillofacial surgery** if the implant is in the canal and I cannot remove it, if there is painful dysaesthesia from the outset, or if there is no meaningful improvement by around **four weeks**. The window in which surgical exploration or repair can help is measured in **weeks to a few months**, not years — so referring late is itself a failure. I would refer with my baseline, my maps and my imaging."

Q What do you say to him at that first review?

> " 'I'm sorry — the feeling in your lip and chin has been affected, and that's the risk we discussed. The scan shows where the implant is and I'm acting on that today. Most of these improve, many completely, but it can take weeks or months and I can't promise you it will fully recover. I'm going to see you weekly to start with, I'm going to record exactly how it changes, and if it isn't improving by a month I'll refer you to a specialist unit.' Then **duty of candour** in writing, and I would notify my indemnifier the same day."

Q What would success look like, and what would you audit?

> "Success is an integrated implant with stable bone levels, a cleansable crown that he can floss and that carries no excursive contacts — and **normal sensation**. For audit I would set a standard that **100% of posterior mandibular implant cases have a recorded pre-operative sensory baseline, a CBCT-derived height at the planned site, and a documented planned clearance**, then measure my own compliance and my incidence of altered sensation, and re-audit after any change. That is a legitimate quality improvement project."

WATCH OUT

Come back to the margin: state the true bone height, the clearance you are protecting and how you will verify it. Nearly every question in this case is really asking how you keep 2 mm between a drill and a nerve, and how you would tell the patient if you failed.

Running this case to time

Time check. The flow above holds 39 questions and about 2,846 words of model answer. At 140 words a minute plus roughly five seconds a question, that runs to nearly **twenty-four minutes** — an overrun of about **eight and a half**. What goes first is the **lingual plate and floor-of-mouth bleed** sub-chain in Domain 3 — real, but this case is about a nerve, and Domain 4 is entirely the nerve injury. If your answers run past **thirty seconds** you will never reach Domain 4, which here means never being asked what you do when he telephones.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the OPT up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A fit 48-year-old wanting a single 36, with an apparent 9 mm above the canal. Two features stated as the case: **9 mm on an OPT is not 9 mm of usable bone** — it magnifies, gives no buccolingual information, and the superior border may not be the true one, so **no length can be planned from it**; and he is a **clarinettist and a teacher**, so altered lip sensation is a career risk, not an inconvenience. Under **Montgomery** that makes nerve injury highly material *for this patient*.
1:20–4:30	Domain 1: the anatomical groundwork, then reading the imaging	Undergraduate anatomy first: the **inferior alveolar nerve** from the posterior division of V3, dividing at about the second premolar into **mental and incisive** branches, and what the mental nerve supplies **unilaterally**; the **anterior loop**, **not reliably visible on a panoramic**, so a foramen-based margin can be falsely reassuring; **bifid and uncorticated canals**; the **submandibular fossa**, and why a drill on the crestal axis perforates lingually. Then the OPT dismissed as a planning film at **20–30% non-uniform magnification**, the CBCT **justified under IR(ME)R** by a named question, and a **baseline sensory examination of both sides recorded before you touch him**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the options, and the groundwork behind them	Every option including **no treatment and the shortened dental arch**. Then **argue against your own implant**: a surgical risk to a nerve that works perfectly, in a man whose livelihood depends on lip sensation, for a tooth he has managed without for two years. The colleague's bridge granted its strength. **Montgomery** stated properly — materiality judged from **this patient's position**, not by a body of doctors. His hobby changing the plan: a **shorter, wider implant**, a **fully guided approach**, and the bridge and shortened arch **re-offered**. **SAC normatively Straightforward, raised to Advanced** by the anatomical and expectation modifiers.
7:30–11:30	Domain 3: designing the safety margin	**At least 2 mm from the superior cortex of the canal to the implant apex**, and why — imaging, technique and drill error, and the fact that a bundle can be **compressed or heated without being touched**. So 9 mm true height means an implant of **no more than about 7 mm**, defended: **6–8 mm implants in the posterior mandible** perform comparably to longer implants with augmentation, protected by a wider diameter, a narrow table and no cantilever. **Fully guided**, with the guide's limitations volunteered — scan, merge and seating error, **apical deviations of a millimetre or more**, false confidence. Articaine **infiltration** rather than a block, then the honest counter: **a warning sensation is not a safety mechanism**.
11:30–14:30	Domain 4: the nerve injury, if it happens	**See him the same day** — the one complication where delay changes the outcome. Anaesthesia, paraesthesia or **painful dysaesthesia**, mapped on a facial diagram and photographed; **light touch, sharp-blunt, two-point, thermal**, both sides, against his baseline, with the effect on his **embouchure** recorded. **Seddon and Sunderland**, used prognostically. Apex in the canal — **remove or back it out within 24 to 36 hours**: decompression before fixture. Clear of it — likelier drill, retraction or oedema, and close review rather than reassurance. **Urgent OMFS** if it is in the canal, if there is dysaesthesia from the outset, or if there is no improvement by **four weeks**; referring late is itself a failure.
14:30–15:00	Closes on what you say and what you write	**Said:** *"I'm sorry — the feeling in your lip and chin has been affected, and that's the risk we discussed. Most improve, many completely, but it can take months and I can't promise full recovery. I'll see you weekly, record how it changes, and refer you if it isn't improving by a month."* **Written:** the options including no treatment; the risk discussed **in the context of his clarinet playing, quoting the words used**; the baseline sensory examination; the CBCT justification, the height at the planned site and the **planned clearance**; the cooling-off period; then **candour in writing, and the indemnifier notified the same day**.

The spine — the twelve answers to have ready if nothing else

1 The inferior alveolar nerve, and what the mental nerve supplies · 2 The anterior loop, and why an OPT hides it · 3 Canal variations that change your margin · 4 The submandibular fossa and lingual perforation · 5 Why the 9 mm cannot be used, and how you justify the CBCT · 6 The baseline sensory examination · 7 All the options, including no treatment · 8 Montgomery, and materiality judged from his position · 9 The 2 mm clearance, and the arithmetic that follows · 10 Short implants, and how you protect them · 11 The limitations of a guide · 12 The telephone call the next day.

Where the marks are lost in this case

Measuring 9 mm off the panoramic and planning a 9 mm implant. The film magnifies by 20 to 30% and not uniformly, gives no buccolingual information and often shows no reliable superior cortex. Say out loud that **you cannot plan a length from this image**, then justify the CBCT by the question it answers, not by habit.

Skipping the baseline. A recorded pre-operative sensory examination of both mental distributions takes minutes and transforms your position if he telephones. Candidates plan a beautiful guided case, never volunteer it, and then have no way to say whether anything is new. In a patient for whom this risk is material, volunteering it is the difference between a 3 and a 5.

Consenting the procedure instead of the man. Under Montgomery the test is what a person **in his position** would attach significance to, and he plays clarinet in an orchestra — so the record must show the risk discussed in those terms and the **null option genuinely re-offered**. The competence line follows: a CBCT-verified single posterior implant is yours, but **a clearance under about 2 mm, a bifid or uncorticated canal, or a patient who wants a guarantee, is a referral** — and a nerve injury not improving at four weeks goes urgently to oral and maxillofacial surgery.

SOE Case 11 — The medically complex patient on antiresorptives

THE CASE

71-year-old woman with **rheumatoid arthritis**, taking **methotrexate**, **prednisolone 5 mg daily** and **denosumab 60 mg every six months for osteoporosis**. Warfarin for atrial fibrillation, **INR 2.8** last week. She wants two implants to stabilise a lower denture. She has **limited manual dexterity**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 71-year-old woman requesting a two-implant mandibular overdenture, with four medical issues and one major functional one. She is on **denosumab for osteoporosis**, so there is an **MRONJ** risk; on **long-term prednisolone**, which both impairs healing and, importantly, changes her MRONJ risk category; on **methotrexate**, which is immunosuppressive and dictates my choice of analgesic; and on **warfarin** with an INR of 2.8, which is manageable but must be verified. Overlying all of that, her rheumatoid arthritis has reduced her manual dexterity, and implants demand lifelong daily cleaning. My position is that the request is entirely reasonable in principle — a two-implant overdenture transforms quality of life — but that I must take each issue in turn, and that the dexterity may prove decisive."

Domain 1 — Clinical interpretation and reasoning

Chain: the pharmacology groundwork

Q How does denosumab work?

"It is a **fully human monoclonal antibody against RANK ligand**. RANK-L, produced by osteoblasts and stromal cells, binds RANK on osteoclast precursors and drives their differentiation, activation and survival. Denosumab sequesters RANK-L, so it acts like an excess of the natural decoy receptor osteoprotegerin, and **osteoclastic resorption is profoundly suppressed**. At 60 mg six-monthly it is used for osteoporosis; at 120 mg monthly for skeletal metastases and giant cell tumour."

Q How does that differ from a bisphosphonate?

"A bisphosphonate **binds to hydroxyapatite and is incorporated into bone**, so its effect persists for years after the last dose and a 'drug holiday' achieves little in the short term. Denosumab is a circulating antibody that is **not incorporated into bone**, so its effect **wanes towards the end of each six-month cycle** and is fully reversible on stopping. Stopping it, though, causes **rebound bone loss and vertebral fracture** — which is why the decision is never mine."

Q Define MRONJ.

"**Exposed bone, or bone that can be probed through an intra-oral or extra-oral fistula, in the maxillofacial region, persisting for more than eight weeks**, in a patient who is taking or has taken an antiresorptive or antiangiogenic drug, and who has **no history of radiotherapy to the jaws** and no obvious metastatic disease at that site. The three limbs all have to be satisfied — the eight weeks and the exclusion of radiotherapy are the parts candidates drop."

Q What risk group is she in?

"This is the trap. The **osteoporosis dose is the lower-risk indication**, and on that alone she would be lower risk. But under **SDCEP** a patient taking an antiresorptive for osteoporosis moves into the **higher-risk group** if they have taken it for more than five years **or if they are also on a systemic glucocorticoid** — and she is on

prednisolone 5 mg daily. So I would treat her as **higher risk**, and I would establish how long she has been on denosumab."

Q What does prednisolone 5 mg daily mean for you otherwise?

"Long-term low-dose steroid causes **adrenal suppression**, impaired wound healing and increased infection risk, and contributes to her osteoporosis. For routine dentistry and minor oral surgery under local anaesthetic, **steroid supplementation is generally not required** — I would ensure she takes her normal morning dose, treat early in the day, and be alert to hypotension or collapse. If in doubt I would ask her rheumatologist rather than guess."

Q Methotrexate — mechanism and dental relevance.

"It is a **folate antagonist**, inhibiting dihydrofolate reductase; in rheumatology it is given weekly at low dose as a disease-modifying immunosuppressant. Dental relevance: **immunosuppression and possible myelosuppression**, so meticulous asepsis, awareness of oral ulceration and infection, and a lower threshold for antibiotic cover. Critically, **NSAIDs reduce its renal excretion and increase methotrexate toxicity**, so they must be avoided."

Q How does warfarin work, and what is the INR?

"Warfarin is a **vitamin K antagonist**, inhibiting vitamin K epoxide reductase and so the synthesis of the vitamin K–dependent clotting factors **II, VII, IX and X**, plus proteins C and S. The **INR** is the patient's prothrombin time divided by a control, raised to the power of the international sensitivity index of the reagent — a standardised ratio, so an INR of 2.8 means her extrinsic pathway takes 2.8 times as long as normal."

Q What features of rheumatoid arthritis itself matter to you?

"**Cervical spine involvement, including atlanto-axial subluxation**, so I must not extend her neck or tip her far back. TMJ involvement limiting opening. Reduced tolerance of long appointments and of maintaining a fixed posture. **Secondary Sjögren's syndrome** giving xerostomia, which affects both caries risk and denture retention. Anaemia of chronic disease. And the hand deformity itself."

Q What would you ask her rheumatologist and GP?

"In writing: the **indication, dose and duration of the denosumab** and when the next dose is due; her disease activity and whether it is stable; her current steroid dose and any advice on cover; her full blood count and renal function; whether her INR has been stable and her target range; and whether they foresee any change of therapy. I would put my own plan and my reasoning in the letter, and file the reply."

Domain 2 — Clinical decision making and treatment planning

Chain: from dexterity to the treatment decision

Q How would you assess her dexterity?

"**Directly, not by assumption.** I would ask her to remove and reinsert her denture in front of me, to demonstrate cleaning it, and to use an interdental brush and a toothbrush on her remaining teeth or the ridge. I would look at her grip, her wrists and her ability to sustain a task. I would also ask who helps her at home, and what her plaque control is like now — **her current oral hygiene is the best predictor of her future oral hygiene**."

Q What are her options?

"A **remade conventional lower denture** with correct extension, a neutral-zone technique and a restored vertical dimension; a **copy denture** preserving what she has adapted to; a **permanent soft liner** for a resorbed or tender ridge; **denture adhesive** and dietary advice, which are legitimate and not a failure; or a **two-implant mandibular overdenture**, which the McGill and York consensus statements describe as the minimum standard of care for an edentulous mandible."

Q Where do you start?

"With the **conventional denture, properly made**. A significant proportion of 'my denture is loose' cases are extension, occlusion or vertical dimension errors rather than a biological limit, and a remake is reversible, cheap and carries no MRONJ risk. If a well-made denture solves it, the medical debate never has to happen. I would only move on when I can say her denture is technically as good as it can be and still fails."

Q Suppose it is not enough. Would you place the implants?

"Not myself. Two implants in the anterior mandible would ordinarily be **Straightforward** work, but a **higher-risk MRONJ categorisation combined with immunosuppression, anticoagulation and reduced dexterity** takes this beyond what I would undertake single-handed. I would refer for a joint opinion — while being clear to her that referral is not refusal, and that the treatment may well be entirely appropriate for her in the right setting."

Q You said the MRONJ risk is small. Argue against treating at all.

"That this is **elective surgery creating a lifelong hygiene burden in a woman who may not be able to meet it**. That MRONJ, though uncommon, is disabling and difficult to treat, and that its risk here is not the reassuring osteoporosis figure but the higher-risk category. That peri-implant mucositis in a patient who cannot clean is close to inevitable. And that a well-made conventional denture with a soft liner and adhesive carries none of these risks."

Q And argue the other way.

"That a lower complete denture on a resorbed ridge is genuinely miserable, that it affects nutrition, speech and confidence in an elderly woman who may have limited years to enjoy the improvement, and that **withholding a well-evidenced, modest intervention because of a small risk is itself a harm**. Two implants and stud attachments are among the least invasive implant treatments there are. The honest answer is that this is her decision to make on properly presented information."

Q If she were on 120 mg denosumab monthly for malignancy?

"I would **decline outright** and explain why. That is the high-risk oncology dosing, MRONJ risk is an order of magnitude greater, and elective implant surgery cannot be justified. I would optimise her denture, keep her under close preventive care, and make sure any necessary extractions were done in a specialist setting."

Q Bar or studs, if implants do go ahead?

"**Studs — Locator-type attachments.** With her dexterity a bar is the wrong answer: it needs a floss threader or superfloss passed underneath, more vertical space, and it is far harder to clean. Two independent studs allow a **single-tufted brush around each abutment** and a denture fit surface she can brush. Simplicity and cleansability outrank theoretical load-sharing here."

Q SAC classification?

"Surgically and prosthodontically a two-implant mandibular overdenture is normatively **Straightforward**. Her **medical risk modifier** — antiresorptive therapy, immunosuppression and anticoagulation — and the **patient-factor modifier** of impaired plaque control take it to **Complex** in my assessment, and that is why it should be delivered with specialist input."

Q What would you say to her, in her own words?

" 'Your osteoporosis injection works by switching off the cells that break bone down, which is what makes your bones stronger. Very occasionally, in people on that injection, the jawbone doesn't heal properly after surgery and a small area of bone stays exposed — that's called osteonecrosis of the jaw. It's uncommon, but because you're also on steroid tablets your risk is a bit higher than average, and it can be difficult to treat. That doesn't mean you can't have implants, but it does mean I'd like to try getting your denture right first, and if we do go ahead I'd want a specialist involved. I would not want you to stop your injection — that would risk your spine.' "

Q What would you record?

"Every drug with **dose, indication and duration** — not just the name. The MRONJ risk category and my reasoning for it. The written liaison with her rheumatologist and the reply. My **direct assessment of her dexterity**, described. The options discussed including no treatment and a conventional remake, the specific risks discussed, and the words I used about MRONJ. Her questions. And the decision, whose it was, and the referral if made."

Domain 3 — Decision making in relation to treatment procedures

Chain: making the surgery as safe as it can be

Q Does the timing within the denosumab cycle matter?

"It is often suggested that elective surgery is best placed in the **later part of the six-month cycle**, when the antibody's effect has waned, and that the next dose is deferred until healing is established. It is biologically plausible and worth discussing with her physician, but I would be honest that **the evidence for it is limited** and I would not present it as established practice."

Q Would you stop the denosumab?

"**No — not my decision and not my drug.** Stopping denosumab causes **rebound-associated vertebral fractures**, which is a far greater harm than the risk I would be trying to avoid. Any alteration is for her rheumatologist, in writing, and I would make clear in my letter that I am not requesting cessation but asking about timing."

Q How do you manage the warfarin?

"Under **SDCEP**: do not interrupt it. Check the **INR within 24 hours** of the procedure, or up to 72 hours if she is stably anticoagulated. If it is **below 4.0**, proceed with **local haemostatic measures**. Treat **early in the day and early in the week**, limit the initial surgical area, and give her written post-operative instructions and a contact number."

Q The INR on the day is 4.2. What now?

"**Postpone and refer back** to her anticoagulation service or GP for adjustment, then re-check and rebook. I would not proceed and I would not adjust her dose myself. I would also use the delay to ask whether her control has been unstable, because that is relevant to whether elective surgery is wise at all."

Q What local haemostatic measures?

"**Atraumatic technique first** — the less trauma, the less bleeding. Then **oxidised regenerated cellulose or a collagen sponge in the socket or under the flap**, **sutures to approximate the wound**, and firm **pressure with a damp gauze pack** for at least twenty minutes. **Tranexamic acid mouthwash** if oozing persists. Avoid vigorous rinsing for 24 hours and no hot drinks or alcohol that day."

Q What analgesia would you prescribe, and what would you avoid?

"**Paracetamol, regularly and by the clock**, with **codeine** added if needed. I would **avoid all NSAIDs** — twice over: they increase methotrexate toxicity, and they potentiate warfarin both pharmacokinetically and by their antiplatelet effect and gastric irritation. I would tell her this explicitly, including that she must not buy ibuprofen over the counter."

Q Antibiotic prophylaxis?

"I would give it here — a single pre-operative dose, for example **amoxicillin 3 g orally one hour before**, or clindamycin if penicillin-allergic. The justification is her **immunosuppression from methotrexate and prednisolone** plus her antiresorptive status, not implant surgery per se. I would avoid **metronidazole and miconazole** for anything else, because both markedly potentiate warfarin."

Q Describe your surgical modifications.

"**Minimise trauma throughout.** A **flapless or minimal-flap** approach if the CBCT allows, or a small mid-crestal incision with careful atraumatic elevation. Avoid unnecessary bone removal and smooth any sharp edges. **Co-**

pious irrigation and sharp drills to avoid thermal injury. **Primary closure** wherever possible, because exposed bone is precisely what I am trying to prevent. Pre-operative chlorhexidine rinse."

Q How would you plan the appointment around her arthritis?

"A **morning appointment**, short, with her usual medication taken. **Do not extend her neck** — treat her close to upright, support the head and neck, and let her change position; if she has known atlanto-axial instability I would be especially careful. Allow time for her to get into and out of the chair. Ask about her hands before handing her anything, and check she can manage the journey."

Q What post-operative instructions would you give her?

"Written as well as spoken, and a contact number. **Do not rinse for 24 hours**, then warm salt-water rinses. **Paracetamol only — no ibuprofen or aspirin.** Leave the denture out for 24 to 48 hours and then only wear it after I have relieved and relined it. Soft diet, no smoking, no alcohol that day. And specifically: **come back if bleeding does not stop with 20 minutes of pressure, or if the site has not healed over**."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: watching for what can go wrong

Q What are you watching for, and when?

"**Bleeding in the first 48 hours. Infection in the first week.** And then, specifically, a **review at eight weeks for delayed healing or exposed bone**, because that is the point at which non-healing meets the definition of MRONJ. In the longer term, peri-implant mucositis driven by her plaque control, and denture-induced trauma over a healing site."

Q She telephones on day two with bleeding. Manage it.

"Reassure, sit her up, and have her **bite firmly on a damp gauze or a clean handkerchief for twenty minutes without checking**. If that fails, see her: identify whether it is oozing from the mucosa or a vessel, clean the site, remove any liver clot, **re-suture and pack with a haemostatic agent**, and use tranexamic mouthwash. Check the INR and ask about ibuprofen. If it still will not settle, or if there is airway swelling, she goes to hospital."

Q At ten weeks there is a 4 mm area of exposed bone. What is your diagnosis and what do you do?

"That is **MRONJ**, and by definition — exposed bone beyond eight weeks in a patient on an antiresorptive without jaw radiotherapy. I would take a radiograph, **not debride it**, prescribe **chlorhexidine 0.2% mouthwash**, treat any secondary infection with antibiotics, remove any sharp bony edge causing trauma only if clearly necessary, keep the denture off the area, and **refer to oral and maxillofacial surgery**. And I would inform her rheumatologist."

Q How is MRONJ staged?

"Broadly: **stage 0** with non-specific symptoms or radiographic change but no exposed bone; **stage 1** exposed or probeable bone that is asymptomatic and uninfected; **stage 2** exposed bone with pain and infection; and **stage 3** with exposed bone plus extension beyond the alveolar bone — pathological fracture, extra-oral fistula, oro-antral communication or osteolysis to the inferior border. Management is conservative at the lower stages and increasingly surgical at stage 3."

Q She develops mucositis around both abutments at six months. What does that tell you and what do you do?

"That **plaque control has failed**, which is exactly the risk I identified at the outset. **Peri-implant mucositis is reversible** — bleeding on gentle probing without bone loss — so I would treat it before it becomes peri-implantitis: professional debridement with instruments that will not damage the abutment surface, chlorhexidine locally, and above all **re-teach the cleaning with adapted aids** — a **powered brush with a wide handle**, single-tufted brushes, and involving whoever helps her at home."

Q What recall would you set, and what would you record at each visit?

"**Three-monthly**, with hygienist support, given her dexterity and immunosuppression. At each visit: **probing at six sites per implant with a plastic probe**, bleeding and suppuration, mucosal health, plaque scores, denture fit, extension and occlusion, attachment retention, and a check of the ridge for any exposed bone. Radiographs at one year against the baseline taken at fitting, then as clinically indicated."

Q How would you know this had gone well?

"By her own account — that she eats a normal diet and does not think about the denture. Objectively: **both implants stable with no bleeding or suppuration**, bone levels matching baseline, **mucosa intact with no exposed bone at any review**, a denture she can insert, remove and clean herself or with help, and a maintenance interval she attends. If any of those fail, the honest conclusion may be that the conventional denture would have served her better."

WATCH OUT

Take the drugs one at a time — denosumab, prednisolone, methotrexate, warfarin — and say for each what it does, what it means for healing or bleeding, and what you would change. It buys a sentence of thinking time and it is almost always the frame the question is really asking for.

Running this case to time

Time check. The flow above holds 38 questions and about 2,646 words of model answer. At 140 words a minute plus roughly five seconds a question, that comes to about **twenty-two minutes** — an overrun of **seven**. What goes first is the **appointment-logistics** material in Domain 3: the morning slot, the atlanto-axial precaution, the post-operative instructions. It is good practice and it will be taken as one question, because the marks here sit in the drugs and in the eight-week review. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where MRONJ is actually diagnosed.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the drug list in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 71-year-old requesting a two-implant mandibular overdenture, presented as **four medical issues and one functional one**, each named with what it does: **denosumab** — MRONJ risk; **prednisolone 5 mg** — impairs healing *and changes her risk category*; **methotrexate** — immunosuppression and the analgesic it forbids; **warfarin at INR 2.8** — manageable but verified. Then the one that may decide it: **rheumatoid hands, and implants demand lifelong daily cleaning**. Working position — the request is reasonable in principle, but each issue is taken in turn and **the dexterity may prove decisive**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the pharmacology groundwork	Mechanisms, not names. **Denosumab as a monoclonal antibody against RANK-L**, acting like excess osteoprotegerin; **not incorporated into bone**, so its effect wanes across the cycle and is reversible — unlike a bisphosphonate bound to hydroxyapatite for years. **MRONJ defined in all three limbs** — exposed or probeable bone, **more than eight weeks**, antiresorptive exposure, **no jaw radiotherapy** — because the eight weeks and the radiotherapy exclusion are what candidates drop. Then the trap: the osteoporosis dose is the lower-risk indication, **but SDCEP moves her to higher risk on the concurrent glucocorticoid**. **Methotrexate as a folate antagonist — and NSAIDs raise its toxicity. Warfarin against II, VII, IX, X**, and what an INR is. Then the **cervical spine and secondary Sjögren's**.
4:30–7:30	Domain 2: from dexterity to the treatment decision	Dexterity assessed **directly, not assumed** — she removes, reinserts and cleans the denture in front of you, and **her present hygiene predicts her future hygiene**. All the options, starting with a **properly remade conventional denture**, because that is reversible, cheap and carries **no MRONJ risk** — and if it works the medical debate never happens. Then the case argued **both ways**: elective surgery creating a hygiene burden she may not meet, against the misery of a lower complete denture and the harm of withholding a well-evidenced intervention. **Studs, not a bar** — cleansability outranks load-sharing in these hands. **SAC Straightforward normatively, Complex here.**
7:30–11:30	Domain 3: making the surgery as safe as it can be	**You do not stop the denosumab** — not your drug, and stopping causes **rebound vertebral fracture**; timing within the cycle goes to her physician, offered honestly as **limited evidence**. Warfarin under **SDCEP: do not interrupt, check the INR within 24 hours (72 if stable), proceed below 4.0** with local measures — and at **4.2, postpone and refer back**, never adjust it yourself. Haemostasis in order, ending in **twenty minutes' pressure** and tranexamic mouthwash. Analgesia: **paracetamol and codeine, no NSAIDs** — twice over, for methotrexate and for warfarin — said explicitly, including over the counter. **Single-dose amoxicillin 3 g**, justified by immunosuppression rather than by implant surgery; **metronidazole and miconazole avoided**. Surgically: **minimal flap, primary closure** — exposed bone is the thing being prevented.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: watching for what can go wrong	The three windows: **bleeding at 48 hours, infection in the first week, and a review at eight weeks for delayed healing** — because eight weeks is where non-healing becomes MRONJ. The day-two bleed managed by **twenty minutes of pressure without checking**, then liver clot, re-suture, haemostatic agent, INR, and a question about ibuprofen. Then: **4 mm of exposed bone at ten weeks is MRONJ by definition** — radiograph, **do not debride**, chlorhexidine 0.2%, denture off the area, **refer to OMFS**, and inform her rheumatologist. **Staging 0 to 3.** Mucositis at six months read as **plaque control having failed** — reversible, so treated now, with **adapted aids and whoever helps her at home**. **Three-monthly recall.**
14:30–15:00	Closes on the patient and the record	What you say to her: the injection switches off the cells that break bone down; **very occasionally the jaw does not heal after surgery**; her risk is higher than average **because of the steroid tablets**; that does not mean no implants, but the denture is optimised first and a specialist is involved; and — plainly — **"I would not want you to stop your injection."** Recorded: every drug with **dose, indication and duration**, the risk category **and the reasoning for it**, the written liaison and reply, and whose decision it was.

The spine — the twelve answers to have ready if nothing else

1 How denosumab works · 2 How it differs from a bisphosphonate, and why that matters · 3 MRONJ defined, all three limbs · 4 Her risk category, and why the steroid moves it · 5 Methotrexate, and the drug you must not prescribe · 6 Warfarin, the factors, and what an INR is · 7 The features of rheumatoid arthritis that change your appointment · 8 How you assess her dexterity · 9 Why the conventional remake comes first · 10 The INR rules, and what you do at 4.2 · 11 Studs rather than a bar, and why · 12 Exposed bone at ten weeks — diagnosis, and what you do not do.

Where the marks are lost in this case

Reading "denosumab" and stopping there. The mark is in the **drug, the indication and the route together**: 60 mg six-monthly for osteoporosis is the lower-risk indication, and the same molecule at 120 mg monthly for skeletal metastases is a case you **decline outright**. Say which one she is on, say what would change if she were on the other, and say how long she has been taking it.

Missing the escalation the case is built on. Under SDCEP she is not lower risk, because she takes **prednisolone 5 mg daily** — a concurrent systemic glucocorticoid moves an osteoporosis patient into the **higher-risk group**. Candidates recite the reassuring osteoporosis figures and never notice that this patient does not qualify for them. Volunteering that unprompted is the difference between a 3 and a 5.

Planning the surgery and forgetting the hands. Rheumatoid dexterity is not a footnote — it decides the attachment choice, the recall interval and arguably whether to treat at all. And the competence line must be said: two anterior mandibular implants would normally be Straightforward, but **higher-risk MRONJ status plus immunosuppression plus anticoagulation plus impaired plaque control is not work to undertake single-handed** — refer for a joint opinion, making clear to her that **referral is not refusal**.

SOE Case 12 — The early failure

THE CASE

You placed a single implant at the **UR5** eight weeks ago in a healthy 52-year-old. At the review the patient reports mild discomfort. The healing abutment is loose and the implant rotates with light finger pressure. Radiograph shows a **peri-implant radiolucency**.

Domain 1

"This is an **early implant failure** — failure to achieve or maintain osseointegration before loading. The implant is mobile and there is a peri-implant radiolucency, so the diagnosis is not in doubt. My tasks are to explain it to the patient, remove the implant, understand why it happened, and decide what to do next."

Q What are the causes of early failure?

"Grouped: **surgical** — inadequate primary stability, overheating during preparation, contamination, excessive insertion torque causing compression necrosis, or placement into an infected site; **patient** — smoking, uncontrolled diabetes, poor bone quality, previous irradiation; **loading** — micromotion from a denture pressing on the site, or premature loading; and **site** — residual infection or placement into non-osseous tissue such as the incisive canal. I would go back through my own operative record: the bone quality recorded, insertion torque, drill sequence and irrigation, whether the denture was relieved, and whether the patient smoked or was unwell."

Domain 2

Q: What do you tell the patient?

"**Duty of candour** applies. Promptly, honestly and without excessive self-flagellation: 'I'm sorry — the implant hasn't attached to the bone as it should have. That happens in a small percentage of cases, around two to five per cent, even when everything is done correctly. It's not dangerous, but the implant has to come out. Once it's out and the area has healed for a couple of months, we can look at putting another one in, and I would expect that to work — though I have to be honest that the chance of a second failure at the same site is a bit higher than the first time. There will be no further charge from me for the replacement implant.' I would then set out the alternatives again, in case she now prefers a bridge or a denture, and record the conversation."

Q Would you charge?

"That is a practice policy and a professional judgement rather than a regulatory rule, but I would have set it out in writing **before** treatment. My own position would be to replace the implant without charging for the fixture and the surgery, while being clear about laboratory and component costs. What matters for the examination is that the arrangement was **agreed and documented in advance**, so that this conversation is not a negotiation at a difficult moment."

Domain 3

Q: Describe removal and re-treatment.

"Removal is straightforward with a mobile implant — often by hand or with a reverse-torque device. Then thorough **curettage of all granulation tissue**, irrigation, and inspection of the bony walls. If there is a contained defect I would graft with a xenograft and a collagen membrane and close primarily; if the walls are intact I might allow spontaneous healing. An interim prosthesis — a bonded pontic or an acrylic partial denture generously relieved. Re-entry at **2 to 3 months**, or 4 to 6 months if I grafted. At re-placement I would use a **wider and/or longer implant** to engage fresh bone, take particular care with irrigation and drill sharpness, and consider a **two-stage submerged** protocol with an extended healing period."

Domain 4

Q: What would you do to prevent recurrence in your practice?

> "Treat it as a **significant event**. Review the operative record against my protocol; check drill usage counts and replacement intervals; check irrigation flow; review my primary stability threshold; review my denture-relief protocol. Then **audit**: my early failure rate over the last two years against a standard — the literature suggests early failure of the order of 2 to 5% — with a re-audit after any change. That would be a legitimate quality improvement project for the entrance requirements."

Running this case to time

Time check. This case is written in the older format — **five long answers, about 563 words**. At 140 words a minute plus five seconds a question that is barely **four and a half minutes**: it does not overrun, it leaves **ten minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and each is really **three or four short questions** in the mouth of an examiner. Expect the causes of early failure alone to be four. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the audit — the easiest mark in the case to earn and the easiest to miss.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiograph in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Name it in the first clause: an **early implant failure — failure to achieve or maintain osseointegration before loading**. Then the evidence, briefly: a loose healing abutment, an implant that **rotates on light finger pressure**, and a **peri-implant radiolucency** at eight weeks, so the diagnosis is not in doubt. Then the four tasks that structure everything that follows — **explain it to the patient, remove the implant, understand why it happened, and decide what to do next** — in that order, because the order is itself the answer.
1:20–4:30	Domain 1: the biology, then the causes, drilled one group at a time	Expect this to open **below the implant material**: osseointegration, how bone heals round a fixture, and **primary versus secondary stability with the dip between them** — the exact window this implant has fallen into at eight weeks. Then the causes, delivered as **four named groups, not a list**: **surgical** — poor primary stability, overheating, contamination, excessive torque causing compression necrosis, an infected site; **patient** — smoking, uncontrolled diabetes, poor bone quality, irradiation; **loading** — micromotion from an unrelieved denture; **site** — residual infection or non-osseous tissue. Then the move that earns the domain: **go back through your own operative record**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: telling her, and the money	**Duty of candour**, named as a duty: promptly, honestly, **without excessive self-flagellation**. The words themselves — the implant has not attached; this happens in **around two to five per cent** of cases even when everything is done correctly; it must come out; a replacement after a couple of months; and the honest qualifier that **a second failure at the same site is more likely than the first**. Then the alternatives **re-offered**, because she may now prefer a bridge or a denture. On charging: a **practice policy set out in writing before treatment**, not a regulatory rule — the mark is that it was **agreed and documented in advance**.
7:30–11:30	Domain 3: removal, the site, and re-treatment	Removal is easy in a mobile implant — **by hand or reverse-torque**. Then the part candidates skip: **thorough curettage of all granulation tissue**, irrigation, and **inspection of the bony walls**, because the walls decide what happens next. A **contained defect** takes a xenograft and a collagen membrane; **intact walls** can heal spontaneously. An interim prosthesis **generously relieved**, since an unrelieved denture is one of the causes you just listed. Re-entry at **two to three months, or four to six if grafted**; a **wider or longer implant into fresh bone**, and a **two-stage submerged protocol**.
11:30–14:30	Domain 4: the significant event and the audit	Treat it as a **significant event**, and say the words. Then the systems review, itemised: the operative record against protocol, **drill usage counts and replacement intervals**, irrigation flow, the **primary stability threshold** you work to, and the denture-relief protocol. Then the audit proper — **your own early failure rate over two years against a standard of roughly 2 to 5%**, with a **re-audit after any change**, offered as a quality improvement project. This is the domain where a prepared candidate simply outperforms an unprepared one.
14:30–15:00	Closes on the patient and the record	What you say: *"I'm sorry — the implant hasn't attached to the bone as it should have… it has to come out… once the area has healed we can look at putting another one in, and I'd expect that to work, though the chance of a second failure at the same site is a bit higher."* What you write: the candour conversation and its date, the words used about the failure rate and the higher second-attempt risk, the alternatives re-offered, the charging arrangement **as previously agreed**, and the significant event logged.

The spine — the twelve answers to have ready if nothing else

1 Early failure defined, and the evidence for it here · 2 Osseointegration, and the primary-to-secondary stability dip · 3 The surgical causes · 4 The patient causes · 5 The loading causes, including the unrelieved denture · 6 The site causes · 7 What you look for in your own operative record · 8 Duty of candour, and the actual words · 9 The two-to-five per cent figure, and the higher second-attempt risk · 10 Charging, agreed and documented in advance · 11 Removal, curettage, and what the bony walls decide · 12 The significant event, the systems review and the audit standard.

Where the marks are lost in this case

Removing the implant and moving on. The examiner is not testing whether you can unscrew a loose fixture — that part is trivial and takes one sentence. The case is about **why it happened and what you change**, and a candidate who spends three minutes on removal technique has spent them in the cheapest domain in the case.

Listing causes instead of grouping them. Say **surgical, patient, loading, site** and then populate each. Then volunteer the thing candidates almost never volunteer: that you would **audit your own operative record first** — bone quality, torque, irrigation, denture relief — before reaching for the patient's smoking. Looking at your own technique before the patient's habits is the professional instinct being marked.

Treating candour as an apology rather than a duty. It is a regulatory duty, discharged promptly, honestly and in writing, with the failure rate quoted and the **higher risk of a second failure** stated rather than hidden. And the line to name is when to stop: after a repeated failure at the same site the answer is not a third fixture but the **bridge or denture, re-offered honestly** — and a defect at re-entry that is not contained goes on for an opinion.

SOE Case 13 — Repeated mechanical failure in a bruxist

THE CASE

44-year-old man with implant crowns at the **36 and 46**, both fitted 2 years ago. He has returned three times with **screw loosening at the 36** and has now **fractured the porcelain** on the 46. **Masseteric hypertrophy**, **linea alba**, marked wear facets on the anterior teeth. He works long hours and reports high stress.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 44-year-old man with recurrent mechanical complications at two posterior implants, two years after fitting, in the presence of clear clinical signs of parafunction — masseteric hypertrophy, linea alba and anterior wear facets. Three episodes of screw loosening at the 36 and now veneer fracture at the 46. I would not read repeated loosening as a nuisance to be re-tightened; it tells me the **screw joint is being overloaded or is not passively fitting**, and untreated it progresses to **screw fracture and ultimately implant fracture**. So my priorities are to establish why the joint is failing — occlusion, fit, component handling or all three — to confirm the implants themselves are osseointegrated and the bone levels stable, and to correct the cause before I repair the effect."

Domain 1 — Clinical interpretation and reasoning

Chain: the groundwork — parafunction and the periodontal ligament

Q Define bruxism.

"**Repetitive masticatory muscle activity** characterised by clenching or grinding of the teeth and by bracing or thrusting of the mandible. The international consensus separates **sleep bruxism** from **awake bruxism**, because they have different aetiologies. In an otherwise healthy person it is a **behaviour rather than a disorder** — but it becomes a risk factor when it damages teeth or restorations, which it clearly has here."

Q How certain is your diagnosis?

"Graded. **Possible** bruxism is self-report alone. **Probable** is self-report plus positive clinical findings — which is where this man sits, with hypertrophy, linea alba and wear facets. **Definite** requires instrumental confirmation, portable electromyography or polysomnography. I would call this probable sleep bruxism with an awake clenching component, and I would say so in the notes rather than overclaim."

Q Describe the periodontal ligament.

"A **collagenous connective tissue** about **0.2 mm wide** between cementum and alveolar bone, with organised principal fibre groups — alveolar crest, horizontal, oblique, apical and interradicular — a rich vascular plexus, and **mechanoreceptors** supplying proprioceptive feedback. It transmits and dissipates occlusal load, allows physiological tooth movement, and remodels in response to force."

Q How does a tooth move under load compared with an implant?

"A tooth moves **25 to 100 micrometres** in two phases — an initial rapid phase within the ligament, then a slower phase of bone deformation. An osseointegrated implant has no ligament, so it moves only by **elastic deformation of bone, some 3 to 5 micrometres**, and only in one phase. Its centre of rotation sits apically, not in the middle third."

Q What follows clinically from that difference?

"Three things. The implant has **no shock absorption**, so load goes straight into the components and the crestal bone. It has **no ligament proprioception**, so the protective reflex is blunted and tactile thresholds are far higher

— he cannot feel that he is overloading it. And connecting an implant rigidly to a natural tooth risks **intrusion of the tooth**, which is why I avoid it."

Q Define osseointegration.

"A **direct structural and functional connection between ordered living bone and the surface of a load-carrying implant**, in Brånemark's terms; clinically, as Albrektsson set out, an implant that is immobile, symptomless, without peri-implant radiolucency and with bone loss within accepted limits. The important word for this case is **load-carrying** — integration is maintained or lost according to how it is loaded."

Q What is preload, and why does it matter here?

"Tightening the screw stretches it elastically; that stored tension is the **preload**, and it clamps the abutment to the implant. The joint stays together while the **separating forces are less than the preload**. Loosening happens when preload is never achieved, when it is lost to **embedment relaxation** as microscopic surface roughness settles, or when eccentric load and bending exceed it — all three are plausible in this man."

Chain: reading this patient

Q What would you assess in the occlusion?

"Intercuspal contacts with **shimstock and two thicknesses of articulating paper**, checking whether the implant crowns hold more heavily than the adjacent teeth. Then all excursions — working, non-working and protrusive — looking specifically for **non-working interferences** on the implants. And the crown form: **occlusal table width, cusp inclines and any cantilever** relative to the implant platform."

Q Why should an implant crown be lighter in contact than the neighbouring teeth?

"Because the natural teeth intrude within their ligaments under firm closure — of the order of **28 micrometres** — while the implant does not. So a contact that is even under light tapping becomes a **premature contact under clenching**. I aim for the implant to hold shimstock only on firm clenching, and to be free in all excursions with the canines providing guidance."

Q What else would you examine, and what radiographs?

"Muscles of mastication and TMJ, tongue scalloping, tooth wear and its distribution, and whether the abutments are fully seated. History of sleep quality, morning jaw ache or headache, stress, caffeine, alcohol, recreational drugs, SSRIs. Radiographically, **parallel-technique periapicals** of both implants to check bone levels against baseline, seating of the abutment, and any radiolucent line suggesting a fractured screw or implant body."

Domain 2 — Clinical decision making and treatment planning

Chain: what is actually causing the failure

Q List the causes of repeated screw loosening.

"**Inadequate initial torque** or no re-torque; a **re-used or damaged screw**; a **non-passive or misfitting** abutment-crown assembly; **excessive occlusal load**, including cantilevers, a wide occlusal table and steep cusps; **eccentric loading** off the implant axis; **a joint design with poor anti-rotation**, such as an external hex; and **wrong or mismatched components**. Bone loss lengthening the lever arm compounds all of them."

Q Is bruxism a contraindication to implants?

"Not an absolute one. The evidence links it convincingly to **mechanical and technical complications** — screw loosening, veneer chipping, component fracture — rather than to loss of osseointegration, where the association is much weaker. So it changes the **design, the material, the occlusal scheme, the consent and the maintenance interval**, rather than the decision to treat."

Q Classify this case on SAC.

"The original treatment — two single posterior implant crowns in a healed site — would be **Straightforward to Advanced**. Managing recurrent mechanical failure in a diagnosed bruxist is **Advanced**, because of the parafunction modifier and the need for occlusal and prosthodontic re-planning. It is still within my scope."

Q What would take it beyond your competence?

"A **fractured implant body** needing explantation, a **fractured screw I could not retrieve**, evidence that the arch needs full occlusal reorganisation at an altered vertical dimension, or a co-existing painful temporomandibular disorder that was not settling. Those I would refer — to a specialist in restorative dentistry or oral surgery as appropriate — and I would say so at the outset rather than after a failed attempt."

Q Give me your plan in order.

"**Correct the cause before repairing the effect.** First, occlusal correction and a definitive parafunction diagnosis. Second, disassemble, inspect and **replace both screws with new ones** at correct torque. Third, remake the fractured 46 restoration in a more fracture-tolerant material. Fourth, a **hard full-coverage occlusal splint** for night wear. Fifth, address the contributing factors — stress, sleep, and screening for sleep-disordered breathing."

Q What material for the new 46, and why?

"**Monolithic zirconia**, screw-retained. The commonest technical failure of implant crowns is **chipping of layered veneering ceramic**, and removing the veneer removes that failure mode. It is strong in thin section, which suits his limited clearance, and screw retention keeps it retrievable so I can service the joint without destroying the crown."

Q Argue against monolithic zirconia.

"It is **unrepairable in the mouth** — a fracture means a remake. Its **wear behaviour against natural enamel** depends heavily on polish, and a rough or adjusted surface will abrade his opposing teeth, which already show facets. It is **stiff and unforgiving**, so it transmits rather than absorbs load. And it hides the very early cracks that a layered ceramic would show as chips. I mitigate by polishing every adjusted surface and reviewing the antagonists."

Q A colleague would provide a metal occlusal surface instead. Why might they be right?

"Because **cast metal is the most forgiving occlusal material we have** — it does not chip, it wears at a rate close to enamel, it is easy to adjust and polish, and in a heavy bruxist in a non-visible molar site aesthetics cost him nothing. In a patient who has already fractured one ceramic, that is a genuinely defensible choice, and I would offer it as an option."

Q Would you splint the two implants together?

"Not here — they are in different quadrants, so splinting is not even possible. Splinting adjacent implants does share load and reduce screw loosening, but it **compromises cleansability and complicates retrievability**, so I reserve it for short implants, poor bone quality or a cantilevered segment. In this case the better load control comes from the occlusion and the splint."

Domain 3 — Decision making in relation to treatment procedures

Chain: servicing the screw joint

Q Take me through disassembly at the 36.

"Confirm which system and screwdriver, isolate with a **throat pack or floss ligature** so nothing is swallowed or inhaled, remove the access filling, clean the screw head thoroughly so the driver seats fully, and unscrew. Then inspect the **abutment, the screw threads and the implant connection** under magnification for wear or deformation, and check the fit surface for a marginal discrepancy."

Q Why must the screw be replaced rather than re-tightened?

"Because it has already been stretched into its plastic range and its threads and head have deformed and abraded. A **fatigued screw cannot generate or hold full preload**, so it will loosen again sooner. A new screw is cheap; a fractured one retained in the implant is not."

Q How exactly do you torque it?

"With a **calibrated torque wrench** to the manufacturer's stated value for that screw and connection — commonly 30 to 35 Ncm for an abutment screw, but system-specific. Then **re-torque after about ten minutes** to compensate for embedment relaxation. Never torque with a hand driver, and never guess."

Q The screw fractures in the implant. What now?

"Stay calm, isolate, and magnify. Try to **rotate the fragment out with an ultrasonic tip or a fine probe** running anticlockwise, sometimes after a drop of lubricant. If that fails, use the manufacturer's **screw retrieval kit** with its guide and reverse-cutting drill. Avoid anything that damages the internal threads. If the threads are destroyed the implant may become unrestorable — which is when I refer."

Q Screw-retained or cement-retained for the new crown here — argue it.

"**Screw-retained.** In a patient who needs repeated access to the joint, retrievability is worth more than anything cementation offers, and it removes any risk of **residual subgingival cement**, which is a recognised trigger for peri-implant disease. Cement retention only wins where the screw channel would emerge through a cusp or a facial surface, or where interocclusal space is very limited — neither applies at the 46."

Q The access channel emerges through the central fossa. Any concerns?

"It removes occlusal material at the point of heaviest loading, so I keep the channel as **narrow as the driver allows**, seal it with **PTFE tape and composite**, and ensure the occlusal contacts sit on sound ceramic away from the channel margin. An angulated screw channel is an alternative if the emergence would otherwise be unacceptable."

Q Describe the splint you would make.

"A **hard heat-cured acrylic full-coverage splint**, usually maxillary, about 1 to 1.5 mm thick at the posteriors, made on articulated casts with a facebow and a centric record at a comfortable vertical dimension. Full coverage is non-negotiable — a partial-coverage appliance permits over-eruption of the uncovered teeth."

Q How do you adjust it at fit and afterwards?

"Check it seats fully and retains without rocking. Adjust to **even, simultaneous point contacts** on all opposing cusp tips in the retruded position, with **shallow canine ramps** giving immediate posterior disclusion in excursion and no working or non-working interferences. Then review it at intervals — the **pattern of shiny wear facets on the acrylic is my objective measure** of how much he is grinding."

Q A colleague would give him a soft vacuum-formed splint. Respond.

"It is quicker and cheaper and he would tolerate it, and for short-term protection that has some value. But a soft splint is **not adjustable to a defined occlusal scheme**, it wears rapidly, and there is a reasonable body of opinion that a resilient surface can **increase muscle activity** in some bruxists — the 'chewing gum' effect. For a diagnosed bruxist with a fractured restoration I want a hard, adjustable appliance."

Q What would you record?

"The parafunction diagnosis and its certainty level. Baseline and current **radiographs and bone levels**. The **component batch and the torque value with date** for every screw placed. Occlusal findings before and after adjustment, with photographs. The splint design, its adjustment and its wear at each review. And the consent discussion — specifically that mechanical complications are likely to recur and carry a cost."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: when it goes wrong again

Q What do you tell him, in his words?

> " 'The implants themselves are healthy. The problem is the force you put through them at night — it is well beyond what a normal crown is built for. I can make the crowns much tougher and even up your bite, but I can't switch off the grinding, so the night guard isn't optional. You should expect parts to need replacing over the years, and I'd rather you planned for that cost than were surprised by it.' And I would put that in writing."

Q Two years on, the screw loosens again despite all of this. Differential?

> "In order: **occlusal load not adequately controlled** — check the splint is being worn and the contacts have not drifted; a **misfitting or non-passive component**; a **damaged internal connection** from the earlier episodes; **progressive crestal bone loss** lengthening the lever arm; a **cracked screw** already partly failed; or the wrong component from a previous repair. I would radiograph, disassemble and inspect rather than simply re-tighten a fourth time."

Q When would you worry the implant body itself will fracture?

> "Implant fracture is rare — of the order of **0.2 to 1%** — but the risk rises with a **narrow-diameter implant under heavy load**, with **progressive bone loss increasing the lever arm**, and with **repeated untreated screw loosening**, because a loose joint transfers bending stress into the implant wall. Screw fracture is often the warning shot before implant fracture."

Q It has fractured. How do you manage it?

> "Explain honestly what has happened and that it is not restorable. A fractured implant almost always has to be **removed — by reverse-torque device or trephine** — or, if removal would be destructive, the apical portion **left submerged as a sleeper** with the coronal part smoothed below the soft tissue. Then reassess whether to replace it, immediately or after healing, and whether the same design would fail again. Removal in a mandibular molar site near the canal is where I would refer."

Q There is 2 mm of bone loss at the 36. Is that overload?

> "I would not assume so. The dominant cause of progressive peri-implant bone loss is **biofilm-associated inflammation**, and I would first probe for bleeding and suppuration, check for residual cement and assess plaque control. **Occlusal overload alone, in the absence of inflammation, is not a proven cause of bone loss**, though it plausibly accelerates it once inflammation is present. So I treat the inflammation and correct the occlusion — both, not one."

Q How and how often would you review him?

> "**Six-monthly**, given his risk. At each visit: peri-implant probing with bleeding and suppuration, plaque control, occlusal check with shimstock, screw and crown integrity, examination of the splint for wear and fit, and the antagonist teeth. Radiographs at one year after any remake and then as clinically indicated, always compared with baseline."

Q How would you know you had succeeded?

> "No further loosening or ceramic fracture over successive reviews; **stable bone levels and healthy peri-implant tissues**; a splint that is worn, wearing evenly and being replaced on schedule; and the man himself reporting that his jaw is more comfortable in the morning. If loosening recurs despite all of it, the honest conclusion is that the **design is wrong for the forces**, not that he needs another screw."

WATCH OUT

Go back to the screw joint: preload against separating force. Ask yourself whether the preload was ever achieved, whether it has been lost, or whether the separating force is too great — one of those three is nearly always the answer in a mechanical failure question.

Running this case to time

Time check. The flow above holds 37 questions and about 2,352 words of model answer. At 140 words a minute plus roughly five seconds a question, that is close to **twenty minutes** — an overrun of about **five**. What goes first is the **splint construction** material in Domain 3: the thickness, the facebow, the adjustment and the soft-splint argument compress into one question, because the marks in this case sit in the **screw joint** and in Domain 4's differential when it loosens again. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the implant fractures.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapicals up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 44-year-old with recurrent mechanical complications at two posterior implants two years after fitting, against **clear signs of parafunction — masseteric hypertrophy, linea alba, anterior wear facets**. Then the interpretation, which is the whole opening: **repeated loosening is not a nuisance to be re-tightened** — it says the screw joint is **overloaded or not passively fitting**, and untreated it progresses to **screw fracture and then implant fracture**. Priorities in order: establish why the joint is failing, confirm the implants are integrated with stable bone, and **correct the cause before repairing the effect**.
1:20–4:30	Domain 1: the groundwork — parafunction and the periodontal ligament — then reading this patient	Undergraduate ground first. **Bruxism defined** as repetitive masticatory muscle activity, sleep and awake separated, a **behaviour rather than a disorder** until it damages something — then graded **possible, probable, definite**, with this man at **probable**, said out loud rather than overclaimed. The **periodontal ligament** at 0.2 mm with its **mechanoreceptors**; a tooth moving **25–100 μm in two phases** against an implant's **3–5 μm in one**; and the three consequences — **no shock absorption, no proprioception, no rigid tooth-to-implant connection**. **Preload**, and its three failure modes: never achieved, lost to **embedment relaxation**, or exceeded. Then **shimstock and two paper thicknesses**, hunting **non-working interferences**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: what is actually causing the failure	The causes listed as a **structured differential** — inadequate torque, a re-used or damaged screw, a **non-passive fit**, excessive or eccentric load, poor anti-rotation, mismatched components — with bone loss lengthening the lever arm compounding all of them. Then the judgement: **bruxism is not a contraindication**, because the evidence links it to **technical complications, not to loss of osseointegration** — so it changes design, material, occlusal scheme, consent and recall, not the decision to treat. **SAC Advanced.** The plan opening with **correct the cause before repairing the effect**. **Monolithic zirconia** argued for — **veneer chipping is the commonest technical failure, so remove the veneer** — then against: unrepairable, wear against enamel, and it hides early cracks. The colleague's **cast metal occlusal surface** granted its strength.
7:30–11:30	Domain 3: servicing the screw joint	Disassembly with the airway secured — **throat pack or floss ligature** — then **inspection under magnification** of abutment, threads and connection. Why the screw is **replaced, not re-tightened**: it has been stretched into its plastic range and **cannot generate or hold full preload**. **A calibrated torque wrench to the manufacturer's value, commonly 30–35 Ncm, then re-torqued at ten minutes** for embedment relaxation — never a hand driver, never a guess. A fractured screw retrieved by **ultrasonic tip anticlockwise**, then the manufacturer's **retrieval kit**, protecting the internal threads. **Screw-retained** argued on retrievability and **no subgingival cement**. Then the splint: **hard, full-coverage, even point contacts, shallow canine ramps** — and its wear facets as **your objective measure of how much he grinds**.
11:30–14:30	Domain 4: when it goes wrong again	The loosening at two years given a **differential in order**: occlusal load uncontrolled, non-passive fit, a damaged connection, bone loss lengthening the lever arm, an already-cracked screw, a wrong component — **radiograph, disassemble and inspect rather than tighten a fourth time**. Implant fracture at **0.2 to 1%**, rising with narrow diameter, bone loss and **repeated untreated loosening — screw fracture is the warning shot**. Managed by **reverse-torque or trephine**, or left as a **sleeper**. Then the trap: 2 mm of bone loss at the 36 is **not assumed to be overload** — probe and check for cement first, because **overload alone is not a proven cause of bone loss** without inflammation. **Six-monthly review**, splint included.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	In his words: *"The implants themselves are healthy. The problem is the force you put through them at night… I can make the crowns tougher and even up your bite, but I can't switch off the grinding, so the night guard isn't optional. Expect parts to need replacing — I'd rather you planned for that cost than were surprised by it."* Recorded: the parafunction diagnosis **with its certainty level**, bone levels against baseline, the **component batch and torque value with date for every screw**, occlusal findings before and after, and consent that **complications are likely to recur and carry a cost**.

The spine — the twelve answers to have ready if nothing else

1 Bruxism defined, and the three certainty grades · 2 The periodontal ligament, and tooth versus implant movement under load · 3 The three clinical consequences of having no ligament · 4 Preload, and the three ways a joint loses it · 5 Why an implant crown is held lighter than its neighbours · 6 The causes of repeated screw loosening · 7 Whether bruxism contraindicates implants · 8 Monolithic zirconia argued both ways, and the metal alternative · 9 Disassembly, and why the screw is replaced not re-tightened · 10 Torque value, torque wrench, and the ten-minute re-torque · 11 The fractured screw, and when the implant becomes unrestorable · 12 Two millimetres of bone loss — inflammation before overload.

Where the marks are lost in this case

Re-tightening the screw a fourth time. The examiner has given you three episodes of loosening precisely to see whether you treat the symptom. Say in the opening that repeated loosening means the **joint is overloaded or not passively fitting**, that it progresses to screw then implant fracture, and that you **correct the cause before you repair the effect**. Candidates who reach for a torque wrench first have lost Domain 1 in the first thirty seconds.

Blaming the bruxism for the bone loss. When the 2 mm at the 36 appears, the volunteered answer is that **progressive peri-implant bone loss is biofilm-driven**, and that occlusal overload alone is not a proven cause in the absence of inflammation — so you probe, check for residual cement, and treat both the inflammation and the occlusion. Reaching straight for "he grinds" is the trap the question is set to catch.

Never naming what you would not attempt. Managing recurrent mechanical failure in a bruxist is **Advanced and yours**. What is not: a **fractured implant body needing explantation**, a **fractured screw you cannot retrieve**, an arch needing **full occlusal reorganisation at an altered vertical dimension**, or a **painful temporomandibular disorder that is not settling** — those go to a specialist in restorative dentistry or oral surgery. Say it at the outset, not after a failed attempt.

SOE Case 14 — Immediate placement in an intact anterior socket

THE CASE

35-year-old woman, non-smoker, **low lip line**. The **11** has a **vertical root fracture** below a post crown and is unrestorable. CBCT shows **intact socket walls**, a **facial wall 1.2 mm thick**, **6 mm of bone apical and palatal** to the socket apex, the socket lying **within the alveolar envelope**, and **no infection**. **Thick phenotype**, no recession on the adjacent teeth.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A healthy 35-year-old non-smoker with an unrestorable upper central incisor — a **vertical root fracture** beneath a post crown, which is not repairable and means the tooth must come out. What makes this case worth discussing is that the site is anatomically about as favourable as an anterior socket gets: the walls are intact, the facial wall is **1.2 mm**, there is 6 mm of bone apical and palatal for stability, the socket sits within the bone envelope, and there is no infection. She has a **thick phenotype**, a **low lip line** and no recession on the neighbours, so her aesthetic risk is low. My working position is that this satisfies the criteria for **immediate placement**, but I need to establish what she wants, what she can accept if the contour changes, and whether the case sits within my own competence."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical and biological groundwork

Q Describe the anatomy of an anterior maxillary extraction socket.

"Four bony walls lined by **bundle bone** — the alveolar bone proper, which is the cribriform plate the periodontal fibres insert into. Buccally it is often the *whole* thickness of the wall; palatally it is backed by thick cancellous and cortical bone. The socket contains the periodontal ligament, its blood supply and the clot. Above the crest sits the supracrestal attachment, and the **incisive canal** lies palatal and medial."

Q What is bundle bone, and why does it matter here?

"It is **tooth-dependent bone**. Its blood supply and its reason for existing are the periodontal ligament, so when the tooth is removed the ligament goes and the bundle bone resorbs — regardless of how atraumatic you were. Since the facial wall in the anterior maxilla is largely bundle bone, that is why the buccal contour collapses. **Immediate implant placement does not prevent this**, which is the single most important thing to say about this case."

Q What happens dimensionally if you extract and do nothing?

"Around **50% of the ridge width is lost in the first year**, and roughly two-thirds of that in the **first three months**. Systematic review figures put horizontal loss at about **3.5 to 4 mm** and vertical loss at about **1 to 1.5 mm** at six months, and the loss is **predominantly buccal**, so the ridge migrates palatally as well as shrinking."

Q Where does the buccal plate get its blood supply from?

"Three sources — the **periodontal ligament**, which is lost with the tooth; the **periosteum** overlying it; and the intra-osseous supply, which in a 1 mm wall is negligible. So once the ligament has gone, the periosteum is essentially all that is left. That is the argument for a **flapless** approach: raising a flap strips the one remaining blood supply from a wall that has just lost its other one."

Q How thick is the facial wall in the anterior maxilla usually?

"**Thin.** In the large majority of anterior maxillary sites it is under **1 mm**, and in around half of them under 0.5 mm. That is precisely why **1.2 mm here is favourable** — she is in the minority, and a wall of that thickness has some chance of surviving as a wall rather than resorbing entirely."

Q How would you assess her periodontal phenotype chairside?

"The simplest reproducible method is to place a **periodontal probe in the sulcus and see whether it shows through the tissue** — if it is visible, the phenotype is thin, under about 1 mm. I would combine that with the gingival appearance and the crown form: thick phenotype tends to give square crowns, flat scallop and a broad band of keratinised tissue, thin gives triangular crowns and high scallop."

Q Why does her lip line matter so much?

"Because it determines **what she will actually see** if the outcome is imperfect. A low lip line means the mid-facial gingival margin and the transition line are hidden at maximum smile, so recession or a visible crown margin is far less consequential. It is one of the strongest modifiers in any aesthetic risk assessment, and it converts this from a high-stakes case into a manageable one."

Chain: reading this patient

Q How would you have diagnosed the vertical root fracture?

"Clinically: a **narrow, isolated, deep probing defect** on an otherwise healthy tooth, often a **sinus tract at or near the gingival margin**, tenderness to bite, and a post-crowned root — post-retained teeth are the classic setting. Radiographically a **J-shaped or halo radiolucency**. CBCT can show a fracture line but the post generates beam-hardening artefact, so a negative scan does not exclude one; **direct vision after flap or on removal** is definitive."

Q What does the CBCT add here, and what are its limits?

"It gives me the facial wall thickness, the apical and palatal bone available for stability, the position of the incisive canal, and whether the socket sits within the envelope — none of which I can get from a periapical. Its limits are **artefact from the post and core**, that it does **not show soft tissue thickness**, and that it must be justified individually under **IR(ME)R**."

Q Give me a formal aesthetic risk assessment on her.

"Using the **ITI ERA** headings: medical status healthy, **non-smoker**, expectations to be established, **low lip line**, **thick phenotype**, tooth shape to be assessed, **no infection**, bone levels on the adjacent teeth normal, adjacent teeth unrestored, a **single-tooth span**, and intact soft tissue anatomy. Almost every parameter is low risk. The only unknown is her expectation, and that is the one I have to explore before I promise anything."

Q What is the SAC classification, and is this within your competence?

"A single implant in the aesthetic zone is **Advanced** at best; **immediate placement with immediate restoration in the aesthetic zone is Complex** on both the surgical and the restorative axis. So the honest answer is that I would **complete the assessment, the wax-up and the options conversation myself and refer the surgical phase**. What I must be able to do at my level is recognise why this site is favourable, understand the protocol well enough to consent her properly, and manage her afterwards."

Domain 2 — Clinical decision making and treatment planning

Chain: the options, and the timing decision

Q What are her options for replacing the 11?

"**No treatment**, accepting a space — rarely acceptable at a central incisor. A **removable partial denture**, which replaces soft tissue as well but is removable. A **resin-bonded bridge** from the 21, minimally invasive and reversible. A **conventional bridge**, which I would not do on unrestored abutments. Or an **implant-retained crown**. Orthodontic space closure is not realistic for a central incisor here."

Q You have chosen an implant. Argue against yourself — why might a resin-bonded bridge be better?

"Because it is **cheap, quick, reversible and does not depend on a healing ridge**. She is 35 with an intact adjacent tooth and thick tissue, so the abutment preparation is minimal or nil and survival at ten years is respectable. It avoids surgery, avoids the lifelong mucositis and peri-implantitis risk, and if it debonds I have lost nothing. If she is at all uncertain about surgery, it is the better first answer — and it does not close the implant door."

Q Classify the timing options for implant placement after extraction.

"**Type 1 — immediate**, into the fresh socket at the time of extraction. **Type 2 — early with soft tissue healing**, at about **4 to 8 weeks**. **Type 3 — early with partial bone healing**, at about **12 to 16 weeks**. **Type 4 — late**, at **six months or more**, into a healed ridge. Loading is classified separately as **immediate, early or conventional**."

Q Take me through the pre-operative criteria for immediate placement.

"**Patient factors** — thick soft tissue phenotype, no recession on the site or the neighbours, and low aesthetic risk. **Radiographic factors on CBCT** — an intact socket with a **facial wall of at least 1 mm**; enough bone **apical and palatal** to achieve primary stability; the socket lying **within the bone envelope**; a horizontal gap of at least **2 mm** anticipated at the implant shoulder; and **no acute infection**. She meets all of them."

Q And the intra-operative criteria?

"Three, checked after the tooth is out and not before. **The socket wall is confirmed intact** on probing. **Primary stability is achievable** in the apical and palatal bone. And **the ideal three-dimensional implant position is achievable**. If any one of those fails, I abandon immediate placement on the table and convert to early placement or ridge preservation — that decision must be made in advance and consented for in advance."

Q What are the loading criteria?

"For an immediate restoration I want an insertion torque above about **35 Ncm** or an **ISQ above 70**, **no direct occlusal contact** in intercuspation or in any excursion, and **no parafunction**. If those are not met I place a healing abutment or a custom healing abutment and load conventionally."

Q How well validated is that protocol?

"Type 1 placement with immediate restoration is graded **clinically documented** — it works in selected hands and selected cases, but it is not **scientifically and clinically validated**. Type 4 late placement with conventional loading is the SCV protocol. That distinction belongs in the consent conversation: I am offering her the faster option and telling her it carries the thinner evidence base."

Q What are the risks specific to immediate placement?

"**Mid-facial recession** is the principal one, driven by a thin wall or a buccally positioned implant. Then difficulty achieving the ideal position because the **drill skids off the palatal wall**; unpredictable bone fill of the gap; **reduced primary stability**; and the fact that it does **not prevent bundle bone resorption**, so the buccal contour may still flatten even when everything is done correctly."

Q Suppose the facial wall were 0.6 mm, the phenotype thin and the smile line high. What changes?

"Immediate placement comes off the table. I would move to **Type 2 early placement at 4 to 8 weeks** with simultaneous contour augmentation, or, if the site were incompatible with Types 1 to 3, **ridge preservation at extraction and Type 4 placement at 4 to 6 months**. And the aesthetic risk assessment would read **high**, so the case is unambiguously **Complex** and referred."

Q Your colleague would extract, preserve the ridge and place at four months anyway. Why might they be right?

"Because they get **the healed soft tissue, a keratinised band and a predictable envelope** to work into, they avoid the risk of misjudging stability on the table, and the evidence base for late placement is the strongest we have. The cost is time, a second surgery and an interim prosthesis. In an aesthetically low-risk mouth like hers, the argument for immediacy is convenience — which is a legitimate reason, but I should say so rather than dress it up as biology."

Q What do you tell her, in her own words?

> " 'This tooth has a crack running down the root, and there's no way of repairing that, so it has to come out. The good news is the bone around it is in unusually good shape, which means we may be able to put the implant in the same day and give you a temporary tooth that looks right straight away. Two things I want you to know. The gum and bone at the front always shrink a little after a tooth is removed — the implant slows that down but doesn't stop it, so the shape may change slightly. And this is the more technically demanding way of doing it, so I'll be arranging the surgery with a colleague who does this routinely.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: the surgical sequence and its detail

Q Take me through the extraction.

> "**Flapless and atraumatic.** Sever the periodontal fibres with a **periotome or fine luxator**, work circumferentially, and deliver with rotational rather than buccal force — no elevation against the buccal plate. Because the root is fractured, I expect it to come out in pieces: **section it and remove the fragments individually**, using a physics forceps or root-removal system rather than levering. The buccal plate is the whole game; I would rather spend twenty minutes than lose it."

Q Flap or flapless — justify it.

> "**Flapless.** The buccal plate has just lost its periodontal ligament supply, and raising a mucoperiosteal flap removes the periosteal supply as well; flap elevation is independently associated with greater crestal resorption. Flapless also preserves the soft tissue architecture and the papillae, which in an aesthetic site I cannot rebuild reliably. I would only raise a flap if I found a wall defect that needed grafting under direct vision."

Q How do you confirm the buccal wall is intact?

> "Debride the socket fully of granulation tissue and periodontal ligament remnants, irrigate, then **probe circumferentially inside the socket** with a blunt probe, feeling for any dehiscence or fenestration, and confirm the crestal height on all four walls. Direct vision with good light and suction. If the wall is breached, the intraoperative criteria have failed and I change plan."

Q Where do you start the osteotomy, and why?

> "Against the **palatal wall**, not in the centre of the socket. A drill placed in the socket follows the path of least resistance and skids buccally, which is exactly how immediate implants end up in the wrong place. I make a purchase point on the palatal wall with a small round bur or precision drill, then follow the palatal wall apically, engaging **3 to 5 mm of native bone apical and palatal** to the socket for stability."

Q Where exactly does the implant shoulder sit, in three dimensions?

> "**Apico-coronally**, about **3 mm apical to the planned mid-facial gingival margin**. **Bucco-palatally**, **1 to 2 mm palatal** to the emergence point of the adjacent teeth, leaving the buccal gap. **Mesio-distally**, at least **1.5 mm from each adjacent root**. That is Buser's comfort zone; outside it — too shallow, too buccal, too close to a neighbour — you get recession, a visible margin or loss of papilla."

Q What is the jumping gap, and what do you do with it?

> "The **horizontal gap between the implant surface and the inner aspect of the buccal wall**. I want it to be **at least 2 mm**, and I graft it with a **slowly resorbing xenograft** — deproteinised bovine bone mineral — condensed but not over-packed, up to the crest. The graft is not there to osseointegrate; it is there to **support the contour** while the bundle bone resorbs."

Q Why a xenograft rather than her own bone?

> "Because I want a material that **resorbs slowly or barely at all**, so it holds the buccal profile for years. Autogenous bone is osteogenic but remodels and shrinks, which is the opposite of what a contour graft needs to do. Deproteinised bovine bone mineral is osteoconductive, dimensionally stable and very well documented in this indication."

Q When would you place an immediate provisional, and how is it designed?

"Only if the loading criteria are met — torque above 35 Ncm, no parafunction. It should be **screw-retained**, so there is no cement anywhere near a fresh socket; **completely free of contact** in intercuspation and in every excursion; with a **concave subcritical contour** to allow tissue thickness and a **critical contour** that supports the mid-facial margin and papillae. It is a soft tissue former that happens to look like a tooth."

Q You achieve only 20 Ncm. What now?

"I do **not** provisionalise. Options in order: a **healing abutment or custom healing abutment** with the socket sealed and grafted and a submerged or transmucosal healing period; a **removable interim prosthesis** or an Essix retainer relieved off the site; or a **resin-bonded temporary pontic** bonded to the neighbour, which is my preferred interim because it loads nothing. Then conventional loading at three to four months."

Q Post-operative instructions and prescribing?

"**Soft diet and absolutely no incising with the front teeth for eight weeks.** No brushing over the site for two weeks — **chlorhexidine 0.2% mouthwash** or gel applied locally instead. Analgesia with ibuprofen and paracetamol. Antibiotics are not routine for me, though there is reasonable evidence for a single pre-operative dose of amoxicillin in implant surgery; I would follow local antimicrobial stewardship guidance. Written instructions and a contact number."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: judging and salvaging the aesthetic result

Q How would you monitor the aesthetic outcome?

"**Standardised clinical photographs** at every stage from baseline. Objectively, the **pink and white aesthetic scores**, or at minimum a papilla index and the **mid-facial margin level measured against the adjacent teeth**. Plus peri-implant probing and bleeding, and radiographic bone levels against the baseline film taken at fitting. Review at one week, three weeks, at each provisional adjustment, at definitive fitting, then three and twelve months."

Q When would you declare success?

"**Not at six months.** Mid-facial recession in immediate anterior implants characteristically appears over the **first one to three years**, so my judgement point is at least a year out, and I tell her that at the start so a good six-month result is not oversold."

Q What actually determines whether the papilla fills?

"The **bone level on the adjacent tooth**, not on the implant — which is why an intact neighbour is such an asset here. **Tarnow's** measurement is the one to quote: where the distance from the **contact point to the interproximal bone crest is 5 mm or less the papilla is present almost always**, at 6 mm about half the time, at 7 mm rarely. So the lever I have is the **contact point position** in the definitive crown, and the provisional is where I shape it."

Q At one year there is 1.5 mm of mid-facial recession. Why?

"Most likely the **implant is slightly too buccal**, or the buccal wall resorbed despite the graft, or the tissue was over-contoured by the provisional and pushed apically. Less commonly, mucositis with tissue loss, or an occlusal overload she has developed. I would examine the emergence profile, probe the site, check the occlusion, and take a periapical against baseline."

Q What would you actually do about it?

"In order of invasiveness: **re-contour the restoration** to reduce a bulky critical contour and let the tissue rebound, and re-shape with a provisional. Then a **connective tissue graft** to thicken and coronally reposition, which is the most reliable single measure. Then remaking the crown with a longer clinical crown or **pink ceramic** to disguise it. Explantation is the last resort and I would refer before that point. With a **low lip line**, watchful monitoring and honest discussion may well be the right answer."

Q What would you record?

"Baseline photographs and the aesthetic risk assessment; the consent discussion including the specific words used about contour change and about the evidence grading of the protocol; the CBCT justification and report; the intra-operative findings — wall intact, insertion torque, gap size, graft used, batch number; the implant make, diameter, length and lot; the loading decision and why; the post-operative instructions given; and at every review, probing, bleeding, margin level and bone level against baseline."

Q How would you know this treatment had succeeded?

"By her measure first — that she smiles without checking the tooth. Objectively: an **osseointegrated, asymptomatic implant** with no bleeding or suppuration, **bone levels stable against baseline**, a mid-facial margin within about 1 mm of the contralateral 21, papillae she does not notice, a cleansable emergence profile, and no prosthetic complication at one year."

WATCH OUT

Go back to the two intra-operative gates — *is the socket wall intact, and can I achieve the ideal position with primary stability?* Almost every question in an immediate placement case is a variation on one of those, and saying it aloud buys you a sentence of thinking time.

Running this case to time

Time check. The flow above holds 40 questions and about 2,818 words of model answer. At 140 words a minute plus roughly five seconds a question, that runs to about **twenty-three and a half minutes** — an overrun of **eight and a half**. What goes first is the **surgical detail** in Domain 3: the extraction technique, the flapless justification and confirming the wall compress into one question, because you are referring that surgery and the marks sit in the **three-dimensional position, the jumping gap and the loading decision**. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is the recession at one year.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A healthy 35-year-old non-smoker; a **vertical root fracture** beneath a post crown, **not repairable**, so the tooth comes out. Then why the case is worth discussing — the site is about as favourable as an anterior socket gets: **walls intact, facial wall 1.2 mm, 6 mm of bone apical and palatal, socket within the envelope, no infection**, with a **thick phenotype and a low lip line**. Working position: this **meets the criteria for immediate placement**, but three things remain — what she wants, what she can accept if the contour changes, and **whether the case is within your own competence**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the anatomical and biological groundwork, then reading this patient	Undergraduate biology first. Four walls lined by **bundle bone** — tooth-dependent, fed by the ligament, therefore resorbing when the tooth goes — and the sentence the whole case turns on: **immediate placement does not prevent this**. The figures: **about 50% of ridge width in the first year, two-thirds of it in three months**, roughly **3.5–4 mm horizontal and 1–1.5 mm vertical at six months, predominantly buccal**. The buccal plate's **three blood supplies**, one already lost — the argument for flapless. Anterior facial walls **under 1 mm in most sites**, so her **1.2 mm is the minority**. Phenotype by **probe transparency**; the **low lip line** as a risk modifier, because it decides what she will see. Then the **ITI aesthetic risk assessment**, with **expectation the only unknown**.
4:30–7:30	Domain 2: the options, and the timing decision	Every option, including **no treatment** and the **resin-bonded bridge argued at its strongest** — cheap, quick, reversible, and it **does not close the implant door**. Then the classification kept clean: **Types 1 to 4 by timing, loading classified separately**. The **pre-operative criteria** matched to her, including the **2 mm horizontal gap**; then the **three intra-operative gates checked after the tooth is out — wall intact, primary stability achievable, ideal position achievable** — with the abandonment plan **consented in advance**. Loading: **>35 Ncm or ISQ >70, no occlusal contact anywhere, no parafunction**. And the evidence grading said out loud — Type 1 with immediate restoration is **clinically documented, not scientifically and clinically validated**.
7:30–11:30	Domain 3: the surgical sequence and its detail	Compressed at the front — **flapless, periotome, sectioned and delivered rotationally, no elevation against the buccal plate**, then the socket debrided and **probed circumferentially** to confirm the wall. Then what the examiner will spend time on: the osteotomy started **against the palatal wall** with a purchase point, because a drill in a socket **skids buccally**, engaging **3–5 mm of native bone apical and palatal**. Position in three dimensions — **3 mm apical to the planned mid-facial margin, 1–2 mm palatal, 1.5 mm from each root**. The **jumping gap at 2 mm or more**, grafted with **deproteinised bovine bone mineral** chosen because it **barely resorbs** — a contour graft, not an osteogenic one. The provisional **screw-retained and out of contact in every excursion**. And at **20 Ncm you do not provisionalise**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: judging and salvaging the aesthetic result	**Standardised photographs from baseline**, pink and white aesthetic scores, and the mid-facial margin measured against the neighbour. Then the timing judgement: **do not declare success at six months** — recession in immediate anterior implants appears over **one to three years**, and she is told that at the start so a good early result is not oversold. **Tarnow** for the papilla, with the point that it depends on the **bone level of the adjacent tooth**, so the lever you hold is the **contact point position**. Then 1.5 mm of recession at a year, with causes in order — **too buccal, the wall resorbed despite the graft, or over-contoured by the provisional** — and treatments in order of invasiveness: **re-contour first**, then a **connective tissue graft**, then pink ceramic, with explantation referred.
14:30–15:00	Closes on the patient and the record	In her words: the crack cannot be repaired, so the tooth comes out; the bone is unusually good, so the implant and a temporary tooth may go in the same day; **the gum and bone at the front always shrink a little, and the implant slows that but does not stop it**; and — plainly — *"this is the more technically demanding way of doing it, so I'll be arranging the surgery with a colleague who does this routinely."* Recorded: baseline photographs and the risk assessment, the **words used about contour change and about the evidence grading**, torque, gap size and graft batch, and the margin and bone level at every review.

The spine — the twelve answers to have ready if nothing else

1 Bundle bone, and why immediate placement does not save the wall · 2 The dimensional change after extraction, with the figures · 3 The three blood supplies of the buccal plate · 4 How thin an anterior facial wall usually is, and why 1.2 mm matters · 5 Phenotype assessed chairside · 6 Why the low lip line changes the case · 7 The resin-bonded bridge argued at its strongest · 8 Types 1 to 4, and loading kept separate · 9 The pre-operative criteria, and the three intra-operative gates · 10 Implant position in three dimensions · 11 The jumping gap, and why a xenograft rather than her own bone · 12 Tarnow, the contact point, and what to do about recession at a year.

Where the marks are lost in this case

Claiming the immediate implant preserves the buccal plate. It does not. The facial wall is largely **bundle bone**, its blood supply left with the ligament, and it resorbs whatever you place in the socket. Say that in the first thirty seconds and then say what immediate placement *does* buy — one surgery, a shorter treatment time and immediate soft tissue support — because that is an honest answer and the other one is not.

Losing the position to the socket. The single technical point this case is built on is that a drill placed in the centre of a socket **skids buccally off the palatal wall**, and a buccally placed implant is the commonest cause of the mid-facial recession that arrives at one to three years. Volunteer the **palatal purchase point**, the **1–2 mm palatal shoulder** and the **2 mm jumping gap** unprompted — that is the difference between a 3 and a 5 in Domain 3.

Planning a Complex case as though it were yours. A single implant in the aesthetic zone is Advanced at best; **immediate placement with immediate restoration in the aesthetic zone is Complex on both axes**, and at this level the surgical phase is referred. What stays yours is the diagnosis, the risk assessment, the wax-up, the options conversation, the consent — including the **evidence grading** — and the lifelong aftercare. Candidates describe the palatal osteotomy beautifully and never say who is holding the drill.

SOE Case 15 — The smoker with treated periodontitis

THE CASE

50-year-old woman, smokes **15 cigarettes a day for 30 years**. History of **generalised periodontitis Stage III Grade C**, treated non-surgically 3 years ago, now with probing depths mostly 4 mm or less, **full-mouth bleeding score 18%**, **plaque score 30%**. Missing **15, 25, 36, 46**. Wants "all of them replaced with implants".

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 50-year-old woman who smokes 15 a day and has a treated history of generalised periodontitis, Stage III Grade C, requesting four implants to replace the 15, 25, 36 and 46. Her pockets have responded, but her **bleeding score of 18% and plaque score of 30% are above the thresholds for periodontal stability**, which are bleeding under 10% and, ideally, plaque under 20%. So the honest interpretation is that she is **improved but not stable**. She also carries the **two strongest patient-level risk factors for peri-implantitis — a history of periodontitis and smoking — operating together**. My priorities are to establish stability, to address the smoking, to reduce the scope of what she is asking for, and to make sure she understands the risk before anything is placed."

Domain 1 — Clinical interpretation and reasoning

Chain: the periodontal groundwork

Q What does Stage III Grade C mean?

"Under the **2017 World Workshop classification**, the **stage** describes severity and complexity at presentation and the **grade** describes the rate of progression and risk of further progression. **Stage III** is severe disease with the potential for additional tooth loss; **Grade C** is rapid progression. The **extent** — generalised — means **30% or more of teeth** are affected."

Q How is the stage determined?

"By the **interdental clinical attachment loss at the worst-affected site**, supported by radiographic bone loss and tooth loss due to periodontitis. Stage III is CAL of **5 mm or more**, bone loss extending to the **middle third of the root and beyond**, and **four or fewer teeth lost**. Complexity factors then may raise the stage — pockets of 6 mm or more, vertical defects of 3 mm or more, **furcation involvement grade II or III**, and ridge defects."

Q And the grade?

"By evidence of progression. **Direct evidence** is radiographic bone loss over five years. **Indirect evidence** is the ratio of **percentage bone loss to age** — under 0.25 is Grade A, 0.25 to 1.0 Grade B, over 1.0 Grade C. Grade is then modified upward by risk factors: **smoking 10 or more cigarettes a day, or HbA1c of 7.0% or above in diabetes**, both push a patient to Grade C. At 15 a day, her smoking alone places her there."

Q Define a successfully treated, stable periodontitis patient.

"**No probing depths of 4 mm or more that bleed on probing**, no sites of 5 mm or greater, and a **full-mouth bleeding score below 10%**. Bone levels stable. The patient remains a **periodontitis patient for life** — the diagnosis is not cured, it is controlled, and she moves into supportive periodontal care rather than back to routine recall."

Q Is she stable?

"No. Her **bleeding score of 18% is nearly twice the threshold**, and her plaque score of 30% tells me the cause of that bleeding is unresolved. Pockets of 4 mm or less is encouraging, but stability is defined by inflammation,

not depth alone. I would describe her as **in remission but not stable**, and that distinction is the entire basis of my treatment plan."

Q How do you record a full-mouth bleeding score and a plaque score?

"**Six sites per tooth**, all teeth, scored dichotomously — present or absent — and expressed as a percentage of sites. Bleeding is recorded after gentle probing to the pocket base with a **light force of about 0.25 N**; plaque is recorded by running a probe along the gingival margin or after disclosing. The same indices repeated at every visit are what let me and the patient see whether anything is changing."

Chain: smoking and the tissues

Q How does smoking affect the periodontium?

"**Local vasoconstriction and reduced gingival blood flow**, so the tissues look fibrous and pale and **bleed less than the underlying inflammation warrants**. Impaired **neutrophil chemotaxis and phagocytosis**. Reduced fibroblast attachment and proliferation, so healing and reattachment are poorer. A shift towards a more anaerobic subgingival microbiota. The net effect is more attachment loss for the same plaque and a **blunted response to treatment**."

Q So does her bleeding score mislead you?

"Almost certainly — it **understates** her inflammation. In a smoker, 18% bleeding probably represents more disease activity than the same figure in a non-smoker, which makes it even less acceptable as a starting point for implants. It also means that if she stops smoking her bleeding score may temporarily rise, and I would warn her of that so she does not read it as failure."

Q And on implants and healing specifically?

"It impairs **early wound healing and osseointegration** — reduced oxygen tension, impaired osteoblast function, nicotine-induced vasoconstriction — and it substantially reduces the predictability of **bone augmentation and membrane procedures**, where wound dehiscence and graft loss are markedly more common. It is a **dose-dependent** effect: heavier smokers do worse, and the maxilla appears more affected than the mandible."

Q What is her overall risk profile in one sentence?

"A **Grade C periodontitis history that is not yet stable, combined with a 30 pack-year smoking habit** — the two risk factors most consistently associated with peri-implantitis, in a patient asking for four implants at once. That is not a refusal, but it is a mandate to change the sequence and the scope."

Domain 2 — Clinical decision making and treatment planning

Chain: what she is told and what is planned

Q What do you tell her?

"I would not say no, and I would not say yes. 'Implants are possible for you, but the conditions aren't right today, and if I placed them now I think there is a real chance of losing them. Two things are working against us. You've had gum disease, and the same disease happens around implants — where it actually moves faster and is harder to treat. And smoking roughly doubles the chance of an implant failing. So let's get your gums to a stable point, help you stop smoking, then reassess — and start with fewer than four.' "

Q What figures would you quote?

"That smoking **roughly doubles to two-and-a-half times** the implant failure rate and increases peri-implantitis risk in a dose-dependent way. That a **history of periodontitis is the strongest patient-level risk factor** for peri-implantitis. And that peri-implantitis affects in the region of **22% of patients** overall — while making clear that her personal risk sits above that, and that these are ranges from the literature, not predictions about her."

Q Give me your plan.

"Four steps. **One**, re-treat to stability. **Two**, smoking cessation. **Three**, reassess and reduce the scope, considering alternatives to implants at every site. **Four**, only then place a small number of implants with cleansable, screw-retained restorations and committed three-monthly supportive care. Each step is a gate — I do not move to the next until the previous one is met."

Q Take step one further.

"**Step 2 of the EFP S3 guideline** — reinforced oral hygiene and risk factor control, then **subgingival instrumentation of the residual pockets**, targeted rather than full-mouth if only a few sites remain. Re-evaluate at **three months**, aiming for plaque under 20% and bleeding under 10%. If she has not responded, **step 3** and consideration of referral to a periodontist rather than repeating the same non-surgical treatment indefinitely."

Q Why reduce the scope? Which teeth would you actually replace?

"Because four implants in a high-risk mouth quadruple both the biological risk and the maintenance burden for a benefit she may not need. I would ask **which spaces actually trouble her**. The **15 and 25** are single premolar units affecting appearance and are the strongest candidates. The **36 and 46** are molar spaces where a **shortened dental arch** — premolar-to-premolar occlusion — is well evidenced as functionally adequate and would avoid two implants altogether."

Q What are the alternatives at the 15 and 25?

"**Nothing**, if she is functioning and the space is not visible. A **resin-bonded bridge**, minimally destructive and reversible, where the abutment teeth are sound and periodontally stable — though her attachment loss may limit crown-to-root ratio. A **conventional bridge**, which I would be reluctant to prescribe on periodontally compromised abutments. Or a **removable partial denture**, though plaque retention in this patient is a genuine objection."

Q Classify this case on SAC.

"The surgery — single implants in healed premolar sites with adequate bone — would be **Straightforward to Advanced**. But **the patient risk profile raises it to Advanced**: SAC explicitly recognises patient-level modifiers, and active or unstable periodontitis plus heavy smoking are exactly that. If augmentation were needed at any site, it becomes **Complex** in a smoker and I would refer."

Q What is beyond your competence here?

"**Periodontal surgery** or regenerative treatment of residual defects if she fails to respond. **Bone augmentation in a heavy smoker. Established peri-implantitis requiring surgical management.** And any suggestion that the whole dentition needs reorganising rather than four spaces restoring. I would refer for those, and I would tell her at the outset which parts of the pathway I would not be doing myself."

Q If she refuses to stop smoking, would you place implants?

"Smoking is **not an absolute contraindication**, and refusing all treatment to every smoker is not a defensible position. I would treat **if she achieved periodontal stability**, but I would reduce the scope, **avoid augmentation entirely**, use maximally cleansable designs, insist on three-monthly maintenance, and **consent her explicitly and in writing** that her failure risk is materially higher and that I have advised cessation. I would record that I re-offered cessation support at every visit."

Q She says her friend smokes 20 a day and has had implants for ten years. Respond.

" 'That's genuinely true for some people, and I'm glad it worked for her. Risk isn't a certainty — it's a change in the odds. What I can't do is tell you which group you're in beforehand, and you have a second risk your friend may not have, which is the gum disease. So I'd rather stack the odds in your favour than rely on you being lucky.' Honest, non-confrontational, and it keeps her engaged rather than defensive."

Domain 3 — Decision making in relation to treatment procedures

Chain: getting her ready, then placing

Q How would you actually deliver smoking cessation advice?

"**Very brief advice** at every visit — **Ask, Advise, Act**. Ask and record smoking status; advise that the best way to stop is with a combination of behavioural support and pharmacotherapy; act by **referring to the local stop-smoking service**, which roughly triples quit rates compared with willpower alone. I would mention **nicotine replacement therapy** and note that prescribing beyond that is her GP's or the service's role, not mine."

Q How long before surgery would you want her to have stopped?

"Ideally **permanently**. The pragmatic protocol frequently cited is stopping at least **one to two weeks before surgery and for eight weeks afterwards**, covering the critical early healing and osseointegration period. I would frame it as an opportunity — implant treatment gives many patients a concrete reason to quit — rather than as a condition I am imposing."

Q How will you decide she is ready to proceed?

"**Objective criteria, agreed in advance and written down**: plaque score consistently under 20%, full-mouth bleeding under 10%, no residual pockets of 5 mm or more, no bleeding at any 4 mm site, and demonstrated attendance at supportive care. Setting the numbers in advance stops the decision becoming an argument, and it makes the target hers rather than mine."

Q What would you do differently at surgery because she is a smoker?

"Nothing exotic, but I would be **meticulous about the soft tissue**: flap design that guarantees tension-free primary closure, careful handling, and **no elective augmentation**. Chlorhexidine rinse pre-operatively and for the healing period, and clear instructions on abstaining around surgery. And I would favour a **two-stage approach** if there were any doubt about soft tissue healing."

Q What implant and site factors would you choose for her?

"A site with **adequate native bone so no grafting is needed**. A **standard-diameter implant with a well-documented surface and a conical connection with platform switching**, on the reasoning that it keeps the microgap away from the crest. Placed at the correct three-dimensional position — not too deep, because deep placement makes a future peri-implantitis defect far harder to manage in a patient likely to develop one."

Q How much keratinised mucosa do you want, and what if it is inadequate?

"At least **2 mm of keratinised mucosa**, ideally with an adequate band of attached tissue. Where it is deficient, plaque control is more painful, tissue recession and inflammation are more common, and this patient in particular cannot afford that. I would consider a **free gingival graft or an apically repositioned flap** at second stage — soft tissue augmentation, unlike bone augmentation, remains reasonable in a smoker."

Q What loading protocol?

"**Conventional loading**, at around three months in the mandible and four in the maxilla, confirmed by clinical stability. In a heavy smoker with a periodontitis history I would not use immediate or early loading — I gain little and I remove my margin for error. I would explain that the wait is deliberate, not a delay."

Q What restoration design follows from her risk profile?

"**Screw-retained, always**, so I can retrieve and inspect and there is **no risk of retained subgingival cement**. **Shallow, cleansable emergence profiles** that admit an interdental brush, with margins at or just below the mucosal level, never deep. **No unnecessary splinting.** Contours that she can actually clean, even where that costs some aesthetic ideal."

Q What would you record?

"Her **diagnosis with stage, grade and extent**, and the smoking history in pack-years. **Serial plaque and bleeding scores** and a six-point pocket chart at each re-evaluation. Every cessation discussion and referral. The consent conversation including the specific elevated risk figures. Implant details, batch numbers, torque, and a **baseline periapical at fitting** against which all future radiographs are compared."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: peri-implant disease, recognised and treated

Q Define peri-implant health, mucositis and peri-implantitis.

"**Health** — no erythema, swelling, bleeding on probing or suppuration, and no bone loss beyond initial crestal remodelling. **Mucositis** — **bleeding on gentle probing**, with or without erythema and swelling, and no additional bone loss; it is reversible and plaque-induced. **Peri-implantitis** — bleeding or suppuration, increased probing depths compared with baseline, and **radiographic bone loss beyond crestal remodelling**."

Q How do you diagnose peri-implantitis without a baseline?

"By the threshold definition: **bone levels 3 mm or more apical to the most coronal portion of the intra-osseous part of the implant**, together with **bleeding or suppuration on probing** and **probing depths of 6 mm or more**. It is a fallback, not a substitute for having taken a baseline radiograph and pocket chart at the time of fitting — which is precisely why I take one."

Q Three years on there is a 7 mm pocket with bleeding and 3 mm of bone loss at the 15. First steps?

"Confirm the diagnosis — probe six sites, check suppuration, take a **parallel-technique periapical and compare it with baseline**. Check the implant is not mobile, that the restoration is not loose, and that there is no retained cement or over-contoured emergence. Reinforce her plaque control, review her smoking status, and begin **non-surgical management**: mechanical debridement with instruments that will not damage the surface, plus antiseptic adjuncts."

Q The non-surgical phase fails. Now what?

"Then **surgical access is indicated** — open flap debridement with decontamination, with either **resective and implantoplasty** management for a horizontal or one-walled defect, or **reconstructive** management with a graft where the defect is contained and intra-bony. In a patient who is still smoking, the outcome of a reconstructive approach is markedly poorer, so I would favour access and resection for cleansability. **This is the point at which I refer.**"

Q When would you remove the implant?

"When it is **mobile**, when bone loss exceeds roughly half to two-thirds of the implant length, when a defect is unmanageable or the implant is malpositioned so that it cannot be cleaned, when the patient has already failed surgical treatment, or when the patient does not want further intervention. Removing an implant early can be the kinder decision — leaving a failing implant to destroy the adjacent site helps nobody."

Q What maintenance would you set, and why three-monthly?

"**Three-monthly supportive care**, at least initially. The rationale is that the periodontal literature shows the reformation of a pathogenic subgingival biofilm within about three months, and her risk profile gives no margin. Each visit: plaque and bleeding scores, **six-point probing of implants and teeth with a light force**, suppuration, mucosal condition, restoration integrity, and reinforcement of cleaning around each implant with a single-tufted brush and interdental brush."

Q How would you know the plan had worked?

"That she is **stable rather than merely improved** — bleeding under 10%, no pockets of 5 mm or more, stable bone levels against baseline, healthy peri-implant tissues, and a maintenance pattern she keeps. If she has stopped smoking, that is a bigger clinical gain than any implant I could place. And by her own measure: that she can eat and smile without thinking about it, having had fewer implants than she originally asked for."

WATCH OUT

Come back to the sequence: **stabilise the disease, control the risk factor, reduce the scope, then treat.** Say which of those four steps the question belongs to. It buys a sentence of thinking time and it is nearly always the right frame in a periodontally compromised patient.

Running this case to time

Time check. The flow above holds 37 questions and about 2,485 words of model answer. At 140 words a minute plus roughly five seconds a question, that is close to **twenty-one minutes** — an overrun of about **six**. What goes first is the **implant and site selection** material in Domain 3: connection design, platform switching, keratinised mucosa and loading protocol compress into one question, because the marks here sit in **readiness criteria and cleansable design**, not hardware. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where she gets peri-implantitis.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the charting in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 50-year-old smoking 15 a day, treated **generalised periodontitis Stage III Grade C**, asking for four implants. Then the number that is the case: the pockets have responded, but **bleeding at 18% and plaque at 30% are above the thresholds for stability** — under 10% and ideally under 20% — so she is **improved but not stable**. And the pairing: **a history of periodontitis and smoking are the two strongest patient-level risk factors for peri-implantitis, operating together**. Priorities — stability, the smoking, **reduce the scope**, and the risk understood before anything is placed.
1:20–4:30	Domain 1: the periodontal groundwork, then smoking and the tissues	Undergraduate ground first, and a lot of it. **2017 classification**: stage is severity and complexity, grade is rate of progression, extent **generalised at 30% or more of teeth**. Stage III by **interdental CAL of 5 mm or more**, bone loss to the middle third, four or fewer teeth lost. Grade by **percentage bone loss over age** — under 0.25, 0.25–1.0, over 1.0 — modified upward by **smoking 10 or more a day or HbA1c ≥7.0%**, so **her smoking alone puts her in Grade C**. **Stability defined**: no 4 mm sites that bleed, nothing 5 mm or more, **bleeding under 10%** — and she fails it. Then smoking: **vasoconstriction, impaired neutrophil function, poorer fibroblast attachment**, and the consequence — her **bleeding score understates her inflammation**, and may *rise* when she quits.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: what she is told and what is planned	Neither yes nor no. The figures quoted as ranges, not predictions: smoking **roughly doubles to two-and-a-half times** failure; **a history of periodontitis is the strongest patient-level risk factor**; peri-implantitis affects around **22% of patients**, with hers above that. Then the plan as **four gates, each blocking the next**: stability, cessation, reassess and **reduce the scope**, then a small number of implants with three-monthly care. **EFP S3 step 2**, re-evaluated at **three months**, with step 3 or a periodontist rather than repeating the same treatment. Scope reduced by asking **which spaces actually trouble her** — the 15 and 25 the candidates, the **36 and 46 answered by a shortened dental arch**. **SAC Advanced on the patient modifier; Complex and referred if augmentation is needed.**
7:30–11:30	Domain 3: getting her ready, then placing	**Very brief advice at every visit — Ask, Advise, Act** — with referral to the stop-smoking service, which **roughly triples quit rates**. Ideally stopped permanently, pragmatically **one to two weeks before and eight weeks after**. Then the part that earns Domain 3: **objective readiness criteria, agreed in advance and written down** — plaque under 20%, bleeding under 10%, no 5 mm pockets, no bleeding at any 4 mm site, attendance demonstrated — so the decision never becomes an argument. Surgically, **nothing exotic**: tension-free primary closure, **no elective augmentation**, two-stage if in doubt. **At least 2 mm of keratinised mucosa**, and **soft tissue grafting stays reasonable in a smoker even though bone grafting does not**. **Conventional loading.** Restoration **screw-retained always**, shallow cleansable emergence, **no unnecessary splinting**.
11:30–14:30	Domain 4: peri-implant disease, recognised and treated	The **2017 triad** defined, then diagnosis **without a baseline** — bone **3 mm or more apical to the most coronal intra-osseous portion**, with bleeding or suppuration and **6 mm depths** — named as a fallback, which is why you took a baseline. The 7 mm pocket at the 15 handled in order: confirm by probing and a **periapical against baseline**, exclude a loose restoration, retained cement or over-contour, then non-surgical therapy. When that fails, **surgical access — and in a continuing smoker, resection for cleansability rather than reconstruction**, whose outcomes are markedly poorer. **Three-monthly supportive care**, justified by biofilm reformation at about three months.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	In her words: *"Implants are possible for you, but the conditions aren't right today… you've had gum disease, and the same disease happens around implants, where it moves faster and is harder to treat… smoking roughly doubles the chance of an implant failing. Let's get your gums stable, help you stop, then reassess — and start with fewer than four."* And to the friend who smokes 20 a day: **"risk isn't a certainty, it's a change in the odds"**, said without confrontation. Recorded: **stage, grade and extent**, pack-years, serial plaque and bleeding scores, **every cessation discussion and referral**, the elevated-risk figures actually quoted, and a **baseline periapical at fitting**.

The spine — the twelve answers to have ready if nothing else

1 What Stage III Grade C means, and how each is determined · 2 The grade modifiers, and why her smoking alone gives Grade C · 3 Stability defined, and why she does not meet it · 4 How a bleeding and plaque score is actually recorded · 5 What smoking does to the periodontium · 6 Why her bleeding score understates her disease · 7 The figures you would quote her, as ranges · 8 The four gates, in order · 9 Why you reduce the scope, and the shortened arch at the 36 and 46 · 10 The written readiness criteria · 11 The restoration design her risk profile dictates · 12 Peri-implantitis diagnosed without a baseline.

Where the marks are lost in this case

Calling 4 mm pockets a success. Stability is defined by **inflammation, not depth** — bleeding under 10%, nothing 5 mm or more, no bleeding at any 4 mm site — and at 18% and 30% she meets none of it. Say **"in remission but not stable"** and say that the whole plan follows from that distinction; a candidate who reads "mostly 4 mm or less" as ready to place has lost Domain 1.

Missing that the bleeding score flatters her. Smoking causes vasoconstriction, so the tissues bleed **less than the underlying inflammation warrants** — 18% in a smoker represents more disease than 18% in a non-smoker. Volunteer that, and volunteer the corollary she must be warned about: **her bleeding score may rise when she quits**, and that is not failure. Unprompted, that is the difference between a 3 and a 5.

Agreeing to four implants. She asked for four; the answer is fewer, and the argument is hers to hear — four implants in a high-risk mouth quadruple both the biological risk and the maintenance burden, and the **36 and 46 are answered by a shortened dental arch**. Then name the competence line at the outset, not after a failure: **periodontal surgery or regeneration if she does not respond, bone augmentation in a heavy smoker, and surgical management of peri-implantitis are all referred** — and if she will not stop smoking, treatment is conditional on stability, scoped down, ungrafted, and **consented in writing that her risk is materially higher**.

SOE Case 16 — The edentulous maxilla and the fixed-prosthesis request

THE CASE

66-year-old man, edentulous upper for 12 years, wearing a complete upper denture against **natural lower teeth**. He wants **"teeth that don't come out"**. Examination: moderate maxillary resorption, well-extended denture with a **full labial flange**, **high smile line**, generous freeway space. CBCT shows adequate anterior bone but **pneumatised sinuses posteriorly with 3 mm of residual height**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 66-year-old man, edentulous in the maxilla for 12 years, wearing a complete upper denture against a **natural lower dentition** — a high-load and therefore high-risk combination — who is asking for a fixed prosthesis. Three findings dominate. He has a **high smile line**, so any transition between prosthesis and mucosa will be visible. He has a **full labial flange** after 12 years of resorption, so his lip support is very probably prosthesis-dependent. And his **posterior maxilla has 3 mm of residual bone height** beneath pneumatised sinuses, so posterior implants would require lateral window sinus elevation. My task is to establish whether a fixed prosthesis is achievable, whether it is within my scope, and whether it is actually the right answer for him."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical and biological groundwork

Q Describe the maxillary sinus.

"A **pyramidal air-filled cavity** in the body of the maxilla, lined by the **Schneiderian membrane** — thin, ciliated, pseudostratified columnar respiratory epithelium. Its base is the lateral nasal wall, its apex extends towards the zygoma, and its floor is the alveolar process. It drains through the **ostium in the middle meatus**, positioned high on the medial wall, so drainage is against gravity. Bony **septa of Underwood** may divide the floor."

Q What vessels matter during a lateral window approach?

"The **posterior superior alveolar artery** and its anastomosis with the **infraorbital artery**, which runs in the lateral wall of the sinus, often within bone, at a variable height above the crest. Cutting it during a window osteotomy produces brisk, obscuring bleeding. It can be identified on CBCT beforehand, which is one of the arguments for cross-sectional imaging in these cases."

Q Why has his sinus pneumatised?

"**Loss of the teeth removes the functional stimulus to the alveolar bone**, so the ridge resorbs from above while the sinus expands from below by osteoclastic activity on the sinus floor. Positive intra-sinus pressure on expiration and the absence of loading both contribute. The result is exactly what the CBCT shows — a thin residual floor with only 3 mm of bone."

Q How would you assess whether the sinus itself is healthy?

"History of **sinusitis, allergy, nasal obstruction or previous nasal surgery**, and smoking. Radiographically I would look for **mucosal thickening, complete opacification, polyps, retention cysts, and patency of the ostium**, plus septa and the position of the artery. A sinus that cannot drain is a contraindication to elevation until it is managed — that is an ENT decision, not mine."

Q What is combination syndrome?

"**Kelly's description** of the changes seen when a complete maxillary denture opposes a natural anterior mandibular dentition with free-end saddles. Five features: **resorption of the anterior maxillary ridge**, **downgrowth of the maxillary tuberosities**, **papillary hyperplasia of the hard palate**, **over-eruption of the lower anterior teeth**, and **loss of bone under the mandibular free-end saddles**. Loss of occlusal vertical dimension and an anteriorly displaced occlusal plane follow."

Q Why does the opposing natural dentition matter mechanically?

"Because a dentate mandible generates **substantially greater and less evenly distributed bite forces** than an opposing complete denture — a complete denture wearer produces a fraction of the force a dentate patient does. So the maxillary prosthesis is loaded harder, more anteriorly, and less predictably. That raises the mechanical demand on any implant-borne restoration and makes occlusal design a central rather than a peripheral issue."

Q What is the bone quality likely to be, and how do you classify it?

"**Lekholm and Zarb type III or IV**, or **D3 to D4 on Misch** — a thin cortex around low-density trabecular bone. The maxilla is characteristically less dense than the mandible, so **primary stability is harder to achieve** and healing takes longer. That drives my surgical technique and my loading protocol rather than my decision to treat."

Q Classify his residual ridge.

"On **Cawood and Howell**, moderate maxillary resorption with adequate anterior bone is around **Class IV to V** — the anterior ridge losing width and then height. Naming it matters because it tells me the deficiency is progressive and three-dimensional, and because in the maxilla resorption is **upwards and inwards**, which moves the ridge palatally away from where the teeth need to sit."

Q What does that resorption pattern mean for his appearance?

"That the **residual ridge lies well palatal and superior to the original tooth position**, so the prosthetic teeth must be set forward and downward of it. The **labial flange fills that space** — it restores the lost alveolus and supports the upper lip, the nasolabial angle and the philtrum. Remove the flange and you remove the tissue replacement, not just an edge of acrylic."

Q How do you assess the smile line and lip support?

"**Smile line** by asking him to smile broadly and speak — a high smile line displays the gingival margins and the whole transition zone. **Lip support** by viewing him in profile and three-quarters with the denture in and out, assessing philtrum length, vermilion display, nasolabial angle and the depth of the labiomental fold, and by supporting the lip manually with and without the flange. I would photograph all of it."

Q What would you measure on the CBCT?

"**Residual bone height beneath each sinus** — here 3 mm — and the ridge **width and height anteriorly** at each proposed site. The position of the **incisive canal** and the nasal floor. Any **septa**, the position of the posterior superior alveolar artery, and the **sinus membrane thickness**. And I would relate all of it to a radiographic guide reproducing the planned tooth position, so I am planning from the prosthesis backwards."

Domain 2 — Clinical decision making and treatment planning

Chain: options, and the decision that actually matters

Q What are the options?

"**Remake the complete denture**, with attention to extension, the posterior palatal seal, the neutral zone and the vertical dimension — this must be offered and may well be adequate. **An implant-retained maxillary overdenture**, typically four or more anterior implants with a bar or studs, which **retains the labial flange** and is cleansable. **A fixed full-arch implant prosthesis**, which is what he is asking for. Or **no treatment**, with denture adhesive and advice."

Q What would a fixed prosthesis actually require here?

"Enough implants with adequate anteroposterior spread — which with 3 mm of posterior bone means **bilateral lateral window sinus elevation**, or tilted implants, or zygomatic implants. And a design that manages the **transition line** between prosthesis and mucosa in a man with a high smile line, either by having enough bone that no flange is needed or by using pink ceramic or acrylic that will be visible when he smiles."

Q What is the single most important thing to establish before he commits?

"**Whether his appearance depends on the labial flange.** With 12 years of resorption and a high smile line it almost certainly does, and this is the classic cause of profound dissatisfaction after a fixed maxillary prosthesis. Everything else — bone, implants, sinuses — is secondary to that finding, because it determines which prosthesis type is correct."

Q How would you establish it?

"**Duplicate his existing denture and remove the flange from the duplicate**, or produce a diagnostic wax-up without a flange, and **try it in so he can see it himself in a mirror**, with photographs taken from the front, in profile and smiling. Let him make the judgement. If his lip support and transition line collapse without the flange, then a fixed prosthesis without a flange will not satisfy him, and an **implant-retained overdenture is the correct answer — not a compromise**."

Q Which do you recommend?

"A **four-implant maxillary overdenture in the anterior maxilla, retaining the labial flange**, with the palate reduced if support permits. It gives him the security he is asking for, preserves the appearance he already has, avoids two lateral window sinus lifts, remains cleansable in a man of 66 who will age with it, and is **within my scope** where a fixed full-arch reconstruction is not."

Q Argue against your own recommendation.

"That he asked for teeth that do not come out and I am offering him something that does — that is a real failure to meet the stated goal, and there is evidence that patients value fixed prostheses highly. A well-planned fixed prosthesis with adequate bone gives excellent function and no daily removal. My defence is that I am not refusing it, I am **making him see the aesthetic consequence before he chooses**, and offering referral if he still wants it."

Q Classify the options on SAC.

"**Remaking the complete denture** — Straightforward. **A four-implant anterior maxillary overdenture — Advanced**, because of the aesthetic zone, the reduced bone density and the multiple-implant prosthodontics. **A fixed full-arch maxillary rehabilitation with bilateral sinus grafting — Complex in both the surgical and the restorative dimension**, and outside the scope of this level of practice."

Q So what is beyond your competence?

"**Lateral window sinus elevation**, particularly bilateral with only 3 mm of residual height. **Tilted or zygomatic implants. Full-arch fixed maxillary reconstruction.** Any reorganisation of his lower occlusion at an altered vertical dimension. I would refer for those, and I would make that clear before he becomes emotionally committed to a plan I cannot deliver."

Q He insists on the fixed option. What do you do?

"Refer him — properly, not reluctantly. I would send the **photographs, the flangeless try-in, the CBCT and my assessment of his lip support**, so that the specialist begins from the same evidence rather than repeating it. I would make sure he understands the surgical burden, the cost, the healing time and the maintenance. Refusing to refer because I disagree would be paternalistic; referring without the aesthetic evidence would be careless."

Q What if he cannot afford implants at all?

"Then a properly made conventional denture is a legitimate and often very successful treatment, and I would say so without apology. **Copy technique** to preserve the polished surfaces he has adapted to, corrected extension

and posterior palatal seal, neutral zone recording, restored vertical dimension, and a review of his lower arch. Many patients in his position are transformed by a well-made denture alone."

Domain 3 — Decision making in relation to treatment procedures

Chain: delivering the overdenture

Q Outline the sequence.

"**Prosthetically driven from the start.** Remake or copy the denture first to establish correct tooth position, vertical dimension and lip support, and prove it clinically. Duplicate that denture as a **radiographic and then surgical guide**. Plan on CBCT. Place four anterior implants. Allow conventional healing. Second stage if needed, then pick up the attachments into the proven denture design."

Q How many implants, and where?

"**Four**, in the anterior maxilla between the sinuses and clear of the incisive canal — broadly the lateral incisor and premolar regions — with the **widest anteroposterior spread the bone allows**. Two unsplinted implants in the maxilla are not comparable to two in the mandible: the bone is softer, the denture-bearing area behaves differently, and outcomes are poorer. Four, and I would consider splinting them."

Q Why avoid the incisive canal?

"Because it contains **fibrous tissue and the nasopalatine neurovascular bundle**, and an implant engaging it will not osseointegrate — it will be surrounded by soft tissue. It also enlarges with age and resorption, so it can occupy much of the midline. Where it obstructs an ideal position, the answer is to reposition the implant, not to drill through it."

Q Bar or studs — how do you decide?

"By **available restorative space, implant divergence and his ability to clean**. **Studs** need about **8 to 10 mm** from platform to occlusal plane, are independent, cheap and easy to clean, and tolerate divergence up to roughly 20 degrees. A **bar** needs **12 to 15 mm**, splints the implants, shares load, and creates a common path of insertion despite divergence — but it is bulkier and harder to clean. In a maxilla with four implants I lean towards a bar for load sharing."

Q How do you measure the available restorative space?

"On the **articulated diagnostic set-up at the correct vertical dimension**, measuring from the planned implant platform level to the occlusal or incisal plane of the wax-up. Not by eye in the mouth. He has a generous freeway space, which suggests his vertical dimension may be under-restored — restoring it correctly will change the space available, which is another reason the denture must be made first."

Q What surgical modifications for D3 to D4 bone?

"An **undersized osteotomy** to improve primary stability, use of **osteotomes or bone condensers** rather than full drilling to compact rather than remove trabecular bone, avoiding countersinking, and engaging the cortical plates — nasal floor or palatal cortex — where possible. Slow speed and irrigation still apply, though overheating is less of a risk than in dense mandibular bone."

Q What loading protocol, and why?

"**Conventional loading at four to six months** in this bone quality, confirmed clinically and, if available, by resonance frequency analysis. I would not use early or immediate loading here — the bone is soft, the opposing dentition is natural, and I have no need to take that risk. His existing denture is relieved generously and relined with a **tissue conditioner** during healing."

Q Can you remove the palate?

"Only if the implants provide enough support and retention to replace what the palate was contributing, which with four splinted implants is usually achievable. I would **reduce rather than eliminate it initially**, keeping

a horseshoe with adequate rigidity, and reinforce the base with a **cast metal framework** — an unreinforced horseshoe denture in a man opposing natural teeth will flex and fracture across the midline."

Q Does his lower arch need any treatment?

"Very possibly. I would assess for **over-eruption of the lower anteriors, an unlevel occlusal plane, posterior tooth loss and the periodontal condition** — the combination syndrome picture. If the lower occlusal plane is disordered, restoring the maxilla against it simply repeats the original problem. Minor equilibration or restoring lower posterior support may be needed first, and a major reorganisation would be a referral."

Q What would you record and consent for?

"The **photographs and the flangeless try-in** and his own reaction to it — that record is what protects both of us. The options offered including no treatment, and the reasons for the recommendation. The CBCT report and its justification under **IR(ME)R**. Implant details, batch, torque and a **baseline periapical or panoramic at fitting**. And explicit consent to the **lifelong maintenance and its cost**."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: maintenance, complications and judging success

Q What do you tell him, in his own words?

" 'I understand you want teeth that stay in. Before we decide, let me show you something.' Then the flangeless try-in. Then: 'What holds your lip out at the moment is this part of the denture. A fixed bridge cannot have that part, so your lip would sit differently and you would see a join where the bridge meets your gum when you smile — and you have quite a high smile line. There is an option that is very secure and keeps your appearance: implants the denture clips onto firmly. It comes out for cleaning but does not move when you eat or talk. The fully fixed option would also need major bone grafting in both sinuses, which is a much bigger operation and one I would refer you for.' "

Q What complications should he expect, and how often?

"**Mechanical maintenance is frequent and biological failure is uncommon.** Replacement of **worn attachment inserts or matrices**, typically annually. **Relining** as the posterior ridge continues to resorb. **Denture base or framework fracture**, particularly if the palate has been reduced. **Attachment or screw loosening. Wear of the acrylic teeth with loss of vertical dimension.** Then the biological ones — mucositis, hyperplasia around abutments and peri-implantitis — which are less frequent but need the same monitoring."

Q One implant fails to integrate at second stage. What now?

"Explain it honestly and without defensiveness — a proportion of implants do not integrate and it is not a disaster. **Remove it, debride, and allow the site to heal.** Then decide whether three implants can carry the design, which with a bar and good spread they often can, or whether to **replace it after two to three months**, possibly with a wider implant or in a slightly different position. I would review why it failed before repeating the same thing."

Q At 18 months the denture is loose and rocks. Differential?

"**Continued posterior ridge resorption** so the denture is no longer tissue-supported — the commonest cause, and the answer is a **reline**, not new inserts. Then **worn inserts**; a **fractured base or housing**; a **loose abutment or bar screw**; **loss of an implant**; **peri-implant disease**; or an occlusal error. I would examine each in turn with pressure-indicating paste, individual implant testing, and radiographs if indicated."

Q He develops papillary hyperplasia of the palate. Manage it.

"Recognise it as **denture-induced hyperplasia with a likely candidal component**. Leave the denture out as much as tolerated, especially at night; scrupulous **denture and palatal hygiene**; disinfect the denture; correct the fit surface with a tissue conditioner and correct the occlusion, because the cause is usually chronic trauma from an ill-fitting or over-loaded base. Antifungal therapy if candida is confirmed. Surgical reduction and biopsy only if it does not resolve — and any non-resolving lesion is referred."

Q What would you monitor, and how often?

> "**Three- to six-monthly initially.** Peri-implant probing at six sites with a light force, bleeding and suppuration, plaque control around the abutments or under the bar, mucosal health of the whole denture-bearing area, denture fit, extension, occlusion and vertical dimension, and attachment retention. Radiographs at one year and then as clinically indicated, always compared with the baseline taken at fitting."

Q How would you know this had succeeded?

> "By his measure first — that he eats, speaks and smiles without thinking about the prosthesis and no longer feels it move. Objectively: **stable implants with no bleeding or suppuration, bone levels stable against baseline**, a retentive comfortable denture, healthy palatal mucosa, an intact lower dentition, and a maintenance pattern he attends and can afford. And that he is content with an outcome different from the one he first asked for — which is the real test of the consent conversation."

WATCH OUT

Come back to the flange: ask whether his appearance depends on it. Almost every decision in this case — fixed against removable, referral against treating, what you say to him — follows from that single finding, and stating it buys you a sentence of thinking time.

Running this case to time

Time check. The flow above holds 39 questions and about 2,698 words of model answer. At 140 words a minute plus roughly five seconds a question, that comes to about **twenty-two and a half minutes** — an overrun of **seven and a half**. What goes first is the **sinus anatomy** sub-chain in Domain 1: the description, the posterior superior alveolar artery and the assessment of sinus health compress into one question, because the plan you will recommend **avoids the sinus altogether**. Combination syndrome, Cawood and Howell and the flange stay. If your answers run past **thirty seconds**, you will never reach Domain 4.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT and the photographs up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 66-year-old edentulous 12 years in the maxilla against **natural lower teeth** — a **high-load, high-risk combination** — asking for a fixed prosthesis. Then three findings, each with its consequence attached: a **high smile line**, so the transition between prosthesis and mucosa will be visible; a **full labial flange** after 12 years of resorption, so his lip support is **very probably prosthesis-dependent**; and **3 mm of posterior height beneath pneumatised sinuses**. Three tasks — is a fixed prosthesis achievable, is it within your scope, and **is it the right answer for him**.
1:20–4:30	Domain 1: the anatomical and biological groundwork	The sinus and its artery taken briskly, then the material that carries the case. **Why it pneumatised** — loss of functional stimulus above, osteoclastic expansion below. **Combination syndrome, Kelly's five features** named in full, and why they matter: a dentate mandible generates **far greater and less evenly distributed force** than an opposing denture, so occlusal design is central. **Lekholm and Zarb III–IV / D3–D4. Cawood and Howell Class IV–V**, with the point that maxillary resorption is **upwards and inwards**, moving the ridge palatal to where the teeth must sit — which is **what the labial flange is replacing**. Then smile line and lip support assessed **with the denture in and out, and photographed**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, and the decision that actually matters	All four options, opening with a **properly remade complete denture**, which must be offered and may be adequate. Then the question that outranks everything else: **does his appearance depend on the labial flange?** Bone, implants and sinuses are all secondary to it. And how you establish it, which is the technique that earns the domain: **duplicate the denture, remove the flange from the duplicate, try it in, and let him judge it in a mirror with photographs**. If lip support collapses without it, an **implant-retained overdenture is the correct answer, not a compromise**. Recommend **four anterior implants retaining the flange** — then **argue against yourself**: he asked for teeth that do not come out. **SAC: remake Straightforward, overdenture Advanced, fixed full-arch with bilateral sinus grafting Complex.**
7:30–11:30	Domain 3: delivering the overdenture	**Prosthetically driven from the start** — remake or copy the denture first to prove tooth position, vertical dimension and lip support, then duplicate it as radiographic and surgical guide. **Four implants between the sinuses, clear of the incisive canal**, with the widest anteroposterior spread the bone allows — the canal avoided because it holds **fibrous tissue and the nasopalatine bundle, so an implant in it will not integrate**. Bar versus studs on **space, divergence and cleaning**: **studs 8–10 mm, a bar 12–15 mm**, measured on the **articulated set-up at the correct vertical dimension, not by eye** — and his freeway space suggests the vertical is under-restored. For D3–D4: **undersized osteotomy, osteotomes rather than drills**. **Conventional loading at four to six months.** Palate **reduced, not eliminated**, with a **cast metal framework** — an unreinforced horseshoe against natural teeth fractures.
11:30–14:30	Domain 4: maintenance, complications and judging success	The honest shape of it: **mechanical maintenance is frequent, biological failure is uncommon** — attachment inserts roughly annually, relines as the ridge resorbs, base or framework fracture, screw loosening, tooth wear with loss of vertical dimension. A failure at second stage handled without defensiveness: remove, debride, heal, then decide whether **three implants can carry the design**. Then the differential that separates candidates: a denture rocking at 18 months is most often **continued posterior ridge resorption needing a reline, not new inserts**. **Papillary hyperplasia** recognised as denture-induced with a candidal component — **any non-resolving lesion referred**. **Three- to six-monthly review.**

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	*"Before we decide, let me show you something."* Then the flangeless try-in, then: **what holds your lip out is this part of the denture; a fixed bridge cannot have it, so your lip would sit differently and you would see a join when you smile — and you have quite a high smile line.** Then the offer: implants the denture clips onto firmly, out for cleaning but not moving when he eats; and the fixed option needing **grafting in both sinuses, and one you would refer him for**. Recorded: **the photographs, the flangeless try-in and his own reaction to it** — that record protects both of you — the options including no treatment, and explicit consent to **lifelong maintenance and its cost**.

The spine — the twelve answers to have ready if nothing else

1 Why his sinus has pneumatised · 2 Combination syndrome, all five features · 3 Why an opposing natural dentition changes the mechanics · 4 Bone quality classified, and what it changes · 5 Cawood and Howell, and resorption upwards and inwards · 6 What the labial flange is actually replacing · 7 How you assess smile line and lip support · 8 The flangeless try-in, and who makes the judgement · 9 The four options, with the remade denture first · 10 Four implants, their position, and why not the incisive canal · 11 Bar versus studs, with the space figures · 12 A denture rocking at 18 months — reline before inserts.

Where the marks are lost in this case

Answering the question he asked instead of the one that matters. He asked for teeth that do not come out; the case is about whether his face depends on a piece of acrylic. Say early that the **single most important thing to establish is flange dependence**, and describe the **flangeless duplicate try-in** as the method — letting him see it and judge it himself. That one manoeuvre is worth more in Domain 2 than any amount of implant planning.

Selling the overdenture as second best. If his lip support collapses without the flange, a fixed prosthesis will not satisfy him however well it is made, and the overdenture is **the correct answer, not a consolation**. Volunteer the counter-argument too — that patients value fixed prostheses highly and you are not meeting his stated goal — because arguing honestly against your own recommendation is what separates a 5 from a 3 here.

Planning a full-arch reconstruction you would not be doing. This is the competence line, and it is named early, before he is emotionally committed: **bilateral lateral window elevation with 3 mm of residual height, tilted or zygomatic implants, full-arch fixed maxillary reconstruction, and any reorganisation of his lower occlusion at an altered vertical dimension are all beyond a general dentist with an interest in implants.** And if he still insists, the answer is **to refer properly** — sending the photographs, the flangeless try-in, the CBCT and your lip-support assessment — not to refuse, and not to hand him on with nothing.

SOE Case 17 — Two adjacent missing central incisors

THE CASE

40-year-old woman, missing **11 and 21** following trauma 6 months ago. **Medium smile line**, **medium phenotype**. CBCT shows a ridge width of **5 mm** and a **vertical deficiency of 3 mm** relative to the adjacent CEJs. She wants implants.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A healthy 40-year-old woman who lost both upper central incisors to trauma six months ago and would like implants. The findings that matter are a **medium smile line**, a medium phenotype, a ridge that is 5 mm wide and, critically, **3 mm vertically deficient** against the adjacent cemento-enamel junctions. So this is a **two-adjacent-tooth aesthetic case with a combined horizontal and vertical defect** — the most demanding situation short of a whole anterior segment. The central biological problem is that **two adjacent implants cannot support an inter-implant papilla predictably**, because the inter-implant bone crest sits lower than an inter-dental crest. A black triangle between two implant crowns is the expected outcome, not the exception. What I need to establish is exactly what she wants to see, what she will accept prosthetically, and whether an implant is the right vehicle at all."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical and biological groundwork

Q Describe the anatomy of the anterior maxilla relevant to implant placement here.

"The **labial plate**, which in the anterior maxilla is thin — under 1 mm in the great majority of central incisor sites and often under 0.5 mm. Behind it the **incisive canal** at the midline, carrying the nasopalatine bundle, which sits directly palatal to the central incisor roots and enlarges with resorption. Superiorly the **nasal floor** and the **anterior nasal spine**. Laterally the canine eminence and the roots of the 12 and 22. And the **labial concavity** above the root apices, which dictates implant angulation."

Q Why does the thinness of the labial plate matter so much?

"Because that plate is largely **bundle bone** — bone whose blood supply depends on the periodontal ligament. When the tooth is lost the bundle bone resorbs. Araújo and Lindhe showed the buccal wall loses height preferentially, and Schropp showed roughly a **50% loss of ridge width in the first year**, two-thirds of it in the first three months. That is exactly what has produced her present ridge form six months on."

Q Describe the dentogingival junction around a natural tooth.

"From the crest upwards: **connective tissue attachment** of about 1 mm, in which the fibres insert perpendicularly into cementum; **junctional epithelium** of about 1 mm; then the **gingival sulcus**. Together the attachment components form the **supracrestal attachment** — what used to be called biologic width — of roughly 2 mm, and the body defends that dimension. Above that sits the free gingival margin and the papilla."

Q How does the peri-implant soft tissue differ?

"There is **no cementum and no periodontal ligament**, so the collagen fibres run **parallel and circular** around the abutment rather than inserting into it. The attachment is longer — commonly around 3 to 4 mm in total — and the **blood supply is poorer**, coming only from the supraperiosteal vessels rather than also from the PDL. The result is a seal that is weaker, less well vascularised and less resistant to inflammatory breakdown."

Q What are the practical consequences of that for this patient?

"First, the tissue is more prone to **recession**, especially with a thin phenotype and a buccally placed implant. Second, the peri-implant tissue takes its vertical position from the **bone crest**, so if the crest is 3 mm low, the soft tissue margin will be low too. Third, an inflammatory lesion spreads faster and further apically around an implant than around a tooth."

Q What determines whether a papilla is present?

"The **distance from the contact point to the bone crest**. Tarnow showed that between two teeth a papilla fills the space essentially always at **5 mm**, about **56% at 6 mm** and about **27% at 7 mm**. Between an implant and a tooth the papilla takes its support from the **tooth's** attachment level, so the figure is roughly 4.5 to 5 mm. Between **two implants** the soft tissue height above the crest is only about **3.5 mm**, which is why this case is difficult."

Q Why is the inter-implant crest lower?

"Because each implant generates its own circumferential remodelling. Tarnow showed that where the **inter-implant distance is under 3 mm** the lateral crestal loss is around **1 mm**, against about **0.45 mm** at 3 mm or more. Two adjacent implants therefore produce overlapping saucers of bone loss, the crest between them drops, and the soft tissue drops with it."

Chain: reading this patient's ridge

Q Is a 5 mm ridge width adequate?

"Marginally, and not for two implants side by side. I want at least **1 mm of bone circumferentially**, and in the aesthetic zone I want **1.5 to 2 mm on the buccal** to resist recession. A 4 mm implant in a 5 mm ridge leaves half a millimetre buccally. So even for a single implant I would plan **contour augmentation**; for two, the width is the lesser of my problems."

Q And the 3 mm vertical deficiency — what does that actually mean for the result?

"It means the soft tissue zenith of any implant crown will sit roughly 3 mm apical to where her natural centrals sat, so the crowns will be **long**, and the papillae will start from a lower base. I cannot restore that with prosthetics alone without pink material. **Vertical ridge augmentation is the least predictable of all augmentation procedures** and carries the highest morbidity and exposure rate."

Q How would you formalise the aesthetic risk?

"With the **ITI Aesthetic Risk Assessment**. In her favour: healthy, non-smoker, medium rather than high smile line, medium rather than thin phenotype. Against her: an **extended edentulous span of two adjacent teeth**, a combined horizontal and vertical soft and hard tissue defect, and a triangular tooth form if that is what she has. That places her firmly in the **high aesthetic risk** group."

Q What else would you want to know before planning?

"Her smile line at full smile and in speech; tooth form — triangular teeth give longer contacts and more visible black triangles; the interproximal bone levels on the **12 and 22**, because those determine the outer papillae and I cannot recreate them; sensibility and any resorption of the adjacent traumatised teeth; occlusion and any parafunction; her expectations in her own words; and standardised photographs plus a diagnostic wax-up."

Domain 2 — Clinical decision making and treatment planning

Chain: the options and the argument between them

Q What are her options?

"**Two implants, two crowns** — the option she has come asking for, and the one most likely to produce a visible black triangle between them.
One implant with a cantilever pontic — an implant at the 11 with a cantilevered 21, or the reverse.
A two-pontic resin-bonded bridge from the 12 and 22, with soft tissue augmentation under ovate pontics.
A conventional fixed bridge — I would not prepare two sound teeth for it.

A removable partial denture with a labial flange — the only option that reliably replaces lost hard and soft tissue volume.
Implants plus pink ceramic, accepting prosthetic replacement of the missing tissue.
No treatment, with the site kept under review."

Q Why would a single implant with a cantilever pontic give a better papilla than two implants?

"Because a papilla between an **implant crown and a pontic** is supported by the ridge tissue under the pontic and by the implant on one side only — it behaves far more predictably than a papilla between two implant platforms with their overlapping remodelling. You trade a difficult biological problem for a manageable **mechanical** one, which is a cantilever in the anterior region."

Q What are the mechanical risks of that cantilever?

"**Screw loosening** first, then abutment or screw fracture, then ceramic chipping. I would mitigate them by keeping the pontic **narrow and out of excursive contact**, screw-retaining where possible so it is retrievable, torquing to the manufacturer's specification, and checking the occlusion so that the cantilever carries no lateral guidance. In a bruxist I would not do it."

Q Make the case for the resin-bonded bridge.

"It replaces two teeth with **no surgery to the ridge, no implant risk and no biological compromise of the adjacent teeth** beyond minimal enamel preparation. With **soft tissue augmentation under ovate pontics** the emergence can look excellent, and a pontic papilla is more predictable than an inter-implant papilla. In a 40-year-old with intact, unrestored 12 and 22 it is a genuinely strong choice, and it is reversible in a way an implant is not."

Q And the case against it?

"**Debonding** is the main issue — it is the commonest complication, though a single retainer per pontic and a cobalt-chromium framework improve on older designs. It is also technique-sensitive for moisture control, the metal can shine through thin incisal edges, and a two-pontic span is a bigger biomechanical ask than a single cantilever."

Q So which do you recommend?

"Given the **3 mm vertical deficiency and a medium smile line**, I would be very cautious about two implants. My preference is either a **single implant with a cantilever pontic**, combined with contour augmentation and a connective tissue graft, or a **resin-bonded bridge with soft tissue augmentation under ovate pontics**. I would present both, with a **mock-up**, and let her decide with her eyes rather than on my description."

Q She insists on two implants. What do you say?

"That I will not refuse her, but that she must choose it knowing the outcome. I would say: **'I can put two implants in and I can make you two very good-looking teeth. What I cannot promise you is the little triangle of gum between them. On the evidence, that space is likely to be dark, and if it is, we would fill it with pink ceramic rather than your own gum.'** Then I would show her that on a mock-up before she commits, and record that I did."

Q What is the SAC classification?

"**Complex** surgically — an extended edentulous span in the aesthetic zone with a combined horizontal and vertical defect. **Complex** prosthodontically — two adjacent units in the aesthetic zone with a soft tissue deficiency. There is no reading of this case that makes it Straightforward."

Q What is within your competence and what is not?

"Within: the full assessment, the risk assessment, the wax-up and mock-up, the options conversation, an interim resin-bonded or removable prosthesis, and long-term maintenance. **Beyond me: vertical ridge augmentation and the two-implant aesthetic reconstruction.** I would complete the assessment and the consent conversation myself and **refer for the definitive surgical and prosthetic treatment**, staying involved as the referring clinician."

Domain 3 — Decision making in relation to treatment procedures

Chain: how you would actually deliver it

Q Where does the planning start?

"**From the teeth backwards.** A diagnostic wax-up of the two centrals in the position her lip and smile need, converted to an intra-oral mock-up, then a **CBCT with a radiographic guide** so I can see the planned tooth position against the available bone. Only then do I know how much bone is missing and in which direction — which is a different question from how much bone is there."

Q Where would you place a single implant, in three dimensions?

"**Mesiodistally**, at least 1.5 mm from the adjacent root and centred in the tooth position. **Buccopalatally**, with the platform about 1.5 to 2 mm palatal to the line joining the adjacent CEJs, leaving 2 mm of buccal bone for contour grafting. **Apicocoronally**, about **3 mm apical to the planned gingival margin** of the future crown — not to the existing bone crest, which is 3 mm low. The comfort zone concept: shallow and buccal is the position that produces recession."

Q How would you augment the buccal contour?

"**Simultaneous guided bone regeneration** at placement, provided the implant is fully within bone and stable: a slowly resorbing **xenograft** over the buccal surface to over-contour it, covered with a **resorbable collagen membrane**, with tension-free closure achieved by periosteal release. I would combine that with a **subepithelial connective tissue graft** from the palate to thicken the phenotype, which is the single most reliable thing I can do to protect the mid-facial margin."

Q Why a connective tissue graft rather than more bone?

"Because **soft tissue thickness predicts mid-facial stability** better than an extra millimetre of graft particulate does. Converting a medium phenotype to a thick one reduces recession, masks any grey show-through from the abutment, and gives the papillae more tissue to work with. It is a lower-risk procedure than vertical augmentation with a much better evidence base."

Q How would you manage the site while it heals?

"With a prosthesis that **does not load the graft**. A **resin-bonded temporary bridge** with ovate pontics relieved off the ridge is my first choice, or an Essix retainer with the ridge fully relieved. I would specifically avoid a removable acrylic denture pressing on a fresh graft — that is a common cause of membrane exposure and graft loss."

Q How would you develop the emergence profile before the definitive crown?

"With a **screw-retained provisional crown** worn for at least two to three months. I would shape it progressively — starting concave subgingivally to allow tissue in, then adding flowable composite to push the tissue into the desired zenith and papilla form. When she and I are both happy with the soft tissue, I **capture it with a customised impression coping** so the laboratory reproduces exactly that emergence."

Q What abutment and crown material?

"In the aesthetic zone with a medium phenotype, a **zirconia abutment or a titanium base with a zirconia superstructure** rather than a plain titanium abutment, to avoid grey show-through. A **screw-retained** crown where the access channel allows, because it is retrievable and avoids subgingival cement. If it must be cemented, I would use a **customised abutment with the margin no more than 1 mm subgingivally** and clean meticulously."

Q Why is subgingival cement such a concern?

"Because residual cement is a well-recognised cause of **peri-implant mucositis and peri-implantitis**, and it is very difficult to detect radiographically or to remove once it is deep. The deeper the margin, the more cement is left. That is the argument for screw retention wherever the trajectory permits."

Q What would you record?

"The **options discussed and declined**, in her own words where possible; the specific risks named — black triangle, long crowns, recession, grey show-through, graft exposure and failure; that a **mock-up was shown, photographed and approved**; the ITI Aesthetic Risk Assessment; the CBCT justification and report under IR(ME)R; implant details, lot numbers, torque values and graft materials; and the **timescale and cost** given. A signed form is not consent — the note of the discussion is."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcomes, disappointment and its management

Q What would you tell her to expect, in your own words to her?

"'**The teeth themselves we can make look excellent. The difficulty is the gum between them — you have lost about three millimetres of bone there and I cannot reliably grow that back. We should plan on the basis that some of what looks like gum will be pink porcelain. If grafting is involved this will take nine to twelve months. There is a real chance a graft becomes exposed and has to be redone. And the mock-up we make is the honest preview — I would far rather you turned it down at that stage than were disappointed at the end.**'"

Q How would you measure the result objectively?

"**Standardised photographs** against baseline; the **Pink and White Aesthetic Score**; the **papilla index**; the mid-facial margin level compared with the 12 and 22; peri-implant probing depths, bleeding and suppuration; radiographic bone levels against the baseline taken at fitting; and, most importantly, **her own rating against what she said she wanted at the start**."

Q At one year the mid-facial margin on the 11 has receded 1 mm. Why?

"Most likely a **thin buccal plate or a buccally positioned implant**, with or without a phenotype that was never thick enough. Contributing factors are an over-contoured crown, a deep cement margin causing chronic inflammation, and traumatic brushing. I would examine for inflammation first, because inflammatory recession may partly resolve once it is controlled."

Q And how would you manage it?

"Control the inflammation, correct an over-contoured or cement-retentive restoration, and reinforce hygiene. If the margin is still unacceptable and the implant is otherwise healthy, a **connective tissue graft** with or without a coronally advanced flap can improve it, though it is less predictable than around a tooth. If the implant is fundamentally malpositioned, no soft tissue procedure will correct it and the honest answer is prosthetic camouflage with pink ceramic, or explantation."

Q What is the commonest long-term disappointment in this specific case?

"A **black triangle between the two units**, and crowns that are longer than her natural centrals were. Both are predictable from the outset, which is precisely why they belong in the consent conversation and the mock-up rather than in the complaint letter."

Q How would you prevent that disappointment?

"By **agreeing the endpoint before the drill**: wax-up, mock-up, photographs, and an explicit agreement about whether pink ceramic is acceptable to her. Then by choosing the design that biology supports — a single implant with a cantilever, or a resin-bonded bridge — rather than the one she first asked for. Prevention here is almost entirely a communication and planning task."

Q How would you know the treatment had succeeded?

"That she smiles without covering her mouth and does not think about the teeth. Objectively: stable implants with no bleeding or suppuration, bone levels stable against baseline, an acceptable Pink and White Aesthetic Score, and a maintenance regime she attends and can manage at home."

WATCH OUT

Come back to the papilla: state the contact-point-to-crest distance and whether the supporting attachment belongs to a tooth, an implant or a pontic. That single frame explains almost every aesthetic decision in this case and buys you a sentence of thinking time.

Running this case to time

Time check. The flow above holds 37 questions and about 2,545 words of model answer. At 140 words a minute plus roughly five seconds a question, that comes to about **twenty-one minutes** — an overrun of **six**. What goes first is the **provisionalisation and abutment-material** sub-chain in Domain 3: emergence shaping, the customised impression coping, zirconia versus titanium and the cement argument compress into one question, because the definitive work here is referred and the marks sit in **position, graft choice and the papilla**. If your answers run past **thirty seconds**, you will never reach Domain 4 — which in this case is the disappointment, and the whole point.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT and photographs up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A healthy 40-year-old, **11 and 21 lost to trauma six months ago**, medium smile line, medium phenotype, a **5 mm ridge with a 3 mm vertical deficiency against the adjacent CEJs**. Named for what it is: a **two-adjacent-tooth aesthetic case with a combined horizontal and vertical defect**. Then the biology, volunteered in the opening: **two adjacent implants cannot support an inter-implant papilla predictably**, because the inter-implant crest sits lower than an interdental one — **a black triangle is the expected outcome, not the exception**. What remains: what she wants to see, what she will accept prosthetically, and whether an implant is the right vehicle at all.
1:20–4:30	Domain 1: the anatomical and biological groundwork, then reading her ridge	Undergraduate ground first. The **labial plate under 1 mm in most central incisor sites**, the **incisive canal** enlarging with resorption, the labial concavity that dictates angulation. Bundle bone, with **Schropp's 50% of width in the first year**. The **dentogingival junction** described properly — 1 mm connective tissue, 1 mm junctional epithelium, a **~2 mm supracrestal attachment** — then the peri-implant version: **no cementum, no ligament, fibres parallel, supraperiosteal supply only**. Then the arithmetic the case turns on: **Tarnow — 5 mm always, 6 mm 56%, 7 mm 27%** between teeth; **4.5–5 mm** implant to tooth; **about 3.5 mm between two implants**; and **crestal loss of ~1 mm under a 3 mm inter-implant distance against 0.45 mm above it**. Her 5 mm ridge judged **marginal**; her 3 mm vertical deficit meaning **long crowns from a lower base**, with **vertical augmentation the least predictable procedure there is**. **ITI ERA: high aesthetic risk.**

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the options and the argument between them	All seven offered, including **no treatment** and the **removable partial denture with a flange as the only option that reliably replaces lost hard and soft tissue volume**. Then the two that matter: **one implant with a cantilever pontic**, because a papilla between an implant crown and a pontic is supported by ridge tissue and one platform rather than two — **trading a difficult biological problem for a manageable mechanical one** — with the mechanical risks named and the rule that in a bruxist you would not do it. And the **resin-bonded bridge with ovate pontics over soft tissue augmentation**, argued at its strongest, then argued against on **debonding**. Recommend both, **shown on a mock-up**, and let her decide **with her eyes rather than on your description**. **SAC: Complex surgically and prosthodontically — no reading of this case makes it Straightforward.**
7:30–11:30	Domain 3: how you would actually deliver it	**From the teeth backwards** — wax-up, intra-oral mock-up, then a **CBCT with a radiographic guide**, because how much bone is missing and in which direction is a different question from how much is there. Position in three dimensions: **1.5 mm from the adjacent root, 1.5–2 mm palatal to the line joining the adjacent CEJs, and 3 mm apical to the planned gingival margin — not to the existing crest, which is 3 mm low**. **Simultaneous GBR** with a slowly resorbing xenograft under a collagen membrane — combined with a **subepithelial connective tissue graft**, and the reason volunteered: **soft tissue thickness predicts mid-facial stability better than another millimetre of particulate**. Healing under something that **does not load the graft** — a bonded temporary with relieved ovate pontics, never an acrylic denture. Then the provisional and abutment, with **screw retention wherever the trajectory allows**.
11:30–14:30	Domain 4: outcomes, disappointment and its management	Measured objectively — **standardised photographs, Pink and White Aesthetic Score, papilla index, mid-facial margin against the 12 and 22** — and, weighted highest, **her own rating against what she said she wanted at the start**. Then 1 mm of mid-facial recession at a year: **a thin plate or a buccal implant**, with over-contour, deep cement and traumatic brushing contributing — **inflammation examined first, because inflammatory recession may partly resolve**. Managed by control, correction, then a **connective tissue graft**, and the honest endpoint: if the implant is malpositioned **no soft tissue procedure will correct it**. And the commonest disappointment named: **a black triangle and crowns longer than her natural centrals** — both predictable from the outset, which is why they belong in the mock-up and not in a complaint letter.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	In her words: *"The teeth themselves we can make look excellent. The difficulty is the gum between them — you have lost about three millimetres of bone there and I cannot reliably grow it back. Plan on the basis that some of what looks like gum will be pink porcelain… and the mock-up is the honest preview. I would far rather you turned it down at that stage than were disappointed at the end."* Recorded: the options **discussed and declined in her own words**, the risks named individually — **black triangle, long crowns, recession, grey show-through, graft exposure** — that the **mock-up was shown, photographed and approved**, the ERA, and the timescale and cost. **A signed form is not consent; the note of the discussion is.**

The spine — the twelve answers to have ready if nothing else

1 The labial plate, the incisive canal, and why the plate resorbs · 2 The dentogingival junction described · 3 How the peri-implant attachment differs, and the three consequences · 4 Tarnow between teeth · 5 The papilla height between two implants, and why the crest is lower · 6 Whether 5 mm of width is enough · 7 What the 3 mm vertical deficiency actually costs her · 8 The ITI aesthetic risk assessment on her · 9 Why one implant and a cantilever beats two implants · 10 The resin-bonded bridge argued both ways · 11 Implant position in three dimensions, measured from the planned margin · 12 What you say when she insists on two implants.

Where the marks are lost in this case

Planning two implants because she asked for two teeth. The inter-implant crest remodels from both sides, the soft tissue above it is only about **3.5 mm**, and the black triangle follows. Volunteer that in the opening presentation — not when the examiner produces it in Domain 4 — and then offer the design biology supports: **one implant with a cantilever pontic, or a resin-bonded bridge with ovate pontics**.

Measuring the implant depth from the bone you can see. Her crest is **3 mm low**, so an implant placed 3 mm below *it* is 6 mm below where the margin needs to be. The position is planned **from the wax-up — 3 mm apical to the planned gingival margin** — which is exactly why the sequence runs mock-up, then CBCT with a guide, then surgery. Candidates who plan from the CBCT alone build the deficiency into the result.

Never saying where you stop. Assessment, wax-up and mock-up, the options conversation, an interim bonded or removable prosthesis and long-term maintenance are yours. **Vertical ridge augmentation and the two-implant aesthetic reconstruction are not** — they are referred, with the consent conversation completed first and you staying involved as the referring clinician. And if she insists on two implants the answer is not refusal: it is **"I can make you two very good-looking teeth; what I cannot promise you is the triangle of gum between them"**, shown on a mock-up, and recorded.

SOE Case 18 — The adolescent with missing lateral incisors

THE CASE

15-year-old girl with **congenitally absent 12 and 22**. Orthodontic treatment has just been completed, creating spaces of **6.5 mm** at each site. Her mother is pressing for implants "before she starts college".

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 15-year-old girl with **hypodontia of both maxillary lateral incisors**, orthodontically created 6.5 mm spaces, and family pressure for a rapid definitive solution. The overriding issue is **skeletal and dento-alveolar growth**. An implant behaves as an ankylosed tooth: it does not move with the developing alveolus, so it becomes progressively **infra-occluded**, which in the anterior maxilla is a conspicuous and effectively irreversible aesthetic failure. Craniofacial growth continues into the late teens in girls and later in boys, and continues at a low level into adulthood. So my position is that implants are the right long-term plan but the wrong treatment today. What I need to establish is whether growth has ceased, whether space opening was the right orthodontic decision at all, what interim prosthesis she will accept, and how to manage her mother's expectations."

Domain 1 — Clinical interpretation and reasoning

Chain: the developmental groundwork

Q Define hypodontia and tell me how common it is.

"**Hypodontia** is the developmental absence of one to six teeth excluding third molars; **oligodontia** is six or more; **anodontia** is complete absence. Excluding third molars, hypodontia affects roughly **5 to 6%** of the population, and it is somewhat commoner in females. Third molars are absent in a further quarter or so."

Q Which teeth are most commonly absent, and what else would you look for?

"Most commonly the **mandibular second premolars**, then the **maxillary lateral incisors**, then the maxillary second premolars. Associated findings are **microdontia and peg-shaped laterals** — often on the contralateral side to an absent one — **delayed eruption and delayed dental development**, **retained deciduous teeth**, **taurodontism**, and **ectopic or palatally displaced canines**. I would also ask about a family history and about hair, nails and sweating, because oligodontia can signal **ectodermal dysplasia**."

Q Why does hypodontia matter for the bone at these sites?

"Because **alveolar bone develops in response to the tooth germ and the erupting tooth**. Where a tooth has never formed, the alveolar process never fully develops, so the ridge is characteristically **narrow buccopalatally** and sometimes deficient vertically. That is quite different from an extraction site, which had normal alveolus and then resorbed. It means I should expect to need augmentation, and it is one reason orthodontic space opening can leave a knife-edge ridge."

Q How does the anterior maxilla grow?

"By **sutural growth** at the circummaxillary sutures displacing the maxilla downwards and forwards, by **surface remodelling** — with resorption on the anterior surface of the premaxilla and deposition posteriorly — and by **alveolar growth with dental eruption**, which continues as long as the teeth continue to erupt. It is that last component, vertical alveolar development with continuous eruption, that keeps changing after skeletal growth has largely finished."

Q So why does an implant behave like an ankylosed tooth?

"Because it is **osseointegrated with no periodontal ligament**. A natural tooth erupts and drifts with the alveolus throughout life because the PDL remodels the socket. An implant is locked to the bone it was placed in, so as the adjacent teeth continue to erupt and the alveolus grows, the implant crown is left behind — **infra-occlusion**, with the margin exposed, the incisal edge short, and the crown appearing to sink."

Q When does growth actually stop?

"Skeletal growth is essentially complete around **17 or 18 in girls and 20 or 21 in boys**, roughly **two years after peak height velocity**, which in girls is usually a year or so after menarche. But **vertical alveolar change continues at a low level into the third and fourth decades**, particularly in patients with a long face pattern or continued eruption, so I cannot promise her that infra-occlusion will never occur — only that the risk falls sharply once growth is complete."

Chain: assessing this patient

Q How would you determine whether growth has ceased?

"Not by chronological age alone. Most practically, a period of **clinical observation with serial study casts and standardised photographs** showing stable facial height and a stable dentition, alongside the history of menarche and when she last grew in height or shoe size. **Serial cephalometric radiographs 6 to 12 months apart showing superimposable growth** are the classical method, but I would have to justify that exposure. **Hand-wrist radiographs** or **cervical vertebral maturation** on an existing cephalogram are the alternatives."

Q What would you assess clinically at the two sites?

"Space **at the crest and at the incisal level** — they are often different — and the mesiodistal width in relation to the contralateral tooth and the golden proportion. **Ridge width and contour** by palpation, the **height and thickness of keratinised tissue**, the **papilla height and bone level on the adjacent teeth**, the **smile line and lip dynamics**, and the **root angulation of the 11, 13, 21 and 23** — orthodontists frequently leave adequate crown space but converging roots."

Q What imaging would you take today?

"Today, **none beyond what orthodontics has already produced**. A periapical or the existing panoramic tells me about root angulation and space. A **CBCT is not justified now** because it would not change management for several years and she is 15 — **IR(ME)R justification and ALARP** apply with particular force in a growing child. I would take the CBCT when she is ready for surgical planning."

Q Classify this case on SAC.

"**Complex.** Anterior maxilla, high aesthetic risk, a growing patient, an expected horizontal ridge deficiency at a site where the alveolus never developed, narrow-diameter implants and thin adjacent roots. Under ITI, a single anterior implant in an aesthetic zone is at best Advanced; adding growth and a probable augmentation makes it **Complex, and outside what I would undertake alone**."

Domain 2 — Clinical decision making and treatment planning

Chain: revisiting the orthodontic decision

Q Was space opening the right decision?

"It may not have been, and I would want that re-opened with the orthodontist before anything is committed. **Canine substitution** — closing the space, bringing the canines into the lateral position, reshaping and recontouring them, and building up the first premolars as canines — avoids a prosthesis altogether and gives a result that grows with her. In a 15-year-old, that is a real advantage."

Q When does canine substitution work best?

"When the canine is **narrow, pale and not markedly bulbous**, when the patient is **Class II or Class I with crowding** so space closure helps the occlusion, when the **smile line is not high** so a slightly gingivally-positioned

canine margin is hidden, and where the first premolar is of reasonable shape and colour to serve as a canine. It works less well in a Class III patient, where the space is needed, or where the canine is dark and heavily built."

Q You have argued for canine substitution — argue against it.

"It converts a **group-function or canine-guided occlusion into premolar guidance**, which some regard as less protective. It requires irreversible enamel reduction and composite or veneer work on four teeth, which is lifelong maintenance in a 15-year-old. The gingival margins can be uneven, and if it fails aesthetically it is very hard to undo. Space opening preserves the option of an ideal-looking lateral incisor later."

Q Your orthodontic colleague says the spaces are already open, so just prosthetically restore them. Why might she be right?

"Because re-treating orthodontically means another 18 months in appliances for a 15-year-old who has just finished, with the burden of demineralisation and root resorption risk that carries. Space opening also gives a **restoration in the correct anatomical position** rather than a reshaped canine, and it keeps the definitive plan simple. If she is happy with the spaces and the occlusion is good, re-opening the debate may cause more harm than good."

Chain: the interim prosthesis

Q What do you offer her in the meantime?

"An interim prosthesis that is aesthetic, reversible and adjustable. **A resin-bonded cantilever bridge from each canine** — excellent aesthetics, minimal or no preparation, adjustable, and it maintains the space; my preferred option. **The orthodontic retainer with prosthetic teeth added** — practical, and she needs a retainer anyway. **A removable acrylic partial denture** — simple and cheap but less well accepted socially. **Composite build-ups or a composite-and-fibre splint** — quick, cheap and easily repaired."

Q Why a cantilever rather than a fixed-fixed resin-bonded bridge?

"Because a **fixed-fixed design fails by debonding at one retainer**, which then goes undetected and caries develops beneath it. A **single-retainer cantilever** either stays on or comes off cleanly, and the evidence for cantilever resin-bonded bridges in the anterior maxilla is good — survival in the region of **80 to 90% at five years**. It also avoids the problem that the abutments have different mobilities and paths of insertion."

Q Which abutment would you choose, and how would you prepare it?

"The **canine**, because it has the largest bonding area and the best enamel. Preparation is **minimal and confined to enamel**: a supragingival chamfer or none at all, a defined path of insertion, cingulum rest seats for support, and adequate wrap-around for resistance form. A **cobalt-chromium or nickel-chromium framework**, sandblasted and cemented with a **10-MDP-containing resin cement** under rubber dam or good isolation."

Q How does the interim prosthesis interact with retention?

"It must not compromise it. **Space maintenance is the priority** — if the space closes even a millimetre, the implant plan is lost. So the retainer regime continues, and either the pontic is incorporated into the retainer or the resin-bonded bridge itself acts as a fixed retainer for those teeth, with a bonded retainer elsewhere as the orthodontist prescribes. I would agree the plan with the orthodontist in writing rather than assume."

Chain: communication and consent

Q What do you say to the mother, in your own words?

" 'If we put an implant in now, it will stay exactly where we put it while the rest of her face and jaw keep growing. In a few years that tooth would look short and high compared with the ones next to it, and the only way to fix it would be to take it out and start again. What I would like to do is give her something that looks good now and can be adjusted as she grows, and place the implant once we are confident she has finished growing — probably in her late teens or early twenties.' "

Q What are the consent issues with a 15-year-old?

"I assess whether she has capacity to consent for herself. At 15 she may well be **Gillick competent**; if she is, her consent is valid and I would still involve her mother with her agreement. If she is not competent, consent comes from someone with **parental responsibility**. Either way, a **competent young person's refusal must be taken very seriously**, and I would never proceed with elective aesthetic treatment because a parent is keen and the patient is reluctant."

Q What would you record?

"The history including growth and menarche, the clinical findings and space measurements, photographs and study casts as the growth baseline, the **options discussed including canine substitution**, the reasons for deferring implants, the specific warning about infra-occlusion, that I spoke to the patient separately and assessed her competence, that her mother was present and what she was told, the agreed interim prosthesis, and the review interval. I would also write to the orthodontist and the GDP."

Domain 3 — Decision making in relation to treatment procedures

Chain: eventually placing the implant

Q At what point would you finally place?

"When **serial records over at least 12 months show no change** in facial height or dental relationships, she is at least in her late teens, and she herself is asking for it rather than her mother. I would rather be a year late than a year early — the cost of waiting is an interim prosthesis, and the cost of being early is an unrestorable aesthetic failure."

Q What is specific about the lateral incisor site dimensionally?

"A 6.5 mm space demands a **narrow-diameter implant**, around **3.3 mm**, ideally in a **titanium-zirconium alloy** for strength at that diameter. I need **1.5 mm of bone between the implant and each adjacent root** and at least **1 mm of buccal bone** — ideally 2 mm. That arithmetic barely works in 6.5 mm, so root angulation must be corrected orthodontically first if the roots converge."

Q Give me the three-dimensional position.

"**Apico-coronally**, the shoulder **3 mm apical to the planned mid-facial gingival margin**. **Bucco-palatally**, within the comfort zone, roughly **1 to 2 mm palatal to the line joining the emergence of the adjacent teeth**, with the buccal plate preserved. **Mesio-distally**, centred, with at least 1.5 mm to each adjacent root. Angulation through the cingulum of the planned crown so the access can be screw-retained."

Q You expect to augment. What and when?

"**Contour augmentation with guided bone regeneration** — a deproteinised bovine bone mineral with a resorbable collagen membrane, placed simultaneously with the implant if the defect is a dehiscence with an intact envelope and the implant has primary stability. If the ridge is a knife edge that will not accept the implant in the correct position, I would **stage it**: graft first, wait four to six months, then place. Soft tissue is often the greater problem — a **connective tissue graft** may be needed for thickness."

Q Where is the boundary of your competence here?

"Placing a straightforward single implant in an adult with adequate bone is within my scope. **This is not that.** A high-aesthetic-risk site in a young patient, needing a narrow implant between converging roots with simultaneous or staged augmentation, is **Complex** — I would **refer to a specialist in restorative dentistry or oral surgery** with an implant practice, and stay involved for the interim prosthesis, the orthodontic liaison and the long-term maintenance."

Q If you were doing it, guided or freehand?

"**Static guided surgery from a CBCT merged with an intra-oral scan and a diagnostic wax-up.** The tolerances here are a millimetre or so between roots, so a fully guided protocol is worth the cost. I would still raise a flap for the augmentation and to visualise the buccal plate, and I would use the guide for the osteotomy rather than trusting it blindly."

Q Immediate, early or conventional restoration?

"**Conventional or early**, not immediate. There is no soft tissue advantage worth the risk of losing a narrow implant in a site that has just been grafted. I would allow **three to four months**, longer if grafted, then a screw-retained provisional crown to develop the emergence profile before the definitive."

Q Why does the provisional matter so much at this site?

"Because the **emergence profile and the papillae are created by the provisional, not by the definitive crown**. I use it to shape the peri-implant mucosa progressively over several weeks, then **capture that contour with a customised impression coping** so the technician reproduces it. Going straight to a definitive crown means the tissue dictates the crown rather than the reverse."

Q Screw-retained or cement-retained?

"**Screw-retained wherever the angulation allows**, because there is no risk of retained subgingival cement — a well-recognised cause of peri-implantitis — and it is retrievable, which matters enormously in a patient who may need the crown remade for infra-occlusion in twenty years. If the access would come through the labial face, I would use an **angulated screw channel** before I resorted to cement."

Q Which material for the definitive crown?

"A **zirconia or lithium disilicate crown on a titanium base**, screw-retained. Ceramic gives the translucency needed at a lateral incisor, and the titanium base preserves a metal-to-metal implant connection rather than ceramic on titanium. I would shade-match with the contralateral central and canine and check it under different lighting, because a single anterior unit is the hardest shade match in dentistry."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: living with the result

Q Ten years on, the implant crown looks short. What has happened and what do you do?

"**Continued vertical alveolar development and eruption of the adjacent teeth** — infra-occlusion. First I quantify it against the baseline photographs and casts. If it is **under about a millimetre**, I would remake the crown with a longer incisal edge and adjusted contours. If it is greater, options are a new crown with modified emergence, composite additions to the adjacent teeth to re-level, orthodontic intrusion of the adjacent teeth, or — as a last resort — **removal and replacement of the implant**. This is exactly why I warned her at consent and why the crown is screw-retained."

Q How much infra-occlusion should she expect?

"It varies too much to promise a figure, but the concept is that it is **progressive and lifelong**, greater the younger the patient at placement and greater in long-face growth patterns. It is measurable in millimetres over decades rather than years. The honest consent line is that the crown will probably need to be remade at least once in her lifetime, and that the implant itself will very likely still be sound when it is."

Q She develops a black triangle mesial to the implant crown. Why, and what can you do?

"Because papilla height at an implant depends on the **bone level on the adjacent natural tooth**, not on the implant — Tarnow's work shows the papilla fills predictably when the contact point is within about **5 mm of the crest**. So I would measure that distance. Management is prosthetic first: **lengthen and lower the contact point** and adjust the emergence. Surgical papilla reconstruction is unpredictable, and composite addition to the adjacent tooth is often the kindest answer."

Q The resin-bonded bridge debonds three times in eighteen months. What are you thinking?

> "Something is wrong with **isolation, design or occlusion**, not with the technique in principle. I would check for a **protrusive or excursive contact on the pontic**, inadequate wrap-around or no defined path of insertion, contamination at cementation, and parafunction. Remedies: re-prepare with better resistance form, remove all excursive contact from the pontic, rebond under rubber dam with an MDP cement, and consider a removable retainer with a pontic if it keeps failing."

Q What would you monitor, and how often?

> "Before placement, **six-monthly review with standardised photographs and casts** to track growth and confirm the space is maintained. After placement, review at three and six months and then annually: **peri-implant probing at six sites**, bleeding and suppuration, radiographic bone level against the baseline taken at fitting, the crown margin and contacts, and — specifically in her — **the incisal and gingival levels compared with the contralateral central and canine**."

Q How would you know this treatment had succeeded?

> "That she reached her early twenties with the space intact, an implant placed once in the right position, stable bone and healthy tissues, and a crown she does not think about when she smiles. And that she understands the maintenance ahead of her — a successful outcome here is measured over decades, not at the fit appointment."

WATCH OUT

Come back to the single organising principle: **implants do not grow**. Say which part of the problem that principle governs — timing, the interim prosthesis, the consent conversation, or the long-term maintenance — and the rest of the answer follows.

Running this case to time

Time check. The flow above holds 38 questions and about 2,730 words of model answer. At 140 words a minute plus roughly five seconds a question, that comes to nearly **twenty-three minutes** — an overrun of **seven and a half**. What goes first is the **restorative detail** in Domain 3: guided versus freehand, provisionalisation, screw versus cement and crown material compress into one question, because that surgery is years away and referred, and the marks in this case sit in **timing, the interim prosthesis and the consent**. If your answers run past **thirty seconds**, you will never reach Domain 4 — which is the infra-occlusion at ten years, and the reason for the whole plan.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the orthodontic records up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 15-year-old with **hypodontia of both maxillary laterals**, orthodontically created **6.5 mm spaces**, and family pressure for a definitive answer. Then the organising principle, stated first: **an implant behaves as an ankylosed tooth** — it does not move with the developing alveolus, so it becomes **progressively infra-occluded**, which anteriorly is conspicuous and effectively irreversible. So: **implants are the right long-term plan and the wrong treatment today**. Four things to establish — has growth ceased, was space opening right at all, what interim prosthesis she will accept, and **how to manage her mother's expectations**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the developmental groundwork, then assessing this patient	Undergraduate ground first. **Hypodontia, oligodontia, anodontia** defined, at **5–6%** excluding third molars; the associated findings — **microdontia and peg laterals, delayed development, retained deciduous teeth, ectopic canines** — with **ectodermal dysplasia** asked about. Then the point specific to this site: **alveolar bone develops in response to the tooth germ**, so where a tooth never formed the ridge is **narrow buccopalatally** — a different problem from a resorbed extraction site. Maxillary growth by **sutural displacement, surface remodelling and alveolar growth with eruption**, and it is the last that keeps changing. Cessation at about **17–18 in girls, 20–21 in boys, two years after peak height velocity** — but **vertical alveolar change continues into the third and fourth decades**. Then how you confirm it: **serial casts and standardised photographs**, cephalometric superimposition, or **cervical vertebral maturation**. And **no CBCT today — IR(ME)R and ALARP** apply with particular force at 15.
4:30–7:30	Domain 2: revisiting the orthodontics, the interim prosthesis, then consent	**Re-open the space-opening decision with the orthodontist before anything is committed. Canine substitution** described, with the conditions under which it works — **narrow, pale canine, Class II or Class I with crowding, not a high smile line** — and its advantage at 15: **it grows with her**. Then argued against: premolar guidance, irreversible enamel reduction on four teeth, hard to undo. And the orthodontist's counter granted — another 18 months in appliances, with demineralisation and root resorption risk. Then the interim: a **resin-bonded cantilever from each canine**, and why **cantilever not fixed-fixed** — a fixed-fixed fails by **undetected debonding at one retainer with caries beneath**, and cantilever survival is **80–90% at five years**. **Space maintenance is the priority**: lose a millimetre and the implant plan is lost, so the retention regime is agreed **in writing**. Then consent: **Gillick competence assessed**, parental responsibility if not, and **a competent young person's refusal taken very seriously**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: eventually placing the implant	The trigger, stated as a rule: **serial records over at least 12 months showing no change**, late teens at least, and **she** asking for it rather than her mother — **"I would rather be a year late than a year early."** Then the arithmetic of the site: 6.5 mm demands a **narrow-diameter implant around 3.3 mm, ideally titanium-zirconium**, needing **1.5 mm to each adjacent root and 1–2 mm buccally** — which **barely works in 6.5 mm**, so converging roots must be corrected orthodontically first. Position: **3 mm apical to the planned mid-facial margin, 1–2 mm palatal, centred, angled through the cingulum**. Augmentation expected: **contour GBR simultaneously** where the defect is a dehiscence with primary stability, **staged** where the ridge is a knife edge — and **soft tissue is often the greater problem**. Then, compressed: guided surgery, a **screw-retained provisional to create the emergence**, and **conventional or early, never immediate**.
11:30–14:30	Domain 4: living with the result	Ten years on and the crown looks short: **continued vertical alveolar development and eruption of the adjacent teeth**, quantified against the **baseline photographs and casts** — managed by scale: under a millimetre remade, more than that by composite re-levelling, orthodontic intrusion, or removal as a last resort. The honest figure refused: it is **progressive and lifelong, worse the younger the patient and worse in long-face patterns**, and the consent line is that **the crown will probably be remade at least once in her lifetime while the implant is still sound**. The black triangle explained by **Tarnow — papilla height depends on the adjacent tooth's bone level** — so it is managed **prosthetically first**. Repeated debonding read as **isolation, design or occlusion**, not a failed technique. Monitoring **six-monthly with photographs and casts before placement**, then annually.
14:30–15:00	Closes on the patient and the record	To the mother, in her words: *"If we put an implant in now, it will stay exactly where we put it while the rest of her face and jaw keep growing. In a few years that tooth would look short and high next to the others, and the only way to fix it would be to take it out and start again. I'd like to give her something that looks good now and can be adjusted as she grows, and place the implant once we're confident she's finished."* Recorded: growth history and menarche, space measurements, **photographs and casts as the growth baseline**, the options **including canine substitution**, the warning about infra-occlusion, that **you spoke to her separately and assessed her competence**, the interim prosthesis and review interval — and letters to the orthodontist and GDP.

The spine — the twelve answers to have ready if nothing else

1 Hypodontia defined, its prevalence and its associated findings · 2 Why the ridge at these sites was never normal · 3 How the anterior maxilla grows, and which component keeps changing · 4 Why an implant behaves as an ankylosed tooth · 5 When growth stops, and why that is not the whole answer · 6 How

you would actually determine growth has ceased · 7 Why no CBCT today · 8 Canine substitution, argued both ways · 9 The interim prosthesis, and cantilever rather than fixed-fixed · 10 Space maintenance as the priority · 11 Gillick competence, and whose refusal counts · 12 The infra-occluded crown at ten years.

Where the marks are lost in this case

Answering the mother's question. She wants implants before college, and the mark is for declining while keeping everyone on side — **something that looks good now and can be adjusted, with the implant when growth is complete**. And the consent point is not decoration: at 15 the patient may be **Gillick competent**, her consent is then valid, and **her refusal must be taken very seriously**. Never proceed with elective aesthetic treatment because the parent is keen.

Never questioning the orthodontics. The spaces exist because someone chose to open them, and **canine substitution** — a result that grows with her and needs no prosthesis — should be re-opened with the orthodontist before anything is committed. Volunteer that unprompted, argue it honestly in both directions, and grant the colleague's point that re-treatment means another 18 months in appliances. Candidates who accept the 6.5 mm spaces as a fixed premise have skipped the best question in the case.

Planning surgery that is not yours. A single anterior implant in an aesthetic zone is Advanced at best; **a narrow implant between converging roots, in a site whose alveolus never developed, with augmentation, in a patient whose growth must first be proved complete, is Complex** — and at this level it is referred to a specialist in restorative dentistry or oral surgery. What stays yours, and matters for a decade: the **interim prosthesis, the space maintenance, the orthodontic liaison, the growth records and the long-term maintenance**. Say where you stop before you describe the osteotomy.

SOE Case 19 — The free-end saddle and the shortened dental arch

THE CASE

58-year-old man, fit and well, non-smoker. Missing **35, 36 and 37** on the lower left; the **34** and the whole upper arch are intact and unrestored. He manages well and has "got used to chewing on the other side", but his dentist has told him he should replace the teeth.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A healthy 58-year-old non-smoker with a **lower left free-end saddle** — a Kennedy Class II — who is functioning satisfactorily and attending because he has been told he should have treatment, not because he wants it. That framing matters, because the first question is **whether he needs anything at all**, and 'no treatment' is a genuine contender here rather than a formality. In his favour: he is medically fit, a non-smoker, has an unrestored opposing arch, and has already adapted. Against: the upper 26 and 27 are unopposed and may over-erupt, and the 34 may drift. What I need to establish is what he actually wants to be able to do, the periodontal and occlusal status of the remaining dentition, and whether the anterior loop of the mental nerve and the lingual concavity would permit implants if he chose them."

Domain 1 — Clinical interpretation and reasoning

Chain: the groundwork — classifying and understanding the saddle

Q Classify this edentulous arch.

"**Kennedy Class II, modification 0** — a unilateral free-end saddle with no other edentulous areas. Applegate's rules apply: classification is determined after any extractions, the most posterior edentulous area determines the class, and additional areas are modifications, of which there are none here. The upper arch is complete."

Q Why does the Kennedy class matter clinically?

"Because it determines the **support**. Classes I and II are **tooth- and mucosa-borne**, so the saddle can move independently of the abutment teeth as the mucosa is displaced. That produces **rotation about the fulcrum line** through the most distal rests, torquing forces on the abutment, and progressive resorption under the saddle. Classes III and IV are entirely tooth-borne and behave predictably. A free-end saddle is the design problem in removable prosthodontics for exactly this reason."

Q Describe the anatomy relevant to implants in the lower left posterior mandible.

"The **inferior alveolar canal** running forwards in the body of the mandible and exiting at the **mental foramen**, usually below and between the premolar apices, often with an **anterior loop** running mesial to the foramen before turning back. The **external oblique ridge** buccally and the **mylohyoid ridge** lingually. Below the mylohyoid ridge, the **submandibular fossa** — a lingual concavity that undercuts the molar region. And the **lingual nerve** running close to the lingual plate at the level of the third molar."

Q What is the anterior loop and why does it matter?

"It is the portion of the inferior alveolar nerve that passes **anterior to the mental foramen before looping back** to exit. It is present in a substantial proportion of patients and can extend several millimetres. It matters because if I measure only to the radiographic foramen on a panoramic, I may place an implant into the loop and produce **altered sensation of the lower lip and chin**. It is one of the reasons a **CBCT is justified** here rather than a panoramic alone."

Q And the lingual concavity?

"The **submandibular fossa** creates a lingual undercut in the molar region, so the ridge is often much narrower than it looks on a panoramic. Drilling along the perceived long axis of the crest can **perforate the lingual plate** — which risks damage to the sublingual or submental vessels and, in the worst case, a **floor-of-mouth haematoma with airway compromise**. It is the reason I want cross-sectional imaging and why I would angle the osteotomy slightly buccally in the molar region."

Q What would you examine clinically?

"The **periodontal status of the whole dentition** with a six-point chart, because the plan is worthless if the remaining teeth are not sound. The **34** for drift, tipping and mobility. The **26 and 27** for over-eruption and whether they still have an occlusal contact. **Ridge form and width by palpation**, keratinised tissue width, and the **inter-arch space** and vertical dimension. His **occlusion** — the scheme in excursions, the guidance, and any signs of parafunction. And plaque control and caries risk, since implants demand more of both."

Q What imaging?

"**Periapicals or a panoramic first** to see the bone levels, the canal, any retained roots and the 26/27 position. If he is seriously considering implants, a **CBCT of the left posterior mandible**, justified under IR(ME)R because it is the only way to assess the canal position, the anterior loop and the lingual concavity in cross-section. I would take it with a **radiographic guide or a scan appliance** so the planned tooth position is visible."

Q What are you measuring on that CBCT?

"**Bone height above the canal** at each planned site, **ridge width at the crest and 5 mm apical**, the **buccolingual angulation** of the ridge and the depth of the lingual concavity, the **position of the foramen and any anterior loop**, and the **bone density**. I would also check the contralateral side and the sinus incidentally, and report the whole volume — I am responsible for the entire field of view."

Q What bone quality do you expect and why does it matter?

"**D2 to D3** on Misch in the posterior mandible — thick to thin cortex with coarse to fine trabecular bone. That is favourable: enough cortex for primary stability without the heat and vascularity problems of D1. If it turns out to be D3 or D4 I would **under-prepare the osteotomy** to improve stability and extend the healing period before loading."

Q Classify this case on SAC.

"Restoratively, a posterior mandibular free-end saddle with an intact opposing arch and no aesthetic demand is **Straightforward to Advanced**. Surgically, it is **Advanced** because of the proximity of the inferior alveolar canal and the lingual concavity — the anatomical risk, not the prosthodontics, is what raises it."

Domain 2 — Clinical decision making and treatment planning

Chain: does he need treatment at all?

Q Explain the shortened dental arch concept.

"That a dentition of about **20 teeth, including four occluding premolar pairs**, provides adequate function, comfort and appearance for most adults without replacement of the molars. It comes from **Käyser's** work and has been supported by long-term follow-up: no measurable increase in temporomandibular disorder, no accelerated periodontal breakdown, and patient satisfaction comparable with restored arches. The **World Health Organization** has endorsed it as an acceptable treatment goal."

Q So is this man a shortened dental arch?

"Not quite — he has a **unilateral** shortening, with occlusion to the 34 on the left and to the molars on the right. That is an asymmetric arch rather than the classical bilateral shortened arch, so the evidence applies less directly. But he is **asymptomatic, has adapted, and has occluding contacts on both sides**, so the principle still holds: accepting it is a **legitimate positive choice, not a failure to treat**."

Q What are the risks of leaving it?

"**Over-eruption of the 26 and 27**, which are unopposed distal to the 34 — this is the real risk and it can eventually make prosthetic replacement impossible without enameloplasty, elective root treatment or extraction. **Drifting or tipping of the 34**. **Occlusal instability** and loss of vertical dimension on that side if the right side then breaks down. And a slow increase in load on the remaining teeth."

Q You have argued for the shortened arch — argue against it.

"He is only 58, so he has perhaps thirty years in which the right side may fail; replacing the left now while the bone is good and he is fit is easier than doing it at 75 with a compromised right side. Unopposed upper molars are a real and progressive problem. And **he has not actually said he is happy** — he came because he was told to, but I have not yet asked him what he cannot do."

Q A colleague says you should simply make him a denture and be done with it. Why might he be right?

"Because it is **reversible, cheap, non-surgical and immediately available**, it restores the occlusion to the upper molars and prevents over-eruption, and if the patient dislikes it nothing has been lost. It also serves as a **diagnostic trial** of whether he actually wants posterior teeth — if he wears it, implants are justified; if he abandons it in a drawer, that answers the question for both of us."

Chain: the options

Q Set out the options.

"**Shortened dental arch with active monitoring** — accept it, review with photographs and study casts. **A removable partial denture** — cheapest, reversible, but a free-end saddle design is the least well tolerated. **Implants** — one, two or three, restoring anything from a single 36 to the full span. **A cantilever bridge from the 34** — which I would not do for a molar span. And **no treatment with no monitoring**, which I would not offer."

Q If he wanted a denture, how would you design it?

"A **cobalt-chromium framework**, not acrylic. **RPI or RPA clasping on the 34** — mesial rest, distal guide plate, gingivally approaching I-bar — so that under load the clasp disengages rather than torquing the abutment. **Indirect retention anterior to the fulcrum line** through a rest on the contralateral side, cross-arch bracing with a lingual bar, and an **altered cast or functional impression** for the free-end saddle so it is supported by displaced mucosa. Broad coverage of the buccal shelf and retromolar pad."

Q Why not a cantilever bridge from the 34?

"Because a **single premolar abutment cannot support a molar cantilever**. The bending moment at the abutment is proportional to the cantilever length, and a molar pontic is both long and heavily loaded. I would expect debonding, or worse, root fracture or loss of the abutment. Cantilevers are for **small pontics on sound abutments in low-load positions** — a lateral incisor from a canine, not a molar from a premolar."

Q If he chooses implants, how many and where?

"I would **not place three implants for three teeth**. My preferred plan is **two implants at the 35 and 36 positions**, restored as two separate crowns, accepting occlusion to the first molar — which is itself a shortened arch and functionally sufficient. If bone dictates otherwise, **two implants supporting a three-unit fixed prosthesis** with an intervening pontic is reasonable. I would site them by the prosthetic plan and then check the anatomy permits it, not the other way round."

Q Why not a distal cantilever to a 37?

"Because a **distal cantilever generates large bending moments** on the terminal implant, and the evidence associates cantilevers with increased screw loosening, framework and porcelain fracture, and marginal bone loss. A 37 adds little function — the second molar contributes least to masticatory efficiency — for a disproportionate mechanical cost. I would rather stop at the 36."

Q How do you have the conversation with him?

> " 'You are managing well and there is nothing here that is harming you. There are three honest options. The first is to leave it and keep an eye on it — a lot of people function perfectly well with teeth back to the premolars, and there is good evidence for that. The second is a removable denture, which is the cheapest but the one people are least likely to wear. The third is implants, which would give you back a fixed chewing surface on that side; it is the most expensive, involves surgery, and commits you to lifelong cleaning and maintenance. Which appeals to you — and what would you like to be able to do that you cannot do now?' "

Q Why is that last question the important one?

> "Because if he **cannot answer it**, the shortened arch is almost certainly right. Implants are justified by a problem to solve, not by the existence of a gap. If he says he avoids steak, or he bites his cheek, or he is worried about the right side failing, that is a reason to treat. If he says nothing, treating him is meeting my expectation rather than his need."

Domain 3 — Decision making in relation to treatment procedures

Chain: doing it safely

Q Where is the boundary of your competence in this case?

> "Two implants in a posterior mandible with **adequate bone height above the canal** and no need for grafting is **within my scope**. If the CBCT shows less than about **2 mm of clearance above the canal**, a marked lingual concavity, or a need for vertical augmentation or nerve lateralisation, that is **beyond me and I would refer** — nerve injury is the complication that does the most harm and generates the most litigation, and it is entirely avoidable by not attempting the case."

Q What would you specifically warn him about at consent?

> "**Altered sensation of the lower lip, chin and gum** — that it is uncommon, usually temporary, but can be permanent; that it is not painful in most cases but is intrusive. Also failure to integrate, infection, the need for lifelong maintenance and hygiene, screw loosening, the cost of maintenance, and peri-implant disease. I would give written information and a cooling-off period, and record that the nerve risk was specifically discussed."

Q Take me through the surgery.

> "Chlorhexidine rinse, aseptic technique, **infiltration rather than a block** so he can report a sharp sensation as an early warning. **Mid-crestal incision** with a distal relieving incision kept away from the mental foramen; full-thickness flap; identify the external oblique ridge and, if it is close, the mental bundle. **Sequential drilling with a drill stop set from the CBCT**, slow speed, copious irrigation, sharp drills, checking depth and angulation with direction indicators and a periapical. Angle slightly **buccally in the molar region** to stay off the lingual plate. Insert at **25 to 45 Ncm**."

Q What is your safety margin above the canal?

> "At least **2 mm** between the implant apex and the superior border of the canal, and I plan the drill length to that, not to the implant length. I add a margin for CBCT measurement error and for drill overshoot beyond the tip. If that arithmetic leaves me a shorter implant than I would like, I use a **short implant** — 6 to 8 mm implants have good evidence in the posterior mandible — rather than augment or take the risk."

Q You perforate the lingual plate with the drill. What do you do?

> "Stop immediately, **do not advance further**, and assess for bleeding. Most perforations are uneventful, but a torn sublingual or submental vessel can produce a **rapidly expanding floor-of-mouth haematoma with airway obstruction**, which is a medical emergency. I would apply pressure, monitor the floor of mouth and the airway actively, abandon or reposition the implant, and if there is any swelling or elevation of the tongue, **call for emergency help and arrange urgent transfer** rather than watch."

Q One stage or two?

"**One stage** with healing abutments, given good primary stability and adequate keratinised tissue — it avoids a second surgical episode and lets the soft tissue mature around the abutment. **Two stage** if primary stability is poor, if I have grafted, or if I cannot achieve tension-free closure over an adequate cuff."

Q When would you load?

"**Conventional loading at about three months** in the mandible, confirmed by clinical stability and, if I have it, ISQ. Early loading at six to eight weeks is defensible in D2 bone with good primary stability, but conventional is what I would do and what I would say. He keeps the arch as it is in the interim — there is no interim prosthesis needed since he is already functioning."

Q Splint the two crowns or leave them separate?

"**Separate** if the implants are well positioned, of adequate diameter and in good bone — it allows flossing between them, permits independent assessment and re-treatment, and simplifies any future repair. **Splint** them if the implants are short, narrow, in poor bone, or if there is parafunction, because splinting distributes load and reduces screw loosening. Here, with D2 bone and a fit non-smoker, I would leave them separate."

Q Screw-retained or cement-retained, and what occlusal scheme?

"**Screw-retained** — retrievable and no risk of retained subgingival cement. Occlusally: a **narrow occlusal table** to reduce the lever arm, **light contact in intercuspation** so the crowns come into contact only under firm closure, given that implants lack the periodontal ligament and its proprioception, **no excursive contacts** with guidance kept on the canines and natural teeth, and no cantilever."

Q Why light contact rather than equal contact?

"Because a natural tooth intrudes about **25 to 100 microns** under load through its periodontal ligament, whereas an implant moves only **3 to 5 microns** by bone deformation. If they are set to equal contact at rest, the implant takes disproportionate load once the natural teeth depress. Leaving the implant crowns fractionally light means they share load only when the whole arch is loaded."

Q What would you record?

"The full history and examination including the periodontal chart, the imaging justification and findings with the canal measurements, the options discussed **including no treatment and the shortened dental arch**, the specific nerve risk discussion, his stated goals in his own words, the consent, and then surgically the implant make, diameter, length, batch number, torque achieved, and any complication. Plus the baseline radiograph at fitting."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcomes and their management

Q What survival would you quote him?

"In the region of **95% at five years and around 90 to 95% at ten years** for single implants in a healthy non-smoker with good bone — the posterior mandible is among the more predictable sites. But I would be careful to distinguish **implant survival from complication-free survival**: the implant staying in the bone is likely, while some prosthetic or biological event over a decade is close to expected."

Q The next day he reports a numb lower lip. Walk me through it.

"Reassure and see him urgently. **Map and record the affected area** with a diagram, test light touch, sharp-blunt and two-point discrimination, and ask whether it is numbness alone or **painful dysaesthesia**. Take a periapical and, if in doubt, a CBCT to see whether the implant impinges on the canal. If it does, **remove or back off the implant within 24 to 36 hours** — that is time-critical. Consider a short course of corticosteroids and NSAIDs. Document everything, review at intervals, and **refer early** if there is no improvement by about a month or if there is dysaesthesia."

Q Why is early referral so important there?

> "Because the window for any surgical intervention is short and the prognosis worsens with delay, and because **persistent dysaesthesia is disabling**. Sensory recovery, if it happens, largely occurs in the first three months and is usually judged permanent by about a year. Beyond that, management becomes symptomatic and psychological support becomes part of it. Delay is the single thing that turns a manageable injury into a permanent one."

Q Two years on, one crown feels loose. What is your differential and your sequence?

> "**Abutment screw loosening** is by far the commonest — check with a periapical and by testing the crown against the abutment. Then **fracture of the screw**, **loss of integration**, **a fractured crown or ceramic chip**, or simply a loose adjacent contact. My sequence is: radiograph, remove the crown through the access channel, inspect the screw and the interface, check for mobility of the implant itself, then re-torque with a **new screw at the manufacturer's torque** and — crucially — **find out why it loosened**, which is usually an occlusal overload or an unsplinted narrow implant."

Q He develops bleeding and 5 mm probing depths around the 36 implant. What is going on and what do you do?

> "That is **peri-implant mucositis** if there is no bone loss beyond initial remodelling, and **peri-implantitis** if there is progressive bone loss against the baseline radiograph. Mucositis is reversible: **non-surgical debridement with appropriate instruments, oral hygiene instruction focused on the implant, correction of any prosthetic contour that prevents cleaning**, and review. Peri-implantitis needs the same first, then reassessment, and surgical access or referral if it does not resolve. And I would look for a cause — plaque, an over-contoured crown, or lack of keratinised tissue."

Q What would you monitor and how often?

> "Annually as a minimum: **peri-implant probing at six sites**, bleeding and suppuration, plaque control, mobility, screw and crown integrity, and the occlusion. Radiographs at one year and thereafter as clinically indicated, compared with the baseline. And if he chose the **shortened arch** instead, I would still review annually with **photographs and study casts** for over-eruption of the 26 and 27 and drift of the 34 — so the decision remains an active one rather than a drift into a worse position."

Q How would you know the treatment had succeeded?

> "By whether he can do the thing he told me he wanted to do. Objectively: stable implants, no bleeding or suppuration, bone levels stable against baseline, an intact prosthesis, and an occlusion that has not changed. And if the right answer turned out to be no treatment, success is that he reaches 70 with the same arch, no over-eruption and no regret."

WATCH OUT

Return to the question the whole case turns on: **what problem am I solving, and for whom?** State whether the driver is the patient's function, the dentist's expectation, or the risk of future deterioration — that names the issue and usually points straight at the answer.

Running this case to time

Time check. The flow above holds 41 questions and about 3,077 words of model answer. At 140 words a minute plus roughly five seconds a question, that is close to **twenty-five minutes** — an overrun of about **ten**, the longest in this section. What goes first is the **removable partial denture design** material in Domain 2: the cobalt-chromium framework, the RPI clasping, the altered cast impression and the argument against a cantilever from the 34 collapse into a single question, because the marks in this case sit in the **shortened dental arch** and in the **canal and lingual concavity**. The Domain 1 anatomy will also be sampled rather than recited. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the numb lip appears.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiographs where you can see them	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	A fit, non-smoking 58-year-old with a **lower left free-end saddle — Kennedy Class II** — who is **functioning satisfactorily and attending because he was told to, not because he wants treatment**. Then the framing that is the whole opening: the first question is **whether he needs anything at all**, and **"no treatment" is a genuine contender here rather than a formality**. In his favour: medically fit, non-smoker, unrestored opposing arch, already adapted. Against: the **26 and 27 unopposed and liable to over-erupt**, and the 34 liable to drift. What you must still establish: what he actually wants to be able to do, the periodontal and occlusal status of the remaining dentition, and whether the **anterior loop and the lingual concavity** would permit implants at all.
1:20–4:30	Domain 1: the groundwork — classifying the saddle — then the anatomy, then reading this patient	Undergraduate prosthodontics first. **Kennedy Class II modification 0**, with **Applegate's rules** named; then why the class matters — Classes I and II are **tooth- and mucosa-borne**, so the saddle rotates about the **fulcrum line** through the distal rests and torques the abutment. Then the anatomy: **inferior alveolar canal, mental foramen, anterior loop, external oblique ridge, mylohyoid ridge, submandibular fossa, lingual nerve**. The **anterior loop** defined and its consequence named — altered lip and chin sensation if you measure only to the radiographic foramen. The **submandibular fossa** as the reason the ridge is narrower than a panoramic suggests, with **lingual plate perforation and a floor-of-mouth haematoma with airway compromise** as the worst case. CBCT justified **under IR(ME)R** with a scan appliance; the measurements listed; **D2 to D3** on Misch.
4:30–7:30	Domain 2: does he need treatment at all — then the options	The **shortened dental arch** stated properly: **about 20 teeth including four occluding premolar pairs**, from **Käyser's** work, endorsed by the **World Health Organization**, with no increase in temporomandibular disorder and no accelerated periodontal breakdown. Then the honesty that scores: he is a **unilateral** shortening, so the evidence applies less directly — but accepting it is a **legitimate positive choice, not a failure to treat**. The risks of leaving it named, **over-eruption of the 26 and 27 first**. The case argued **against** yourself — he is only 58, and **he has not actually said he is happy**. The denture granted its due as a **reversible diagnostic trial**. Then the options, including **no treatment with no monitoring, which you would not offer**; **two implants at 35 and 36, not three for three teeth**; **no distal cantilever to a 37**. **SAC: Straightforward to Advanced restoratively, Advanced surgically** — the anatomy, not the prosthodontics, raises it.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: doing it safely	The competence line **volunteered first**: two implants with adequate height above the canal are yours; **less than about 2 mm of clearance, a marked lingual concavity, vertical augmentation or nerve lateralisation is not** — refer, because nerve injury does the most harm and is entirely avoidable by not attempting the case. Consent with **altered sensation named specifically and recorded**. Then the surgery in sequence: **infiltration rather than a block** so he can report a sharp sensation; mid-crestal incision with the relieving incision **away from the foramen**; drill stop set from the CBCT; **angle slightly buccally in the molar region**; insertion at **25 to 45 Ncm**. **At least 2 mm above the canal**, planned to the drill length not the implant length, and a **6 to 8 mm short implant in preference to augmenting**. The lingual perforation managed: **stop, pressure, watch the floor of mouth and the airway actively, call for help rather than watch**. One stage; **conventional loading at three months**; crowns **separate, not splinted**, in D2 bone; **screw-retained, narrow occlusal table, light contact, no excursive contacts**, with the **25–100 μm against 3–5 μm** arithmetic given for why.
11:30–14:30	Domain 4: outcomes and their management	Survival quoted honestly — **around 95% at five years, 90 to 95% at ten** — with **implant survival separated from complication-free survival**. The numb lip the next day worked through as a sequence: **map and record with a diagram**, light touch, sharp-blunt, two-point discrimination, **numbness versus painful dysaesthesia**, periapical then CBCT, and **remove or back off the implant within 24 to 36 hours** — time-critical. Why early referral matters: recovery largely happens in the **first three months**, is judged permanent by about a year, and **delay is the single thing that turns a manageable injury into a permanent one**. Then screw loosening at two years with its sequence, and **find out why it loosened**. Bleeding and 5 mm pockets separated into **mucositis and peri-implantitis against the baseline film**. Annual review — and, if he chose the shortened arch, **annual photographs and study casts for over-eruption**, so the decision stays active.

14:30–15:00	Closes on the patient and the record	In his words: *"You are managing well and there is nothing here that is harming you… a lot of people function perfectly well with teeth back to the premolars, and there is good evidence for that… What would you like to be able to do that you cannot do now?"* — and then why that last question is the important one: **if he cannot answer it, the shortened arch is almost certainly right**. Recorded: the periodontal chart, the imaging justification and canal measurements, **the options including no treatment and the shortened dental arch**, the **specific nerve risk discussion**, his goals in his own words, and at surgery the make, diameter, length, batch and torque.

The spine — the twelve answers to have ready if nothing else

1 Kennedy Class II modification 0, and Applegate's rules · 2 Why the class matters — tooth- and mucosa-borne, and the fulcrum line · 3 The anterior loop, and what it costs you if you miss it · 4 The submandibular fossa, and the floor-of-mouth haematoma · 5 What you measure on the CBCT, and why it is justified · 6 The shortened dental arch, with Käyser and the WHO · 7 Why this man is not quite a classical shortened arch, and why it still holds · 8 The risks of leaving it — over-eruption of the 26 and 27 first · 9 Two implants at 35 and 36, and no cantilever to a 37 · 10 The 2 mm margin above the canal, and the short implant instead of a graft · 11 The numb lip — map it, image it, back the implant out within 36 hours · 12 What you say to him, and the one question you ask him.

Where the marks are lost in this case

Treating the gap instead of the patient. The examiner has told you he came because his dentist sent him, and that he has adapted. A candidate who opens with implant options has answered a question nobody asked. **Say in the opening that no treatment is a genuine contender**, name the **shortened dental arch** with Käyser and the WHO behind it, and offer it as a positive plan — then let the over-eruption of the 26 and 27 be the argument that reopens it.

Under-reading the mandible. This is an **Advanced surgical site** and the reason is anatomical, not prosthodontic. Volunteer the **anterior loop** and the **submandibular fossa** before you are asked, say that a panoramic will not show either, and give the **2 mm clearance above the canal** as an arithmetic you plan the drill length to. Candidates who talk about implant length rather than drill length have told the examiner they have never planned one.

Never naming what you would not attempt. Two implants in adequate native bone with a healthy non-smoker is **yours**. What is not: **less than 2 mm above the canal, nerve lateralisation, vertical augmentation, or a lingual concavity you cannot drill around**. Say it before the surgery question rather than after it, and say where it goes. Reaching for a short implant instead of a graft is the mark of judgement here; reaching for a graft you cannot do is the mark of a fail.

SOE Case 20 — The dissatisfied patient and the governance response

THE CASE

A patient returns **4 months** after you fitted an implant crown at the **11**. She is unhappy: the crown "looks longer and greyer" than her own tooth and there is a **black triangle mesially**. She says she "was never told this could happen" and is threatening to complain. Your records show a **signed consent form but no detailed note of the discussion**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A patient four months after fitting an implant crown at the 11, dissatisfied with an aesthetic outcome she says she did not expect — a crown that looks **longer and greyer** than the contralateral tooth, and a **black triangle mesially** — and threatening to complain. My records contain a signed consent form but **no contemporaneous note of the discussion**. There are two entirely separate problems here and I must not let them contaminate each other. The **clinical** problem is establishing the cause of each of the three complaints and whether each is remediable. The **professional** problem is how I respond to her dissatisfaction, how I handle a records deficiency I cannot retrospectively repair, and what I change in my practice so it does not recur. My first action is neither of those: it is to listen properly."

Domain 1 — Clinical interpretation and reasoning

Chain: the groundwork — why implant crowns look different

Q Describe the peri-implant soft tissue and how it differs from that around a tooth.

"There is **no periodontal ligament and no cementum**, so the collagen fibres run **parallel and circular** around the abutment rather than inserting into it. The supracrestal attachment is longer than the roughly 2 mm around a tooth — commonly 3 to 4 mm — and the **blood supply is poorer**, arising only from the supraperiosteal vessels. It is a weaker seal, and its vertical position is dictated by the bone crest beneath it."

Q What determines the presence of a papilla?

"The **contact point to bone crest distance**. Tarnow showed that between two teeth the papilla fills the embrasure essentially **always at 5 mm**, about **56% at 6 mm** and about **27% at 7 mm**. Between an implant and a tooth the papilla is supported by the **tooth's** attachment level, so the critical distance is roughly 4.5 to 5 mm, and it is the adjacent tooth's interproximal bone that governs it — not the implant."

Q So what has probably produced her black triangle?

"Either the **contact point is too far incisal** relative to the interproximal crest on the 12, or the interproximal bone on that adjacent tooth was already reduced before I started, or there has been crestal remodelling around the implant. It may also simply be that the papilla is still maturing — some papillary rebound occurs over the first one to two years, so four months is early."

Q Why might the crown look grey?

"Three candidates. **Show-through of a titanium abutment** through thin buccal mucosa — the commonest and the most easily fixed. **Loss of buccal bone or a buccally placed implant**, so there is no white tissue behind the mucosa. Or the **restorative material itself** — a metal-ceramic or high-opacity core with poor translucency at the cervical third, or a shade mismatch against a natural tooth that has its own internal translucency."

Q How would you distinguish those clinically?

"**Transillumination and blanching** — greyness that changes with light angle and tissue tension suggests show-through rather than an intrinsic shade problem. **Probing the buccal tissue thickness**, and probing the implant to assess the depth of the margin. Radiographs and, if needed, CBCT to see the implant's buccopalatal position and any buccal plate. And direct comparison of the crown against the 21 with a shade guide and standardised photographs."

Q And why might the crown be longer than her natural tooth?

"Because the **soft tissue margin sits apical to the contralateral zenith** — from crestal bone loss after extraction, a buccally or deeply placed implant, or recession over the four months. The restoration is then made longer to reach the tissue. Sometimes it is not the implant crown that is wrong at all but a discrepancy in the **gingival zenith levels** that was there before treatment and was never pointed out."

Chain: examining this patient

Q What exactly would you examine?

"Crown length against the 21; the **mid-facial gingival margin level** against the 21 and 12; papilla height and the **contact point to crest distance**; soft tissue phenotype and thickness; the abutment material and margin depth; **peri-implant probing at six sites with bleeding and suppuration**, to exclude inflammation as a cause of recession; the implant's three-dimensional position on radiograph and, if needed, CBCT; the occlusion; and standardised photographs compared with any pre-operative images."

Q Why does inflammation matter to an aesthetic complaint?

"Because inflamed tissue is oedematous, red and prone to further recession, and because some apparent recession **resolves once inflammation is controlled**. If I remake a crown against an inflamed baseline I will get the tissue level wrong. Inflammation also raises the possibility of **retained cement**, which would be both a cause and something I am responsible for."

Q What would you look for on the radiograph?

"**Bone levels mesially and distally** compared with the film taken at fitting; the interproximal bone level on the **adjacent 12**, because that determines the papilla; any radiopaque **residual cement**; the abutment–implant interface for a gap or misfit; and the implant's angulation. A single periapical with a paralleling technique, not a panoramic."

Q She has no pre-operative photographs in her record. Does that change your position?

"Yes — it weakens it considerably. Without a baseline I cannot demonstrate what the zeniths and papillae looked like before, so I cannot show that a discrepancy pre-existed. That is a **records failure of the same family as the missing consent note**, and I would treat it as such rather than argue around it."

Domain 2 — Clinical decision making and treatment planning

Chain: the clinical options

Q If the greyness is titanium show-through, what do you do?

"Replace the abutment with a **zirconia abutment, or a titanium base with a zirconia superstructure**, and remake the crown. It is a straightforward change with a large visual effect. I would combine it with a **connective tissue graft** to thicken the buccal tissue if it is thin, because that both masks the underlying metal and protects the margin long term."

Q And for the black triangle?

"In order of invasiveness: **modify the crown to move the contact point apically**, which shortens the contact-to-crest distance and is often enough on its own; **add composite or ceramic to the mesial of the 12** to close the embrasure from the other side; **orthodontic movement** to approximate the roots; or a **connective tissue graft**,

which is far less predictable for papillae than for mid-facial recession. I would also allow time, since papillary maturation continues."

Q And if the crown is simply too long?

"Remake it with a modified emergence profile and, if the tissue cannot be brought coronally, **pink ceramic at the cervical margin** to restore the apparent length. That is a legitimate solution, not a failure — but it has to be shown to her and agreed, not sprung on her at the fit."

Q What if the implant is fundamentally malpositioned?

"Then I say so plainly: **no restoration will fully correct a malpositioned implant**. The options become prosthetic camouflage with pink ceramic, or **explantation and re-treatment** with grafting and a new implant, or removal and conversion to a resin-bonded bridge. I would set out the time, the cost, who would do it, and who would pay for it — and I would take advice from my indemnity provider before discussing the financial side."

Q What is the one thing you would do before committing to any of these?

"Make a **provisional or a mock-up first**, so she can see the proposed improvement in her own mouth before she commits. It converts an argument about my description into a decision she makes with her eyes. It also protects both of us: if the mock-up does not satisfy her, we have learned that before spending her money."

Q What is the SAC classification of what is now in front of you?

"The original single implant in the aesthetic zone was **Advanced** prosthodontically at best. The **corrective** treatment is **Complex** — revision in the aesthetic zone with a soft tissue deficiency and a dissatisfied patient — and if explantation and regrafting are involved it is unequivocally Complex. Re-treatment is always harder than treatment."

Q What is within your competence here and what is not?

"Within: the assessment, the abutment and crown revision, contact point modification, hygiene and inflammation management, the complaint handling and the governance response. **Beyond me: explantation with regrafting and re-implantation in the aesthetic zone.** If that is where this goes, I would **refer**, and I would say so early rather than after another failed attempt. Referring a case I have already restored is not an admission of incompetence — persisting is."

Chain: the professional response

Q What do you do first, on the day she comes in?

"I **listen**. A proper appointment, not a doorstep conversation — private, unhurried, no one waiting — and I let her tell the whole story without interrupting. Then I acknowledge it genuinely: **'I'm sorry you're unhappy with how this looks. That matters to me, and I want to understand exactly what's bothering you.'** Only then do I examine. Most complaints escalate because the patient did not feel heard, not because of the clinical outcome."

Q How do you handle the complaint formally?

"Follow the **practice complaints procedure**: acknowledge in writing within the stated timescale, offer a meeting, investigate properly, and respond in writing with what I found, what I propose, and what she can do if she remains dissatisfied — naming the relevant ombudsman or regulator. I would **inform my indemnity provider early**, before I write anything formal, and take their help with the wording."

Q What about the records problem?

"I **cannot and would not alter them**. A signed form is not consent — consent is the discussion, and my note of that discussion is not as detailed as it should have been. I would tell her that honestly and apologise for it. If I add anything now it is **clearly dated and signed as a late entry recording only today's conversation**. Retrospectively backfilling a consent note would be dishonest and, if discovered, far more serious than the original omission."

Q Does duty of candour apply to an aesthetic complaint?

> "Yes. The professional duty of candour applies when something goes wrong and a patient is harmed or distressed. It requires me to **tell her promptly, apologise, explain what happened as far as I know it, and put right what can be put right**. An **apology is not an admission of liability** — that is explicit in the Compensation Act and in the regulators' guidance, and it is the single most useful thing to know in this room."

Q You said the signed form is not consent. Argue against yourself — what use is it?

> "It is evidence that a process took place and that documentation was given, and it usually lists the material risks, which is worth something. But it is a **record of an event, not of an exchange**. It cannot show what she asked, what mattered to her, or what she understood — and those are precisely what **Montgomery** makes the test."

Q State the Montgomery standard.

> "That the clinician must take reasonable care to ensure the patient is aware of any **material risk** in the proposed treatment and of any **reasonable alternative**. Materiality is judged from the **patient's** perspective — what a reasonable person in her position would attach significance to, or what this particular patient would. In aesthetic implant dentistry, a black triangle and a colour mismatch are unambiguously material."

Domain 3 — Decision making in relation to treatment procedures

Chain: doing the corrective work properly

Q Take me through the abutment and crown revision.

> "Remove the crown, retrieve the abutment, and **assess the tissue at rest** before deciding anything. Fit a **screw-retained provisional** with a modified emergence and let the tissue respond for at least eight to twelve weeks. Refine the provisional with flowable composite until she and I are both satisfied. Then capture that contour with a **customised impression coping** so the laboratory reproduces exactly what she approved, and only then make the definitive crown."

Q How would you shape the emergence to help the papilla?

> "Keep the subgingival contour **concave or flat rather than over-contoured**, so tissue has room to occupy the embrasure, and place the **contact point more apically and slightly longer**. Over-contouring the cervical third pushes tissue apically and is a common self-inflicted cause of exactly this appearance."

Q If you graft connective tissue, where from and how?

> "A **subepithelial connective tissue graft from the palate**, taken with a single-incision or trap-door technique from the premolar region, staying anterior to the greater palatine bundle and away from the rugae. Placed through a tunnel or an envelope flap buccal to the implant, sutured coronally, and left undisturbed. I would warn her about **donor site discomfort** and the need for a palatal stent."

Q What could go wrong with the corrective treatment?

> "Graft necrosis or shrinkage; further recession; donor site bleeding or discomfort; failure of the tissue to fill the black triangle despite everything; a shade that still does not match; and — the risk that matters most here — **spending her money and her goodwill for a small improvement**. So I would set the expectation deliberately low and demonstrate it on a provisional first."

Q What would you record from now on in this case?

> "Every contact **contemporaneously and factually**: what she said, in her words where I can; what I examined and found; the options offered and declined; the risks named; that a provisional was shown, photographed and approved; and the financial arrangement. **No opinion about her character and no defensive commentary** — the record will be read by others."

Q Who else should know about this?

"My **indemnity provider**, early. My practice owner or clinical lead, under the practice's complaints and significant event procedures. The laboratory, if the technical work is part of the problem. And the whole team, through a **significant event analysis** — because the systems failures here are the practice's, not only mine."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: closing the loop

Q What would you change in your consent process?

"A **written, implant-specific consent document** naming the material risks — recession, grey show-through, black triangles, crown length discrepancy, long-term aesthetic change, biological and mechanical complications — given at a **separate visit with a cooling-off period**, and, crucially, a **contemporaneous note of the discussion**, recording what was discussed, what was given, what she asked, and what she said mattered to her."

Q What would you change in your aesthetic case protocol?

"A **mock-up or provisional shown, photographed and approved before the definitive restoration** in every anterior case, with the approval recorded. **Standardised pre-operative photographs** as a mandatory record. And an **ITI Aesthetic Risk Assessment** completed and shared with the patient for every anterior implant, so the risk conversation is structured rather than remembered."

Q Design the audit.

"**Standard:** 100% of implant cases have a contemporaneous note of the consent discussion and a pre-operative photographic record. **Sample:** my last 20 implant cases. **Measure** against the standard, identify the gap, **implement** the new proforma and photographic protocol, then **re-audit the next 20 cases at six months** to close the loop. That is the shape of a quality improvement project, and it is what the entrance requirements for this qualification are asking for."

Q What is the difference between that audit and a significant event analysis?

"**Audit** measures practice against a defined standard over a sample and closes a loop with re-audit. **Significant event analysis** examines a single episode in depth — what happened, why, what went well, what did not, and what will change — with the team, and is recorded and reviewed. Both belong here: the SEA for this case, the audit for the system."

Q How would you know your changes had worked?

"The **re-audit** reaching the standard is the process measure. The outcome measures are fewer aesthetic complaints, patients approving provisionals before definitive work, and — the one I would actually watch — patients arriving at the fit appointment with **no surprises**, because everything they see they have already seen."

Q What do you say to her at the end of the first appointment?

"'**I've heard what you're unhappy about, and I can see it too. I want to find out exactly why each of those three things has happened before I promise you anything. I'll examine, take some photographs and an X-ray, and I'll see you again with a proper explanation and some options. In the meantime, if you want to make a formal complaint, that's absolutely your right and it won't affect how I look after you.**'"

Q What is the single most important lesson from this case?

> "That in aesthetic implant dentistry the outcome is judged against the **patient's expectation**, not against my technical standard. The work that determines whether a case succeeds is done **before the drill** — establishing what she wanted, showing her honestly what was achievable, and **recording that conversation**."

WATCH OUT

Separate the two threads and say which one you are answering: the clinical cause and its remedy, or the professional response — listen, apologise, investigate, be candid, involve indemnity, change the system. Naming the thread buys you a sentence and is almost always the frame the examiner wants.

Running this case to time

Time check. The flow above holds 37 questions and about 2,410 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty minutes** — an overrun of about **five**. What goes first is the **connective tissue graft** material in Domain 3: the donor site, the single-incision technique, the greater palatine bundle and the palatal stent compress into one question, because this case is not marked on soft tissue surgery. The marks sit in the **professional response** — listening, candour, Montgomery, the records, the indemnity provider and the audit — and an examiner will protect those. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the audit and the closed loop live.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs and the record in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A patient four months after an implant crown at the **11**, dissatisfied with an outcome she says she did not expect — **longer, greyer, and a black triangle mesially** — and threatening to complain, against a record holding **a signed consent form but no contemporaneous note of the discussion**. Then the interpretation, which is the whole opening: there are **two entirely separate problems and they must not contaminate each other**. The **clinical** problem is the cause of each of the three complaints and whether each is remediable. The **professional** problem is how you respond, how you handle a records deficiency **you cannot retrospectively repair**, and what you change so it does not recur. And the first action is neither: **it is to listen properly**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the groundwork — why implant crowns look different — then examining this patient	Undergraduate biology first. **No periodontal ligament and no cementum**, so fibres run **parallel and circular**, the supracrestal attachment is **3 to 4 mm rather than 2**, and the blood supply is **supraperiosteal only** — a weaker seal whose position is dictated by the bone crest. Then **Tarnow**: papilla essentially always at **5 mm**, about **56% at 6 mm**, about **27% at 7 mm** — and between implant and tooth the papilla is supported by **the tooth's attachment**, so the adjacent 12 governs it, not the implant. Greyness given **three candidates — titanium show-through, lost buccal bone or a buccal implant, or the restorative material** — and then distinguished by **transillumination and blanching**, tissue thickness, and standardised photographs. Length explained by a **soft tissue margin apical to the contralateral zenith**. Probing at six sites to exclude inflammation, because **some apparent recession resolves once inflammation is controlled** and because retained cement would be both a cause and your responsibility. And the honest answer on the missing pre-operative photographs: **it weakens your position considerably, and it is a records failure of the same family**.
4:30–7:30	Domain 2: the clinical options, then the professional response	Each complaint matched to a remedy in order of invasiveness — **zirconia abutment or titanium base with a connective tissue graft** for the greyness; **move the contact point apically**, add to the mesial of the 12, orthodontics, or time for the black triangle; **pink ceramic shown and agreed, not sprung at the fit** for the length. Then the sentence that must be said out loud: **no restoration will fully correct a malpositioned implant**. **A provisional or mock-up before committing to anything**, so it becomes a decision she makes with her eyes. **SAC: the corrective treatment is Complex** — re-treatment is always harder than treatment. Then the professional thread: **listen first, a proper private appointment, no doorstep conversation**; the practice **complaints procedure** with written acknowledgement, a meeting, an investigation and a written response naming the ombudsman; **indemnity informed early, before anything formal is written**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the records, the standard, and doing the corrective work properly	The records answer, which is the highest-value thirty seconds in the case: **a signed form is not consent — consent is the discussion**; you **cannot and would not alter the record**; anything added now is a **clearly dated and signed late entry recording only today's conversation**; and **retrospective backfilling would be dishonest and far more serious than the original omission**. **Duty of candour** applies — tell her promptly, apologise, explain, put right what can be put right — and **an apology is not an admission of liability**, which is explicit in the **Compensation Act** and the regulators' guidance. **Montgomery** stated properly: reasonable care that the patient is aware of any **material risk** and any **reasonable alternative**, materiality judged **from the patient's perspective**. Then the corrective sequence: crown off, **assess the tissue at rest**, **screw-retained provisional for eight to twelve weeks**, refine with flowable, capture with a **customised impression coping**, and only then the definitive crown, with the subgingival contour **concave or flat rather than over-contoured**.
11:30–14:30	Domain 4: closing the loop	The system changes, volunteered as changes rather than confessed as failures: an **implant-specific written consent document** naming recession, grey show-through, black triangles and crown length; a **separate visit with a cooling-off period**; a **contemporaneous note of what was discussed, given, asked and what mattered to her**; **standardised pre-operative photographs as a mandatory record**; an **ITI Aesthetic Risk Assessment** completed and shared; and a **mock-up photographed and approved before every definitive anterior restoration**. Then the audit designed properly — **standard 100%, sample the last 20 cases, measure, implement, re-audit the next 20 at six months** — and the distinction that separates a 3 from a 5: **audit measures practice against a standard over a sample and closes a loop; significant event analysis examines a single episode in depth with the team**. Both belong here — **the SEA for this case, the audit for the system**.
14:30–15:00	Closes on the written response and the duty of candour	What you say to her: *"I've heard what you're unhappy about, and I can see it too… I'll examine, take some photographs and an X-ray, and I'll see you again with a proper explanation and some options. In the meantime, if you want to make a formal complaint, that's absolutely your right and it won't affect how I look after you."* And what goes in writing: the **acknowledgement within the practice timescale**, what you found, what you propose, **an apology for the record-keeping**, the escalation route named, and thereafter **every contact recorded contemporaneously and factually — no opinion about her character and no defensive commentary**, because the record will be read by others.

The spine — the twelve answers to have ready if nothing else

1 The two separate problems, and that you listen first · 2 The peri-implant attachment, and why it is a weaker seal · 3 Tarnow, and who actually owns the papilla · 4 Three causes of greyness, and how you tell them apart · 5 Why inflammation must be excluded before you judge any tissue level · 6 The remedies for each complaint, in order of invasiveness · 7 A provisional or mock-up before committing to anything · 8 A signed form is not consent — and you do not alter records · 9 Duty of candour, and that an apology is not an admission of liability · 10 Montgomery, and materiality from the patient's perspective · 11 Audit versus significant event analysis, and that both belong here · 12 What you say to her, and what you put in writing.

Where the marks are lost in this case

Defending the dentistry instead of answering the complaint. The examiner has given you a records deficiency deliberately. Candidates who explain that the black triangle was unavoidable, or that she signed the form, have lost the case in the first minute. **Say in the opening that there are two problems and that the first action is to listen** — then handle the clinical cause and the professional response as separate threads, naming which one you are in.

Trying to make the record better than it is. The volunteered answer is that **consent is the discussion, not the signature**, that your note was inadequate, that you would **tell her so and apologise for it**, and that anything added now is a **dated late entry about today only**. Candidates who hedge — "I would clarify my notes" — have described the thing that ends careers. Say **retrospective alteration is dishonest and more serious than the omission** before you are pushed to it.

Never naming the escalation line. The competence line in this case is not surgical. Within your scope: the assessment, the abutment and crown revision, contact point modification, inflammation control, and **the complaint handling and governance response**. Beyond it: **explantation with regrafting and re-implantation in the aesthetic zone**, which is referred — and, separately, **anything touching liability or the financial settlement, which goes to your indemnity provider before you speak to her about money**. Referring a case you have already restored is not an admission of incompetence; persisting is.

SOE Case 21 — Failed endodontics in the aesthetic zone

THE CASE

38-year-old teacher, fit and well, non-smoker. The **21** has had two root canal treatments and an apicectomy; there is a **buccal sinus**, a post crown, and a **6 mm buccal probing depth**. CBCT shows a **large periapical lesion with buccal plate loss over 8 mm**. **Medium smile line, medium phenotype**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A fit, non-smoking 38-year-old teacher with a **failed, apicected upper left central incisor**. It has had two root canal treatments and an apicectomy, it has a chronic **buccal sinus**, a post crown, and a 6 mm buccal probing depth. The CBCT is the decisive finding: a large periapical lesion with **loss of the buccal plate over 8 mm**. The tooth is not salvageable — three endodontic attempts have failed and the supporting wall is gone. So the decision is not whether to extract, but **how to manage a site that has already lost its buccal wall** in a patient with a medium smile line and an occupation that puts her in front of people all day. My immediate questions are how much interproximal bone remains on the adjacent 11 and 22, what her smile actually shows, and what she is expecting."

Domain 1 — Clinical interpretation and reasoning

Chain: the endodontic and periapical groundwork

Q What is the likely periapical diagnosis?

"**Chronic apical periodontitis with a sinus tract** — previously called a chronic apical abscess — arising from a persistently infected root canal system. It is chronic, so it is asymptomatic; the sinus is the reason she has no pain. There is no pulp to test: the tooth is **previously treated**, so pulpal diagnosis is by history, not by sensibility testing."

Q Why do endodontic treatments fail?

"**Persistent intraradicular infection** — untreated canals, an inadequately disinfected apical third, or resistant organisms such as *Enterococcus faecalis* — is the commonest by far. Then **coronal leakage** past a restoration or post. Then **extraradicular infection** or a true apical cyst that will not resolve non-surgically. And **non-endodontic causes**: a vertical root fracture, a perforation, or a foreign body reaction. After two orthograde attempts and an apicectomy I would be considering a **root fracture or a true cyst**."

Q How would you confirm a vertical root fracture?

"A **narrow, isolated, deep probing defect** on one aspect — which is what a 6 mm buccal pocket could represent; a **J-shaped or halo radiolucency** tracking along the root; a sinus at or near the gingival margin rather than at the apex; and direct visualisation after raising a flap or after extraction. CBCT can suggest it but metal artefact from a post makes it unreliable."

Q What does the 6 mm buccal probing actually represent here?

"Most probably a **sinus tract draining along the buccal root surface through the sulcus**, rather than true periodontal attachment loss. I would confirm by **tracing the sinus with a gutta-percha point** and radiographing it. The distinction matters: a resolving endodontic lesion leaves a site with better regenerative potential than genuine periodontal attachment loss, and it changes what I expect from the adjacent papillae."

Q What is the significance of the buccal plate in this region normally?

"In the anterior maxilla the buccal plate is **thin — under 1 mm in the large majority of central incisor sites**, and often under 0.5 mm. It is largely **bundle bone**, dependent on the periodontal ligament for its blood supply, so it resorbs once the tooth is lost. Here she has already lost 8 mm of it to disease, which means she starts the extraction with the deficiency most patients only develop afterwards."

Q What happens to a socket after extraction?

"Remodelling that is greatest on the **buccal** aspect. Schropp showed roughly a **50% reduction in ridge width in the first year**, about two-thirds of it in the first three months. Araújo and Lindhe showed the buccal wall loses height preferentially because of bundle bone resorption. Superimposed on an existing 8 mm dehiscence, that is a very substantial defect."

Q What actually determines the papillae around the future restoration?

"The **interproximal bone height on the adjacent teeth** — the 11 and 22 — not the implant. Tarnow's work gives the rule: a papilla fills the embrasure essentially always at a **5 mm contact-to-crest distance**, about **56% at 6 mm**, about **27% at 7 mm**. Between an implant and a tooth the papilla is supported by the tooth's attachment. So if the 11 and 22 have intact interproximal bone I can still get papillae; if they do not, **I cannot recreate it**."

Chain: reading this patient

Q What further information do you want?

"Her **smile line at full smile and during speech**; the exact vertical and horizontal extent of the buccal defect and whether the **palatal wall** is intact; the interproximal bone levels on the **11 and 22**; sensibility of the 11 and 22, since the 21 may not have been the only tooth traumatised; her expectations in her own words and what her job demands aesthetically; occlusion and any parafunction; and standardised **photographs** plus a diagnostic wax-up."

Q Why does the palatal wall matter?

"Because it determines whether an implant can be placed at all in the correct prosthetic position and whether it will have primary stability. In the anterior maxilla the implant is engaged in the **palatal wall and the apical bone**, not the buccal plate. If the palatal wall is intact I have a scaffold to graft against; if it is lost as well, the defect becomes a **two-wall or non-contained** one and predictability falls sharply."

Q What is her aesthetic risk?

"On the **ITI Aesthetic Risk Assessment**: in her favour, healthy, non-smoker, **medium** rather than high smile line, medium phenotype, and a single tooth space with adjacent teeth present. Against her, **acute or chronic infection at the site** and a **major hard tissue defect**. Those two alone move her to **high aesthetic risk**."

Q Why does the CBCT change your management here, when a periapical would show the lesion?

"Because the periapical shows the lesion in two dimensions but tells me nothing about the **buccal plate**, which is superimposed on the root. The buccal loss is precisely the finding that rules out an immediate implant and dictates a staged approach. The exposure is **justified under IR(ME)R** because it changes management, and I would report the whole volume, not just the region of interest."

Domain 2 — Clinical decision making and treatment planning

Chain: choosing between the options

Q What are her options?

"**No treatment** — not realistic with a discharging sinus.
Retreatment or re-apicectomy — it has failed twice already, so the prognosis is poor.
Extraction plus a removable prosthesis — reasonable as an interim, not as the endpoint for a 38-year-old teacher.
Extraction plus a resin-bonded cantilever bridge from the 11, with soft tissue augmentation under an ovate

pontic.
Extraction plus ridge preservation, then a delayed implant at 4 to 6 months with simultaneous or staged GBR.
Immediate implant placement — which I would rule out."

Q Why do you rule out an immediate implant?

"Because the **ITI decision tree for immediate placement requires an intact socket with a facial wall of at least 1 mm, adequate apical and palatal bone for primary stability, a thick phenotype and no acute infection**. This site has an 8 mm buccal dehiscence and a chronic draining sinus. Placing into it risks failure of osseointegration, and if it does integrate it will do so in a position with no buccal support and a predictable mid-facial recession."

Q What would you actually do, and why?

"**Extract atraumatically without a flap**, curette the lesion and the sinus tract thoroughly, and perform **ridge preservation** — recognising that with 8 mm of buccal plate missing this is an **Elian Type 3 socket** where preservation can only be partially effective. Provide a bonded pontic immediately. **Reassess at four months with a CBCT**: if I can place an implant in the correct prosthetic position with simultaneous contour GBR, I will; if a staged augmentation is needed, I stage it or refer."

Q Explain the socket classification you have just used.

"**Elian: Type 1** — intact buccal plate and intact soft tissue. **Type 2** — soft tissue present but the buccal plate is partly or wholly missing beneath it. **Type 3** — both the buccal plate and the overlying soft tissue are deficient. It is useful because it tells you immediately whether you are managing a hard tissue problem, a soft tissue problem, or both. Here it is both."

Q Make the case for the resin-bonded bridge instead.

"It avoids implanting into a compromised site altogether. With **soft tissue augmentation under an ovate pontic** the emergence can be excellent, and a **pontic papilla is more predictable than an implant papilla**. The preparation on the 11 is minimal and essentially reversible. In a site that has lost 8 mm of buccal wall, it may well give a better aesthetic result for less biological risk — and I would say that honestly rather than assume she has come for an implant."

Q Argue against it.

"**Debonding** is the principal complication, and a teacher who talks all day will notice it at the worst moment; a single-retainer cantilever design in cobalt-chromium reduces but does not eliminate it. It also commits the 11 to being part of the restoration, it can grey the incisal third of the abutment tooth, and it does not maintain the ridge. It is a good option, not a free one."

Q Your colleague says he would place an immediate implant with simultaneous grafting. Why might he be right?

"Because immediate placement with simultaneous contour grafting preserves the soft tissue architecture, reduces the number of surgical episodes, and in **experienced hands** with an intact palatal wall and good primary stability it has respectable outcomes even in compromised sockets. Extraction site infection of endodontic origin is not an absolute contraindication if the site is thoroughly debrided. My answer is that the evidence supports it in **selected** cases with an intact facial wall — and this site has none — and that it sits beyond my level of practice."

Q What is the SAC classification?

"Surgically **Complex** — aesthetic zone, an extensive buccal defect, and infection at the site. Prosthodontically **Advanced to Complex** for a single anterior unit with a soft tissue deficiency. The extraction and ridge preservation on their own are **Advanced**."

Q What is within your competence and what is not?

"Within: the diagnosis, the CBCT interpretation, the atraumatic extraction, thorough debridement, **ridge preservation**, the immediate bonded pontic, and the maintenance. **Beyond me: staged vertical augmentation, and**

implant placement into a site that still has a major defect at reassessment. I would do the extraction and preservation myself and **co-manage or refer for the reconstruction**, having had the options conversation first."

Q What would you tell her, in your words to her?

"'**This tooth has been treated three times and it is not going to get better — the infection has eaten away the bone at the front of it. I need to take it out, clean the area properly and put a graft in to hold the shape. You will not be without a tooth at any point. In about four months I will scan it again and we will decide together whether an implant is sensible or whether a bridge would give you a better-looking result. What I cannot promise you is that the gum will look identical to the other side, and I would rather tell you that now than at the end.**'"

Domain 3 — Decision making in relation to treatment procedures

Chain: the surgery and the graft

Q Describe the extraction.

"**Flapless wherever possible**, to preserve what soft tissue blood supply remains. Remove the post crown first if it improves access. Sever the ligament with a **periotome or fine luxators**, section if necessary, and deliver with rotational movements — no buccal luxation against a wall that is already missing. Then **curette all the granulation tissue and the sinus tract thoroughly**, irrigate copiously with saline, and inspect and probe the defect to map its walls."

Q Why is complete removal of the sinus tract important?

"Because epithelialised tract tissue left behind will not heal, it maintains a route of contamination into the graft, and it can produce a persistent soft tissue depression or fistula. The graft has no chance in an incompletely debrided site — **debridement is the step that determines whether everything after it works**."

Q What do you graft with, and why?

"A **slowly resorbing xenograft** — deproteinised bovine bone mineral — because in a site I want to hold dimensionally for four months, slow substitution is the point. It maintains ridge contour better than a rapidly resorbing material. I would cover it with a **resorbable collagen membrane** tucked under the buccal soft tissue, and seal the coronal aspect."

Q How do you seal the socket without primary closure?

"With a **free gingival punch graft from the palate or a collagen matrix** over the entrance, secured with a figure-of-eight or crossed mattress suture. I avoid advancing a flap for primary closure because it **drags keratinised tissue coronally and flattens the vestibule**, which then has to be undone before the definitive restoration. Keeping the mucogingival line where it is saves a procedure later."

Q What are the principles of successful guided bone regeneration?

"The **PASS** principles: **P**rimary wound closure or a stable seal; **A**ngiogenesis, so decortication of the recipient bed and preservation of blood supply; **S**pace maintenance, so a graft or membrane that will not collapse; and **S**tability of the clot and the graft, meaning fixation and no micromovement. If any one of those four fails, the graft fails."

Q What temporary restoration would you provide?

"An immediate **bonded pontic** — her own crown or a denture tooth bonded to the 11 and 22 — with the tissue surface **fully relieved off the graft**. It is aesthetic from day one, it does not load the site, and it avoids the acrylic denture pressing on a fresh graft, which is a common cause of membrane exposure and graft loss. An Essix retainer with the ridge relieved is the alternative."

Q At four months the ridge is still deficient buccally. What then?

"**Staged horizontal augmentation**: decortication of the recipient bed, a **composite graft of autogenous bone and xenograft**, a collagen membrane, and **tension-free primary closure**, with 6 to 9 months of healing before implant placement. That is at the upper limit of what I would undertake. If the defect were **vertical as well as horizontal**, I would refer — vertical augmentation is the least predictable procedure in the field and has the highest complication rate."

Q The membrane becomes exposed at two weeks. What do you do?

"Do **not** attempt to re-cover it by advancing a flap over infected tissue. Manage it conservatively: **chlorhexidine gel applied locally**, no brushing over the site, weekly review, and antibiotics only if there is spreading infection. A small exposure of a resorbable membrane over a xenograft often heals with partial graft loss. If it becomes purulent and the graft is mobile, remove the graft, let the site heal, and re-plan."

Q What would you record?

"The **diagnosis and the reasoning for extraction**; the CBCT justification and full report under IR(ME)R; the **options discussed and declined**, including the bridge, in her words; the specific risks named — graft exposure and failure, a shorter crown, a flatter contour, a black triangle, the possibility that an implant proves impossible; the **materials, lot numbers and batch details** of graft and membrane; photographs; and the timescale and costs given."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: judging and salvaging the result

Q What will you measure at reassessment?

"**Resolution of the sinus** first — that is the primary indicator that the infection is gone. Then radiographic healing of the periapical area; **ridge width and height at four months against the pre-extraction CBCT**; soft tissue quality and the width of keratinised tissue; and, after restoration, the mid-facial margin level and papilla index against the 11 and 22, standardised photographs, and **peri-implant probing and bone levels** against a baseline taken at fitting."

Q The sinus has not resolved at four months. What is going on?

"**Incomplete debridement** of the tract or of granulation tissue, most likely. Then a **retained root fragment or foreign body**, an infected graft, or a pathology I have misdiagnosed — so I would consider whether this was ever purely endodontic. I would re-image, explore and re-debride, and I would not place an implant into a site with an unresolved sinus under any circumstances."

Q What is the most likely long-term disappointment?

"A **shorter clinical crown and a flatter buccal contour** than the contralateral central, and possibly a **black triangle** mesially or distally. Those follow directly from 8 mm of lost buccal plate — they are predictable, which is why they belong in the consent conversation."

Q How do you prevent that becoming a complaint?

"By showing her at the **mock-up stage** and agreeing **in advance** whether pink ceramic is acceptable, rather than discovering her view at the fit appointment. Photograph the mock-up, record her approval, and give her the option of stopping at that point. The mock-up is the honest preview — I would far rather she rejected the plan there than be disappointed at the end."

Q At two years the implant crown has 1 mm of mid-facial recession. Your approach?

"Exclude **inflammation** first, with probing, bleeding and a radiograph, and treat it if present — some apparent recession resolves. Check for an **over-contoured crown or residual cement**. If it persists and the implant is otherwise healthy, a **connective tissue graft** with a coronally advanced flap may improve it, though less predictably than around a tooth. If the implant is malpositioned, the honest answer is prosthetic camouflage or explantation."

Q How would you know this treatment had succeeded?

> "That she teaches all day without thinking about the tooth. Objectively: the **sinus resolved and never recurred**, the implant stable with no bleeding or suppuration, bone levels stable against baseline, an acceptable Pink and White Aesthetic Score, and her own rating measured against what she told me she wanted at the very first visit."

WATCH OUT

Go back to the defect: name which walls are missing, whether the deficiency is horizontal, vertical or both, and whether the problem is hard tissue, soft tissue or infection. That frame answers most questions in this case and buys you a sentence of thinking time.

Running this case to time

Time check. The flow above holds 37 questions and about 2,590 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty-one minutes** — an overrun of about **six and a half**. What goes first is the **staged augmentation** material in Domain 3: decortication, the composite autograft-and-xenograft, tension-free primary closure and the 6-to-9-month wait compress into one question, because that procedure sits **above your competence line anyway** and the examiner would rather hear you say so than describe it. The endodontic aetiology in Domain 1 gets sampled rather than recited. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the unresolved sinus is.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT slices up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A fit, non-smoking 38-year-old teacher with a **failed, apicected 21** — two root treatments, an apicectomy, a chronic **buccal sinus**, a post crown and a 6 mm buccal pocket — and the decisive CBCT finding: a large periapical lesion with **loss of the buccal plate over 8 mm**. Then the interpretation: **the tooth is not salvageable**, so the decision is not whether to extract but **how to manage a site that has already lost its buccal wall**, in a **medium smile line** and an occupation that puts her in front of people all day. What you must still establish: **the interproximal bone on the 11 and 22**, what her smile actually shows, and what she expects.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the endodontic and periapical groundwork, then reading this patient	**Chronic apical periodontitis with a sinus tract**, and the point that scores — **the sinus is why she has no pain**, and there is no pulp to test because the tooth is **previously treated**. Failure causes ranked: **persistent intraradicular infection first**, then coronal leakage, then extraradicular infection or a true cyst, then non-endodontic causes — and after two orthograde attempts and an apicectomy you are thinking **root fracture or true cyst**. The 6 mm pocket reinterpreted: most probably a **sinus tract draining along the root through the sulcus, not attachment loss** — confirmed by **tracing with a gutta-percha point** — and why the distinction changes what you expect of the papillae. Then the buccal plate: **under 1 mm in most central incisor sites**, largely **bundle bone**, and **Schropp's 50% width loss in the first year, two-thirds of it in three months** superimposed on an existing 8 mm dehiscence. And **Tarnow again**: the papilla belongs to the 11 and 22, **and if their interproximal bone is gone you cannot recreate it**.
4:30–7:30	Domain 2: choosing between the options	The options listed with **no treatment named and dismissed for a reason**, retreatment given its poor prognosis, and the two real contenders set against each other. **Immediate placement ruled out** against the **ITI decision tree** — an intact socket with **at least 1 mm of facial wall**, adequate apical and palatal bone, thick phenotype, no acute infection — none of which she has. The plan: **flapless atraumatic extraction, thorough curettage of the lesion and the tract, ridge preservation recognised as only partially effective in an Elian Type 3 socket, a bonded pontic the same day, and reassessment at four months with a CBCT**. **Elian** given properly — **Type 1 intact, Type 2 plate missing beneath intact soft tissue, Type 3 both deficient**. Then the honesty that separates a 3 from a 5: the **resin-bonded cantilever with an ovate pontic argued for on its merits** — a **pontic papilla is more predictable than an implant papilla** — before it is argued against on debonding. The colleague's immediate implant granted its evidence **in selected cases with an intact facial wall**, and refused because this site has none. **SAC: surgically Complex.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the surgery and the graft	The competence line **volunteered here, not extracted**: yours are the diagnosis, the CBCT interpretation, the **atraumatic flapless extraction, thorough debridement, ridge preservation, the immediate bonded pontic** and the maintenance; **beyond you are staged vertical augmentation and implant placement into a site still holding a major defect at reassessment** — do the extraction and preservation yourself and **co-manage or refer the reconstruction**. Then the sequence with the numbers: **periotome or fine luxators, no buccal luxation against a wall that is already missing**; complete removal of the tract, because **debridement is the step that determines whether everything after it works**; a **slowly resorbing xenograft** chosen precisely because you want the site held dimensionally for four months; a **resorbable collagen membrane**; and the socket sealed with a **free gingival punch or collagen matrix rather than an advanced flap**, because advancing drags keratinised tissue coronally and **flattens the vestibule you will need later**. **PASS** named and expanded. The provisional a **bonded pontic fully relieved off the graft**, not a denture pressing on it.
11:30–14:30	Domain 4: judging and salvaging the result	What is measured at reassessment, **resolution of the sinus first** — that is the primary indicator the infection is gone. Then the unresolved sinus given a differential in order: **incomplete debridement most likely**, then a retained fragment or foreign body, an infected graft, or a misdiagnosis — and the absolute: **you would not place an implant into a site with an unresolved sinus under any circumstances**. The membrane exposed at two weeks managed conservatively — **chlorhexidine gel, no brushing, weekly review, antibiotics only for spreading infection, and no attempt to re-cover it by advancing a flap over infected tissue**. The most likely long-term disappointment named before you are asked: **a shorter clinical crown, a flatter buccal contour, and possibly a black triangle** — predictable, which is exactly why they belong in the consent conversation.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	In her words: *"This tooth has been treated three times and it is not going to get better — the infection has eaten away the bone at the front of it… You will not be without a tooth at any point. In about four months I will scan it again and we will decide together whether an implant is sensible or whether a bridge would give you a better-looking result. What I cannot promise you is that the gum will look identical to the other side, and I would rather tell you that now than at the end."* Recorded: the diagnosis and reasoning, the **IR(ME)R justification and a report of the whole volume**, the options **including the bridge** in her words, the specific risks — graft exposure, a shorter crown, a flatter contour, a black triangle, **the possibility that an implant proves impossible** — the **lot numbers of graft and membrane**, photographs, timescale and costs.

The spine — the twelve answers to have ready if nothing else

1 Chronic apical periodontitis with a sinus tract, and why she has no pain · 2 Why root treatments fail, and what two failures plus an apicectomy suggest · 3 What the 6 mm buccal pocket actually is, and how you prove it · 4 The buccal plate — thickness, bundle bone, and what happens after extraction · 5 Tarnow, and why the papilla belongs to the 11 and 22 · 6 Why the palatal wall decides whether an implant is possible at all · 7 The ITI criteria for immediate placement, and which ones she fails · 8 Elian Type 1, 2 and 3 · 9 The resin-bonded cantilever argued for, then against · 10 Debridement of the tract, and why it decides everything after it · 11 PASS, and why you seal rather than advance a flap · 12 The sinus that has not resolved at four months.

Where the marks are lost in this case

Racing to the implant. The examiner has put 8 mm of missing buccal plate and a discharging sinus in front of you to see whether you notice that this is a **grafting and infection problem before it is an implant problem**. Volunteer that immediate placement is ruled out **against the ITI criteria**, name **Elian Type 3**, and say that the four-month reassessment is a genuine decision point, not a formality — because it may still end in a bridge.

Never offering the bridge honestly. The **resin-bonded cantilever with soft tissue augmentation under an ovate pontic** is a real answer here, and in a site that has lost its buccal wall it may give a **better aesthetic result for less biological risk** — a pontic papilla is more predictable than an implant papilla. Candidates who list it and move on have missed the mark; argue it properly, then argue it down on debonding in a teacher who talks all day.

Never naming what you would not attempt. Extraction, debridement, ridge preservation and the bonded pontic are **yours**. **Staged vertical augmentation is not** — it is the least predictable procedure in the field with the highest complication rate — nor is placing into a site that still has a major defect at four months, nor into one where the **sinus has not resolved**. Say it at the outset and say who it goes to. Describing an augmentation you would never perform is worse than admitting you would refer it.

SOE Case 22 — The furcation-involved molar

THE CASE

55-year-old, non-smoker, well-controlled hypertension. The **46** has a **grade III buccal and lingual furcation involvement**, 9 mm pockets, grade II mobility and a failed crown. The rest of the mouth is periodontally stable (BPE 1/1/1 elsewhere).

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 55-year-old non-smoker with well-controlled hypertension and a single failing lower first molar. The 46 has a through-and-through furcation, 9 mm pockets, grade II mobility and a failed crown, while the rest of the mouth charts at BPE 1. That mismatch is the striking feature: **advanced localised destruction in an otherwise healthy mouth is not the pattern of generalised periodontitis**, so my working diagnosis is a local cause — an endodontic-periodontal lesion, a vertical root fracture or an iatrogenic perforation — and I would establish which before condemning the tooth. Whatever the cause, a grade III furcation with grade II mobility carries a **poor long-term prognosis**. The questions I need to answer are why this tooth alone has failed, whether it is restorable at all, and what he wants done about the space."

Domain 1 — Clinical interpretation and reasoning

Chain: the periodontal groundwork

Q What is a furcation involvement?

"Loss of attachment and bone in the **inter-radicular area** of a multi-rooted tooth, such that a probe passes into the space between the roots. It is not a separate disease — it is periodontitis, or a local lesion, reaching an anatomical site that cannot be instrumented properly and that the patient cannot clean. That is why furcation involvement is treated as a **prognostic marker** and not simply as another measurement."

Q How do you classify it?

"**Hamp** classifies it horizontally: Degree I is horizontal loss of less than a third of the tooth width, Degree II is more than a third but not through-and-through, and Degree III is **through-and-through**. Glickman's classification is comparable, with Grade IV being a through-and-through that is clinically visible because the tissue has receded. I measure with a **Nabers probe**, buccally and lingually, and record both."

Q Why is a lower first molar so vulnerable to this?

"The **furcation entrance is frequently narrower than a curette blade** — commonly under 1 mm — so it cannot be debrided properly. The roof of the furcation lies only 3 to 4 mm below the amelocemental junction, so little attachment loss is needed to reach it. Add root concavities, **cervical enamel projections or enamel pearls**, and accessory canals in the furcation floor, and it becomes the classic site for both periodontal and endodontic involvement."

Q Grade II mobility — define the grades.

"On **Miller**, Grade I is horizontal movement up to 1 mm, Grade II is horizontal movement of more than 1 mm, and Grade III is horizontal movement plus **vertical depressibility**. Mobility is a sign, not a diagnosis. I would separate mobility from irreversible attachment loss, which will not improve, from mobility due to an inflamed periodontium or occlusal trauma, which often does improve with treatment."

Q How would you diagnose him under the 2017 World Workshop?

"With a single affected tooth in a BPE 1 mouth, I would first exclude a non-periodontitis cause, because periodontitis requires **interproximal attachment loss at two or more non-adjacent teeth**. If periodontitis were confirmed, a grade III furcation and the anticipated tooth loss put it at **Stage III or IV**, graded by rate of progression against age and risk factors, with an extent descriptor of localised."

Q Does the hypertension matter?

"Not to the diagnosis. To the treatment in three ways: a **blood pressure reading before any surgery** and a threshold above which I defer; awareness that some antihypertensives cause xerostomia or postural hypotension; and that **calcium channel blockers cause gingival overgrowth**. I would confirm the actual drug, the last recorded reading and who monitors him, and record all three."

Chain: reading this particular tooth

Q What is your differential for a single tooth behaving like this?

"A **primary endodontic lesion draining through the furcation**, which is common in molars with accessory canals in the furcation floor. A **vertical root fracture** beneath the crown. An **iatrogenic furcation perforation** from a post or crown preparation. Or genuinely localised periodontitis driven by a plaque-retentive factor — an overhang or an open margin on that failed crown."

Q What investigations?

"**Sensibility testing** with cold and electric pulp test, compared against adjacent and contralateral teeth. **Periapicals at two horizontal angles**. Six-point charting with Nabers probing buccally and lingually, and a check for a narrow isolated defect suggesting fracture. Assessment for a crack under magnification and transillumination, with the crown removed if need be. And a **small field-of-view CBCT** if fracture or perforation is suspected, which also gives me the implant planning data."

Q If it proves non-vital and the lesion is endodontic in origin, does that change things?

"Potentially. A **primary endodontic lesion with secondary periodontal involvement** can respond dramatically to root canal treatment, and I would reassess at three months before condemning the tooth. But a genuine through-and-through furcation with grade II mobility rarely resolves. I would be honest that this is buying time rather than curing, and only attempt it if he understood that."

Domain 2 — Clinical decision making and treatment planning

Chain: can the tooth be kept?

Q What are the options for the 46?

"**Retain and treat** — non-surgical therapy, then possibly furcation surgery, root resection or a tunnel preparation. **Extract and leave the space**, accepting a shortened arch. **Extract and place an implant. Extract and bridge** from the 45 to the 47. **Extract and provide a partial denture.** Each has to be set against the fact that this tooth's prognosis is already poor before I start."

Q Talk me through root resection and tunnelling.

"**Root resection** removes one root and its part of the crown, leaving a restorable remainder — but survival figures are modest, root fracture and recurrent caries are common, and it demands root canal treatment plus a complex restoration. **Tunnel preparation** opens the furcation for an interdental brush, at high risk of root caries. Both are **Complex procedures**, both are declining in use, and neither is what I would offer here."

Q Tell me about the shortened dental arch.

"The concept that a dentition of **around 20 teeth, with premolar-to-premolar occlusion**, gives adequate function, comfort and stability without replacing molars. There is good long-term evidence for it. Here, if the 47 is sound and in occlusion and the arch is stable, extracting the 46 and leaving the space is a **legitimate definitive plan**, not a failure — and it must be offered."

Q You said the shortened arch is legitimate. Argue against it in this patient.

"At 55 he has a long time to live with the consequences. If the 47 is unopposed or already tilted, the space may not stay stable — I would expect **over-eruption of the 16 and mesial tipping of the 47**. If he is a heavy function patient the load redistributes onto fewer units. And if he later wants the tooth replaced, the ridge will have resorbed further and the site will be harder."

Q What do you recommend?

"Extraction with **ridge preservation**, given the volume of interradicular bone likely already lost, then reassessment at four months and **a single implant at the 46** if the site allows. I would section the tooth to preserve the plates. If the residual height or width is inadequate at reassessment, I would revert to the shortened arch rather than escalate to grafting."

Q Would you place immediately into the molar socket?

"No. Immediate placement in a molar site relies on the **inter-radicular septum** for primary stability, and in a grade III furcation the septum is precisely what has been destroyed. There is also active infection. **Immediate placement is contraindicated here**, and forcing it risks a malpositioned implant with poor stability."

Q A colleague would still place immediately into the septum with a wide implant and graft the gaps. Why might they be right?

"Because in a molar socket with a substantial intact septum and no suppuration, immediate placement shortens treatment, preserves the socket walls, and there is reasonable evidence for it in experienced hands. Their argument is that waiting guarantees resorption. My answer is that it is **Advanced on SAC and dependent on a septum this patient does not have** — the technique is defensible, the case selection is not."

Chain: scope, consent and the conversation

Q What is the SAC classification here?

"The **extraction with ridge preservation and a delayed single implant in a healed posterior mandible is Straightforward to Advanced** — Straightforward prosthodontically in a non-aesthetic site, Advanced surgically because of proximity to the inferior alveolar canal and an unpredictable healed volume. Immediate placement, block grafting or vertical augmentation at this site would be **Complex and beyond my competence** — I would refer."

Q What would you tell him, in your own words to him?

" 'The bone between the roots of this tooth has been destroyed, and the tooth is loose. I don't think we can keep it reliably. My plan is to take it out carefully, put a graft into the socket to hold the shape, and let it heal for about four months. Then we look again. If there's enough bone, an implant is a good option. If there isn't, leaving the gap is a perfectly reasonable place to stop — plenty of people function very well without that tooth.' "

Q What would you record at this appointment?

"The six-point chart and the Nabers readings buccally and lingually, the mobility grade, the sensibility results, the radiographs and their report, the blood pressure and medication, my **diagnosis and prognosis for the 46**, every option discussed including no treatment and the shortened arch, the risks I named, the costs quoted, and that he was given time and written information before consenting."

Domain 3 — Decision making in relation to treatment procedures

Chain: what happens to a socket

Q What happens to an extraction socket if you do nothing?

"A clot organises into granulation tissue, then provisional woven bone, then remodels to lamellar bone over months. Externally the ridge resorbs — most of the dimensional change occurs in the **first three to six months**, with horizontal loss of the order of half the ridge width and greater loss buccally than lingually. The socket is radiographically filling long before the ridge stops shrinking."

Q Why does the buccal plate resorb the most?

"Because in most sites it is thinner, and because a significant part of it is **bundle bone** — the bone into which the periodontal ligament fibres insert. Bundle bone is tooth-dependent; when the tooth goes, its blood supply and its function go with it, so it resorbs. That is the mechanism behind the classic loss of buccal contour."

Q So what does ridge preservation actually achieve?

"It does not prevent resorption — it **reduces and slows it**, typically limiting horizontal and vertical loss by a few millimetres compared with unassisted healing, and it makes the volume more predictable. That is enough to change whether a site can take an implant without further grafting. It is a **socket management technique, not an augmentation**, and I would not oversell it to him."

Q Describe the extraction.

"Local anaesthetic, atraumatic technique. I would **section the tooth at the furcation with a bur and deliver each root separately** using periotomes and root forceps, rather than trying to lever out a mobile, furcated molar whole — sectioning protects the buccal and lingual plates, which are what I need. Then thorough curettage of all granulation tissue and irrigation, and I would inspect the socket walls directly."

Q And the ridge preservation itself.

"Once the socket is clean and the walls assessed, I graft with a slowly resorbing **xenograft** placed to the level of the crest without over-packing, and seal it — a resorbable **collagen membrane** tucked under the flap margins, or a collagen plug or free gingival socket seal graft. Secure with a figure-of-eight suture. No primary closure that drags keratinised tissue across, and no flap advancement that shifts the mucogingival line."

Q How long would you wait?

"A molar socket is large, so **at least four months, and I would prefer five to six**. I want the graft consolidated enough to drill without it crumbling. I would reassess clinically and with a CBCT at that point rather than assuming the healed volume — a furcation lesion may leave a defect well below the level of the surrounding crest."

Q Implant selection at reassessment?

"From the CBCT: a **wide-diameter implant of around 5 mm** if the width allows 1.5 to 2 mm of bone buccally and lingually, and a length that leaves at least **2 mm of clearance above the inferior alveolar canal**. A molar occlusal table on a narrow implant is a mechanical mistake — I want diameter here rather than length. Bone is likely D2, so I would countersink and watch the insertion torque."

Q If that leaves only 7 mm of usable height?

"Then I use a **short, wide implant**. The evidence for short implants in the posterior mandible is now good — survival comparable to longer implants with augmentation, and fewer complications, because the graft is what carries the risk. I would keep the crown-to-implant ratio in mind, keep the occlusal table narrow, and accept a short implant in preference to vertical augmentation, which I would refer."

Q How do you keep away from the nerve?

"Plan from the CBCT, not the panoramic, and measure from the crest to the **superior border of the canal** at the exact planned site. Build in a **2 mm safety margin** above the canal for measurement error and drill overshoot. Identify the mental foramen if the incision runs forward. Use drill stops and a surgical guide, check with a direction indicator and periapical, and use infiltration rather than a block so he can report a sharp sensation."

Q One stage or two?

"**One stage with a healing abutment** if primary stability is good, there is adequate keratinised tissue and I have not grafted at placement. Two stage if primary stability is poor, if I have had to graft simultaneously, or if soft tissue closure would be compromised. In a posterior mandible with a healed preserved ridge, one stage is usually achievable and saves him a second procedure."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: what can go wrong, and what you do about it

Q What are the main risks in this case?

> "**Nerve injury**, because it is a molar site whose height may be reduced by the furcation destruction. **Inadequate primary stability** if the grafted socket has not consolidated. **A residual defect** at reassessment forcing simultaneous grafting or abandonment. And restoratively, the **mechanical complications of molar loading** — screw loosening, ceramic chipping — on a single implant carrying a wide occlusal table."

Q The day after surgery he reports numbness of the lower lip. What do you do?

> "See him the same day. Map and record the affected area, test with light touch, sharp-blunt and two-point discrimination, and document it — including whether it is anaesthesia, paraesthesia or painful dysaesthesia. Take a **radiograph or CBCT to see whether the implant is impinging on the canal**; if it is, remove or back it out promptly, within about 36 hours. Reassure, review, prescribe as indicated, and refer early if it is not settling."

Q How would you restore it?

> "Conventional loading at about three months, implant-level impression or intraoral scan with a scan body. A **screw-retained monolithic zirconia crown** with a **narrow occlusal table**, light contact in intercuspation, no working or non-working excursive contact, a cleansable emergence profile, and no cement subgingivally. Screw retention because retrievability matters most in exactly the site where mechanical complications happen."

Q At three years the crown feels loose. Differential and management?

> "**Abutment screw loosening** is much the commonest and is what I expect. Then a fractured screw, a fractured abutment, a failed cement seal if it were cemented, or a **failing implant**. I would take a periapical, remove the crown, inspect the components, check the occlusion and the excursive contacts for the cause, replace the screw and torque it to the manufacturer's value. Repeated loosening means the occlusion or the design is wrong, not the screw."

Q How would you monitor him?

> "Six-monthly, given a stable periodontal condition. **Six-point peri-implant probing with bleeding and suppuration**, plaque control, marginal bone level against the baseline periapical taken at fitting, occlusal review and screw integrity, and the periodontal status of the remaining dentition. Radiographs at one year and then as clinically indicated."

Q How would you know this had succeeded?

> "That he chews comfortably on that side and has stopped thinking about the tooth. Objectively: the implant stable with no bleeding or suppuration, bone levels stable against baseline, the crown intact with a maintained occlusion, the adjacent 45 and 47 unharmed, and the rest of the mouth still at BPE 1 — because in this patient the **untouched periodontal health of the other teeth is the real measure**."

WATCH OUT

Fall back to four questions about the failing tooth — is it restorable, is it endodontically treatable, is it periodontally maintainable, and what does the patient actually want — and say out loud which of the four fails. It buys a sentence and it is nearly always the right frame for a questionable tooth.

Running this case to time

Time check. The flow above holds 36 questions and about 2,412 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty minutes** — an overrun of about **five**. What goes first is the **root resection and tunnelling** material in Domain 2: both are declining in use, both are Complex, and neither is what you would offer, so they compress into one sentence of acknowledgement. The marks here sit in the **differential for a single failing tooth in a BPE 1 mouth**, in the **shortened dental arch offered as a definitive plan**, and in the **canal**. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the numb lip appears.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the chart and the periapicals in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 55-year-old non-smoker with **well-controlled hypertension** and a single failing **46** — through-and-through furcation, **9 mm pockets, grade II mobility, a failed crown** — in a mouth that charts **BPE 1 elsewhere**. Then the interpretation, which is the whole opening: **that mismatch is the striking feature**. **Advanced localised destruction in an otherwise healthy mouth is not the pattern of generalised periodontitis**, so the working diagnosis is a **local cause — an endodontic-periodontal lesion, a vertical root fracture, or an iatrogenic perforation** — and you would establish which **before condemning the tooth**. Whatever the cause, a grade III furcation with grade II mobility carries a **poor long-term prognosis**. The three questions: why this tooth alone has failed, whether it is restorable at all, and what he wants done about the space.
1:20–4:30	Domain 1: the periodontal groundwork, then reading this particular tooth	Undergraduate periodontology first. **Furcation involvement defined** as attachment and bone loss in the inter-radicular area — **not a separate disease, but periodontitis reaching a site that cannot be instrumented and that the patient cannot clean**, which is why it is a **prognostic marker**. **Hamp** given horizontally, **Grade III through-and-through**, measured with a **Nabers probe buccally and lingually**. Why a lower first molar: the **furcation entrance is frequently narrower than a curette blade**, the roof lies **3 to 4 mm below the amelocemental junction**, plus root concavities, **cervical enamel projections** and accessory canals in the furcation floor. **Miller** grades, with the point that **mobility is a sign, not a diagnosis**. The **2017 World Workshop** applied honestly — periodontitis needs **interproximal attachment loss at two or more non-adjacent teeth**, so you exclude a non-periodontitis cause first. **Hypertension** dealt with in three lines: **a reading before surgery with a threshold to defer, xerostomia and postural hypotension, and calcium channel blocker gingival overgrowth**. Then the differential and its investigations — **sensibility testing, two horizontal angles, small field-of-view CBCT**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: can the tooth be kept, and what replaces it	The options for the 46 given **with retention first**, and the honest note that the prognosis is poor before you start. **Root resection and tunnelling** acknowledged and set aside — modest survival, root fracture and recurrent caries, both **Complex**, neither offered. Then the answer this case is built around: a **primary endodontic lesion with secondary periodontal involvement can respond dramatically to root canal treatment**, so you would **reassess at three months before condemning the tooth** — while being honest that with a genuine through-and-through and grade II mobility you are **buying time rather than curing**. Then the **shortened dental arch** stated properly — **around 20 teeth, premolar-to-premolar occlusion**, good long-term evidence — and offered as what it is: **a legitimate definitive plan, not a failure, and it must be offered**. Argued against in this man: **he is 55**, the **16 may over-erupt and the 47 tip**, and the ridge will be worse if he changes his mind. The recommendation: **extraction with ridge preservation, reassess at four months, a single implant if the site allows — and revert to the shortened arch rather than escalate to grafting if it does not**. **Immediate placement contraindicated**: it relies on the **inter-radicular septum, which is precisely what a grade III furcation has destroyed**, and there is active infection. **SAC: Straightforward to Advanced; immediate placement, block grafting or vertical augmentation here is Complex and referred.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the socket and the implant	What happens to a socket if you do nothing — **most dimensional change in the first three to six months**, horizontal loss of the order of half the ridge width, **greater buccally**, and the trap named: **the socket is radiographically filling long before the ridge stops shrinking**. Why the buccal plate goes: **bundle bone is tooth-dependent**. Then the sentence that scores: **ridge preservation does not prevent resorption — it reduces and slows it**, it is a **socket management technique, not an augmentation**, and **you would not oversell it to him**. The extraction described properly: **section at the furcation and deliver each root separately** rather than levering out a mobile furcated molar whole, because **the plates are what you need**. The graft to the crest **without over-packing**, sealed, **no flap advancement that shifts the mucogingival line**. Waiting **at least four months, preferring five to six**, with a **CBCT at reassessment rather than an assumption** — a furcation lesion may leave a defect well below the crest. Implant selection: **wide, around 5 mm, with 1.5 to 2 mm of bone buccally and lingually, and at least 2 mm above the canal — diameter here rather than length**, because a molar occlusal table on a narrow implant is a mechanical mistake. At 7 mm of height, a **short wide implant in preference to vertical augmentation, which you refer**. Nerve safety: **CBCT not panoramic, 2 mm margin, drill stops and a guide, infiltration rather than a block**.
11:30–14:30	Domain 4: what can go wrong, and what you do about it	The risks named in order — **nerve injury**, inadequate primary stability in an unconsolidated graft, **a residual defect at reassessment**, and the **mechanical complications of molar loading on a single implant**. The numb lip the next day: **see him the same day, map and record, anaesthesia versus paraesthesia versus painful dysaesthesia, radiograph or CBCT, and back the implant out within about 36 hours**, then **refer early if it is not settling**. The restoration: conventional loading at three months, **screw-retained monolithic zirconia, narrow occlusal table, light contact, no excursive contact, cleansable emergence, no subgingival cement** — screw retention **because retrievability matters most in exactly the site where mechanical complications happen**. The loose crown at three years given a differential, with the volunteered line: **repeated loosening means the occlusion or the design is wrong, not the screw**. Six-monthly review, **six-point probing, bone level against the baseline periapical taken at fitting**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	In his words: *"The bone between the roots of this tooth has been destroyed, and the tooth is loose. I don't think we can keep it reliably… If there's enough bone, an implant is a good option. If there isn't, leaving the gap is a perfectly reasonable place to stop — plenty of people function very well without that tooth."* Recorded: the six-point chart and **the Nabers readings buccally and lingually**, mobility grade, sensibility results, radiographs and their report, **the blood pressure and the medication**, the diagnosis and prognosis for the 46, **every option including no treatment and the shortened arch**, the risks, the costs, and that he had time and written information before consenting. And the measure of success: **the rest of the mouth still at BPE 1** — in this patient, the untouched periodontal health of the other teeth is the real outcome.

The spine — the twelve answers to have ready if nothing else

1 Why one tooth failing in a BPE 1 mouth changes the diagnosis · 2 Furcation involvement defined, and why it is a prognostic marker · 3 Hamp, the Nabers probe, and both aspects recorded · 4 Why a lower first molar is the classic site · 5 Miller mobility, and mobility as a sign not a diagnosis · 6 The differential — endo-perio, vertical fracture, perforation, plaque-retentive factor · 7 Hypertension in three lines · 8 The shortened dental arch offered as a definitive plan, then argued against · 9 Why immediate placement fails here — the septum is what has been destroyed · 10 Ridge preservation reduces, it does not prevent · 11 Wide and short in preference to augmenting, with 2 mm above the canal · 12 The numb lip, and the 36-hour window.

Where the marks are lost in this case

Extracting the tooth in the first sentence. The examiner has deliberately given you **advanced destruction at one tooth in a BPE 1 mouth**. Say in the opening that this pattern is **not generalised periodontitis** and name the local differential — **endo-perio lesion, vertical root fracture, iatrogenic perforation** — then say you would **sensibility-test and reassess at three months** if it proves endodontic. Candidates who condemn the 46 before testing it have failed Domain 1 in thirty seconds.

Treating the gap as a foregone conclusion. The **shortened dental arch is a legitimate endpoint here and it must be offered**, in Käyser's terms, as a positive plan rather than the option of last resort — and it is also your **fallback if the reassessment CBCT is disappointing**, which is the judgement the case is really testing. Say out loud that you would **revert to the shortened arch rather than escalate to vertical augmentation**.

Never naming what you would not attempt. Sectional extraction, ridge preservation and a **delayed single implant in adequate native bone** are yours. What is not: **immediate placement into this socket** — the septum has gone — **block grafting, vertical augmentation**, or any plan that needs less than **2 mm above the inferior alveolar canal**. Say it before the surgical chain begins, and say that a **short wide implant is the answer you would reach for instead of a graft**.

SOE Case 23 — The failing three-unit bridge

THE CASE

62-year-old woman with an upper right **14–16 conventional bridge** placed 18 years ago. The **16 abutment has recurrent caries to the gingival level and is non-vital**; the **14 abutment is sound**. She is anxious about "losing more teeth" and wants "something permanent".

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 62-year-old woman with an 18-year-old three-unit bridge from the 14 to the 16 that has failed at the distal abutment. The 16 has **recurrent caries to gingival level and is non-vital**; the 14 appears sound. Eighteen years is respectable service, and the mode of failure — caries at the margin of an abutment — is the commonest way conventional bridges end. The decision turns on two things: whether the 16 is **restorable**, which I cannot judge until the bridge is off, and what the 14 needs once it loses its retainer. She is anxious about losing more teeth and wants something permanent, so **expectation management is as much a part of this case as the restorative decision**. My immediate need is to remove the bridge, assess both abutments directly, and give her a provisional so she is never without teeth."

Domain 1 — Clinical interpretation and reasoning

Chain: fixed prosthodontic groundwork

Q Name the components of a conventional fixed bridge.

"The **abutment** is the tooth; the **retainer** is the restoration cemented to it; the **pontic** replaces the missing tooth; and the **connector** joins pontic to retainer, either rigid or non-rigid. This one is a fixed-fixed design with rigid connectors and two abutments, which means a failure at one end effectively condemns the whole unit."

Q What are the commonest reasons a conventional bridge fails?

"**Caries at an abutment margin** is the commonest, followed by **loss of vitality and periapical disease**, then loss of retention or decementation, then periodontal breakdown of an abutment, then mechanical failure — porcelain fracture or connector fracture. Around **90% survive to ten years**, so failure at eighteen is not a treatment error; it is the natural history."

Q What is Ante's law, and do you believe it?

"That the root surface area of the abutments should equal or exceed that of the teeth being replaced. It is a useful instinct rather than a rule — the evidence does not support it as a hard predictor of survival, and periodontally reduced but stable abutments can carry bridges successfully. I use it as one factor alongside mobility, crown-to-root ratio and occlusal load."

Q How do you decide whether the 16 is restorable?

"Four things: how far the caries extends **subgingivally and whether I can reach a sound margin**; how much **ferrule** remains; the **root and furcation anatomy**; and whether it is endodontically treatable. I would remove all caries first and then judge — deciding restorability with the bridge still on is guesswork."

Q What is the ferrule effect, and how much do you need?

"A circumferential band of the crown that grips **sound tooth structure coronal to the preparation finish line**, so that occlusal load is transferred to the root rather than levering the post. I want at least **1.5 to 2 mm of height with a 1 mm wall thickness, circumferentially**. Ferrule is a better predictor of survival of a post-retained tooth than the post design itself."

Q If the margin is subgingival, what is the biological constraint?

"The **supracrestal tissue attachment** — what used to be called biologic width — roughly 2 mm of junctional epithelium and connective tissue attachment above the bone crest. I need about **3 mm from the restoration margin to the alveolar crest** to allow for that plus a sulcus. Violating it gives chronic inflammation, bleeding and unpredictable recession."

Q So crown lengthening or orthodontic extrusion?

"**Extrusion** preserves bone and the interdental papillae and is kinder to the adjacent 15, but it is slow and needs a period of stabilisation. **Surgical crown lengthening** is quicker, but on an upper first molar removing 3 mm of bone circumferentially will often **expose the furcation**, and a tooth with an exposed furcation carrying a bridge retainer is a poor long-term bet. That single fact usually decides this tooth."

Q How would you classify a furcation involvement?

"**Hamp**: Class I is horizontal loss of less than a third of the tooth width, Class II more than a third but not through-and-through, Class III is through-and-through. The 2017 World Workshop also grades furcation involvement as part of periodontitis staging. Anything beyond Class I in an abutment I am about to root-treat and crown makes me much more inclined to extract."

Q What radiographs would you take?

"**Periapicals of the 14, 16 and 17** with a paralleling technique — I need bone levels, root morphology, the periapical status of the 16 and the depth of the caries. If an implant becomes the plan I would then consider a **CBCT of the 16 site** specifically to measure residual bone height beneath the sinus, justified separately under IR(ME)R."

Q What does her anxiety about "losing more teeth" tell you?

"That the consent conversation is the difficult part of this case, not the dentistry. She may be assuming an implant is permanent and maintenance-free. If I do not correct that gently and **before** treatment, every future complication will feel to her like a betrayal rather than an expected event."

Domain 2 — Clinical decision making and treatment planning

Chain: generating and testing the options

Q Give me the options.

"First, **remove the bridge and assess** — nothing is decidable until then. If the **16 is restorable**: root canal treatment, core and adequate ferrule, then either two separate crowns or a new bridge. If the **16 is unrestorable**: extract it, restore the 14 separately, and for the space either a **cantilever from the 14**, a **new 14–17 bridge**, an **implant at the 16**, a **partial denture**, or **no replacement**."

Q What would you recommend?

"Assuming the 16 is unrestorable: **extract with ridge preservation, crown the 14 as a single unit, and then give her a genuine choice** between an implant at the 16 and accepting the space with monitoring. She is 62, medically well and wants something fixed, so an implant is reasonable — but I would confirm the residual height beneath the sinus first, because an 18-year-old failing molar is usually accompanied by a **pneumatised sinus**."

Q You have not offered a new 14–17 bridge. Argue for it.

"It avoids surgery, is quicker and cheaper, and she already accepts fixed work. But it requires me to **prepare a sound, unrestored 17** — destroying tissue on a healthy tooth to replace one that has already failed — and it puts three teeth at risk instead of one. In a patient with a sound 14 and a sound 17, that is the argument that defeats it."

Q A colleague would place a cantilever pontic from the 14. Why might they be right?

"Because it involves only a tooth that already needs crowning, avoids surgery entirely, and cantilever bridges do have acceptable survival — around **80% at ten years** — particularly with a single premolar-sized pontic. My hesitation is that a **molar-sized pontic cantilevered from a single premolar** creates a substantial lever arm on an abutment that is already compromised, and the failure mode is decementation or abutment fracture."

Q Make the case for leaving the space.

"**Shortened dental arch** thinking — Käyser's work showed that an arch of premolars and one molar pair gives adequate function, comfort and stability without progressive deterioration. If she has a sound 17 in occlusion, she has a functional arch. That is a positive, evidence-based plan, not a failure to treat, and I would present it that way rather than as the option of last resort."

Q What must you warn her about if she accepts the space?

"**Over-eruption of the opposing 46**, drifting or tilting of the 15 and 17, and food packing. I would take study models or a scan as a baseline, review at six and twelve months, and be explicit that if drifting begins, the implant option becomes harder — so the decision is not indefinitely reversible."

Q Classify this case on SAC.

"The 16 site is **Advanced** if the sinus is pneumatised and grafting is needed, and **Straightforward to Advanced** if there is adequate native bone — it is a single posterior unit, non-aesthetic, adjacent teeth intact. Prosthodontically **Straightforward**. The whole case is **complex in decision-making rather than in execution**."

Q What if there were only 4 mm below the sinus?

"Then the realistic options narrow to a **short implant**, a **lateral window sinus elevation**, or **no replacement**. A lateral window graft is **beyond my competence and I would refer it** — as is a transcrestal lift at that residual height, where the gain required is large and the membrane is at real risk. Given a sound 14 and 17, I would present the shortened arch positively rather than steer her into a major graft."

Q What do you know about short implants?

"Implants of **6 mm or so** placed in native bone have shown outcomes comparable with longer implants placed alongside sinus augmentation at medium-term follow-up, with **fewer complications and lower morbidity**. The caveats are a shorter track record, sensitivity to crown-to-implant ratio and occlusal overload, and the need for a wide diameter and a narrow occlusal table. It is a legitimate option to discuss, not a compromise to apologise for."

Q How would you explain "permanent" to her?

" 'Nothing in dentistry is permanent, and I'd rather be honest now than have you disappointed later. An implant will very likely still be there in ten years — around nine in ten are. But the crown on top is a mechanical thing: screws can loosen, porcelain can chip, and parts get replaced. And it needs cleaning and check-ups just like a tooth. What I can promise is that I'll tell you what's likely to happen and what it will cost.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: taking the bridge off and getting to a stable position

Q How would you remove the bridge?

"I would try **conservative removal first** — a crown-and-bridge remover or ultrasonic scaler on the retainer margins — because the 14 retainer may then be reusable as a provisional. If that fails, **section the connector and section over the 14 retainer**, cutting through the metal and lifting it off rather than levering, so I preserve every millimetre of tooth structure on an abutment I still want."

Q What do you do immediately after removal?

"Assess and stabilise. **Caries removal on both abutments**, a vitality test and a periapical for the 14, and an honest look at the 16. Then a **provisional** — a temporary bridge or a temporary crown on the 14 with the space held — cemented with temporary cement. She must not leave with a visible gap or an unprotected preparation."

Q Describe the extraction of the 16.

"**Atraumatic and section-based**: raise a minimal or no flap, section the tooth into its three roots with a fine bur, and elevate each root individually along its own path. Periotomes and physics-style forceps rather than crude luxation. The aim is to **preserve the buccal plate and the interproximal bone**, because those determine what I can do next."

Q What is ridge preservation, and what does it actually achieve?

"Grafting the socket — typically a slowly resorbing xenograft with a collagen membrane or a resorbable dressing over it — to **limit the dimensional change of the ridge**. It does not prevent resorption; it reduces it, typically preserving 1 to 2 mm of width and height compared with unassisted healing. In a molar site with an intact septum I would still do it, because the sinus floor drops as the socket resorbs."

Q How long before implant placement?

"**Four months** for a grafted molar socket — enough for soft tissue closure and organisation of the graft, before significant ridge collapse. Immediate placement in a three-rooted molar socket with an infected apex is **not something I would attempt**; the primary stability is unpredictable and the site is contaminated."

Q Give me the treatment sequence.

"Remove bridge and assess → provisional on the 14 → extract the 16 with ridge preservation if unrestorable → interim provisional bridge or denture → **re-evaluate at four months, clinically and radiographically** → definitive crown on the 14 → implant at the 16 if indicated and possible → healing and definitive implant crown → maintenance programme."

Q Describe your ideal three-dimensional implant position at the 16.

"**Mesiodistally** centred in the space with at least 1.5 mm to each adjacent root and enough room for a cleansable emergence. **Buccolingually** with 1.5 to 2 mm of bone buccally, positioned so the screw access emerges through the central fossa. **Apico-coronally** about 3 mm apical to the intended gingival margin. A wide-platform implant, and **at least 2 mm of clearance from the sinus floor**."

Q You catch the Schneiderian membrane during the osteotomy. What now?

"If I am simply preparing an osteotomy and I perforate, I stop, reassess with a **Valsalva test**, and decide honestly whether I can still achieve primary stability in the remaining bone without displacing anything into the sinus. If I cannot, I **abandon and close**, cover with antibiotics and decongestant advice, warn the patient not to blow their nose, and refer. A perforation is not a disaster; **improvising a repair I am not trained for is**."

Q Screw-retained or cement-retained crown here?

"**Screw-retained**, because it is retrievable and because it eliminates **residual subgingival cement**, which is a well-recognised cause of peri-implant disease. At the 16 the access hole falls in the central fossa where it is neither aesthetic nor occlusally critical. I would only cement if the angulation made the access emerge through a cusp, and then I would use a custom abutment with a **supragingival or just-subgingival margin**."

Q How would you design the occlusion on that crown?

"**Narrow occlusal table**, cusp inclines reduced, **light contact in maximum intercuspation** — holding shimstock only under firm closure — and **no working or non-working excursive contact**. Implants have no periodontal ligament and so no proprioceptive protection, and the mesial component of force is what loosens screws."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcomes, records and learning from the failure

Q What survival figures would you quote her?

> "**Conventional fixed bridges: around 90% at ten years**, with about a fifth having had some complication. **Single implant crowns: around 95% survival at ten years**, but with a comparable rate of technical complication — screw loosening, ceramic chipping — and a significant rate of biological complication. I would say plainly that the implant is **not more permanent, it is differently maintained**."

Q What would you record for this case?

> "The pre-operative findings and photographs, the restorability assessment and my reasoning, the options discussed **including no treatment and the shortened arch**, the risks and costs given in writing, her questions and her decision, the implant system, diameter, length and lot number, the insertion torque, the baseline periapical at fitting, and the maintenance interval agreed."

Q She returns at five years with a loose implant crown. Differential and approach?

> "Most likely a **loose abutment screw** — I would remove the crown, inspect the screw and the internal connection, replace the screw, torque to the manufacturer's value and re-check the occlusion, because an occlusal error is usually why it loosened. If the **implant itself is mobile**, that is loss of integration and the implant is removed. I would radiograph and probe before I touch anything."

Q And if the 14 fractures in ten years?

> "Then I assess it exactly as I assessed the 16 — caries extent, ferrule, restorability — and if it is unrestorable I have a **single implant at the 16 that stands alone**, which is precisely why I restored the 14 as an individual crown rather than splinting it into a new bridge. Keeping units independent limits the size of the next failure."

Q What is within your competence here, and what is not?

> "**Within**: removal and assessment of the bridge, endodontics and restoration of the 14, sectional extraction with ridge preservation, and a single posterior implant in adequate native bone with a screw-retained crown. **Beyond**: lateral window sinus elevation, significant vertical augmentation, and management of a displaced implant or a large perforation. I would refer those, and I would say so to her before starting rather than mid-procedure."

Q What would you audit from a case like this?

> "**The longevity and mode of failure of conventional bridges in my own practice.** If mine are failing at the margins, that is a lesson about margin placement, moisture control at cementation and the maintenance advice I give. It is a legitimate quality improvement project: set a standard, take a sample, measure, change something specific, and **re-audit**."

Q How would you know this treatment had succeeded?

> "That at five years she has a stable, cleansable arch she can eat with, that she has not needed unplanned treatment, and that when something does need replacing she is **not surprised by it** — because that was the actual clinical problem she presented with."

WATCH OUT

Go back to restorability: caries extent, ferrule, furcation, endodontic prognosis. Deciding whether the tooth stays or goes is the hinge of this whole case, and saying so out loud buys you a sentence of thinking time.

Running this case to time

Time check. The flow above holds 38 questions and about 2,406 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty minutes** — an overrun of about **five**. What goes first is the **bridge removal** material at the head of Domain 3: the crown-and-bridge remover, sectioning the connector and the temporary cement compress into one question, because the marks in this case sit in **restorability** — ferrule, supracrestal attachment, furcation — in the **argument between the**

options, and in the conversation about the word **"permanent"**. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the survival figures and the audit are.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapicals up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 62-year-old woman with an **18-year-old 14–16 fixed-fixed bridge** that has failed at the distal abutment: the **16 has recurrent caries to gingival level and is non-vital**, the **14 appears sound**. Then the interpretation: **eighteen years is respectable service**, and **caries at the margin of an abutment is the commonest way conventional bridges end** — this is the natural history, not a treatment error. The decision turns on two things: whether the 16 is **restorable, which you cannot judge until the bridge is off**, and what the 14 needs once it loses its retainer. And the framing that scores: she is anxious about losing more teeth and wants "something permanent", so **expectation management is as much a part of this case as the restorative decision**. Immediate need: remove the bridge, assess both abutments directly, **and give her a provisional so she is never without teeth**.
1:20–4:30	Domain 1: fixed prosthodontic groundwork, then judging the 16	Undergraduate prosthodontics first. **Abutment, retainer, pontic, connector** named separately, and this one identified as **fixed-fixed with rigid connectors, so a failure at one end condemns the whole unit**. Failure modes ranked — **caries at an abutment margin first**, then loss of vitality, then decementation, then periodontal breakdown, then mechanical failure — against **around 90% survival at ten years**. **Ante's law** stated and then handled with judgement: **a useful instinct rather than a rule**, not supported as a hard predictor. Then restorability in four parts — **subgingival extent of caries, ferrule, root and furcation anatomy, endodontic treatability** — with the crucial method: **remove all the caries first and then judge; deciding restorability with the bridge still on is guesswork**. **Ferrule** at **1.5 to 2 mm of height with a 1 mm wall, circumferentially**, and the point that **ferrule predicts survival better than the post design**. **Supracrestal tissue attachment** — about 2 mm, needing **3 mm from margin to crest**. Then the sentence that usually decides this tooth: **surgical crown lengthening on an upper first molar will often expose the furcation**, and a furcated tooth carrying a bridge retainer is a poor long-term bet. **Hamp** given for the grading.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: generating and testing the options	The options built around the hinge — **remove and assess first, nothing is decidable until then** — then split by whether the 16 is restorable. The recommendation: **extract with ridge preservation, crown the 14 as a single unit, and then give her a genuine choice** between an implant at the 16 and accepting the space, with the caveat volunteered: an 18-year-old failing molar usually comes with a **pneumatised sinus**, so confirm residual height first. Then the two arguments that carry Domain 2. **The new 14–17 bridge argued for and then defeated on its own ground**: it requires **preparing a sound, unrestored 17 — destroying tissue on a healthy tooth to replace one that has already failed** — and puts three teeth at risk instead of one. **The cantilever from the 14 granted its 80% at ten years**, then refused because a **molar-sized pontic on a single premolar abutment** is a lever arm. And **the space presented positively** — **Käyser**, premolars and one molar pair, **a functional arch** — as an evidence-based plan, with the warning that it is **not indefinitely reversible**: over-eruption of the 46, drifting of the 15 and 17, baseline models and review at six and twelve months. **SAC: complex in decision-making rather than in execution.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: getting to a stable position, then the implant	The referral line **volunteered where it belongs — at the sinus question**. With only **4 mm below the sinus** the options narrow to **a short implant, a lateral window elevation, or no replacement**, and **the lateral window is beyond you and referred — as is a transcrestal lift at that residual height**, where the gain required is large and the membrane is at real risk. Given a sound 14 and 17, **you present the shortened arch positively rather than steer her into a major graft**. **Short implants** stated with the evidence: **around 6 mm in native bone, comparable outcomes to longer implants with sinus augmentation, fewer complications and lower morbidity — a legitimate option to discuss, not a compromise to apologise for**. Then the sequence given as a sequence, and the surgery: **sectional extraction into three roots, each elevated along its own path**, to preserve the plates; ridge preservation that **limits but does not prevent** change, and matters here because **the sinus floor drops as the socket resorbs**; **four months**; **no immediate placement into a three-rooted socket with an infected apex**. The 3D position with its numbers — **1.5 mm to each adjacent root, 1.5 to 2 mm buccally, 3 mm apical to the gingival margin, screw access through the central fossa, 2 mm of clearance from the sinus floor**. Then the perforation answer: **stop, Valsalva, decide honestly whether stability is achievable — and if not, abandon and close, antibiotics, decongestant advice, do not blow the nose, refer. A perforation is not a disaster; improvising a repair you are not trained for is.**
11:30–14:30	Domain 4: outcomes, records and learning from the failure	The survival figures given as a comparison rather than a sales pitch: **conventional bridges around 90% at ten years with about a fifth having had a complication; single implant crowns around 95% at ten years with a comparable technical complication rate** — and then the line the whole case is built to elicit: **the implant is not more permanent, it is differently maintained**. The loose crown at five years worked as a sequence — **radiograph and probe before you touch anything**, screw first, and **mobility of the implant itself means loss of integration and removal**. The 14 fracturing in ten years answered with the design logic: **you restored it as an individual crown rather than splinting it into a new bridge, so a single implant at the 16 stands alone — keeping units independent limits the size of the next failure**. Then the audit: **the longevity and mode of failure of conventional bridges in your own practice**, set as a standard, sampled, measured, changed and **re-audited**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	In her words: *"Nothing in dentistry is permanent, and I'd rather be honest now than have you disappointed later. An implant will very likely still be there in ten years — around nine in ten are. But the crown on top is a mechanical thing: screws can loosen, porcelain can chip, and parts get replaced… What I can promise is that I'll tell you what's likely to happen and what it will cost."* Recorded: the pre-operative findings and photographs, **the restorability assessment and the reasoning**, the options **including no treatment and the shortened arch**, risks and costs **in writing**, her questions and her decision, the system, diameter, length and lot number, insertion torque, the **baseline periapical at fitting**, and the maintenance interval agreed. Success is that at five years **she is not surprised by anything** — because that was the clinical problem she actually presented with.

The spine — the twelve answers to have ready if nothing else

1 Why a bridge failing at eighteen years is natural history, not error · 2 Abutment, retainer, pontic, connector — and why fixed-fixed condemns the unit · 3 The commonest modes of bridge failure, in order · 4 Ante's law, and why you do not treat it as a rule · 5 The four tests of restorability, judged after caries removal · 6 Ferrule — the numbers, and why it beats post design · 7 Supracrestal attachment, and why crown lengthening exposes the furcation here · 8 The new 14–17 bridge argued for, then defeated on the sound 17 · 9 The cantilever from the 14, and the lever arm that kills it · 10 The shortened arch presented positively, and why it is not indefinitely reversible · 11 Four millimetres under the sinus — short implant, referral, or no replacement · 12 What "permanent" actually means, in her words.

Where the marks are lost in this case

Deciding the 16 before the bridge is off. The examiner will push you to condemn or save the tooth on the history alone. The answer that scores is that **restorability is judged after complete caries removal, not through a retainer** — and then the four tests, with the specific reason this tooth usually loses: **3 mm of crown lengthening on an upper first molar exposes the furcation**. Candidates who reach a verdict in the opening have skipped the whole of Domain 1.

Selling the implant as permanent. She has asked for "something permanent" and the examiner is watching what you do with the word. Volunteer that **an implant is not more permanent, it is differently maintained**, give both survival figures side by side, and say that the complication rates are comparable. Correcting that expectation **before** treatment is the difference between an expected event and a betrayal — and it is the answer the closing question is set up to reward.

Never naming the referral line at the sinus. A single posterior implant in **adequate native bone** is yours. What is not: **lateral window sinus elevation, significant vertical augmentation, a transcrestal lift where the required gain is large, and management of a displaced implant or a large perforation** — all referred, and said **to her before starting rather than mid-procedure**. The candidate who describes a lateral window at this level has told the examiner they do not know where the line is.

SOE Case 24 — Well-controlled diabetes and a single posterior implant

THE CASE

57-year-old man, **type 2 diabetes, HbA1c 47 mmol/mol**, on metformin. Non-smoker, regular attender, excellent oral hygiene. Missing **36** for 6 months following extraction for a vertical root fracture. Bone height 12 mm above the canal, width 8 mm.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 57-year-old man missing the 36 for six months after extraction for a **vertical root fracture**, with a single systemic factor — type 2 diabetes on metformin. His **HbA1c of 47 mmol/mol is below the diagnostic threshold for diabetes**, so this is excellent control and places him at essentially the same risk as a non-diabetic. He is a non-smoker, attends regularly and cleans well. The site has **12 mm of height above the canal and 8 mm of width**, which is generous in both dimensions, and the adjacent teeth are sound. This is the profile of a patient who does well with implants. My job is to confirm that impression with a proper medical and site assessment, to plan his appointment around his diabetes rather than around his diagnosis, and **not to over-complicate a straightforward case**."

Domain 1 — Clinical interpretation and reasoning

Chain: the medical groundwork

Q What is HbA1c and what does it actually measure?

"**Glycated haemoglobin** — the fraction of haemoglobin that has been non-enzymatically glycated in proportion to the ambient glucose concentration. Because red cells live around 120 days, it reflects average glycaemia over roughly the **preceding two to three months**, weighted toward the most recent weeks. It is not affected by whether he ate breakfast, which is why it is the measure we use."

Q Interpret his figure of 47.

"In IFCC units, **48 mmol/mol is the diagnostic threshold for diabetes** and equates to 6.5% in the old DCCT units. At **47 he is below that threshold** — his glycaemia is currently in the non-diabetic range. A typical treatment target is around 48 to 53 mmol/mol, and above about **69 mmol/mol** I would regard control as poor. He is doing better than most people without a diagnosis."

Q What is HbA1c unreliable in?

"Anything that changes red cell lifespan or haemoglobin: **anaemia, haemoglobinopathies, recent transfusion, haemolysis, chronic kidney disease and pregnancy**. In those situations the figure can mislead in either direction, and I would ask his physician what they are using instead rather than treat the number as gospel."

Q How does diabetes affect implant outcomes?

"**Well-controlled diabetes shows implant survival close to that of non-diabetics.** Poor control is associated with delayed healing, higher early failure and a higher incidence of peri-implantitis. The consistent message across the literature is that it is **control, not the diagnosis, that matters** — and that a well-controlled diabetic is not a contraindication."

Q What is the biological mechanism?

"Chronic hyperglycaemia produces **advanced glycation end products** which stiffen collagen and alter the inflammatory response through the RAGE receptor. There is impaired **neutrophil chemotaxis and phagocytosis**, reduced collagen synthesis and turnover, and **microvascular disease** reducing perfusion and oxygen delivery to the wound. The net effect is slower, less reliable healing and reduced resistance to infection."

Q Tell me about metformin.

"A **biguanide** — it reduces hepatic gluconeogenesis and improves peripheral insulin sensitivity. It is first line in type 2 diabetes. Importantly for me, **metformin alone rarely causes hypoglycaemia** because it does not stimulate insulin secretion; the drugs that do are sulfonylureas and insulin. Its rare serious adverse effect is lactic acidosis, usually in renal impairment."

Q How would you recognise and manage a hypoglycaemic episode in the chair?

"Sweating, tremor, tachycardia, confusion, aggression, drowsiness. **Stop treatment, sit him up, check capillary blood glucose** — below 4 mmol/L is hypoglycaemia. If conscious and cooperative, **15 to 20 g of oral glucose** repeated after 10 to 15 minutes, then a longer-acting carbohydrate. If unconscious or unable to swallow, **1 mg glucagon intramuscularly**, oxygen, and call an ambulance."

Q What else would you confirm about his diabetes?

"**Who manages it and when he was last reviewed**; whether he has microvascular complications — retinopathy, nephropathy, neuropathy — since those indicate longer-standing disease regardless of his current figure; his full drug list; his renal function, which matters for my analgesic choice; and whether he has ever had a hypoglycaemic episode. I would **write to his GP** setting out the plan."

Chain: the site

Q Describe the anatomy you must respect at the 36.

"The **inferior alveolar canal** and its neurovascular bundle running below the site; the **mental foramen** and its **anterior loop** further forward, which I must identify if my incision or flap extends mesially; the **submandibular fossa**, a lingual concavity below the mylohyoid ridge that can cause a straight drill to perforate the lingual plate; the **lingual nerve**, superficial to the periosteum lingual to the third molar region; and the **external oblique ridge** buccally."

Q What would perforation of the lingual plate cause?

"At best a benign perforation. At worst **haemorrhage into the floor of the mouth** from the sublingual or submental vessels, which can produce a rapidly expanding haematoma and airway compromise. That is why I angle the osteotomy to follow the lingual concavity, palpate the lingual plate through the flap, and consider a CBCT where the cross-sectional form is uncertain."

Q Interpret his bone dimensions.

"**12 mm of height above the canal** lets me place a 10 mm implant with a **2 mm safety margin**, which is what I want as a minimum. **8 mm of width** allows a 4.5 to 5 mm diameter implant with **1.5 to 2 mm of bone buccally and lingually**. Both dimensions are adequate without augmentation — this is native-bone surgery."

Q Why did the tooth fracture vertically, and does it matter now?

"**Vertical root fracture** is typically a root-filled, post-restored or heavily loaded tooth, with the classic signs of a narrow deep isolated probing defect, a sinus tract near the gingival margin and a J-shaped or halo radiolucency. It matters now for two reasons: it usually means **chronic infection was present**, so I want the site properly healed and radiographically clear before placing; and it tells me something about his **occlusal load**, which I will build into the crown design."

Domain 2 — Clinical decision making and treatment planning

Q Give me the options.

"**No treatment with monitoring** for over-eruption of the 26 and tilting of the 35 and 37; a **partial denture**; a **conventional bridge from the 35 and 37**, which means preparing two sound unrestored teeth; a **resin-bonded cantilever**, which is unreliable at a molar span; or an **implant**. Given intact adjacent teeth and generous bone, the implant is clearly the option of choice."

Q Why is the bridge the wrong answer here specifically?

"Because the biological cost falls on **two sound, unrestored teeth**. Preparation removes a substantial volume of dentine and carries a real risk of pulpal death over time, and if either abutment later fails I have converted a one-tooth problem into a three-tooth problem. When the adjacent teeth are unrestored, that is the strongest single argument for an implant."

Q Your colleague would defer until the HbA1c is confirmed by the GP within three months. Why might they be right?

"Because a figure I have been told is not a figure I have verified, and HbA1c can drift. Confirming a **current, GP-sourced value** costs one letter and a few weeks, documents that I checked, and protects the patient if control has slipped since. I would in fact do exactly that — I would obtain written confirmation before surgery while proceeding with planning."

Q At what level of control would you decline?

"I would be reluctant above about **69 mmol/mol**, and would decline elective implant surgery in someone poorly controlled or with unstable disease until they had been optimised with their physician. That is not a permanent refusal — it is **a deferral with a route back**, which I would frame to the patient as working together with their doctor."

Q Classify this case on SAC.

"Surgically **Straightforward** — a single posterior site, adequate height and width, healed ridge, no augmentation, non-aesthetic. Prosthodontically **Straightforward** — a single unit between two sound teeth with adequate restorative space. His diabetes is a **general modifier**, but at this level of control it does not raise the classification. **Well within my competence.**"

Q What would raise it beyond your competence?

"Inadequate height requiring **vertical augmentation or nerve lateralisation**; a severely resorbed site needing block grafting; poorly controlled or complicated diabetes; a bisphosphonate or antiresorptive history with MRONJ risk; or head and neck radiotherapy. Any of those and I refer — and I would say so at the outset rather than discover it mid-surgery."

Q What would you consent for specifically?

"The standard risks — failure to integrate, infection, need for further procedures, cost. Plus site-specific ones: **altered sensation of the lip and chin**, though with 12 mm of clearance the risk is low; **the need for a morning appointment having eaten and taken his metformin**; that healing may be marginally slower; the possibility of screw loosening or ceramic chipping; and the **lifelong maintenance commitment**. Written estimate and a cooling-off period."

Q How would you explain the diabetes issue to him, in his words?

"'Your diabetes is very well controlled — your last blood test was actually in the normal range, and at that level your healing should be just like anyone else's. What I'd ask is that we do this in the morning, that you eat normally and take your tablet as usual beforehand, and that you keep an eye on your sugars for a few days afterwards, because surgery and eating less can unsettle them. If your control slips in future, tell me, because it changes how closely I watch the implant.'"

Domain 3 — Decision making in relation to treatment procedures

Q How do you schedule and prepare the appointment?

"A **morning appointment**, not last thing in the afternoon, so he is not fasting and I am not rushing. Confirm on arrival that he has **eaten and taken his metformin**. A capillary blood glucose if he is unwell, symptomatic or unsure. Keep it short. Have oral glucose available. Chlorhexidine rinse and aseptic technique."

Q Would you give antibiotic prophylaxis?

"Yes — the evidence supports a **single pre-operative dose of amoxicillin 2 to 3 g** an hour before, with a number needed to treat of around 25 to prevent one early failure. **Well-controlled diabetes does not by itself justify a post-operative course**, and I would not give one. Prolonged courses add resistance and side effects without benefit."

Q What local anaesthetic, and why?

"**Buccal and lingual infiltration** with articaine rather than an inferior alveolar block, so that he **retains protective sensation** and can report a sharp shooting pain if I approach the bundle. With 12 mm of clearance the risk is low, but the principle holds. I would avoid excessive adrenaline load only if there were a cardiac indication; his diabetes is not one."

Q Take me through the surgery.

"**Mid-crestal incision** over keratinised tissue with a sulcular extension, full-thickness flap raised atraumatically, mental foramen identified if I extend mesially. **Graded drilling with copious sterile saline, sharp drills, slow speed and intermittent pressure**, direction indicator checked in two planes, depth gauge to confirm. Bone here is likely **D2**, so I would watch torque and countersink if needed."

Q Implant dimensions and torque?

"A **4.5 to 5.0 mm diameter, 10 mm long** implant — wide enough for a molar occlusal table, leaving **2 mm above the canal and 1.5 to 2 mm of bone buccally and lingually**. Insertion torque **25 to 45 Ncm**; I would not chase high torque in dense bone because over-compression causes crestal necrosis and early bone loss."

Q One stage or two?

"**One stage with a healing abutment**, given good primary stability, adequate keratinised tissue and no graft. It avoids a second surgical episode and lets the soft tissue mature around the emergence profile from the start. Two stage if stability were poor, if I had grafted, or if I could not achieve tension-free closure."

Q Closure and post-operative care?

"**4/0 monofilament interrupted sutures**, removed at 7 to 10 days. Written post-operative instructions, ice, no rinsing for 24 hours then warm saline, chlorhexidine, and analgesia. And one instruction specific to him — to **monitor his blood glucose over the next few days**, because surgery, soreness and reduced intake can disturb control in either direction."

Q What analgesia, and is there a caution?

"**Paracetamol with ibuprofen** is my default. The caution is that NSAIDs plus metformin in someone who becomes dehydrated can precipitate **renal impairment and, rarely, lactic acidosis** — so I would check his renal function is normal, keep the NSAID course short, and tell him to keep his fluid intake up. If renal function were impaired I would use paracetamol alone."

Q How and when would you restore it?

"**Conventional loading at about three months**, confirmed by clinical stability and ISQ if available. An **implant-level impression** — closed tray in polyether, or a digital scan with a scan body. A **screw-retained monolithic zirconia crown** on a titanium base, with a narrow occlusal table, a cleansable emergence profile and no subgingival cement."

Q How would you set the occlusion?

"**Light contact in maximum intercuspation** — shimstock held only under firm closure, so that under normal load the implant carries less than the adjacent teeth, which have a periodontal ligament and intrude slightly. **No working or non-working excursive contacts.** Given his history of a vertical root fracture, I would look hard for parafunction and consider a **night guard**."

Domain 4 — Interpretation and problem solving of clinical outcomes

Q What parameters would you monitor, and how often?

"Survival; **marginal bone level against the baseline periapical taken at fitting**; six-point probing depths, bleeding and suppuration with a plastic probe; plaque score; the **prosthesis** for screw integrity, contour and occlusion; and his **HbA1c trend**, which I would ask about at every recall. Six-monthly, moving to four-monthly if control deteriorated or if I saw bleeding on probing."

Q He returns at 4 years with bleeding on probing and 4 mm pockets, no bone loss. Diagnosis?

"**Peri-implant mucositis** — bleeding on gentle probing, with or without increased probing depth, but **no bone loss beyond initial crestal remodelling**. It is reversible, it is caused by biofilm, and it is the precursor of peri-implantitis. That is why the baseline radiograph at fitting earns its keep."

Q Manage it.

"**Remove the crown if retrievable** to assess and clean properly. Debride with **titanium, PEEK or plastic instruments and glycine air-polishing** — never steel on the implant surface. Review the crown contour and whether an interdental brush can actually pass. **Re-instruct with a correctly sized interdental brush and have him demonstrate it**. Check his glycaemic control. **Review at one month**, with resolution of bleeding as the target."

Q It does not resolve at one month. Now what?

"I look for the reason rather than repeating the same treatment. **Is the prosthesis cleansable** — over-contoured, deep margin, residual cement? **Is his technique actually working** — disclose and watch him. **Has his control slipped** — recheck the HbA1c. Repeat non-surgical debridement, and if bleeding persists with deepening probing depths and bone loss on radiograph, I have moved to **peri-implantitis** and a different pathway."

Q If it were peri-implantitis, what would you do?

"Diagnose it properly — **probing depth increase, bleeding or suppuration, and radiographic bone loss beyond initial remodelling**. Non-surgical debridement and risk factor control first, then referral for surgical management if it did not respond, because **regenerative or resective peri-implant surgery is beyond my competence**. And I would be honest with him about the prognosis rather than repeatedly re-cleaning a losing site."

Q His HbA1c comes back at 80 mmol/mol at the next recall. What changes?

"My **recall interval shortens to three or four months**, my threshold for radiographs falls, and I reinforce hygiene actively rather than passively. I would **write to his GP** flagging the periodontal and peri-implant risk, because there is a bidirectional relationship — treating the inflammation can help his glycaemic control as well as the other way round."

Q What would you record for this case?

"The medical history and the **source and date of the HbA1c**, the GP correspondence, the options discussed including no treatment, the consent conversation and written estimate, the implant system, diameter, length and lot number, insertion torque, the surgical protocol and the antibiotic given, the **baseline radiograph at fitting**, and the agreed maintenance interval."

Q How would you know this treatment had succeeded?

"That at ten years he still has a **stable, symptom-free, cleansable implant**, bone levels unchanged against baseline, no bleeding or suppuration, an intact crown he has not needed unplanned appointments for, and two

adjacent teeth still sound and unprepared. That last point is the real measure — I solved the problem **without spending any healthy tissue**."

WATCH OUT

Come back to the principle that in diabetes it is **control, not diagnosis**, that determines risk — then say what you would check, what you would modify, and what would make you defer. It buys a sentence and it is almost always the right frame.

Running this case to time

Time check. The flow above holds 39 questions and about 2,389 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty minutes** — an overrun of about **five**. What goes first is the **generic surgical sequence** in Domain 3: the incision, the flap, the graded drilling and the closure compress into one question, because none of it is specific to this patient. The marks sit in the **HbA1c arithmetic**, in the modifications that are actually about his diabetes — **the morning appointment, the metformin, the glucose monitoring, the NSAID caution** — and in Domain 4's mucositis pathway. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the HbA1c comes back at 80.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiograph and the medical history in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 57-year-old missing the **36** for six months after extraction for a **vertical root fracture**, with a single systemic factor — type 2 diabetes on metformin. Then the sentence the whole case turns on: **an HbA1c of 47 mmol/mol is below the diagnostic threshold for diabetes**, so this is **excellent control and places him at essentially the same risk as a non-diabetic**. Non-smoker, regular attender, cleans well. The site has **12 mm of height above the canal and 8 mm of width** — generous in both dimensions — and the adjacent teeth are sound. **This is the profile of a patient who does well with implants.** Your job: confirm the impression with a proper medical and site assessment, **plan the appointment around his diabetes rather than around his diagnosis**, and **not over-complicate a straightforward case**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the medical groundwork, then the site	Undergraduate medicine first, and it is marked. **HbA1c defined** — glycated haemoglobin, reflecting average glycaemia over the **preceding two to three months** because red cells live about 120 days, and unaffected by whether he ate breakfast. Then the numbers: **48 mmol/mol is the diagnostic threshold**, equal to 6.5% in DCCT units; a target of **48 to 53**; **above about 69, control is poor**. **What HbA1c is unreliable in** — anaemia, haemoglobinopathies, transfusion, haemolysis, chronic kidney disease, pregnancy. The evidence stated as evidence: **well-controlled diabetes shows survival close to non-diabetics; poor control raises early failure and peri-implantitis; it is control, not the diagnosis, that matters**. Then the mechanism — **advanced glycation end products and RAGE, impaired neutrophil chemotaxis and phagocytosis, reduced collagen turnover, microvascular disease**. **Metformin** as a **biguanide that rarely causes hypoglycaemia because it does not stimulate insulin secretion** — sulfonylureas and insulin do. Hypoglycaemia recognised and managed: **glucose below 4 mmol/L, 15 to 20 g oral glucose, or 1 mg glucagon intramuscularly if unconscious**. Then the site: **canal, mental foramen and anterior loop, submandibular fossa, lingual nerve, external oblique ridge**, and **12 mm giving a 10 mm implant with a 2 mm margin, 8 mm giving 4.5 to 5 mm with 1.5 to 2 mm of bone either side**.
4:30–7:30	Domain 2: the options and the threshold	The options with **no treatment named and monitored**, and the bridge refused for a specific reason: **the biological cost falls on two sound, unrestored teeth**, and **when the adjacent teeth are unrestored that is the strongest single argument for an implant**. The colleague who would defer for a GP-confirmed figure **granted the point and then agreed with — a figure you have been told is not a figure you have verified**, it costs one letter, and you would obtain written confirmation before surgery while proceeding with planning. The threshold said out loud: **reluctant above about 69 mmol/mol, declining in poorly controlled or unstable disease — but a deferral with a route back, framed as working with his doctor, not a refusal**. **SAC: Straightforward surgically and prosthodontically; his diabetes is a general modifier that at this level does not raise the classification.** Then what would raise it beyond you: **vertical augmentation or nerve lateralisation, block grafting, poorly controlled or complicated diabetes, an antiresorptive history with MRONJ risk, head and neck radiotherapy** — refer, and **say so at the outset rather than discover it mid-surgery**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the appointment planned around the diabetes	The modifications that are actually specific to him, volunteered rather than extracted: a **morning appointment, not last thing in the afternoon**; confirm on arrival he has **eaten and taken his metformin**; capillary glucose if unwell or unsure; keep it short; **oral glucose available**. Antibiotics answered with the evidence: **a single pre-operative dose of amoxicillin 2 to 3 g an hour before, NNT around 25** — and the discipline that scores, **well-controlled diabetes does not by itself justify a post-operative course, and you would not give one**. **Buccal and lingual infiltration with articaine rather than a block**, so he **retains protective sensation**. Then the numbers: **4.5 to 5.0 mm diameter, 10 mm long, 2 mm above the canal, 1.5 to 2 mm buccally and lingually, insertion torque 25 to 45 Ncm — and you would not chase high torque in dense bone, because over-compression causes crestal necrosis**. **One stage** with a healing abutment. Post-operatively, the instruction that is his alone: **monitor blood glucose over the next few days, because surgery, soreness and reduced intake can disturb control in either direction**. And the analgesia caution: **NSAIDs plus metformin in a dehydrated patient risk renal impairment and, rarely, lactic acidosis** — check renal function, keep the course short, push fluids, **paracetamol alone if renal function is impaired**. Restoration: **conventional loading at three months, screw-retained monolithic zirconia on a titanium base, narrow occlusal table, light contact, no excursive contacts** — and, given a **vertical root fracture in the history, look hard for parafunction and consider a night guard**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: monitoring, and what happens when control slips	What is monitored, with **HbA1c trend asked about at every recall** alongside bone level against **the baseline periapical taken at fitting**. Bleeding with 4 mm pockets and no bone loss named correctly as **peri-implant mucositis** — reversible, biofilm-driven, the precursor of peri-implantitis, **and the reason the baseline radiograph earns its keep**. Managed with **titanium, PEEK or plastic instruments and glycine air-polishing, never steel**, crown contour reviewed, **an interdental brush she can actually pass**, glycaemic control checked, and **review at one month with resolution of bleeding as the target**. If it does not resolve, **look for the reason rather than repeat the same treatment** — cleansability, technique, control. If it becomes peri-implantitis: diagnose it properly, non-surgical therapy and risk-factor control first, then **refer, because regenerative or resective peri-implant surgery is beyond your competence** — and **be honest about prognosis rather than repeatedly re-cleaning a losing site**. Then the HbA1c at 80: **recall to three or four months, a lower threshold for radiographs, active hygiene reinforcement, and a letter to the GP flagging the bidirectional relationship** — treating the inflammation can help his control as well as the other way round.
14:30–15:00	Closes on the patient and the record	In his words: *"Your diabetes is very well controlled — your last blood test was actually in the normal range, and at that level your healing should be just like anyone else's. What I'd ask is that we do this in the morning, that you eat normally and take your tablet as usual beforehand, and that you keep an eye on your sugars for a few days afterwards… If your control slips in future, tell me, because it changes how closely I watch the implant."* Recorded: the medical history with **the source and date of the HbA1c**, the **GP correspondence**, the options including no treatment, the consent and written estimate, the system, diameter, length and lot number, insertion torque, **the antibiotic given**, the baseline radiograph, and the maintenance interval. Success at ten years is a stable cleansable implant **and two adjacent teeth still sound and unprepared** — the problem solved **without spending any healthy tissue**.

The spine — the twelve answers to have ready if nothing else

1 What HbA1c measures, and over what period · 2 Forty-eight is the threshold — so 47 is not diabetic control, it is normal glycaemia · 3 What HbA1c is unreliable in · 4 Control, not diagnosis, is what the evidence says determines risk · 5 The mechanism — AGEs, neutrophil function, microvascular disease · 6 Metformin, and why it rarely causes hypoglycaemia · 7 Hypoglycaemia in the chair — the number, the glucose, the glucagon · 8 Why the bridge is wrong: two sound unrestored teeth · 9 The level at which you would defer, and that it is a deferral with a route back · 10 The appointment modifications that are actually his · 11 The NSAID and metformin caution · 12 Mucositis at four years — instruments, air-polishing, one-month review.

Where the marks are lost in this case

Treating the diagnosis instead of the control. The examiner has chosen 47 mmol/mol deliberately, because it sits **below the diagnostic threshold**. Candidates who launch into diabetic risk, deferral and

antibiotic cover have failed to read the number and have over-complicated a **Straightforward** case. Say the arithmetic out loud — **48 is the threshold, he is at 47, this is normal glycaemia** — and then say that **it is control, not the diagnosis, that the evidence says matters**.

Missing the modifications that are genuinely his. Having said the case is straightforward, you must still show what changes: a **morning appointment with his metformin taken and a meal eaten**, glucose available in the surgery, **glucose monitoring for a few days afterwards**, and the **NSAID-with-metformin renal caution**. Those four, volunteered, are what separate a candidate who knows the disease from one who has memorised a threshold. Equally, volunteer that you **would not give a post-operative antibiotic course** — restraint scores here.

Never naming the two lines. There are two, and both must be said before you are asked. **Medically**: above about **69 mmol/mol**, or in unstable or complicated disease, you defer and optimise with his physician. **Surgically**: **vertical augmentation, nerve lateralisation, block grafting, MRONJ risk and irradiated bone** are referred — and so is **surgical management of peri-implantitis** if it ever comes to that. The candidate who never names a line has not shown the examiner they know where theirs is.

SOE Case 25 — Early peri-implant mucositis in a new patient

THE CASE

49-year-old woman registers at your practice. She has **four implants** — 15, 25, 36, 46 — placed 4 years ago. She is asymptomatic. On examination: **bleeding on probing at all four implants**, probing depths 4 mm, **no suppuration**, plaque score 45%, no radiographic bone loss beyond the first thread. She has not seen a hygienist for 3 years.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 49-year-old woman new to the practice with four implants placed four years ago, **asymptomatic**, but with **bleeding on gentle probing at all four sites**, probing depths of 4 mm, **no suppuration**, and **no radiographic bone loss beyond initial crestal remodelling**. Her plaque score is 45% and she has had no professional maintenance for three years. On the 2017 case definitions that is **peri-implant mucositis at all four implants** — a plaque-induced, reversible inflammation of the peri-implant mucosa. The generalised pattern and the plaque score point to a **maintenance and plaque-control failure rather than four separate local problems**, though I must still check the cleansability of each restoration. What matters is that this has been caught at the reversible stage: mucositis is the precursor of peri-implantitis, and my whole plan is aimed at making sure I never have that second conversation with her."

Domain 1 — Clinical interpretation and reasoning

Chain: the peri-implant tissues from first principles

Q Describe the soft tissue attachment around an implant.

"A **junctional epithelium** of around 2 mm attached by hemidesmosomes to the titanium oxide layer, and below it a zone of **connective tissue about 1 to 1.5 mm deep** in contact with the surface. Together they form a supracrestal seal of roughly **3 to 4 mm** — somewhat longer than around a tooth."

Q How does it differ from the attachment to a natural tooth?

"Three critical differences. There is **no cementum and no periodontal ligament**, so no inserting Sharpey's fibres — the collagen runs **parallel to the implant surface** rather than perpendicular into it. The connective tissue is **more collagen-rich and less vascular**, more scar-like. And the blood supply comes only from the **supraperiosteal vessels**, without the periodontal plexus."

Q What is the clinical consequence of that?

"The seal is **adherent rather than attached**, so it resists apical spread of biofilm less well, and the reduced vascularity means a **less effective inflammatory and reparative response**. Once inflammation is established it spreads more rapidly and further apically than around a tooth, and peri-implantitis lesions are typically larger, non-self-limiting and closer to the bone."

Q Should you probe implants at all?

"Yes. Probing is the **only practical way to detect inflammation early**, and the concern about damaging the seal is unfounded — the epithelium re-adheres within days. I use a **plastic or blunt probe with light force, around 0.25 N**, at **six sites per implant**, recording depth, bleeding and suppuration. Depth alone means little without a baseline; **change** is what I am watching."

Q Define peri-implant health.

"**Absence of erythema, swelling, suppuration and bleeding on gentle probing.** Health can exist with reduced bone support — a previously treated site can be healthy at a lower bone level. Probing depths in health vary with the depth of the tissue at placement, so there is no single normal number; **absence of bleeding is the criterion, not a millimetre value**."

Q Distinguish mucositis from peri-implantitis here.

"**Mucositis**: bleeding on gentle probing, with or without increased probing depth from swelling, but **no bone loss beyond initial crestal remodelling**. **Peri-implantitis** additionally requires **increased probing depth compared with a previous examination and bone loss beyond initial remodelling** — or, without baseline data, bleeding or suppuration plus a probing depth of **6 mm or more** plus a bone level **3 mm or more apical to the coronal intraosseous part** of the implant. She meets none of those."

Q What is "initial crestal remodelling", and how much is acceptable?

"The physiological bone loss in the **first year of function** as the biological width establishes — conventionally accepted up to about **1.5 to 2 mm**, and thereafter no more than around **0.2 mm per year**, which are essentially the Albrektsson success criteria. Her radiographs show loss no further than the first thread, which is consistent with normal remodelling and not with disease."

Q Do you have a baseline?

"Not yet — and I would work hard to get one. With her consent I would **write to the placing practice** for the surgical records, the implant system, diameters and lot numbers, and any radiographs taken at placement and at fitting. Without a baseline I am relying on surrogate thresholds and my own records start today, which makes future interpretation harder."

Q What radiographs would you take now?

"**Long-cone paralleling periapicals of each implant**, angled so the threads are clearly resolved on both sides — if the threads are blurred the geometry is wrong and I cannot measure bone level. These become **her baseline**, dated and stored. A CBCT is not justified for this diagnosis; it adds dose and suffers scatter artefact from the implants."

Q What else would you assess?

"**Full periodontal charting and a BPE of her natural teeth** — mucositis with a 45% plaque score usually sits alongside gingivitis or periodontitis. Smoking, medical history including diabetes, medication, dexterity, and what cleaning aids she actually uses. And critically, the **cleansability of each restoration**: contour, embrasure size, margin position, whether they are cemented, and whether an interdental brush can physically pass."

Q Her natural teeth show 5 mm pockets and 30% bone loss. Classify that.

"On the 2017 classification: bone loss in the coronal third at 30% and interdental attachment loss of that order puts her at **Stage II**, generalised in extent. **Grade** comes from the bone loss percentage divided by age — 30 over 49 is about 0.6, so **Grade B**, moderate progression, modified upward if she smokes or is diabetic. That diagnosis is not incidental: **a history of periodontitis is the strongest risk factor for peri-implantitis**."

Q Name the other risk factors.

"**Poor plaque control** and **no supportive maintenance** — both present here. **Smoking**. **Poorly controlled diabetes**. **Residual submucosal cement**. **Restorations that cannot be cleaned** — over-contoured or with deep margins. **Lack of keratinised mucosa**. And **occlusal overload** as a contributory rather than initiating factor."

Domain 2 — Clinical decision making and treatment planning

Q How common is what you are looking at?

"Common. Meta-analysis of the prevalence puts **peri-implant mucositis at around 43% of implants** and **peri-implantitis at around 22%** — so roughly one implant in five eventually develops the bone-losing form. That

figure is worth quoting to her, because it reframes maintenance from an optional expense into the thing that determines which group she joins."

Q Give me your management plan.

"**Cause-related therapy** — explain the disease and that it is reversible, assess and modify risk factors, **assess each prosthesis for cleaning access and modify contours where needed**, and assess her ability to perform plaque control. **Non-surgical therapy** — professional biofilm removal, treatment of the periodontitis on her natural teeth, and individualised oral hygiene instruction. **Re-assess at one month.** Then **supportive therapy at 3 to 5 monthly intervals** if resolved."

Q Why treat her natural teeth at the same time?

"Because periodontal pockets are a **reservoir of pathogens** that recolonise the peri-implant sulcus, and because the two conditions share the same cause and the same behavioural fix. Treating the implants while leaving 5 mm pockets untouched is treating half a mouth. I would follow the **EFP S3 stepwise approach** — behaviour change, then subgingival instrumentation, then re-evaluation, then supportive care."

Q What if the crowns are over-contoured and cannot be cleaned?

"Then the restorations are the cause and no amount of instruction will fix it. I would **recontour where the material allows**, or discuss **remaking them as screw-retained, cleansable designs** with the cost explained honestly. With a patient who has just registered, I would present that as a **finding about the shape of the crown**, not as a criticism of the previous clinician."

Q How would you check whether they are cemented, and why does it matter?

"Look for a **screw access channel** under a composite plug, transilluminate, and take a **well-angled periapical** which often shows the abutment–crown junction and sometimes the cement itself. It matters because **residual submucosal cement is a recognised cause of peri-implant disease**, and it is a cause I can actually remove — which makes it worth ten minutes of looking."

Q Does she need more keratinised tissue?

"Not on this presentation. **Less than 2 mm of keratinised mucosa** is associated with more plaque, more brushing discomfort and more recession, and a graft can be justified where a patient genuinely cannot clean because it hurts. But her problem is **45% plaque and three years without maintenance**, and I would not offer surgery for what is a hygiene and access problem."

Q Your colleague would prescribe a course of antibiotics. Why might they be right, and would you?

"They might argue that four inflamed sites represent an established infection. But **mucositis is a biofilm disease and biofilm is removed mechanically, not chemically** — systemic antibiotics show no benefit in mucositis, and prescribing them here risks resistance and side effects and, worse, teaches her that a tablet solves it. **I would not prescribe.** I would use a **chlorhexidine mouthwash as a short-term adjunct at most.**"

Q Classify this case on SAC, and where does your competence end?

"As a maintenance and non-surgical case it is **Straightforward** and entirely within my competence — diagnosis, debridement, hygiene instruction, contour assessment and supportive care. **Beyond my competence**: surgical management of peri-implantitis, whether resective or regenerative, implant surface decontamination protocols at open flap, implantoplasty, and explantation. If she progressed to established peri-implantitis unresponsive to non-surgical care, **I would refer**."

Q How would you explain this to her, given she has no symptoms?

" 'Your gums are bleeding around all four implants when I test them gently. That's an infection in the gum — it doesn't hurt, which is exactly why people don't notice it. The important thing is that at this stage it's completely reversible. If it's left, it can spread to the bone that holds the implants in, and that stage is much harder to treat and can end with losing them. So what I'd like to do is clean them properly, show you how to clean around them, and see you again in a month.' "

Q She says she cannot afford three-monthly hygienist visits. What do you do?

"I do not treat that as a refusal. I would find the **minimum viable interval she will actually attend**, make her home care carry more of the load with a disclosed, supervised technique, and be explicit that a longer interval means a higher risk which I will monitor for. And I would **record the conversation**, because a patient who cannot afford maintenance needs to have understood the trade-off, not just been told it."

Domain 3 — Decision making in relation to treatment procedures

Q Describe the professional cleaning.

"**Titanium or PEEK curettes** and an ultrasonic with a **plastic-sheathed tip** on the abutment and any exposed implant surface. **Glycine or erythritol air-polishing delivered submucosally**, which reaches thread and abutment surfaces far better than a curette can. Polish with a non-abrasive paste. Then the part that actually determines the outcome — **disclose, size an interdental brush to each embrasure, and have her demonstrate it in a mirror while I watch and correct her**."

Q Why never steel?

"Because a steel instrument **scratches the titanium oxide layer**, leaving a roughened surface that retains biofilm more readily than the original machined or polished collar, and it can smear metal onto the surface. The damage is permanent and it makes every subsequent clean less effective. On the **crown or the abutment above the mucosa** conventional instruments are acceptable; on the implant surface they are not."

Q Why is air-polishing better than curettage on an implant?

"Because an implant surface is **threaded and convex, not a flat root surface** — a curette blade cannot adapt to it and physically cannot reach between threads or into a concave transmucosal contour. A **low-abrasive glycine or erythritol powder** follows the surface hydrodynamically, disrupts biofilm effectively, and does not roughen titanium. It is also more comfortable, which matters for a treatment she must accept repeatedly."

Q What about lasers, local antimicrobials and probiotics?

"For **mucositis, the evidence does not show a consistent added benefit over thorough mechanical biofilm removal with good oral hygiene**. Local antimicrobials have some role as adjuncts in peri-implantitis; lasers and probiotics remain unproven. I would **not spend her money on adjuncts before I had proved that the mechanical and behavioural basics were being achieved**."

Q Take me through the oral hygiene instruction in detail.

"**Disclose first**, so she sees the plaque rather than hears about it. Demonstrate in her own mouth, in a mirror. A **powered toothbrush** angled at the mucosal margin. **Interdental brushes sized individually to each embrasure and written down by colour**, passed straight through rather than forced. **Superfloss or a floss threader** under a splinted or bridge-form restoration. Then have her do it back to me — **instruction she has not performed is instruction she has not received**."

Q How would you recontour an over-contoured crown intra-orally?

"**Fine diamonds under copious water** to reduce the emergence and open the embrasure, working from the mucosal margin outward, then a **graded polishing sequence to a high lustre** — a rough surface is worse than the original contour. On zirconia I would use zirconia-specific polishers. If the required reduction would perforate to the abutment or destroy the contact point, **the crown needs remaking, not adjusting**."

Q You need to remove a cemented crown to assess a site. How?

"Ideally I do not — I try to manage it in situ. If I must, I would **attempt a crown remover or an ultrasonic on the margin**, accepting that a definitively cemented crown may have to be **sectioned and remade**, and that must be consented and costed **before** I start. That risk is one of the strongest practical arguments for screw retention in the first place."

Q What do you do at the one-month re-assessment?

"**Re-probe six sites per implant and record bleeding**, re-disclose and re-score plaque, re-check the periodontal charting on her natural teeth, and ask her to demonstrate her cleaning. Compare directly with the figures recorded today. **Resolution of bleeding is the endpoint**; if plaque control has improved but bleeding has not, the prosthesis or the anatomy is the problem, not her."

Q How would you plan her supportive care?

"A defined **3 to 5 monthly recall**, tailored to her risk — and with a history of periodontitis, a 45% plaque score and four implants, I would start at **three months**. Each visit: plaque score, six-point peri-implant probing and bleeding, periodontal charting, professional biofilm removal, re-instruction, and **radiographs when clinically indicated rather than routinely**."

Domain 4 — Interpretation and problem solving of clinical outcomes

Q What would you record today?

"**Six-point probing and bleeding at each implant and each tooth**, the plaque score, the prosthetic assessment site by site, the periapical radiographs as baseline, the **diagnosis in the words of the 2017 classification**, the advice given, the interdental brush sizes prescribed, the recall interval agreed, and the letter requesting the original records."

Q How do you define success at one month?

"**Absence of bleeding on gentle probing** at all four implants, with no increase in probing depth. I would tell her that plainly at the outset — that this is our shared target and it is a target she controls more than I do. Reduced but persistent bleeding is a partial response and means further non-surgical therapy and another look at the prostheses."

Q At the one-month review three sites have resolved but the 46 still bleeds. What is going on?

"Something **local to the 46**. My differential in order: an **over-contoured or unreachable emergence profile**; **residual cement**; an **open or deficient margin**; **inaccessibility because of the tongue or a shallow vestibule**; or that she genuinely cannot get the brush there. I would re-examine the site with a radiograph and disclose it specifically, rather than simply repeating the debridement."

Q Two years on, that site has 6 mm pockets, suppuration and 3 mm of bone loss. Diagnosis and action?

"**Peri-implantitis at the 46.** I would confirm against my own baseline, restate the risk factors, remove or modify the prosthesis if it is contributory, and carry out **non-surgical debridement with air-polishing and risk-factor control**. If it did not respond, **surgical management is beyond my competence and I would refer**. And I would be honest with her about the prognosis rather than re-cleaning a losing site indefinitely."

Q She asks whether the implants will have to come out. What do you say?

"'Not at this stage, and I don't want to alarm you. What you have now is treatable, and most implants in this situation settle down completely with cleaning. If it ever did get to the bone, we'd treat that too and I'd get a specialist involved. Implants only come out when the bone loss can't be stopped, and we are a long way from that — which is why I'd like to see you regularly rather than in three years' time."'

Q What would you audit from a case like this?

"The **proportion of my own implant patients who are actually in a defined maintenance programme**, and their bleeding and plaque scores. If patients are drifting out of recall, that is a systems failure in my practice, not a patient failure — and the change I would test is a recall protocol with active follow-up, then **re-audit** the attendance and the bleeding scores."

Q What is the single most important thing in this case?

"**That it has been caught at the reversible stage.** Mucositis is the last point at which the disease can be turned round completely. Everything I do — the debridement, the contour modification, the instruction and, above all, getting her into a **3- to 5-monthly maintenance programme** — is aimed at making sure the peri-implantitis conversation never happens."

WATCH OUT

Go back to the case definitions — bleeding, probing depth, bone loss — say which of the three are present and which are absent, and the diagnosis and the plan follow from it.

Running this case to time

Time check. The flow above holds 39 questions and about 2,535 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty-one minutes** — an overrun of about **six**. What goes first is the **restoration-handling** material in Domain 3: intra-oral recontouring with fine diamonds and the removal of a definitively cemented crown compress into one question, and the lasers, local antimicrobials and probiotics get sampled rather than listed. The marks sit in the **2017 case definitions**, in **probing an implant properly**, in the **risk factors and the prevalence figures**, and in the **one-month reassessment**. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the 46 is still bleeding.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the four periapicals and the chart in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 49-year-old new patient with four implants placed four years ago, **asymptomatic**, with **bleeding on gentle probing at all four**, **4 mm depths**, **no suppuration**, **no bone loss beyond initial crestal remodelling**, a **45% plaque score** and **three years without professional maintenance**. Then the diagnosis, given in the language of the classification: **peri-implant mucositis at all four implants** — plaque-induced, reversible. Then the interpretation that scores: the **generalised pattern and the plaque score point to a maintenance and plaque-control failure rather than four separate local problems**, though you must still check the cleansability of each restoration. And the framing: this has been **caught at the reversible stage**, mucositis is the precursor of peri-implantitis, and **the whole plan is aimed at never having the second conversation**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the peri-implant tissues from first principles, then this patient	Undergraduate biology first. **Junctional epithelium of about 2 mm attached by hemidesmosomes to the titanium oxide layer, connective tissue of 1 to 1.5 mm, a supracrestal seal of 3 to 4 mm.** Three differences from a tooth: **no cementum and no periodontal ligament, so fibres run parallel rather than inserting; more collagen-rich, less vascular, more scar-like; blood supply from supraperiosteal vessels only**. Then the consequence — the seal is **adherent rather than attached**, so lesions spread further and faster and are **larger and non-self-limiting**. **Yes, you probe** — the only practical way to detect inflammation early, the epithelium re-adheres in days, **plastic or blunt probe at around 0.25 N, six sites** — and **change, not a millimetre value, is what you watch**. **Health** defined as **absence of erythema, swelling, suppuration and bleeding**, and it can exist at a reduced bone level. The case definitions given cleanly, including the **no-baseline threshold — bleeding or suppuration plus 6 mm or more plus bone 3 mm or more apical to the coronal intraosseous part** — and **she meets none of them**. **Initial remodelling of 1.5 to 2 mm in the first year, then 0.2 mm a year**, the Albrektsson criteria. The absent baseline chased: **write to the placing practice with her consent** for records, system, lot numbers and films. **Long-cone paralleling periapicals with the threads clearly resolved on both sides — if the threads are blurred the geometry is wrong**, and no CBCT for this diagnosis. Then her natural teeth: **Stage II, generalised, Grade B — 30 over 49 is about 0.6** — and the line that must be volunteered: **a history of periodontitis is the strongest risk factor for peri-implantitis**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the plan, and what you refuse	The prevalence quoted as a lever, not a statistic: **mucositis around 43% of implants, peri-implantitis around 22% — roughly one implant in five** — and **worth quoting to her because it reframes maintenance from an optional expense into the thing that decides which group she joins**. The plan in its three named stages: **cause-related therapy, non-surgical therapy, reassessment at one month, then supportive therapy at 3 to 5 monthly intervals**. Her natural teeth treated at the same time, and why: **periodontal pockets are a reservoir that recolonises the peri-implant sulcus**, and **treating the implants while leaving 5 mm pockets is treating half a mouth** — the **EFP S3 stepwise approach** named. The over-contoured crown answered as a **finding about the shape of the crown, not a criticism of the previous clinician**. Residual cement hunted for, **because it is a cause you can actually remove — which makes it worth ten minutes of looking**. Keratinised tissue refused for now: **her problem is 45% plaque and three years without maintenance, and you would not offer surgery for a hygiene and access problem**. And the antibiotic refused with the reason: **mucositis is a biofilm disease and biofilm is removed mechanically, not chemically** — no benefit in mucositis, and prescribing **teaches her that a tablet solves it**. **SAC: Straightforward.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the treatment, and the instruction	The competence line said plainly: **diagnosis, debridement, hygiene instruction, contour assessment and supportive care are yours; surgical management of peri-implantitis — resective or regenerative — open-flap decontamination protocols, implantoplasty and explantation are not.** Then the cleaning with its materials: **titanium or PEEK curettes, a plastic-sheathed ultrasonic tip, glycine or erythritol air-polishing delivered submucosally**, non-abrasive polish. **Never steel**, and the reason — it **scratches the titanium oxide layer**, leaving a surface that **retains biofilm more readily than the original collar**, and **the damage is permanent**; above the mucosa on the crown or abutment, conventional instruments are fine. Why air-polishing beats a curette: an implant surface is **threaded and convex, not a flat root surface**, so a blade **physically cannot reach between threads**. Then the part that actually determines the outcome — **disclose first, demonstrate in her own mouth in a mirror, a powered brush at the mucosal margin, interdental brushes sized individually and written down by colour, superfloss under bridge-form work — and then have her do it back to you**, because **instruction she has not performed is instruction she has not received**. The one-month reassessment specified as a **direct comparison against today's recorded figures**, with the diagnostic sentence: **if plaque control has improved but bleeding has not, the prosthesis or the anatomy is the problem, not her**. Supportive care **starting at three months** given her risk.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: outcomes, and the site that does not settle	What is recorded today — six-point probing and bleeding at **every implant and every tooth**, plaque score, **a prosthetic assessment site by site**, the periapicals **as baseline**, the diagnosis **in the words of the 2017 classification**, the brush sizes, the interval, and the letter requesting the original records. Success at one month defined as **absence of bleeding on gentle probing with no increase in depth**, stated to her at the outset as **a shared target she controls more than you do**. The 46 still bleeding given a differential **local to that site, in order**: over-contoured or unreachable emergence, **residual cement**, an open or deficient margin, inaccessibility from the tongue or a shallow vestibule, or she genuinely cannot get the brush there — **re-examine and disclose that site specifically rather than repeat the debridement**. Two years on, **6 mm, suppuration and 3 mm of bone loss** named as **peri-implantitis at the 46**, confirmed **against your own baseline**, treated non-surgically with risk-factor control, and **referred if it does not respond** — with prognosis given honestly **rather than re-cleaning a losing site indefinitely**. Then the audit: **the proportion of your own implant patients actually in a defined maintenance programme**, because patients drifting out of recall is **a systems failure in your practice, not a patient failure**.
14:30–15:00	Closes on the patient and the record	In her words: *"Your gums are bleeding around all four implants when I test them gently. That's an infection in the gum — it doesn't hurt, which is exactly why people don't notice it. The important thing is that at this stage it's completely reversible… So what I'd like to do is clean them properly, show you how to clean around them, and see you again in a month."* And when she asks whether the implants will come out: *"Not at this stage… Implants only come out when the bone loss can't be stopped, and we are a long way from that — which is why I'd like to see you regularly rather than in three years' time."* If she cannot afford three-monthly care, **find the minimum interval she will actually attend**, load her home care, say plainly that a longer interval means higher risk, **and record the conversation** — she needs to have understood the trade-off, not just been told it.

The spine — the twelve answers to have ready if nothing else

1 The peri-implant seal — the layers, and the three differences from a tooth · 2 Why the seal is adherent rather than attached, and what that costs · 3 Yes you probe, and with what force at how many sites · 4 Peri-implant health defined without a millimetre value · 5 Mucositis versus peri-implantitis, including the no-baseline thresholds · 6 Initial crestal remodelling — 1.5 to 2 mm, then 0.2 mm a year · 7 Staging and grading her natural teeth, and why that diagnosis is not incidental · 8 The risk factors, with previous periodontitis first · 9 The prevalence figures, and why you quote them to her · 10 Why no antibiotic for mucositis · 11 Never steel — and why air-polishing beats a curette on a thread · 12 The one-month endpoint, and what persistent bleeding at one site means.

Where the marks are lost in this case

Calling it peri-implantitis. Every element the examiner has given you — **bleeding, 4 mm, no suppuration, no bone loss beyond the first thread** — is chosen so that the case definitions are the answer. Name **peri-implant mucositis in the words of the 2017 classification**, say what she would need to meet to be peri-implantitis **including the no-baseline thresholds**, and say that **this is the last point at which the disease can be turned round completely**. A candidate who overcalls the diagnosis has thrown away the whole frame of the case.

Treating four implants and ignoring the mouth. She has **5 mm pockets and 30% bone loss** on her natural teeth, and that is not background detail: **a history of periodontitis is the strongest risk factor for peri-implantitis**, and untreated pockets are a **reservoir that recolonises the peri-implant sulcus**. Stage and grade her, treat both together under the **EFP S3 approach**, and volunteer that **treating the implants alone is treating half a mouth**.

Never naming what you would not attempt — and what you would not prescribe. The competence line here is unusually clean: **non-surgical management, contour assessment, instruction and supportive care are yours; resective or regenerative peri-implantitis surgery, open-flap decontamination, implantoplasty and explantation are not** — refer. Say it before Domain 4 forces it. And there is a second line of the same kind: **you would not prescribe antibiotics for mucositis, and you would not spend her money on lasers, local antimicrobials or probiotics before proving the mechanical and behavioural basics were being achieved.** Restraint, said out loud, is what a 5 looks like here.

SOE Case 26 — Progressive bone loss at five years

THE CASE

60-year-old male, ex-smoker (stopped 3 years ago, 25 pack-years). Implant at the **26** placed 5 years ago with a **cement-retained crown**. Baseline radiograph available. Current: **bleeding and suppuration**, probing 8 mm mesially, **bone loss of 5 mm — approximately 50% of the implant length**. Implant is not mobile.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 60-year-old ex-smoker with a five-year-old cement-retained crown on an implant at the 26. He has bleeding and suppuration, an 8 mm pocket mesially, and 5 mm of bone loss against a documented baseline radiograph — roughly half the implant length. The implant is not mobile. My diagnosis is **advanced peri-implantitis**, and with a baseline available I can say that confidently rather than by inference. The likely drivers are **residual submucosal cement**, a **25 pack-year smoking history** and the absence of supportive care. The implant is still integrated, so the central question is whether it is **rational to keep it**, and that decision needs to be made explicitly, with an endpoint agreed in advance, rather than drifting into open-ended treatment."

Domain 1 — Clinical interpretation and reasoning

Chain: the peri-implant tissues

Q Describe the soft tissue attachment around an implant.

"A **junctional epithelium** of around 2 mm attached by hemidesmosomes, and below it a connective tissue zone of about 1 to 1.5 mm in contact with the implant surface. Together they form the **supracrestal attachment** — what used to be called the biologic width. The collagen fibres in that connective tissue run **parallel or circular** to the implant surface; they do not insert into it."

Q How does that differ from a tooth, and why does it matter?

"Around a tooth, Sharpey's fibres insert **perpendicularly into cementum**, and the **periodontal ligament** supplies a second vascular plexus, proprioception and progenitor cells. The implant has none of that. Its blood supply is **supraperiosteal only**, and the tissue is more collagenous and less cellular — closer to scar. So the seal is weaker mechanically and the defence is poorer."

Q What is the clinical consequence of those differences?

"Peri-implant lesions **extend further apically and closer to bone** than periodontal lesions, they are less well contained by the tissues, and untreated peri-implantitis tends to progress in a **non-linear, accelerating** pattern. There is also no proprioception, so occlusal overload is not signalled, and the patient feels nothing until suppuration or mobility appear."

Q How do you probe an implant, and what is normal?

"Gently — around **0.2 to 0.25 N** — with a plastic or a fine conventional probe at six sites. A depth of **3 to 4 mm** can be entirely healthy, because depth reflects mucosal thickness and prosthetic emergence as much as disease. What matters is **change from the baseline recorded at fitting**, together with bleeding and suppuration."

Q Define peri-implant health, mucositis and peri-implantitis using the 2017 World Workshop.

"**Health** — no erythema, swelling, suppuration or bleeding on gentle probing, and no bone loss beyond initial crestal remodelling. **Mucositis** — bleeding on probing with no bone loss beyond that remodelling; it is **reversible**. **Peri-implantitis** — bleeding or suppuration, increased probing depth compared with baseline, and **progressive bone loss beyond crestal remodelling**."

Q And if you have no baseline records?

"Then the case definition is **probing depth of 6 mm or more, bleeding or suppuration, and bone levels 3 mm or more apical to the most coronal intra-osseous part of the implant**. I would use that threshold and note in the record that the diagnosis was made without baseline data."

Q Which do you have here?

"**Peri-implantitis, and advanced.** I have suppuration, an 8 mm pocket and 5 mm of bone loss measured against a documented baseline. The absence of mobility does not soften the diagnosis — mobility means complete loss of integration and is a late, terminal sign, not a screening test."

Q How do you quantify the bone loss?

"Against the baseline film, measuring from a fixed reference — the **implant shoulder** — to the first bone-to-implant contact, mesially and distally, using the **thread pitch as an internal scale**. Fifty per cent of a 10 mm implant is 5 mm, and that figure is what will drive the retain-or-remove decision."

Q What radiographic technique, and would you take a CBCT?

"A **long-cone paralleling periapical** with a holder, reproducing the original geometry so the two films are comparable. I would only add a **CBCT** if the buccal or palatal defect morphology would change my surgical plan, and I would justify that exposure under IR(ME)R — scatter from the implant degrades the image, so it is not a free decision."

Q What has caused this?

"In order of likelihood: **residual submucosal cement**; his **smoking history**; the absence of any structured maintenance; and a prosthesis or implant position that cannot be cleaned. I would look at the film specifically for radiopaque cement at the margin, and assess the depth of the crown margin and the emergence profile once the crown is off."

Domain 2 — Clinical decision making and treatment planning

Chain: risk factors and the periodontal groundwork

Q List the recognised risk factors for peri-implantitis.

"A **history of periodontitis** — the strongest — **poor plaque control**, and **absence of supportive maintenance**. Then **smoking**, poorly controlled **diabetes**, **submucosal cement**, prostheses that cannot be cleaned, implant malposition, and a lack of keratinised mucosa. Most of them are present or plausible in this man."

Q He stopped smoking three years ago. Does 25 pack-years still count against him?

"Yes, but less than it did. Smoking impairs the microvasculature, neutrophil function and collagen turnover, and it is dose-related. **Three years of cessation is a genuinely favourable prognostic factor** and I would say so to him, because it makes him more likely to engage. But his tissues carry the history, and I would still class him as higher risk than a lifelong non-smoker."

Q If he had residual periodontitis on his own teeth, how would you stage and grade it?

"**Stage** by severity and complexity from interproximal bone loss or attachment loss at the worst site — I to IV, Stage III and IV involving tooth loss. **Grade** by rate of progression — A slow, B moderate, C rapid — modified by **smoking of ten or more a day and by HbA1c of 7% or above, which push you to Grade C**. His history would have graded him C while he smoked."

Q Why is submucosal cement so damaging?

"It is a **rough, retentive foreign body** sitting below the mucosa where neither the patient nor the hygienist can reach it. It holds a biofilm directly against the implant surface and provokes a sustained inflammatory response. The deeper the crown margin, the more cement is left and the less likely it is to be detected — which is why the damage is often extensive before anything is noticed."

Q How should it have been avoided?

"Keep the **margin at or no more than 1 mm below the mucosa**, use minimal cement, seat the crown first on a **replica abutment or a cementation jig** to extrude the excess extra-orally, clean thoroughly with floss and a probe, and take a **post-cementation radiograph**. Or avoid the problem entirely with a **screw-retained restoration**, which is what I would do at a posterior site."

Chain: retain or remove

Q Is it rational to keep this implant?

"It is borderline, and I would set out criteria rather than guess. **Against** — 50% bone loss, suppuration, a possibly uncleanable restoration, and a smoking history. **For** — it is still integrated, and the site is **posterior and non-aesthetic**, so resective therapy with recession is an acceptable outcome. Removal means either grafting and a new implant, or no molar at all."

Q So what is your position?

"I would attempt treatment with a **clearly defined endpoint agreed in advance and put in writing**: if suppuration and bleeding persist after surgical therapy, or if bone loss progresses at review, the implant comes out. Setting that threshold now prevents it being renegotiated in eighteen months when we are both invested in having tried."

Q A colleague would remove it today. Why might they be right?

"Because at 50% bone loss the published outcomes of surgical therapy are **unreliable — recurrence is common** — and every month of treatment costs him money and destroys more bone, which makes the eventual grafting harder. Removing now, while the surrounding bone is better, gives the cleanest site for a replacement. That is a defensible position, and if he did not want a treatment attempt I would support it."

Q What is the SAC classification, and what is within your competence?

"The non-surgical and prosthodontic management is **Straightforward to Advanced**, and I would do all of it — diagnosis, removal of the crown and cement, debridement, and conversion to a cleansable screw-retained restoration. **Surgical management of a 50% defect is Complex** on SAC: a compromised site, reduced bone, a systemic risk factor, uncertain outcome. **That phase I would refer**, having completed and documented the cause-related therapy first."

Q Tell him what is happening, in his words.

" 'The gum and bone around this implant are infected. You've lost about half the bone that was supporting it, and there's pus coming from the gum. The most likely cause is a small amount of cement left under the gum when the crown was fitted, and the smoking history hasn't helped. We can try to stop it getting worse, but we can't grow the bone back, and the gum will shrink so more of the implant shows. If it doesn't settle, the implant has to come out.' "

Q What must the consent cover?

"That treatment aims to **arrest** the disease, not regenerate bone; that **recession and exposure of the implant surface are expected consequences of successful treatment**, not complications; the total cost including remaking the crown; that treatment may fail and removal may still be needed; the alternative of removing it now; and what replacement would involve and cost."

Domain 3 — Decision making in relation to treatment procedures

Chain: cause-related and non-surgical therapy

Q What is the first phase of treatment?

> "**Cause-related therapy.** Remove the crown, find and remove all residual cement, and assess whether the restoration and the implant position are cleansable. Reinforce his smoking cessation. Detailed **oral hygiene instruction with interdental brushes and a single-tufted brush** aimed specifically at this site, and full treatment of any periodontitis elsewhere, because the mouth is one ecosystem."

Q The crown is cemented and will not come off. What then?

> "I would attempt controlled removal with a **crown remover or by sectioning the crown**, accepting that it will be destroyed — the crown is not the asset here, the implant is. If the abutment screw channel can be located from the original records or a radiograph, I would cut down to it. If I genuinely cannot retrieve it, I would treat around it and factor that uncleanable margin into the prognosis I give him."

Q What would the replacement restoration look like?

> "**Screw-retained**, so there is no cement at all. A **cleansable emergence profile** — flat or slightly concave, no ridge-lap — with the margin at mucosal level, a narrow occlusal table, and enough embrasure space for an interdental brush. In a compromised site I would accept a slightly less pretty contour in exchange for access."

Q Describe non-surgical debridement of an implant.

> "Mechanical disruption of the biofilm with **titanium or carbon-fibre curettes or a PEEK-tipped ultrasonic**, avoiding steel on the machined surface, plus **glycine or erythritol air-polishing** submucosally, which is the most effective way of reaching threads. Then irrigation and chlorhexidine, and reinforcement of home care."

Q Would you use antibiotics or local antimicrobials?

> "Not routinely. **Systemic antibiotics have no reliable role in non-surgical peri-implantitis therapy** and I would not prescribe them simply because there is pus — antimicrobial stewardship applies. Local delivery agents may give a small additional benefit as an adjunct to mechanical debridement, not instead of it. I would reserve systemic antibiotics for spreading infection or as a peri-operative adjunct at surgery."

Q When do you reassess, and what counts as resolution?

> "At **six to eight weeks**. Resolution would be absence of suppuration, no bleeding on gentle probing, and probing depths reduced. With an 8 mm pocket and suppuration I expect **non-surgical therapy alone will not resolve it** — its realistic function here is to reduce inflammation and prove his engagement before I commit him to surgery."

Chain: surgical therapy

Q What are the surgical options, and how do you choose between them?

> "**Access surgery** alone for debridement; **resective therapy** with osseous recontouring, implantoplasty and an apically positioned flap; or **reconstructive therapy** with a bone substitute and possibly a membrane. The choice is driven by **defect morphology and aesthetics**: contained intrabony defects with three or four walls suit reconstruction; supracrestal, horizontal or one-wall defects suit resection."

Q Which applies here?

> "This is a **posterior, non-aesthetic site**, so recession is tolerable and **resective therapy with implantoplasty** is my default. If the defect proves to be a contained circumferential crater once the flap is raised, I would consider a reconstructive approach — but I would not promise regeneration in advance."

Q Describe the resective surgery.

"Intrasulcular and relieving incisions preserving keratinised tissue; full-thickness flap; **complete removal of granulation tissue**; surface decontamination mechanically with a titanium brush and chemically with saline or chlorhexidine; **osseous recontouring** to eliminate the residual craters; **implantoplasty** of the exposed threads; then an **apically positioned flap** so the smoothed surface is cleansable. Sutures, chlorhexidine post-operatively, no brushing at the site for two weeks."

Q What are the risks of implantoplasty?

"It **weakens the implant** by reducing its wall thickness, which matters if the implant is narrow or the loss is extensive, and it disperses **titanium particles** into the tissues, whose long-term significance is uncertain. So I use copious irrigation and high-volume suction, protect the flap, and I would think hard before doing it on a narrow implant."

Q What determines whether reconstruction has a chance?

"**Defect containment above everything** — a circumferential intrabony crater with intact walls has the blood supply and space maintenance to fill; a dehiscence does not. Then a decontaminated surface, primary closure without tension, no smoking, and a patient who will attend maintenance. Even in ideal conditions I would describe it as **defect fill, not re-osseointegration**."

Q Post-operative regime?

"**Chlorhexidine 0.2% mouthwash twice daily for two weeks** in place of brushing at the site, analgesia, and no denture or removable appliance pressure. Peri-operative systemic antimicrobials only if clinically indicated. Suture removal at **7 to 14 days**, and no probing for **three months** so the healing attachment is not disrupted."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcome, failure and what happens next

Q How will you judge success at one year?

"**Absence of suppuration**, absence of bleeding on gentle probing, probing depths no greater than the post-treatment baseline, and **stable bone levels on a new baseline radiograph** taken after surgery. Recession and a visibly longer crown are expected, not failures. If bone loss has progressed, the treatment has failed."

Q How likely is that, honestly?

"Less likely than I would want. Surgical therapy for advanced peri-implantitis achieves disease resolution in only a **proportion** of cases, and **recurrence over the following years is common** — so I would present this as a treatment attempt with real uncertainty, not a cure. That is exactly why the endpoint is agreed in writing beforehand."

Q What is his maintenance regime afterwards?

"**Three-monthly supportive care indefinitely.** At each visit: plaque and bleeding scores, six-point probing with a gentle force, check for suppuration, check the abutment screw and the restoration, professional biofilm removal with glycine air-polishing, and reinforcement of interdental cleaning. Radiographs at one year and then as indicated. **Failure to attend maintenance is itself a reason to revisit the retain-or-remove decision.**"

Q What would you record?

"The full six-point chart with bleeding and suppuration, the measured bone loss with the reference points used, both radiographs and their dates, the diagnosis in World Workshop terms, the risk factors, the options discussed including removal, **the agreed endpoint for abandoning treatment**, the words used with the patient, the consent, and the review interval."

Q If it progresses, how do you remove an integrated implant?

"**Counter-torque with a reverse-torque device first**, because it is the most bone-conserving method and often works in a compromised implant. If that fails, **piezo-surgery** around the neck, and only then the **smallest**

trephine that will pass, under copious irrigation because trephining generates a great deal of heat. Then debride, irrigate and assess the residual defect."

Q Would you place another implant in that site?

"Only after **healing and grafting**, with him fully informed that an implant placed in a previously infected site in a former smoker carries a **higher risk of the same outcome**. The honest alternative is a **shortened dental arch to the 25**, which for a 60-year-old with an intact opposing arch and this history may be the wiser long-term answer."

Q How would you present that choice to him?

"Neutrally, with the numbers I have. 'Once the implant is out we can rebuild the bone and try again, which takes about a year, costs this much, and carries a higher-than-normal risk of the same problem. Or we can leave the gap — most people chew perfectly well on the teeth in front of it. There isn't a wrong answer here, and I'd rather you chose than I did.' "

WATCH OUT

Come back to the three questions in order — what is the diagnosis in World Workshop terms, what is the cause, and is it rational to keep this implant. Almost every question in this case is a sub-question of one of those three.

Running this case to time

Time check. The flow above holds 41 questions and about 2,438 words of model answer. At 140 words a minute plus roughly five seconds a question, that is close to **twenty-one minutes** — an overrun of about **six**. What goes first is the **surgical technique** chain in Domain 3: the resective sequence, the implantoplasty risks, the containment criteria for reconstruction and the post-operative regime compress into one question — for the good reason that **you have already said you would refer that phase**, so the examiner would rather hear the referral than the flap design. The marks sit in the **diagnosis against a baseline**, the **cement**, and the **retain-or-remove decision with an endpoint agreed in advance**. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the implant comes out.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the baseline and current periapicals side by side	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 60-year-old **ex-smoker of three years with 25 pack-years**, a five-year-old **cement-retained** crown at the 26, **bleeding and suppuration, 8 mm mesially, and 5 mm of bone loss — about half the implant length — against a documented baseline**. The implant is **not mobile**. Diagnosis: **advanced peri-implantitis**, and **with a baseline available you can say it confidently rather than by inference**. Likely drivers named in order: **residual submucosal cement, the smoking history, and the absence of supportive care**. Then the framing that is the whole opening: the implant is still integrated, so the central question is **whether it is rational to keep it** — and that decision must be **made explicitly, with an endpoint agreed in advance, rather than drifting into open-ended treatment**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the peri-implant tissues, then measuring this defect	Undergraduate biology first. **Junctional epithelium of about 2 mm on hemidesmosomes, connective tissue of 1 to 1.5 mm, fibres parallel or circular and not inserting.** Against a tooth: **Sharpey's fibres into cementum, a periodontal ligament with a second vascular plexus, proprioception and progenitor cells** — the implant has none of it, and its supply is **supraperiosteal only**. The consequences volunteered: lesions **extend further apically and closer to bone**, progress in a **non-linear, accelerating** pattern, and **there is no proprioception, so he felt nothing until there was pus**. Probing at **0.2 to 0.25 N**, six sites, with the discipline that **3 to 4 mm can be entirely healthy and change from baseline is what matters**. The **2017 definitions** given cleanly, **including the no-baseline thresholds** — then applied: **advanced peri-implantitis**, and **the absence of mobility does not soften it, because mobility is a late, terminal sign, not a screening test**. Bone loss quantified properly: **from the implant shoulder to first bone-to-implant contact, using the thread pitch as an internal scale**, on a **long-cone paralleling film reproducing the original geometry**. CBCT only if defect morphology would change the surgical plan, **justified under IR(ME)R, with scatter acknowledged**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: risk factors, the cement, and retain or remove	The risk factors listed with **previous periodontitis first**, and then the honest reading of this man: **three years of cessation is a genuinely favourable prognostic factor and you would say so to him, because it makes him more likely to engage** — while his tissues still carry 25 pack-years. Staging and grading given, with the modifiers — **ten a day or HbA1c of 7% and above push you to Grade C**. Then the cement answered mechanistically: **a rough, retentive foreign body below the mucosa where neither he nor the hygienist can reach it**, and **the deeper the margin, the more cement is left and the less likely it is to be found** — which is why the damage is extensive before anything is noticed. How it should have been avoided: **margin at or no more than 1 mm below the mucosa, minimal cement, a replica abutment or cementation jig, a post-cementation radiograph — or a screw-retained restoration, which is what you would do at a posterior site**. Then the decision made as a decision: criteria **for and against**, and the position — **attempt treatment with a clearly defined endpoint agreed in advance and put in writing**, because **setting the threshold now prevents it being renegotiated in eighteen months when you are both invested in having tried**. The colleague who would remove it today **granted the point** — at 50% loss the outcomes are unreliable, recurrence is common, and removing now gives the cleanest site. **SAC: non-surgical and prosthodontic management Straightforward to Advanced and yours; surgical management of a 50% defect is Complex and referred, having completed and documented the cause-related therapy first.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: cause-related and non-surgical therapy — the phase that is actually yours	**Remove the crown, find and remove all residual cement, and judge whether the restoration and the implant position are cleansable.** If it will not come off: **section it and destroy it — the crown is not the asset here, the implant is**; and if you genuinely cannot retrieve it, **treat around it and factor that uncleanable margin into the prognosis you give him**. The replacement described by its function: **screw-retained so there is no cement at all, a flat or slightly concave emergence with no ridge-lap, the margin at mucosal level, a narrow occlusal table, and embrasure space for an interdental brush — in a compromised site you accept a less pretty contour in exchange for access**. Debridement with **titanium or carbon-fibre curettes or a PEEK-tipped ultrasonic, never steel, plus glycine or erythritol air-polishing submucosally**. Antibiotics refused with the reason: **systemic antibiotics have no reliable role in non-surgical peri-implantitis therapy, and you would not prescribe them simply because there is pus** — stewardship applies, and they are reserved for spreading infection or as a peri-operative adjunct. Reassessment at **six to eight weeks**, with the honesty that scores: **you expect non-surgical therapy alone will not resolve an 8 mm pocket with suppuration — its realistic function is to reduce inflammation and prove his engagement before you commit him to surgery**. Then the surgical options named and the choice justified by **defect morphology and aesthetics — posterior, non-aesthetic, so resection with implantoplasty is the default; contained circumferential craters may suit reconstruction, but you would not promise regeneration in advance, and even then it is defect fill, not re-osseointegration**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: outcome, failure and what happens next	Success at one year defined precisely: **no suppuration, no bleeding on gentle probing, depths no greater than the post-treatment baseline, and stable bone on a new baseline film taken after surgery** — with the line that must be volunteered, **recession and a visibly longer crown are expected consequences of successful treatment, not failures**. Then the honesty: **resolution in only a proportion of cases, recurrence over the following years common — a treatment attempt with real uncertainty, not a cure**, which is exactly why the endpoint was written down. Maintenance at **three months indefinitely**, with **failure to attend maintenance itself a reason to revisit the retain-or-remove decision**. Removal described in the bone-conserving order: **reverse-torque device first, then piezo-surgery around the neck, and only then the smallest trephine that will pass, under copious irrigation because trephining generates a great deal of heat**. And replacement answered without a sales pitch: only after **healing and grafting**, with him told that an implant in a previously infected site in a former smoker carries **a higher risk of the same outcome** — and **the honest alternative of a shortened dental arch to the 25**, which for a 60-year-old with an intact opposing arch may be the wiser long-term answer.
14:30–15:00	Closes on the patient and the record	In his words: *"The gum and bone around this implant are infected. You've lost about half the bone that was supporting it, and there's pus coming from the gum. The most likely cause is a small amount of cement left under the gum when the crown was fitted, and the smoking history hasn't helped. We can try to stop it getting worse, but we can't grow the bone back, and the gum will shrink so more of the implant shows. If it doesn't settle, the implant has to come out."* And on replacement, **neutrally**: *"There isn't a wrong answer here, and I'd rather you chose than I did."* Recorded: the six-point chart with bleeding and suppuration, **the measured bone loss with the reference points used**, both radiographs and their dates, the diagnosis **in World Workshop terms**, the risk factors, the options **including removal**, **the agreed endpoint for abandoning treatment**, the words used with him, the consent and the review interval.

The spine — the twelve answers to have ready if nothing else

1 The peri-implant attachment, and the three things a tooth has that it does not · 2 Why peri-implant lesions progress faster and are felt later · 3 Probing force, six sites, and why depth alone means nothing · 4 The 2017 definitions, with and without a baseline · 5 Why the absence of mobility does not soften the diagnosis · 6 How you measure 5 mm of loss — shoulder to first contact, thread pitch as scale · 7 The risk factors, and how you score three years of cessation · 8 Why submucosal cement is so damaging, and how it should have been avoided · 9 Retain or remove — the criteria, and the endpoint agreed in writing · 10 Non-surgical therapy, and why you do not prescribe antibiotics for pus · 11 What success at one year looks like, recession included · 12 How you remove an integrated implant, in the bone-conserving order.

Where the marks are lost in this case

Starting treatment before making the decision. The examiner has given you **50% bone loss and an implant that is not mobile** precisely to see whether you will drift. The answer that scores is to say, in the opening, that **the central question is whether it is rational to keep this implant**, and that you would **agree an endpoint in writing before you begin** — if suppuration and bleeding persist after surgical therapy, or bone loss progresses at review, **it comes out**. Candidates who describe debridement without ever making that decision have answered a different case.

Promising bone. Two things must be volunteered and are routinely not. First, **treatment aims to arrest the disease, not regenerate bone** — even a reconstructive result is **defect fill, not re-osseointegration**. Second, **recession and a longer-looking crown are expected consequences of successful treatment, not complications** — and that must be in the consent, with the cost of remaking the crown, before anything starts. A patient who is surprised by the recession will experience a successful outcome as a failure.

Never naming the split. The competence line here divides the case in half, and the split is the answer. **Yours**: the diagnosis against baseline, crown removal, **finding and removing the cement**, debridement, hygiene, smoking-cessation reinforcement, and conversion to a **cleansable screw-retained restoration**. **Not yours**: **surgical management of a 50% defect — resective or regenerative — implantoplasty, and the reconstruction of the site afterwards**, all referred, having completed and documented the cause-related therapy first. Say it before Domain 3 asks for a flap design.

SOE Case 27 — Sinus membrane perforation during a transcrestal elevation

THE CASE

You are performing a **transcrestal sinus floor elevation at the UL6** with **7 mm of residual bone height**. After the third osteotome, the **Valsalva manoeuvre produces bubbling in the osteotomy**. The patient is comfortable and unaware.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"This is an intra-operative complication. I am performing a transcrestal sinus floor elevation at the UL6 with 7 mm of residual height, and bubbling in the osteotomy on Valsalva tells me I have a **Schneiderian membrane perforation** — the commonest complication of sinus elevation. The patient is comfortable and unaware, so I have time to think rather than react. What I need to establish is the size and containment of the tear, whether any graft has entered the sinus, and whether I can still achieve primary stability in the residual bone. My working position is that with 7 mm of native bone I do not need this elevation at all — a **short implant** converts the complication into an acceptable and well-evidenced treatment. The governing principle is that if I am uncertain, I abandon the elevation rather than push graft against a torn membrane."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical groundwork

Q Describe the maxillary sinus.

"A **pyramidal air-filled cavity** in the body of the maxilla, the largest paranasal sinus, with its base at the lateral nasal wall and its apex towards the zygoma. The floor is the alveolar process, often dipping between the roots of the premolars and molars; the roof is the orbital floor carrying the infra-orbital nerve; the anterior wall carries the anterior superior alveolar nerve. It drains through the **ostium** into the **middle meatus**, high on the medial wall — which is why drainage is against gravity and why it obstructs so readily."

Q What lines it?

"The **Schneiderian membrane** — **pseudostratified ciliated columnar epithelium** with goblet cells on a thin lamina propria and periosteum. It is typically **0.8 to 1.5 mm** thick. The cilia beat towards the ostium, which is the sinus's self-clearance mechanism, and that is precisely what is impaired when you fill it with graft or blood."

Q What is the blood supply, and why does it matter surgically?

"Chiefly from the **posterior superior alveolar** and **infra-orbital arteries**, which anastomose in the **lateral wall** as an intraosseous vessel — the alveolar antral artery. It matters most for a lateral window, where cutting it causes brisk bleeding that obscures the field; on a CBCT I look for it and site my window below it. In a transcrestal approach it is less relevant, but the anastomosis explains why sinus membrane bleeding can be surprisingly vigorous."

Q What anatomy would you have assessed on the CBCT before starting?

"**Residual bone height and width** at the site, the **contour of the sinus floor** — flat, oblique or with a concavity — the presence of **septa** and their orientation, membrane thickness and any existing pathology, patency of the ostiomeatal complex, and the position of the lateral wall artery. Septa and an oblique floor are the two findings that most raise perforation risk with osteotomes."

Q What membrane thickness concerns you?

"A very **thin membrane** — under about half a millimetre — tears easily. A **thickened membrane** over about 2 mm suggests chronic inflammation or an obstructed ostium and, although it is more robust, it signals that the sinus may not tolerate grafting; that is a case for ENT assessment before surgery rather than a surgical challenge on the day."

Chain: reading this complication

Q What exactly does bubbling on Valsalva tell you?

"That there is a **communication between the sinus and the osteotomy** — the membrane is breached. It is a highly specific sign. Its limitation is that it tells me nothing about the **size** of the tear, and a negative Valsalva does not exclude a perforation, because a small tear can be sealed by blood or by the graft already placed."

Q How do you assess it further?

"**Direct vision** with good illumination down the osteotomy; **gentle probing with a blunt instrument** to feel for the defect rather than to enlarge it; repeat Valsalva watching for **membrane movement** — an intact membrane billows, a torn one does not; and an assessment of **how much graft I have already placed and whether any has entered the sinus**."

Q How would you classify the perforation?

"Pragmatically by size and containment: **small**, under about 5 mm and contained within the osteotomy, where the membrane still tents and the tear can be covered; **large or uncontained**, where I cannot see the margins, the membrane has collapsed, or graft has passed into the antrum. A transcrestal approach gives poor visibility, so my threshold for calling it 'uncertain' is deliberately low — and uncertain is managed as large."

Q Why is a perforation a problem at all — what actually goes wrong?

"Three things. Graft particles enter the sinus and act as a **foreign body**, risking chronic sinusitis or a fungal ball. The graft **loses containment**, so it disperses and the elevation fails. And **ciliary clearance and drainage are impaired**, so a low-grade sinusitis can establish. The perforation itself, if left clean and small, usually heals — the membrane has good regenerative capacity."

Q What is the reported incidence?

"It is the **commonest complication of sinus floor elevation**, and it is more frequent with the lateral window than with the transcrestal approach — although with the transcrestal technique it is more often missed rather than genuinely absent, because you cannot see it. Risk rises with a thin membrane, septa, an oblique or narrow sinus floor, and previous sinus surgery."

Domain 2 — Clinical decision making and treatment planning

Chain: the decision on the day

Q What are your options right now?

"**Seal and proceed** — if the perforation is small and contained, fold a **resorbable collagen membrane** into the osteotomy to cover the tear, then elevate cautiously with reduced graft volume, or place the implant with no further elevation.
Abandon — if it is large, if I cannot see it, or if graft has entered the sinus: remove loose graft, close, and re-plan at three to six months.
Place a short implant into the residual 7 mm with no elevation at all — which turns a complication into a perfectly acceptable definitive treatment."

Q Which do you choose, and why?

"A **6 to 7 mm short implant** in the residual bone with no further elevation. Seven millimetres of native bone is enough for primary stability; the evidence for short implants in the posterior maxilla is good; it stops me

compounding an error; and it removes the graft-in-sinus risk entirely. **I would not push more graft against a torn membrane to rescue a plan.**"

Q You said short implants. Argue against yourself.

"The counter-argument is **crown-to-implant ratio** and the poor bone quality of the posterior maxilla — a short implant in **D3 or D4** bone carrying a molar crown is loaded unfavourably. The mitigation is a wider implant if the ridge allows, a **narrow occlusal table with no cusp interferences in excursion**, a cemented or screw-retained single crown rather than a distal cantilever, and unsplinted only if stability is good. The systematic reviews nonetheless show survival comparable to longer implants in grafted sinuses, with fewer complications."

Q Your colleague would seal the tear and carry on with the graft. Why might they be right?

"Because a small contained perforation covered with collagen genuinely does heal, and completing the elevation gives a **longer implant in more bone**, which is a better long-term biomechanical position — and it avoids a second procedure if the short implant later fails. If they had **clear direct vision** of a small tear, minimal graft placed and good experience of the technique, that is a defensible decision. My reservation is that transcrestally I usually do not have that vision."

Q What is the SAC classification here?

"The original plan — transcrestal elevation with simultaneous implant placement in the posterior maxilla — is **Advanced** surgically. A **short implant in 7 mm of residual bone** is **Advanced** too, but the moment there is a perforation with graft displacement or a need for lateral window repair, it becomes **Complex** and I would refer."

Q So what is within your competence and what is beyond it?

"Within: transcrestal elevation of a few millimetres with a flat floor and adequate residual height, recognising and managing a small perforation, and placing a short implant. **Beyond: lateral window elevation, repair of a large perforation, and retrieval of graft or an implant from the antrum** — those go to an oral surgeon or a colleague with a specialist interest, and displaced material with sinus symptoms goes to ENT."

Q What do you tell the patient?

"Immediately and honestly under **duty of candour**, but calmly and after the procedure rather than mid-operation while she is comfortable. 'The lining of your sinus is a very thin membrane, and during the procedure a small tear occurred. That's a recognised complication and it usually heals without trouble. Because of it I've changed the plan slightly and used a shorter implant instead of lifting the sinus floor, which is a well-proven approach. I'll give you some antibiotics and some specific instructions about not blowing your nose, and I'll see you in a week.' "

Q Was this covered in your original consent?

"It should have been — **sinus membrane perforation, sinusitis, the possibility of abandoning the elevation and the possibility of a different implant or none at all** are all foreseeable and material risks that a reasonable patient would want to know. Under **Montgomery** the test is materiality to this patient, not what I consider significant, so 'the plan may change on the day' has to be said in advance, not explained afterwards."

Domain 3 — Decision making in relation to treatment procedures

Chain: technique, repair and aftercare

Q Take me through the transcrestal technique you were using.

"Crestal incision and flap, a pilot osteotomy stopping **1 to 2 mm short of the sinus floor**, then progressive **osteotomes** used with light malleting to greenstick-fracture the floor and elevate the membrane hydraulically or with graft acting as a cushion. Small increments of graft, each one lifting the membrane a little further, with Valsalva and radiographic checks. It is a **blind technique** — you are working by feel, which is exactly why the perforation happened and why my elevation ambition should be modest."

Q How much elevation is realistic transcrestally?

"Around **3 to 4 mm** reliably, and I would want at least **6 mm of residual height with a flat floor** before attempting it, which is why the case selection here was borderline-acceptable but the ambition must stay small. If I need more than that, the answer is a lateral window — and that is a referral."

Q If you had decided to repair it, how?

"A **resorbable collagen membrane**, trimmed and folded, introduced through the osteotomy and unfolded to lie against the sinus floor covering the tear, with the graft then contained beneath it. Collagen is the material of choice because it is haemostatic, adheres to the wet membrane and needs no fixation. Fibrin sealant or a platelet-rich fibrin membrane are alternatives."

Q What graft material would you have used, and why does the choice matter here?

"**Deproteinised bovine bone mineral** is the best-evidenced sinus graft — slow-resorbing, good space maintenance, excellent long-term data. It matters here because a **particulate xenograft that escapes into the sinus does not resorb**; it sits there as a foreign body. Autogenous bone or an allograft would at least remodel. That asymmetry is another reason to abandon rather than persist."

Q What post-operative management?

"**Sinus precautions** — no nose-blowing, sneeze with the mouth open, no straws, no smoking, avoid flying and diving, sleep with the head elevated. **Antibiotics** — amoxicillin, or clarithromycin if penicillin-allergic. A **decongestant** such as xylometazoline nasal drops for a few days plus steam inhalation. Analgesia. Review at one week."

Q Justify the antibiotic.

"The sinus is now in communication with the mouth, cilial clearance is impaired and there is a foreign body — an implant — at the interface, so this is one of the genuine indications for **therapeutic prophylaxis** rather than routine cover. I would prescribe it as a defined short course, document the indication, and be clear that this is a specific-risk situation, not a habit — antimicrobial stewardship still applies."

Q What specifically do you ask at the one-week review?

"**Nasal discharge, facial pain or pressure, a blocked nose, a bad taste or smell, nosebleed, and altered sensation.** Plus the usual wound review: healing, dehiscence, swelling, and whether the healing abutment or cover screw is undisturbed. I would also check she has actually managed the sinus precautions."

Q What would make you refer, and to whom?

"**Acute sinusitis not settling** — purulent discharge, facial pain, fever — goes to **ENT**. **Displacement of the implant or a significant volume of graft into the antrum** is an urgent referral for retrieval, because retained foreign material risks chronic sinusitis or a **fungal ball**. A large perforation needing lateral window repair goes to an oral surgeon. I would make the referral the same day where there are systemic signs."

Q The implant disappears into the sinus during placement. What do you do?

"Stop. Do not fish for it blindly through the osteotomy — that enlarges the defect and pushes it further. Obtain **imaging to localise it**, close the site, cover with antibiotics and sinus precautions, inform the patient under duty of candour, and arrange **urgent referral for retrieval**, usually via a lateral antrostomy or endoscopically. Retrieval is not something I would attempt at this level."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: outcomes, records and learning

Q How would you record this?

"Contemporaneously: **what happened, when and how it was identified**; the size and appearance of the perforation; what I did and why; the graft volume placed and whether any was retrieved; **what I said to the patient and**

when; the implant actually placed with its **lot and batch number**; medication prescribed; instructions given; and the review arranged. This is exactly the kind of entry that protects both the patient and me."

Q Does this need reporting beyond the notes?

"I would treat it as a **significant event**, log it in the practice's significant event system, and discuss it at a team meeting. It is a recognised complication managed appropriately rather than an incident of harm, but the duty of candour conversation and the change of plan should both be visible in the record and in the practice's governance."

Q What is the prognosis for a perforation managed this way?

"Good. The Schneiderian membrane **heals reliably** when the tear is left clean and the sinus is not loaded with displaced graft. The literature does not show a consistent reduction in implant survival where a perforation is recognised and appropriately managed — the poor outcomes cluster around **unrecognised perforations with graft displacement and subsequent sinusitis**. Recognition is the variable that matters."

Q At two weeks she has a blocked nose, a bad taste and unilateral facial pain. Differential?

"**Acute maxillary sinusitis**, most likely — secondary to the perforation, graft material or impaired drainage. Also consider a **wound infection or dehiscence** over the implant, an **oro-antral communication** if the crestal wound has broken down, and less likely a failing implant. I would examine the site, test for air passage, take a periapical or CBCT if I suspect displaced material, and refer to ENT if it is not settling on antibiotics and decongestants."

Q How would you distinguish an oro-antral communication from simple sinusitis?

"Air or fluid passing between mouth and nose — **regurgitation of liquids nasally, a change in voice, inability to inflate the cheek, and a positive Valsalva at the wound**. Sinusitis alone gives pain, congestion and discharge without air passage. A communication over about 5 mm or persisting beyond three weeks needs surgical closure, which is a referral."

Q What would you change in your practice?

"Review my **case selection** — transcrestal elevation is indicated for a flat floor with adequate residual height, so I would recheck how I assess **floor contour and septa** on CBCT. I would consider whether a **short implant should have been my first plan** rather than my rescue plan at 7 mm. And if I found a pattern across my cases I would **audit** my sinus elevations against those criteria."

Q How would you know, a year on, that this had ended well?

"The implant osseointegrated and restored, with stable bone levels against the baseline radiograph taken at fitting; **no sinus symptoms** and no ENT involvement needed; the patient chewing comfortably; and a record that shows the complication was recognised, disclosed and managed. Success here is measured by the absence of the sinus becoming a problem, not by whether I completed the operation I planned."

WATCH OUT

Come back to the three questions this complication always reduces to: how big is the tear, has anything gone into the sinus, and do I still need the elevation at all? Saying them out loud buys thinking time and is almost always the right frame.

Running this case to time

Time check. The flow above holds 35 questions and about 2,399 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **twenty minutes** — an overrun of about **five**. What goes first is the **sinus blood supply** material in Domain 1: the posterior superior alveolar and infra-orbital anastomosis matters chiefly to a **lateral window, which you have already said you would refer**. What must not be compressed is the **sequence on the day** — this is an intra-operative complication, and the examiner is marking whether you act in a safe order, not whether you can recite the sinus. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the sinusitis and the oro-antral communication are.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, tells you you are mid-procedure	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Name it as what it is: **an intra-operative complication**. A transcrestal elevation at the UL6 with **7 mm of residual height**, and **bubbling on Valsalva after the third osteotome — a Schneiderian membrane perforation, the commonest complication of sinus elevation**. Then the sentence that buys you the whole case: **the patient is comfortable and unaware, so you have time to think rather than react**. What must be established: **the size and containment of the tear, whether any graft has entered the sinus, and whether primary stability is still achievable in the residual bone**. The working position: **with 7 mm of native bone you do not need this elevation at all — a short implant converts the complication into an acceptable and well-evidenced treatment**. And the governing principle, said out loud: **if you are uncertain, you abandon the elevation rather than push graft against a torn membrane**.
1:20–4:30	Domain 1: the anatomical groundwork, then reading this complication	The one anatomical fact that explains everything downstream: the **ostium sits high on the medial wall and drains into the middle meatus, so drainage works against gravity and depends entirely on ciliary transport**. The lining named — **Schneiderian membrane, pseudostratified ciliated columnar epithelium with goblet cells, 0.8 to 1.5 mm thick, cilia beating towards the ostium** — and **that self-clearance is precisely what is impaired when you fill it with graft or blood**. What the CBCT should have shown: residual height and width, floor contour, **septa**, membrane thickness, ostiomeatal patency — with **septa and an oblique floor as the two findings that most raise perforation risk with osteotomes**. Thickness read both ways: **under half a millimetre tears easily; over 2 mm signals chronic inflammation and is a case for ENT assessment before surgery, not a challenge on the day**. Then the sign with its limits: bubbling is **highly specific for a communication but says nothing about size, and a negative Valsalva does not exclude a perforation**. Assessed by **direct vision, gentle blunt probing to feel the defect rather than enlarge it, and repeat Valsalva — an intact membrane billows, a torn one does not**. Classified **small and contained versus large or uncontained**, with the judgement that scores: **transcrestal visibility is poor, so your threshold for calling it uncertain is deliberately low — and uncertain is managed as large**. Why it matters: **graft as a foreign body risking sinusitis or a fungal ball, loss of containment, impaired ciliary clearance** — while a **clean, small tear usually heals**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the decision on the day	Three options — **seal and proceed, abandon, or place a short implant with no elevation at all** — then the choice made with its reasons: **a 6 to 7 mm implant; 7 mm of native bone is enough for primary stability; the evidence for short implants in the posterior maxilla is good; it stops you compounding an error; and it removes the graft-in-sinus risk entirely**. Then the sentence the case exists to elicit: **you would not push more graft against a torn membrane to rescue a plan**. Argued against yourself on **crown-to-implant ratio and D3/D4 bone**, with the mitigations — **wider if the ridge allows, narrow occlusal table, no excursive interferences, no distal cantilever**. The colleague who would seal and carry on **granted their ground** — a small contained tear covered with collagen does heal — **with the reservation that transcrestally you usually do not have the vision to know it is small**. **SAC: the original plan Advanced; a short implant Advanced; but with graft displacement or a need for lateral window repair it is Complex.** Then the competence line in full: **within — a few millimetres of transcrestal elevation with a flat floor, recognising and managing a small perforation, and a short implant. Beyond — lateral window elevation, repair of a large perforation, and retrieval of graft or an implant from the antrum: to an oral surgeon, and displaced material with sinus symptoms to ENT.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: technique, aftercare, and the second complication	The technique described honestly, including its weakness: pilot osteotomy **1 to 2 mm short of the floor**, progressive osteotomes greenstick-fracturing it, **small graft increments each lifting the membrane a little further**, Valsalva and radiographic checks — and then **it is a blind technique, you are working by feel, which is why the perforation happened and why your ambition should be modest**. The limits as numbers: **about 3 to 4 mm reliably, and at least 6 mm of residual height with a flat floor before attempting it — if you need more, the answer is a lateral window, and that is a referral**. The repair, had you chosen it: **a resorbable collagen membrane, trimmed, folded and unfolded against the floor**, chosen because it is **haemostatic, adheres to a wet membrane and needs no fixation**. The graft answer with its sting: **deproteinised bovine bone mineral is the best-evidenced sinus graft — and a particulate xenograft that escapes into the sinus does not resorb; it sits there as a foreign body**, which is **another reason to abandon rather than persist**. **Sinus precautions** as instructions to a patient. The antibiotic **justified rather than assumed**: the sinus is in communication with the mouth, **clearance is impaired and there is a foreign body at the interface — a defined short course with the indication documented, a specific-risk situation and not a habit**. Then the second complication as a sequence: **the implant disappears into the sinus — stop; do not fish for it blindly, which enlarges the defect and pushes it further; image to localise; close; antibiotics and precautions; tell the patient under duty of candour; urgent referral for retrieval, which you would not attempt at this level.**

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: outcomes, records and learning	The one-week review specified by **what you ask**: **nasal discharge, facial pain or pressure, a blocked nose, a bad taste or smell, nosebleed, altered sensation** — plus the wound, and whether she has actually managed the precautions. The prognosis with its variable named: the membrane **heals reliably** when the tear is clean and the sinus is not loaded, and **the poor outcomes cluster around unrecognised perforations with graft displacement — recognition is the variable that matters**. At two weeks: **acute maxillary sinusitis** first, then wound infection or dehiscence, then an **oro-antral communication**, then a failing implant — the last two distinguished by **air passage**: fluid to the nose, a change in voice, inability to inflate the cheek, a positive Valsalva at the wound. **A communication over about 5 mm or persisting beyond three weeks needs surgical closure, which is a referral.** Then governance: recorded **contemporaneously — what happened and how it was identified, the size of the tear, what you did and why, the graft volume placed, what you said to the patient and when, and the implant actually placed with its lot number** — logged as a **significant event** and discussed with the team. The practice change: **recheck how you assess floor contour and septa, and consider whether a short implant should have been the first plan rather than the rescue plan at 7 mm**.
14:30–15:00	Closes on the patient and the record	Under **duty of candour**, calmly, **after the procedure rather than mid-operation**: *"The lining of your sinus is a very thin membrane, and during the procedure a small tear occurred. That's a recognised complication and it usually heals without trouble. Because of it I've changed the plan slightly and used a shorter implant instead of lifting the sinus floor, which is a well-proven approach. I'll give you some antibiotics and some specific instructions about not blowing your nose, and I'll see you in a week."* And the consent point conceded rather than defended: **perforation, sinusitis, abandoning the elevation and the possibility of a different implant or none at all are foreseeable and material risks** — under **Montgomery**, so **"the plan may change on the day" has to be said in advance, not explained afterwards**. Success a year on is **the absence of the sinus becoming a problem, not whether you completed the operation you planned**.

The spine — the twelve answers to have ready if nothing else

1 Name it: a Schneiderian perforation, and the patient is comfortable so you have time · 2 The ostium in the middle meatus, and why drainage is against gravity · 3 The membrane — epithelium, thickness, and the cilia you are about to disable · 4 Septa and an oblique floor, the two CBCT findings that raise the risk · 5 What bubbling tells you, and the two things it does not · 6 How you assess the tear without enlarging it · 7 Small and contained versus large — and uncertain is managed as large · 8 The three options, and why the short implant is the answer · 9 The limits of transcrestal elevation: 3 to 4 mm, and 6 mm of floor · 10 Why the antibiotic is justified here specifically · 11 The implant in the antrum — stop, image, close, refer, do not fish · 12 Sinusitis versus oro-antral communication, distinguished by air passage.

Where the marks are lost in this case

Trying to finish the operation you planned. The examiner has engineered a moment where persisting is tempting and wrong. The safe sequence beats the complete one: **stop, assess without enlarging the tear, establish whether anything has entered the sinus, then abandon the elevation and place a short implant in the 7 mm you already have**. Say the governing principle in the opening, and say that **uncertain is managed as large**, because transcrestally you cannot see. Candidates who fold in a collagen membrane and carry on have chosen completeness over safety in front of an examiner watching for exactly that.

Failing to tell her, and failing to have told her. Two candour points, both routinely missed. **After** the procedure she is told plainly and calmly that a recognised complication occurred and the plan changed — not mid-operation, and not never. **Before** it, the consent should already have named **perforation, sinusitis, abandonment of the lift, and the possibility of a different implant or none at all**, because under **Montgomery** those are material to her. Conceding that scores far better than defending the record.

Never naming the line the case is built across. It is crossed twice. **Yours**: a modest transcrestal lift with a flat floor and adequate height, recognising a perforation, and a short implant. **Not yours**: **lateral window elevation, repair of a large perforation, and retrieval of graft or an implant from the antrum** — to an oral surgeon; and **sinusitis that is not settling, or displaced material with sinus symptoms — to ENT, the same day where there are systemic signs**. **An implant in the antrum is an ENT and oral surgery problem, not yours.** Say it before you are asked, and say that you would not fish for it.

SOE Case 28 — An implant displaced into the maxillary sinus

THE CASE

A colleague refers a patient urgently. During placement at the **16**, the implant was **lost into the maxillary sinus**. The patient is well, with **mild epistaxis** and no other symptoms. A **CBCT** shows the implant lying free on the sinus floor.

Pace: one structured opening of about 60 seconds, then roughly 34 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"This is an urgent referral following an implant **displaced into the maxillary sinus** during placement at the 16 — an uncommon but recognised complication, usually the result of inadequate residual bone height, overzealous elevation, or attempting placement into an elevated site without primary stability. The patient is systemically well with only mild epistaxis, so this is **urgent but not an emergency**. A free foreign body in the sinus cannot be left: it predisposes to chronic sinusitis, oro-antral communication and, in some cases, a **fungal ball**, and it may migrate. My role is to establish the implant's position and mobility, look for an oro-antral communication, start interim medical management and sinus precautions, and make a **prompt, complete referral to oral and maxillofacial surgery** — not to attempt retrieval myself. Alongside that sit the professional issues of candour and record-keeping."

Domain 1 — Clinical interpretation and reasoning

Chain: the anatomical and physiological groundwork

Q Describe the maxillary sinus and its drainage.

"A **pyramidal air-filled cavity** in the body of the maxilla, the largest paranasal sinus. Its floor is the alveolar process, its roof the orbital floor carrying the infra-orbital nerve, its medial wall the lateral wall of the nose. It drains through the **ostium** into the **middle meatus** via the **ostiomeatal complex** — and critically the ostium sits **high on the medial wall**, so drainage works against gravity and depends entirely on ciliary transport."

Q What lines it, and why does that matter with a foreign body inside?

"**Pseudostratified ciliated columnar epithelium** with goblet cells — the Schneiderian membrane. The **mucociliary escalator** sweeps mucus towards the ostium. A foreign body disrupts that transport locally, mucus stagnates, the ostium can obstruct, and you get the classic sequence of **stasis, obstruction and infection**. That is the mechanism by which a retained implant produces chronic sinusitis."

Q What is a fungal ball, and why does an implant predispose to one?

"An **aspergilloma** — a dense non-invasive tangle of fungal hyphae, usually *Aspergillus*, colonising a chronically obstructed sinus. Any retained foreign material provides a nidus and a stagnant environment; dense radiopaque foreign bodies in particular are classically associated. It is non-invasive in an immunocompetent patient but needs surgical clearance, which is another reason retrieval is the standard of care."

Q Why is she bleeding from the nose?

"Because the **Schneiderian membrane and the sinus floor have been breached** and blood has drained through the ostium into the nasal cavity. Mild epistaxis in this context is expected and reassuring. Persistent or heavy bleeding would make me think of the **posterior superior alveolar and infra-orbital anastomosis** in the lateral wall, and that would change this from urgent to emergency."

Q How would you classify the sinus as healthy or not on the CBCT?

"I look at **membrane thickness** — a few millimetres of thickening is common and not necessarily pathological, but generalised thickening over about 2 to 3 mm, an **air-fluid level**, complete opacification, or an **obstructed ostiomeatal complex** all indicate disease. I would also check the contralateral sinus for comparison and look for septa and any retention cyst."

Chain: reading this patient's situation

Q What exactly do you need to establish?

"The **exact position and mobility** of the implant on the CBCT; whether it is free in the antrum or still contained beneath an intact membrane; whether there is an **oro-antral communication** at the crest; whether there is **pre-existing sinus disease or ostium obstruction**; her symptoms and any systemic signs; and what the referring colleague has already done and already told her."

Q Does it matter whether the implant is free in the antrum or sub-antral?

"Yes. If it is **beneath an intact membrane** it is technically extra-sinus, contamination risk is lower, and there is more time. If it is **free in the antral cavity** it will move with head position and mucociliary flow, it can migrate towards the ostium or into other sinuses, and the contamination risk is immediate. Free means urgent retrieval."

Q How would you check for an oro-antral communication clinically?

"Direct inspection of the crest with good light and suction; a **gentle Valsalva** watching for air or bubbling at the socket; asking about **fluid passing into the nose on drinking** and any change in voice; and the nose-blowing test only cautiously, because it can enlarge a small defect. I would **not probe** it, because probing converts a small communication into a larger one."

Q Why can this not simply be left and watched?

"Because a free foreign body in the sinus predisposes to **chronic sinusitis, oro-antral communication and fungal ball formation**, and it may migrate to a site that is harder to retrieve. There is no evidence that leaving it is safe, and the standard of care is **retrieval**. Watchful waiting is only entertained where retrieval carries a greater risk than the foreign body, which is not the situation here."

Domain 2 — Clinical decision making and treatment planning

Chain: management and referral

Q What is your management?

"**Refer urgently to an oral and maxillofacial surgery unit for retrieval.** In the interim: **antibiotics and a decongestant**, sinus precautions, suture or close the crestal site if it is open, analgesia, and clear safety-netting instructions to attend urgently with facial swelling, purulent discharge, visual disturbance or fever. I would speak to the unit by telephone as well as writing, so the urgency is understood."

Q Why not attempt retrieval yourself?

"Because blind retrieval through the osteotomy **risks pushing the implant further**, creating a larger communication, damaging the membrane or the lateral wall vessel, and converting a straightforward endoscopic retrieval into a bigger operation. I have neither the visibility nor the instrumentation. **Recognising the limit of my competence is the correct answer here**, and it is the answer the patient is best served by."

Q What is the SAC classification?

"Placement at the 16 with a deficient sinus floor was **Advanced** at best. Once the implant is displaced into the antrum, management is **Complex** — it sits outside the scope of a general dentist with an interest in implant dentistry and belongs in a specialist unit."

Q What goes in the referral letter?

"**What happened and when**; the implant **system, diameter and length**; the **CBCT** itself, not just a report; the current symptoms and findings including the epistaxis and the state of the crestal wound; her **medical history and medication**, including any anticoagulant or bisphosphonate; what has been **prescribed** and when; **what she has already been told**; and my contact details. An incomplete referral is the commonest reason for delay."

Q How would the retrieval be performed?

"Typically by a **transnasal endoscopic approach** through the middle meatus, or by an **intraoral antrostomy** — a Caldwell-Luc type lateral window — under local or general anaesthesia, with the implant retrieved under direct or endoscopic vision and the sinus irrigated. The endoscopic route is generally preferred where the anatomy allows because it is less morbid."

Q What are the professional issues?

"For the referring colleague: **duty of candour** — the patient must be told promptly and honestly, with an apology and an explanation, and it must be recorded; their indemnity provider should be informed. For me: manage her immediate care and referral, and describe my findings **factually without speculating about the colleague's technique** in front of her. If she asks how to raise concerns, I explain the route neutrally."

Q She asks you directly whether the colleague did something wrong. What do you say?

" 'This is a recognised complication of implant treatment in the upper back jaw — it can happen even when things are done carefully. I wasn't there, so I can't tell you what happened in your particular case, and it wouldn't be fair for me to guess. What I can do is make sure you're looked after now, and your dentist should explain to you exactly what happened. If you want to take it further afterwards, I can tell you how.' Honest, neutral and it does not close down her options."

Q What do you tell her about what happens next, in her words?

" 'The implant has moved into the sinus — the air space above your back teeth. It can't stay there, because over time it would cause infection, so it needs to be taken out. That's a routine procedure for a hospital surgical team, usually done through the nose with a camera or through a small window in the bone. In the meantime I'm giving you antibiotics and a nasal spray, and I need you not to blow your nose. You're not in danger, but this needs doing soon rather than later.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: the oro-antral communication and interim care

Q If there is also an oro-antral communication, how do you manage it?

"**Assess the size.** Under about **2 mm** may close spontaneously with a suture and sinus precautions. Larger defects need surgical closure — most commonly a **buccal advancement flap**, sometimes a **buccal fat pad** or a **palatal rotation flap**. Given she also needs retrieval, I would let the receiving unit manage both **at the same operation** rather than close it and have them reopen it."

Q What is the difference between a communication and a fistula?

"A **communication** is the acute, unepithelialised defect. If it persists — conventionally beyond about **two to three weeks** — it epithelialises and becomes an **oro-antral fistula**, which will not close spontaneously and requires excision of the epithelial tract before flap closure. That time window is why prompt referral matters."

Q Describe a buccal advancement flap.

"A **broad-based trapezoidal buccal mucoperiosteal flap** raised over the defect, with the **periosteum scored horizontally at its base** to release it so the flap advances passively over the defect and is sutured tension-free to the palatal margin, on sound bone. Its drawback is loss of buccal sulcus depth, which matters if implants or a denture are planned later — a relevant consideration for this patient."

Q What antibiotic and decongestant would you prescribe, and why?

"**Amoxicillin**, or **clarithromycin or doxycycline** if penicillin-allergic, as a defined short course — the indication is a contaminated communication with a foreign body in a sinus with impaired clearance, which is a genuine therapeutic indication rather than routine cover. A **decongestant** such as xylometazoline nasal spray for no more than five to seven days, to reduce mucosal oedema and keep the ostium patent; longer use causes rebound congestion. Steam inhalation as an adjunct."

Q What sinus precautions do you give?

"**No nose-blowing; sneeze with the mouth open; no straws, no smoking, no playing wind instruments; avoid flying and diving; sleep with the head elevated; soft diet and avoid vigorous rinsing for the first day.** I would give these in writing, because verbal instructions in a distressed patient are not retained."

Q What are the red flags you would safety-net?

"**Facial swelling, purulent nasal discharge, fever, severe or increasing pain, visual disturbance or diplopia, and heavy bleeding.** Periorbital swelling with any eye sign is an emergency — orbital involvement — and goes straight to hospital rather than back to me."

Q How could this have been prevented?

"Adequate **CBCT assessment** of residual height, floor contour and septa; respecting the thresholds for transcrestal elevation — a **flat floor with adequate residual height**, otherwise a lateral window and a referral; **never seating an implant that lacks primary stability** into an elevated site; and abandoning rather than persisting when the anatomy does not cooperate. A **short implant** is usually the honest alternative in a deficient posterior maxilla."

Q What if the patient refuses referral?

"I would establish why — usually fear, cost or loss of trust in dentistry after this experience. I would explain the specific consequences of leaving it in plain terms: chronic sinusitis, possible fungal infection, and a bigger operation later. I would document the discussion and the refusal, give written safety-netting, offer to arrange the appointment myself, and **keep the door open** with a defined review rather than discharging her."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: aftermath, records and the future

Q What would you record?

"Contemporaneously and factually: the **history as given**, my clinical findings including the state of the crest and the presence or absence of a communication, the **CBCT findings**, the medication prescribed with doses, the instructions and written advice given, the referral made and the time it was made, **what I told the patient and in what words**, and her questions. No opinion about the colleague's competence in the clinical record."

Q Does this need reporting beyond the notes?

"It should be logged as a **significant event** by the practice where it occurred, with a team discussion and a review of case selection and scope of practice. The referring colleague should notify their **indemnity provider**. If there were a pattern of similar events, that becomes a concern about a colleague's practice and there is a professional duty to act on it — but on a single recognised complication, appropriately disclosed and referred, it is significant event learning."

Q The unit retrieves it. What is the expected outcome?

"Generally very good — a single implant retrieved promptly from an otherwise healthy sinus, with the communication closed, usually resolves completely. Persistent problems cluster around **delayed retrieval, pre-existing sinus disease, or an unclosed fistula**. She should expect the sinus to settle over a few weeks and to be discharged from ENT or maxillofacial follow-up."

Q Six weeks later she has facial pain and a foul-tasting discharge. What has happened?

"Most likely **chronic maxillary sinusitis**, from residual contamination, an unclosed **oro-antral fistula**, retained graft material, or obstructed drainage. I would examine the site, test for air passage, take imaging, and refer back to the treating unit rather than manage repeated antibiotic courses myself. A dense opacity with flecks of calcification would raise the possibility of a **fungal ball**."

Q What will you tell her about the future of that site?

"That the priority is **retrieving the implant and settling the sinus**, and that replacement should not be considered until the sinus is healthy — typically **several months**. Then the options are a **lateral window elevation by a specialist**, a **short implant** if enough bone remains, or a **non-implant solution** such as a bridge or a partial denture. **I would not promise her an implant in that site.**"

Q What is within your competence in her ongoing care?

"Interim medical management, wound care, oral hygiene support, and the eventual **prosthodontic conversation** — including a conventional solution if she chooses one. **Beyond me: retrieval, sinus surgery, flap closure of a significant communication, and any subsequent lateral window elevation.** I would also be honest that if she wants a second implant attempt there, it should be done by someone doing sinus surgery routinely."

Q How would you judge that you handled this well?

"That she was **retrieved promptly and without complication**, that she understood at every stage what was happening and why, that the referral was complete enough that nothing had to be repeated, that my record would stand up to scrutiny — and that she still has confidence in dental care, which after an event like this is not a trivial outcome."

WATCH OUT

Come back to the frame: is she safe now, is the foreign body coming out and who by, and is the mouth-to-sinus pathway closed? Saying those three aloud buys thinking time and is almost always the right structure here.

Running this case to time

Time check. The flow above holds 33 questions and about 2,158 words of model answer. At 140 words a minute plus roughly five seconds a question, that is a little over **eighteen minutes** — an overrun of about **three**, the shortest in this section, because the case is built on judgement rather than technique. What goes first is the **flap closure** material in Domain 3: the buccal advancement flap and the periosteal scoring compress into one question, since you have already said you would let **the receiving unit close the defect at the same operation as the retrieval**. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the candour, the records and the future of the site are.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT up	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	An urgent referral: an implant **displaced into the maxillary sinus** during placement at the 16 — uncommon but recognised, usually from **inadequate residual bone height, overzealous elevation, or placement into an elevated site without primary stability**. She is systemically well with only mild epistaxis, so — and this is the calibration the opening turns on — **this is urgent but not an emergency**. Then the absolute: **a free foreign body in the sinus cannot be left** — it predisposes to **chronic sinusitis, oro-antral communication and, in some cases, a fungal ball**, and it may migrate. Your role as four tasks: **establish the implant's position and mobility, look for an oro-antral communication, start interim medical management and sinus precautions, and make a prompt, complete referral to oral and maxillofacial surgery — not to attempt retrieval yourself.** And alongside that: **candour and record-keeping**.
1:20–4:30	Domain 1: the anatomical and physiological groundwork, then this situation	The sinus given with the fact that explains everything downstream: the **ostium sits high on the medial wall and drains into the middle meatus through the ostiomeatal complex, so drainage works against gravity and depends entirely on ciliary transport**. The lining — **pseudostratified ciliated columnar epithelium with goblet cells** — then the mechanism spelled out: **a foreign body disrupts the mucociliary escalator, mucus stagnates, the ostium obstructs, and you get stasis, obstruction and infection**. The **fungal ball** defined properly — **an aspergilloma, a dense non-invasive tangle of hyphae colonising a chronically obstructed sinus**, non-invasive in an immunocompetent patient **but needing surgical clearance, which is another reason retrieval is the standard of care**. The epistaxis explained and calibrated: **expected and reassuring**, but **persistent or heavy bleeding points at the posterior superior alveolar and infra-orbital anastomosis and turns this from urgent into an emergency**. The CBCT read for **membrane thickening, an air-fluid level, opacification, an obstructed ostiomeatal complex, and the contralateral sinus for comparison**. And the distinction that changes urgency: **beneath an intact membrane is technically extra-sinus with lower contamination risk and more time; free in the antrum moves with head position and mucociliary flow, can migrate towards the ostium, and means urgent retrieval**. The communication checked by **inspection, gentle Valsalva, and asking about fluid passing into the nose — and not by probing, because probing converts a small communication into a larger one**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: management and referral	The management as a list you could act on tonight: **urgent referral to an oral and maxillofacial unit for retrieval**; in the interim **antibiotics and a decongestant, sinus precautions, closure of the crestal site if open, analgesia, and explicit safety-netting** — with **a telephone call as well as a letter, so the urgency is understood**. Then the sentence the case exists for: **you would not attempt retrieval yourself**, because blind retrieval through the osteotomy **risks pushing the implant further, creating a larger communication, damaging the membrane or the lateral wall vessel, and converting a straightforward endoscopic retrieval into a bigger operation — recognising the limit of your competence is the correct answer here, and it is the answer the patient is best served by**. **SAC: placement was Advanced at best; once displaced, management is Complex and belongs in a specialist unit.** The referral letter itemised — **what happened and when, the system, diameter and length, the CBCT itself and not just a report, current findings including the epistaxis and the crestal wound, medical history and medication, what has been prescribed, what she has already been told, and your contact details** — because **an incomplete referral is the commonest cause of delay**. And how retrieval is done: **transnasal endoscopic through the middle meatus, or an intraoral antrostomy, with the endoscopic route preferred where the anatomy allows**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: interim care, the communication, and prevention	The communication **sized** rather than described: **under about 2 mm may close with a suture and sinus precautions; larger needs surgical closure** — with the judgement that scores, **let the receiving unit manage both at the same operation rather than close it and have them reopen it**. The time window: a communication persisting beyond **two to three weeks** epithelialises into a **fistula** that will not close spontaneously — **which is why prompt referral matters**. The antibiotic justified rather than assumed: **a contaminated communication with a foreign body in a sinus with impaired clearance is a genuine therapeutic indication, not routine cover** — a defined short course, with an alternative if penicillin-allergic. The decongestant limited: **no more than five to seven days, because longer use causes rebound congestion**. Sinus precautions given **in writing, because verbal instructions in a distressed patient are not retained**. The red flags named as red flags — **facial swelling, purulent discharge, fever, severe or increasing pain, visual disturbance or diplopia, heavy bleeding** — with the escalation: **periorbital swelling with any eye sign is an emergency and goes straight to hospital, not back to you**. Then prevention, said as your own practice: **CBCT assessment of residual height, floor contour and septa; respecting the thresholds for transcrestal elevation; never seating an implant that lacks primary stability into an elevated site; abandoning rather than persisting — and a short implant as the honest alternative in a deficient posterior maxilla**. If she refuses referral: **establish why, explain the consequences plainly, document the refusal, offer to arrange the appointment yourself, and keep the door open with a defined review rather than discharging her.**

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: aftermath, records and the future of the site	Records **contemporaneous and factual** — the history as given, the state of the crest, the presence or absence of a communication, the CBCT findings, medication with doses, written advice, **the referral and the time it was made**, and **what you told her and in what words** — with the discipline being marked: **no opinion about the colleague's competence in the clinical record**. Governance: logged as a **significant event by the practice where it occurred**, with the **referring colleague notifying their indemnity provider**; a single recognised complication, disclosed and referred, is significant event learning — **a pattern would be a concern about a colleague's practice, and there is a professional duty to act on that**. Outcome **generally very good** after prompt retrieval from an otherwise healthy sinus, with poor results clustering around **delayed retrieval, pre-existing sinus disease, or an unclosed fistula**. At six weeks with facial pain and foul discharge: **chronic maxillary sinusitis** from residual contamination, an unclosed fistula, retained graft or obstructed drainage — **referred back to the treating unit rather than managed with repeated antibiotic courses**, with **flecks of calcification in a dense opacity raising a fungal ball**. Then the future of the site, answered without a promise: **nothing until the sinus is healthy, typically several months**, then a **lateral window by a specialist, a short implant if enough bone remains, or a non-implant solution — and you would not promise her an implant in that site**.
14:30–15:00	Closes on the patient and the record	To her: *"The implant has moved into the sinus — the air space above your back teeth. It can't stay there, because over time it would cause infection, so it needs to be taken out… In the meantime I'm giving you antibiotics and a nasal spray, and I need you not to blow your nose. You're not in danger, but this needs doing soon rather than later."* And when she asks whether the colleague did something wrong: *"This is a recognised complication… I wasn't there, so I can't tell you what happened in your particular case, and it wouldn't be fair for me to guess. What I can do is make sure you're looked after now, and your dentist should explain to you exactly what happened. If you want to take it further afterwards, I can tell you how."* Honest, neutral, and **it does not close down her options**.

The spine — the twelve answers to have ready if nothing else

1 Urgent but not an emergency — and a free foreign body cannot be left · 2 The ostium high on the medial wall, and drainage against gravity · 3 The mucociliary escalator, and stasis, obstruction, infection · 4 The fungal ball, and why it makes retrieval the standard of care · 5 Why she is bleeding, and what would make it an emergency · 6 Free in the antrum versus beneath an intact membrane · 7 Checking for an oro-antral communication — and not probing it · 8 Why you do not attempt retrieval yourself · 9 What goes in the referral letter, the CBCT included · 10 The antibiotic and decongestant, each justified and time-limited · 11 Communication versus fistula, and the two-to-three-week window · 12 What you say when she asks whether her dentist did something wrong.

Where the marks are lost in this case

Reaching into the sinus. The single thing that fails this case is a candidate who describes retrieving the implant. **Blind retrieval through the osteotomy pushes it further, enlarges the communication, and can turn an endoscopic procedure into an antrostomy.** Say plainly that **an implant in the antrum is an ENT and oral surgery problem** and that **recognising the limit of your competence is the correct answer here** — then show what you *do* do: position and mobility, the communication check, interim medication and precautions, and a **complete, telephoned referral**. Doing the safe things in the right order beats knowing the retrieval technique.

Getting the urgency wrong in either direction. She is well with mild epistaxis: **urgent, not an emergency** — but a **free** implant is more urgent than a sub-antral one, and **periorbital swelling or any eye sign goes straight to hospital**. Volunteer the calibration and the red flags together, and volunteer that **a communication left beyond two to three weeks becomes a fistula that will not close on its own**. Candidates who neither escalate nor de-escalate have shown no judgement at all.

Talking about the colleague. The professional line here is as marked as the surgical one. **Describe findings factually, do not speculate about another clinician's technique in front of the patient or in the record, and do not write an opinion about their competence in the notes.** Say that **duty of candour and the indemnity notification belong to the referring colleague**, that you would log this as a **significant event**, and that if she wishes to raise concerns you explain the route **neutrally** — advocating neither way. The answer that scores is honest, neutral, and leaves her options open.

SOE Case 29 — Fracture of a narrow implant in a molar site

THE CASE

A **58-year-old man** with a **3.3 mm narrow implant at the 46**, restored **seven years ago**. He reports the crown feels loose. On examination the **crown moves together with the abutment**; the radiograph shows a **fracture line across the implant body at the level of the third thread**, with **bone loss to that level**.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 58-year-old man with a **fractured implant body** at the 46, seven years after restoration. The crown and abutment move as one unit and the radiograph shows a fracture across the implant at the third thread with bone loss to that level. Implant fracture is uncommon — of the order of a fraction of a per cent — and it is **always the end of that implant's life**; there is no repair. The important part of this case is understanding **why** it fractured, because the same forces will act on whatever I place next. My working explanation is a **narrow 3.3 mm implant in a molar position**, crestal bone loss lengthening the lever arm, and very probably a history of repeated screw loosening. What I need to establish is his occlusion, any parafunction, and what he actually wants for that site."

Domain 1 — Clinical interpretation and reasoning

Chain: the mechanical groundwork

Q The patient says the crown is loose. What is your differential, before you look?

"**Loosening of the abutment or prosthetic screw** — much the commonest. **Decementation** of a cemented crown. **Fracture of the abutment or the screw**. **Loss of osseointegration**, so the whole implant is mobile. And **fracture of the implant body** itself. They are distinguished by what moves relative to what, and by the radiograph."

Q How do you distinguish them clinically?

"If the **crown alone** moves off a stable abutment, it is decementation. If the **crown and abutment move together but the implant is stable**, it is screw loosening or screw fracture. If the **whole assembly including the implant moves**, it is either loss of integration or a fracture below the crest. Here the crown and abutment move together, which pointed me at the screw — and the radiograph corrected me."

Q How does a screw joint actually work?

"Tightening the screw stretches it elastically and generates **preload** — a clamping force holding the abutment to the implant. The joint is stable while the **functional load stays below the preload**. Repeated micromovement, settling of machining irregularities in the first hours, and any misfit all reduce preload, and once it is exceeded the joint opens, micromoves and loosens."

Q So how do you maintain preload in practice?

"Torque to the **manufacturer's specification with a calibrated torque driver**, then **retorque after about ten minutes** to compensate for the settling effect. Ensure a passive, accurately fitting component, clean threads, and no interposed debris. And design the occlusion so the functional load stays axial and modest — the occlusion is what protects the screw joint."

Q Why is repeated screw loosening a warning rather than an inconvenience?

"Because it means the loading exceeds the joint's capacity, and the **escalation is predictable: loosening, then screw fracture, then abutment or implant fracture**. A loose joint also transfers bending stress from the screw

into the implant body, so retightening without investigating the cause is actively harmful. Every episode should trigger review of the occlusion, the fit and the crown design."

Q What is the implant made of, and why does that matter here?

"**Commercially pure titanium** — usually grade 4 — or a **titanium-zirconium or Ti-6Al-4V alloy**. Grade 4 cp titanium has good biocompatibility but modest strength, so in a **narrow-diameter implant the wall thickness at the internal connection becomes the weak point**. Titanium-zirconium alloys were developed precisely to allow narrow implants with better fatigue strength, and they matter when diameter is constrained."

Q Explain the mechanics of why this one failed.

"Bending stress rises with the **cube of the reduction in radius**, so diameter dominates strength — a 3.3 mm implant is far weaker than a 4.8. Molar occlusal loads are the highest in the arch. **Crestal bone loss to the third thread** removed the coronal support, lengthening the **lever arm** and shifting peak stress to an unsupported section of a thin wall. Add cyclic loading over seven years and you get **fatigue fracture** — which is why it happened at seven years and not at one."

Q What else would you check?

"His history of **previous problems with this crown**, especially loosening; **parafunctional signs** — masseteric hypertrophy, wear facets, linea alba, a scalloped tongue; the **occlusion including excursive contacts** on this crown; the width of the occlusal table and any cantilever; the **opposing dentition** and whether it is natural, restored or implant-borne; and the general periodontal and peri-implant status."

Q How would you classify the bone loss you can see?

"Radiographically I would state it against the **baseline film at fitting** if I can get one, and describe it as bone loss to the level of the third thread. If the implant had been intact, loss of that magnitude with bleeding and probing depth would meet the **2017 World Workshop** definition of **peri-implantitis**. Here it is academic for this implant, but it matters because peri-implant bone loss is what set up the fracture, and the same host and plaque factors will apply to a replacement."

Domain 2 — Clinical decision making and treatment planning

Chain: what to do about it

Q What are the options?

"The implant must come out; **there is no repair**.
Remove and replace with a wider implant after healing and grafting, accepting that there will be a defect at the fracture level.
Remove and leave a sleeper — if the apical fragment cannot be retrieved without excessive bone destruction, it may be left submerged provided it is asymptomatic and fully covered.
Remove and do not replace, accepting a **shortened dental arch** to the 45 if the occlusion permits.
Remove and bridge from the 45 and 47 — destructive if those teeth are sound."

Q What do you recommend?

"Removal — ideally by **counter-torque** if the coronal fragment can be engaged, otherwise piezo-surgery or a small trephine — then a healing period and reassessment. If I replace, it must be a **wide-diameter implant appropriate to a molar**, with the occlusion corrected and a night guard if there is parafunction. I would also put the **shortened dental arch** genuinely on the table, because this site has already failed once."

Q Make the case for the shortened arch.

"**Käyser's** work supports the shortened dental arch as a legitimate endpoint: a functional dentition of the anterior teeth and premolars gives acceptable masticatory function, comfort and stability for most patients, and the second molar contributes little. Here it avoids re-treating a site that has already failed mechanically, avoids grafting, and costs him nothing further. If he chews adequately on the 45 and the contralateral side, it is the lowest-risk answer."

Q You have argued for the shortened arch. Now argue against it.

"He is 58, not 80, and he has already invested in this site. Leaving the space risks **overeruption of the opposing tooth and drifting of the 45 and 47**, and a single wide implant in a healed, grafted site with corrected occlusion has a good prognosis. The mechanical failure was a **design error, not a biological one**, and correcting the design corrects the risk. If his function or his expectations require the molar, replacement is reasonable."

Q What is the SAC classification?

"**Removal of a fractured implant is Complex** — the fragment may be unretrievable without significant bone loss, and the outcome is unpredictable. **Replacement into a healed, grafted molar site with a vertical and horizontal defect is Advanced to Complex.** A straightforward counter-torque removal of an engageable fragment is within my scope; a trephine removal with a large resulting defect and simultaneous grafting is where I would refer."

Q So when would you refer?

"If the fragment cannot be engaged and removal would require **substantial bone sacrifice near the inferior alveolar canal**; if the resulting defect needs **vertical augmentation**, which is beyond me; if there is a large **non-contained defect**; or if I am uncertain about the mandibular canal proximity on the CBCT. Removal near the canal in a resorbed posterior mandible is exactly the situation where a second opinion is cheaper than a nerve injury."

Q What do you tell him about cost and responsibility?

"I would first establish **what he was told at the outset** and whether there was any written arrangement about failure. Then be honest: 'The implant has fractured. That's uncommon, and I think it happened because a narrow implant was used in a position that takes heavy chewing forces and because the bone level dropped. It has to come out. Here's what replacing it involves and what it would cost, and here's the alternative of leaving the space.' I would **not blame the previous clinician**, but I would not pretend the implant choice was ideal either — I describe the mechanics factually."

Q He asks whether he should complain. What do you say?

"That a fractured implant is a recognised complication and I was not there, so I cannot say what was or was not reasonable in his case. I would encourage him to raise it with the treating practice first, since they may well wish to contribute, and explain the formal routes neutrally if he wants them. **Factual, no speculation, no advocacy either way, and all of it recorded.**"

Q What would you record?

"The **history including any previous loosening**, the mobility findings and what moved relative to what, the **radiographic findings with the film dated**, the diagnosis, the options I discussed and their risks and costs, his questions and his decision, **what I said about causation and in what words**, and the plan with review dates. If I retrieve a fragment I record the **implant system, diameter and length**, and whether anything was left in situ."

Domain 3 — Decision making in relation to treatment procedures

Chain: removal and rebuilding

Q Describe the removal.

"**Flap access.** Attempt to engage the coronal fragment with a **counter-torque device**; if the internal connection is damaged, engage the outer surface with a fragment-removal device. If neither works, **piezo-surgery circumferentially**, and only then a **trephine of the smallest diameter that will pass**. **Copious irrigation throughout**, because trephining generates significant heat in dense bone. Then debride, irrigate and assess the residual defect."

Q Why is counter-torque preferred?

"Because it removes the implant **without sacrificing surrounding bone** — the reverse torque breaks the interface and the fragment unscrews, leaving a socket rather than a crater. A trephine removes a cylinder of host bone with

the implant, which enlarges the defect, may approach the **inferior alveolar canal**, and turns a straightforward replacement into a grafting case. Counter-torque first, trephine last, always."

Q What do you do about the defect?

"Assess containment. If the defect is **contained**, I would graft at the same visit with a particulate graft and a resorbable membrane. If it is **non-contained or the soft tissue is compromised**, allow **six to eight weeks of soft tissue healing** and then graft as a staged procedure, which gives a keratinised soft tissue envelope to close over. **Vertical defects are a referral.**"

Q What interim prosthesis?

"For a lower first molar, often **nothing is needed** — he can function on the remaining teeth for the healing period. Otherwise a simple **acrylic partial denture** relieved over the surgical site, or an adhesive bridge if aesthetics demanded it, which here they do not. I would warn about **overeruption of the opposing tooth** if the interval is going to be long."

Q If you replace it, what would you do differently?

"A **wide-diameter implant of 5 mm or more**, correctly positioned in three dimensions; a **screw-retained monolithic zirconia crown** with a **narrow occlusal table**, light centric contact and **no excursive contact**; a **night guard** if there is parafunction; and a maintenance regime that includes **checking screw integrity at every recall**, so that loosening is investigated rather than repeatedly retightened."

Q Why screw-retained rather than cemented here?

"**Retrievability**, chiefly — this is a site with a history of mechanical problems, and I want to be able to remove the crown to inspect and retorque without destroying it. It also eliminates **residual excess cement**, which is a recognised cause of peri-implant disease and would compound the bone loss risk in a patient who has already lost crestal bone."

Q Describe the occlusal scheme you want on the replacement crown.

"**Light contact in maximum intercuspation** — noticeably lighter than the adjacent teeth on shimstock, so that under load it shares rather than leads, because the implant has no periodontal ligament and no proprioceptive protection. **No contact in lateral or protrusive excursion** — guidance on the canines and anterior teeth. **Narrow occlusal table**, cusp inclines reduced, contacts centred over the implant axis, and **no cantilever**."

Q Why does the absence of a periodontal ligament matter?

"A tooth has around **25 to 100 microns** of physiological mobility and rich proprioception; an implant has roughly **3 to 5 microns** of viscoelastic bone deformation and far poorer tactile sensitivity. So an implant does not intrude to share load and the patient does not feel that they are overloading it. That is the whole rationale for **implant-protected occlusion** — deliberately lightening the contact."

Q Does a night guard prevent implant fracture?

"It **reduces the risk rather than eliminating it** — it distributes load, protects the ceramic and gives me a diagnostic record of the wear pattern. The evidence base for splints preventing implant mechanical complications is not strong, so I would present it as a sensible precaution rather than a guarantee, and be clear that **correct diameter and correct occlusion do far more work** than the splint does."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: prognosis, monitoring and learning

Q How common is implant fracture?

> "**Uncommon** — well under one per cent of implants, and typically a late event after five years or more, since it is a fatigue phenomenon. It clusters in **narrow implants, molar sites, bruxists, cantilevered or long-span prostheses, and implants with prior crestal bone loss or repeated screw loosening**. That risk profile is essentially a description of this patient."

Q What is the prognosis for a replacement implant in this site?

> "Reasonable, but **poorer than a virgin site**. Implants placed into previously failed sites have a measurably higher failure rate, and the site will have a healing defect and altered bone quality. That said, the failure here was **mechanical and explicable**, so correcting the diameter and the occlusion addresses the actual cause — which is a better position than a site that failed for unexplained biological reasons."

Q You leave an apical fragment as a sleeper. What do you tell him and what do you monitor?

> "That a small piece of the implant remains in the bone because removing it would have destroyed more bone than leaving it, that it is inert titanium fully covered by bone and gum, and that it is expected to cause no trouble — but that I will **review it and radiograph it periodically**, and that it must be flagged to any future clinician. I record its position clearly and monitor for **symptoms, exposure through the mucosa, or radiolucency** around it."

Q If a sleeper becomes exposed or symptomatic, what then?

> "Then it is no longer a sleeper — an exposed fragment is a **plaque-retentive foreign body** that cannot be maintained. It needs removal, and if that is difficult it goes to an oral surgeon. That possibility is part of the consent for leaving it in the first place."

Q What would you monitor on the replacement, and how often?

> "Six-monthly initially: **peri-implant probing at six sites, bleeding and suppuration**, plaque control, mucosal health, **radiographs at one year against a baseline taken at fitting** and thereafter as indicated, and — specifically for this patient — **occlusal check on shimstock, inspection for wear facets, and confirmation that the screw is tight** at each recall."

Q What is the wider lesson from this case?

> "Two. First, **implant diameter must match the loading demands of the site** — narrow implants belong in lateral incisor and lower incisor positions, not molar sites. Second, **repeated screw loosening is a warning, not an inconvenience**: each episode should prompt investigation of the occlusion and the fit, because the escalation runs loosening, screw fracture, implant fracture."

Q How would you know you had handled this well?

> "That he understood what had happened and why, without being handed either a blame narrative or a false reassurance; that removal was achieved with minimal bone loss; that whichever endpoint he chose — replacement or a shortened arch — he is chewing comfortably and can maintain it; and that the mechanical cause was **corrected rather than repeated**."

WATCH OUT

Come back to the mechanics: what force, on what lever arm, through what diameter? Naming those three buys you a sentence of thinking time and is almost always the right frame for a mechanical implant failure.

Running this case to time

Time check. The flow above holds 35 questions and about 2,391 words of model answer. At 140 words a minute plus roughly five seconds a question, that is **twenty minutes** — an overrun of about **five**. What goes first is the **interim prosthesis and night guard** pair in Domain 3: for a lower first molar the honest answer is that **often nothing is needed**, and the splint **reduces rather than eliminates** the risk, so the two compress into one question. The marks sit in the **mechanics** — diameter, lever arm, fatigue — in the **escalation from loosening to fracture**, and in the **shortened arch offered against a site that has already**

failed. If your answers run past **thirty seconds**, you will never reach Domain 4 — and Domain 4 is where the sleeper is.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapical up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	A 58-year-old with a **fractured implant body at the 46**, seven years after restoration: crown and abutment move as one, and the film shows a **fracture across the implant at the third thread with bone loss to that level**. Then the two facts that frame everything: implant fracture is **uncommon — a fraction of a per cent** — and it is **always the end of that implant's life; there is no repair**. So the important part of the case is **why it fractured, because the same forces will act on whatever you place next**. The working explanation, volunteered: a **narrow 3.3 mm implant in a molar position**, **crestal bone loss lengthening the lever arm**, and **very probably a history of repeated screw loosening**. Still to establish: his occlusion, any parafunction, and what he actually wants for that site.
1:20–4:30	Domain 1: the mechanical groundwork, then this implant	The differential given **before** looking and then resolved by **what moves relative to what**: crown alone on a stable abutment is **decementation**; crown and abutment together with a stable implant is **screw loosening or fracture**; the whole assembly moving is **loss of integration or a fracture below the crest — here the clinical picture pointed at the screw and the radiograph corrected you**, which is worth saying out loud. Then the screw joint from first principles: **tightening stretches the screw and generates preload**, the joint is stable **while functional load stays below preload**, and preload is lost to **micromovement, settling in the first hours, and misfit**. Maintained by **a calibrated torque driver to the manufacturer's value, a retorque at about ten minutes, passive components, clean threads — and an occlusion designed to keep load axial and modest, because the occlusion is what protects the screw joint**. Then the warning: **repeated loosening means loading exceeds the joint's capacity, and the escalation is predictable — loosening, screw fracture, then abutment or implant fracture** — so **retightening without investigating the cause is actively harmful**. Materials: **grade 4 commercially pure titanium**, with **titanium-zirconium alloys developed precisely to allow narrow implants with better fatigue strength**. And the mechanics of this failure: **bending stress rises with the cube of the reduction in radius, so diameter dominates strength**; molar loads are the highest in the arch; **bone loss to the third thread removed coronal support and lengthened the lever arm**; and cyclic loading gave a **fatigue fracture — which is why it happened at seven years and not at one**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: what to do about it	The options with the constraint first — **the implant must come out; there is no repair** — then **replace with a wider implant after healing and grafting**, **leave a sleeper** if the apical fragment cannot be retrieved without excessive bone destruction, **accept a shortened dental arch to the 45**, or **bridge from the 45 and 47, which is destructive if those teeth are sound**. The recommendation with its sequence: **counter-torque if the coronal fragment can be engaged, otherwise piezo-surgery or a small trephine**, then healing and reassessment — and if replacing, **a wide implant appropriate to a molar with the occlusion corrected**. Then the honesty the case is built on: **the shortened dental arch put genuinely on the table, because this site has already failed once — Käyser** cited, the second molar contributing little, and the argument that it **avoids re-treating a site that has failed mechanically, avoids grafting, and costs him nothing further**. Argued against with equal force: **he is 58, not 80**, over-eruption and drifting are real, and **the failure was a design error, not a biological one — correcting the design corrects the risk**. **SAC: removal of a fractured implant is Complex; replacement into a healed, grafted molar site is Advanced to Complex.** The referral line: **counter-torque removal of an engageable fragment is yours; a trephine removal with a large defect, substantial bone sacrifice near the inferior alveolar canal, a non-contained defect, vertical augmentation, or uncertainty about canal proximity is not — removal near the canal in a resorbed posterior mandible is exactly where a second opinion is cheaper than a nerve injury**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: removal and rebuilding	The removal in strict order with the reason attached: **counter-torque first, then a fragment-removal device on the outer surface, then piezo-surgery circumferentially, and only then the smallest trephine that will pass, under copious irrigation because trephining generates significant heat in dense bone**. Why the order matters: **counter-torque removes the implant without sacrificing bone, leaving a socket rather than a crater**, whereas **a trephine takes a cylinder of host bone with it, enlarges the defect, may approach the canal, and turns a straightforward replacement into a grafting case — counter-torque first, trephine last, always**. The defect judged by **containment**: contained, graft at the same visit with particulate and a resorbable membrane; **non-contained or with compromised soft tissue, allow six to eight weeks of soft tissue healing and stage it, so you have a keratinised envelope to close over — vertical defects are a referral**. Then what you would do differently, which is the answer to the whole case: **a wide-diameter implant of 5 mm or more**, a **screw-retained monolithic zirconia crown** for **retrievability in a site with a history of mechanical problems** and **no residual cement**, a **narrow occlusal table with light centric contact and no excursive contact**, and a recall that includes **checking screw integrity every time, so loosening is investigated rather than repeatedly retightened**. The physiology given with its numbers: **25 to 100 microns and rich proprioception against 3 to 5 microns of bone deformation and poor tactile sensitivity** — the implant does not intrude to share load, **and he cannot feel that he is overloading it**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: prognosis, monitoring and learning	Fracture quantified — **well under one per cent, typically late, after five years or more, because it is a fatigue phenomenon** — then the risk profile: **narrow implants, molar sites, bruxists, cantilevered or long-span prostheses, and implants with prior crestal bone loss or repeated screw loosening**. The line that scores: **that risk profile is essentially a description of this patient**. Replacement prognosis given honestly — **reasonable but poorer than a virgin site** — with the qualification that matters: **the failure here was mechanical and explicable, so correcting the diameter and the occlusion addresses the actual cause, which is a better position than a site that failed for unexplained biological reasons**. The sleeper handled as a consent issue rather than a shortcut: **explained to him plainly, recorded with its position, flagged to any future clinician, reviewed and radiographed periodically, and monitored for symptoms, exposure or radiolucency** — and **an exposed fragment is no longer a sleeper but a plaque-retentive foreign body that must come out**, a possibility that **belongs in the consent for leaving it**. Monitoring specified for this patient: **shimstock occlusal check, inspection for wear facets, and confirmation that the screw is tight at each recall**.
14:30–15:00	Closes on the patient and the record	To him, factually and without a blame narrative: *"The implant has fractured. That's uncommon, and I think it happened because a narrow implant was used in a position that takes heavy chewing forces and because the bone level dropped. It has to come out. Here's what replacing it involves and what it would cost, and here's the alternative of leaving the space."* On whether to complain: **a recognised complication, you were not there, raise it with the treating practice first since they may wish to contribute, and the formal routes explained neutrally — no speculation, no advocacy either way, and all of it recorded**. Recorded: the history **including any previous loosening**, the mobility findings and **what moved relative to what**, the dated radiograph, the diagnosis, the options with risks and costs, his decision, **what you said about causation and in what words**, and if you retrieve a fragment, the **system, diameter and length, and whether anything was left in situ**.

The spine — the twelve answers to have ready if nothing else

1 The differential for a loose crown, resolved by what moves relative to what · 2 Preload, and the three ways a joint loses it · 3 Torque value, torque driver, and the ten-minute retorque · 4 Why repeated loosening is a warning: loosening, screw fracture, implant fracture · 5 Grade 4 titanium, and why titanium-zirconium exists · 6 Bending stress and the cube of the radius — why 3.3 mm failed here · 7 How bone loss to the third thread lengthened the lever arm · 8 Why it happened at seven years — fatigue, not overload on the day · 9 The shortened arch offered against a site that has already failed · 10 Counter-torque first, trephine last, always · 11 What you would do differently: 5 mm, screw-retained, narrow table, no excursive contact · 12 The sleeper — what he is told, what you record, what you monitor.

Where the marks are lost in this case

Explaining the fracture as bad luck. The examiner has specified **3.3 mm**, **a molar site**, **seven years** and **bone loss to the third thread** because together they are a complete mechanical explanation. Volunteer it: **bending stress rises with the cube of the reduction in radius**, **crestal bone loss lengthened the lever arm**, and **cyclic loading produced a fatigue fracture — which is why it failed then and not at one year**. And volunteer the history you have not been given: **ask about repeated screw loosening**, because **the escalation runs loosening, screw fracture, implant fracture**, and **retightening without investigating the cause is actively harmful**.

Replacing it without changing anything. The site has already failed once. A candidate who says "remove and place another implant" and stops has learned nothing from the case. Say what changes — **5 mm or more, screw-retained for retrievability and no cement, narrow occlusal table, light centric contact, no excursive contact, screw integrity checked at every recall** — and then say the harder thing: **the shortened dental arch to the 45 is a legitimate endpoint here** and must be offered, because it avoids re-treating a site that has already failed. Offering it, then arguing honestly against it, is what a 5 looks like.

Never naming the removal line. Removal is where this case can do real harm. **Counter-torque on an engageable fragment is yours.** What is not: **a trephine removal that would sacrifice substantial bone near the inferior alveolar canal, a large non-contained defect, vertical augmentation, or any uncertainty about canal proximity on the CBCT** — all referred. Say the sequence with its reason — **counter-torque leaves a socket, a trephine leaves a crater** — and say that near the canal in a resorbed posterior mandible, **a second opinion is cheaper than a nerve injury**.

SOE Case 30 — Aesthetic failure: the implant crown that looks too long

THE CASE

44-year-old woman, 8 months after fitting of an implant crown at the **11** by another dentist. She is distressed: the crown is **2 mm longer than the 21**, there is a **black triangle mesially**, and the mucosal margin is 2 mm apical to that of the 21. She has a **high smile line**. The implant is healthy — no bleeding, no bone loss.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 44-year-old woman with a **biologically successful but aesthetically failed** implant restoration at the 11, eight months after fitting elsewhere. The implant is healthy — no bleeding, no suppuration, no bone loss — so this is entirely a problem of tissue position and crown proportion: a crown 2 mm longer than the 21, a mid-facial margin 2 mm apical to its neighbour, and a mesial black triangle. Her **high smile line** makes every millimetre of that visible, so her distress is proportionate and I would not minimise it. My tasks are to establish **why** it looks like this — most importantly whether the implant is malpositioned — to work out what is genuinely recoverable, and to be honest with her about the difference between improvement and correction."

Domain 1 — Clinical interpretation and reasoning

Chain: the aesthetic groundwork

Q Describe the normal gingival architecture around an upper central incisor.

"A **scalloped** margin following the CEJ, with the **gingival zenith** — the most apical point of the margin — sitting slightly **distal to the long axis** of a central incisor and on the axis for a canine. The margins of the two centrals should be level with each other and with the canines, with the laterals about **1 mm coronal**. Between the teeth the **interdental papilla** fills the embrasure to the contact point, and there is a band of firm **keratinised attached tissue** apical to a free gingival groove."

Q What is the supracrestal attachment, and what are its dimensions?

"It is the soft tissue attachment between the bone crest and the base of the sulcus — what used to be called biologic width, renamed **supracrestal attached tissue** at the **2017 World Workshop**. Gargiulo's figures are roughly **1 mm of junctional epithelium and 1 mm of connective tissue attachment**, with a sulcus of about 0.7 mm above it. It re-establishes itself at a constant dimension, which is why a margin placed too deep produces either inflammation or recession."

Q How does the peri-implant equivalent differ?

"It is **longer and weaker**. The junctional epithelium is around 2 mm and the connective tissue zone 1 to 1.5 mm, so the total is greater than around a tooth. Crucially the collagen fibres run **parallel to the implant surface**, not inserted perpendicularly, because there is no cementum and no Sharpey's fibres. Vascularity is reduced, coming only from the supraperiosteal vessels. That is precisely why soft tissue around implants is less stable and less correctable than around teeth."

Q What decides whether a papilla fills the embrasure?

"The **distance from the contact point to the underlying bone crest**. On **Tarnow's** data, at 5 mm the papilla is present almost always, at 6 mm about half the time, and at 7 mm only about a quarter. Around an implant the critical bone is the **crest on the adjacent tooth**, not the implant, so I cannot create a papilla by anything I do to the implant — I can only shorten the distance by lowering the contact point."

Q Classify smile lines.

"On **Tjan's** classification: a **low** smile shows less than three-quarters of the upper anterior teeth; an **average** smile shows three-quarters to all of the clinical crown plus the interdental papillae; a **high** smile shows the full crown length plus a continuous band of gingiva. Roughly a fifth of patients have a high smile line, more commonly women. It is the single biggest determinant of aesthetic risk, and it is why this restoration is a problem rather than an observation."

Q What are the normal proportions of an upper central incisor?

"A **width-to-length ratio of about 80%**, typically around 8.5 mm wide by 10.5 to 11 mm long. Her crown is 2 mm longer than the 21, so the ratio has dropped to roughly two-thirds and the eye reads it as a long tooth even before it registers the margin discrepancy. Proportion, symmetry with the contralateral, and the level of the gingival margins are what an observer actually notices."

Chain: reading this particular failure

Q So what has gone wrong here?

"Most likely the **implant is malpositioned** — too shallow, too buccal, or both — so the mucosal margin sits apically and the crown had to be lengthened to reach the contact. Contributing factors would be a **thin phenotype** with recession over the first year, a **contact point more than 5 mm from the adjacent crest** producing the black triangle, loss of interproximal bone on the 12, and a pre-existing tissue deficit that was never augmented."

Q Which single cause is most likely, and why?

"**Three-dimensional implant position.** A crown that is 2 mm too long with a margin 2 mm too apical, in a healthy implant with no bone loss, is the signature of a platform placed too far coronally or too far buccally — the restoration is simply following where the implant told it to go. Recession alone would not usually give both findings together to that degree."

Q What would you measure and record at that first visit?

"Crown length at the 11 and 21 and the mid-facial margin discrepancy in millimetres; the papilla level against **Jemt's index**; the **contact point to crest distance** radiographically; phenotype and the width of keratinised tissue; full peri-implant probing depths, bleeding and suppuration; the smile line on video, at rest and animated. Plus **standardised photographs**, study casts, and — with her consent — the previous clinician's records."

Q What would a CBCT add?

"The **buccolingual and apicocoronal position of the platform** and the thickness of any remaining buccal bone — neither of which I can determine clinically or from a periapical. That is the finding that decides everything: if the implant sits within a restorable envelope I can work prosthetically and surgically; if it does not, no crown and no graft will fix it. I would justify the exposure on that basis under IR(ME)R."

Q Define the envelope a central incisor implant should sit in.

"**Mesiodistally**, at least 1.5 to 2 mm from the adjacent root. **Orofacially**, the shoulder about **1 to 2 mm palatal** to the emergence line drawn between the adjacent teeth, so the restoration emerges rather than the implant. **Apicocoronally**, about **3 mm apical to the planned mucosal margin** of the definitive crown. Buser describes these as comfort and danger zones — outside them, the prosthesis cannot compensate."

Domain 2 — Clinical decision making and treatment planning

Chain: the conversation before the treatment plan

Q She is in tears. What do you do first?

"**Listen, uninterrupted.** I would let her say the whole of it before I look in her mouth, and then acknowledge it explicitly rather than moving straight to examination. Distress that is dismissed becomes a complaint; distress that is heard usually becomes a working relationship. Only then would I examine, and I would tell her at the outset that I am going to need records before I can say anything useful."

Q Tell me in your own words what you would say to her.

> " 'I can see this is really affecting you, and I don't think you're being unreasonable — with the way you smile, this shows. What I can tell you already is that the implant itself is healthy, so we're not dealing with an infection or something failing. The problem is where the gum sits and how long the tooth looks. Before I promise you anything I need a three-dimensional scan, because the position of the implant under the gum decides what's realistically possible.' "

Q She asks whether the previous dentist did something wrong. Answer her.

> " 'I wasn't there, I haven't seen the planning records, and it wouldn't be fair for me to judge a colleague's work from the outside. What I can do is tell you exactly what I find and what your options are now.' I would not criticise him — but I would not defend the result either, and I would not tell her it looks fine when it does not."

Q Do you have a duty of candour here?

> "The statutory duty falls on the clinician whose care caused the harm, so it sits with him, not me, for anything that happened before I met her. My professional duty is to be **open and honest with her about what I find**, to support her if she wishes to raise it with him or with the GDC, and to apply candour fully to anything that arises from my own subsequent treatment."

Chain: the options

Q Set out her options.

> "**Accept and monitor** with a full explanation — legitimate, but unlikely to satisfy a high smile line. **Prosthetic modification alone** — a new crown with a lowered contact point, altered proportions and cervical pink ceramic. **Soft tissue augmentation** with a connective tissue graft, alone or with a new crown. **Orthodontic levelling** of the adjacent gingival margins. **Explantation and reconstruction** if the implant is malpositioned. Or **removal and a resin-bonded bridge** with an ovate pontic over a grafted site."

Q Which do you recommend?

> "It depends entirely on the CBCT. Within a restorable envelope, I would stage **prosthetic modification plus a connective tissue graft**, working through a **screw-retained provisional** so she can see the result before committing to anything definitive. Grossly malpositioned, I would tell her plainly that no crown will fix it and set out explantation or removal and a bonded bridge, with honest timescales and cost."

Q You said prosthetic modification first — argue against it.

> "Fairly: a longer crown with pink ceramic at the neck is **camouflage, not correction**, and in a high smile line at conversational distance pink ceramic can read as false. It also commits her to a crown she may still dislike, having spent money. The counter-argument is that it is reversible, cheap relative to the alternatives, carries no surgical morbidity, and can be trialled in a provisional first — which is why I would still start there."

Q A colleague would go straight to explantation. Why might they be right?

> "Because if the implant is buccally positioned, every intermediate step is delay: the graft will relapse, the crown will still be long, and she will have spent a year and considerable money arriving at the option she could have started with. If the CBCT shows the platform outside the envelope, they are right and I would say so — the mistake is grafting a position problem."

Q How predictable is soft tissue grafting around an implant compared with a tooth?

> "**Considerably less.** Around a tooth with an intact interproximal attachment, root coverage is highly predictable. Around an implant there is no cementum, no perpendicular fibre insertion and reduced blood supply, so the realistic aim is **thickening the phenotype and gaining perhaps 1 mm**, not complete coverage. Complete correction of a 2 mm mid-facial discrepancy is unlikely, and I would say that in those words before she consents."

Q What is the SAC classification of what you are proposing?

> "The original case was **Complex** — a single central incisor in a high smile line always is. Correcting an established aesthetic failure is **Complex** as well, and arguably harder than the primary case, because I am working against a fixed implant position and scarred tissue. Explantation of an osseointegrated anterior implant with reconstruction is beyond straightforward practice."

Q Where do your own limits sit?

> "**Within my competence**: the assessment, the CBCT interpretation, the provisional, the new definitive crown, the maintenance, and the conversation. **Beyond it**: explantation of an integrated anterior implant, and advanced mucogingival surgery or contour augmentation in the aesthetic zone. I would refer those, and I would tell her where the boundary is at the first appointment rather than discovering it mid-treatment."

Domain 3 — Decision making in relation to treatment procedures

Chain: the prosthetic route

Q Take me through remaking the crown.

> "Remove the definitive crown, place a **screw-retained provisional with a concave subgingival contour** so the tissue is not being pushed apically, and let it rebound for several weeks. Then adjust it incrementally to develop the emergence profile and the margin position I want, review it against photographs and with her, and only when both of us are satisfied convert it with a **customised impression coping** into a definitive screw-retained crown."

Q How do you lower the contact point without making a bulbous crown?

> "By moving the contact **apically and slightly palatally** and lengthening it vertically rather than thickening it buccally, so the embrasure closes from beneath while the labial contour and the width at the incisal third stay correct. If the contact-to-crest distance is 7 mm I cannot close a black triangle with proportion alone, and I would be honest that a small triangle may remain."

Q What subgingival contour do you want, and why?

> "**Concave**, or at least flat, in the critical zone immediately below the margin — a convex or over-contoured emergence displaces the mucosa apically and is a common iatrogenic cause of the very recession she has. Below that, in the subcritical zone, the contour is freer. The principle is that the tissue is supported, not pushed."

Q When would you use pink ceramic, and what are its drawbacks?

> "Where the vertical tissue deficit cannot be closed surgically and the alternative is an unacceptably long tooth — so, potentially here. Drawbacks: the **transition line between ceramic and mucosa is visible in a high smile**, the colour match is difficult and does not change as her own tissue does, and it is unforgiving if further recession exposes the join. I would trial it in the provisional before committing."

Q Screw or cement retained, and why?

> "**Screw-retained** wherever the access hole allows it. It is retrievable, which matters enormously in a case I may need to revise repeatedly, and it eliminates **subgingival cement**, which is a well-recognised cause of peri-implant disease. If the angulation forces a cingulum-emerging channel I would use an angulated screw channel system rather than accept cement in the aesthetic zone."

Q What abutment material?

> "A **titanium base with a zirconia or ceramic superstructure**. Titanium alone shows through thin mucosa as a grey band, which in her phenotype and smile line would compound the problem. Zirconia bonded to a titanium base gives the optical result without the risk of a full ceramic abutment fracturing at the connection."

Chain: the surgical route

Q If you graft, describe the sequence.

"Provisional in place with a concave contour and the tissue rebounded. Then a **subepithelial connective tissue graft** placed through a split-thickness envelope or tunnel with a **coronally advanced flap**, secured with sling sutures, and the provisional relieved so it does not compress the graft. Allow **3 to 6 months** of maturation, adjusting the provisional incrementally, then a customised impression coping and the definitive crown."

Q Where would you harvest the graft from?

"The **palate**, between the distal of the canine and the mesial of the first molar, staying about 2 mm from the gingival margin and superficial enough to avoid the **greater palatine neurovascular bundle**, which lies further apically and closer in a shallow vault. The **maxillary tuberosity** is an alternative with denser tissue and less morbidity. A xenogeneic collagen matrix avoids a donor site but gains less thickness."

Q What would you consent her for, specifically?

"Donor site pain and bleeding, and the possibility of a palatal bleed needing a stent; graft shrinkage and **incomplete correction**; colour or texture mismatch of the grafted tissue; relapse over the following one to three years; the number of visits and the total timescale of roughly nine to twelve months; the cost; and, most importantly, that the realistic aim is **improvement rather than symmetry with the 21**."

Q How long before you would judge the result?

"Not before **six months**, and I would tell her that at the start so she does not panic at three weeks when it looks over-corrected and swollen. Peri-implant soft tissue continues to remodel and mid-facial change typically declares itself over the **first one to three years**, so the six-month appearance is an interim reading, not the verdict."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: judging and recording the outcome

Q How would you measure the outcome objectively?

"With **standardised photographs** taken at the same distance, magnification and lighting throughout, and a numerical score — the **pink and white aesthetic scores**, or at minimum **Jemt's papilla index**, the mid-facial margin level measured against the 21 in millimetres, and the crown length ratio. Alongside that, peri-implant probing, bleeding and bone levels against baseline, and her own rating of how she feels about it."

Q What would you promise her at the outset?

"**Improvement, not perfection** — in those words, and recorded in those words. I would say that I expect to make it noticeably better, that I cannot make it identical to the 21, and that if she needs it to be identical then the honest answer is that this implant is not the route to that. Promising symmetry in this situation is how a difficult case becomes a complaint."

Q At six months the mid-facial margin has relapsed 1 mm. What now?

"First, compare against the standardised photographs rather than her recollection or mine, and confirm it is genuine relapse and not remodelling. Then check the implant is still healthy and that the crown contour is not over-contoured and pushing the tissue. Options are a second graft with diminishing returns, prosthetic recontouring with cervical pink ceramic, or acceptance — and I would present them in that honest order."

Q When would you tell her to stop?

"When the marginal gain from each further intervention has become smaller than its cost, morbidity and risk of making things worse — usually after one graft and one crown remake. **Recognising the point of diminishing returns is a clinical skill**, and continuing to operate on an unhappy aesthetic patient rarely ends well. At that point I would offer a second opinion rather than another procedure."

Q What would you record throughout?

> "Her presenting concern in **her own words**; all measurements and the CBCT findings; the options discussed including doing nothing, with the risks and realistic outcomes of each; the explicit statement that the aim is improvement not perfection; that a mock-up or provisional was used as the decision tool and that she saw and approved it; costs; the limits of my competence and any referral; and standardised photographs at every stage."

Q What is the lesson for your own practice?

> "That anterior implant aesthetics are decided **before the drill** — by the three-dimensional position, by phenotype modification at placement or second stage, and by a mock-up agreed with the patient in advance. Every corrective option here is less predictable than getting it right at placement. I would make sure my own aesthetic consent explicitly covers recession, black triangles and long-term change, with a contemporaneous note, and I would audit that."

WATCH OUT

Fall back to the question "is this a **position** problem, a **tissue** problem or a **prosthetic** problem?" — say which, and say what evidence would tell you. That is the frame the whole case hangs on, and it buys you a sentence of thinking time.

Running this case to time

Time check. The flow above holds 39 questions and about 2,723 words of model answer. At 140 words a minute plus roughly five seconds a question, that is close to **twenty-three minutes** — an overrun of **seven to eight**. What goes first is the front of Domain 1: **normal gingival architecture and the supracrestal attachment collapse into a single question**, and the **harvest site** and **abutment material** questions in Domain 3 disappear entirely. What never goes is **Tarnow**, the **three-dimensional envelope**, the **CBCT that decides everything**, and **improvement not perfection**. If your answers run past **thirty seconds** you will never reach Domain 4 — and Domain 4 here is where the case is actually won.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Name it in the first clause: a **biologically successful but aesthetically failed** implant restoration at the 11. Then the evidence in numbers — a crown **2 mm longer than the 21**, a mid-facial margin **2 mm apical** to its neighbour, a mesial **black triangle**, and **no bleeding, no suppuration, no bone loss**, so this is entirely tissue position and crown proportion. Then the fact that sets the stakes: a **high smile line**, which makes every millimetre visible, so **her distress is proportionate and you would not minimise it**. Then the three tasks — establish **why** it looks like this and above all **whether the implant is malpositioned**, work out what is genuinely recoverable, and be honest about the difference between **improvement and correction**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the aesthetic groundwork, then reading this failure	Expect it to open **below the implant material**. The **scalloped margin** with the **zenith slightly distal to the long axis** of a central, centrals level with the canines, laterals **1 mm coronal**. The **supracrestal attachment**, renamed at the **2017 World Workshop** — roughly **1 mm junctional epithelium and 1 mm connective tissue** — and how the peri-implant version differs: **longer and weaker**, about **2 mm of epithelium and 1 to 1.5 mm of connective tissue**, with fibres running **parallel** to the surface because there is **no cementum and no Sharpey's fibres**, and reduced vascularity from supraperiosteal vessels only. Then **Tarnow**: papilla presence is decided by **contact point to bone crest — 5 mm almost always, 6 mm about half, 7 mm about a quarter** — and the crest that matters is **on the adjacent tooth, not the implant**, so you cannot create a papilla, only lower the contact. **Tjan's** smile line classification with the figure — about a fifth of patients, more often women. **Width-to-length about 80%.** Then the resolution: the single most likely cause is **three-dimensional implant position — too shallow, too buccal, or both** — because a healthy implant with no bone loss producing both a long crown and an apical margin is the signature of the platform, not of recession. And the **envelope**: **1.5 to 2 mm from the adjacent root**, shoulder **1 to 2 mm palatal** to the emergence line, **3 mm apical to the planned mucosal margin** — Buser's comfort and danger zones.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the conversation, then the options	The first answer is not clinical: **listen, uninterrupted, before you look in her mouth**, then acknowledge it explicitly. On the previous dentist: **"I wasn't there and I haven't seen the planning records"** — no criticism, but **no defence of the result either, and you do not tell her it looks fine when it does not**. On candour, the precise point: the statutory duty sits with **the clinician whose care caused the harm**, so not you for what preceded you; your duty is to be **open about what you find**, to support her if she wishes to raise it, and to apply candour fully to **your own** subsequent treatment. Then the six options named cleanly — **accept and monitor; prosthetic modification alone; connective tissue graft; orthodontic levelling; explantation and reconstruction; removal and a resin-bonded bridge with an ovate pontic**. The recommendation with its condition attached: **it depends entirely on the CBCT** — within the envelope, prosthetic modification plus a graft, worked through a **screw-retained provisional**; grossly malpositioned, say plainly that **no crown will fix it**. Then the two honesty questions: grafting round an implant is **considerably less predictable than round a tooth** — realistic aim **thickening the phenotype and perhaps 1 mm**, not coverage of a 2 mm discrepancy — and the colleague who goes straight to explantation **may be right**, because **grafting a position problem is the mistake**. **SAC: Complex, and arguably harder than the primary case.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the prosthetic route, then the surgical one	The provisional first, and why: **screw-retained with a concave subgingival contour**, tissue allowed to rebound over several weeks, adjusted incrementally, converted with a **customised impression coping** only when both of you are satisfied — the provisional is the **decision tool**, not a placeholder. The contact lowered **apically and slightly palatally and lengthened vertically**, not thickened buccally, with the honest rider that **at 7 mm you cannot close the triangle with proportion alone**. The contour principle in one line: **concave or flat in the critical zone — the tissue is supported, not pushed** — because over-contour is a common **iatrogenic** cause of the very recession she has. **Screw-retained wherever the access hole allows**, for retrievability in a case you may revise repeatedly and to eliminate **subgingival cement**; an **angulated screw channel** rather than cement in the aesthetic zone. Pink ceramic named as **camouflage, not correction**, with its drawbacks — a **visible transition line in a high smile**, a colour that does not change as her tissue does — and trialled in the provisional first. Then the graft: **subepithelial connective tissue through a tunnel or envelope with a coronally advanced flap**, provisional relieved off it, **3 to 6 months of maturation**. Consent for **shrinkage and incomplete correction**, relapse over **one to three years**, **nine to twelve months** of treatment, and that the aim is **improvement, not symmetry**. Judge nothing before **six months**, and tell her that at the start.
11:30–14:30	Domain 4: judging the outcome, and knowing when to stop	Objective measurement, named: **standardised photographs at the same distance, magnification and lighting**, plus **pink and white aesthetic scores** or at minimum **Jemt's papilla index**, the mid-facial level in millimetres against the 21, and the crown length ratio — alongside peri-implant probing and bone levels, and **her own rating**. Then the promise, in the exact words that are also the recorded words: **improvement, not perfection**. The 1 mm relapse at six months handled by **comparing against the standardised photographs, not recollection**, confirming health and that the crown is not over-contoured, then three options **in honest order** — second graft with diminishing returns, prosthetic recontouring, or acceptance. Then the answer that separates candidates: **when to stop** — when the marginal gain is smaller than the cost, morbidity and risk, usually **after one graft and one crown remake**; **recognising diminishing returns is a clinical skill**, and the offer at that point is **a second opinion, not another procedure**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	What you say: *"The implant itself is healthy — this isn't an infection or something failing. The problem is where the gum sits and how long the tooth looks. I expect to make this noticeably better; I can't make it identical to the tooth next door, and if it has to be identical then this implant isn't the route to that."* What you write: her concern **in her own words**; every measurement and the CBCT findings; all options **including doing nothing**, with realistic outcomes; the explicit statement that the aim is **improvement not perfection**; that a provisional was used as the decision tool and **she saw and approved it**; costs; **the limits of your competence and any referral**; and standardised photographs at every stage.

The spine — the twelve answers to have ready if nothing else

1 Biologically successful, aesthetically failed — said in the first clause · 2 The supracrestal attachment, and how the peri-implant version differs · 3 Tarnow: contact point to crest, and why the crest is on the tooth · 4 Tjan's high smile line, and why it makes this a problem not an observation · 5 Why position, not recession, explains both findings · 6 The three-dimensional envelope, in millimetres · 7 What the CBCT decides, and the IR(ME)R justification · 8 Candour — whose duty it is, and what yours actually is · 9 The six options, and the one the CBCT would rule out · 10 Grafting round an implant: 1 mm, not coverage · 11 Concave contour, screw retention, and the provisional as decision tool · 12 Improvement not perfection — and when to stop.

Where the marks are lost in this case

Treating it as a crown problem. The examiner has given you a **healthy implant with no bone loss**, a crown **2 mm long** and a margin **2 mm apical** — that combination is the signature of **three-dimensional implant position**, and a candidate who goes straight to remaking the crown has missed the question the case is built on. Say the frame out loud: **is this a position problem, a tissue problem or a prosthetic problem** — and say that the **CBCT decides it**, because within the envelope you can work and outside it no crown and no graft will fix it.

Over-promising the graft. Volunteer, unprompted, that soft tissue grafting round an implant is **considerably less predictable than round a tooth — no cementum, no perpendicular fibre insertion, reduced blood supply** — and that the realistic gain is **thickening the phenotype and perhaps a millimetre**, not correction of a 2 mm mid-facial discrepancy. Then say the harder thing: the colleague who would go straight to explantation **may be right**, because **grafting a position problem** simply spends her year and her money arriving at the option she could have started with.

Never naming the limit. Within your competence: the assessment, the CBCT interpretation, the provisional, the definitive crown, the maintenance and the conversation. Not within it: **explantation of an integrated anterior implant**, and **advanced mucogingival or contour augmentation in the aesthetic zone** — both referred, and both named **at the first appointment rather than discovered mid-treatment**. The second limit is temporal: knowing when to stop treating an unhappy aesthetic patient and offer **a second opinion instead of another procedure**.

SOE Case 31 — The severely anxious patient who asks for an implant

THE CASE

42-year-old woman, dental phobic, has avoided dentistry for 12 years. Presents with pain from a grossly carious **36**, which is unrestorable. She is otherwise fit and well. She has heard that implants are "done under sedation" and asks for one.

Pace: one structured opening of about 60 seconds, then roughly 35 questions in the remaining 14 minutes. Nothing after the opening earns you more than a few seconds of thought.

Opening — the 60-second presentation

Q Present this case.

"A 42-year-old dental phobic woman who has avoided dentistry for twelve years and now presents in pain from an unrestorable 36, asking for an implant because she believes it can be done under sedation. My immediate priority is **relief of pain**, delivered in a way that begins to rebuild rather than confirm her fear. The implant question is a much later one. Implant treatment is a long, multi-visit surgical undertaking followed by **lifelong maintenance**, so it is one of the least suitable ways to begin treating a phobic patient, and I would say that to her kindly rather than simply refusing. What I need to establish is the nature and severity of her anxiety, the state of the rest of her mouth, whether the underlying disease is controlled, and whether she can realistically sustain implant maintenance."

Domain 1 — Clinical interpretation and reasoning

Chain: the undergraduate groundwork — her pain and her tooth

Q She is in pain. How do you establish the diagnosis?

"**History first** — site, onset, character, radiation, duration, exacerbating and relieving factors, whether it wakes her, whether hot or cold provokes it and whether it lingers, and whether she can bite on it. Then examination and **special tests**: sensibility testing with cold and electric pulp test on the suspect and control teeth, percussion, palpation, mobility, periodontal probing, and a periapical radiograph."

Q Distinguish irreversible pulpitis from apical periodontitis on those findings.

"**Irreversible pulpitis** — spontaneous, poorly localised, often worse lying down, an exaggerated response to cold or heat that **lingers beyond ten seconds**, and typically not tender to percussion until the inflammation reaches the periodontal ligament. **Apical periodontitis** — well localised because the periodontal ligament is proprioceptive, **tender to percussion and biting**, the tooth may feel high, sensibility testing is usually negative, and there may be a widened periodontal ligament space or a periapical radiolucency."

Q What makes a tooth unrestorable?

"Insufficient sound coronal tissue to obtain a **ferrule of 1.5 to 2 mm circumferentially** on sound dentine; caries extending well subcrestally or into the furcation; an unfavourable crown-to-root ratio; a vertical root fracture; inaccessible or unnegotiable canals; or a periodontal prognosis that is hopeless regardless. It is also a judgement about **strategic value** and about what the patient can sustain — a heroic restoration in someone who has not attended for twelve years is not a kindness."

Q How would you anaesthetise that tooth?

"An **inferior alveolar nerve block**, needle entering at the coronoid notch level with the barrel over the contralateral premolars, about 1 cm above the mandibular occlusal plane, advancing 20 to 25 mm to bone contact and withdrawing 1 mm before aspirating. Plus a **long buccal infiltration** for the buccal soft tissue. Blocks fail more often in irreversible pulpitis, so I would plan supplementary techniques from the start."

Q Why do blocks fail in irreversible pulpitis, and what do you do about it?

"Because the **inflamed tissue is acidic**, so less of the anaesthetic exists in the un-ionised form that crosses the nerve membrane; because inflamed nerves show altered, **tetrodotoxin-resistant sodium channels**; and because of central sensitisation. The answers are a **buccal infiltration of 4% articaine** as a supplement, then intraligamentary, intraosseous, or as a last resort intrapulpal anaesthesia — and telling her in advance that I have those options, because unexpected pain is what created the phobia in the first place."

Q Maximum safe dose of lidocaine?

"**4.4 mg/kg** with adrenaline, to a usual adult maximum of around 300 mg. A 2.2 ml cartridge of **2% lidocaine contains 44 mg**, so for a 70 kg adult that is about seven cartridges — far more than I would need. The practical point in a phobic patient is not the ceiling; it is warming the cartridge, applying topical properly, and injecting slowly and out of her line of sight."

Chain: assessing the anxiety itself

Q What do you need to establish about her anxiety?

"Its **origin and its trigger** — needles, the drill, gagging, loss of control, a specific previous traumatic experience, or a fear of judgement about the state of her mouth, which is very common after long avoidance. Whether she has had sedation or general anaesthesia before and how she found it. Her medical and psychiatric history and medication. Who supports her, and what she actually wants."

Q How would you measure it?

"With a validated instrument — the **Modified Dental Anxiety Scale**. Five questions, each scored one to five, giving a total of five to twenty-five, with **nineteen or above** indicating severe anxiety or dental phobia. It takes a minute, it gives me a baseline I can re-measure, and it demonstrates to her that I take it seriously as a clinical finding rather than a personality trait."

Q Is anxiety a contraindication to implant treatment?

"**No — but untreated anxiety that prevents maintenance is.** The implant will integrate regardless of how she feels about it. What it will not survive is a patient who cannot attend for peri-implant monitoring, cannot tolerate professional debridement, and presents years later with advanced peri-implantitis. So the question I am asking is not 'can she cope with the surgery' but 'can she cope with the next twenty years'."

Q What else about a twelve-year absence concerns you?

"That the 36 is unlikely to be the only problem. I would expect **active caries elsewhere**, undiagnosed periodontal disease, and possibly oral mucosal lesions that have gone unexamined. I would want a full mouth examination with a **basic periodontal examination**, bitewings and appropriate radiographs, and a soft tissue and cancer screening — she is in a group where late presentation is the norm."

Domain 2 — Clinical decision making and treatment planning

Chain: the pharmacology and the escalation ladder

Q Define conscious sedation.

"A technique in which one or more drugs produce a depression of the central nervous system sufficient to allow treatment, but during which **verbal contact with the patient is maintained throughout**. The drugs and techniques used must carry a margin of safety wide enough to render loss of consciousness unlikely. If verbal contact is lost, that is general anaesthesia, and it must be managed as such."

Q Set out the anxiety management options from least to most invasive.

"**Behavioural techniques** first — calm environment, tell-show-do, an agreed stop signal, giving her control over pace, relaxation and distraction, careful appointment timing and a longer appointment so nothing is rushed. Then **inhalation sedation**. Then **intravenous conscious sedation with midazolam**. **General anaesthesia** only

in a hospital setting and very rarely for elective treatment. Running alongside all of it, referral for **cognitive behavioural therapy**."

Q Tell me about inhalation sedation.

"**Nitrous oxide and oxygen**, titrated, typically at 30 to 40%, delivered through a nasal hood with a minimum 30% oxygen and active scavenging. It is anxiolytic and mildly analgesic, has a very wide margin of safety, and recovers in minutes so she needs **no escort**. It is excellent for needle phobia. Contraindications are nasal obstruction, first-trimester pregnancy, recent middle ear or intraocular surgery, and previous bleomycin."

Q And intravenous midazolam?

"Titrated to effect — typically 2 mg over 30 seconds, then **1 mg increments at 60-second intervals**, most adults settling around 5 to 7 mg, with much smaller doses and slower titration in the elderly or medically compromised. End points are slurred speech, relaxation, delayed response and **Verrill's sign**. It gives anterograde amnesia, which is a genuine advantage in a phobic patient. She needs a **responsible escort** and cannot drive or work for the rest of the day."

Q What monitoring and what standards?

"Continuous **pulse oximetry** and blood pressure, clinical observation of colour, respiration and responsiveness, and a trained second appropriate person present throughout. Compliance with the **IACSD standards** — appropriate training and continuing education, a designated sedationist, documented pre-operative assessment and consent, emergency drugs and equipment immediately available, and audit."

Q What is flumazenil and how do you use it?

"A **competitive benzodiazepine antagonist** for reversal of over-sedation or respiratory depression. **200 micrograms intravenously over 15 seconds, then 100 micrograms at one-minute intervals**, to a usual maximum of 1 mg. It must be immediately available whenever midazolam is used. The critical point is that its **half-life is shorter than midazolam's**, so re-sedation can occur and the patient must be observed, not discharged on reversal."

Q When would general anaesthesia be appropriate?

"For emergency treatment where sedation has genuinely failed, for severe learning disability, or where the surgical requirement is beyond what sedation can cover — and always in a hospital setting with an anaesthetist. It is **almost never appropriate for elective implant placement**, because a patient who requires general anaesthesia to accept a single visit cannot sustain the twenty years of maintenance that follow."

Chain: the plan for this patient

Q What is your plan, in phases?

"**Phase one, relief of pain** — extraction of the 36 with whatever anxiety management she needs today. **Phase two, stabilisation and acclimatisation** — treat the remaining disease by graded exposure across several short appointments, and address the cause: diet, fluoride, hygiene. **Phase three, re-evaluation** — only once she is attending reliably and the disease is controlled do I raise replacing the 36. **Phase four, replacement options**, of which an implant is one."

Q What are the options for replacing the 36?

"**No replacement** — a shortened arch to the 35 if the occlusion is stable, requiring no surgery at all. A **removable partial denture**. A **bridge from 35 to 37** if both are sound, though that is destructive of unrestored teeth. Or an **implant**. Given her anxiety I would present the shortened arch first and genuinely, not as a token gesture before steering her elsewhere."

Q You've put the shortened arch first — argue against it.

"Fairly: she is 42, so she may live another forty years without a first molar; the 37, if present, may tip and the 26 over-erupt; and there is an argument that a patient should not be denied the definitive option simply because

she finds dentistry frightening — that risks being paternalistic. My counter is that the evidence for functional adequacy of a shortened arch is good, and I would monitor drifting and over-eruption rather than assume they will not happen."

Q She insists on an implant today. What do you actually say to her?

" 'I can hear how much you want this dealt with in one go, and I'm not saying no. But an implant isn't one appointment — it's the surgery, then three or four months of healing, then impressions, then the crown, and then a check-up every few months for the rest of your life. Sedation makes each visit easier; it doesn't make there be fewer of them. Let's get you out of pain today, and let's find out together whether coming here is something you can do. Then we'll talk about the implant properly.' "

Domain 3 — Decision making in relation to treatment procedures

Chain: delivering treatment to this patient

Q How would you modify today's extraction appointment?

"**First appointment of the day**, minimal waiting, a quiet surgery. Explain everything before starting and agree a **stop signal** that I actually honour. Topical anaesthetic left in place long enough to work, a warmed cartridge, slow injection out of her sight. Inhalation sedation if she accepts it. Headphones and music, a companion if she wants one, continuous narration of what is happening, and no time pressure."

Q Would you socket-preserve at the extraction?

"I would extract **as atraumatically as possible** — periotomes, sectioning rather than levering, preserving the buccal and lingual plates — because that keeps the implant option open at no extra burden to her. Formal ridge preservation with a graft I would only do if she had already engaged with the idea of an implant; performing an extra procedure on a phobic patient for a treatment she may never have is hard to justify."

Q If she does proceed to an implant, how would you modify the surgical visit?

"A **pre-operative visit** to see the room and the equipment with nothing done. Then: first appointment of the day, sedation as agreed with escort arranged, the same stop signal, instruments kept out of her line of sight, drapes positioned so she is not covered, warning her before every noise and every vibration, and generous time. **A phone call from me that evening** — which does more for a phobic patient than almost anything else."

Q Would you place an implant under intravenous sedation?

"Yes, if she needs it and she is suitable — sedation and implant surgery combine well. But I would make two things explicit to her first: sedation makes the **appointment** tolerable, it does not shorten the treatment or make maintenance optional; and a patient who requires sedation for every visit will find a three- to six-monthly lifelong recall very difficult. That is a genuine clinical argument for the simpler solution."

Q What must be in place before you sedate her intravenously?

"A documented **pre-operative assessment** including ASA status, medical history, medication, allergies, airway and body mass index, and a plan for the treatment itself. Written consent for both sedation and the procedure, taken at a **separate prior visit** and confirmed on the day. Fasting advice per local policy, an escort confirmed, escort and post-operative instructions given in writing beforehand, and cannulation site checked."

Q What are your discharge criteria?

"She is **fully conscious and orientated**, vital signs stable and back to baseline, able to stand and walk unaided, no bleeding, pain controlled, and in the care of her **responsible escort** who has had the written instructions explained. No driving, no operating machinery, no alcohol, no important decisions or legal documents for the rest of the day, and a contact number for the evening."

Q Halfway through, she becomes distressed and wants to stop. What do you do?

"**Stop.** Immediately, and completely — because honouring the stop signal is the single most important thing I will do for her long-term relationship with dentistry. Make the site safe, remove instruments, sit her up, and let her recover. Then find out what specifically triggered it, and either resume on her terms or stage the rest to another visit. Overriding a stop signal once destroys everything I have built."

Q What is the SAC classification of a single implant at the 36?

"Surgically **Straightforward** in adequate bone with good access, and prosthodontically **Straightforward** for a single posterior crown. But the SAC classification recognises **patient modifiers**, and severe dental phobia with a twelve-year absence and unproven maintenance capacity is a significant one — so in practice I would treat this case as **Advanced** in its management, even though the drilling is simple."

Q Where does the limit of your competence sit here?

"The extraction, the acclimatisation, the restorative care and the implant itself are within my scope. **Beyond it**: intravenous sedation unless I hold current IACSD-compliant training and work within a compliant team, and any general anaesthetic, which needs a hospital referral. Managing the **phobia itself** is also beyond me — I would refer for cognitive behavioural therapy, which is the only intervention that treats it rather than working around it."

Domain 4 — Interpretation and problem solving of clinical outcomes

Chain: judging whether this was the right decision

Q How would you judge whether implant treatment was right for her?

"**Not by whether the implant integrates**, but by whether she is attending, cleaning and maintaining it at two and five years, and by whether the experience improved or worsened her relationship with dentistry. If a patient completes implant treatment and then never returns, that is a failure regardless of what the radiographs show. I would also re-measure her anxiety score and expect it to have fallen."

Q At one year she has not attended for ten months and there is bleeding around the implant. Assessment?

"Probe six sites with light force, record bleeding and suppuration, and take a **periapical against baseline**. Bleeding with no bone loss is **peri-implant mucositis** — reversible, and managed by professional debridement and re-establishing her oral hygiene. Bleeding with progressive bone loss beyond initial remodelling is **peri-implantitis**, which is much harder to arrest and which non-attendance predicts."

Q How would you have prevented that?

"By making **maintenance part of the consent, not an afterthought** — agreeing the recall interval before placing anything, booking the appointments in advance rather than relying on recall letters, keeping the visits short and predictable, and using the same behavioural approach for hygiene visits as for surgery. And by choosing a **cleansable design** — screw-retained, no residual cement, accessible embrasures."

Q She now says she wishes she had never had it done. How do you respond?

"By listening first, then reviewing honestly what was discussed and recorded at the outset. If the consent conversation genuinely covered maintenance and she has decided she cannot sustain it, the discussion is about how to make it manageable, not about blame. If something I did fell short, I would say so and apply **duty of candour**. Either way I would offer a second opinion rather than become defensive."

Q What would you record?

"Her **anxiety score and its date**; the discussion of all options **including no treatment and the shortened arch**; the specific conversation that maintenance is lifelong and her stated ability to attend, in her own words; the sedation assessment, consent taken at a prior visit, drugs and doses titrated, monitoring, recovery and discharge; her escort's name; and the written instructions given."

Q What is the lesson you take from this case?

> "That **the hardest clinical decision here is not surgical**. The implant is straightforward; the judgement is whether this particular person can carry it for the next twenty years, and that judgement is made by watching her attend rather than by asking her whether she will. Acclimatisation first, disease control second, implants last — and sometimes never, which is a legitimate outcome."

WATCH OUT

Fall back to the phased frame — pain relief, stabilisation, re-evaluation, then replacement — and say which phase the question belongs to. It buys you a sentence and it is almost always the right ordering for a neglected, anxious mouth.

Running this case to time

Time check. The flow above holds 37 questions and about 2,557 words of model answer. At 140 words a minute plus roughly five seconds a question that is a little over **twenty-one minutes** — an overrun of about **six**. What goes first is the pharmacology detail: the **lidocaine maximum dose** and the **flumazenil regimen** compress into one question, and the **discharge criteria** and **pre-sedation requirements** merge. What never goes is **conscious sedation defined**, the **escalation ladder**, the **phased plan**, and the sentence the whole case turns on — **sedation makes each visit easier; it does not make there be fewer of them**. If your answers run past **thirty seconds** you will never reach Domain 4, which here is where the judgement is marked.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	The order is the answer: **relief of pain first**, delivered in a way that **begins to rebuild rather than confirm her fear**. Then the position on the implant, stated without refusing her: implant treatment is a **long, multi-visit surgical undertaking followed by lifelong maintenance**, so it is **one of the least suitable ways to begin treating a phobic patient** — and you would say that **kindly**, not as a refusal. Then the four things still to establish: **the nature and severity of her anxiety**, **the state of the rest of her mouth after twelve years**, **whether the underlying disease is controlled**, and **whether she can realistically sustain implant maintenance**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the undergraduate groundwork on her pain, then the anxiety itself	Expect it to open **well below the implant material**. History with the full pain screen, then **sensibility testing on the suspect and a control tooth**, percussion, palpation, mobility, probing, periapical. **Irreversible pulpitis** — spontaneous, poorly localised, worse lying down, **lingering beyond ten seconds**, not tender to percussion until the ligament is involved — against **apical periodontitis**, well localised because **the ligament is proprioceptive**, tender to percussion and biting, sensibility usually negative. Unrestorability defined by **a ferrule of 1.5 to 2 mm circumferentially on sound dentine**, subcrestal or furcation caries, vertical root fracture, unnegotiable canals — plus the judgement clause that scores: **strategic value, and what the patient can sustain; a heroic restoration in someone who has not attended for twelve years is not a kindness**. The block described properly, then the physiology: blocks fail in pulpitis because **inflamed tissue is acidic** so less drug is un-ionised, because of **tetrodotoxin-resistant sodium channels**, and because of central sensitisation — answered by **buccal 4% articaine**, then intraligamentary, intraosseous, intrapulpal, and by **telling her in advance that you have those options, because unexpected pain is what created the phobia**. Then the anxiety: its **origin and trigger** — needles, drill, gagging, loss of control, or **fear of judgement about the state of her mouth, very common after long avoidance** — measured with the **Modified Dental Anxiety Scale**, five items, five to twenty-five, **nineteen or above** severe. And the reframe: **anxiety is not a contraindication; untreated anxiety that prevents maintenance is**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the pharmacology, the ladder, and the plan	**Conscious sedation defined** with the load-bearing clause — **verbal contact maintained throughout**, and a margin of safety wide enough that loss of consciousness is unlikely; **if verbal contact is lost that is general anaesthesia**. The ladder in order: **behavioural techniques**, then **inhalation sedation**, then **intravenous midazolam**, then **general anaesthesia in hospital and very rarely**, with **referral for cognitive behavioural therapy running alongside all of it**. Nitrous oxide **titrated at 30 to 40%**, minimum **30% oxygen**, scavenged, recovery in minutes and **no escort** — excellent for needle phobia; contraindications named. Midazolam **2 mg over 30 seconds then 1 mg at 60-second intervals**, most adults **5 to 7 mg**, smaller and slower in the elderly, endpoints including **Verrill's sign**, **anterograde amnesia a genuine advantage here**, escort essential. **Pulse oximetry and blood pressure, a trained second appropriate person, IACSD standards. Flumazenil 200 micrograms then 100 at one-minute intervals**, with the point that matters: **its half-life is shorter than midazolam's, so re-sedation can occur and she is observed, not discharged**. Then the plan in four phases — **pain relief, stabilisation and acclimatisation, re-evaluation, replacement** — and the options for the 36 with the **shortened arch to the 35 presented first and genuinely**, argued against fairly (she is 42; the 37 may tip, the 26 over-erupt; refusing the definitive option because she finds dentistry frightening risks **paternalism**) and then defended.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: delivering treatment to this particular patient	The modified appointment, itemised: **first appointment of the day**, minimal waiting, a quiet surgery, **an agreed stop signal you actually honour**, topical left long enough to work, **warmed cartridge, slow injection out of her sight**, inhalation sedation if she accepts, narration throughout, no time pressure. Extraction **as atraumatically as possible — periotomes, sectioning rather than levering, plates preserved** — because that keeps the implant option open at no extra burden; but **formal ridge preservation only if she has already engaged with the idea of an implant**, since **an extra procedure on a phobic patient for treatment she may never have is hard to justify**. If she proceeds: a **pre-operative visit to see the room with nothing done**, instruments out of her line of sight, drapes so she is not covered, warning before every noise, and **a phone call from you that evening**. Sedated placement is legitimate — **but say the two qualifiers first**: sedation makes the appointment tolerable, **not the treatment shorter or the maintenance optional**; and a patient who needs sedation every visit **will find a lifelong three- to six-monthly recall very difficult — which is a genuine clinical argument for the simpler solution**. Pre-sedation requirements and **discharge criteria** given crisply. Then the answer that must be instant: she becomes distressed mid-procedure — **stop, immediately and completely**, make the site safe, sit her up, because **overriding a stop signal once destroys everything you have built**. **SAC: Straightforward surgically and prosthodontically — but the patient modifier makes the management Advanced.**

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: whether this was the right decision at all	The criterion, stated against the obvious wrong answer: success is **not whether the implant integrates** but whether she is **attending, cleaning and maintaining at two and five years**, and whether the experience **improved or worsened her relationship with dentistry — a patient who completes treatment and never returns is a failure regardless of the radiographs**. Re-measure the anxiety score and expect it to have fallen. At one year, ten months of non-attendance with bleeding: probe six sites, **periapical against baseline**, and separate **peri-implant mucositis — bleeding, no bone loss, reversible** from **peri-implantitis — progressive loss beyond initial remodelling, much harder to arrest, and predicted by non-attendance**. Prevention volunteered: **maintenance as part of the consent, not an afterthought**, appointments **booked in advance rather than relying on recall letters**, kept short and predictable, the same behavioural approach for hygiene visits as for surgery, and a **cleansable screw-retained design with no residual cement**. If she says she wishes she had never had it done: **listen first**, review what was recorded, and if something fell short, **duty of candour** — otherwise a discussion about making it manageable, **not about blame**, with a second opinion offered rather than defensiveness.
14:30–15:00	Closes on the patient and the record	What you say: *"I can hear how much you want this dealt with in one go, and I'm not saying no. But an implant isn't one appointment — it's the surgery, then three or four months of healing, then impressions, then the crown, and then a check-up every few months for the rest of your life. Sedation makes each visit easier; it doesn't make there be fewer of them. Let's get you out of pain today, and let's find out together whether coming here is something you can do."* What you write: her **anxiety score and its date**; all options **including no treatment and the shortened arch**; the maintenance conversation and **her stated ability to attend, in her own words**; the sedation assessment, **consent taken at a prior visit**, drugs and doses titrated, monitoring, recovery and discharge; **the escort's name**; and the written instructions given.

The spine — the twelve answers to have ready if nothing else

1 Pain relief first, and why the order is the answer · 2 Irreversible pulpitis against apical periodontitis on the findings · 3 Unrestorable — ferrule, and what she can sustain · 4 Why blocks fail in pulpitis, and the three supplementary techniques · 5 The Modified Dental Anxiety Scale, and the score that matters · 6 Anxiety is not the contraindication; unmaintainable anxiety is · 7 Conscious sedation defined — verbal contact throughout · 8 The ladder: behavioural, inhalation, intravenous, GA, with CBT alongside · 9 Midazolam titration, and why flumazenil does not permit discharge · 10 The four phases, and the shortened arch offered first and genuinely · 11 The stop signal, honoured absolutely · 12 Success measured at five years by attendance, not by integration.

Where the marks are lost in this case

Answering the sedation question instead of the case. The examiner is not asking whether you can titrate midazolam — that is one question and it is worth one answer. The case is about **whether this woman can**

carry an implant for twenty years, and the sentence that scores is that **sedation makes the appointment tolerable, not the treatment shorter or the maintenance optional**. A patient who needs sedation for every visit will struggle with a **lifelong three- to six-monthly recall**, and saying so is the clinical argument, not a soft one.

Skipping the undergraduate third. The first three minutes are pulp diagnosis, unrestorability and local anaesthetic physiology, and they are marked. Volunteer the part candidates never do: **blocks fail more often in irreversible pulpitis**, for named reasons, so you **plan supplementary techniques from the start and tell her you have them** — because **unexpected pain is what created the phobia in the first place**. And volunteer the **shortened arch to the 35 first and genuinely**, then argue against it fairly, rather than producing it as a token before steering her to an implant.

Never naming the limit. The extraction, the acclimatisation, the restorative care and the implant itself are yours. Not yours: **intravenous sedation unless you hold current IACSD-compliant training and work in a compliant team**, and **any general anaesthetic**, which is a hospital referral. And the limit candidates miss entirely — **the phobia itself**, which is treated by **cognitive behavioural therapy**, the only intervention that treats it rather than working around it. Acclimatisation first, disease control second, implants last, **and sometimes never — which is a legitimate outcome**.

SOE Case 32 — High smile line, thin phenotype, immediate implant request

THE CASE

29-year-old model. The **11** has a horizontal root fracture below the crest following trauma 3 weeks ago and is unrestorable. **High smile line, thin scalloped phenotype, triangular crown forms.** CBCT: **buccal plate 0.6 mm thick, intact**. She has read about "same-day teeth" and wants the tooth out and an implant placed with a temporary crown at the same visit.

Domain 1

"This is the highest-risk aesthetic scenario in implant dentistry: an upper central incisor in a young patient with a **high smile line, thin scalloped phenotype, triangular crowns and a 0.6 mm buccal plate**, in a profession where appearance is her livelihood. She is asking for the protocol that carries the greatest aesthetic risk in exactly the situation where risk is least acceptable."

Q Work through the ITI immediate placement decision tree.

"The **pre-operative assessment** requires a thick soft tissue phenotype, absence of recession and low aesthetic risk. She fails all three. It then requires CBCT confirmation of an intact socket with a **facial wall of at least 1 mm** — hers is 0.6 mm — enough apical and palatal bone, the socket within the bone envelope, a horizontal defect of at least 2 mm, and no infection. She therefore falls into the **second branch**: facial wall under 1 mm. The tree directs to **Type II–III placement at 4 to 16 weeks**, or, if the situation is incompatible with Type I to III placement, **ridge preservation or a formal graft** followed by **Type IV late placement at 4 to 6 months**. On the SAC classification this is **Complex** surgically and prosthodontically."

Q What else would you assess?

"Exact smile line and lip dynamics on video, not just at rest; papilla heights and interproximal bone levels on the 12 and 21, which determine the papillae and which I cannot recreate; sensibility of the adjacent teeth after trauma; the position of the fracture relative to the crest; her occlusion, particularly overbite and protrusive contacts; her expectations in her own words; and comprehensive photographic records."

Domain 2

Q: What do you tell her?

"Honestly and without condescension. 'I understand why same-day teeth appeal, and in the right mouth they work well. Yours isn't the right mouth for it, and I'd be doing you a disservice if I pretended otherwise. The bone on the front of your tooth is about half a millimetre thick — thinner than a fingernail — and your gum is thin too. If I place an implant today, the most likely result over the next couple of years is that the gum shrinks back and you see a grey line or a longer tooth. In your job that's not a small thing. What I'd like to do instead is take the tooth out very carefully, preserve the socket, give you a temporary tooth that looks good and that nobody will notice, and place the implant once the site is right.' "

Q What is your plan?

"**Atraumatic extraction without a flap**, preserving the buccal plate absolutely — periotomes, sectioning if needed, no buccal levering. **Ridge preservation** with a slowly resorbing xenograft and a socket seal. An immediate **resin-bonded provisional bridge or a bonded pontic** — not a denture, which would press on the site and is unacceptable to her professionally. Reassess at **8 weeks** clinically and, if needed, with CBCT. Then either **early placement** (**Type 2–3**) with simultaneous contour augmentation and a connective tissue graft, or **late placement at 4 to 6 months** if the ridge needs staged augmentation.
Two-stage submerged healing, then a screw-retained provisional to develop the emergence over 8 to 12 weeks, then a customised impression coping and a definitive screw-retained restoration on a Ti-base with a ceramic superstructure."

Q Would you treat her yourself?

> "This is **Complex** in both dimensions. I would undertake the extraction and ridge preservation and the restorative planning, and I would refer or co-manage the implant placement with contour augmentation and connective tissue grafting in this aesthetic zone. Saying that to her early is better than discovering my limits mid-treatment."

Q What must the consent cover, given who she is?

> "Under *Montgomery*, materiality is judged from her position. So: **mid-facial recession**; **grey show-through**; **black triangles**, which her triangular crown form makes more likely; that the appearance may change over years; the total treatment time of nine to twelve months and what she will wear during it — which for a model is a material fact; the possibility that the result will not match the 21 exactly; the cost including maintenance; and that no one can guarantee an aesthetic outcome. I would use a **mock-up** so she can see the achievable result before committing, and record all of it."

Domain 3

Q: How does the thin phenotype change your surgical plan?

> "It makes **phenotype modification part of the plan, not a rescue procedure**. I would perform a **connective tissue graft** at placement or at second stage — the two windows when it is most predictable — to convert a thin phenotype to a thicker one. I would also position the implant **palatally**, use a **narrow-diameter implant** to leave more buccal bone, over-contour the buccal graft slightly to allow for resorption, and use a **Ti-base with a ceramic superstructure** rather than a titanium abutment to avoid grey show-through."

Domain 4

Q: How would you monitor the aesthetic outcome?

> "Standardised photographs at every stage; **pink and white aesthetic scores**, or at minimum the papilla index and the mid-facial margin level measured against the 21; peri-implant probing, bleeding and bone levels. Reviews at 1 week, 3 weeks, second stage, provisional adjustments, fitting, 3 months, 12 months, then maintenance. I would counsel her that **mid-facial recession, if it occurs, typically appears over the first one to three years**, so I would not declare success at six months."

Q What if, despite everything, she develops recession at two years?

> "I would have prepared her for that possibility. Management would be a connective tissue graft, a change of abutment material, prosthetic recontouring with cervical pink ceramic, or — if the implant position is the problem — explantation and reconstruction. I would be honest, apply duty of candour if anything I did contributed, and involve a colleague for a second opinion. And I would use the photographs to have that conversation objectively."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 903 words**. At 140 words a minute plus five seconds a question that is barely **seven and a quarter minutes**: it does not overrun, it leaves **almost eight minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions** in the mouth of an examiner. The decision tree alone is four: the pre-operative criteria, which ones she fails, which branch that puts her in, and what the branch directs you to. If your answers run past **thirty seconds** you will still never reach Domain 4 — and Domain 4 here is the one-to-three-year recession window, which is the easiest mark in the case to earn.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT slice up	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	Name the risk in the first clause: **the highest-risk aesthetic scenario in implant dentistry** — an upper central in a young patient with a **high smile line, thin scalloped phenotype, triangular crown forms and a 0.6 mm buccal plate** — and then the sentence that frames the whole case: **she is asking for the protocol that carries the greatest aesthetic risk in exactly the situation where risk is least acceptable**. Add the modifier that makes it material: **appearance is her livelihood**.
1:20–4:30	Domain 1: the decision tree, drilled one criterion at a time. Expect the single printed answer to become **four or five questions**	*What does the ITI tree require pre-operatively?* — a **thick soft tissue phenotype, absence of recession, low aesthetic risk**. *Which does she fail?* — **all three**. *What does it require on the CBCT?* — an **intact socket with a facial wall of at least 1 mm**, sufficient apical and palatal bone, the socket **within the bone envelope**, a **horizontal defect of at least 2 mm**, and **no infection**. *Hers?* — **0.6 mm**. *So which branch?* — the **second: facial wall under 1 mm**, directing to **Type II–III placement at 4 to 16 weeks**, or, if incompatible with Type I to III, **ridge preservation or a formal graft then Type IV late placement at 4 to 6 months**. *And the classification?* — **Complex surgically and prosthodontically**. Then the second printed answer, which is really the assessment list broken into three: **smile line and lip dynamics on video, not at rest**; **papilla heights and interproximal bone on the 12 and 21, which decide the papillae and which you cannot recreate**; and the rest — **sensibility of adjacent teeth after the trauma**, the **fracture level relative to the crest**, overbite and protrusive contacts, **her expectations in her own words**, and comprehensive photographic records.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: what you say to her, the plan, and who does it — three printed answers that become eight or nine	*How do you say no without saying no?* — **honestly and without condescension**, and in her language: the bone is **"about half a millimetre thick — thinner than a fingernail"**, the gum is thin too, and the likely result of placing today is that **"the gum shrinks back and you see a grey line or a longer tooth"** — **"in your job that's not a small thing."** Then the offer, so the refusal arrives attached to a plan. *What is the plan?* — **atraumatic extraction without a flap**, buccal plate preserved absolutely, **periotomes, sectioning if needed, no buccal levering**; **ridge preservation with a slowly resorbing xenograft and a socket seal**; and the provisional that matters professionally — a **resin-bonded provisional bridge or bonded pontic, not a denture**, which would press on the site. *When do you reassess?* — **8 weeks**, clinically and with CBCT if needed. *Then what?* — **early placement (Type 2–3) with simultaneous contour augmentation and a connective tissue graft**, or **late placement at 4 to 6 months** if staged augmentation is needed; **two-stage submerged healing**, a screw-retained provisional to develop emergence over **8 to 12 weeks**, then a **customised impression coping** and a definitive screw-retained restoration on a **Ti-base with a ceramic superstructure**. *Would you do it yourself?* — the referral line, below. *And the consent?* — under ***Montgomery*, materiality judged from her position**: **mid-facial recession**, **grey show-through**, **black triangles made more likely by triangular crown form**, change over years, **nine to twelve months and what she wears during it — a material fact for a model**, that the result **may not match the 21**, cost including maintenance, and that **no one can guarantee an aesthetic outcome** — with a **mock-up** used so she sees the achievable result before committing.
7:30–11:30	Domain 3: the thin phenotype. One printed answer, and it is really five	*What changes surgically?* — **phenotype modification becomes part of the plan, not a rescue procedure**. *When?* — a **connective tissue graft at placement or at second stage**, the **two windows when it is most predictable**. *Where do you put the implant?* — **palatally**. *What diameter?* — **narrow, to leave more buccal bone**. *What about the graft volume and the abutment?* — **over-contour the buccal graft slightly to allow for resorption**, and a **Ti-base with a ceramic superstructure rather than a titanium abutment**, because in this phenotype titanium shows through as a **grey band**. This is the domain with the most room in it, so expect the examiner to fill the time here with the practicalities the printed answer compresses — and answer each in a sentence rather than reciting the paragraph.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: monitoring, and what happens if it goes wrong anyway	*How do you measure the outcome?* — **standardised photographs at every stage**, **pink and white aesthetic scores**, or at minimum **the papilla index and the mid-facial margin level against the 21**, with peri-implant probing, bleeding and bone levels. *At what intervals?* — **1 week, 3 weeks, second stage, provisional adjustments, fitting, 3 months, 12 months, then maintenance**. *When do you declare success?* — **not at six months**: **mid-facial recession, if it occurs, typically appears over the first one to three years**, and saying that in advance is the mark. *Recession at two years?* — **you prepared her for it**; then **connective tissue graft, change of abutment material, prosthetic recontouring with cervical pink ceramic**, or, **if the implant position is the problem, explantation and reconstruction** — with **duty of candour if anything you did contributed**, a colleague's second opinion, and **the photographs used to have that conversation objectively**.
14:30–15:00	Closes on the patient and the record	What you say: *"Same-day teeth work well in the right mouth. Yours isn't the right mouth for it, and I'd be doing you a disservice if I pretended otherwise. I'd like to take the tooth out very carefully, protect the socket, give you a temporary tooth nobody will notice, and place the implant once the site is right."* What you write: the **0.6 mm measurement and the three phenotype findings**; the **decision-tree reasoning and which criteria she failed**; her request and your explanation in **her own words**; the *Montgomery* consent list **including timescale and what she wears**; that a **mock-up** was used and approved; and the **referral or co-management arrangement, agreed at the outset**.

The spine — the twelve answers to have ready if nothing else

1 The highest-risk aesthetic scenario, named in the first clause · 2 The three pre-operative criteria, and that she fails all three · 3 The 1 mm facial wall rule, and her 0.6 mm · 4 The second branch: Type II–III at 4 to 16 weeks · 5 Type IV at 4 to 6 months if augmentation is staged · 6 SAC: Complex in both dimensions · 7 The papillae are decided by the 12 and 21, and you cannot recreate them · 8 Atraumatic flapless extraction, ridge preservation, bonded provisional — never a denture · 9 *Montgomery* materiality judged from her position, including what she wears for nine months · 10 Phenotype modification at placement or second stage, palatal position, narrow implant, Ti-base · 11 Recession declares itself at one to three years, not six months · 12 The referral line, named at the first visit.

Where the marks are lost in this case

Refusing her instead of redirecting her. "No, that would be too risky" is a 2. The mark is in **saying no with a plan attached** and in her own words — **half a millimetre, thinner than a fingernail**, a gum that is thin too, and the honest prediction of **a grey line or a longer tooth in a couple of years, which in her job is not a small thing**. Then immediately: careful extraction, socket preserved, a **bonded provisional nobody will notice**, and the implant when the site is right.

Reciting the tree instead of applying it. Do not deliver the ITI criteria as a list and stop. Take them one at a time and say **which she fails and by how much** — thick phenotype, no recession, low aesthetic risk, then the **1 mm facial wall against her 0.6 mm** — and then name the branch and what it directs to. Then volunteer the thing candidates leave out: **the papillae are decided by the interproximal bone on the 12 and 21, and nothing you do to the implant can create them**. And volunteer that **phenotype modification is part of the plan, not a rescue** — a connective tissue graft **at placement or second stage**, because those are the two predictable windows.

Taking the whole case on. This is **Complex in both dimensions** and it sits **beyond a general dentist with an interest in implants**. The extraction, the ridge preservation, the provisional and the restorative planning are yours. **Immediate or early placement in a thin phenotype with a high smile line, with simultaneous contour augmentation and connective tissue grafting in the aesthetic zone, is not** — that is **referred or co-managed with someone who does it regularly**, and, as the case says, **saying so early is better than discovering your limits mid-treatment**.

SOE Case 33 — Multiple missing posterior teeth on a limited budget

THE CASE

51-year-old builder. Missing **16, 17, 26, 27, 36, 37, 46, 47** — all four posterior sextants. He has an intact anterior dentition from 15 to 25 and 35 to 45, all sound. He has marked wear on the anterior teeth and masseteric hypertrophy. He can afford approximately the cost of two implants.

Domain 1

"A patient with a **shortened dental arch to the second premolars in all four quadrants**, intact and sound anterior teeth, clear signs of parafunction, and a limited budget. This is a case where the correct answer may be to do very little, and where the temptation to fill every space is the trap."

Q What is his functional status?

"I would ask rather than assume: can he eat what he wants, does he have pain or discomfort, does he have any TMJ symptoms, and what actually brought him in? A patient with premolar occlusion and no complaint may not need any posterior replacement at all."

Q What is the significance of the anterior wear?

"It tells me he is loading his anterior teeth heavily, partly because of parafunction and partly because he has no posterior support. That is an argument **for** some posterior replacement — to protect the anterior teeth — and simultaneously a warning about the mechanical risk of anything I place."

Domain 2

Q: What are the options?

"**Shortened dental arch** — accept the current situation. Evidence from Käyser onwards indicates that around 20 teeth with four occluding premolar pairs gives adequate function and satisfaction, without increased TMD or periodontal breakdown. He already has that. This is the **default and the cheapest option**, and I would present it as a positive choice.
Removable partial dentures — bilateral free-end saddles, the least well tolerated design; in a builder with parafunction, unlikely to be worn.
Implants — but eight units is far beyond his budget. The question becomes: **if I can only place two, where?**
A combination — implants in the two most valuable positions, accepting the shortened arch elsewhere."

Q If you place two implants, where and why?

"I would place them at the **first molar positions in the arch where he has the greatest functional deficit**, most likely the **36 and 46** — restoring lower first molars gives the greatest functional return, mandibular bone is more favourable, and it re-establishes posterior support to protect the anterior teeth. I would not place single implants at the 17 or 27 positions, which contribute least. But I would say honestly that **the marginal functional gain over a well-managed shortened arch may be small**, and that spending his money on a protective splint, occlusal management of the anterior wear, and prevention may serve him better. That is a legitimate recommendation, not a cop-out."

Q What about his parafunction?

"It must be addressed regardless of what else I do. A **hard, full-coverage occlusal splint** for night-time wear, adjusted to give even contacts and canine guidance. Discussion of stress and sleep, and screening for **obstructive sleep apnoea** if there are symptoms, with referral to his GP. Anything I place will be loaded heavily, so I would design for it: wide implants, monolithic zirconia, narrow occlusal tables, no cantilevers, and consent to a higher mechanical complication rate."

Q SAC?

> "Individual posterior implants in adequate bone would be **Straightforward** surgically. Prosthodontically, the parafunction is a modifier that raises it to **Advanced**. The overall case management — deciding what not to do — is the harder part."

Domain 3

Q: What would you check before placing at the 36 and 46?

> "CBCT for **residual height above the canal** and ridge width; the position of the mental foramina and any anterior loop; the lingual concavity; and the **restorative space**, because unopposed upper teeth may have over-erupted, though here the uppers are also missing so the space is likely generous. Also mouth opening and access — he is a large man and posterior access matters."

Q Surgical and restorative plan?

> "Standard placement with 2 mm canal clearance, wide-diameter implants if width allows, conventional loading at three months, **screw-retained monolithic zirconia crowns**, narrow occlusal tables, light centric contacts, no excursive contacts, and a night guard fabricated after the crowns are fitted and adjusted to include them."

Domain 4

Q: How would you judge success in this case?

> "By his function and comfort, not by the number of teeth. I would ask him at each review what he can eat now that he could not before. I would monitor the **anterior wear** with photographs and casts, because arresting that is arguably the main clinical objective. And I would monitor the implants conventionally."

Q What is the risk of the alternative approach — filling every space?

> "Eight implants he cannot afford, in a heavy parafunctional loader, with a high mechanical complication rate and a maintenance burden he may not sustain — a worse outcome than a well-managed shortened arch. **Knowing when not to treat is an examinable clinical skill**, not a failure of ambition."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 691 words**. At 140 words a minute plus five seconds a question that is under **six minutes**: it does not overrun, it leaves **nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The options answer alone is four: the shortened arch and its evidence, the free-end saddle denture, the arithmetic that eight implants do not fit the budget, and the combination. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the question the whole case exists for — **the risk of filling every space**.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the charting and photographs up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Say what he actually has, not what he is missing: a **shortened dental arch to the second premolars in all four quadrants**, with an **intact and sound anterior dentition from 15 to 25 and 35 to 45**. Then the three modifiers — **marked anterior wear**, **masseteric hypertrophy**, and a budget of **about two implants**. Then the sentence that frames everything: **this is a case where the correct answer may be to do very little, and the temptation to fill every space is the trap**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: function first, then the wear. Two printed answers that become six or seven	*What is his functional status?* — and the mark is in **asking rather than assuming**: can he eat what he wants, is there pain, are there TMJ symptoms, and **what actually brought him in**. Then the conclusion said out loud: **a patient with premolar occlusion and no complaint may not need any posterior replacement at all**. *What does the anterior wear tell you?* — that he is **loading his anterior teeth heavily**, from **parafunction** and from **absent posterior support**. *So is that an argument for or against treatment?* — **both, and say both**: it is an argument **for** some posterior replacement to **protect the anterior teeth**, and simultaneously **a warning about the mechanical risk of anything you place**. Expect the examiner to fill the rest of this domain with the undergraduate groundwork the printed case assumes: the signs of parafunction you would look for — **wear facets, linea alba, tongue scalloping, masseteric hypertrophy, a history of morning jaw ache** — how you would record the wear objectively, and the periodontal and endodontic status of the standing teeth.
4:30–7:30	Domain 2: the options and the money — one printed answer that is really five	*Shortened dental arch?* — the **default and the cheapest option, presented as a positive choice**, with the evidence named: **Käyser onwards, around 20 teeth with four occluding premolar pairs gives adequate function and satisfaction, without increased TMD or periodontal breakdown** — and the point that lands, **he already has it**. *Dentures?* — **bilateral free-end saddles, the least well tolerated design**, and **in a builder with parafunction, unlikely to be worn**. *Implants?* — **eight units is far beyond his budget**, so the question becomes **if I can only place two, where**. *Combination?* — implants in the **two most valuable positions**, shortened arch elsewhere. Then the answer the case is built round: **36 and 46 — lower first molars give the greatest functional return**, mandibular bone is **more favourable**, and it **re-establishes posterior support to protect the anterior teeth**; **not the 17 or 27**, which contribute least. And then the honesty: **the marginal functional gain over a well-managed shortened arch may be small**, and spending his money on **a splint, occlusal management of the wear and prevention** may serve him better — **a legitimate recommendation, not a cop-out**. *And the parafunction?* — addressed **regardless of what else you do**: a **hard, full-coverage occlusal splint** for night wear with **even contacts and canine guidance**, discussion of stress and sleep, and **screening for obstructive sleep apnoea** with GP referral if symptomatic. *SAC?* — **Straightforward surgically, Advanced prosthodontically because of the parafunction**, and **the overall case management — deciding what not to do — is the harder part**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: what you check and what you place — two printed answers that become six	*What do you check before placing at the 36 and 46?* — **CBCT for residual height above the canal and ridge width**; the **mental foramina and any anterior loop**; the **lingual concavity**; and the **restorative space**, likely generous here because the uppers are also missing. Then the practical one candidates omit: **mouth opening and posterior access — he is a large man**. *Surgical plan?* — standard placement with **2 mm canal clearance**, **wide-diameter implants if width allows**, **conventional loading at three months**. *Restorative plan?* — **screw-retained monolithic zirconia crowns**, **narrow occlusal tables, light centric contacts, no excursive contacts, no cantilevers**. *And the splint?* — **fabricated after the crowns are fitted and adjusted to include them**, which is the detail that separates a prepared candidate. *And the consent?* — design for heavy load and **consent explicitly to a higher mechanical complication rate**.
11:30–14:30	Domain 4: judging success, and the risk of the alternative	*How do you judge success here?* — **by his function and comfort, not by the number of teeth**, asking at each review **what he can eat now that he could not before**. *What are you actually monitoring?* — the **anterior wear, with photographs and study casts, because arresting that is arguably the main clinical objective**, alongside conventional implant monitoring. *And the risk of the alternative?* — the closing argument: **eight implants he cannot afford, in a heavy parafunctional loader, with a high mechanical complication rate and a maintenance burden he may not sustain — a worse outcome than a well-managed shortened arch**. Then the sentence to say in exactly these terms: **knowing when not to treat is an examinable clinical skill, not a failure of ambition**.
14:30–15:00	Closes on the patient and the record	What you say: *"You've actually got a working bite — you chew on your premolars, and the evidence says that's enough for most people. What worries me more than the gaps is the wear on your front teeth and how hard you're grinding. If you've got money for two implants, the honest question is whether two lower molars will change your day more than a night guard and looking after what you've got. Let's decide that together rather than filling every space."* What you write: **his stated function and what brought him in, in his own words**; the wear recorded with **photographs and casts**; the options **including doing nothing, with the shortened arch presented first**; the **budget discussion**; the **parafunction consent to a higher mechanical complication rate**; and the splint provision and review interval.

The spine — the twelve answers to have ready if nothing else

1 He has a shortened arch already — say that before you say what is missing · 2 Ask his function; a premolar occlusion with no complaint may need nothing · 3 What the anterior wear means — and that it argues both ways · 4 The shortened arch evidence: Käyser, four occluding premolar pairs, 20 teeth · 5 Free-end saddle dentures in a parafunctional builder · 6 Two implants: 36 and 46, and why not the 17 and 27 · 7 That the

marginal gain over a shortened arch may be small — volunteered · 8 The splint, the OSA screen, and the GP referral · 9 SAC: Straightforward surgically, Advanced prosthodontically · 10 What the CBCT must show: canal height, foramen and loop, lingual concavity · 11 Wide implants, monolithic zirconia, narrow tables, no cantilevers · 12 Knowing when not to treat, said in those words.

Where the marks are lost in this case

Planning eight implants he cannot have. The candidate who starts costing a full posterior reconstruction has failed the case in the first minute. He has **four occluding premolar pairs and sound anterior teeth**, and **Käyser's** evidence says that is functionally adequate. Present the **shortened dental arch first, as a positive choice**, name the evidence, and only then move to what two implants could add — **36 and 46, for functional return and posterior support**, and never the **17 or 27**.

Ignoring the wear. The **marked anterior wear and masseteric hypertrophy** are not scenery; arresting that wear is **arguably the main clinical objective of the case**. Volunteer the **hard full-coverage splint with even contacts and canine guidance**, the discussion of stress and sleep, the **screen for obstructive sleep apnoea with GP referral**, and the design consequences — **wide implants, monolithic zirconia, narrow occlusal tables, no cantilevers, and explicit consent to a higher mechanical complication rate**. And volunteer that the splint is made **after** the crowns and adjusted to include them.

Treating the cheaper option as a lesser one. The competence line here is not surgical — a single posterior implant in adequate bone is within any implant dentist's scope. The line is **honest sequencing**: saying out loud that **a splint, occlusal management and prevention may serve him better than two implants**, and that **a well-made partial denture or an accepted shortened arch is a legitimate endpoint, not a consolation**. Refer if the parafunctional occlusal management runs beyond you — but **say the cheaper option genuinely, or the examiner will read it as a token**.

SOE Case 34 — Adult hypodontia

THE CASE

26-year-old woman with **hypodontia**: congenitally absent **12, 22, 35, 45**, with retained but infra-occluded deciduous **75 and 85**. She has completed orthodontic treatment, with spaces opened at the lateral incisor sites (6 mm each) and the deciduous molars retained. She wants "everything sorted".

Domain 1

"An adult hypodontia case with four missing permanent units and two retained, infra-occluded deciduous molars. This is a **multidisciplinary case**, and my first job is to establish who else needs to be involved and what the endpoint is, rather than to start treating the easiest bit."

Q What are the specific issues in a hypodontia patient?

"The **alveolus never developed fully** around teeth that never formed, so the sites are typically narrow buccolingually and may be reduced in height — augmentation is more likely than in an acquired edentulous site. **Infra-occluded deciduous teeth** are ankylosed and take the surrounding alveolar bone down with them, so their sites are often vertically deficient. There is often a **microdont or unusual crown form** in the remaining teeth requiring restorative harmonisation. And there are psychological and long-term maintenance dimensions in a young patient who will need this care for sixty years."

Q What would you assess?

"Smile line and lip dynamics; the exact space at each site and whether it matches the ideal tooth width; the crown form and shade of the adjacent teeth; the vertical position of the deciduous molars and the extent of infra-occlusion; occlusion and vertical dimension; CBCT for ridge dimensions at all four sites; periodontal status; and her expectations. I would also want the **orthodontist's records and retention plan**, because implants cannot be placed until the position is stable and retained."

Domain 2

Q: What are the options at the lateral incisor sites?

"**Implants** — 6 mm of space allows a 3.3 mm narrow implant with 1.5 mm each side, which is feasible, but hypodontia sites are often narrow and may need contour augmentation.
Resin-bonded cantilever bridges from the canines — minimally invasive, well evidenced for single anterior units, reversible in principle, and avoiding surgery in a 26-year-old. Often the better choice.
Composite build-ups or veneers on the canines with orthodontic space closure — the canine substitution route, though the orthodontics is complete and reopening that discussion has costs.
Removable prosthesis — interim only."

Q And at the deciduous molar sites?

"First, decide the fate of the **75 and 85**. A well-retained, minimally infra-occluded deciduous molar in an adult can serve for many years and is often best retained and monitored, sometimes with onlay build-up to restore the occlusal plane. If they are significantly infra-occluded, ankylosed, submerging further, or damaging the adjacent periodontium, they should be extracted — but I must warn her that **extraction reveals a vertically deficient site** that may need augmentation. If they are retained, no replacement is needed at all."

Q What would you recommend overall?

"A staged, conservative plan:
1. Multidisciplinary discussion with the orthodontist and, given the complexity, a restorative specialist.
2. Retain the deciduous molars if sound, monitoring with annual photographs and radiographs; restore the occlusal plane with composite if needed.
3. At the lateral incisor sites, offer **resin-bonded cantilever bridges** as first line and **implants** as the alternative,

with a mock-up to show her the achievable aesthetics either way.
4. Defer any decision about the deciduous molar sites until they actually fail. This preserves options, avoids irreversible surgery in a 26-year-old, and can be revisited as she ages."

Q SAC?

"The lateral incisor implants: surgically **Advanced** — narrow sites likely needing contour augmentation, aesthetic zone. Prosthodontically **Advanced** — crown form harmonisation across the anterior segment. The overall case is **Complex** by virtue of being multidisciplinary and lifelong, and I would want it managed within a team."

Domain 3

Q: If you place implants at the 12 and 22, describe the key considerations.

"**Space:** 6 mm allows a 3.3 mm implant with 1.5 mm to each root; I would confirm root positions and divergence on CBCT and, if the roots converge, ask the orthodontist for root divergence before placement.
Bone: hypodontia sites are often narrow; I would expect to need **simultaneous contour augmentation** and would plan for a two-stage submerged protocol.
Position: shoulder 3 mm apical to the planned mid-facial margin, 1 to 2 mm palatal, angled through the cingulum.
Phenotype: consider a connective tissue graft at placement or second stage.
Retention: the orthodontic retainer must be designed to allow the implant crowns, and she must remain in retention.
Restoration: screw-retained on Ti-bases, with the crown form harmonised to the canines — which may itself require composite or ceramic modification of the canines."

Domain 4

Q: What is the lifelong picture you must explain to her?

"That she will be maintaining these restorations for sixty years; that implant crowns will need replacing more than once in that time; that **continued eruption of the natural dentition relative to an ankylosed implant** can produce infra-occlusion and open contacts even in adults, requiring modification; and that she will need regular review with photographs to detect these changes. I would rather she heard that now than at 45."

Q What would you monitor specifically?

"Standardised photographs and study casts at intervals, to detect relative infra-occlusion and open proximal contacts; the deciduous molars for progressive submergence and root resorption; peri-implant health; and the periodontal condition of the canines carrying any bonded bridge."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 767 words**. At 140 words a minute plus five seconds a question that is a little over **six minutes**: it does not overrun, it leaves **nearly nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The hypodontia issues answer alone is four: the undeveloped alveolus, the infra-occluded deciduous teeth, the microdont crown forms, and the sixty-year horizon. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the lifelong picture — the domain most candidates never get to and the one that distinguishes them.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiographs and casts out	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	The inventory in one clause — **congenitally absent 12, 22, 35 and 45, with retained infra-occluded deciduous 75 and 85**, orthodontics complete and **6 mm opened at each lateral site**. Then the framing sentence: this is a **multidisciplinary case**, and the first job is **to establish who else needs to be involved and what the endpoint is, rather than to start treating the easiest bit**. Then the age, said as a clinical fact: she is **26**, and whatever is placed she will be **maintaining for sixty years**.
1:20–4:30	Domain 1: what makes hypodontia different, then the assessment. Two printed answers that become seven	*Why are hypodontia sites different?* — **the alveolus never developed fully around teeth that never formed**, so sites are **narrow buccolingually and may be reduced in height**, and **augmentation is more likely than in an acquired edentulous site**. *And the deciduous molars?* — **infra-occluded deciduous teeth are ankylosed and take the surrounding alveolar bone down with them**, so those sites are **often vertically deficient**. *What else?* — **microdont or unusual crown forms** in the remaining teeth needing **restorative harmonisation**, and the **psychological and long-term maintenance dimensions in a young patient**. *What would you assess?* broken into three or four: **smile line and lip dynamics**; **the exact space at each site and whether it matches the ideal tooth width**; **crown form and shade of the adjacent teeth**; **the vertical position of the deciduous molars and the extent of infra-occlusion**; **occlusion and vertical dimension**; **CBCT ridge dimensions at all four sites**; and the one that scores because candidates forget it — **the orthodontist's records and retention plan, because implants cannot be placed until the position is stable and retained**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the options, site by site — three printed answers that become eight or nine	*Options at the lateral sites?* — **implants**, since **6 mm allows a 3.3 mm narrow implant with 1.5 mm each side**, feasible but often needing **contour augmentation**; **resin-bonded cantilever bridges from the canines — minimally invasive, well evidenced for single anterior units, reversible in principle, avoiding surgery in a 26-year-old, and often the better choice**; **composite build-ups or veneers on the canines with orthodontic space closure**, with the honest cost of reopening a completed course; and a **removable prosthesis, interim only**. *And the deciduous molars?* — decide their fate first: **a well-retained, minimally infra-occluded deciduous molar in an adult can serve for many years and is often best retained and monitored**, sometimes with **onlay build-up to restore the occlusal plane**; **extract if significantly infra-occluded, ankylosed, submerging further, or damaging the adjacent periodontium** — but **warn her that extraction reveals a vertically deficient site that may need augmentation**. And the line that scores: **if they are retained, no replacement is needed at all**. *What do you recommend?* — the staged, conservative plan in four steps: **multidisciplinary discussion with the orthodontist and a restorative specialist**; **retain the deciduous molars if sound, with annual photographs and radiographs**; **resin-bonded cantilever bridges first line at the lateral sites with implants as the alternative, and a mock-up either way**; **defer any decision about the deciduous molar sites until they actually fail** — because this **preserves options, avoids irreversible surgery in a 26-year-old, and can be revisited as she ages**. *SAC?* — lateral incisor implants **Advanced surgically and prosthodontically**; the **overall case Complex by virtue of being multidisciplinary and lifelong**, and **managed within a team**.
7:30–11:30	Domain 3: if you place at the 12 and 22 — one printed answer, and it is really six	*Space?* — **6 mm allows a 3.3 mm implant with 1.5 mm to each root**; **confirm root positions and divergence on CBCT** and, **if the roots converge, ask the orthodontist for root divergence before placement**. *Bone?* — narrow sites, so **expect simultaneous contour augmentation and plan a two-stage submerged protocol**. *Position?* — **shoulder 3 mm apical to the planned mid-facial margin, 1 to 2 mm palatal, angled through the cingulum**. *Phenotype?* — **connective tissue graft at placement or second stage**. *Retention?* — **the orthodontic retainer must be designed to allow the implant crowns, and she must remain in retention** — the answer candidates never give. *Restoration?* — **screw-retained on Ti-bases**, with **crown form harmonised to the canines**, which **may itself require composite or ceramic modification of the canines**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: the sixty-year conversation — two printed answers that become five	*What is the lifelong picture?* — that she is **maintaining these restorations for sixty years**; that **implant crowns will need replacing more than once in that time**; and the mechanism candidates almost never name: **continued eruption of the natural dentition relative to an ankylosed implant can produce infra-occlusion and open contacts even in adults**, requiring modification. *When do you tell her?* — **now, not at 45**, and say that in those words. *What do you monitor?* — **standardised photographs and study casts at intervals, to detect relative infra-occlusion and open proximal contacts**; the **deciduous molars for progressive submergence and root resorption**; **peri-implant health**; and the **periodontal condition of the canines carrying any bonded bridge**.
14:30–15:00	Closes on the patient and the record	What you say: *"I know you want everything sorted, and I understand that. But you're 26, and whatever we do here you'll be looking after for sixty years. The baby teeth are doing a job at the moment — I'd rather keep them and watch them than take them out and find I've created a bigger problem. At the front, the least destructive option is a bonded bridge on each side, and implants are a real alternative. Let me show you both on a mock-up before you choose."* What you write: the **multidisciplinary discussion and who is involved**; the **orthodontist's retention plan and confirmation of stability**; the option discussion **including retaining the deciduous molars and deferring their sites**; that a **mock-up** was used and approved; the explicit **sixty-year maintenance conversation including crown replacement and relative infra-occlusion**; and the **monitoring intervals with baseline photographs and casts**.

The spine — the twelve answers to have ready if nothing else

1 Multidisciplinary first — who else is involved and what the endpoint is · 2 The alveolus never developed, so sites are narrow and often need augmentation · 3 Infra-occluded deciduous teeth are ankylosed and take the bone down with them · 4 The orthodontist's retention plan, and why implants wait for stability · 5 6 mm allows a 3.3 mm implant with 1.5 mm each side · 6 Resin-bonded cantilever bridges — minimally invasive, well evidenced, often better here · 7 Retaining the 75 and 85, and what extraction would reveal · 8 If they are retained, no replacement is needed at all · 9 SAC: Advanced at the lateral sites, Complex overall · 10 Root divergence requested before placement · 11 The retainer redesigned around the implant crowns · 12 Sixty years, crown replacement, and relative infra-occlusion from continued eruption.

Where the marks are lost in this case

Filling the spaces because they were opened. The orthodontist opened **6 mm** at each lateral site, and the candidate who reads that as an instruction to place two implants has skipped the decision. In a **26-year-old**, the **resin-bonded cantilever bridge from the canine is first line** — minimally invasive, well evidenced for single anterior units, reversible in principle, and it **avoids irreversible surgery in a young patient whose options should be preserved**. Implants are the genuine alternative, shown alongside it on a **mock-up**, not the default.

Extracting the 75 and 85 on sight. A **well-retained, minimally infra-occluded deciduous molar in an adult can serve for many years** and is often best **retained, monitored with annual photographs and radiographs, and built up in composite to restore the occlusal plane**. Volunteer the consequence of the alternative — **extraction reveals a vertically deficient site that may need augmentation** — and then

volunteer the conclusion candidates never reach: **if they are retained, no replacement is needed at all, and the decision on those sites can be deferred until they actually fail**.

Talking about the treatment and not the sixty years. She is 26. Say now, not at 45, that **implant crowns will need replacing more than once**, that **continued eruption of the natural dentition relative to an ankylosed implant produces infra-occlusion and open contacts even in adults**, and that this is why she needs **standardised photographs and casts at intervals**. And name the line: this is **Advanced at the lateral sites and Complex overall by virtue of being multidisciplinary and lifelong** — it belongs in a team with the **orthodontist and a restorative specialist**, and the placement of narrow implants into unaugmented hypodontia sites in the aesthetic zone is not solo work.

SOE Case 35 — The patient who will not commit to maintenance

THE CASE

47-year-old sales director who travels internationally 40 weeks a year. Missing **36 and 46**. Medically fit, non-smoker, moderate plaque control, no periodontal disease. He wants implants but says frankly that he "cannot commit to appointments every few months" and is "away too much".

Domain 1

"A medically and dentally suitable candidate whose limiting factor is entirely behavioural and practical: he is telling me, honestly, that he cannot attend maintenance. That honesty is valuable, and I should treat it as clinical information rather than as an obstacle to be talked around."

Q Why does this matter so much?

"Because the strongest predictor of long-term peri-implant health is **supportive maintenance combined with daily plaque control**. Peri-implantitis affects around 22% of patients, and the risk is substantially higher in those without regular supportive care. An implant placed in a patient who never returns is an implant I am likely to be removing in ten years."

Q What would you assess?

"His actual plaque control, disclosed and scored, not his description of it. His periodontal status and BPE. His dexterity and technique. What his travel pattern really is — whether he is home at predictable intervals, which may be perfectly workable. Whether he has a dentist wherever he spends most time. And what he actually wants replaced and why."

Domain 2

Q: How do you approach the conversation?

"Not as a refusal. 'Thank you for being straight with me — that helps. Let me be equally straight. Implants don't decay, but they can get gum disease, and that's the main way people lose them. The single thing that protects them is being seen regularly and cleaning them properly every day. If you genuinely can't be seen every few months, that changes the risk substantially, and it changes what I'd recommend. Let's work out what is actually possible for you.' Then problem-solve: could he be seen twice a year when he is home? Could he see a dentist in the city he most often works in, with a shared care arrangement and my records sent? Could his hygiene be made robust enough with a simpler prosthesis design?"

Q What are the options?

"**No treatment** — a shortened arch to the second premolars, which requires no maintenance beyond his normal dentistry. Given he is missing only the first molars, this is genuinely reasonable and I would present it first.
Removable partial denture — low biological risk if he cleans it, easily replaced.
Bridges from 35–37 and 45–47 if those teeth are sound — but destructive, and they also require maintenance.
Implants with a modified plan — fewer implants, maximally cleansable screw-retained designs, and a documented shared-care maintenance arrangement."

Q Would you place implants?

"Yes, potentially — but only with three things in place: a **realistic, agreed maintenance arrangement**, even if that is twice yearly rather than quarterly, possibly with a second clinician abroad; **demonstrated improvement in his plaque control** at a review before I commit; and **explicit written consent** that his risk of peri-implant disease is higher because of reduced maintenance, with the consequences spelled out. If his plaque score does not improve at review, I would decline and say why."

Q How do you design for a low-maintenance patient?

> "**Screw-retained, always** — retrievable, and no cement. **Wide embrasures and a cleansable emergence profile** that an interdental brush passes through easily. **No deep subgingival margins. Separate crowns rather than splinted units**, so he can clean between them. Adequate **keratinised mucosa**, grafted if necessary, since it makes brushing more comfortable and is associated with less inflammation. And a **tissue-level implant** in a non-aesthetic posterior site, which keeps the microgap supracrestal and simplifies cleaning."

Domain 3

Q: What would you actually do at the placement and restoration stages differently?

> "Nothing exotic — the differences are in design rather than technique. I would ensure at least 2 mm of keratinised mucosa, using an apically positioned flap at second stage if needed. I would use a **custom abutment shaped for cleansability rather than for maximum tissue support**, and I would spend disproportionate time on the hygiene instruction appointment, sizing brushes, watching him use them, and giving written instructions with the sizes noted."

Domain 4

Q: How would you structure his maintenance realistically?

> "Agree a fixed pattern he can actually keep — for example, two long appointments a year timed to when he is home, each including full peri-implant probing, professional biofilm removal and re-instruction, with radiographs at one year and then as indicated. Provide him with a written record and his implant card so that any clinician anywhere can pick up his care. Set a **recall reminder system** that reaches him, and record every failed or cancelled appointment."

Q If he stops attending altogether and returns at five years with peri-implantitis, what then?

> "I would treat him without recrimination, following the decision tree, and I would use the consent record to have an honest conversation about how we got here and what needs to change. Blaming him achieves nothing; but a clear, documented record of what was agreed protects the therapeutic relationship as well as me."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 721 words**. At 140 words a minute plus five seconds a question that is under **six minutes**: it does not overrun, it leaves **nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The design answer alone is five: screw retention, embrasures, margin depth, splinting, keratinised mucosa and the tissue-level implant. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the realistic maintenance schedule and the five-year return — the part of the case where the earlier consent conversation either saves you or does not.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Say what is and is not the problem: **medically and dentally suitable — fit, non-smoker, no periodontal disease, missing only the 36 and 46** — so **the limiting factor is entirely behavioural and practical**. Then the sentence that sets the register for the whole case: **he is telling me honestly that he cannot attend maintenance, and that honesty is valuable — I should treat it as clinical information rather than as an obstacle to be talked around**. Then what you still need: **his real plaque control**, **his actual travel pattern**, and **what he wants replaced and why**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: why maintenance decides this case, then the assessment. Two printed answers that become six	*Why does maintenance matter this much?* — because **the strongest predictor of long-term peri-implant health is supportive maintenance combined with daily plaque control**. *Give me a figure.* — **peri-implantitis affects around 22% of patients**, and the risk is **substantially higher without regular supportive care**. *Say the consequence plainly.* — **an implant placed in a patient who never returns is an implant I am likely to be removing in ten years**. *What would you assess?* broken into four: **his plaque control disclosed and scored, not his description of it**; **periodontal status and BPE**; **dexterity and technique**; and **what his travel pattern really is — whether he is home at predictable intervals, which may be perfectly workable**, and **whether he has a dentist wherever he spends most time**. Expect the examiner to fill the remaining time here with the undergraduate groundwork: how you record a plaque score, what a BPE code obliges you to do, and the distinction between **peri-implant mucositis and peri-implantitis** — because the case rests on it.
4:30–7:30	Domain 2: the conversation, the options, and whether you would proceed — three printed answers that become nine	*How do you open?* — **not as a refusal**: thank him for being straight, then be **equally straight** — *"Implants don't decay, but they can get gum disease, and that's the main way people lose them. If you genuinely can't be seen every few months, that changes the risk substantially, and it changes what I'd recommend." Then what?* — **problem-solve rather than refuse**: **twice a year when he is home**; **a dentist in the city he works in most, with a shared-care arrangement and records sent**; **a simpler prosthesis design that makes hygiene robust**. *Options?* — **no treatment, a shortened arch to the second premolars**, which **requires no maintenance beyond his normal dentistry** and, since he is **missing only the first molars, is genuinely reasonable — presented first**; **a removable partial denture, low biological risk if he cleans it and easily replaced**; **bridges from 35–37 and 45–47 if sound, but destructive and also requiring maintenance**; **implants with a modified plan**. *Would you place them?* — **yes, potentially, but only with three things in place**: a **realistic, agreed maintenance arrangement, even if twice yearly, possibly with a second clinician abroad**; **demonstrated improvement in his plaque control at a review before you commit**; and **explicit written consent that his risk of peri-implant disease is higher because of reduced maintenance, with the consequences spelled out**. *And if the plaque score does not improve?* — **you decline, and you say why** — the answer that separates a 5 from a 3.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: designing for a low-maintenance patient — two printed answers that become seven	*Retention?* — **screw-retained, always: retrievable, and no cement**. *Emergence?* — **wide embrasures and a cleansable emergence profile an interdental brush passes through easily**. *Margins?* — **no deep subgingival margins**. *Splinting?* — **separate crowns rather than splinted units, so he can clean between them**. *Soft tissue?* — **adequate keratinised mucosa, grafted if necessary**, since it **makes brushing more comfortable and is associated with less inflammation — at least 2 mm, using an apically positioned flap at second stage if needed**. *Implant type?* — a **tissue-level implant in a non-aesthetic posterior site**, which **keeps the microgap supracrestal and simplifies cleaning**. *What is different technically?* — and the honest answer scores: **nothing exotic — the differences are in design rather than technique**; a **custom abutment shaped for cleansability rather than for maximum tissue support**, and **disproportionate time on the hygiene appointment — sizing brushes, watching him use them, written instructions with the sizes noted**.
11:30–14:30	Domain 4: the realistic schedule, and the five-year return	*What does maintenance actually look like for him?* — **a fixed pattern he can keep**: **two long appointments a year timed to when he is home**, each with **full peri-implant probing, professional biofilm removal and re-instruction**, **radiographs at one year and then as indicated**. *What travels with him?* — **a written record and his implant card, so any clinician anywhere can pick up his care**. *And the administration?* — a **recall reminder system that reaches him**, and **every failed or cancelled appointment recorded**. *He returns at five years with peri-implantitis — what now?* — **treat him without recrimination, following the decision tree**, and use **the consent record to have an honest conversation about how we got here and what needs to change**. Then the line to say in these words: **blaming him achieves nothing; but a clear, documented record of what was agreed protects the therapeutic relationship as well as me**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	What you say: *"Thank you for being straight with me — that helps. Here's my side of it: the one thing that protects an implant is being seen regularly and cleaning it properly every day. Twice a year when you're home, plus someone I can share your records with in the city you work in, I can work with. What I can't do is put these in and pretend the risk is the same as for someone I see every three months. Let's see your cleaning at a review first, and then decide."* What you write: his **stated travel pattern and attendance capacity in his own words**; the **disclosed plaque score at assessment and at review**; **all options including no treatment and the shortened arch, presented first**; the **agreed maintenance arrangement and any shared-care clinician**; the **explicit written consent that his risk is higher because of reduced maintenance, with the consequences spelled out**; and **every cancelled or failed appointment**.

The spine — the twelve answers to have ready if nothing else

1 The limiting factor is behavioural, and his honesty is clinical information · 2 Supportive maintenance plus daily plaque control is the strongest predictor · 3 Peri-implantitis in around 22% of patients, higher without supportive care · 4 Plaque control disclosed and scored, not described · 5 What his travel pattern actually is — home at predictable intervals may be workable · 6 The shortened arch to the second premolars, offered first and genuinely · 7 The three conditions before you would place · 8 Declining, and saying why, if the plaque score does not improve · 9 Screw-retained, wide embrasures, no deep margins, unsplinted · 10 Keratinised mucosa and the tissue-level implant · 11 Two long appointments a year, shared care, the implant card · 12 The five-year return handled without recrimination, on the consent record.

Where the marks are lost in this case

Talking him round, or turning him away. Both are failures. The candidate who says "I'd explain that maintenance is essential and hope he complies" has not made a decision, and the one who refuses outright has ignored that he is **missing only two first molars and is otherwise fit and disease-free**. The mark is in **problem-solving**: **twice a year when he is home**, a **shared-care clinician in the city he works in with records sent**, and a **design robust enough to survive imperfect attendance** — with the **shortened arch put on the table first and genuinely**.

Consenting without conditions. Volunteer the three things that must be in place before you place anything: a **realistic, agreed maintenance arrangement**, **demonstrated improvement in his plaque control at a review before you commit**, and **explicit written consent that his risk of peri-implant disease is higher because of reduced maintenance, with the consequences spelled out**. Then volunteer the harder half — that **if his plaque score does not improve at review, you decline and tell him why**. Conditional consent that is never tested is not consent.

Designing as though he were an ordinary patient. The competence line in this case is not surgical; a single implant at the 36 or 46 is Straightforward. The line is **whether you will proceed at all with reduced maintenance**, and it is drawn by design as much as by conversation: **screw-retained always, no cement, wide embrasures, no deep subgingival margins, separate crowns rather than splinted units, at least 2 mm of keratinised mucosa grafted if needed, and a tissue-level implant to keep the microgap supracrestal**. A patient you will see twice a year needs a prosthesis that forgives — and if he will not attend at all, **the shortened arch or a well-made denture is the right answer, recorded as a considered decision**.

SOE Case 36 — Failing lower denture in an elderly patient with early dementia

THE CASE

81-year-old woman with **mild Alzheimer's dementia**, living with her daughter. Edentulous for 20 years. Her lower complete denture is loose and she is losing weight. Her daughter asks whether implants would help. She takes donepezil and an antihypertensive; she is otherwise well and mobile.

Domain 1

"An 81-year-old with early dementia, a loose lower denture and weight loss — which makes this a **nutritional and quality-of-life problem, not merely a prosthetic one**. The clinical question of whether a two-implant overdenture would help is straightforward; the harder questions are capacity, the trajectory of her dementia, and who will maintain the prosthesis in five years."

Q What do you need to establish about her dementia?

"Its severity and rate of progression; whether she has **decision-specific capacity** for this decision, assessed by the four-part functional test — can she understand, retain, weigh and communicate a decision about implant treatment; her insight; whether she can cooperate for a surgical appointment; and her manual dexterity and ability to manage a prosthesis. I would seek information from her general medical practitioner and her daughter, while remembering that **the patient, not the daughter, is my patient**."

Q What about the weight loss?

"It must be taken seriously and not assumed to be dental. I would ask about appetite, swallowing, dry mouth, taste changes, mood and depression, and refer to her GP for assessment. Dental causes — an unstable denture, sore spots, reduced masticatory efficiency — may be contributory but are rarely the sole cause. Referral to a dietitian may help regardless of what I do."

Domain 2

Q: What are the options?

"**Optimise the existing prosthesis first**: assess extension, occlusion, vertical dimension and fit; consider a **reline, a copy denture** — which preserves the tooth position and polished surfaces she has adapted to over twenty years — or a remake using neutral zone technique. **Denture adhesive** and dietary advice. This is low-risk, reversible and may be sufficient.
Two-implant mandibular overdenture — the McGill and York consensus position that this is the minimum standard of care for the edentulous mandible, and the intervention with the best evidence for improving satisfaction and, in some studies, nutritional outcomes.
No further intervention with supportive dietary care."

Q How does the dementia change the decision?

"It does not automatically preclude implants — but it changes the calculus in three ways.
Capacity: she must be able to consent. If she lacks capacity, elective implant treatment in her best interests would be very difficult to justify, and I would be extremely cautious.
Trajectory: dementia is progressive. In five years she may be unable to clean around abutments, may not tolerate examination, and may be in residential care. A prosthesis that becomes unmaintainable is a liability — peri-implant disease in a patient who cannot cooperate with treatment is a real problem.
Simplicity: if I proceed, the design must be as simple and cleanable as possible — **two implants with stud attachments, not a bar**, so that a carer can clean around two abutments with a single-tufted brush."

Q What would you recommend?

> "I would **optimise the conventional denture first**, using a copy technique, and see whether that resolves the problem — many do. If it does not, and if she has capacity and her daughter can support daily cleaning, I would offer a **two-implant overdenture with Locator-type attachments** as a modest, well-evidenced intervention, undertaken sooner rather than later while she can still consent and cooperate. I would be explicit that the argument for acting **now rather than in two years** is precisely her dementia — the window for a treatment that requires her cooperation is closing."

Q How do you handle the daughter?

> "With her mother's agreement, involve her fully — she is likely to be the person supporting daily cleaning and attendance. But I would **address my questions and explanations to the patient**, check the patient's own understanding, and ensure the decision is hers while she has capacity. If mother and daughter disagree, the patient's own decision prevails while she has capacity."

Domain 3

Q: Describe the surgical modifications.

> "Short, calm appointments at a time of day when she is at her best — usually the morning. A familiar accompanying person present. Clear, simple explanations repeated as needed. Two implants in the canine regions, at least 5 mm anterior to the mental foramina; a simple mid-crestal incision; a **one-stage protocol with healing abutments** to avoid a second surgery; minimal grafting; and conventional loading at about three months. Careful denture relief and soft relining afterwards."

Domain 4

Q: What is the maintenance plan?

> "Written, simple and shared with her daughter: remove the denture at night; clean around each abutment with a single-tufted brush; clean the housings in the denture fit surface; review every three to six months. I would demonstrate to the daughter as well as to the patient, and give written instructions with pictures. I would record who is providing daily care."

Q When would you decide not to proceed?

> "If she lacked capacity; if she could not tolerate a surgical appointment; if her dementia were progressing rapidly; if there were no reliable daily support; or if optimising the conventional denture solved the problem. Any of those would make a conventional denture the right answer — and I would document the reasoning so that it is clear this was a considered decision, not neglect."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 778 words**. At 140 words a minute plus five seconds a question that is a little over **six minutes**: it does not overrun, it leaves **nearly nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The capacity answer alone is four: the four-part functional test, decision-specific capacity, who you take information from, and whose patient she is. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the carer-delivered maintenance plan and the decision not to proceed — the two easiest marks in the case to earn.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	Reframe the referral question in the first clause: an 81-year-old with **early Alzheimer's dementia**, a loose lower complete denture and **weight loss** — which makes this a **nutritional and quality-of-life problem, not merely a prosthetic one**. Then the honest split: **the clinical question of whether a two-implant overdenture would help is straightforward; the harder questions are capacity, the trajectory of her dementia, and who will maintain the prosthesis in five years**. And the sentence to plant early because it governs everything: the daughter asked the question, but **the patient, not the daughter, is my patient**.
1:20–4:30	Domain 1: capacity, then the weight loss. Two printed answers that become seven	*What do you need to know about the dementia?* — **its severity and rate of progression**, her **insight**, whether she can **cooperate for a surgical appointment**, and her **manual dexterity and ability to manage a prosthesis**. *And capacity?* — **decision-specific capacity**, assessed by the **four-part functional test: can she understand, retain, weigh and communicate a decision about implant treatment**. *Where do you get the information?* — from her **general medical practitioner and her daughter**, while remembering **whose patient she is**. *And the weight loss?* — the answer that scores by refusing the easy attribution: **take it seriously and do not assume it is dental**. *So what do you ask?* — **appetite, swallowing, dry mouth, taste changes, mood and depression**. *And do?* — **refer to her GP for assessment**, and consider **a dietitian regardless of what you do**, because **dental causes may be contributory but are rarely the sole cause**. Expect the examiner to fill the rest of this domain with the undergraduate groundwork the case assumes: **why a lower complete denture loses stability over twenty years of residual ridge resorption**, the **landmarks that give it support**, and **support, retention and stability defined separately**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the options, and how the dementia changes them — three printed answers that become nine	*Options?* — **optimise the existing prosthesis first**: assess **extension, occlusion, vertical dimension and fit**, and consider a **reline, a copy denture — which preserves the tooth position and polished surfaces she has adapted to over twenty years — or a remake using neutral zone technique**, with **adhesive and dietary advice**; this is **low-risk, reversible and may be sufficient**. Then the **two-implant mandibular overdenture**, with **McGill and York** quoted in a clause as **the minimum standard of care for the edentulous mandible** and the evidence for **satisfaction and, in some studies, nutritional outcomes**. Then **no further intervention with supportive dietary care**. *How does the dementia change it?* — **not automatically a preclusion**, but in **three named ways**. *Capacity:* she must be able to consent, and **if she lacks capacity, elective implant treatment in her best interests would be very difficult to justify**. *Trajectory:* **dementia is progressive** — in five years she may be **unable to clean around abutments, unable to tolerate examination, and in residential care**, and **peri-implant disease in a patient who cannot cooperate with treatment is a real problem**. *Simplicity:* **two implants with stud attachments, not a bar**, so **a carer can clean around two abutments with a single-tufted brush**. *What do you recommend?* — **optimise the conventional denture first, using a copy technique**, and if that fails and she has capacity and daily support, offer a **two-implant Locator overdenture** — with the argument stated explicitly: **the reason to act now rather than in two years is precisely her dementia, because the window for a treatment that requires her cooperation is closing**. *And the daughter?* — **involve her fully with her mother's agreement**, since she is likely to be **supporting daily cleaning and attendance**, but **address the questions and explanations to the patient**, check **the patient's own understanding**, and **if mother and daughter disagree, the patient's decision prevails while she has capacity**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the surgical modifications — one printed answer, and it is really six	*When do you see her?* — **short, calm appointments at a time of day when she is at her best, usually the morning**, with **a familiar accompanying person present** and **clear, simple explanations repeated as needed**. *Where do the implants go?* — **two in the canine regions, at least 5 mm anterior to the mental foramina**. *Flap?* — a **simple mid-crestal incision**, **minimal grafting**. *One stage or two?* — **one-stage with healing abutments, to avoid a second surgery** — and say why, because avoiding a second operation in this patient is the whole point. *Loading?* — **conventional at about three months**. *Afterwards?* — **careful denture relief and soft relining**. Expect the examiner to spend the spare minutes here on the prosthetic detail the printed answer omits — **Locator versus bar and the vertical space each needs**, **acrylic thickness over the housing**, and **why attachment wear and relines are inevitable**.
11:30–14:30	Domain 4: maintenance by somebody else, and when not to proceed — two printed answers that become six	*What is the maintenance plan?* — **written, simple and shared with her daughter**: **remove the denture at night**; **clean around each abutment with a single-tufted brush**; **clean the housings in the denture fit surface**; **review every three to six months**. *How is it taught?* — **demonstrated to the daughter as well as to the patient**, with **written instructions with pictures**. *What is recorded?* — **who is providing daily care** — the detail candidates omit and the one that makes the plan real. *When would you decide not to proceed?* — five conditions, given as a list and not hedged: **if she lacked capacity; if she could not tolerate a surgical appointment; if her dementia were progressing rapidly; if there were no reliable daily support; or if optimising the conventional denture solved the problem**. *And then?* — **a conventional denture is the right answer, and you document the reasoning so that it is clear this was a considered decision, not neglect**.
14:30–15:00	Closes on the patient and the record	What you say — **to her, with her daughter present**: *"Your bottom denture is loose, and I think that's part of why eating has become hard work. The first thing I'd like to do is make you a new one copied from this one, so it feels familiar — that alone helps a lot of people. If it doesn't, two small implants would hold it in, and it's a straightforward operation. I'd want to do that sooner rather than later while it's easier for you. And I'm going to ask your doctor to look at the weight loss too, because that may not all be down to the denture."* What you write: the **capacity assessment against the four-part test and its date**; that **explanations were addressed to the patient and her understanding checked**; the **daughter's involvement with the patient's agreement**; the **GP referral for the weight loss**; the option discussion **including doing nothing**; **who provides daily cleaning**; and, if you decline, **the reasoning in full**.

The spine — the twelve answers to have ready if nothing else

1 A nutritional and quality-of-life problem, not merely a prosthetic one · 2 Decision-specific capacity and the four-part functional test · 3 The patient, not the daughter, is my patient · 4 The weight loss referred to the GP, not assumed to be dental · 5 Optimise the conventional denture first — copy technique, neutral zone, soft liner · 6 McGill and York in one clause · 7 The three ways dementia changes the calculus: capacity, trajectory, simplicity · 8 Two studs, not a bar, so a carer can clean it · 9 Why acting now rather than in two years is the argument · 10 One-stage with healing abutments, canine regions, 5 mm anterior to the foramina · 11 The written carer maintenance plan, and recording who delivers it · 12 The five circumstances in which you would not proceed.

Where the marks are lost in this case

Answering the daughter's question. The daughter asked whether implants would help, and the candidate who answers her has already lost the domain. **Address the explanations to the patient, check her own understanding, and say that if mother and daughter disagree the patient's decision prevails while she has capacity.** Then give the framework by name: **decision-specific capacity**, assessed on the **four-part functional test — understand, retain, weigh, communicate** — and the consequence, that **if she lacks capacity, elective implant treatment in her best interests would be very difficult to justify**.

Treating the weight loss as a denture problem. It is the most attractive trap in the case, because it makes the implants look indicated. Volunteer that **weight loss must not be assumed to be dental**: ask about **appetite, swallowing, dry mouth, taste changes, mood and depression**, **refer to her GP**, and consider **a dietitian regardless of what you do**. Then volunteer the first-line dental answer anyway — **optimise the existing denture with a copy technique, which preserves twenty years of adaptation** — because **many are resolved by that alone**.

Planning for today and not for five years. Dementia is progressive, and the prosthesis must be designed for the person she will be: **two implants with stud attachments, not a bar**, so **a carer can clean two abutments with a single-tufted brush**; a **written plan with pictures, demonstrated to the daughter**; and **who provides daily care recorded in the notes**. Then name the line in both directions — the argument for **acting now while she can still consent and cooperate**, and the five circumstances in which **a conventional denture is the right answer**, documented **so that it is clear this was a considered decision, not neglect**.

SOE Case 37 — The severely atrophic mandible

THE CASE

74-year-old man, edentulous for 30 years, wearing complete dentures. He cannot tolerate the lower denture. Panoramic radiograph and CBCT show a **severely atrophic mandible of 7 mm height in the symphysis**, with the mandibular canal at the crest posteriorly. He is medically fit.

Domain 1

"A severely atrophic mandible — 7 mm of height in the anterior region and a posteriorly superficial canal — in a patient with genuine functional difficulty. This is a **Complex** case in which the principal risks are **iatrogenic mandibular fracture and nerve injury**, and in which the temptation to help must be balanced against the potential to do serious harm."

Q How would you classify this ridge?

"On the Lekholm and Zarb quantity classification this is **Class D or E** — advanced resorption of basal bone. Cawood and Howell would classify it as Class V or VI. The key figures are the residual height and the position of the canal, and 7 mm in the symphysis is at the threshold where standard implant placement becomes hazardous."

Q What else would you assess?

"The existing dentures critically — extension, occlusion, vertical dimension, polished surface form, and whether the neutral zone has been respected, because a poorly made denture on an atrophic ridge is a common and correctable cause of intolerance. Floor of mouth height relative to the crest, mentalis and mylohyoid attachments, tongue position and size, saliva, mucosal quality, and whether there is denture-induced hyperplasia. And what he actually finds intolerable — pain, looseness, or both."

Domain 2

Q: What are the options?

"**Remake the conventional denture** using a **neutral zone technique** and careful attention to extension, with a soft liner. On a severely atrophic ridge this is frequently the single most effective intervention and it carries no surgical risk. I would do this first regardless.
Two-implant overdenture in the symphysis — the standard-of-care option, but with 7 mm of height it carries a genuine risk of fracture and would require short implants and careful case selection.
Four implants with a bar, distributing load — more surgery, more risk.
Augmentation before implants — vertical augmentation of an atrophic mandible is Complex and unpredictable.
Specialist options — nerve lateralisation, or extraoral bone grafting — firmly outside my scope.
Continue with conventional dentures and adhesive, with dietary support."

Q What would you recommend?

"**Remake the lower denture properly first**, with a neutral zone technique and a soft liner, and reassess. If he remains intolerant and wants to proceed, I would **refer for a specialist opinion** rather than place implants myself, because with 7 mm of height the risk of mandibular fracture is real, the case is Complex, and I would want it assessed by someone who manages atrophic mandibles regularly. I would explain that clearly and not present referral as abandonment."

Q Why is mandibular fracture such a concern?

"Because in a mandible of this height, an osteotomy removes a substantial proportion of the remaining bone, and the atrophic mandible has poor vascularity and poor healing capacity. A fracture in this situation is a serious

injury, often requiring open reduction and internal fixation, sometimes with non-union. Patients can be warned, but the risk is best avoided by not creating it."

Q What would you consent for if implants were placed?

"Beyond the standard risks: **mandibular fracture, both during surgery and afterwards under function**, and what its management would involve; **mental and inferior alveolar nerve injury**, with the canal superficial posteriorly; **failure to integrate** in poorly vascularised atrophic bone; and the maintenance burden of an overdenture. I would give this in writing."

Domain 3

Q: If a specialist placed two short implants, what would your restorative plan be?

"**Stud attachments rather than a bar** if the implants are parallel and space permits, because a bar demands 12 to 15 mm of vertical space and adds a further stress concentration; but with divergent implants in atrophic bone a bar may distribute load better and I would take the surgeon's view. The overdenture must be **implant-retained and tissue-supported**, with full extension so the posterior saddles continue to carry load — an implant-supported design would concentrate all the load on two short implants in fragile bone. Conventional loading, careful occlusal balance, and a soft liner if needed."

Domain 4

Q: What are you watching for at review?

"Any sign of mandibular fracture — pain, altered occlusion, mobility; peri-implant health; attachment wear; and continued resorption requiring relines. I would review him more frequently than usual in the first year."

Q What is the lesson of this case?

"That the best intervention is often the least invasive one, and that **recognising a case as beyond one's competence is a clinical skill in its own right**. A well-made conventional denture in a severely atrophic mandible is not a failure of ambition — it may be the safest and most effective thing I can offer."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 687 words**. At 140 words a minute plus five seconds a question that is under **six minutes**: it does not overrun, it leaves **over nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The options answer alone is six, one per option, each with its own follow-up on risk. If your answers run past **thirty seconds** you will still never reach Domain 4 — and Domain 4 here is one line, that **recognising a case as beyond your competence is a clinical skill in its own right**, which is the mark the whole case is built to award.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the panoramic and CBCT up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	The two numbers first: **7 mm of height in the symphysis** and **the mandibular canal at the crest posteriorly**, in a man edentulous for thirty years with **genuine functional difficulty**. Then the classification and the risks in one clause: this is a **Complex** case whose principal risks are **iatrogenic mandibular fracture and nerve injury**. Then the sentence that governs the whole fifteen minutes: **the temptation to help must be balanced against the potential to do serious harm**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: classification, then what else you assess. Two printed answers that become seven	*Classify the ridge.* — **Lekholm and Zarb Class D or E — advanced resorption of basal bone**; **Cawood and Howell Class V or VI**. *What are the key figures?* — **residual height and the position of the canal**, and **7 mm in the symphysis is at the threshold where standard implant placement becomes hazardous**. *What else would you assess?* broken into four: **the existing dentures critically — extension, occlusion, vertical dimension, polished surface form, and whether the neutral zone has been respected**, because **a poorly made denture on an atrophic ridge is a common and correctable cause of intolerance**; **floor of mouth height relative to the crest, mentalis and mylohyoid attachments, tongue position and size**; **saliva, mucosal quality, and denture-induced hyperplasia**; and the question candidates forget to ask — **what he actually finds intolerable: pain, looseness, or both**. Expect the spare minutes to be filled with the undergraduate groundwork underneath: **why residual ridge resorption is progressive and greater in the mandible**, **support, retention and stability defined separately**, and **the landmarks that give a lower denture its support**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the options, the recommendation, and why fracture is the whole argument — three printed answers that become ten	*Options?* — one question each. **Remake the conventional denture using a neutral zone technique with careful extension and a soft liner — frequently the single most effective intervention on a severely atrophic ridge, and it carries no surgical risk**. **Two-implant overdenture in the symphysis** — the standard-of-care option, **but with 7 mm of height it carries a genuine risk of fracture and would require short implants and careful case selection**. **Four implants with a bar — more surgery, more risk**. **Augmentation before implants — vertical augmentation of an atrophic mandible is Complex and unpredictable**. **Nerve lateralisation or extraoral bone grafting — firmly outside my scope**, and say it in those words. **Conventional dentures and adhesive with dietary support.** *What do you recommend?* — **remake the lower denture properly first, with a neutral zone technique and a soft liner, and reassess — do this first regardless**. *And if he remains intolerant?* — **refer for a specialist opinion rather than place implants yourself**, because **with 7 mm the risk of mandibular fracture is real, the case is Complex, and it should be assessed by someone who manages atrophic mandibles regularly — explained clearly and not presented as abandonment**. *Why is fracture such a concern?* — because **an osteotomy removes a substantial proportion of the remaining bone**, and **the atrophic mandible has poor vascularity and poor healing capacity**; a fracture here is **a serious injury, often requiring open reduction and internal fixation, sometimes with non-union**, and **patients can be warned, but the risk is best avoided by not creating it**. *What would you consent for?* — beyond the standard risks: **mandibular fracture, both during surgery and afterwards under function, and what its management would involve**; **mental and inferior alveolar nerve injury, with the canal superficial posteriorly**; **failure to integrate in poorly vascularised atrophic bone**; and **the maintenance burden of an overdenture — in writing**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: your restorative plan if a specialist places — one printed answer, and it is really six	*Studs or a bar?* — **stud attachments if the implants are parallel and space permits**, because **a bar demands 12 to 15 mm of vertical space and adds a further stress concentration**; **but with divergent implants in atrophic bone a bar may distribute load better, and I would take the surgeon's view** — the deference is itself part of the answer. *How is the overdenture supported?* — **implant-retained and tissue-supported, with full extension so the posterior saddles continue to carry load**, because **an implant-supported design would concentrate all the load on two short implants in fragile bone**. *Loading?* — **conventional**. *Occlusion?* — **careful balance**. *Anything else?* — **a soft liner if needed**. Expect the spare minutes here to go on the prosthodontic detail: **acrylic thickness over the housing on a thin denture base**, **denture fracture risk in an overdenture on an atrophic ridge**, and **attachment wear and the reline interval**.
11:30–14:30	Domain 4: what you watch for, and the lesson — two printed answers that become five	*What are you watching for at review?* — **any sign of mandibular fracture: pain, altered occlusion, mobility** — named first because it is the one that matters — then **peri-implant health**, **attachment wear**, and **continued resorption requiring relines**, with **more frequent review than usual in the first year**. *And if the denture keeps fracturing?* — the load has nowhere to go; reassess extension, occlusion and base thickness before adding anything. *What is the lesson?* — **that the best intervention is often the least invasive one**, and that **recognising a case as beyond one's competence is a clinical skill in its own right**. *Say the corollary.* — **a well-made conventional denture in a severely atrophic mandible is not a failure of ambition; it may be the safest and most effective thing I can offer.**
14:30–15:00	Closes on the patient and the record	What you say: *"Your lower jaw has shrunk a long way over thirty years — there's about seven millimetres of bone left at the front, and the nerve runs near the surface at the back. Implants can be done in a jaw like yours, but the risk of breaking the jaw is real, so that's not something I'd take on myself. What I'd like to do first is make you a properly moulded lower denture with a soft lining, because on ridges like yours that often makes more difference than people expect. If it doesn't, I'll refer you to someone who does these jaws all the time — that's not me giving up on you."* What you write: the **measured residual height and the canal position**; **the assessment of the existing dentures against extension, occlusion, vertical dimension and neutral zone**; **all options including the ones outside your scope, named as such**; **the fracture and nerve injury risks discussed, and that consent would be given in writing**; **the referral, its reason and to whom**; and the **decision to remake conventionally first, with the reasoning**.

The spine — the twelve answers to have ready if nothing else

1 7 mm in the symphysis and a crestal canal — the two numbers, said first · 2 Lekholm and Zarb D or E; Cawood and Howell V or VI · 3 The existing denture assessed critically — a correctable cause of intolerance · 4 The neutral zone remake with a soft liner, done first regardless · 5 Two implants in the symphysis, and why 7 mm makes it hazardous · 6 Why mandibular fracture is the governing risk — bone volume, vascularity, non-union · 7 Nerve lateralisation and extraoral grafting, named as outside your scope · 8 Refer for a specialist opinion, and say it is not abandonment · 9 The consent list: fracture, nerve injury, failure to integrate, maintenance · 10 Studs versus bar, and taking the surgeon's view · 11 Implant-retained and tissue-supported, fully extended · 12 Recognising a case as beyond your competence is a clinical skill.

Where the marks are lost in this case

Placing implants because he asked. This is the case that examines restraint. With **7 mm of symphysis-seal height** an osteotomy removes **a substantial proportion of the remaining bone** in a jaw with **poor vascularity and poor healing capacity**, and a fracture there is **a serious injury, often requiring open reduction and internal fixation, sometimes with non-union**. Say the line the case gives you: **patients can be warned, but the risk is best avoided by not creating it**.

Skipping the denture. The single most effective intervention on a severely atrophic ridge is frequently **a properly made conventional denture**, and it carries **no surgical risk**. Volunteer the assessment before the options — **extension, occlusion, vertical dimension, polished surface form, and whether the neutral zone has been respected** — and volunteer the technique by name: **a neutral zone remake with a soft liner, done first regardless of what follows**. Then ask the question candidates never ask: **what does he actually find intolerable — pain, looseness, or both?**

Never naming the referral, or naming it as defeat. The competence line here is explicit: **assessment, the conventional remake and the restorative plan are yours; placement in a 7 mm atrophic mandible is not**, and **nerve lateralisation and extraoral bone grafting are firmly outside your scope**. Refer for **a specialist opinion from someone who manages atrophic mandibles regularly**, explain it clearly, and **do not present it as abandonment** — then say the closing sentence in these words: **a well-made conventional denture in a severely atrophic mandible is not a failure of ambition; it may be the safest and most effective thing you can offer.**

SOE Case 38 — Failure to integrate in a previously grafted site

THE CASE

You placed an implant at the **UR5** six months after a staged guided bone regeneration procedure. At the 4-month review the implant is **mobile**, with a peri-implant radiolucency. The patient is a 55-year-old non-smoker with no medical history.

Domain 1

"An implant that has failed to integrate in a previously grafted site. The immediate management is clear — the implant must come out. The valuable part of this case is working out **why**, because a failure in grafted bone has a different differential from a failure in native bone."

Q What are the possible causes?

"**Graft-related:** the regenerated bone was immature or of low density, with a lower bone-to-implant contact and slower remodelling than native bone; the healing period after the graft may have been too short — staged GBR usually requires **6 to 9 months**; or the graft may have been largely residual xenograft particles with little vital bone.
Surgical: inadequate primary stability at placement; overheating; contamination.
Loading: micromotion from a denture pressing on the site.
Patient: an undisclosed factor — smoking, undiagnosed diabetes, medication.
Site: residual infection from the original pathology or a retained membrane fragment."

Q How would you investigate?

"Review my own operative records: the healing period allowed after grafting, the insertion torque recorded, the drilling protocol, whether primary stability was documented, and whether a denture was worn. Re-take the medical history including a specific question about smoking and any new diagnoses. Take a periapical, and consider a **biopsy or histological assessment of the removed tissue** if there is any suggestion of unusual pathology."

Domain 2

Q: What do you tell the patient?

"**Duty of candour.** Promptly, honestly and without excessive self-criticism: 'I'm sorry — the implant hasn't attached to the bone. That's more likely in an area where we've had to rebuild bone first, and it happens in a small proportion of cases even when everything is done properly. It has to come out. Once the area has healed we can look at whether to try again, and I need to be honest that a second attempt in a site that's already needed grafting carries a higher risk than the first.' I would then set out the alternatives afresh, in case she now prefers a bridge or a denture."

Q What about the fee?

"That should have been agreed **in writing before treatment** — my own position would be to replace the implant without charging for the fixture and the surgery, while being clear about laboratory and component costs. What matters is that the arrangement was documented in advance rather than negotiated now."

Q What is your plan?

"**Remove** the implant — with a mobile implant this is usually simple. **Curette all granulation tissue** thoroughly and inspect the bony walls; send tissue for histology if anything is unexpected. Assess the residual defect. If contained, graft with a composite graft and a membrane; if the walls are intact, allow spontaneous healing. Provide an interim prosthesis that does not press on the site — a **bonded pontic** in preference to a denture. Re-entry at **4 to 6 months**, since this is a grafted site rather than a simple socket, with reassessment by CBCT. At replacement, use a **wider and/or longer implant** to engage the maximum available bone, take particular care

with drill sharpness and irrigation, and use a **two-stage submerged protocol with an extended healing period**. I would also consider whether the right answer is **not** to re-implant — a resin-bonded bridge in this premolar site may be the more predictable option after a failure."

Domain 3

Q: What would you do differently at the second attempt?

"Allow a **longer healing period after grafting** — closer to 9 months than 6. **Document insertion torque and ISQ** at placement, and use serial ISQ during healing so that a falling trend gives early warning. **Under-prepare** the osteotomy to maximise primary stability in low-density regenerated bone. **Submerge** the implant. **Avoid any denture pressure** on the site. And extend the healing period before loading."

Domain 4

Q: What governance actions follow?

"Treat it as a **significant event**. Review my grafting protocol, my healing intervals, my drill replacement policy and my irrigation. Then **audit** my implant survival, separating implants in native bone from those in grafted sites, against a standard drawn from the literature, and re-audit after any change. That would be a legitimate quality improvement project. I would also inform my indemnity provider if the patient raised a concern, and record everything contemporaneously."

Q How would you decide whether to keep doing GBR at all?

"By looking at my own outcomes honestly. If my grafted-site failure rate were materially higher than published figures, the right responses would be more mentoring, a narrower case selection, or referring augmentation cases — not doing more of the same and hoping. **Working within one's competence includes knowing one's own results.**"

Running this case to time

Time check. This case is written in the older format — **eight long answers, about 704 words**. At 140 words a minute plus five seconds a question that is under **six minutes**: it does not overrun, it leaves **over nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The causes answer alone is five, one per group, and the plan answer is six. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the audit that separates grafted from native sites — the easiest mark in the case to earn and the easiest to miss.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapical up	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Name it in the first clause: an implant that has **failed to integrate in a previously grafted site** — mobile at four months with a peri-implant radiolucency, so **the diagnosis is not in doubt and the immediate management is clear: it must come out**. Then the sentence that tells the examiner you have understood which case this is: **the valuable part is working out why, because a failure in grafted bone has a different differential from a failure in native bone**. Then the fact you must own: **you placed it**, and the record you will interrogate is your own.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the causes by group, then the investigation. Two printed answers that become eight	*Graft-related causes?* — and this group comes first because it is what makes the case different: **the regenerated bone was immature or of low density, with lower bone-to-implant contact and slower remodelling than native bone**; **the healing period after grafting may have been too short — staged GBR usually requires 6 to 9 months**; or **the graft may have been largely residual xenograft particles with little vital bone**. *Surgical?* — **inadequate primary stability at placement, overheating, contamination**. *Loading?* — **micromotion from a denture pressing on the site**. *Patient?* — **an undisclosed factor: smoking, undiagnosed diabetes, medication**. *Site?* — **residual infection from the original pathology, or a retained membrane fragment**. *How would you investigate?* — **review your own operative records first**: **the healing period allowed after grafting, the insertion torque recorded, the drilling protocol, whether primary stability was documented, and whether a denture was worn**. *And the patient?* — **re-take the medical history with a specific question about smoking and any new diagnoses**. *Anything else?* — **a periapical, and consider biopsy or histological assessment of the removed tissue if there is any suggestion of unusual pathology**. Expect the spare minutes to be filled below the implant material: **osseointegration and how bone heals round a fixture**, and **primary versus secondary stability with the dip between them** — the exact window this implant has fallen into.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: candour, money, and the plan — three printed answers that become nine	*What do you tell her?* — **duty of candour, named as a duty**: **promptly, honestly and without excessive self-criticism**, and then the actual words — *"I'm sorry — the implant hasn't attached to the bone. That's more likely in an area where we've had to rebuild bone first, and it happens in a small proportion of cases even when everything is done properly. It has to come out." And the second attempt?* — said, not hidden: **a second attempt in a site that has already needed grafting carries a higher risk than the first**. *Then what?* — **set out the alternatives afresh, in case she now prefers a bridge or a denture**. *The fee?* — **agreed in writing before treatment**; the stated position is **to replace the implant without charging for the fixture and the surgery, while being clear about laboratory and component costs** — and the mark is that **the arrangement was documented in advance rather than negotiated now**. *The plan?* — **remove**, usually simple in a mobile implant; **curette all granulation tissue thoroughly and inspect the bony walls**, with **histology if anything is unexpected**; **assess the residual defect — contained takes a composite graft and a membrane; intact walls can heal spontaneously**; an interim prosthesis that **does not press on the site — a bonded pontic in preference to a denture**. *When do you re-enter?* — **4 to 6 months, since this is a grafted site rather than a simple socket**, with **CBCT reassessment**. *And at replacement?* — **a wider and/or longer implant to engage the maximum available bone**, **particular care with drill sharpness and irrigation**, and **a two-stage submerged protocol with an extended healing period**. *Is re-implanting even right?* — the answer candidates skip: **consider that it may not be — a resin-bonded bridge in this premolar site may be the more predictable option after a failure**.
7:30–11:30	Domain 3: what changes at the second attempt — one printed answer, and it is really six	*Healing period?* — **closer to 9 months than 6 after grafting**. *Measurement?* — **document insertion torque and ISQ at placement, and use serial ISQ during healing so that a falling trend gives early warning** — the single most useful change and the one candidates never volunteer. *Osteotomy?* — **under-prepare, to maximise primary stability in low-density regenerated bone**. *Healing protocol?* — **submerge the implant**. *Prosthesis?* — **avoid any denture pressure on the site**. *Loading?* — **extend the healing period before loading**. Expect the spare minutes to go on the practical detail: **which graft material and membrane, and why a composite graft in a contained defect**; **how you judge containment at re-entry**; and **what a non-contained or vertical defect would make you do instead**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: governance, and knowing your own results — two printed answers that become five	*What follows?* — treat it as a **significant event**, and say the words. *What do you review?* — **your grafting protocol, your healing intervals, your drill replacement policy and your irrigation**. *And the audit?* — the answer the case is built to reward: **audit your implant survival, separating implants in native bone from those in grafted sites**, against **a standard drawn from the literature**, with **re-audit after any change** — offered as **a legitimate quality improvement project**. *Anything else?* — **inform your indemnity provider if she raised a concern**, and **record everything contemporaneously**. *How would you decide whether to keep doing GBR at all?* — **by looking at your own outcomes honestly**: if your grafted-site failure rate were **materially higher than published figures**, the responses are **more mentoring, a narrower case selection, or referring augmentation cases — not doing more of the same and hoping**. Then the closing line, said in these words: **working within one's competence includes knowing one's own results**.
14:30–15:00	Closes on the patient and the record	What you say: *"I'm sorry — the implant hasn't attached to the bone, and it has to come out. That's more likely in an area where we had to rebuild the bone first, and it does happen even when everything is done properly. Once the site has healed we can look at trying again, and I have to be honest that a second attempt there carries a higher risk than the first — so I also want to go back over the alternatives with you, because a bridge may now be the more predictable answer."* What you write: the **candour conversation, its date and the words used**; the **failure rate and the higher second-attempt risk as stated to her**; the **alternatives re-offered**; the **charging arrangement as previously agreed in writing**; the **operative findings at removal, the state of the bony walls, and any histology**; and the **significant event logged with the systems review and the audit that follows**.

The spine — the twelve answers to have ready if nothing else

1 Failure to integrate in grafted bone — a different differential from native bone · 2 The graft-related causes, named first · 3 6 to 9 months for staged GBR, and what a short healing period does · 4 The surgical, loading, patient and site groups · 5 Your own operative record read before the patient's habits · 6 Duty of candour, and the actual words · 7 The higher risk of a second attempt at the same site, volunteered · 8 The fee agreed in writing before treatment, not negotiated now · 9 Removal, curettage, and what the bony walls decide · 10 Re-entry at 4 to 6 months with CBCT, submerged, wider or longer · 11 Insertion torque and serial ISQ as early warning · 12 The audit that separates grafted sites from native ones.

Where the marks are lost in this case

Giving the generic failure list. This implant failed in **regenerated bone**, and the examiner has specified that on purpose. Lead with the **graft-related group — immature or low-density regenerated bone with lower bone-to-implant contact and slower remodelling, a healing period shorter than the 6 to 9 months staged GBR usually requires**, and **a graft that may have been largely residual xenograft particles with little vital bone** — and only then give the surgical, loading, patient and site groups. A list that could have been recited for any failure earns a 3.

Reaching for the patient before your own record. You placed this implant. The professional instinct being marked is to **review your own operative records first — the healing period allowed after grafting, the**

insertion torque, the drilling protocol, whether primary stability was documented, whether a denture was worn — before asking about her smoking. Then volunteer the two things candidates never do: that **the right answer may be not to re-implant at all**, since **a resin-bonded bridge in this premolar site may be more predictable after a failure**; and that at any second attempt you would **document insertion torque and use serial ISQ during healing so a falling trend gives early warning**.

Stopping at the re-treatment. Domain 4 is the cheapest domain in the case and the one most candidates never reach. Say **significant event**, review **grafting protocol, healing intervals, drill replacement policy and irrigation**, and then **audit your implant survival with grafted sites separated from native ones against a published standard, re-auditing after any change**. And name the competence line the case names for you: if your grafted-site failure rate is **materially higher than published figures**, the answer is **mentoring, narrower case selection, or referring augmentation cases** — because **working within one's competence includes knowing one's own results**.

SOE Case 39 — Recurrent ceramic fracture on a screw-retained bridge

THE CASE

56-year-old male with a **three-unit screw-retained implant bridge at the 45–47** on two implants (45, 47), placed 4 years ago in layered zirconia. The **veneering ceramic has fractured three times**. He has no pain and the implants are healthy. He grinds his teeth and has an intact, heavily worn natural dentition elsewhere.

Domain 1

"A recurring technical complication — chipping of veneering ceramic — on an otherwise healthy implant bridge in a parafunctional patient. Three fractures in four years is not bad luck; it is a **design and load mismatch** that will continue until I change something."

Q Why does veneering ceramic chip on implant restorations?

"Several reasons, and usually more than one: **absence of a periodontal ligament**, so the load is transmitted without the damping and proprioceptive protection a tooth would give; **parafunctional loading**; **inadequate or uneven framework support** for the veneer — the ceramic must be supported in even thickness, and unsupported ceramic over a poorly designed connector or a flat framework is where fractures start; **residual thermal stresses** from an inappropriate cooling protocol during firing; and **occlusal interferences**, particularly in excursion."

Q What would you examine?

"The occlusion in detail: contacts in intercuspation with shimstock and articulating paper of two thicknesses, and **all excursive movements including non-working interferences**. Whether the implant bridge is contacting more heavily than the adjacent natural teeth. The **framework design** on a radiograph and, if the bridge is removed, directly — is the veneer evenly supported? Whether there is a **cantilever**. Whether the screws have loosened previously, which would indicate overload. Signs of parafunction — masseteric hypertrophy, linea alba, wear facets, and whether he has ever had or worn a splint."

Domain 2

Q: What are the options?

"**Repair with composite** — a temporary measure at best, and it has already failed as a strategy.
Polish and monitor — appropriate only for a very minor chip.
Remake in monolithic zirconia — removing the weak veneering layer entirely. This is the definitive answer for a parafunctional patient and is what I would recommend.
Remake in metal-ceramic with reduced veneer or a metal occlusal surface — an alternative, less aesthetic but very robust.
Redesign — reconsider the span, connector dimensions and whether a cantilever is present."

Q What is your plan?

"**Correct the cause before remaking**, otherwise I will remake it again. 1. Adjust the occlusion so the bridge holds shimstock only under heavy clenching and has **no excursive contacts**, with guidance on the natural canine. 2. Confirm the implants and screws are sound; replace the screws with new ones and torque correctly, re-torquing at ten minutes. 3. **Remake in monolithic zirconia**, screw-retained, with a narrow occlusal table, reduced cusp inclination, generous connector dimensions and no cantilever. 4. Provide a **hard, full-coverage occlusal splint** for night-time wear, adjusted after the new bridge is fitted, and review its wear as an objective measure of his parafunction. 5. Discuss the contributing factors — stress, sleep, caffeine, alcohol — and **screen for obstructive sleep apnoea**, referring to his GP if there are symptoms, since sleep bruxism and OSA are associated."

Q How would you discuss the cost?

> "Honestly and early. I would establish what was originally agreed, explain that the material choice was reasonable at the time but has not withstood his bite, and set out what a remake in monolithic zirconia would cost. I would also make clear that even monolithic zirconia is not immune — he generates forces beyond what restorations are designed for — and that the splint is **not optional**. I would put the maintenance expectation in writing so future component replacement is not a surprise."

Domain 3

Q: What would you specify to the laboratory?

> "Monolithic zirconia, screw-retained, with the implant system, platform and connection specified; connector dimensions adequate for the span; a narrow occlusal table with reduced cusp inclination; occlusal contacts light in intercuspation and absent in excursion; a cleansable emergence with open embrasures; and a request for a bisque or design preview before final milling. I would also send photographs and my occlusal instructions, and ask the technician to advise if he sees a design problem — a good technician will often identify the reason a restoration keeps failing."

Q How would you adjust and finish the zirconia occlusally?

> "With fine diamonds under copious water, then **polish thoroughly with a dedicated zirconia polishing system**. An adjusted but unpolished zirconia surface causes rapid wear of the opposing dentition, which in a bruxist with an already worn dentition would be a serious harm."

Domain 4

Q: How would you monitor?

> "Six-monthly review with occlusal assessment, screw integrity checks, examination of the splint for wear as a proxy for his parafunctional activity, and photographs of the restoration. Radiographs at one year and then as clinically indicated. I would record the torque values and dates of any component replacement, so I can see whether the pattern is improving."

Q What would you audit?

> "The incidence of technical complications in my own implant restorations, separated by material and by whether the patient has diagnosed parafunction. If layered restorations in bruxists are failing in my hands, the change is to specify monolithic materials for that group as a matter of protocol — and re-audit."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 748 words**. At 140 words a minute plus five seconds a question that is a little over **six minutes**: it does not overrun, it leaves **nearly nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The five reasons ceramic chips are five questions on their own, and the plan is five more, one per numbered step. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the audit by material and by parafunction — a straightforward mark that most candidates never arrive at.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs and radiograph up	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	Name it as a **technical complication, not a biological one**: **chipping of veneering ceramic** on a **three-unit screw-retained bridge at the 45–47 in layered zirconia**, four years old, with **healthy implants and no pain**. Then the modifiers: **he grinds**, and he has an **intact but heavily worn natural dentition elsewhere**. Then the sentence the whole case rests on: **three fractures in four years is not bad luck; it is a design and load mismatch that will continue until I change something**.
1:20–4:30	Domain 1: why ceramic chips, then what you examine. Two printed answers that become nine	*Why does veneering ceramic chip on implant restorations?* — one question per reason, and **usually more than one operates**. **Absence of a periodontal ligament**, so load is transmitted **without the damping and proprioceptive protection a tooth would give**. **Parafunctional loading. Inadequate or uneven framework support — the ceramic must be supported in even thickness, and unsupported ceramic over a poorly designed connector or a flat framework is where fractures start**. **Residual thermal stresses from an inappropriate cooling protocol during firing. Occlusal interferences, particularly in excursion.** *What would you examine?* — the occlusion in detail: **contacts in intercuspation with shimstock and articulating paper of two thicknesses**, and **all excursive movements including non-working interferences**; **whether the implant bridge is contacting more heavily than the adjacent natural teeth**. *And the prosthesis itself?* — the **framework design on a radiograph and, if the bridge is removed, directly — is the veneer evenly supported?** — and **whether there is a cantilever**. *And the history?* — **whether the screws have loosened previously, which would indicate overload**. *And the patient?* — **masseteric hypertrophy, linea alba, wear facets, and whether he has ever had or worn a splint**. Expect the spare minutes here to go on the groundwork underneath: **the difference in proprioception and intrusion between a tooth and an implant**, and **why that difference matters to every occlusal decision you go on to make**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the options, the sequence, and the money — three printed answers that become ten	*Options?* — one question each. **Repair with composite — a temporary measure at best, and it has already failed as a strategy. Polish and monitor — appropriate only for a very minor chip. Remake in monolithic zirconia, removing the weak veneering layer entirely — the definitive answer for a parafunctional patient and the recommendation. Remake in metal-ceramic with reduced veneer or a metal occlusal surface — less aesthetic but very robust. Redesign — span, connector dimensions, and whether a cantilever is present.** *What is your plan?* — and the first clause is the mark: **correct the cause before remaking, otherwise I will remake it again**. Then the five steps as five short answers: **adjust the occlusion so the bridge holds shimstock only under heavy clenching and has no excursive contacts, with guidance on the natural canine**; **confirm implants and screws are sound, replace the screws with new ones, torque correctly and re-torque at ten minutes**; **remake in monolithic zirconia, screw-retained, with a narrow occlusal table, reduced cusp inclination, generous connector dimensions and no cantilever**; **provide a hard, full-coverage occlusal splint for night wear, adjusted after the new bridge is fitted, with its wear reviewed as an objective measure of his parafunction**; **discuss stress, sleep, caffeine and alcohol, and screen for obstructive sleep apnoea with GP referral, since sleep bruxism and OSA are associated**. *And the cost?* — **honestly and early**: establish **what was originally agreed**, explain that **the material choice was reasonable at the time but has not withstood his bite**, and set out the remake cost — with two things said plainly: **even monolithic zirconia is not immune, because he generates forces beyond what restorations are designed for**, and **the splint is not optional**. **Put the maintenance expectation in writing so future component replacement is not a surprise.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the laboratory prescription and the finish — two printed answers that become seven	*What do you specify?* — **monolithic zirconia, screw-retained**, with **the implant system, platform and connection specified**; **connector dimensions adequate for the span**; **a narrow occlusal table with reduced cusp inclination**; **contacts light in intercuspation and absent in excursion**; **a cleansable emergence with open embrasures**; and **a bisque or design preview before final milling**. *What else do you send?* — **photographs and your occlusal instructions**. *And the technician?* — the answer candidates omit and examiners like: **ask him to advise if he sees a design problem — a good technician will often identify the reason a restoration keeps failing**. *How do you adjust the zirconia?* — **fine diamonds under copious water**. *And then?* — **polish thoroughly with a dedicated zirconia polishing system**. *Why does that matter here specifically?* — because **an adjusted but unpolished zirconia surface causes rapid wear of the opposing dentition**, which **in a bruxist with an already worn dentition would be a serious harm**.
11:30–14:30	Domain 4: monitoring and audit — two printed answers that become five	*How do you monitor?* — **six-monthly review with occlusal assessment**; **screw integrity checks**; **examination of the splint for wear as a proxy for his parafunctional activity** — the objective measure, and the one that makes the recall meaningful; and **photographs of the restoration**. *Radiographs?* — **at one year and then as clinically indicated**. *What do you record?* — **torque values and the dates of any component replacement, so you can see whether the pattern is improving**. *What would you audit?* — **the incidence of technical complications in your own implant restorations, separated by material and by whether the patient has diagnosed parafunction**. *And the action?* — **if layered restorations in bruxists are failing in your hands, the change is to specify monolithic materials for that group as a matter of protocol — and re-audit**.

14:30–15:00	Closes on the patient and the record	What you say: *"Your implants are fine — this is the porcelain, and it's chipped three times because you generate more force than the material was designed for. Replacing it the same way will give you the same result. What I'd like to do is adjust how it meets your other teeth, remake it in solid zirconia with no porcelain layer to chip, and make you a night guard — and I need to be straight with you: the night guard isn't optional, and even solid zirconia isn't unbreakable in a bite like yours."* What you write: the **occlusal findings including excursive and non-working contacts**; **the signs of parafunction**; **any history of screw loosening**; **what was originally agreed on cost and what is agreed now**; the **statement that even monolithic zirconia is not immune and that the splint is a condition of the remake**; **torque values and screw replacement dates**; and the **OSA screening discussion and any GP referral**.

The spine — the twelve answers to have ready if nothing else

1 Three fractures in four years is a design and load mismatch, not bad luck · 2 No periodontal ligament — no damping, no proprioceptive protection · 3 Unsupported veneer over a poor framework is where fractures start · 4 Firing and cooling protocol as a cause · 5 Excursive and non-working interferences, checked properly · 6 Whether the bridge contacts more heavily than the adjacent natural teeth · 7 Previous screw loosening as evidence of overload · 8 Correct the cause before remaking · 9 Monolithic zirconia, screw-retained, narrow table, generous connectors, no cantilever · 10 The splint, adjusted after fitting, and its wear as an objective measure · 11 Polish adjusted zirconia, or wear the opposing dentition · 12 Audit by material and by diagnosed parafunction.

Where the marks are lost in this case

Remaking it and expecting a different result. The first clause of the plan is the mark: **correct the cause before remaking, otherwise I will remake it again**. That means the **occlusion adjusted so the bridge holds shimstock only under heavy clenching with no excursive contacts and guidance on the natural canine**, **new screws torqued correctly and re-torqued at ten minutes**, the **framework and any cantilever reconsidered**, and only then a **monolithic zirconia remake**. A candidate who names the material and stops has answered a laboratory question, not a clinical one.

Leaving the splint as an afterthought. He grinds, he has **masseteric hypertrophy** and a **heavily worn natural dentition**, and he has broken the same restoration three times. Volunteer the **hard, full-coverage night splint, fabricated after the new bridge is fitted and adjusted to include it**, and then volunteer the part that makes it examinable: **its wear is reviewed as an objective measure of his parafunction**. Volunteer too the **screen for obstructive sleep apnoea with GP referral**, since **sleep bruxism and OSA are associated** — and say to him that **the splint is not optional**.

Forgetting what zirconia does to the opposing teeth. The competence line in this case is not surgical. It is in the finishing and the honesty. **Adjusted but unpolished zirconia causes rapid wear of the opposing dentition**, and in a bruxist whose remaining teeth are already worn **that is a serious harm you would have caused** — so **fine diamonds under water, then a dedicated zirconia polishing system**, every time. And be honest on cost before you start: **the original material choice was reasonable but has not withstood his bite**, **even monolithic zirconia is not immune**, and **the maintenance expectation goes in writing so future component replacement is not a surprise**.

SOE Case 40 — Residual cement discovered incidentally

THE CASE

50-year-old woman attends for a routine examination. She has an implant crown at the **24**, cemented 3 years ago elsewhere. She is asymptomatic. A bitewing taken for caries screening shows an **obvious radiopaque mass at the implant–abutment junction mesially**. Probing at that site is **5 mm with bleeding, no suppuration**, and there is **1.5 mm of bone loss** compared with a radiograph she has brought from the time of fitting.

Domain 1

"An incidental finding of residual cement with early peri-implant disease in an asymptomatic patient. The diagnosis is **peri-implantitis** — she has bleeding on probing, increased probing depth and, against her own baseline, bone loss beyond initial remodelling. Because I have a baseline and the disease is early, this is a case where intervention now is likely to be effective."

Q Confirm the diagnosis against the 2017 criteria.

"The criteria are bleeding and/or suppuration on gentle probing, increased probing depth compared with previous examinations, and bone loss beyond crestal remodelling. She has bleeding, 5 mm probing depth, and 1.5 mm of loss against a documented baseline. She does not meet the surrogate thresholds used when no baseline exists — probing depth of 6 mm or more and 3 mm of bone loss — but I do not need those, because I have her baseline. That is exactly why the baseline radiograph matters."

Q Would you have found this without the bitewing?

"Probably, on probing — which is why **peri-implant probing at six sites is part of every implant recall**, not an optional extra. The lesson is that an asymptomatic implant is not a healthy implant until it has been probed and compared with a baseline."

Domain 2

Q: What is your management?

"**Cause-related therapy first**, and here the cause is identified and removable. 1. Explain the finding and obtain **consent to remove the crown**, including the real possibility of damaging or destroying it and the cost of a remake. Without that consent I cannot treat the cause. 2. Remove the crown — ultrasonic first, then a crown remover, sectioning only if necessary. 3. **Remove all residual cement** under magnification with titanium or PEEK instruments, knotted floss drawn circumferentially, and glycine air-polishing. Verify radiographically, noting that non-radiopaque cement would not show. 4. Assess the abutment: is it fully seated? Is the margin excessively subgingival? Is the emergence over-contoured? 5. Debride the peri-implant tissues, deliver **individualised oral hygiene instruction** with a correctly sized interdental brush. 6. **Replace the restoration with a screw-retained, cleansable design** — this removes the cause permanently. 7. **Re-assess at one month.** With early disease and the cause removed, I would expect resolution of bleeding. 8. **Supportive therapy at 3 to 5 monthly intervals**, with a new baseline radiograph."

Q What if she declines to have the crown removed because of the cost?

"I would explain the consequences clearly and without pressure: that the cement is the cause, that non-surgical treatment without removing it is unlikely to resolve the disease, and that continued bone loss risks losing the implant, which would be more costly. I would offer to stage the work. If she still declines, I would record the **informed refusal**, treat as well as I can non-surgically, monitor closely at three-monthly intervals with radiographs when indicated, and revisit the discussion. Her refusal is her right; my job is to make sure it is informed and documented."

Q Would you tell her the previous dentist made an error?

"No. I would describe my findings factually — 'there is cement material left under the gum, which is a recognised complication of the cemented technique and a known cause of this kind of gum infection' — explain what I propose, and tell her how she can raise concerns formally if she wishes. I would not put a critical opinion of a colleague into her notes as fact."

Domain 3

Q: How, practically, do you remove submucosal cement you cannot see?

"Magnification and good illumination first. Then **knotted dental floss** drawn around the abutment and worked in a see-saw motion, which engages set cement in the sulcus remarkably effectively. **Titanium or PEEK curettes and scalers** worked circumferentially — never steel. **Glycine or erythritol air-polishing** delivered subgingivally to reach threads and residual particles. If access remains inadequate and cement persists, a **small flap for direct vision** is justified. Then a verification radiograph."

Domain 4

Q: What is the prognosis?

"Good, precisely because it has been caught early and the cause is removable. Studies of cement-associated peri-implant disease show that removal of the cement resolves the inflammation in a substantial proportion of cases. My success criteria at re-assessment are **absence of bleeding on gentle probing, no further increase in probing depth, and stable bone levels** against a new baseline."

Q What does this case change in your own practice?

"Three things. **Screw retention by default** wherever angulation allows. If cementing is unavoidable, **margins at or within 1 mm of the mucosal margin, minimal radiopaque cement and an extra-oral replica technique, with a verification radiograph**. And **peri-implant probing at six sites at every recall**, with a baseline radiograph taken and recorded at every prosthesis fitting — I would audit my compliance with that last point, since it is the single measure that made this diagnosis possible."

Running this case to time

Time check. This case is written in the older format — **eight long answers, about 721 words**. At 140 words a minute plus five seconds a question that is under **six minutes**: it does not overrun, it leaves **over nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three or four short questions**. The management answer is eight questions on its own, one per numbered step, and the diagnosis answer is three. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the prognosis and the three changes to your own practice — the cheapest marks in the case.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the bitewing and her old radiograph side by side	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Give the diagnosis, not the finding: **peri-implantitis at the 24, with residual cement as the identified cause**, found **incidentally in an asymptomatic patient**. Then the evidence in her numbers: **bleeding on probing, 5 mm probing depth, and 1.5 mm of bone loss against a radiograph she has brought from the time of fitting**. Then the two things that make this a good case rather than a bad one: **the cause is identified and removable**, and **it has been caught early — so intervention now is likely to be effective**. And say the thing that makes it possible: **she has a baseline**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the 2017 criteria, then why it was found at all. Two printed answers that become six	*Confirm the diagnosis against the 2017 criteria.* — **bleeding and/or suppuration on gentle probing**; **increased probing depth compared with previous examinations**; **bone loss beyond crestal remodelling**. *Does she meet them?* — **yes: bleeding, 5 mm, and 1.5 mm of loss against a documented baseline**. *What about the surrogate thresholds?* — **probing depth of 6 mm or more and 3 mm of bone loss, used when no baseline exists — she does not meet those, and does not need to, because I have her baseline**. Then the sentence to say out loud: **that is exactly why the baseline radiograph matters**. *Would you have found it without the bitewing?* — **probably, on probing — peri-implant probing at six sites is part of every implant recall, not an optional extra**. *And the lesson?* — **an asymptomatic implant is not a healthy implant until it has been probed and compared with a baseline**. Expect the spare minutes here to be filled underneath: **peri-implant mucositis against peri-implantitis**, **what constitutes initial crestal remodelling**, **probing force and why gentle probing round implants**, and **how cement gets subgingivally in the first place**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: management in order, and the two conversations — three printed answers that become eleven	*What is your management?* — **cause-related therapy first, and here the cause is identified and removable** — then the steps as separate short answers. **Explain the finding and obtain consent to remove the crown, including the real possibility of damaging or destroying it and the cost of a remake — without that consent I cannot treat the cause. Remove the crown — ultrasonic first, then a crown remover, sectioning only if necessary. Remove all residual cement. Assess the abutment: is it fully seated? is the margin excessively subgingival? is the emergence over-contoured? Debride the peri-implant tissues and deliver individualised oral hygiene instruction with a correctly sized interdental brush. Replace the restoration with a screw-retained, cleansable design — this removes the cause permanently. Re-assess at one month, expecting resolution of bleeding in early disease with the cause removed. Supportive therapy at 3 to 5 monthly intervals, with a new baseline radiograph.** *If she declines because of the cost?* — **explain the consequences clearly and without pressure**: **the cement is the cause, non-surgical treatment without removing it is unlikely to resolve the disease**, and **continued bone loss risks losing the implant, which would be more costly**; **offer to stage the work**; and if she still declines, **record the informed refusal, treat as well as you can non-surgically, monitor at three-monthly intervals with radiographs when indicated, and revisit the discussion** — because **her refusal is her right; my job is to make sure it is informed and documented**. *Would you tell her the previous dentist made an error?* — **no**, and the reason is the answer: describe your findings **factually** — *"there is cement material left under the gum, which is a recognised complication of the cemented technique and a known cause of this kind of gum infection"* — explain what you propose, **tell her how she can raise concerns formally if she wishes**, and **do not put a critical opinion of a colleague into her notes as fact**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: removing cement you cannot see — one printed answer, and it is really six	*First?* — **magnification and good illumination**. *Then?* — **knotted dental floss drawn around the abutment and worked in a see-saw motion, which engages set cement in the sulcus remarkably effectively**. *Instruments?* — **titanium or PEEK curettes and scalers worked circumferentially — never steel**, and say why: steel scores the titanium surface and creates a plaque-retentive roughness. *Anything else?* — **glycine or erythritol air-polishing delivered subgingivally, to reach threads and residual particles**. *If access is still inadequate?* — **a small flap for direct vision is justified**. *And afterwards?* — **a verification radiograph** — with the caveat that scores the point: **non-radiopaque cement would not show**, so absence on the film is not proof of removal. Expect the spare minutes here to go on the practicalities: **how you protect the tissues while sectioning a crown**, **what you do if the abutment screw is seized**, and **what an over-contoured emergence profile would make you change in the replacement**.
11:30–14:30	Domain 4: prognosis, criteria, and what you change — two printed answers that become five	*Prognosis?* — **good, precisely because it has been caught early and the cause is removable**, and **studies of cement-associated peri-implant disease show that removal of the cement resolves the inflammation in a substantial proportion of cases**. *Your success criteria?* — three, named: **absence of bleeding on gentle probing, no further increase in probing depth, and stable bone levels against a new baseline**. *What does this change in your own practice?* — **three things**. **Screw retention by default wherever angulation allows. If cementing is unavoidable: margins at or within 1 mm of the mucosal margin, minimal radiopaque cement, an extra-oral replica technique, and a verification radiograph. Peri-implant probing at six sites at every recall, with a baseline radiograph taken and recorded at every prosthesis fitting.** *And which do you audit?* — **the last one, since it is the single measure that made this diagnosis possible**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	What you say: *"Your implant looks and feels fine to you, but the gum around it is bleeding and there's a bit of bone loss compared with the picture you brought me. The cause is some cement left under the gum when the crown was fitted — that's a recognised problem with cemented crowns, and it's fixable. To fix it I have to take the crown off, and I have to be honest that it may not survive coming off, so there may be a cost to replacing it. If we deal with it now the outlook is good; if we leave it, you could eventually lose the implant."* What you write: her **baseline radiograph and the comparison**; the **six-site probing chart**; the **diagnosis and the criteria it meets**; the findings **stated factually, with no opinion about a colleague recorded as fact**; the **consent to crown removal including the risk to the crown and the cost of a remake**; **how she may raise concerns formally, if she asks**; any **informed refusal**; and the **new baseline radiograph and supportive interval**.

The spine — the twelve answers to have ready if nothing else

1 Peri-implantitis with an identified, removable cause — said before the finding · 2 The 2017 criteria, and which of hers meet them · 3 The surrogate thresholds, and why she does not need them · 4 Why the baseline radiograph is the whole case · 5 Six-site peri-implant probing at every recall, not an optional extra · 6 Cause-related therapy first · 7 Consent to remove the crown, including that it may not survive · 8 Knotted floss, titanium or PEEK instruments, glycine air-polishing · 9 The verification radiograph, and that non-radiopaque cement will not show · 10 Replace with a screw-retained, cleansable design · 11 The informed refusal, recorded, with three-monthly monitoring · 12 The three changes to your own practice, and which one you audit.

Where the marks are lost in this case

Treating the tissues and leaving the cement. This is the trap the case exists to set. Non-surgical debridement without removing the cement **is unlikely to resolve the disease**, so the sequence is **cause-related therapy first**, and the first clinical step is not an instrument but a **consent — to remove the crown, including the real possibility of damaging or destroying it and the cost of a remake**. Say it in that order, and say the consequence: **without that consent I cannot treat the cause**.

Blaming the previous dentist. She will ask, and the answer is **no**. Describe the finding **factually — "cement material left under the gum, a recognised complication of the cemented technique and a known cause of this kind of gum infection"** — explain what you propose, and **tell her how she can raise concerns formally if she wishes**. Then volunteer the record-keeping point that scores: **you do not put a critical opinion of a colleague into her notes as fact**. Candour about what you have found is not the same as a judgement about who caused it.

Missing the two lessons the case is actually testing. First, that **the baseline radiograph is what made this diagnosis possible** — without it she would have had to reach **6 mm and 3 mm of bone loss** before the surrogate criteria applied. Second, that the fix is **prevention in your own practice**: **screw retention by default wherever angulation allows**; if cementing is unavoidable, **margins at or within 1 mm of the mucosal margin, minimal radiopaque cement, an extra-oral replica technique and a verification radiograph**; and **six-site probing with a baseline radiograph at every prosthesis fitting — the point you would audit**. Surgical management of established peri-implantitis, if the disease does not resolve, is beyond this level and goes on for an opinion.

SOE Case 41 — Post-orthodontic lateral incisor spaces

THE CASE

19-year-old female, orthodontic treatment completed 3 months ago. Congenitally absent **12 and 22**; spaces opened to **6.2 mm** each. She is in fixed retainers bonded 13 to 23. Medium smile line, medium phenotype, canines of normal shade and shape. She is at university and will be abroad for a year from next September.

Domain 1

"A recently completed orthodontic case with bilateral lateral incisor spaces in a 19-year-old. Three issues determine the plan: **whether growth has ceased**, **whether the roots are adequately divergent**, and **the practical constraint of a year abroad**, which affects timing and continuity of care."

Q Has growth ceased at 19?

"In a female, very probably. But I would confirm rather than assume — a history of stable height and shoe size for at least two years, stable dentition, and if there were real doubt, serial cephalometric radiographs a year apart or hand-wrist films, though I would need to justify the exposure. Craniofacial growth continues at a low level into adulthood, and **late infra-occlusion of anterior implants is a recognised long-term problem** that I would warn her about regardless."

Q What do you need from the orthodontist?

"Confirmation that treatment is complete and the position is stable; the retention plan and its duration; the **root positions and divergence** at the 12 and 22 sites — I need at least 1.5 mm from each adjacent root, and converging roots are a common reason an apparently adequate space cannot take an implant; and whether the space was distributed to the ideal tooth width, checked against the golden proportion and the contralateral side."

Q What else would you assess?

"Smile line and lip dynamics on video; papilla heights and the interproximal bone levels on the 11, 13, 21 and 23; phenotype; CBCT for ridge width at both sites, since **hypodontia sites are typically narrow because the alveolus never developed around a tooth**; and photographs and study casts with a diagnostic wax-up."

Domain 2

Q: Options?

"**Resin-bonded cantilever bridges** from the canines — minimally invasive, excellent published survival for single anterior units, no surgery, and readily repaired or replaced if she debonds while abroad. My first-line option for a 19-year-old.
Implants — definitive and preserving the adjacent teeth, but committing a 19-year-old to a restoration she will maintain for sixty years, in sites likely to need contour augmentation, with a real risk of late infra-occlusion.
Canine substitution — the orthodontics is complete, so reopening that would mean further treatment; and her canines are of normal shade and shape, which makes substitution less attractive aesthetically. Worth mentioning but not pursuing.
Composite build-ups on the canines with orthodontic space closure — as above.
Removable prosthesis — interim only."

Q What would you recommend, and how does the year abroad affect it?

"The year abroad is decisive for **timing**. Implant treatment would take six to twelve months if augmentation is needed, and would leave her mid-treatment when she leaves — with a provisional, no local clinician, and no continuity of care. That is not acceptable. So I would recommend **resin-bonded cantilever bridges now**, which can be delivered in two or three visits, look excellent, and are robust enough to survive a year abroad. I would tell her explicitly that this is **not a compromise but a legitimate long-term option**, and that if she later wants implants, the bridges have preserved the option entirely. If she strongly prefers implants, I would either

complete the whole treatment before she leaves — which is only realistic if no augmentation is needed and I can use a conventional protocol with time in hand — or defer until she returns."

Q SAC?

"Implants here would be surgically **Advanced** — aesthetic zone, narrow hypodontia sites likely requiring contour augmentation, 6.2 mm spaces requiring narrow implants with tight tolerances. Prosthodontically **Advanced**. Resin-bonded bridges are Straightforward and within my competence comfortably."

Domain 3

Q: If she chooses resin-bonded bridges, describe the key points.

"**Cantilever design from the canine**, single retainer — the evidence favours cantilevers over fixed-fixed designs, which fail by debonding at one retainer with undetected caries beneath. Minimal or no preparation, confined to enamel, with a defined path of insertion and a supragingival margin. A **cobalt-chromium or nickel-chromium framework** sandblasted and treated with a metal primer, or a zirconia framework, bonded with a resin cement under good moisture control. An **ovate pontic** shaped to support the tissue, with **soft tissue augmentation beneath it** if the ridge is deficient — a connective tissue graft under a pontic is far more predictable than around an implant. Occlusion: **no contact in intercuspation or excursion on the pontic**, particularly important with her fixed retainer in place. I would also coordinate with the orthodontist about the **bonded retainer**, since it may need modifying to accommodate the retainers."

Q If she chooses implants, what would you plan?

"**3.3 mm narrow-diameter implants** in a titanium-zirconium alloy, with 1.5 mm to each adjacent root — 6.2 mm makes this feasible but tight, and I would confirm root divergence first. Position: shoulder 3 mm apical to the planned mid-facial margin, 1 to 2 mm palatal, angled through the cingulum. **Expect to need simultaneous contour augmentation** and a connective tissue graft. **Two-stage submerged healing**, then a screw-retained provisional to develop the emergence over 8 to 12 weeks, then a definitive screw-retained restoration on a Ti-base."

Domain 4

Q: What would you warn her about for the long term?

"That whichever option she chooses, this is **maintenance for life, not a one-off procedure**. For bridges: debonding, and the need for periodic replacement. For implants: that **continued eruption of the natural dentition relative to an ankylosed implant** can produce relative infra-occlusion and open contacts over decades, that crowns will need replacing more than once, and that recession may occur. I would arrange standardised photographs and study casts at intervals so those changes are detected rather than discovered."

Q How would you handle continuity of care while she is abroad?

"Give her a written summary of what has been done, photographs, and my contact details; advise her to register with a dentist wherever she is; and, if she has implants, give her the implant card with the system, diameter, length and lot numbers so any clinician can service them. I would set a recall reminder for her return."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 905 words**. At 140 words a minute that is **six and a half minutes**, and ten questions at five seconds each adds **fifty seconds**: call it **seven and a half minutes**. It does not overrun; it leaves **the better part of eight minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three to six short questions**. The options answer alone is five; the resin-bonded bridge answer is six. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the warning about late infra-occlusion and the continuity of care while she is abroad.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs and radiographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Name the case, then name the three things that decide it, because the case itself does: **bilateral lateral incisor spaces of 6.2 mm in a 19-year-old, orthodontics completed three months ago**, where the plan turns on **whether growth has ceased**, **whether the roots are adequately divergent**, and **the year abroad from next September**, which affects timing and continuity of care. Then the working position, volunteered: **congenitally absent 12 and 22, medium smile line and phenotype, canines of normal shade and shape, fixed retainer 13 to 23** — and **the timing constraint is likely to be decisive**.
1:20–4:30	Domain 1: growth, the orthodontist, and the assessment — three printed answers that become nine	*Has growth ceased at 19?* — **in a female, very probably** — then the qualifier that is the mark: **I would confirm rather than assume**. *How?* — **stable height and shoe size for at least two years, a stable dentition**, and **only if there were real doubt, serial cephalometric films a year apart or hand-wrist films, and I would have to justify the exposure**. *And even then?* — **craniofacial growth continues at a low level into adulthood, so late infra-occlusion of anterior implants is a recognised long-term problem I would warn her about regardless**. *From the orthodontist?* — three things: **confirmation that treatment is complete and stable**; **the retention plan and its duration**; and the one that decides feasibility — **root positions and divergence, because I need at least 1.5 mm from each adjacent root, and converging roots are a common reason an apparently adequate space cannot take an implant**. Add **whether the space was distributed to the ideal tooth width, against the golden proportion and the contralateral side**. *Your own assessment?* — **smile line and lip dynamics on video; papilla heights and interproximal bone levels on the 11, 13, 21 and 23; phenotype; CBCT for ridge width, because hypodontia sites are typically narrow — the alveolus never developed around a tooth; and photographs, casts and a diagnostic wax-up**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, recommendation, SAC — three printed answers that become eight	*Options?* — five, a sentence each: **resin-bonded cantilever bridges from the canines — minimally invasive, excellent published survival for single anterior units, no surgery, readily repaired if she debonds abroad; my first-line option for a 19-year-old**. **Implants — definitive and tooth-preserving, but committing a 19-year-old to a restoration she maintains for sixty years, in sites likely to need contour augmentation, with a real risk of late infra-occlusion. Canine substitution — the orthodontics is complete and her canines are of normal shade and shape; mention it, do not pursue it. Composite build-ups with space closure — as above. A removable prosthesis — interim only.** *What decides it?* — **the year abroad, and it decides timing**: implants **take six to twelve months if augmentation is needed** and would leave her **mid-treatment when she leaves, with a provisional, no local clinician and no continuity of care — that is not acceptable**. *So?* — **bonded bridges now, in two or three visits**, presented in these words: **not a compromise but a legitimate long-term option, and if she later wants implants the bridges have preserved the option entirely**. *If she insists on implants?* — **complete everything before she goes, realistic only if no augmentation is needed, or defer until she returns**. *SAC?* — **implants surgically Advanced — aesthetic zone, narrow hypodontia sites, tight tolerances — and prosthodontically Advanced; bonded bridges Straightforward and comfortably within my competence**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the bridge, then the implant plan — two printed answers that become ten	*Design?* — **cantilever from the canine, single retainer**, because **fixed-fixed designs fail by debonding at one retainer with undetected caries beneath**. *Preparation?* — **minimal or none, confined to enamel, defined path of insertion, supragingival margin**. *Framework?* — **cobalt-chromium or nickel-chromium, sandblasted and metal-primed, or zirconia, bonded with resin cement under good moisture control**. *Pontic?* — **ovate, shaped to support the tissue**, with **soft tissue augmentation beneath if the ridge is deficient** — and say why: **a connective tissue graft under a pontic is far more predictable than around an implant**. *Occlusion?* — **no contact in intercuspation or excursion on the pontic**, which matters **with her fixed retainer in place**. *The retainer?* — **coordinate with the orthodontist; it may need modifying**. Then the implant plan: **3.3 mm narrow-diameter titanium-zirconium implants with 1.5 mm to each root — 6.2 mm makes this feasible but tight**; **shoulder 3 mm apical to the planned mid-facial margin, 1 to 2 mm palatal, angled through the cingulum**; **expect simultaneous contour augmentation and a connective tissue graft**; **two-stage submerged healing, a screw-retained provisional over 8 to 12 weeks, then a definitive screw-retained restoration on a Ti-base**.
11:30–14:30	Domain 4: the long-term warning and the year abroad — two printed answers that become six	*What do you warn her about?* — the sentence first: **whichever option she chooses, this is maintenance for life, not a one-off procedure**. *For bridges?* — **debonding and periodic replacement**. *For implants?* — **continued eruption of the natural dentition relative to an ankylosed implant produces relative infra-occlusion and open contacts over decades**; **crowns replaced more than once**; **recession**. *How would you detect it?* — **standardised photographs and study casts at intervals, so those changes are detected rather than discovered** — that verb pair is the mark. *Abroad?* — **a written summary, photographs and my contact details**; **register with a dentist wherever she is**; **the implant card with system, diameter, length and lot numbers if she has implants**. *Administratively?* — **a recall reminder set for her return**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	What you say: *"You're 19 and about to spend a year abroad, and whatever we do has to last decades. My advice is bonded bridges from your canines now — two or three visits, no surgery, and they look excellent. That isn't me fobbing you off with a temporary: it's a proper long-term option, and it keeps the implant door completely open for when you're back and settled."* What you write: **her wishes in her own words**; **what you asked the orthodontist about completion, retention and root divergence**; **all options, including no treatment and canine substitution**; **the reasoning about the year abroad and continuity of care**; **the infra-occlusion warning given regardless of option**; **the written summary and photographs handed over**; and **the recall set for her return**.

The spine — the twelve answers to have ready if nothing else

1 The three deciding issues: growth, root divergence, the year abroad · 2 Growth probably ceased, but confirmed rather than assumed · 3 Late infra-occlusion warned about regardless of option · 4 The 1.5 mm to each adjacent root, and converging roots as the common blocker · 5 Hypodontia sites are narrow because the alveolus never developed · 6 Resin-bonded cantilever bridges as first line, and why · 7 The year abroad decides timing, not principle · 8 "Not a compromise but a legitimate long-term option" · 9 SAC: implants Advanced/Advanced, bridges Straightforward · 10 Cantilever not fixed-fixed, ovate pontic, no occlusal contact · 11 A connective tissue graft under a pontic beats grafting round an implant · 12 The written summary, the implant card, and the recall for her return.

Where the marks are lost in this case

Treating the bonded bridge as the consolation prize. The commonest failure is a candidate who describes implants lovingly and adds bonded bridges as an afterthought for a patient who cannot wait. The case says the opposite: **cantilever bridges are first line for a 19-year-old, with excellent published survival**, and the words to volunteer are that this is **not a compromise but a legitimate long-term option that preserves the implant option entirely**. Say it before it is drawn out of you.

Forgetting that 6.2 mm is not the only measurement. Candidates recite that a 3.3 mm implant fits a 6.2 mm space and stop. What decides feasibility is **root divergence — at least 1.5 mm from each adjacent root** — and **converging roots are a common reason an apparently adequate space cannot take an implant**. Volunteer that you would ask the orthodontist for the root positions, and that **hypodontia sites are typically narrow buccopalatally**, so a CBCT and **planned contour augmentation** are expected rather than a surprise.

Missing that the competence line here is drawn by timing. Implants at these sites are **surgically and prosthodontically Advanced**, and a candidate at this level should say so and consider referral or co-management. But the sharper line is temporal: **starting a six-to-twelve-month treatment on a patient who leaves the country mid-course, with a provisional and no local clinician, is not acceptable care**. Say that plainly, then close the loop she will live with — **the written summary, the photographs, the implant card if relevant, and a recall set for her return**.

SOE Case 42 — Parafunction and an anterior implant

THE CASE

45-year-old male, implant at the **11** restored 2 years ago. He presents with the **screw access composite lost and the crown loose**. He has an **edge-to-edge incisal relationship**, marked incisal wear on all upper and lower anteriors, and reports "clenching at work". The implant is healthy radiographically.

Domain 1

"A loose screw-retained anterior crown in a patient with an **edge-to-edge incisal relationship and heavy anterior wear** — that is, an implant placed in the position that receives the greatest excursive load in the mouth, in a patient with parafunction. The looseness is a symptom; the anterior guidance is the disease."

Q Why is an anterior implant particularly vulnerable in this patient?

"Because an edge-to-edge relationship means the upper incisors take substantial load in protrusion and in incising, and the implant has **no periodontal ligament** to damp that load or to provide protective proprioceptive feedback. Implant axial movement is 3 to 5 micrometres against 25 to 100 for a tooth. Combined with clenching, that concentrates fatigue at the screw joint."

Q What would you examine?

"Whether it is the **screw or the implant** that is loose — I would remove the crown and test the implant separately. The screw and internal connection for damage, and the implant body for fracture. **Occlusion in detail**: contacts in intercuspation with shimstock, protrusive and lateral excursions, and specifically whether the implant crown carries protrusive guidance. Signs of parafunction — masseteric hypertrophy, linea alba, scalloped tongue, wear facets that match in excursion. Whether he has ever had a splint. Peri-implant probing and bone levels. And I would ask about sleep quality, snoring and daytime somnolence."

Domain 2

Q: What is your management?

"**Never simply retighten.** 1. Remove the crown; inspect all components and the implant; **replace the screw with a new one** — a screw that has loosened has been fatigued. 2. **Correct the occlusion.** The implant crown should hold shimstock only under heavy clenching in intercuspation, and — critically here — should carry **as little protrusive guidance as possible**. In an edge-to-edge relationship that is difficult, so I would aim to distribute protrusive guidance across as many natural incisors as possible and minimise the implant's share, adjusting the crown's palatal contour. 3. **Re-torque to the manufacturer's value and re-torque at ten minutes.** 4. Reseal with **PTFE tape and composite**, then re-check the occlusion because the composite adds height. 5. **Provide a hard, full-coverage occlusal splint** for night-time wear, and consider a daytime habit-awareness programme given he clenches at work. 6. Consider whether the crown should be **remade in a more robust design** — monolithic material, modified palatal contour — if loosening recurs. 7. Screen for **obstructive sleep apnoea** and refer to his GP if there are symptoms."

Q What if it loosens again despite this?

"Then I would escalate the investigation rather than the tightening. Check for a **non-passive fit**, an unrecognised **implant fracture**, or a **damaged internal connection**. Consider remaking with a different abutment design. And I would have a frank conversation about the mechanical limits of an anterior implant in his occlusion, including the possibility that repeated loosening will progress to **screw fracture and eventually implant fracture** — which is the natural history if the cause is not corrected."

Q What should have been done differently at the planning stage?

"The edge-to-edge relationship and the wear would have been visible at assessment. Ideally the plan would have included **occlusal management before the implant** — assessment of the vertical dimension, possible

reorganisation of the anterior guidance, and a splint from the outset — and consent that explicitly covered a higher mechanical complication rate. The wear pattern should also have prompted an assessment of the whole dentition, not just the missing 11."

Domain 3

Q: Describe the splint you would provide.

"A **hard, heat-cured or milled acrylic full-coverage splint**, usually maxillary, covering all teeth so none can over-erupt, with **even bilateral contacts in intercuspation** and **smooth canine guidance in excursion** so the posterior teeth disclude. It must be adjusted at fit and reviewed, and its wear pattern monitored at each recall as an objective measure of his parafunctional activity. A soft splint is not appropriate — it can increase muscle activity."

Domain 4

Q: What would you monitor?

"Screw integrity at every recall — I would check it deliberately rather than wait for him to report looseness; the occlusion; the splint for wear; peri-implant health and bone levels against baseline; and the wear of his natural anterior teeth with photographs and study casts. Six-monthly review."

Q What would you record?

"The cause identified, the occlusal adjustments made, the screw replacement with date and torque value, the splint provided and its review, the discussion about mechanical risk and the possibility of eventual implant fracture, and the maintenance expectation. If loosening recurs a third time, that record becomes the basis for a considered decision about redesign or removal."

Running this case to time

Time check. This case is written in the older format — **eight long answers, about 699 words**. At 140 words a minute that is **five minutes**, and eight questions at five seconds each adds **forty seconds**: call it **five and three-quarter minutes**. It does not overrun; it leaves **more than nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three to seven short questions** — the management answer alone is seven, printed as a numbered list an examiner will take one item at a time. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the deliberate screw check at every recall and the record that eventually justifies removal.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiograph and clinical photographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Give the diagnosis and the disease in one breath: **a loose screw-retained crown at the 11 in a 45-year-old with an edge-to-edge incisal relationship and marked anterior wear** — that is, **an implant placed in the position that receives the greatest excursive load in the mouth, in a patient with parafunction**. Then the sentence that governs the whole case and should be volunteered, not extracted: **the looseness is a symptom; the anterior guidance is the disease**. Then the working position: **the implant is healthy radiographically, so this is a mechanical problem in a biologically sound site**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: why this implant is vulnerable, then the examination — two printed answers that become eight	*Why is an anterior implant particularly vulnerable in him?* — the undergraduate groundwork first, because the examiner will open below the implant material: **an edge-to-edge relationship means the upper incisors take substantial load in protrusion and in incising; an implant has no periodontal ligament to damp that load or to give protective proprioceptive feedback**; and the numbers — **implant axial movement is 3 to 5 micrometres against 25 to 100 micrometres for a tooth**. *Consequence?* — **combined with clenching, that concentrates fatigue at the screw joint**. Then the examination, broken into five: *Is it the screw or the implant?* — **remove the crown and test the implant separately**, and **inspect the screw and internal connection for damage and the implant body for fracture**. *Occlusion?* — **contacts in intercuspation with shimstock, protrusive and lateral excursions, and specifically whether the implant crown carries protrusive guidance**. *Signs of parafunction?* — **masseteric hypertrophy, linea alba, scalloped tongue, wear facets that match in excursion**, and **whether he has ever had a splint**. *Peri-implant?* — **probing and bone levels**. And the one candidates never volunteer: **sleep quality, snoring and daytime somnolence**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the management, the recurrence, and the planning failure — three printed answers that become eleven	Open with the two words that carry the domain: **never simply retighten**. Then the list, one item per question: **remove the crown, inspect all components and the implant, and replace the screw with a new one — a screw that has loosened has been fatigued**. **Correct the occlusion: the implant crown should hold shimstock only under heavy clenching in intercuspation and should carry as little protrusive guidance as possible** — and the honest qualifier that earns the mark, **in an edge-to-edge relationship that is difficult, so I would distribute protrusive guidance across as many natural incisors as possible and minimise the implant's share by adjusting the crown's palatal contour**. **Re-torque to the manufacturer's value, and re-torque at ten minutes. Reseal with PTFE tape and composite, then re-check the occlusion because the composite adds height. Provide a hard full-coverage occlusal splint for night-time wear, and a daytime habit-awareness programme, since he clenches at work. Consider remaking the crown in a more robust design — monolithic material, modified palatal contour — if loosening recurs. Screen for obstructive sleep apnoea and refer to his GP if there are symptoms.** *And if it loosens again?* — the principle first: **escalate the investigation rather than the tightening**, then the three things to look for — **a non-passive fit, an unrecognised implant fracture, a damaged internal connection** — then **a frank conversation about the mechanical limits of an anterior implant in his occlusion, including that repeated loosening progresses to screw fracture and eventually implant fracture, which is the natural history if the cause is not corrected**. *What should have been done at planning?* — **the edge-to-edge relationship and the wear were visible at assessment**: the plan should have included **occlusal management before the implant — assessment of the vertical dimension, possible reorganisation of the anterior guidance, and a splint from the outset** — with **consent explicitly covering a higher mechanical complication rate**; and **the wear should have prompted assessment of the whole dentition, not just the missing 11**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the splint, in specification-level detail — one printed answer that becomes six	*Material and coverage?* — **hard, heat-cured or milled acrylic, full coverage, usually maxillary**. *Why full coverage?* — **so no tooth can over-erupt**. *Contacts?* — **even bilateral contacts in intercuspation**. *Excursions?* — **smooth canine guidance so the posterior teeth disclude**. *Aftercare?* — **adjusted at fit, reviewed, and its wear pattern monitored at each recall as an objective measure of his parafunction** — the clause that scores, because it turns the splint into a diagnostic instrument. *Why not a soft splint?* — **it is not appropriate; it can increase muscle activity**. Expect the rest of this block to go to the groundwork beneath it: **what an edge-to-edge relationship does to protrusive guidance, and how you would decide whether the vertical dimension needs reorganising at all** — answered honestly, because **a full occlusal reorganisation across a worn dentition is beyond a general dentist with an interest in implants and belongs with a restorative colleague**.
11:30–14:30	Domain 4: what you monitor, and what you write — two printed answers that become seven	*What do you monitor?* — **screw integrity at every recall, checked deliberately rather than waiting for him to report looseness** — the most volunteerable line in the domain; **the occlusion**; **the splint for wear**; **peri-implant health and bone levels against baseline**; and **the wear of his natural anterior teeth, with photographs and casts**. *At what interval?* — **six-monthly review**. *What do you record?* — **the cause identified**, not just the treatment done; **the occlusal adjustments made**; **the screw replacement with date and torque value**; **the splint provided and its review**; **the discussion about mechanical risk and the possibility of eventual implant fracture**; and **the maintenance expectation**. *Why does that record matter?* — **if loosening recurs a third time, that record becomes the basis for a considered decision about redesign or removal** — a decision, documented, rather than a reaction.
14:30–15:00	Closes on the patient and the record	What you say: *"The crown coming loose isn't the problem — it's the sign. Your front teeth meet edge to edge and you clench, so this implant takes more sideways load than any implant is designed for. I'll fit a new screw, adjust the bite so your own teeth do more of the work when you slide forward, and make you a hard night splint. If we don't change the loading, the next thing that happens is the screw breaking, and after that the implant itself."* What you write: **the cause identified as occlusal overload in an edge-to-edge relationship with parafunction**; **the adjustments and the new screw with date and torque value**; **the splint and its review date**; **the frank discussion about screw and implant fracture**; **the OSA screening and any referral to his GP**; and **the maintenance expectation, six-monthly, with a deliberate screw check each time**.

The spine — the twelve answers to have ready if nothing else

1 The looseness is a symptom; the anterior guidance is the disease · 2 No periodontal ligament: 3–5 μm against 25–100 μm · 3 Test screw and implant separately; inspect the connection and body · 4 Shimstock, protrusive and lateral excursions, and whether the crown guides · 5 Masseteric hypertrophy, linea alba, scalloped tongue, matching wear facets · 6 Sleep quality, snoring, daytime somnolence · 7 Never simply retighten — always a new screw · 8 Distribute protrusive guidance across the natural incisors · 9 Re-torque to the manufacturer's value and again at ten minutes · 10 Hard full-coverage splint, canine guidance, never soft · 11 Escalate investigation, not torque: non-passive fit, implant fracture, damaged connection · 12 Deliberate screw check at every recall, and the record that justifies removal.

Where the marks are lost in this case

Retightening and calling it management. A candidate who says "I'd remove the crown, clean it, retighten to torque and reseal" has described the trivial half of the case. Say **never simply retighten**, say **a screw that has loosened has been fatigued and is replaced**, and spend your words where the marks are — **correcting the occlusion so the implant carries as little protrusive guidance as possible**, and **the hard full-coverage splint with canine guidance**. Treating the joint without treating the load is why this crown will be loose again by the next recall.

Never mentioning sleep. The case asks about **sleep quality, snoring and daytime somnolence**, and about a **daytime habit-awareness programme**, because he clenches at work and may be clenching at night for a reason. Almost nobody volunteers **obstructive sleep apnoea screening and referral to his GP** — a free mark that also shows you are treating a patient rather than a screw joint.

Not knowing when to stop tightening. The competence line here has two limbs. First: **after a second episode you escalate the investigation, not the torque — non-passive fit, implant fracture, a damaged internal connection** — saying plainly that **the natural history of uncorrected overload is screw fracture and then implant fracture**. Second: **rebuilding the vertical dimension and the anterior guidance across a worn dentition is beyond this level and belongs with a restorative colleague**. And take the retrospective mark the case hands you: **this should have been managed before the implant was placed, with consent that named the higher mechanical complication rate** — professional honesty, not self-incrimination.

SOE Case 43 — Head and neck radiotherapy and xerostomia

THE CASE

61-year-old man, 3 years after **chemoradiotherapy (66 Gy) for a right tonsillar squamous cell carcinoma**, in remission. He has severe xerostomia, rampant cervical caries, and has lost several teeth. He wears an upper partial denture he cannot tolerate because of mucosal soreness. He asks about implants.

Domain 1

"A patient three years after high-dose head and neck radiotherapy with severe xerostomia, radiation caries and denture intolerance. The 66 Gy dose is well above the level at which the **risk of osteoradionecrosis** becomes significant, and I need to know precisely **which structures were in the field**."

Q What determines whether implants are possible?

"**The radiation field and dose to the specific jaw site.** A right tonsillar tumour field will have included the right posterior mandible at or near full dose, while the anterior mandible and the maxilla may have received considerably less. I would obtain the **radiotherapy planning records** from his oncology centre — the dose-volume data for the mandible and maxilla — rather than guess. That single piece of information changes the answer more than anything I can determine clinically."

Q What else is relevant?

"Time since radiotherapy — the risk of osteoradionecrosis persists lifelong but the tissues are most hostile in the first year, and three years is more favourable. Whether he had a neck dissection. Current oncological status and surveillance. Smoking and alcohol, which are both risk factors for ORN and for second primaries. Trismus, which affects access. Xerostomia severity and whether saliva substitutes or pilocarpine have been tried. His caries risk and prevention regime. And his nutritional status and weight."

Q Why is his denture causing soreness?

"Irradiated, xerostomic mucosa is atrophic, poorly lubricated and fragile, and a denture that would be comfortable in a normal mouth causes ulceration. This is a common and very difficult problem, and it is one of the strongest arguments **for** implant support in irradiated patients — an implant-retained prosthesis can reduce mucosal loading substantially."

Domain 2

Q: What is your approach?

"**This is specialist care and I would refer**, but I would do so having completed a proper assessment and having managed what is within my scope.
Within my scope: intensive prevention — high-fluoride toothpaste 5000 ppm, fluoride varnish, dietary advice, saliva substitutes and stimulants, meticulous caries management, and treatment of any candidal infection; optimisation of the existing denture with a soft liner and careful extension; and liaison with his oncology team.
Referral: to a restorative consultant or a hospital-based implant service with experience of irradiated patients, for assessment of implant provision. The decision will depend on the site doses. I would **not** place implants in a field that received 66 Gy in general practice."

Q What are the specific risks you would explain?

"**Osteoradionecrosis** — the most serious, potentially requiring major surgery; **higher implant failure rates** in irradiated bone, particularly the mandible and particularly at higher doses; impaired soft tissue healing; and that any implant will be harder to maintain in a xerostomic mouth with a high caries and infection risk. I would also explain that **the alternative — continuing with an intolerable denture — is not benign either**, and that this is a balance of risks rather than a simple prohibition."

Q What about hyperbaric oxygen?

> "It has been used to improve tissue oxygenation before surgery in irradiated bone, but the evidence is **contested and it is not routinely recommended**. It is a specialist decision, not a general practice one, and I would neither promise it nor dismiss it — I would say that the receiving unit would advise."

Q If implants were provided, what design would you favour?

> "Wherever possible, in the site that received the **lowest dose** — often the anterior mandible. **Fewest implants that will do the job**, since each osteotomy is a risk. An **overdenture rather than a fixed prosthesis**, because it is removable, cleansable, allows mucosal inspection, and can be relined or converted if an implant is lost. Maximum cleansability, and a rigorous preventive and maintenance regime."

Domain 3

Q: What would your role be after specialist placement?

> "Shared care: intensive prevention, three-monthly recall, peri-implant monitoring, denture maintenance and relining, and prompt referral of any non-healing mucosal lesion — because in a patient with a previous head and neck malignancy, **any ulcer persisting beyond two weeks requires urgent referral to exclude recurrence or a second primary**. That surveillance responsibility is arguably the most important part of my role."

Domain 4

Q: What would you monitor?

> "Peri-implant health and bone levels; mucosal condition, with a low threshold for referral; caries and the effectiveness of the prevention regime; xerostomia and its management; his weight and nutrition; trismus; and his oncological follow-up. Three-monthly review, and I would document every mucosal examination."

Q What is the overriding principle in this case?

> "That his cancer surveillance and his prevention matter more than his implants, and that any implant decision belongs in a team that manages irradiated patients regularly. My most valuable contributions are the referral, the prevention, and the vigilance."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 706 words**. At 140 words a minute that is **five minutes**, and ten questions at five seconds each adds **fifty seconds**: call it **six minutes**. It does not overrun; it leaves **nine minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three to five short questions**. "What else is relevant?" alone is eight separate risk factors an examiner will take one at a time. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the three-monthly surveillance and the two-week ulcer rule — the part of the case where the marks are highest and the clinical stakes are highest too.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiographs and the medical summary in front of you	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	Lead with the dose, because the dose is the case: **a 61-year-old man three years after chemoradiotherapy at 66 Gy for a right tonsillar squamous cell carcinoma, in remission, with severe xerostomia, rampant cervical caries and denture intolerance**. Then the sentence that sets your position before any option is discussed: **66 Gy is well above the level at which the risk of osteoradionecrosis becomes significant, and I need to know precisely which structures were in the field**. Then the working position: **this is specialist care and my contribution is assessment, prevention, surveillance and referral** — said in the first minute rather than extracted in the eighth.
1:20–4:30	Domain 1: dose and field, the wider history, and the sore denture — three printed answers that become eleven	*What determines whether implants are possible?* — **the radiation field and the dose to the specific jaw site**, then the anatomy that follows from a right tonsillar tumour: **the right posterior mandible at or near full dose, while the anterior mandible and maxilla may have received considerably less**. *How would you find out?* — the answer that separates a 5 from a 3: **obtain the radiotherapy planning records — the dose-volume data for the mandible and maxilla — rather than guess**, because **that single piece of information changes the answer more than anything I can determine clinically**. *What else is relevant?* — eight items, one per question: **time since radiotherapy — the risk persists lifelong but the tissues are most hostile in the first year, and three years is more favourable**; **a neck dissection**; **current oncological status and surveillance**; **smoking and alcohol, risk factors for ORN and for second primaries**; **trismus, which affects access**; **xerostomia severity, and whether saliva substitutes or pilocarpine have been tried**; **caries risk and prevention regime**; and **nutritional status and weight**. *Why is the denture sore?* — the undergraduate biology: **irradiated, xerostomic mucosa is atrophic, poorly lubricated and fragile, so a denture comfortable in a normal mouth causes ulceration**. Then the inversion that earns the domain: **this is one of the strongest arguments *for* implant support in irradiated patients, because an implant-retained prosthesis substantially reduces mucosal loading**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: your approach, the risks, hyperbaric oxygen, and the design — four printed answers that become twelve	*What is your approach?* — the line first, unhedged: **this is specialist care and I would refer — having completed a proper assessment and managed what is within my scope**. *What is within your scope?* — **intensive prevention: 5000 ppm fluoride toothpaste, fluoride varnish, dietary advice, saliva substitutes and stimulants**; **meticulous caries management**; **treatment of any candidal infection**; **optimising the existing denture with a soft liner and careful extension**; and **liaison with his oncology team**. *Refer to whom?* — **a restorative consultant or a hospital implant service experienced with irradiated patients; the decision will depend on the site doses**. *The limit, in one sentence* — **I would not place implants in a field that received 66 Gy in general practice**. *What risks would you explain?* — **osteoradionecrosis, the most serious, potentially requiring major surgery**; **higher implant failure rates in irradiated bone, particularly the mandible and at higher doses**; **impaired soft tissue healing**; and **greater difficulty maintaining an implant in a xerostomic mouth**. Then the balancing clause candidates omit: **continuing with an intolerable denture is not benign either — this is a balance of risks, not a simple prohibition**. *Hyperbaric oxygen?* — **used to improve tissue oxygenation before surgery in irradiated bone, but the evidence is contested and it is not routinely recommended; a specialist decision — I would neither promise it nor dismiss it, and the receiving unit would advise**. *If implants were provided, what design?* — four rules: **the site that received the lowest dose, often the anterior mandible**; **the fewest implants that will do the job, since each osteotomy is a risk**; **an overdenture rather than a fixed prosthesis — removable, cleansable, allowing mucosal inspection, and relinable or convertible if an implant is lost**; and **maximum cleansability with a rigorous maintenance regime**.
7:30–11:30	Domain 3: your role in shared care — one printed answer that becomes six	*Your role after specialist placement?* — say **shared care** and itemise it, because the examiner will take four minutes over three printed lines: **intensive prevention**; **three-monthly recall**; **peri-implant monitoring**; **denture maintenance and relining**; and the one that matters most — **prompt referral of any non-healing mucosal lesion**. *The rule?* — **in a patient with a previous head and neck malignancy, any ulcer persisting beyond two weeks needs urgent referral to exclude recurrence or a second primary**. *And say where it ranks* — **that surveillance responsibility is arguably the most important part of my role**. Expect the rest of this block to go on groundwork: **why xerostomia produces cervical rather than occlusal caries, why an irradiated mandible heals badly, and what a soft liner does and does not solve**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: monitoring and the overriding principle — two printed answers that become eight	*What do you monitor?* — seven items: **peri-implant health and bone levels**; **mucosal condition, with a low threshold for referral**; **caries and the effectiveness of prevention**; **xerostomia and its management**; **weight and nutrition**; **trismus**; and **his oncological follow-up**. *At what interval?* — **three-monthly, with every mucosal examination documented** — the documentation clause is the mark, because an examination that is not recorded cannot be shown to have happened. *The overriding principle?* — worth saying in almost the case's own words: **his cancer surveillance and his prevention matter more than his implants, and any implant decision belongs in a team that manages irradiated patients regularly. My most valuable contributions are the referral, the prevention and the vigilance.**
14:30–15:00	Closes on the patient and the record	What you say: *"I can help you a great deal with the dryness, the decay and the sore denture, and I want to start there. Implants might well be possible, but that depends on how much radiation reached each part of your jaw, and I need those records from your cancer centre before anyone can answer it — and that decision belongs with a hospital team who look after irradiated patients regularly. One thing to remember: if any ulcer or sore patch lasts more than two weeks, ring me straight away — don't wait for your check-up."* What you write: **the dose, tumour site and date of treatment**; **the request to the oncology centre for the dose-volume data**; **the prevention regime and the denture modifications**; **the discussion of ORN, higher failure rates and impaired healing, with the honest statement that an intolerable denture is not risk-free either**; **the referral and to whom**; **the two-week ulcer rule given and understood**; and **three-monthly recall with the mucosal examination recorded each visit**.

The spine — the twelve answers to have ready if nothing else

1 66 Gy is above the threshold at which ORN risk becomes significant · 2 The field: right posterior mandible at or near full dose, anterior mandible and maxilla likely less · 3 Obtain the dose-volume planning records rather than guess · 4 Three years is more favourable, but the risk is lifelong · 5 Smoking and alcohol — ORN and second primaries · 6 Why irradiated xerostomic mucosa ulcerates under a denture · 7 Denture intolerance as an argument *for* implant support · 8 Prevention: 5000 ppm, varnish, saliva substitutes, candidal infection, diet · 9 I would not place implants in a 66 Gy field in general practice · 10 Hyperbaric oxygen — contested, not routine, the receiving unit advises · 11 Lowest-dose site, fewest implants, overdenture not fixed · 12 Three-monthly recall, and any ulcer beyond two weeks referred urgently.

Where the marks are lost in this case

Answering the implant question before establishing the dose to the site. Candidates hear "implants after radiotherapy" and recite failure rates. The examinable insight is narrower: **a right tonsillar field irradiates the right posterior mandible at or near full dose while the anterior mandible and maxilla may have received considerably less**, so **the answer is site-specific**, and **the dose-volume planning records change it more than anything you can determine clinically**. Ask for them, out loud, before offering an opinion.

Presenting it as a simple prohibition. The easy failure is "he has had 66 Gy, so no implants". The case rejects that: **his denture intolerance is one of the strongest arguments for implant support**, and

continuing with an intolerable denture is not benign either. The mark is in framing it as **a balance of risks assessed by a team that does this regularly** — with the design rules volunteered: **lowest-dose site, fewest implants, overdenture rather than fixed, maximum cleansability**.

Missing that the surveillance is the job. The competence line is firm and easy to state — **I would not place implants in a field that received 66 Gy in general practice; this goes to a restorative consultant or a hospital implant service experienced with irradiated patients, and the decision is made jointly with his oncology team**. The mark most often left behind is what you do *instead*: **intensive prevention, three-monthly recall, weight, trismus and xerostomia monitored, every mucosal examination documented**, and **any ulcer persisting beyond two weeks referred urgently to exclude recurrence or a second primary**.

SOE Case 44 — Nerve injury caused by another clinician

THE CASE

A patient attends your practice 6 weeks after implant placement at the **36** by a dentist she has since stopped attending. She has **persistent numbness and burning pain in the left lower lip and chin**. A periapical shows the implant apex **apparently within the mandibular canal**. She is angry and asks you to "sort it out" and to write a report for her solicitor.

Domain 1

"A patient with a persistent neurosensory disturbance six weeks after implant placement, with radiographic evidence suggesting the implant is within the mandibular canal, presenting with both a clinical problem and a medico-legal one. My priorities, in order, are: **her clinical care**, an accurate independent assessment, and appropriate professional conduct regarding the report."

Q What is the clinical significance of six weeks?

"It is late. The evidence favours **decompression within 24 to 36 hours** where an implant is impinging on the nerve, and the window for any surgical intervention is generally measured in weeks. At six weeks the prognosis for full recovery is substantially reduced, and the presence of **burning pain — dysaesthesia — is a poor prognostic sign** suggesting neuropathic pain rather than simple conduction block."

Q How would you assess her?

"A structured neurosensory assessment. **Map the affected area onto a diagram with measurements** and photograph it. Characterise the disturbance: anaesthesia, hypoaesthesia, paraesthesia or dysaesthesia; whether there is allodynia or hyperalgesia. Perform mechanoceptive tests — light touch with cotton wool, static two-point discrimination, brush stroke direction — and nociceptive tests — sharp/blunt discrimination and thermal. Compare with the contralateral side. Ask about functional impact: eating, drinking, speech, shaving, dribbling, and psychological effect. And obtain a **small field-of-view CBCT** to establish definitively whether the implant is within or adjacent to the canal."

Domain 2

Q: What would you do?

"**Refer urgently** to an oral and maxillofacial surgery unit with experience in trigeminal nerve injury — today, not by routine letter. The decisions about whether to remove the implant now, and about any nerve exploration, belong there. In the meantime: explain honestly what I have found and what I do not know; discuss management of **neuropathic pain** — which may require medication such as a tricyclic or gabapentinoid, prescribed by the appropriate clinician, and which is not managed by removing the implant alone; and provide support and clear information. I would **not remove the implant myself** at six weeks without specialist advice, since removal at this stage may not improve the sensory outcome and may cause further injury."

Q Would you remove the implant if it is clearly in the canal?

"That is a specialist decision at this stage. Early removal — within 24 to 36 hours — is well supported. At six weeks, the injury is established; removal may still be indicated if the implant is causing ongoing compression, but it should be done on specialist advice and with the patient's understanding that it may not restore sensation. I would not act unilaterally."

Q She asks you to write a report for her solicitor. How do you respond?

"Carefully and professionally. I would explain that I am happy to provide a **factual report of my own findings** — what she told me, what I examined, what I measured, what the imaging showed, and what I did — and that this is a **factual report, not an expert opinion on another clinician's care**. I would explain that a report commenting on whether the previous treatment fell below an acceptable standard requires an **appropriately**

instructed independent expert, and that I am not in that role. I would **contact my own indemnity provider before writing anything**, obtain her **written consent** for disclosure, and ensure the report is dated, signed, and based only on my own records. I would not speculate, would not use emotive language, and would not put an opinion about a colleague into her clinical record as if it were a finding."

Q She asks you directly whether the previous dentist was negligent.

" 'I understand why you're asking, and I'm sorry this has happened to you. I can tell you what I've found: the implant appears to be very close to or within the nerve canal, and you have a significant sensory disturbance. What I can't do is give you an opinion about another dentist's decisions — that isn't my role, and I don't have their records or their reasoning. What I can do is make sure you're seen quickly by the right specialist, give you a factual report of what I've found, and tell you how to raise a formal concern if you want to.' "

Domain 3

Q: What information would you give her about prognosis?

"Honestly and without false reassurance: that some sensory disturbances improve over weeks to months; that recovery after six weeks with dysaesthesia is less likely to be complete; that the specialist assessment will give a better estimate; and that even where sensation does not fully recover, neuropathic pain can often be managed. I would avoid both minimising and catastrophising."

Domain 4

Q: What would you record?

"Everything, contemporaneously and factually: her account in her own words, the date of the original surgery, my examination findings with the mapped diagram and measurements, the imaging and my report of it, my advice, the referral made and its urgency, what I said about the report and about raising concerns, and her consent to disclosure. This record may be scrutinised, and its value lies in its being contemporaneous and neutral."

Q What is the wider lesson for your own practice?

"That posterior mandibular implants demand **CBCT planning with a documented 2 mm safety margin, drill stops, direction indicators with radiographic verification, and a recorded pre-operative baseline sensory assessment** — and that any report of post-operative numbness must be seen **the same day**, not at a routine review. I would audit my own compliance with those standards."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 818 words**. At 140 words a minute that is **five minutes fifty**, and nine questions at five seconds each adds **forty-five seconds**: call it **six and a half minutes**. It does not overrun; it leaves **eight and a half minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything you would say out loud, and **each is really three to six short questions** — the neurosensory assessment alone is six, the solicitor's report five. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the contemporaneous record and the audit of your own posterior mandibular protocol.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the periapical in front of you	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	State the problem, then your priorities in order, because the order is the answer: **a persistent neurosensory disturbance six weeks after implant placement at the 36 by another clinician, with radiographic evidence suggesting the implant is within the mandibular canal, and burning pain as well as numbness**. Then: **she presents with both a clinical problem and a medico-legal one, and my priorities in order are her clinical care, an accurate independent assessment, and appropriate professional conduct regarding the report**. Say those three in that order in the first minute and the case runs itself. Then: **six weeks is late, and I need to know today whether anything is time-critical**.
1:20–4:30	Domain 1: the significance of six weeks, then the assessment — two printed answers that become nine	*The significance of six weeks?* — **it is late**, then the numbers: **decompression is favoured within 24 to 36 hours where an implant is impinging on the nerve**, and **the window for surgical intervention is measured in weeks**. *Prognosis?* — **at six weeks the prospect of full recovery is substantially reduced**. *And the burning pain?* — volunteer it: **burning pain is dysaesthesia, a poor prognostic sign suggesting neuropathic pain rather than simple conduction block**. Then the assessment, printed as one paragraph and examined as six: **map the area onto a diagram with measurements and photograph it**; **characterise it — anaesthesia, hypoaesthesia, paraesthesia or dysaesthesia, with or without allodynia or hyperalgesia**; **mechanoceptive tests — cotton wool, static two-point discrimination, brush stroke direction**; **nociceptive tests — sharp/blunt and thermal**; **compare with the contralateral side**; **functional impact — eating, drinking, speech, shaving, dribbling, and the psychological effect**. *Imaging?* — **a small field-of-view CBCT to establish definitively whether the implant is within or adjacent to the canal**: the periapical raises the question, the CBCT answers it.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: referral, removal, and the report — four printed answers that become twelve	*What would you do?* — **refer urgently to an oral and maxillofacial surgery unit with experience in trigeminal nerve injury — today, not by routine letter**, because **the decisions about removal and nerve exploration belong there**. *In the meantime?* — **explain honestly what I have found and what I do not know**; **discuss neuropathic pain, which may need a tricyclic or a gabapentinoid prescribed by the appropriate clinician and is not managed by removing the implant alone**; **provide support and clear information**. *Would you remove it yourself?* — **no, not at six weeks without specialist advice; removal now may not improve the sensory outcome and may cause further injury**. *But it is clearly in the canal* — hold the line: **at six weeks the injury is established; removal may still be indicated for ongoing compression, but on specialist advice and with her understanding that it may not restore sensation. I would not act unilaterally.** *The report for her solicitor?* — four answers, each scoring separately: **a factual report of my own findings — what she told me, what I examined and measured, what the imaging showed and what I did — not an expert opinion on another clinician's care**; **an opinion on whether the previous treatment fell below an acceptable standard requires an appropriately instructed independent expert, and I am not in that role**; **contact my indemnity provider before writing anything, and obtain her written consent for disclosure**; **dated, signed, based only on my own records**. *The prohibitions?* — **no speculation, no emotive language, and no opinion about a colleague in her clinical record as if it were a finding**. *Was the previous dentist negligent?* — have the words: *"I'm sorry this has happened. I can tell you what I've found: the implant appears to be very close to or within the nerve canal, and you have a significant sensory disturbance. What I can't do is give an opinion about another dentist's decisions — that isn't my role, and I don't have their records or their reasoning."*
7:30–11:30	Domain 3: prognosis, honestly — one printed answer that becomes five	*Prognosis?* — **honestly and without false reassurance**, in four parts: **some sensory disturbances improve over weeks to months**; **recovery after six weeks with dysaesthesia is less likely to be complete**; **the specialist assessment will give a better estimate**; and **even where sensation does not fully recover, neuropathic pain can often be managed**. *The register?* — say the phrase: **avoid both minimising and catastrophising**. Expect the rest of the block to go on groundwork: **the course of the inferior alveolar nerve at the 36**, **neurapraxia, axonotmesis and neurotmesis and what each means for recovery**, and **why a periapical cannot resolve a buccolingual relationship that a CBCT can**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: the record, and the lesson for your own practice — two printed answers that become seven	*What would you record?* — **everything, contemporaneously and factually**: **her account in her own words**; **the date of the original surgery**; **the findings, with the mapped diagram and measurements**; **the imaging and my report of it**; **my advice**; **the referral and its urgency**; **what I said about the report and about raising concerns**; and **her consent to disclosure**. *Why does the manner matter?* — **this record may be scrutinised, and its value lies in being contemporaneous and neutral**. *The wider lesson?* — five standards offered as your own protocol: **CBCT planning with a documented 2 mm safety margin**; **drill stops**; **direction indicators with radiographic verification**; **a recorded pre-operative baseline sensory assessment**; and **any post-operative numbness seen the same day, not at a routine review**. *And then?* — **audit my own compliance**, which turns a reflection into governance.
14:30–15:00	Closes on the patient and the record	What you say: *"I'm sorry this has happened. The implant looks to be very close to or inside the nerve canal, and the numbness and burning are real and significant. The most useful thing I can do today is get you seen urgently by a maxillofacial team who deal with nerve injuries — not a routine letter, today. I'll write you a factual report of exactly what I found and did, and tell you how to raise a formal concern. What I can't do is give an opinion on another dentist's treatment, because I wasn't there and I don't have their records."* What you write: **her account verbatim and the date of the original surgery**; **the mapped diagram, measurements and photographs**; **the CBCT and your report of it**; **the urgent referral, to whom and when**; **the advice on neuropathic pain**; **what you said about the scope of the report and about raising concerns**; **her written consent**; and **that you offered no opinion on the previous care** — with no adjective anywhere in the entry.

The spine — the twelve answers to have ready if nothing else

1 Priorities in order: her care, an independent assessment, professional conduct on the report · 2 Six weeks is late — decompression is supported within 24 to 36 hours · 3 Burning pain is dysaesthesia and a poor prognostic sign · 4 Map, measure and photograph the affected area · 5 Mechanoceptive and nociceptive testing against the contralateral side · 6 Small field-of-view CBCT to establish the relationship definitively · 7 Urgent referral to an OMFS unit experienced in nerve injury — today · 8 Do not remove the implant unilaterally at six weeks · 9 Neuropathic pain is not treated by removing the implant · 10 A factual report of my own findings, not an expert opinion on a colleague · 11 Indemnity provider first, written consent, dated and signed · 12 Your own standards: a 2 mm margin, drill stops, baseline sensory testing, same-day review of numbness.

Where the marks are lost in this case

Rushing to remove the implant. It is the instinctive answer and it is wrong here. **Early removal within 24 to 36 hours is well supported; at six weeks the injury is established, removal may not improve the sensory outcome and may cause further injury.** The competence line is explicit: **a specialist decision, referred urgently today to an OMFS unit experienced in trigeminal nerve injury** — and the point candidates never volunteer is that **neuropathic pain needs managing in its own right**, not by unscrewing a fixture.

Blurring what you owe her with what you must not do. She is angry and asks directly about negligence. What you owe **this patient**: **a full independent neurosensory assessment, mapped, measured and photographed; a small field-of-view CBCT; an urgent referral; honest prognostic information; and a factual report of your own findings with her written consent**. What you must not do: **speculate about a colleague, give an expert opinion you were never instructed to give, or record an opinion about another clinician in her notes as though it were a finding**. Have the sentence ready — saying it fluently is the mark.

Writing anything before ringing your indemnity provider. Candidates offer the report willingly and forget the sequence: **indemnity provider first, then her written consent, then a dated, signed report based only on your own records**. And do not stop there — the last domain hands you the governance mark: **CBCT planning with a documented 2 mm safety margin, drill stops, direction indicators with radiographic verification, a recorded pre-operative baseline sensory assessment, and any post-operative numbness seen the same day** — followed by **an audit of your own compliance**.

SOE Case 45 — The patient who wants a healthy implant crown removed

THE CASE

35-year-old woman with a screw-retained implant crown at the **21**, fitted 18 months ago. It is biologically and mechanically healthy — no bleeding, no bone loss, correct occlusion. She dislikes it: she says it is "half a shade too yellow" and "doesn't quite match". You and two colleagues consider the match good. She has had two previous crowns remade elsewhere for the same reason. She has a low smile line.

Domain 1

"A biologically and mechanically successful restoration with which the patient is dissatisfied, in the context of **a low smile line and a history of two previous remakes elsewhere for the same complaint**. That pattern is the significant finding here, and it changes the case from a technical problem into one about expectation, perception and possibly psychological factors."

Q What concerns you about the history?

"Serial dissatisfaction with successive clinicians, a focus on a defect that others cannot perceive, a discrepancy between the objective finding and her distress, and a low smile line meaning the tooth is barely visible in function. These are recognised features that should prompt consideration of **body dysmorphic disorder** or another underlying dissatisfaction. I would not diagnose that myself, but I would recognise the pattern and behave accordingly."

Q How would you assess this properly rather than dismissively?

"I would take her complaint seriously and examine objectively. Standardised photographs in controlled lighting; shade assessment under a colour-corrected light with the teeth hydrated, comparing cervical, middle and incisal thirds; cross-polarised photographs if available; assessment of value, chroma, hue, translucency and surface texture separately, since a mismatch is often in **value or texture rather than hue**; assessment of the emergence, contour and the mucosal margin; and a genuine attempt to identify what specifically she is seeing. Sometimes the patient is right and three clinicians are wrong — a difference in surface lustre or incisal translucency can be real and hard to articulate."

Q What would you ask her?

"What specifically bothers her, and in what circumstances — is it in a mirror, in photographs, in particular lighting? What happened with the previous two remakes and why they did not satisfy her. What she is hoping will be different this time. And, gently, how much she thinks about it and whether it affects her daily life — because the answer to that last question is the one that matters most."

Domain 2

Q: What are the options?

"**Do nothing**, with an honest explanation and support.
Modify — external staining and reglazing, or surface texture adjustment and repolishing, which is minimally invasive and occasionally solves it.
Remake the crown, potentially with the technician present at a shade appointment or with a custom shade tab.
Whiten the adjacent natural teeth — if she perceives the crown as too yellow, and the natural teeth are the reference, bleaching the natural teeth and then remaking the crown to the new shade may be the actual answer she is looking for.
Decline further treatment and, with her agreement, seek an opinion from a colleague or suggest she discuss her concerns with her general medical practitioner."

Q What would you do?

"I would **not proceed straight to a third remake**. My approach would be: 1. Take proper records and demonstrate them to her — showing her the photographs and the shade analysis is often more powerful than telling her. 2. Offer a **reversible trial**: a **provisional crown made to the shade she believes she wants**, worn for a few weeks. If it satisfies her, I have a clear target for the definitive restoration. If it does not, that is important information for both of us, gathered without destroying anything. 3. Discuss bleaching the natural teeth if the mismatch is directional. 4. Be honest about the possibility that a further remake may not resolve her dissatisfaction, given the history — and say that kindly, not as an accusation. 5. If the pattern is that of body dysmorphic concern, **decline elective aesthetic re-treatment**, explain why, and suggest, carefully and non-judgementally, that a discussion with her doctor may help."

Q How do you have that last conversation without offending her?

" 'I want to be honest with you, because I think that's more use to you than another crown. Three of us have looked at this and we can't see what you're seeing, and you've had two remade already without it feeling right. That tells me that another remake might not solve it either, and I'd rather not take your money and your time for something that may leave you feeling the same. What I'd like to do is try a temporary version first so we can test it without committing. And I'd gently say — if this is on your mind a lot, it might be worth mentioning to your doctor, because that's a very common thing and there is help for it.' "

Q Is she entitled to insist you remake it?

"She is entitled to request it. **Patient autonomy allows a patient to refuse treatment, not to compel a clinician to provide treatment that is not in their interest.** If I judge that a further remake is unlikely to help and may reinforce the preoccupation, I may decline — provided I explain my reasoning, offer to help her obtain another opinion, and record the discussion. Declining must never be dismissive."

Domain 3

Q: If you do remake it, how would you maximise the chance of success?

"Involve the **technician directly**, ideally with the patient present for shade selection. Take shade early in the appointment with hydrated teeth under corrected lighting, recording value, chroma and hue separately plus translucency, texture and lustre. Photograph with the tab in the same plane, with a grey card, and cross-polarised. Communicate the **stump shade**. Use a **provisional at the target shade first**. Request a **bisque try-in** so adjustments precede glazing. And agree in writing, in advance, what would constitute an acceptable result."

Domain 4

Q: What would you record?

"Her complaint in her own words; my objective findings and those of my colleagues; the photographs and shade analysis; the options offered including no treatment; the reversible trial proposed; my reasoning if I decline further remakes; the suggestion of a medical discussion and how it was received; and her response. This record protects her — by making the reasoning visible to any future clinician — as much as it protects me."

Q What is the general principle?

> "That the outcome of an aesthetic restoration is judged against the patient's perception, and that **when perception and objective finding diverge repeatedly, more dentistry is rarely the answer**. Recognising that is a clinical skill, and handling it with kindness rather than dismissal is a professional one."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 955 words**. At 140 words a minute that is **six minutes fifty**, and ten questions at five seconds each adds **fifty seconds**: call it **seven and a half minutes**. It does not overrun; it leaves **seven and a half minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything said out loud, and **each is really three to six short questions**: the shade assessment alone is six. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the record that protects her as much as you, and the principle that more dentistry is rarely the answer.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Name what is and is not the problem: **a biologically and mechanically successful screw-retained crown at the 21, eighteen months old — no bleeding, no bone loss, correct occlusion — with which the patient is dissatisfied**. Then the finding that reframes the case, and it must come from you: **a low smile line, so the tooth is barely visible in function, and two previous crowns remade elsewhere for the same complaint — that pattern is the significant finding, and it changes this from a technical problem into one about expectation, perception and possibly psychological factors**. Then the working position: **I would take her complaint seriously and assess it objectively before deciding anything** — because a candidate who opens by dismissing her has lost Domain 2 before it begins.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the pattern, the assessment, the questions — three printed answers that become eleven	*What concerns you about the history?* — four features named as features, not as a diagnosis: **serial dissatisfaction with successive clinicians**; **a focus on a defect others cannot perceive**; **distress out of proportion to the finding**; **a low smile line**. *And what do they suggest?* — **recognised features that should prompt consideration of body dysmorphic disorder or another underlying dissatisfaction** — with the qualifier that earns the mark: **I would not diagnose that myself, but I would recognise the pattern and behave accordingly**. *How do you assess it properly rather than dismissively?* — **standardised photographs in controlled lighting**; **shade under a colour-corrected light, teeth hydrated, comparing cervical, middle and incisal thirds**; **cross-polarised photographs if available**; **value, chroma, hue, translucency and texture assessed separately, since a mismatch is often in value or texture rather than hue**; **emergence, contour and the mucosal margin**; and **a genuine attempt to identify what she is seeing**. *The honest caveat?* — **sometimes the patient is right and three clinicians are wrong: a difference in lustre or incisal translucency can be real and hard to articulate**. *What would you ask her?* — **what specifically bothers her, and when — mirror, photographs, particular lighting**; **what happened with the previous two remakes**; **what she hopes will be different**; and, **gently, how much she thinks about it and whether it affects her daily life — because that last answer matters most**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, the plan, the conversation, autonomy — four printed answers that become thirteen	*Options?* — five: **do nothing, honestly explained**; **modify — external staining and reglazing, or texture adjustment and repolishing**; **remake, with the technician present at a shade appointment**; **whiten the adjacent natural teeth — if they are the reference, bleaching them and remaking to the new shade may be the actual answer**; **decline, and with her agreement seek a colleague's opinion or suggest she discuss it with her GP**. *What would you do?* — open with the refusal that is not a dismissal — **I would not proceed straight to a third remake** — then five steps: **take proper records and demonstrate them to her, because showing is more powerful than telling**; **offer a reversible trial — a provisional at the shade she believes she wants, worn for a few weeks — if it satisfies her I have a target; if not, that is important information, gathered without destroying anything**; **discuss bleaching the natural teeth if the mismatch is directional**; **be honest that a further remake may not resolve her dissatisfaction — kindly, not as an accusation**; **and if the pattern is that of body dysmorphic concern, decline elective aesthetic re-treatment, explain why, and suggest a discussion with her doctor may help**. *How do you say it without offending her?* — have the words: *"I want to be honest with you, because that's more use to you than another crown. Three of us can't see what you're seeing, and you've had two remade already without it feeling right — so another remake might not solve it either, and I'd rather not take your money and your time for that. Let's try a temporary version first and test it without committing."* *Is she entitled to insist?* — the distinction, crisply: **she is entitled to request it. Autonomy allows a patient to refuse treatment, not to compel a clinician to provide treatment that is not in their interest.** *So you may decline?* — **yes, provided I explain my reasoning, help her obtain another opinion, and record the discussion; declining must never be dismissive**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: if you do remake it — one printed answer that becomes eight	*How do you maximise the chance of success?* — eight answers, and this is where an unprepared candidate runs dry: **the technician involved directly, ideally with the patient present for shade selection**; **shade taken early, teeth hydrated, under corrected lighting**; **value, chroma and hue recorded separately, plus translucency, texture and lustre**; **photographs with the tab in the same plane, a grey card, cross-polarised**; **the stump shade communicated**; **a provisional at the target shade first**; **a bisque try-in so adjustments precede glazing**; and the one that is as much consent as technique — **agree in writing, in advance, what would count as an acceptable result**. Expect the rest of the block to go on groundwork: **why value is perceived before hue, and why emergence and the mucosal margin can make a correct shade look wrong**.
11:30–14:30	Domain 4: the record and the principle — two printed answers that become six	*What would you record?* — **her complaint in her own words**; **your findings and those of your colleagues**; **the photographs and shade analysis**; **the options offered, including no treatment**; **the reversible trial**; **your reasoning if you decline further remakes**; **the suggestion of a medical discussion and how it was received**; and **her response**. *Why does it matter here?* — the clause that scores: **this record protects her — by making the reasoning visible to any future clinician — as much as it protects me**. *The general principle?* — **an aesthetic outcome is judged against the patient's perception**, and **when perception and objective finding diverge repeatedly, more dentistry is rarely the answer — recognising that is a clinical skill; handling it with kindness rather than dismissal is a professional one**.
14:30–15:00	Closes on the patient and the record	What you say: *"I don't think another crown is the first thing to try. Let me take proper photographs and a shade analysis and show you what I'm seeing. Then let's make you a temporary at exactly the shade you think you want, and you wear it for a few weeks. If that's right, I know what to aim for; if it isn't, we've lost nothing and we've both learned something."* What you write: **her complaint verbatim**; **the findings, including the two colleagues' assessments**; **the photographic and shade record**; **every option, including no treatment and declining**; **the reversible trial**; **the agreed written definition of an acceptable result**; **your reasoning if you decline**; **the suggestion of a discussion with her doctor and how it was received**; and **the offer of a further opinion**.

The spine — the twelve answers to have ready if nothing else

1 Biologically and mechanically successful; the dissatisfaction is the finding · 2 Two previous remakes plus a low smile line — the pattern · 3 Recognise body dysmorphic concern; do not diagnose it · 4 Value, chroma, hue, translucency, texture and lustre assessed separately · 5 Sometimes the patient is right and three clinicians are wrong · 6 How much does she think about it · 7 Whitening the natural teeth if the mismatch is directional · 8 The reversible provisional trial at her target shade · 9 Autonomy permits refusal,

not compulsion · 10 Declining is legitimate — with reasoning, another opinion, and a record · 11 Technician present, stump shade, bisque try-in, criteria agreed in writing · 12 When perception and finding diverge repeatedly, more dentistry is rarely the answer.

Where the marks are lost in this case

Dismissing her, or capitulating to her. Both fail. "The crown is fine and I'd explain that" ignores that **a difference in surface lustre or incisal translucency can be real and hard to articulate**; agreeing to a third remake ignores **two previous remakes elsewhere for the same complaint**. The mark is in the middle: **objective records demonstrated to her**, then **a reversible provisional at the shade she believes she wants**, which turns an argument about perception into an experiment that destroys nothing.

Never asking the question that matters. Candidates ask about lighting and mirrors and stop. The decisive question is **how much she thinks about it and whether it affects her daily life** — asked gently, and asked because the answer changes the management. Volunteer alongside it that **serial dissatisfaction, a defect others cannot perceive, distress out of proportion, and a low smile line should prompt consideration of body dysmorphic disorder** — with the boundary in the same breath: **recognise the pattern; do not diagnose it**.

Not knowing that you are allowed to say no. The competence line here is not surgical; it is the point at which further elective aesthetic treatment becomes harmful. **Autonomy allows a patient to refuse treatment, not to compel a clinician to provide treatment that is not in their interest**, so **you may decline a third remake — provided you explain the reasoning, help her obtain another opinion, suggest non-judgementally that a conversation with her doctor may help, and record all of it**. That record **protects her as much as you**, because otherwise the next clinician starts the cycle again.

SOE Case 46 — The second opinion

THE CASE

A 58-year-old man attends for a second opinion. Another practice has proposed **bilateral lateral window sinus elevation with block grafting and eight implants** to restore his upper posterior teeth, at considerable cost. He has an intact upper anterior dentition 15 to 25, a stable lower arch with all teeth present to the second molars, and no complaints — he attended for a routine check-up and was offered the plan. CBCT shows 4 mm of bone bilaterally below the sinuses.

Domain 1

"A man with no symptoms and no complaint who has been offered extensive, expensive and irreversible surgery. My first task is to establish **what problem the proposed treatment is meant to solve**, because if he has a stable, functional shortened arch and no complaint, the starting point is not which technique but whether any treatment is needed at all."

Q What would you assess?

"His **function and symptoms** in his own words — can he eat what he wants, is there pain, is anything changing? The **occlusal stability** — over-eruption of the lower molars into the upper spaces, drifting, wear, and whether the anterior teeth are being overloaded. Periodontal status and the prognosis of the remaining teeth. The CBCT, independently. His expectations and what he was told. And what has changed recently, if anything, that led to this proposal."

Q How do you approach giving a second opinion without criticising a colleague?

"By confining myself to my own assessment and by describing options rather than passing judgement. 'I can't comment on another dentist's reasoning because I don't have their records or their discussion with you. What I can do is examine you myself, tell you what I find, and set out the options as I see them, including doing nothing. Then you'll have two views and can decide.' That is both professionally correct and more useful to him than an opinion about someone else."

Domain 2

Q: What are the options?

"**No treatment with monitoring.** He has occlusion to the second premolars bilaterally — a **shortened dental arch**, for which the evidence indicates adequate function, comfort and satisfaction in most people, without increased TMD or periodontal breakdown. He is asymptomatic. This is the baseline against which everything else must justify itself, and I would present it first and seriously.
Limited treatment — for example, managing over-eruption of the lower molars if present, or replacing one first molar per side if there is a genuine functional deficit.
The proposed plan — bilateral lateral window elevation, block grafting and eight implants. It is technically feasible but represents major surgery, substantial cost, months of treatment, significant morbidity, and a lifelong maintenance burden, to treat a situation causing him no trouble.
Intermediate options — short implants where residual height allows, though at 4 mm that is marginal; or a removable prosthesis, which he is unlikely to want or need."

Q What would you recommend?

"That he does nothing for now, with **active monitoring**: photographs and study casts to detect over-eruption and drift, periodontal maintenance, and review. I would tell him that if his function deteriorates or the occlusion becomes unstable, the options remain open and we can revisit them. I would be explicit that this is a **positive recommendation based on evidence**, not a refusal to treat — and that the fact a treatment is possible does not make it necessary."

Q How would you explain the risk-benefit balance?

> " 'The plan you've been offered would work, technically. But it's a big undertaking — two sinus operations, bone grafting, eight implants, the best part of a year, a lot of money, and a commitment to cleaning and check-ups for the rest of your life. And at the moment, you have no symptoms and you can eat what you like. There's good evidence that a set of teeth back to the premolars gives most people all the function they need. So my honest advice is to leave it alone, keep an eye on it, and treat if something changes. If you want the fuller reconstruction, it's available — but I'd want you to choose it because you want it, not because you've been told you need it.' "

Q What if he says he does want it?

> "Then I would take him seriously, and make sure he is choosing it fully informed: the number of procedures, the risks of lateral window elevation and block grafting including membrane perforation, sinusitis, graft failure and donor-site morbidity, the timescale, the total cost including maintenance, and the alternatives. **Bilateral lateral window sinus elevation with block grafting is Complex and firmly outside my scope**, so I would refer him — to a specialist unit, with a full assessment and my own records — rather than provide it. I would also suggest he ask the receiving clinician the same question: what problem is this solving?"

Domain 3

Q: If treatment were eventually undertaken, what would your role be?

> "Assessment and planning, the restorative phase if agreed with the surgical team, and long-term maintenance. I would want the shared-care arrangement documented from the outset — who does what, who provides maintenance, and who manages complications."

Domain 4

Q: What would you record?

> "That he attended for a second opinion; his symptoms and function in his own words; my independent findings including the CBCT; the options I presented **including no treatment and the evidence for the shortened arch**; my recommendation and reasoning; that I made no comment on the other practice's plan; and his decision. I would also give him a written summary, so that whatever he decides, he decides with the reasoning in front of him."

Q What is the underlying principle?

> "That **the ability to perform a treatment is not an indication for it**. The examination is explicitly about safe, risk-assessed, harmless overall patient care — and the most important word in that phrase is 'harmless'. Recommending no treatment, with monitoring and a clear rationale, is a defensible and often correct clinical decision."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 837 words**. At 140 words a minute that is **six minutes**, and nine questions at five seconds each adds **forty-five seconds**: call it **six and three-quarter minutes**. It does not overrun; it leaves **eight and a quarter minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything said out loud, and **each is really three to five short questions**: the options answer alone is four. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the written summary you hand him and the principle that ability is not indication.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT in front of you	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	Lead with the absence of a complaint, because it is the whole case: **a 58-year-old man with no symptoms who attended elsewhere for a routine check-up and was offered bilateral lateral window sinus elevation with block grafting and eight implants, at considerable cost**. Then the anatomy: **an intact upper dentition 15 to 25, a stable lower arch complete to the second molars, and 4 mm of bone bilaterally below the sinuses**. Then the sentence that structures everything: **my first task is to establish what problem the proposed treatment is meant to solve, because if he has a stable, functional shortened arch and no complaint, the starting point is not which technique but whether any treatment is needed at all**. Volunteer in the same breath that **you will not be commenting on another practice's reasoning**.
1:20–4:30	Domain 1: the assessment, and how to give a second opinion — two printed answers that become nine	*What would you assess?* — six items, one per question: **his function and symptoms in his own words — can he eat what he wants, is there pain, is anything changing?**; **occlusal stability — over-eruption of the lower molars, drifting, wear, anterior overload**; **periodontal status and the prognosis of the remaining teeth**; **the CBCT, read independently**; **his expectations and what he was told**; and **what changed recently, if anything, that led to this proposal**. *How do you give a second opinion without criticising a colleague?* — the method first: **confine myself to my own assessment and describe options rather than pass judgement**. Then the words: *"I can't comment on another dentist's reasoning because I don't have their records or their discussion with you. What I can do is examine you myself, tell you what I find, and set out the options as I see them, including doing nothing."* Then the justification, which is the mark: **that is both professionally correct and more useful to him than an opinion about someone else**. Expect the rest of the block to go below the implant material: **the shortened dental arch concept and the evidence behind it, and how you read 4 mm of subantral bone**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, recommendation, the conversation, and what if he insists — four printed answers that become twelve	*Options?* — four, and the order is itself an answer: **no treatment with monitoring — he has occlusion to the second premolars bilaterally, a shortened dental arch, for which the evidence indicates adequate function, comfort and satisfaction in most people without increased TMD or periodontal breakdown; the baseline against which everything else must justify itself, presented first and seriously**; **limited treatment — managing over-eruption of the lower molars, or replacing one first molar per side for a genuine functional deficit**; **the proposed plan — feasible, but major surgery, substantial cost, months of treatment, significant morbidity and a lifelong maintenance burden, to treat a situation causing him no trouble**; **intermediate options — short implants where height allows, marginal at 4 mm, or a removable prosthesis he is unlikely to want**. *Recommendation?* — **nothing for now, with active monitoring: photographs and casts to detect over-eruption and drift, periodontal maintenance, review; if function deteriorates or the occlusion becomes unstable, the options remain open**. *And name it* — **a positive recommendation based on evidence, not a refusal to treat: the fact that a treatment is possible does not make it necessary**. *If he does want it?* — **take him seriously and consent him fully: the number of procedures; the risks of lateral window elevation and block grafting — membrane perforation, sinusitis, graft failure, donor-site morbidity; the timescale; the total cost including maintenance; the alternatives**. *Your own position?* — **bilateral lateral window elevation with block grafting is Complex and firmly outside my scope, so I would refer to a specialist unit with a full assessment and my own records rather than provide it** — and **suggest he asks the receiving clinician the same question: what problem is this solving?**
7:30–11:30	Domain 3: your role if treatment is eventually undertaken — one printed answer that becomes five	*What would your role be?* — **assessment and planning**; **the restorative phase, if agreed with the surgical team**; **long-term maintenance**. *And what must be settled first?* — **the shared-care arrangement documented from the outset**: **who does what**, **who provides maintenance**, and **who manages complications** — the third being the one always left undecided and always the one that matters. This is a short printed answer in a long block, so expect the time to go on the procedures themselves: **why a lateral window is raised, how a membrane perforation is managed, and what 4 mm of residual height does to the choice between a lateral window and a short implant** — answered accurately, then closed with **and none of that is work I would undertake myself**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: the record and the principle — two printed answers that become seven	*What would you record?* — **that he attended for a second opinion**; **his symptoms and function in his own words**; **your independent findings, including your reading of the CBCT**; **the options, including no treatment and the shortened-arch evidence**; **your recommendation and reasoning**; **that you made no comment on the other practice's plan**; and **his decision**. *Anything else?* — the item candidates omit: **a written summary given to him, so that whatever he decides, he decides with the reasoning in front of him**. *The underlying principle?* — **the ability to perform a treatment is not an indication for it**. *Tie it to the standard* — **this examination is about safe, risk-assessed, harmless overall patient care, and the most important word is "harmless": recommending no treatment, with monitoring and a clear rationale, is a defensible and often correct decision**.
14:30–15:00	Closes on the patient and the record	What you say: *"You came in for a check-up and were offered a very big operation. I can't comment on why — I don't have their notes. What I can tell you is what I find: you're comfortable, you eat what you like, and you have teeth meeting back to your premolars, which for most people is all the function they need. My advice is to do nothing for now, take photographs and models so we can see if anything shifts, and treat only if something changes. That's a recommendation, not a refusal — and if you want the fuller reconstruction, choose it because you want it, not because you were told you need it."* What you write: **the second-opinion context**; **his symptoms and function verbatim**; **your findings and CBCT report**; **every option, with no treatment and the shortened-arch evidence recorded first**; **your recommendation and reasoning**; **the explicit note that you offered no comment on the other plan**; **the written summary handed over**; and **his decision, with the monitoring interval and what would trigger a review**.

The spine — the twelve answers to have ready if nothing else

1 What problem is the proposed treatment meant to solve? · 2 His function and symptoms in his own words · 3 Over-eruption, drift, wear and anterior overload as the objective test of stability · 4 The shortened dental arch, and the evidence behind it · 5 Presenting no treatment first, and seriously · 6 Active monitoring: photographs, casts, periodontal maintenance, review · 7 A positive recommendation, not a refusal to treat · 8 The words for the risk-benefit conversation · 9 If he wants it: procedures, risks, timescale, total cost, alternatives · 10 Lateral window with block grafting is Complex and outside scope — refer · 11 Shared care documented: who does what, who maintains, who manages complications · 12 The ability to perform a treatment is not an indication for it.

Where the marks are lost in this case

Debating the technique instead of the indication. The CBCT is interesting — 4 mm bilaterally, sinus elevation, short implants — and a candidate can spend eight minutes on lateral windows in a man with no complaint. The examinable move is the first one: **establish what problem the treatment is meant to solve**, then **present no treatment first and seriously**, with **the shortened dental arch evidence — adequate function, comfort and satisfaction in most people, without increased TMD or periodontal breakdown** — named as the baseline everything else must justify itself against.

Being drawn into judging the other practice. He will invite it and the examiner will let him. Neither criticise nor defend: **confine yourself to your own assessment and describe options rather than pass judgement**, with the sentence ready — *"I can't comment on another dentist's reasoning because I don't have their records or their discussion with you."* Then volunteer the two things candidates forget: **that you recorded explicitly that you made no comment on the other plan**, and **that you would suggest he put the same question to the receiving clinician**.

Turning a recommendation into a refusal. If he wants the reconstruction after a proper discussion, that is his decision. **Consent him fully — membrane perforation, sinusitis, graft failure, donor-site morbidity, the timescale, the total cost including maintenance** — then draw the competence line plainly: **bilateral lateral window sinus elevation with block grafting is Complex in both dimensions and firmly outside the scope of a general dentist with an interest in implants, so he goes to a specialist unit with your full assessment and the shared-care arrangement documented from the outset**. The principle to say before the bell: **the ability to perform a treatment is not an indication for it**.

SOE Case 47 — Maxillary overdenture versus fixed prosthesis

THE CASE

68-year-old woman, edentulous maxilla for 8 years, dentate mandible (all teeth present, 35 to 45 plus first molars). Her upper complete denture is unstable and she gags. CBCT: adequate bone anteriorly (10–12 mm height, 6–7 mm width from 15 to 25), **pneumatised sinuses posteriorly with 3 mm**. Medium smile line. She wants "teeth fixed in so I don't gag".

Domain 1

"An edentulous maxilla opposing a **complete natural mandibular dentition** — a high-load situation — with adequate anterior bone but severe posterior pneumatisation, in a patient whose principal complaint is **gagging and instability** rather than appearance. That distinction matters: her problem may be solvable by removing palatal coverage, which does not require a fixed prosthesis."

Q Why does the opposing natural dentition matter?

"Because a maxillary prosthesis opposing a full natural mandibular dentition generates substantially higher and less evenly distributed loads than one opposing a denture. It increases the mechanical demand on any implant prosthesis and is a recognised setting for accelerated maxillary ridge resorption."

Q What would you assess?

"Her existing denture critically — extension, post-dam, occlusion, vertical dimension, and whether the gagging is related to palatal coverage, over-extension or an incorrect vertical dimension, because many gaggers are made worse by a poorly fitting denture. Smile line and lip dynamics. **Whether her appearance depends on the labial flange** — the single most important planning question. Interarch space. The condition and prognosis of her lower teeth. Her expectations, and what"fixed" means to her — many patients mean "doesn't move", not "cannot be taken out"."

Domain 2

Q: What are the options?

"**Remake the conventional denture** with attention to extension, post-dam and vertical dimension, and desensitisation techniques for gagging. Low risk and may be sufficient.
Implant-retained maxillary overdenture, typically four to six implants in the anterior maxilla with a bar or studs. This can be **palate-free**, which directly addresses her gagging, retains the labial flange for lip support, and is removable and cleansable. **This is my leading option.**
Fixed full-arch implant prosthesis — would require bilateral sinus elevation given 3 mm posteriorly, or tilted implants; it is Complex, expensive, harder to clean, and cannot provide a labial flange.
No change with adhesive and dietary advice."

Q How do you decide between the overdenture and the fixed option?

"By testing the **flange question** objectively. I would duplicate her denture, remove the flange from the duplicate, and try it in — with photographs — so she can see for herself what a fixed prosthesis without a flange would look like. After eight years of resorption and with a medium smile line, the transition line and lip support will very likely be compromised. I would also explain that her actual complaint — gagging — is caused by **palatal coverage**, and that a **palate-free implant overdenture solves that completely**, whereas a fixed prosthesis solves it at far greater surgical and biological cost."

Q What would you say to her?

" 'I think I can solve the gagging without needing to graft your sinuses. The reason you gag is the plate across the roof of your mouth. If we put four implants in the front of your upper jaw, we can make a denture that clips on very firmly and has **no palate at all** — it won't move when you eat or speak, and you take it out to clean it.

> The fully fixed option would mean two sinus operations and much bigger surgery, and it also can't have the little lip that supports your top lip, which I think you'd notice. Let me show you what that would look like first.' "

Q SAC?

> "Four anterior maxillary implants for an overdenture: surgically **Advanced**, prosthodontically **Advanced**. A fixed full-arch maxillary rehabilitation with bilateral sinus grafting is **Complex** in both dimensions and **outside the scope of this level of practice** — I would refer if she pursued it after understanding the alternatives."

Domain 3

Q: Outline the treatment plan for the overdenture.

> "1. **Remake or copy the upper denture first**, establishing the correct tooth position, vertical dimension, occlusion and lip support. That denture becomes the radiographic and surgical guide and the interim prosthesis. 2. Duplicate in clear acrylic with radiopaque markers; CBCT; plan **four implants between the 15 and 25 regions**, avoiding the **incisive canal**, with adequate anteroposterior spread. 3. Surgery: crestal incision, full-thickness flap, graded drilling with copious irrigation; the anterior maxilla is typically **D3**, so I would under-prepare to improve primary stability. Two-stage submerged healing given the denture will be worn over the site. 4. **Conventional loading at 4 to 6 months** in maxillary D3 bone. Relieve and soft-line the denture throughout. 5. Second stage, then attachment selection: **studs need 8 to 10 mm, a bar needs 12 to 15 mm** of vertical space. With divergent implants or if maximum stability is needed against a natural lower dentition, a **bar** distributes load better; if space is limited and cleaning ability is a concern, studs. 6. **Palate-free design** with a cast metal framework for strength, since removing the palate weakens an acrylic base considerably. 7. Occlusion: balanced or lingualised, carefully developed against her natural lower teeth."

Domain 4

Q: What maintenance would you set out?

> "Attachment inserts replaced as they wear, typically annually or more often; the framework and base checked for fracture; relines as the ridge resorbs; peri-implant probing at every recall; and daily cleaning of the abutments with a single-tufted brush and of the fit surface including the housings. I would give this in writing **with costs**, because unexpected maintenance charges are the commonest source of later dissatisfaction."

Q What would you monitor in the opposing arch?

> "Her natural lower teeth — for wear, mobility, and periodontal change under the increased load — and the occlusion, since changes in the lower arch will alter the loading on the maxillary prosthesis. That surveillance is as much a part of the plan as the implants themselves."

Running this case to time

Time check. This case is written in the older format — **nine long answers, about 850 words**. At 140 words a minute that is **six minutes**, and nine questions at five seconds each adds **forty-five seconds**: call it **just under seven minutes**. It does not overrun; it leaves **eight minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything said out loud, and **each is really three to seven short questions**: the treatment plan alone is seven, printed as a numbered sequence an examiner will take one item at a time. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the maintenance schedule given in writing **with costs**, and the surveillance of her natural lower teeth.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the CBCT and photographs in front of you	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	Name the maxillomandibular relationship first, because it is the mechanical heart of the case: **an edentulous maxilla of eight years opposing a complete natural mandibular dentition — a high-load situation** — with **adequate anterior bone, 10 to 12 mm by 6 to 7 mm from 15 to 25, but severe posterior pneumatisation leaving 3 mm**. Then her complaint, correctly interpreted: **her principal complaint is gagging and instability, not appearance**. Then the sentence that decides the case, volunteered: **that distinction matters, because her problem may be solvable by removing palatal coverage, which does not require a fixed prosthesis**. Then the working position: **a palate-free implant overdenture is my leading option, and I would test the lip support question objectively before anyone commits to anything fixed**.
1:20–4:30	Domain 1: the opposing dentition, then the assessment — two printed answers that become nine	*Why does the opposing natural dentition matter?* — the biomechanics: **it generates substantially higher and less evenly distributed loads than an opposing denture**; **it increases the mechanical demand on any implant prosthesis**; and **it is a recognised setting for accelerated maxillary ridge resorption** — the mechanism behind the eight years of loss you are looking at. *What would you assess?* — the denture first, because the undergraduate groundwork is examined first: **extension, post-dam, occlusion and vertical dimension, and whether the gagging is related to palatal coverage, over-extension or an incorrect vertical dimension — many gaggers are made worse by a poorly fitting denture**. Then **smile line and lip dynamics**; then the one to name as such — **whether her appearance depends on the labial flange, the single most important planning question**; then **interarch space**; **the condition and prognosis of her lower teeth**; and **her expectations, and what "fixed" means to her — many patients mean "doesn't move", not "cannot be taken out"**. That last clause is worth a mark on its own, because it reframes her request rather than accepting it.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, the flange test, the conversation, SAC — four printed answers that become twelve	*Options?* — four, in order of biological cost: **remake the conventional denture — extension, post-dam, vertical dimension, plus desensitisation for gagging; low risk and may be sufficient**; **an implant-retained overdenture, four to six implants anteriorly with a bar or studs** — for four reasons: **it can be palate-free, addressing her gagging directly; it retains the labial flange; it is removable and cleansable; it is my leading option**; **a fixed full-arch prosthesis — requiring bilateral sinus elevation given 3 mm posteriorly, or tilted implants; Complex, expensive, harder to clean, and unable to carry a labial flange**; **no change, with adhesive and dietary advice**. *How do you decide between them?* — the method, not the opinion: **duplicate her denture, remove the flange from the duplicate, try it in and photograph it, so she sees for herself what a fixed prosthesis without a flange looks like**. *Your prediction?* — **after eight years of resorption and with a medium smile line, the transition line and lip support will very likely be compromised**. *And the gagging?* — **her complaint is caused by palatal coverage; a palate-free overdenture solves it completely, a fixed prosthesis at far greater surgical and biological cost**. *SAC?* — **four anterior maxillary implants for an overdenture: surgically Advanced, prosthodontically Advanced. A fixed full-arch maxillary rehabilitation with bilateral sinus grafting is Complex in both dimensions and outside the scope of this level of practice — I would refer if she pursued it after understanding the alternatives.**
7:30–11:30	Domain 3: the overdenture plan in sequence — one printed answer that becomes eight	*Step one?* — the step candidates skip: **remake or copy the upper denture first, establishing tooth position, vertical dimension, occlusion and lip support — it becomes the radiographic and surgical guide and the interim prosthesis**. *Planning?* — **duplicate in clear acrylic with radiopaque markers; CBCT; four implants between the 15 and 25 regions, avoiding the incisive canal, with adequate anteroposterior spread**. *Surgery?* — **crestal incision, full-thickness flap, graded drilling with copious irrigation**; **the anterior maxilla is typically D3, so under-prepare for primary stability**; **two-stage submerged healing, since the denture is worn over the site**. *Loading?* — **conventional, at 4 to 6 months in maxillary D3 bone**, with the denture **relieved and soft-lined throughout**. *Attachments?* — the numbers matter: **studs need 8 to 10 mm, a bar 12 to 15 mm of vertical space**; **with divergent implants, or where maximum stability is needed against a natural lower dentition, a bar distributes load better; where space is limited and cleaning a concern, studs**. *Base?* — **palate-free with a cast metal framework, since removing the palate weakens an acrylic base considerably**. *Occlusion?* — **balanced or lingualised, developed against her natural lower teeth**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: maintenance and the opposing arch — two printed answers that become seven	*What maintenance would you set out?* — five items: **attachment inserts replaced as they wear, typically annually or more often**; **framework and base checked for fracture**; **relines as the ridge resorbs**; **peri-implant probing at every recall**; **daily cleaning of the abutments with a single-tufted brush and of the fit surface including the housings**. *How is it given?* — **in writing, with costs** — and say why, because the reason is the mark: **unexpected maintenance charges are the commonest source of later dissatisfaction**. *The opposing arch?* — **her natural lower teeth monitored for wear, mobility and periodontal change under the increased load**, and **the occlusion, since changes there alter the loading on the maxillary prosthesis**. *How would you rank it?* — **that surveillance is as much a part of the plan as the implants themselves**.
14:30–15:00	Closes on the patient and the record	What you say: *"What's making you gag is the plate across the roof of your mouth, and I can take that away. Four implants at the front, a denture that clips on so firmly it won't move when you eat or talk, and no palate at all — and you take it out at night to clean it. The completely fixed version means two sinus operations, much bigger surgery, harder to keep clean, and it can't have the little lip that holds your top lip out, which after eight years I think you'd notice. Let me show you that on a copy of your own denture first."* What you write: **her complaint verbatim and what "fixed" means to her**; **the flange test — the duplicate with the flange removed, tried in, photographed, and her reaction**; **all options including remaking the denture and no change**; **the SAC assessment and the statement that a fixed full-arch reconstruction with bilateral sinus grafting is outside your scope and would be referred**; **the interarch space and the attachment chosen**; and **the written maintenance schedule with costs, signed and dated**.

The spine — the twelve answers to have ready if nothing else

1 An edentulous maxilla opposing a complete natural mandibular dentition is a high-load situation · 2 A recognised setting for accelerated maxillary ridge resorption · 3 The complaint is gagging and instability, not appearance · 4 Palatal coverage causes the gagging; a palate-free design removes it · 5 Whether her appearance depends on the labial flange — the single most important planning question · 6 The duplicate-denture flange test, tried in and photographed · 7 What "fixed" means to her — "doesn't move", not "can't be taken out" · 8 Remake the denture first; it becomes the guide and the interim prosthesis · 9 Four implants 15 to 25, avoiding the incisive canal, D3 bone, under-prepared, loaded at 4 to 6 months · 10 Studs 8 to 10 mm, bar 12 to 15 mm; a bar against a natural lower dentition · 11 Palate-free with a cast metal framework · 12 Maintenance in writing with costs, and the lower teeth monitored under the increased load.

Where the marks are lost in this case

Taking "teeth fixed in" literally. She said it, but the case tells you what she means: **her complaint is gagging and instability, not appearance**, and **many patients mean "doesn't move", not "cannot be taken out"**. A candidate who plans a fixed full-arch reconstruction against 3 mm of subantral bone has accepted a request instead of making a diagnosis. The winning move is the logic in one sentence: **the gagging is caused by palatal coverage; a palate-free implant overdenture removes it completely; a fixed prosthesis removes it at far greater surgical and biological cost.**

Arguing the flange question instead of testing it. Lip support after eight years of resorption is not settled by opinion. **Cut the flange off a duplicate of her denture, try it in and photograph it** — so **she** sees what a fixed prosthesis without a flange looks like, and the transition line is discussed before anything irreversible happens. Volunteer the honest comparison rather than a sales pitch: the overdenture is **removable, cleansable, relinable, palate-free and flanged**; the fixed option is **two sinus operations, greater cost, harder cleaning, and no flange**.

Naming the competence line in the wrong place. Four anterior maxillary implants for an overdenture are Advanced in both dimensions — demanding, but within reach at this level with proper planning. **A fixed full-arch maxillary rehabilitation with bilateral sinus elevation, where resorption has left 3 mm posteriorly, is Complex in both dimensions and beyond a general dentist with an interest in implants: it is referred, and she is told so at the outset, not mid-treatment.** And do not let the surgery crowd out Domain 4, where the cheapest marks are: **the maintenance schedule in writing with costs**, and **surveillance of her natural lower teeth under the increased load**.

SOE Case 48 — "Teeth in a day"

THE CASE

55-year-old businessman with a **failing upper dentition**: generalised periodontitis Stage IV Grade B, several mobile teeth, a failing anterior bridge, and caries. He has researched "All-on-4" online, wants his upper teeth removed and **fixed teeth fitted the same day**, and has booked a holiday in six weeks. He is a non-smoker, medically fit.

Domain 1

"A patient with a failing upper dentition requesting immediate full-arch fixed rehabilitation, with a fixed deadline. Two things are immediately clear: **full-arch fixed rehabilitation is explicitly outside the scope of my practice at this level**, and the six-week deadline is driving the request rather than the clinical need. My job is to give him a proper assessment and an honest account of the options, not to accommodate the timetable."

Q What would you assess?

"Full periodontal charting and radiographs; the prognosis of every tooth individually, because 'failing' does not necessarily mean 'hopeless'; caries and restorability; the occlusion and vertical dimension; smile line and lip support; the opposing arch; CBCT for bone volume and anatomy; and his medical history, parafunction and smoking status. And critically, his **expectations** — what he thinks 'teeth in a day' involves, and what he understands about maintenance."

Q What does Stage IV Grade B mean here?

"Stage IV: interdental attachment loss with bone loss to the apical third, tooth loss due to periodontitis of five or more teeth, and complexity such as masticatory dysfunction, secondary occlusal trauma, bite collapse or drifting. Grade B: progression at a moderate rate, bone loss to age ratio between 0.25 and 1.0. The Grade B is relatively reassuring — this is not the galloping destruction of Grade C — and it raises a genuine question about whether **some teeth could be saved** rather than clearing the arch."

Domain 2

Q: What are the options?

"**Periodontal treatment with strategic extractions and a partial denture or partial fixed prosthesis** — preserving what can be saved. This must be assessed properly before anything is cleared.
Full clearance and a conventional complete denture, with or without later implant support.
Full clearance with an implant-retained overdenture — fewer implants, removable, cleansable, and it retains the labial flange for lip support.
Full clearance with a fixed full-arch implant prosthesis, immediately loaded or conventionally loaded.
No treatment — not viable given active disease."

Q What would you tell him about the six-week deadline?

"That it cannot drive the clinical decision. 'I understand the holiday matters, and I'll make sure you're not without teeth. But taking out a whole arch and fixing teeth in on the same day is a big, irreversible decision, and it isn't one to make because of a date in the diary. What I can do in six weeks is get you comfortable, stable and looking presentable. What I'd rather not do is clear an arch that might not all need clearing.' "

Q What are the specific concerns about immediate full-arch loading?

"First, whether it is **necessary** — some of his teeth may be salvageable, and a full-arch prosthesis is irreversible. Second, the **loss of the labial flange**: after clearance and alveoloplasty, a fixed prosthesis without a flange may not support the lip, and with a medium or high smile line the transition line may show. This must be tested with a trial denture before he commits. Third, **immediate loading criteria** must be met — adequate primary stability across the arch, no parafunction, and cross-arch splinting; and the ITI evidence grading rates immediate

protocols as less well validated than conventional ones. Fourth, **maintenance**: a fixed full-arch prosthesis is difficult to clean and requires lifelong professional maintenance and periodic removal. Fifth, **scope** — this is Complex in both surgical and prosthodontic dimensions and outside what I would undertake."

Q What is your plan?

"**Phase 1, within six weeks:** full assessment; emergency treatment; extraction of genuinely hopeless teeth with immediate additions to a provisional prosthesis so he is never without teeth; non-surgical periodontal therapy; oral hygiene instruction. He goes on holiday comfortable and presentable.
Phase 2: re-evaluation at three months against the EFP stability criteria — probing depths of 4 mm or less with no bleeding at those sites, and full-mouth bleeding under 10%.
Phase 3: with a stable periodontal condition and a clear picture of which teeth have a good prognosis, present the definitive options properly — including a trial denture to test the flange question.
Phase 4: refer for the surgical phase if he chooses a full-arch implant solution, having done the assessment, the periodontal stabilisation and the prosthodontic planning myself."

Domain 3

Q: If he proceeds with a fixed full-arch prosthesis elsewhere, what is your continuing role?

"Shared care: periodontal maintenance of the opposing arch, peri-implant monitoring at three- to six-monthly intervals, hygiene reinforcement, and prompt referral of complications. I would want the responsibilities documented at the outset, and I would want to know who removes the prosthesis for maintenance and how often."

Q What would you insist on before any clearance?

"A **trial denture set-up** tried in the mouth, photographed and approved by him, testing tooth position, lip support, smile line, speech and the transition line **with and without a flange**. Clearing an arch without that is planning blind, and it is the commonest route to an unrecoverable aesthetic disappointment."

Domain 4

Q: What would you record?

"The full assessment and the individual prognosis of every tooth; the options presented including preserving teeth; the discussion about the deadline and why it should not determine the plan; the trial denture and his approval; the scope discussion and the referral; and what I told him about maintenance and cost. I would give him a written summary."

Q What is the principle underlying this case?

"That **irreversible treatment should never be driven by a timetable**, and that a request for a specific technique is not a diagnosis. The examination's stated aim is 'harmless, independent overall patient care' — and the harm most available in this case is clearing an arch that did not all need clearing."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 853 words**. At 140 words a minute that is **six minutes**, and ten questions at five seconds each adds **fifty seconds**: call it **seven minutes**. It does not overrun; it leaves **eight minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything said out loud, and **each is really three to five short questions**: the concerns about immediate loading are five, and the plan is four phases. If your answers run past **thirty seconds** you will still never reach Domain 4, which here is the record of the deadline conversation and the principle that irreversible treatment is never driven by a timetable.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiographs and charting in front of you	Nothing. Do not talk into this.

Clock	The examiner does	What earns the mark
0:20–1:20	**Lets you present, uninterrupted**	State the request, then the two things the case says are immediately clear: **a 55-year-old man with a failing upper dentition — periodontitis Stage IV Grade B, several mobile teeth, a failing anterior bridge, caries — requesting immediate full-arch fixed rehabilitation, with a holiday in six weeks**. Then: **full-arch fixed rehabilitation is explicitly outside the scope of my practice at this level**, and **the deadline is driving the request rather than the clinical need**. Then the working position, in the first minute: **my job is a proper assessment and an honest account of the options, not to accommodate the timetable — and "failing" does not necessarily mean "hopeless"**.
1:20–4:30	Domain 1: the assessment, and the classification — two printed answers that become nine	*What would you assess?* — eight items the examiner will drill one at a time: **full periodontal charting and radiographs**; **the prognosis of every tooth individually, because "failing" does not mean "hopeless"**; **caries and restorability**; **occlusion and vertical dimension**; **smile line and lip support**; **the opposing arch**; **CBCT for bone volume and anatomy**; **medical history, parafunction and smoking**. Then the one to volunteer as critical: **his expectations — what he thinks "teeth in a day" involves, and what he understands about maintenance**. *Stage IV Grade B?* — the undergraduate groundwork, examined first, so have it exact: **Stage IV — interdental attachment loss with bone loss to the apical third, tooth loss from periodontitis of five or more teeth, and complexity such as masticatory dysfunction, secondary occlusal trauma, bite collapse or drifting; Grade B — moderate progression, bone loss to age ratio 0.25 to 1.0**. *And clinically?* — the inference that scores: **Grade B is relatively reassuring — not the galloping destruction of Grade C — and it raises a genuine question about whether some teeth could be saved rather than clearing the arch**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, the deadline, the five concerns, the plan — four printed answers that become fifteen	*Options?* — five, preservation first: **periodontal treatment with strategic extractions and a partial denture or partial fixed prosthesis, preserving what can be saved — assessed properly before anything is cleared**; **clearance and a conventional complete denture, with or without later implant support**; **clearance with an implant-retained overdenture — fewer implants, removable, cleansable, and it keeps the labial flange**; **clearance with a fixed full-arch implant prosthesis, immediately or conventionally loaded**; **no treatment — not viable given active disease**. *The deadline?* — the principle, then the words: **it cannot drive the clinical decision**; *"I'll make sure you're not without teeth. But taking out a whole arch and fixing teeth in on the same day is a big, irreversible decision, and it isn't one to make because of a date in the diary." The concerns about immediate full-arch loading?* — five answers: **whether it is necessary at all — some teeth may be salvageable and a full-arch prosthesis is irreversible**; **loss of the labial flange — after clearance and alveoloplasty a fixed prosthesis may not support the lip, and the transition line may show, so it is tested with a trial denture first**; **the immediate loading criteria — adequate primary stability across the arch, no parafunction, cross-arch splinting — with the ITI evidence grading rating immediate protocols less well validated than conventional ones**; **maintenance — difficult to clean, needing lifelong professional maintenance and periodic removal**; **scope — Complex in both dimensions and outside what I would undertake**. *Your plan?* — four phases, given as phases: **Phase 1, within six weeks — assessment, emergency treatment, extraction of genuinely hopeless teeth with immediate additions to a provisional so he is never without teeth, non-surgical periodontal therapy, hygiene instruction; he goes on holiday comfortable and presentable**; **Phase 2 — re-evaluation at three months against the EFP stability criteria: probing depths of 4 mm or less with no bleeding at those sites, full-mouth bleeding under 10%**; **Phase 3 — with the periodontium stable and a clear picture of which teeth have a good prognosis, present the definitive options, including a trial denture to test the flange question**; **Phase 4 — refer the surgical phase if he chooses a full-arch implant solution, having done the assessment, stabilisation and planning myself**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: shared care, and the non-negotiable before clearance — two printed answers that become six	*If he proceeds elsewhere, your continuing role?* — **shared care**: **periodontal maintenance of the opposing arch**; **peri-implant monitoring three- to six-monthly**; **hygiene reinforcement**; **prompt referral of complications**. *What must be settled first?* — **the responsibilities documented at the outset**, and the specific question candidates never ask: **who removes the prosthesis for maintenance, and how often**. *What would you insist on before any clearance?* — one answer, said with force: **a trial denture set-up tried in the mouth, photographed and approved by him**, testing **tooth position, lip support, smile line, speech and the transition line — with and without a flange**. *Why?* — **clearing an arch without that is planning blind, and it is the commonest route to an unrecoverable aesthetic disappointment**. Expect the rest of the block to go below the implant material: **how you judge an individual tooth's prognosis, and what alveoloplasty does to the ridge and the lip**.
11:30–14:30	Domain 4: the record and the principle — two printed answers that become six	*What would you record?* — **the assessment and the individual prognosis of every tooth**; **the options, including preserving teeth**; **the deadline discussion and why it should not determine the plan**; **the trial denture and his approval**; **the scope discussion and the referral**; **what you told him about maintenance and cost**; and **a written summary given to him**. *The principle?* — two clauses, both worth saying: **irreversible treatment should never be driven by a timetable**, and **a request for a specific technique is not a diagnosis**. *Tie it to the standard* — **the examination's stated aim is harmless, independent overall patient care, and the harm most available here is clearing an arch that did not all need clearing**.
14:30–15:00	Closes on the patient and the record	What you say: *"Here's what I can promise for the holiday: you won't be without teeth, you'll be comfortable, and you'll look presentable. Here's what I won't do in six weeks — take out a whole arch that might not all need to come out. Your gum disease is progressing at a moderate rate, not a galloping one, and some of these teeth may well be savable. So: settle the infection now, temporary teeth for the trip, and in three months we look properly at what's worth keeping. If you still want the fixed option then, I'll plan it and send you to someone who does that surgery — but you'd be choosing it with the facts, not because of a date in the diary."* What you write: **the individual prognosis of every tooth**; **all options, with preservation recorded first**; **the deadline conversation and your reasoning**; **the trial denture, the flange test and his written approval**; **the immediate-loading concerns, including the evidence grading and the maintenance burden**; **the scope statement and the referral**; **the costs**; and **the written summary handed over**.

The spine — the twelve answers to have ready if nothing else

1 Full-arch fixed rehabilitation is outside my scope at this level · 2 The deadline is driving the request, not the clinical need · 3 "Failing" does not mean "hopeless" — every tooth prognosed individually · 4 Stage IV defined, Grade B as moderate progression, 0.25 to 1.0 · 5 Grade B raises a genuine question about saving teeth · 6 The options, preservation first · 7 The words for the deadline conversation · 8 The five concerns about immediate full-arch loading · 9 Loss of the labial flange and the transition line · 10 EFP criteria at three months: $\leq$4 mm with no bleeding, full-mouth bleeding <10% · 11 A trial denture, photographed and approved, with and without a flange, before clearance · 12 Irreversible treatment is never driven by a timetable.

Where the marks are lost in this case

Accepting the request as the diagnosis. He has researched All-on-4 and asks for it, and the weak answer discusses All-on-4. **A request for a specific technique is not a diagnosis.** What scores is prognosing **every tooth individually** and then saying the inference out loud: **Stage IV Grade B is moderate progression, not the galloping destruction of Grade C, and it raises a genuine question about whether some teeth could be saved**. The harm most available here is **clearing an arch that did not all need clearing** — name it.

Negotiating with the six weeks. Candidates either squeeze a plan into the deadline or refuse outright. Do neither: **the deadline cannot drive the clinical decision, and it does not have to be defied either — Phase 1 delivers emergency treatment, extraction of the genuinely hopeless teeth with immediate additions to a provisional, non-surgical therapy and hygiene instruction**, and he goes away comfortable. Then **re-evaluate at three months against the EFP criteria — 4 mm or less with no bleeding at those sites, full-mouth bleeding under 10%** — before any irreversible decision.

Consenting to a clearance without testing the face. The competence line is in the case and must be stated by you: **a fixed full-arch implant prosthesis is Complex in both dimensions and outside what you would undertake — you do the assessment, the periodontal stabilisation and the prosthodontic planning, and refer the surgical phase**. But the point candidates never volunteer is the one immediately before it: **nothing is cleared until a trial denture set-up has been tried in, photographed and approved by him, testing tooth position, lip support, smile line, speech and the transition line with and without a flange** — because **after clearance and alveoloplasty a fixed prosthesis cannot carry a flange**.

SOE Case 49 — Imminent explantation in the aesthetic zone

THE CASE

47-year-old woman with an implant at the **11**, placed 6 years ago. She presents with **suppuration, 9 mm probing, and radiographic bone loss to the apical third — approximately 70% of the implant length**. There is 3 mm of mid-facial recession with visible threads. She has a **high smile line**. The adjacent 12 and 21 are sound with good bone levels.

Domain 1

"An implant in the most aesthetically critical position with **70% bone loss, suppuration and visible threads**, in a patient with a high smile line. This implant is not retainable — no treatment will produce a maintainable, restorable, aesthetically acceptable result — and the clinical work now is about **how to remove it and what to replace it with**, and about managing a conversation that will be difficult."

Q How do you determine that it is "irrational to keep"?

"The ITI decision tree makes this an explicit decision point. Against retention: bone loss beyond half the implant length; suppuration; exposed threads in the aesthetic zone; and a high smile line, so that even successful resective treatment — which produces recession and exposes more metal — would leave an unacceptable appearance. Reconstructive therapy in a 70% defect is not predictable. Taken together, **removal is the correct recommendation**, and I would state that clearly rather than offer treatment I do not believe will work."

Q What would you establish first?

"Whether the implant is mobile; the exact defect morphology on CBCT; the bone levels and periodontal status of the **12 and 21**, because the interproximal bone on those teeth determines the papillae in whatever replaces the implant and I cannot recreate it; the phenotype; her smile line on video; the cause — cement, position, maintenance, smoking, systemic factors — because whatever I place next will face the same environment; and her expectations."

Domain 2

Q: What are the replacement options after removal?

"**A new implant** after ridge reconstruction. This is the option patients expect, but the site will have a substantial combined horizontal and vertical defect after removal, and **vertical augmentation in the aesthetic zone is the least predictable of all augmentation procedures**. It would mean staged grafting, a year or more, and a real chance of an inferior aesthetic result.
A resin-bonded cantilever bridge from the 21, with soft tissue augmentation under an ovate pontic. Frequently the **most aesthetically reliable outcome** after an anterior implant failure: a pontic site can be sculpted with connective tissue grafting far more predictably than peri-implant tissue can be regenerated, and the papillae are supported by the natural adjacent teeth.
A conventional bridge from the 12 and 21 — destructive to two sound teeth.
A removable prosthesis with a flange — the only option that replaces lost hard and soft tissue volume reliably, and worth mentioning honestly.
No replacement — not realistic in this position."

Q What would you recommend?

"Removal by **counter-torque** to preserve as much bone as possible, thorough debridement, and then — after healing — a **resin-bonded cantilever bridge with soft tissue augmentation under an ovate pontic** as my first recommendation, with a new implant as the alternative if she wants it and accepts the longer, less predictable route. I would be explicit that I am recommending the bridge **not as a compromise but because it is more likely to give her a good result**, and I would show her the reasoning."

Q How do you have this conversation?

> "Sit down, allow time, and lead with the finding rather than the solution. 'I'm sorry — I have some difficult news. The bone around this implant has been lost to about two-thirds of its length and there's an active infection. I don't think it can be saved, and I don't want to put you through treatment that I don't believe will work. It needs to come out.' Then pause, and let her respond, before moving to options. Then: 'The good news is that the teeth on either side are healthy, and there are good ways to replace it. The one I'd recommend might surprise you, because it isn't another implant…' and explain why. I would give written information, allow her time, and offer a second opinion."

Q SAC and referral?

> "Explantation with ridge reconstruction in the aesthetic zone is **Complex** in both dimensions. I would undertake the diagnosis, the options discussion and, if she chooses it, the resin-bonded bridge with soft tissue grafting — the last only if within my experience. For explantation and any implant re-treatment I would refer, having done the assessment and the conversation myself."

Domain 3

Q: Describe the removal and the immediate aftermath.

> "**Counter-torque device** first — least destructive and preserves bone for reconstruction. Escalate to piezo-surgery if needed, and only as a last resort to a trephine. Thorough removal of granulation tissue, irrigation, and assessment of the residual defect. Then either allow **6 to 8 weeks of soft tissue healing** before staged grafting, or graft immediately if the defect is contained. The interim prosthesis matters enormously to her: a **resin-bonded provisional bridge or a bonded pontic** rather than a denture, so nothing presses on the site and so she is never without a tooth in the front of her mouth. I would arrange this **before** the removal appointment, not after."

Q If she chooses a new implant, what would you tell her about the sequence?

> "Removal; 6 to 8 weeks of soft tissue healing; staged horizontal and probably vertical augmentation; 6 to 9 months of graft maturation; implant placement, possibly with further contour augmentation; 3 to 6 months of integration; second stage with a connective tissue graft; 8 to 12 weeks of provisional soft tissue conditioning; then the definitive crown. **Fourteen to twenty months**, with several surgical episodes, and no guarantee that the papillae or the mid-facial margin will match the 21. She should hear that timeline before she chooses."

Domain 4

Q: How would you monitor whichever route she takes?

> "Standardised photographs at every stage; the papilla index and mid-facial margin level against the 21; peri-implant or periodontal parameters as appropriate; and her own satisfaction recorded at intervals. For a bonded bridge, the abutment tooth's vitality and the bond integrity."

Q What is the principle you would take from this case?

> "That **recognising an unsalvageable situation early and saying so plainly is better care than optimistic treatment** — and that after an anterior implant failure, the reflex to place another implant should be questioned. The aim is a result she can live with for twenty years, not a particular technique."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 932 words**. At 140 words a minute that is **six minutes forty**, and ten questions at five seconds each adds **fifty seconds**: call it **seven and a half minutes**. It does not overrun; it leaves **seven and a half minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything said out loud, and **each is really three to six short questions**: the replacement options are five, and the re-implant sequence nine consecutive stages. If your answers run past **thirty seconds** you will still never reach Domain 4 — the papilla index against the 21, and the principle that recognising an unsalvageable situation early is better care than optimistic treatment.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the radiograph and smile photographs in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Give the finding and the verdict together, because hedging here loses the case: **an implant at the 11 placed six years ago, with suppuration, 9 mm probing, bone loss to the apical third — about 70% of the implant length — 3 mm of mid-facial recession with visible threads, and a high smile line**. Then, unqualified: **this implant is not retainable — no treatment will produce a maintainable, restorable, aesthetically acceptable result**. Then: **the work now is how to remove it and what to replace it with, and managing a difficult conversation**. Add the good news you will need in four minutes: **the adjacent 12 and 21 are sound with good bone levels**.
1:20–4:30	Domain 1: why it is irrational to keep, and what you establish first — two printed answers that become nine	*How do you determine it is "irrational to keep"?* — name the framework: **the ITI decision tree makes this an explicit decision point**. Then the four findings against retention: **bone loss beyond half the implant length**; **suppuration**; **exposed threads in the aesthetic zone**; **a high smile line, so even successful resective treatment — which produces recession and exposes more metal — leaves an unacceptable appearance**. *Reconstruction?* — **not predictable in a 70% defect**. *So?* — in these words: **removal is the correct recommendation, and I would state it clearly rather than offer treatment I do not believe will work**. *What would you establish first?* — **whether the implant is mobile**; **the defect morphology on CBCT**; **the bone levels and periodontal status of the 12 and 21** — with the highest-value sentence in the domain: **the interproximal bone on those teeth determines the papillae in whatever replaces the implant, and I cannot recreate it**; **the phenotype**; **her smile line on video**; **the cause — cement, position, maintenance, smoking, systemic factors — because whatever I place next faces the same environment**; **her expectations**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, recommendation, the conversation, SAC — four printed answers that become thirteen	*Replacement options after removal?* — five, each with its reasoning: **a new implant after ridge reconstruction — the option patients expect, but the site will have a combined horizontal and vertical defect, and vertical augmentation in the aesthetic zone is the least predictable of all augmentation procedures: staged grafting, a year or more, and a real chance of an inferior result**; **a resin-bonded cantilever bridge from the 21 with soft tissue augmentation under an ovate pontic — frequently the most aesthetically reliable outcome after an anterior implant failure**, for two reasons — **a pontic site can be sculpted with connective tissue grafting far more predictably than peri-implant tissue can be regenerated**, and **the papillae are supported by the natural adjacent teeth**; **a conventional bridge from the 12 and 21 — destructive to two sound teeth**; **a removable prosthesis with a flange — the only option that reliably replaces lost tissue volume**; **no replacement — not realistic here**. *Your recommendation?* — **removal by counter-torque to preserve bone, thorough debridement, and after healing a resin-bonded cantilever bridge with soft tissue augmentation under an ovate pontic**, with **a new implant as the alternative if she wants it and accepts the longer, less predictable route** — framed as **not a compromise, but because it is more likely to give her a good result**. *The conversation?* — **sit down, allow time, lead with the finding rather than the solution**, then **pause and let her respond before moving to options** — the pause is examinable — then **written information, time, and the offer of a second opinion**. *SAC and referral?* — **explantation with ridge reconstruction in the aesthetic zone is Complex in both dimensions. I would undertake the diagnosis, the options discussion and, if she chooses it, the resin-bonded bridge with soft tissue grafting — the last only if within my experience — and refer the explantation and any implant re-treatment, having done the assessment and the conversation myself.**

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: removal, the interim, and the re-implant sequence — two printed answers that become eleven	*How do you remove it?* — **a counter-torque device first — least destructive, and it preserves bone for reconstruction**; **piezo-surgery if needed**; **a trephine only as a last resort**. *Then?* — **thorough removal of granulation tissue, irrigation, assessment of the residual defect**; then **either 6 to 8 weeks of soft tissue healing before staged grafting, or immediate grafting if the defect is contained**. *The interim prosthesis?* — the answer that shows you are treating a person: **a resin-bonded provisional or bonded pontic rather than a denture, so nothing presses on the site and she is never without a tooth in the front of her mouth — arranged before the removal appointment, not after**. *If she chooses a new implant, the sequence?* — count it off: **removal; 6 to 8 weeks of soft tissue healing; staged horizontal and probably vertical augmentation; 6 to 9 months of graft maturation; placement, possibly with further contour augmentation; 3 to 6 months of integration; second stage with a connective tissue graft; 8 to 12 weeks of provisional conditioning; the definitive crown**. *Total?* — **fourteen to twenty months, several surgical episodes, and no guarantee that the papillae or the mid-facial margin will match the 21 — heard before she chooses, not during**.
11:30–14:30	Domain 4: monitoring and the principle — two printed answers that become six	*How would you monitor whichever route she takes?* — **standardised photographs at every stage**; **the papilla index and mid-facial margin level measured against the 21**; **peri-implant or periodontal parameters as appropriate**; and the one candidates omit — **her own satisfaction, recorded at intervals**. *For a bonded bridge?* — **the abutment tooth's vitality and the bond integrity**. *The principle?* — **recognising an unsalvageable situation early and saying so plainly is better care than optimistic treatment**, and **after an anterior implant failure the reflex to place another implant should be questioned**. *The aim?* — **a result she can live with for twenty years, not a particular technique**.

Clock	The examiner does	What earns the mark
14:30–15:00	Closes on the patient and the record	What you say: *"I'm sorry — this is difficult news. The bone around this implant has gone to about two-thirds of its length and it's infected. It can't be saved, and I don't want to put you through treatment I don't believe will work. It needs to come out. The better news is that the teeth either side are healthy. What I'd recommend might surprise you — it isn't another implant. A bridge bonded to the tooth next door, with the gum contour built up underneath, will very likely look better and take a fraction of the time. Another implant is possible, but that's fourteen to twenty months and several operations."* What you write: **the findings — suppuration, 9 mm probing, 70% bone loss, 3 mm recession with visible threads, high smile line**; **the reasoning that it is irrational to retain, referenced to the decision tree**; **all options with timelines, including no replacement**; **your recommendation and why the bridge is a first choice rather than a compromise**; **the interim arranged before the removal appointment**; **the SAC assessment and what you referred**; **the written information and second opinion offered**; and **her expectations verbatim**.

The spine — the twelve answers to have ready if nothing else

1 Not retainable — say so plainly rather than offer treatment you do not believe in · 2 The four findings against retention, including the high smile line · 3 Resective treatment exposes more metal; reconstruction in a 70% defect is unpredictable · 4 The interproximal bone on the 12 and 21 decides the papillae · 5 Identify the cause, because the replacement faces the same environment · 6 Vertical augmentation in the aesthetic zone is the least predictable procedure there is · 7 The resin-bonded cantilever with an ovate pontic and connective tissue graft, first · 8 Not a compromise — more likely to give a good result · 9 Lead with the finding, then pause · 10 Counter-torque, then piezo, trephine last · 11 The bonded provisional arranged before the removal appointment · 12 Fourteen to twenty months for the implant route.

Where the marks are lost in this case

Offering treatment you do not believe will work. The instinct is to be seen to try — decontamination, resective surgery, a regenerative attempt — and here that is wrong. With **70% bone loss, suppuration, exposed threads and a high smile line**, **resective therapy exposes more metal and reconstruction in a defect that size is not predictable**. The mark is for **stating the recommendation clearly**, and saying it as a diagnosis rather than an apology.

Reflexively replacing an implant with an implant. She will expect it, and the examiner is watching to see whether you question it. Volunteer the reasoning: **after removal the site has a combined horizontal and vertical defect, and vertical augmentation in the aesthetic zone is the least predictable of all augmentation procedures**; whereas **a pontic site can be sculpted with connective tissue grafting far more predictably, and the papillae are supported by her own healthy 12 and 21**. Then make it a recommendation rather than a consolation — **not a compromise, but more likely to give her a good result** — with the honest counterweight of **fourteen to twenty months** for the implant route.

Managing the tissue and forgetting the person. Two details separate a 5 from a 3, and neither is surgical. First, **the interim prosthesis is arranged before the removal appointment, not after** — a bonded provisional or pontic, never a denture pressing on the site, so **she is never without a tooth in the front of her mouth**. Second, the conversation has a shape: **lead with the finding, then pause and let her respond before moving to options**, with **written information and a second opinion offered**. And name the competence line: **explantation with ridge reconstruction in the aesthetic zone is Complex in both dimensions and is referred — you keep the diagnosis, the options discussion, and the bonded bridge if it is within your experience.**

SOE Case 50 — Governance, audit and the quality improvement project

THE CASE

You are asked to discuss your own clinical governance arrangements for implant dentistry, including your quality improvement project, your audit activity, and how you assure the quality of your implant care. (*The Regulations require a completed quality improvement project as an entry condition, so a case of this kind is entirely foreseeable — it must be prepared, not improvised.*)

Domain 1 — Interpretation and reasoning

Q: What does clinical governance mean in the context of implant dentistry?

"A framework through which I am accountable for the quality of the implant care I provide and for continuously improving it. In practice it has several strands: **evidence-based practice**; **audit and quality improvement**; **risk management**, including significant event analysis and duty of candour; **education and training**, with verifiable CPD and reflection, mentoring, and working within competence under GDC Principle 7; **patient involvement**, including consent, information and complaints handling; **information governance**, covering records, confidentiality and data protection; and **staffing and team training**, including medical emergencies and decontamination."

Q How do you know whether your implant treatment is any good?

"By measuring it, not by assuming it. The parameters are **implant survival**; **marginal bone level change** against a baseline taken at prosthesis fitting; **peri-implant health** — probing depths, bleeding, suppuration; **prosthetic complication events**, such as screw loosening, ceramic fracture and component replacement; **early failure rate**; and **patient-reported outcomes**, using a validated measure such as OHIP-14 alongside satisfaction with function and appearance. Without a baseline radiograph and a recall system, none of that can be measured — which is why those two things are foundational."

Domain 2 — Decision making

Q: Describe your quality improvement project.

(*Candidates must give their own. A model structure:*) "**The problem:** I noticed that when patients returned for maintenance, I could not always find a baseline radiograph or a recorded insertion torque in the notes, which made it impossible to apply the 2017 diagnostic criteria for peri-implantitis.
The standard: 100% of implant cases should have a contemporaneous surgical record including system, diameter, length, lot number and insertion torque, and a standardised baseline periapical radiograph taken at prosthesis fitting with a written report and recorded bone level measurements. The standard is drawn from the FGDP training standards, GDC Standards on record keeping, and the 2017 World Workshop case definitions.
The method: retrospective review of the records of 30 consecutive implant cases over 18 months, against a data collection sheet.
The finding: surgical records were complete in 63%; insertion torque was recorded in 40%; a baseline radiograph existed in 77% but bone level measurements were recorded in only 13%.
The change: I introduced a structured implant surgical proforma and a fitting proforma, both integrated into the practice software, with the baseline radiograph and its measurements as mandatory fields; and I briefed the whole team.
The re-audit at 6 months: 28 consecutive cases — surgical record completeness 100%, insertion torque 100%, baseline radiograph with recorded measurements 96%.
The reflection: the omissions were not knowledge failures but system failures, and the fix was structural rather than educational. It also improved my ability to diagnose peri-implant disease, because I now have comparators."

Q Why is re-audit essential?

"Because without it there is data collection but no improvement. **Closing the loop** is what distinguishes audit from a survey, and it is the part most often omitted."

Q Give three other implant audit topics you could undertake.

"**Radiographic justification and reporting** — the proportion of implant-related exposures with a documented justification and a written report of the whole image, against a standard of 100% under IR(ME)R.
Consent documentation — the proportion of implant cases with a contemporaneous note of the consent discussion, not merely a signed form, including the risks material to that patient and the provision of written information with a cooling-off period.
Antibiotic prescribing — the proportion of implant placements with a recorded indication for any antibiotic, against SDCEP and antimicrobial stewardship standards."

Domain 3 — Procedures and systems

Q: What systems assure quality in your surgical practice?

"**Decontamination** compliant with HTM 01-05, with validated cycles, retained records and periodic testing. **Medical emergency** drugs and oxygen checked and in date, with annual whole-team training in medical emergencies and immediate life support. **Radiography quality assurance** — image quality rating, analysis of repeats, equipment testing, and dose audit. **Medical device traceability** — implant, graft and membrane lot numbers recorded and a patient implant card issued. **Torque wrench calibration** at the manufacturer's recommended intervals, with a record. **Surgical safety checks** — patient, site, consent, implant availability including back-up sizes, allergies, and instrument and swab counts. And **structured recall** with risk-based intervals."

Q How do you handle an adverse event?

"**Duty of candour** first: tell the patient promptly, apologise, explain what happened and what I will do, and record it. Notify my **indemnity provider** early. Then **significant event analysis** with the team — what happened, why, what went well, what should change, and who is responsible for implementing it, with a review date. Where a systemic change results, follow it with an **audit** to confirm it has worked."

Domain 4 — Outcomes

Q: How do you keep your knowledge current?

"Verifiable CPD with **written reflection**, which the entry requirements specify — at least 30 hours of implant-related CPD with reflection in the preceding 18 months. Structured reading of the primary literature and of the ITI Treatment Guides and consensus statements, rather than relying on courses alone. A **mentor** for cases at the edge of my experience. Peer review and case discussion. And I would say that the most useful learning comes from **reviewing my own outcomes**, including the failures."

Q How do you decide whether to add a new technique to your practice?

"Against GDC Principle 7.1 — good quality care based on current evidence and authoritative guidance. I would ask: what is the **evidence base**, and is it more than manufacturer-sponsored case series? Is the technique **within my competence** or would it require formal training and mentoring? Is the **manufacturer stable** and will components be available in ten years — the FGDP standards require me to satisfy myself of that. What are the **failure modes**, and can I manage them? And can I **explain it honestly to a patient**, including how well documented it is? If a technique is only 'clinically insufficiently documented' on the ITI grading, I would say so to the patient rather than present it as established."

Q Final question — what does good governance actually achieve for the patient?

> "It converts good intentions into reliable care. The consent discussion happens because the proforma prompts it; the baseline radiograph exists because the system requires it; the peri-implantitis is caught at the mucositis stage because the recall interval is risk-based and probing is routine. **Almost every serious implant complication I have described in this examination is either preventable or detectable early by a system rather than by an individual remembering.** That is what governance is for."

Running this case to time

Time check. This case is written in the older format — **ten long answers, about 972 words**. At 140 words a minute that is **seven minutes**, and ten questions at five seconds each adds **fifty seconds**: call it **just under eight minutes**. It does not overrun; it leaves **seven minutes unaccounted for**. Read that the right way round — the printed answers are far longer than anything said out loud, and **each is really three to seven short questions**: the quality improvement project alone is seven — problem, standard, method, finding, change, re-audit, reflection — asked in exactly that order. If your answers run past **thirty seconds** you will never reach Domain 4. And this is the one case that **must be prepared rather than improvised**: the project is an entry condition, and the examiner knows you have one.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you and confirms that the discussion is about your own governance arrangements	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	This is the one opening with no patient in it, so give a framework and a headline. The definition first: **clinical governance is the framework through which I am accountable for the quality of the implant care I provide and for continuously improving it**. Then the strands, rapidly: **evidence-based practice; audit and quality improvement; risk management, including significant event analysis and duty of candour; education and training, with verifiable CPD, reflection, mentoring and working within competence under GDC Principle 7; patient involvement; information governance; and team training, including medical emergencies and decontamination**. Then your own project in one sentence — **the problem, the standard, the change, and what the re-audit showed** — because everything after this is the examiner asking you to expand it.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: what governance is, and how you know your work is any good — two printed answers that become nine	Expect the strands to be taken one at a time — *what does risk management mean in your practice?* — so have a concrete example for each rather than the noun alone. Then the better question: *How do you know whether your implant treatment is any good?* — the principle first: **by measuring it, not by assuming it**. Then the parameters, one per question: **survival**; **marginal bone level change against a baseline taken at fitting**; **peri-implant health — probing, bleeding, suppuration**; **prosthetic complications — screw loosening, ceramic fracture, component replacement**; **early failure rate**; **patient-reported outcomes, on a validated measure such as OHIP-14**. Then the sentence that ties the domain together: **without a baseline radiograph and a recall system none of that can be measured — which is why those two are foundational**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: the audit cycle, in your own project — three printed answers that become eleven	The heart of the case, examined as a cycle, one stage per question. *The problem?* — **on maintenance visits I could not always find a baseline radiograph or a recorded insertion torque, making it impossible to apply the 2017 diagnostic criteria for peri-implantitis**. *The standard, and its source?* — **100% of cases to have a contemporaneous surgical record — system, diameter, length, lot number, insertion torque — and a standardised baseline periapical at fitting with a written report and recorded bone levels** — drawn from **the FGDP training standards, GDC Standards on record keeping, and the 2017 World Workshop case definitions**. *Method and sample?* — **a retrospective review of 30 consecutive implant cases over 18 months, against a data collection sheet**. *The findings?* — the numbers, exactly: **surgical records complete in 63%; insertion torque in 40%; a baseline radiograph in 77%, but bone level measurements in only 13%**. *The change?* — **a surgical proforma and a fitting proforma in the practice software, with the baseline radiograph and its measurements as mandatory fields, and the team briefed**. *The re-audit?* — **at 6 months, 28 consecutive cases: record completeness 100%, insertion torque 100%, baseline radiograph with measurements 96%**. *The reflection?* — the insight that lifts the mark: **the omissions were not knowledge failures but system failures, and the fix was structural rather than educational** — with the dividend, **I can now diagnose peri-implant disease against comparators**. *Why is re-audit essential?* — **without it there is data collection but no improvement; closing the loop is what distinguishes audit from a survey, and it is the part most often omitted**. *Three other topics?* — each with its standard: **radiographic justification and reporting, against 100% under IR(ME)R; consent documentation — a contemporaneous note of the discussion, not merely a signed form, with the risks material to that patient; antibiotic prescribing — a recorded indication, against SDCEP and stewardship standards**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the systems, and the adverse event — two printed answers that become eleven	*What systems assure quality in your surgical practice?* — seven, each with its standard or its record: **decontamination compliant with HTM 01-05, validated cycles, retained records, periodic testing**; **emergency drugs and oxygen checked and in date, with annual whole-team emergency and immediate life support training**; **radiography quality assurance — image quality rating, analysis of repeats, equipment testing, dose audit**; **device traceability — implant, graft and membrane lot numbers recorded, a patient implant card issued**; **torque wrench calibration at the manufacturer's intervals, with a record**; **surgical safety checks — patient, site, consent, implant availability including back-up sizes, allergies, instrument and swab counts**; **structured recall at risk-based intervals**. *How do you handle an adverse event?* — in sequence, because the sequence is the answer: **duty of candour first — tell the patient promptly, apologise, explain, record it**; **notify my indemnity provider early**; **significant event analysis with the team — what happened, why, what went well, what should change, who implements it, a review date**; and **where a systemic change results, audit it to confirm it has worked** — the same loop, closed in a different context.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: currency, new techniques, and what it is all for — three printed answers that become nine	*How do you keep your knowledge current?* — **verifiable CPD with written reflection, as the entry requirements specify — at least 30 hours of implant-related CPD with reflection in the preceding 18 months**; **structured reading of the primary literature and the ITI Treatment Guides, not courses alone**; **a mentor for cases at the edge of my experience**; **peer review**; and the honest one — **the most useful learning comes from reviewing my own outcomes, including the failures**. *How do you decide whether to add a new technique?* — anchor it in **GDC Principle 7.1**, then five questions of the technique: **is the evidence more than manufacturer-sponsored case series?**; **is it within my competence, or does it need formal training and mentoring?**; **is the manufacturer stable, and will components be available in ten years — the FGDP standards require me to satisfy myself of that**; **what are the failure modes, and can I manage them?**; **can I explain it honestly, including how well documented it is — saying so if it is only "clinically insufficiently documented" on the ITI grading, rather than presenting it as established?** *What does good governance achieve for the patient?* — **it converts good intentions into reliable care: the consent discussion happens because the proforma prompts it, the baseline radiograph exists because the system requires it, the peri-implantitis is caught at the mucositis stage because the recall is risk-based** — then finish on **almost every serious implant complication is either preventable or detectable early by a system rather than by an individual remembering**.
14:30–15:00	Closes on the patient and on what changed as a result	The patient is not in the room here, so put her there. What you say to the patient the audit changed: *"When we fit your crown I'll take a radiograph and measure the bone level, and that goes in your notes as the starting point. Every time you come back I'll measure it again against that. It means if anything starts to go wrong around the implant, I'll see it while it's still gum inflammation and easy to treat, rather than years later when there's bone missing."* What you write, and what changed as a result: **the completed audit cycle with its dates, standard, sample, findings, change and re-audit figures**; **the proformas now mandatory in the software**; **the team briefing and who owns the change**; **the re-audit date already set**; and — the point of the exercise — **that the 13% of records carrying a measurable baseline is now 96%, so peri-implant disease here is diagnosed against a comparator rather than guessed at**.

The spine — the twelve answers to have ready if nothing else

1 Governance defined, and its strands named · 2 Measure it, don't assume it: survival, bone level change, peri-implant health, prosthetic complications, early failures, PROMs · 3 A baseline radiograph and a recall system are foundational · 4 Your project's problem, in one sentence · 5 The standard, and its three sources

· 6 The sample — 30 consecutive cases over 18 months · 7 The findings, with the actual percentages · 8 The change — proformas with mandatory fields, and the team briefed · 9 The re-audit figures, and why closing the loop is the point · 10 Three further audit topics, each with its standard · 11 The systems: HTM 01-05, emergencies, radiography QA, traceability, torque calibration, safety checks, risk-based recall · 12 Candour, indemnity, significant event analysis, then audit the change.

Where the marks are lost in this case

Improvising a project you were required to have finished. This case is entirely foreseeable — **a completed quality improvement project is an entry condition** — and it is asked for by its stages. A candidate who describes a vague "look at my records" exercise, or who cannot give **the standard, the sample size, the percentages before and after, and the re-audit date**, has failed the domain in the one case where preparation is guaranteed to pay. Know your own numbers, and know where your standard came from — **FGDP training standards, GDC Standards on record keeping, the 2017 World Workshop case definitions** — because "best practice" is not a standard.

Stopping before the loop closes. The commonest omission is the second half: **without re-audit there is data collection but no improvement — closing the loop is what distinguishes audit from a survey**. Do not stop at the numbers either: volunteer the reflection that shows you understood your own result — **the omissions were not knowledge failures but system failures, and the fix was structural rather than educational** — and apply the same reflex to adverse events, where **a systemic change is followed by an audit to confirm it has worked**.

Treating governance as paperwork rather than as competence. The competence line here is not a referral threshold but **GDC Principle 7 — working within your competence** — drawn explicitly when a new technique is offered: **is the evidence more than manufacturer-sponsored case series; is it within my competence or does it need formal training and mentoring; will the components still exist in ten years; can I manage its failure modes; and can I say honestly that a technique is only "clinically insufficiently documented" on the ITI grading?** Then close where the case closes — on the patient: **almost every serious implant complication is either preventable or detectable early by a system rather than by an individual remembering**, and the audit's real product is **a measurable baseline in 96% of records where it existed in 13%**.

SOE Case 51 — The lawyer: replacement of the 21, six weeks after extraction

THE CASE

44-year-old **solicitor**. The **21** was extracted **six weeks ago** because of a failed root canal treatment with a persistent apical infection; it had been root-treated twice and had a post crown. He is currently wearing a **spoon denture**, which he dislikes and says "clicks when I speak in court". Medical history clear; non-smoker; regular attender; good oral hygiene. Medium smile line. The adjacent 11 and 22 are unrestored and vital. The ridge is healing but there is a visible buccal concavity. He wants "the best long-term solution" and asks specifically what the risks are and what happens if it fails.

Domain 1 — Clinical interpretation and reasoning

Q Summarise this case.

"A 44-year-old solicitor, six weeks after extraction of an upper central incisor for failed endodontic treatment, wearing an interim spoon denture he finds unsatisfactory, with unrestored adjacent teeth, a medium smile line and a buccal concavity in a healing ridge. He is articulate, engaged and asking directly about risk and about failure. Three things stand out: the **timing**, which places him exactly at the Type 2 early placement window; the **spoon denture**, which may be loading the healing site; and his **occupation**, which tells me that the consent discussion and my records need to be exemplary."

Q Why does his occupation matter clinically?

"Two reasons, and both are legitimate clinical reasons rather than defensive ones. First, under **Montgomery**, a risk is material if a reasonable person in *this* patient's position would attach significance to it, or if I should be aware that this particular patient would. A solicitor who speaks professionally will attach significance to **anything affecting speech and appearance in public** — so a visible interim prosthesis that clicks, the aesthetic outcome, and the treatment duration are all material to him in a way they might not be to someone else. Second, he is asking precise questions about risk and failure, which tells me he wants to make an informed decision. That is not a threat — it is exactly the patient I should find easiest to consent properly. I would answer fully, give written information, allow a cooling-off period, and **record the discussion contemporaneously in detail** — not because he is a lawyer, but because that is the standard for everyone and he has made it easy to demonstrate."

Q What do you make of the spoon denture?

"It is doing three things I would want to change. It is **unretentive** — that is inherent to the design, since it relies on the palatal vault with no clasps — which explains the clicking in speech and is why he dislikes it. It has **palatal coverage**, which affects speech and taste. And most importantly, **it may be pressing on the healing ridge**. Pressure on a healing extraction site or a graft causes soft tissue trauma and, if a graft is present, can cause dehiscence. I would check it with **pressure-indicating paste**, relieve it generously, and consider replacing it. My preferred interim option here is a **resin-bonded cantilever bridge or a bonded pontic**, because it is fixed, it does not click, the aesthetics are far better, and — decisively — **it applies no load to the healing ridge**. An Essix retainer with a pontic is the quick alternative, but he cannot wear it while eating and it would not solve the speech problem."

Q What else do you need to know and examine?

"**History:** why the endodontic treatment failed — a vertical root fracture, a missed canal, a perforation, or persistent apical periodontitis; whether the extraction was atraumatic and whether ridge preservation was performed; whether there was a flap; and whether he had any post-operative infection.
Examination: smile line at full smile and in speech, since he speaks professionally; lip dynamics; the **soft tissue phenotype**; papilla heights and the **interproximal bone levels on the 11 and 22**, which determine the papillae and which I cannot recreate; the ridge width and the extent of the buccal concavity; keratinised tissue; the shade, form and translucency of the 11 and 22; the occlusion, particularly overbite, overjet and protrusive contacts; and sensibility testing of the adjacent teeth, since they were adjacent to a chronic apical infection.
Investigations: standardised photographs; mounted study casts and a **diagnostic wax-up**; a periapical of the site and the adjacent teeth; and a **small field-of-view CBCT**, justified because I need the ridge width, the extent

of the buccal concavity and the position of the incisive canal, and those will determine whether augmentation is needed and whether the case is within my competence.
Risk assessment: I would complete an **ITI Aesthetic Risk Assessment** and share it with him."

Domain 2 — Clinical decision making and treatment planning

Q What are the options?

"I would give him all of them, including no treatment.
No treatment / continue with the removable prosthesis — he has rejected this in practice, but it must be offered.
A new, better interim removable prosthesis — an Essix or an improved acrylic partial. Interim only.
Resin-bonded cantilever bridge from the 11 — minimally invasive, no surgery, excellent published survival for a single anterior unit, and **it preserves every future option**. Its weakness here is that a pontic over a concave ridge may look poor unless the ridge contour is augmented; a connective tissue graft under an ovate pontic addresses that and is more predictable than augmenting around an implant.
Conventional bridge from the 11 and 22 — I would advise against it: both are unrestored and vital, and preparing two sound anterior teeth in a 44-year-old is a substantial and irreversible biological cost.
Implant-retained crown — preserves the adjacent teeth, best long-term function, and the option he is most likely expecting. It requires management of the buccal concavity.
Orthodontic space closure — not realistic for a central incisor."

Q What would you recommend, and what is the timing?

"**An implant, provided the site allows the implant to be placed in the correct prosthetic position** — with the resin-bonded bridge presented as a genuine alternative rather than a consolation. On timing, six weeks post-extraction places him in the **Type 2 window — early placement with soft tissue healing, four to eight weeks**. That is often the ideal compromise in the aesthetic zone: the soft tissue has healed and can be advanced for tension-free closure, any residual infection has resolved, but the major phase of ridge resorption — about two-thirds of the horizontal loss occurs in the first three months — has not yet completed. The decision at surgery is whether I can achieve the **correct three-dimensional position with at least 2 mm of buccal bone**, using **simultaneous contour guided bone regeneration**. If I can, I place now and graft simultaneously, submerged. If the concavity means I cannot achieve position or primary stability, I **stop and stage** — graft first, place at six to nine months. Placing an implant where the bone allows rather than where the crown needs it is the commonest cause of an unrecoverable aesthetic failure, and I would rather abandon and stage than compromise the position. I would consent him for both possibilities **in advance**, so that a staged approach is a planned contingency rather than a disappointment."

Q SAC classification?

"**Surgically Advanced**, potentially Complex: aesthetic zone, a buccal concavity requiring simultaneous or staged augmentation, and a post-infection site. **Prosthodontically Advanced**: a single central incisor between two unrestored, vital adjacent teeth, in a patient with high expectations and a medium smile line. The modifying factors that raise it are his aesthetic expectations and the bone deficiency. If his smile line proved high, or if a vertical component to the defect emerged, it would move to **Complex** and I would refer or co-manage — and I would say so to him at the outset rather than discovering it mid-treatment."

Q He asks you directly: "What are the risks, and what happens if it fails?" Answer him.

" 'Let me take those separately, and I'll give you all of this in writing as well.

The risks during and just after surgery are pain, swelling, bruising, bleeding and infection — all common and manageable. Because this is your upper jaw, there is no significant nerve risk. There is a small risk of damage to the roots of the teeth either side, which I minimise by planning from a scan and using a guide.

The risk that it does not attach to the bone is around two to five in a hundred. That usually shows up in the first few weeks.

The risks specific to the front of the mouth are the ones I would want you to weigh most carefully. The gum can shrink back over the years, which can show as a longer tooth or a grey line at the margin. There may be a small dark triangle between the new tooth and its neighbour — that depends on the bone level on your own teeth either side, which I can measure but cannot change. And the appearance can drift over ten or twenty years as your gums and face change, even if it looks perfect on the day.

There is also a risk in the other direction: that when I open the site, the bone at the front is too thin to put the implant exactly where the tooth needs to be. If that happens I will stop, rebuild the bone, and place the implant six to nine months later. I would rather do that than put it in the wrong place.

If it fails — and I mean if the implant itself is lost — I would remove it, let the site heal for two to three months, rebuild the bone if needed, and we would decide together whether to try again or to use a bonded bridge instead. I have to be honest that a second implant in the same site has a somewhat higher risk of failing than the first. Here is what that would cost and what is included in my original fee.

And the alternative that carries none of the surgical risk is a bonded bridge from the tooth next door, which in a site like yours can look extremely good, particularly if we build the gum contour underneath it. It is not a second-best option; it is a different set of trade-offs.'

Then I would give him written information, ask him to take it away, and **ask him to tell me back in his own words what we have agreed and what could go wrong**, so I can be sure I have explained it clearly. And I would record all of that."

Domain 3 — Decision making in relation to treatment procedures

Q Take me through the surgery.

"**Restoratively driven throughout.** I start from the diagnostic wax-up of the 21, duplicated into a surgical guide, and I would use a CBCT-planned tooth-supported guide.

Flap: a **palatally displaced crestal incision** over the edentulous site, with **papilla-sparing** intrasulcular extensions to the 11 and 22 to protect the papillae, and a single **vertical relieving incision at the mesial line angle of the 23**, away from where the graft and membrane will sit. Full-thickness, **atraumatic elevation** with a sharp periosteal elevator working against bone, retracting on bone and never crushing the flap margin, keeping the flap moist.

Position: the shoulder about **3 mm apical to the planned mid-facial gingival margin**, **1 to 2 mm palatal** to the point of emergence of the adjacent teeth, at least **1.5 mm from each adjacent root**, angled so the screw channel emerges through the cingulum.

Preparation: graded drilling with copious sterile saline, intermittent pressure, sharp drills, direction indicator checked mesiodistally and buccolingually. Bone here is likely **D3**, so I would under-prepare to raise primary stability. I would aim for **25 to 45 Ncm**.

Augmentation: I expect a buccal dehiscence over the concavity. I would **decorticate** the buccal plate, place **autogenous chips collected from the osteotomy directly against the implant surface**, layer a **deproteinised bovine bone mineral** over it for volume stability, and cover with a **native collagen membrane** extending 2 to 3 mm onto sound bone.

Closure: a **periosteal releasing incision** for tension-free primary closure, tested by checking the flap passively reaches beyond the closure line. **Two-stage submerged** healing because of the membrane. **5/0 or 6/0 monofilament** — horizontal mattress plus interrupted — removed at 10 to 14 days.

Interim prosthesis: a **bonded pontic**, not the spoon denture, so nothing presses on the graft."

Q And the restorative phase?

"Second stage at four to six months with a **palatally displaced or roll flap** to preserve and reposition keratinised tissue, and a **connective tissue graft** at that point if the phenotype is thin. Then a **screw-retained provisional** to develop the emergence profile over 8 to 12 weeks, adjusted incrementally, with a **concave subgingival contour**. Then a **customised impression coping** copying the provisional's subgingival form, an implant-level impression, and a definitive **screw-retained restoration on a titanium base with a milled ceramic superstructure**. Shade taken early with photographs and a stump shade, ideally with the technician present. Occlusion: light contact in intercuspation, **no contact in protrusion or excursion** — the canines guide."

Domain 4 — Interpretation and problem solving of clinical outcomes

Q How will you know whether it has succeeded?

"Against parameters, and against what he told me at the start. **Standardised photographs** at every stage, compared with the contralateral 11; the **mid-facial mucosal margin level** and **papilla index** measured against the 11 and 22; peri-implant probing, bleeding and suppuration; **marginal bone level against the baseline periapical taken at fitting**; and his own rating of the appearance and of speech. Reviews at 1 week, 2 weeks, second stage, provisional adjustments, fitting, 3 months, 12 months, then risk-based maintenance."

Q At two years he has 1 mm of mid-facial recession. What do you do?

> "Diagnose the cause first — is there inflammation, is the phenotype thin, is the implant too buccal, has the abutment been repeatedly disconnected? Then the options are: treat any inflammation; **change the abutment** to a ceramic or Ti-base-plus-ceramic to remove any greyness; a **connective tissue graft** to thicken the tissue, which is more reliable at arresting further recession than at achieving coverage; prosthetic recontouring with cervical pink ceramic; or, if the implant is fundamentally malpositioned, explantation and reconstruction. I would be honest that soft tissue procedures around an established implant are less predictable than around teeth, and I would use the photographs from the start to discuss it objectively. **And I would have warned him at consent that this could happen**, which is why the consent record matters."

Q What would you record throughout?

> "Everything, contemporaneously: his expectations in his own words; the options presented including no treatment and the bonded bridge; the risks discussed, **specifically identifying the ones material to him as someone who speaks professionally**; the written information given and the cooling-off period; his questions and my answers; the aesthetic risk assessment; the surgical record with implant system, diameter, length, lot number and insertion torque; the graft and membrane with lot numbers; the abutment, torque value and occlusal scheme; the baseline radiograph and its measurements; and the maintenance plan, with his implant record card. If a complication arises, I would apply **duty of candour** — tell him promptly, apologise, explain, and put right what I can."

Running this case to time

Time check. This is the outlier of the set — **thirteen questions, about 2,298 words**, roughly two and a half times the length of its neighbours. At 140 words a minute that is **sixteen and a half minutes**, and thirteen questions at five seconds each adds **just over a minute**: call it **seventeen and a half minutes**. Read out in full it **overruns by about two and a half minutes** — the only case in this group that does. But it is not too long: it is **already close to rapid-fire**, and **the printed answers still exceed what would be said aloud**. What an examiner drops is predictable — **the suturing detail, the risk answer delivered as a monologue, and the second half of the restorative sequence**. What an examiner never drops is **the timing decision, the stop-and-stage contingency, and the consent record**. If your answers run past **thirty seconds** you will not reach Domain 4 — where the consent conversation from Domain 2 either saves you or does not.

The fifteen-minute run sheet

Clock	The examiner does	What earns the mark
0:00–0:20	Settles you, confirms the case, puts the photographs and periapical in front of you	Nothing. Do not talk into this.
0:20–1:20	**Lets you present, uninterrupted**	Compress the case, then name the three things that stand out: **a 44-year-old solicitor, six weeks after extraction of the 21 for failed endodontic treatment, in an interim spoon denture he dislikes, with unrestored vital 11 and 22, a medium smile line, and a visible buccal concavity in a healing ridge**. Then: **the timing, which places him exactly at the Type 2 early placement window; the spoon denture, which may be loading the healing site; and his occupation, which tells me the consent discussion and my records need to be exemplary**. Then the working position: **an implant if the site allows the correct prosthetic position, with a resin-bonded cantilever bridge presented as a genuine alternative rather than a consolation**.

Clock	The examiner does	What earns the mark
1:20–4:30	Domain 1: the occupation, the spoon denture, the assessment — four printed answers that become twelve	*Why does his occupation matter clinically?* — the framing must be right in the first clause: **two reasons, both clinical rather than defensive**. **Montgomery — a risk is material if a reasonable person in this patient's position would attach significance to it, and a solicitor who speaks professionally attaches significance to anything affecting speech and appearance in public**, so **the clicking interim, the aesthetic outcome and the treatment duration are all material to him**. And **he asks precise questions about risk, which makes him the patient it is easiest to consent properly**. *The record?* — **exemplary not because he is a lawyer, but because that is the standard for everyone**. *The spoon denture?* — **unretentive by design, relying on the palatal vault with no clasps, which explains the clicking; palatal coverage affecting speech and taste**; and, most importantly, **it may be pressing on the healing ridge, where pressure causes trauma and, over a graft, dehiscence** — so **pressure-indicating paste, generous relief, and preferably a bonded pontic instead**. *What else do you need?* — **why the endodontic treatment failed; whether the extraction was atraumatic and whether ridge preservation was done**; then **smile line in speech, phenotype, the interproximal bone levels on the 11 and 22 — which determine the papillae and which I cannot recreate — ridge width and the concavity, the shade and form of the adjacent teeth, the occlusion, and sensibility testing of teeth that sat beside a chronic apical infection**; then **photographs, mounted casts and a wax-up, a periapical, and a small field-of-view CBCT justified by ridge width, the concavity and the incisive canal** — plus **an ITI Aesthetic Risk Assessment shared with him**.

Clock	The examiner does	What earns the mark
4:30–7:30	Domain 2: options, timing, SAC, and his two questions — four printed answers that become fourteen	*Options?* — six, including no treatment: **continuing with the removable prosthesis, offered even though rejected in practice**; **a better interim prosthesis**; **a resin-bonded cantilever from the 11 — minimally invasive, no surgery, excellent published survival for a single anterior unit, and it preserves every future option**, with the honest weakness that **a pontic over a concave ridge looks poor unless the contour is augmented — and a connective tissue graft under an ovate pontic is more predictable than augmenting around an implant**; **a conventional bridge from two unrestored vital teeth, advised against**; **an implant-retained crown, requiring management of the concavity**; **orthodontic space closure — not realistic for a central incisor**. *Timing?* — **six weeks is the Type 2 window, early placement with soft tissue healing at four to eight weeks**, ideal because **the soft tissue has healed and can be advanced for tension-free closure, residual infection has resolved, and the major phase of resorption — two-thirds of the horizontal loss in the first three months — has not completed**. *The decision at surgery?* — the sentence the case turns on: **whether I can achieve the correct three-dimensional position with at least 2 mm of buccal bone using simultaneous contour guided bone regeneration; if I can, I place now and graft simultaneously, submerged; if not, I stop and stage — graft first, place at six to nine months** — because **placing an implant where the bone allows rather than where the crown needs it is the commonest cause of unrecoverable aesthetic failure** — and **he is consented for both possibilities in advance**. *SAC?* — **surgically Advanced, potentially Complex; prosthodontically Advanced — and if the smile line proved high or a vertical defect emerged it becomes Complex and I would refer or co-manage, said at the outset**. *"What are the risks, and what happens if it fails?"* — not a speech; five answers: **surgical — pain, swelling, bruising, bleeding, infection; no significant nerve risk in the upper jaw; a small risk to the adjacent roots, minimised by scan-based planning and a guide**; **failure to integrate — two to five in a hundred, usually in the first few weeks**; **the aesthetic-zone risks, the ones to weigh most carefully — recession showing as a longer tooth or a grey margin, a dark triangle governed by the bone level on his own teeth, and drift over decades**; **the risk in the other direction — stopping and staging if the buccal bone is too thin**; **and if it fails — remove, heal, rebuild if needed, and decide together between a second implant and a bonded bridge, with the honest statement that a second implant in the same site carries higher risk, and what it would cost**. *Then?* — **written information, a cooling-off period, and asking him to tell me back in his own words what we have agreed and what could go wrong**.

Clock	The examiner does	What earns the mark
7:30–11:30	Domain 3: the surgery and the restorative phase — two printed answers that become twelve	*Take me through the surgery.* — the governing principle first: **restoratively driven throughout, from the wax-up of the 21 duplicated into a CBCT-planned tooth-supported guide**. *Flap?* — **a palatally displaced crestal incision, papilla-sparing intrasulcular extensions to the 11 and 22, one vertical relieving incision at the mesial line angle of the 23; full-thickness, atraumatic elevation against bone, never crushing the flap margin**. *Position?* — the four numbers: **shoulder 3 mm apical to the planned mid-facial margin; 1 to 2 mm palatal; at least 1.5 mm from each adjacent root; angled so the screw channel emerges through the cingulum**. *Preparation?* — **graded drilling with copious saline, sharp drills, direction indicator checked in both planes; likely D3, so under-prepare; aim for 25 to 45 Ncm**. *Augmentation?* — **expect a buccal dehiscence: decorticate; autogenous chips from the osteotomy against the implant surface; deproteinised bovine bone mineral over it for volume stability; a native collagen membrane 2 to 3 mm onto sound bone**. *Closure?* — **a periosteal releasing incision for tension-free primary closure, two-stage submerged healing, sutures out at 10 to 14 days; interim, a bonded pontic, not the spoon denture**. *The restorative phase?* — **second stage at four to six months with a palatally displaced or roll flap, plus a connective tissue graft if the phenotype is thin; a screw-retained provisional developing the emergence over 8 to 12 weeks with a concave subgingival contour; a customised impression coping copying that form, then a definitive screw-retained restoration on a titanium base; occlusion light in intercuspation, no contact in protrusion or excursion**.

Clock	The examiner does	What earns the mark
11:30–14:30	Domain 4: outcomes, the recession at two years, the record — three printed answers that become nine	*How will you know whether it has succeeded?* — **against parameters, and against what he told me at the start**: **standardised photographs compared with the contralateral 11**; **the mid-facial mucosal margin level and papilla index measured against the 11 and 22**; **peri-implant probing, bleeding and suppuration**; **marginal bone level against the baseline periapical taken at fitting**; **his own rating of appearance and speech**. *Reviews?* — **1 week, 2 weeks, second stage, provisional adjustments, fitting, 3 months, 12 months, then risk-based maintenance**. *At two years, 1 mm of mid-facial recession* — **diagnose the cause first: inflammation, a thin phenotype, an implant too buccal, or repeated abutment disconnection**. *Then?* — **treat any inflammation; change the abutment to ceramic or a Ti-base with ceramic to remove greyness; a connective tissue graft, more reliable at arresting recession than achieving coverage; recontouring with cervical pink ceramic; or, if the implant is fundamentally malpositioned, explantation and reconstruction**. *The honesty?* — **soft tissue procedures around an established implant are less predictable than around teeth**, discussed **against the photographs taken from the start** — and the line that closes the loop with Domain 2: **I would have warned him at consent that this could happen, which is why the consent record matters**. *What would you record?* — **everything, contemporaneously** — the list is in the closing row — with **duty of candour applied promptly if a complication arises**.
14:30–15:00	Closes on the patient and the record	What you say: *"Two things before you decide. The timing is good — six weeks is the window where the gum has healed but the bone hasn't finished shrinking. And hear this now rather than on the day: if I open the site and the bone at the front is too thin to put the implant exactly where the tooth needs to be, I will stop, rebuild the bone, and place it six to nine months later. That isn't a complication, it's a plan. Meanwhile I'd like to take that spoon denture out of service and bond a temporary tooth in instead. Take the written information away, and when you come back, tell me in your own words what we've agreed and what could go wrong."* What you write: **his expectations verbatim; all options including no treatment; the risks material to a man who speaks professionally; the stop-and-stage contingency consented in advance; the cooling-off period and the aesthetic risk assessment; the full surgical and prosthetic record with lot numbers and torque; the baseline radiograph with measurements; and the implant card**.

The spine — the twelve answers to have ready if nothing else

1 Six weeks is the Type 2 window — early placement with soft tissue healing · 2 Two-thirds of horizontal ridge loss occurs in the first three months · 3 Montgomery, and why speech and appearance are material to him specifically · 4 The spoon denture may be loading the healing ridge — paste, relieve, replace with a

bonded pontic · 5 The interproximal bone on the 11 and 22 decides the papillae · 6 CBCT justified by ridge width, the concavity and the incisive canal · 7 The resin-bonded cantilever as a genuine alternative, not a consolation · 8 The correct three-dimensional position with 2 mm of buccal bone, or stop and stage · 9 Consent for both possibilities in advance · 10 Shoulder 3 mm apical, 1 to 2 mm palatal, 1.5 mm from each root, cingulum emergence · 11 Autogenous chips against the implant, xenograft over, collagen membrane, tension-free closure · 12 The 1 mm recession at two years — diagnose the cause, then the options, on the consent record.

Where the marks are lost in this case

Getting the reason for the care wrong. Candidates hear "solicitor" and start practising defensive dentistry out loud. The case gives the correct framing and it must be yours: **his occupation matters for two legitimate clinical reasons — under Montgomery a risk is material if this particular patient would attach significance to it, and a man who speaks professionally attaches significance to anything affecting speech and appearance** — and **he asks precise questions about risk and failure, which makes him the patient it is easiest to consent properly**. Then the sentence that settles it: **the records need to be exemplary not because he is a lawyer, but because that is the standard for everyone**.

Promising an implant before the flap is raised. The commonest unrecoverable aesthetic failure here is **placing the implant where the bone allows rather than where the crown needs it**. The examinable answer is conditional, and volunteered before the surgery is described: **I place at six weeks only if I can achieve the correct three-dimensional position with at least 2 mm of buccal bone using simultaneous contour guided bone regeneration; if the concavity prevents that, I stop and stage — graft first, place at six to nine months** — with **both possibilities consented in advance**. Note the competence line the case draws: **Advanced, potentially Complex — and if the smile line proved high or a vertical defect emerged, it becomes Complex and is referred or co-managed, said at the outset**.

Leaving the spoon denture in the mouth. It is the detail candidates skip and it costs twice: the denture **clicks because it is inherently unretentive with no clasps**, and **it may be pressing on the healing ridge** — so the answer is **pressure-indicating paste, generous relief, and replacement with a bonded pontic or resin-bonded cantilever that loads the site not at all**. And do not let the surgical detail crowd out the last domain, where the case actually closes: **the mid-facial margin and papilla index measured against the 11 and 22, the baseline radiograph and its measurements, and the consent record that lets you discuss 1 mm of recession at two years as a warned-of outcome rather than a surprise**.